STATISTICAL AND COMPUTATIONAL EFFICIENCY IN SEQUENTIAL DECISION MAKING

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Supervisor of Dissertation	Co-Supervisor of Dissertation
Giles Hooker	Yuting Wei
Professor of Statistics and Data Science	Associate Professor of Statistics and Data Science

Graduate Group Chairperson

Nancy Zhang, Ge Li and Ning Zhao Professor, Professor of Statistics and Data Science

Dissertation Committee

Enric Boix-Adserà, Assistant Professor of Statistics and Data Science

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[RESOLVE: REPLACE PLACEHOLDER DEDICATION WITH YOUR PERSONAL DEDICATION.] *To my family, mentors, and collaborators.*

ACKNOWLEDGMENT

[RESOLVE: REPLACE PLACEHOLDER ACKNOWLEDGMENT AND VERIFY COMMITTEE CHAIR / MEMBERS ON TITLE PAGE (GILES HOOKER, YUTING WEI, ENRIC BOIX-ADSERA, GRADUATE GROUP CHAIR NANCY ZHANG).]

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ABSTRACT

STATISTICAL AND COMPUTATIONAL EFFICIENCY IN SEQUENTIAL DECISION MAKING

Kevin Tan

Giles Hooker

Yuting Wei

[RESOLVE: REVIEW ABSTRACT.] Sequential decision making and dynamical systems inference must balance the statistical cost of collecting informative observations against the computational cost of representing, updating, and optimizing models, beliefs, and policies. Treating either budget in isolation masks the dominant error source: a statistically optimal method that maintains an implicitly growing policy class or searches an exponential belief tree may be intractable, whereas a convenient approximation in policy parameterization, ensembles, or particle filtering introduces structural bias and optimization variance that alter the statistical guarantee. This dissertation studies how information representation governs statistical and computational efficiency across seven research chapters. Chapter 2 shows how hybrid reinforcement learning directs online exploration toward under-covered state–action regions and breaks purely offline and online sample-size barriers in linear Markov decision processes without global coverage. Chapter 3 recovers shared low-dimensional structure from short heterogeneous trajectories in linear latent contextual bandits via a moment-corrected spectral estimator. Chapter 4 resolves policy-class covering growth in actor–critic methods via rare-switching optimistic critics with uniform actor resets, and achieves constant-time policy evaluation via logit matching on a G-optimal coreset with Langevin Monte Carlo exploration. Chapter 5 analyzes continuous-action diffusion policies under denoising policy gradients, reparameterized natural policy gradients, and optimistic critic cloning. Chapter 6 establishes central limit theorems and valid inference for boosted regression and explainable boosting machines. Chapters 7 and 8 address partial observability via weight-discounted differentiable particle filtering, warm-started optimization, and finite-depth belief-tree control.

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CHAPTER 1

INTRODUCTION

[RESOLVE: REVIEW NARRATIVE ARC AND CHAPTER TRANSITIONS, ESPECIALLY PLACEMENT OF CHAPTER 6 (BOOSTING INFERENCE).]

1.1. Statistical and Computational Costs in Sequential Learning

Many problems in science and engineering require learning from observations that arrive sequentially over time. A robotic controller selects motor commands and observes the resulting motion of a physical system. A recommendation platform observes how users respond to suggested items and adapts future recommendations as more responses arrive. An epidemiological model assimilates weekly reported case counts to infer the unobserved spread of an infectious disease and evaluate candidate interventions. Across these settings, a learning system faces two distinct resource constraints: the statistical cost of collecting informative data from the environment, and the computational cost of representing, updating, and optimizing models, beliefs, and policies once those data are in hand.

In sequential decision making, actions play a dual role because they determine both immediate outcomes and the observations available for future learning. A policy trained only on historical logs may have little evidence about actions that the logging policy rarely selected. Collecting new trajectories through online interaction can supply the missing information, yet every physical trial takes time and incurs real-world cost. Even after data are collected, turning past observations into a decision or a parameter estimate requires computation. Evaluating a policy class whose description grows with past value estimates, optimizing over continuous multimodal action distributions, quantifying uncertainty across sequentially constructed tree ensembles, integrating over unobserved state trajectories with particle filters, or branching over future observation histories in a belief tree can each become the primary bottleneck in practice.

Analyzing statistical sample complexity while treating computation and representation as exact, or designing fast heuristics without tracking the approximation errors they introduce, can obscure the

dominant source of error. A statistically optimal reinforcement learning method that stores an ever-growing sum of past value functions or solves an intractable nonconvex program at every episode cannot be deployed at scale. Conversely, replacing an exact policy update, an exact posterior belief, or an exact lookahead tree with a tractable approximation introduces projection bias, Monte Carlo variance, or structural lookahead error that directly enters the performance guarantee. The central question of this dissertation is how the representation and use of information determine both the statistical cost of data collection and the computational cost of inference and decision making.

1.2. Overview of the Dissertation

This dissertation is organized around seven research chapters that progress from learning policies under observed states and contexts, to statistical inference for sequentially constructed regression ensembles, and finally to parameter estimation and control when the underlying state is only partially observed. Throughout the dissertation, we number algorithms by chapter and state the exact assumptions, sample-size regimes, and approximation errors attached to each theoretical guarantee.

1.2.1. Learning from Offline and Online Data

We begin in Chapter 2 with the most direct question about combining information sources in reinforcement learning: what can an existing offline dataset tell us, and what must an agent still learn through online interaction? This chapter is based on joint work with Ziping Xu (Tan and Xu, 2024) and joint work with Wei Fan and Yuting Wei (Tan et al., 2024b).

Purely offline reinforcement learning avoids new environment interaction by fitting a pessimistic value function to logged transitions, but standard guarantees require the behavior policy to cover the state–action distribution of an optimal policy. When the offline dataset lacks full coverage on part of the state–action space, purely offline methods can fail arbitrarily badly, whereas purely online methods ignore the logged transitions altogether and explore the entire state–action space from scratch. Existing hybrid reinforcement learning algorithms either assume that the offline dataset already satisfies single-policy concentrability or require knowing an explicit partition of the state–action space in advance.

In the first part of Chapter 2, we study a natural extension of optimistic online reinforcement learning under general function approximation (Algorithm 2.1), which simply appends the offline dataset to the empirical Bellman-error constraints used to construct the confidence set. Without knowing where the offline data are informative, the analysis partitions the state–action space into a region where performance is controlled by a partial offline concentrability coefficient and a complementary region where performance is controlled by an online Bellman eluder dimension. Taking the infimum over all such analytical partitions shows that optimism automatically directs online exploration toward regions where offline coverage is weak.

In the second part of Chapter 2, we ask whether combining offline and online data can strictly improve upon the minimax sample complexity barriers of both purely offline and purely online learning in linear Markov decision processes (MDPs), where transition kernels and rewards are linear in a known d-dimensional feature map $\phi_h(s,a) \in \mathbb{R}^d$ over horizon H. We develop two computationally tractable algorithms targeting different evaluation criteria. Algorithm 2.2 takes an online-to-offline perspective for final-policy probably approximately correct (PAC) sample complexity: it uses reward-agnostic online exploration to collect samples in feature directions that are poorly covered by the offline covariance matrix, and then fits a variance-weighted pessimistic value function on the combined dataset. This replaces the offline information matrix $\Sigma^*_{\mathrm{off},h}$ in the instance-dependent minimax suboptimality bound with the combined matrix $\Sigma^*_{\mathrm{off},h} + \Sigma^*_{\mathrm{on},h}$ without requiring full-rank offline coverage. Algorithm 2.3 takes an offline-to-online perspective for cumulative regret over T online episodes given N_{off} offline episodes, achieving a high-probability regret bound of order

$$\mathrm{Reg}(T) \le \widetilde{O}\left(\inf_{\mathcal{X}_{\mathrm{off}},\, \mathcal{X}_{\mathrm{on}}} \left\{ c_{\mathrm{off}} T \sqrt{\frac{dH^3}{N_{\mathrm{off}}}} + \sqrt{d_{\mathrm{on}} d H^3 T} \right\} \right) \tag{1.1}$$

once T and N_{off} exceed the large-sample threshold $\widetilde{\Omega}(d^{13}H^{14})$, where c_{off} measures partial offline coverage on $\mathcal{X}_{\mathrm{off}}$ and $d_{\mathrm{on}} \le d$ is the dimension of the subspace spanned by $\mathcal{X}_{\mathrm{on}}$. When the offline dataset is large and covers all but a d_{on}-dimensional subspace of features, this bound improves upon the online minimax rate $\widetilde{\Theta}(\sqrt{d^2H^3T})$ by replacing one factor of $\sqrt{d}$ with $\sqrt{d_{\mathrm{on}}}$.

1.2.2. Learning Shared Structure from Heterogeneous Data

Chapter 3 examines a second way historical data can reduce the cost of online learning, based on joint work with Chinmaya Kausik and Ambuj Tewari (Kausik et al., 2024). Rather than supplying additional transitions from the same MDP, historical data in recommendation and personalization systems often consist of logged interactions from many heterogeneous users or tasks, each governed by its own unknown reward parameter $\theta_i \in \mathbb{R}^d$. When the user parameters lie near a shared r-dimensional subspace with $r \ll d$, offline data across users can identify this lower-dimensional representation so that online learning for a new user only needs to explore an r-dimensional space.

The central obstacle is that historical logs for each user are collected by unknown, potentially history-dependent behavior policies over very short trajectories, sometimes as short as two actions per user. Under adaptive action selection and non-spherical feature covariances, naive empirical second-moment matrices of feature–reward products are biased by within-trajectory dependencies and observation noise, and existing meta-learning or representation-learning guarantees require trajectory lengths to grow polynomially with the ambient dimension d.

We construct a moment-corrected spectral estimator (Algorithm 3.1) that pairs two observations per offline user through importance-weighted cross-moments to cancel the confounding from adaptive action selection and observation noise. We prove that the subspace estimation error in the sine-theta distance decays at the rate $\widetilde{O}(\sqrt{d/N_{\text{off}}})$ in the number of offline users N_{off} even with a constant number of observations per user. We then pass the estimated r-dimensional subspace to two online contextual bandit algorithms for a new user: a subspace-constrained confidence-set bandit method (Algorithm 3.2) and a computationally efficient projected confidence-ball variant (Algorithm 3.3), both achieving cumulative regret of order $\widetilde{O}(r\sqrt{T} + T\sqrt{d/N_{\text{off}}})$.

1.2.3. Statistically and Computationally Efficient Actor–Critic Methods

Chapters 2 and 3 rely on value-based or confidence-set policies that select actions by maximizing an optimistic value estimate. In Chapter 4, we turn to actor–critic methods, which maintain an explicit policy (the actor) alongside a value estimate (the critic). This chapter is based on joint

work with Wei Fan and Yuting Wei (Tan et al., 2025) and joint work with Max Qiushi Lin, Reza Asad, Haque Ishfaq, Csaba Szepesvári, and Sharan Vaswani (Lin et al., 2026).

Actor–critic methods are widely used in practice because incremental policy updates are convenient to implement, yet existing theoretical guarantees under general function approximation either require global coverage assumptions or suffer suboptimal sample complexity exponents such as $\widetilde{O}(T^{2/3})$ or $\widetilde{O}(T^{3/4})$ regret. In the first part of Chapter 4, we identify the exact statistical mechanism behind this gap. When the actor performs natural multiplicative weight updates with step size η using a sequence of critics $\widehat{Q}_1, \ldots, \widehat{Q}_t$, unrolling the recursion from a uniform initial policy yields

$$\pi_{t+1}(a \mid s) \propto \exp\left(\eta \sum_{\tau=1}^{t} \widehat{Q}_\tau(s,a)\right). \tag{1.2}$$

Each new critic update adds another function from the critic class $\mathcal{F}$ to the exponent, so after K distinct critic updates the log-covering number of the induced policy class Π_T grows linearly with K. If the critic is updated at every episode ($K = T$), uniform concentration over the policy-dependent Bellman target class breaks down and yields a vacuous linear regret bound unless Π_T happens to have a small cover by special structure (the case treated by Algorithm 4.1).

One might hope to control K simply by updating the critic rarely, only when its empirical Bellman loss exceeds a confidence threshold. However, if a frozen critic targets the current policy value Q^{π_t}, the regression target $\mathcal{T}_h^{\pi_t}\widehat{Q}_{h+1}$ drifts whenever the actor updates π_t, which destroys optimism and forces frequent critic switches anyway. Algorithm 4.2 resolves this obstacle with two coupled modifications: the critic targets the optimal action-value function Q^* via the Bellman optimality operator $(\mathcal{T}_h f)(s,a) = \mathbb{E}[r_h + \max_{a'} f(s',a') \mid s,a]$ so the target operator no longer depends on the changing actor, and whenever the rare-switching test triggers a critic refit, the actor resets to a uniform distribution before resuming mirror-descent updates toward the new critic. Because the maximum operator is 1-Lipschitz in the supremum norm, the target class covering number is controlled by $\mathcal{F}$ alone, while uniform resets cap the Kullback–Leibler divergence to the new greedy policy at $\log|\mathcal{A}|$ on each of the $K = O(d_{\mathrm{BE}} H \log T)$ switching intervals. This achieves $\widetilde{O}(\sqrt{T})$ regret and $\widetilde{O}(\epsilon^{-2})$ sample complexity under general function approximation without global coverage.

Although Algorithm 4.2 is statistically optimal in ϵ, evaluating an implicit policy of the form (1.2) still requires summing over all critics accumulated within an interval, and global optimistic confidence-set optimization remains computationally intractable. The second part of Chapter 4 addresses both computational bottlenecks in linear MDPs (Algorithms 4.6–4.8). Instead of storing past critics, Algorithm 4.7 parameterizes the actor with fixed-dimensional log-linear logits $z_w(s,a) = \langle w, \psi(s,a)\rangle$ and approximates the ideal logit update $z_{w_t} + \eta\widehat{Q}_t$ via weighted least-squares regression on a compact G-optimal design coreset of policy features, while Algorithm 4.8 replaces confidence-set optimization in the critic with noisy gradient updates via Langevin Monte Carlo. This keeps per-step policy evaluation cost constant in T and yields $\widetilde{O}(\epsilon^{-4})$ sample complexity under on-policy sampling and $\widetilde{O}(\epsilon^{-2})$ sample complexity when off-policy critic data reuse is permitted.

1.2.4. Learning Continuous-Action Diffusion Policies

In Chapter 5, based on joint work with Zhihan Huang and Yuting Wei (Tan et al., 2026), we extend the study of policy representations to continuous action spaces $\mathcal{A} \subseteq \mathbb{R}^{d_a}$, where optimal or near-optimal policies may be multimodal and poorly approximated by unimodal Gaussian or log-linear families. Diffusion models represent complex continuous distributions by transforming an initial Gaussian noise draw $z_K \sim \mathcal{N}(0, I_{d_a})$ through K reverse-time denoising transitions $z_K \to z_{K-1} \to \cdots \to z_0 = a$ conditioned on the state s.

While expressive, diffusion policies introduce a fundamental obstacle for reinforcement learning: the terminal action density $\pi_\theta(a \mid s) = \int p_\theta(z_{0:K} \mid s)\, dz_{1:K}$ integrates over all unobserved denoising paths and cannot be evaluated in closed form, so standard policy-gradient and natural-policy-gradient updates cannot be applied directly to $\log \pi_\theta(a \mid s)$. Chapter 5 develops and analyzes three algorithmic routes that bypass this intractability in different ways:

1. **Vanilla diffusion policy gradient (Algorithm 5.1):** Using either an unbiased denoising-path score estimate for stochastic denoising diffusion probabilistic model (DDPM) samplers or the exact endpoint score for deterministic denoising diffusion implicit model (DDIM) samplers, we establish $O(T^{3/4})$ expected regret under reset or fixed-state mixing access, up to comparison-policy and compatible-approximation errors.

2. **Monte Carlo diffusion natural policy gradient (Algorithm 5.2):** Under a deterministic DDIM flow, we evaluate the exact endpoint score via the change-of-variables formula and perform natural policy gradient updates under reset, mixing, or hybrid logged-trajectory access, achieving $O(T^{3/4})$ high-probability regret up to comparison-policy and advantage-transfer errors without explicit density-ratio dependence.

3. **Diffusion policy behavior cloning of an optimistic critic (Algorithm 5.3):** To avoid reset-coverage assumptions in finite-horizon reinforcement learning, we construct an optimistic action-value critic (or backward fitted-Q critic with offline data) and train the diffusion policy by denoising score matching on the critic's greedy actions, achieving $\widetilde{O}(\sqrt{T})$ regret up to cumulative score-matching and sampler errors.

1.2.5. Statistical Inference for Boosted Regression and Additive Models

Every method in Chapters 2 through 5 relies on sequential function approximation to estimate values or policies, where each iterate depends on the residuals or targets produced by earlier iterates. In Chapter 6, based on joint work with Haimo Fang and Giles Hooker (Fang et al., 2025) and joint work with Haimo Fang, Jonathan Pipping, and Giles Hooker (Fang et al., 2026), we examine this sequential function-fitting dynamics directly in supervised regression, asking how to conduct rigorous frequentist statistical inference—confidence intervals, prediction intervals, and hypothesis tests—for gradient boosted decision trees and explainable boosting machines.

Standard gradient boosting fits each new regression tree to the empirical residuals of all preceding trees. Because every tree depends on the full sequence of earlier adaptive splits and leaf values, the final ensemble is a highly coupled functional of the response vector, and classical U-statistic arguments used to establish asymptotic normality for random forests do not apply. The moving-average regularization of Zhou and Hooker (2022a) achieves a central limit theorem (CLT) by rescaling the ensemble at step b by $(1-\lambda)/b + \lambda$, which causes the ensemble to converge to the fixed point of a contraction mapping. However, that moving average shrinks the asymptotic prediction to $\boldsymbol{S}(\boldsymbol{I} + \frac{1-\lambda}{\lambda}\boldsymbol{S})^{-1}\boldsymbol{y}$, equivalent to kernel ridge regression with penalty $(1-\lambda)/\lambda > 0$, which introduces substantial shrinkage bias toward zero and degrades predictive accuracy unless

$\lambda = 1$.

In the first part of Chapter 6, we show that integrating moving-average regularization with either random tree dropout (Algorithm 6.1) or parallel tree growing with structured dropout (Algorithm 6.2) changes the contraction fixed point itself. Increasing the dropout probability p or the number of parallel trees K weakens the effective ridge penalty on the signal, allowing the ensemble to interpolate smoothly between regularized boosting and random forests while preserving an explicit asymptotic normal distribution for point predictions, prediction intervals, and chi-squared variable-importance tests. In the second part of Chapter 6, we extend this moving-average contraction framework to cyclic univariate boosting in Generalized Additive Models (Algorithm 6.3), establishing feature-wise asymptotic normality that equips glass-box additive shape functions $f_k(x_k)$ with pointwise confidence bands and formal hypothesis tests for feature significance and nonlinearity.

1.2.6. Transition to Partial Observability: Parameter Estimation and Control

In Chapters 2 through 5, the decision maker observes the Markov state s_h or context x_t directly at each step, so the representation problem concerns how to parameterize value functions and policies over an observed state space. In many scientific and physical domains, however, the underlying Markov state X_t is hidden and can only be inferred through a sequence of noisy, incomplete observations $Y_{1:t}$. As a simple physical illustration, consider applying a braking force to stop a moving cart when the sensor measures only noisy position and not velocity: two carts at the exact same measured position moving in opposite directions require opposite braking forces, so no memoryless policy of the current observation can control the system. Inferring velocity requires tracking changes in position across time and maintaining a posterior belief distribution over the latent state.

Chapters 7 and 8 make hidden-state inference itself part of the statistical and computational problem. Sequential Monte Carlo (particle filtering) approximates the latent state distribution using an ensemble of J weighted particles simulated forward from the transition dynamics and resampled according to observation likelihoods. While particle filters are plug-and-play—requiring only a forward simulator for the latent dynamics rather than closed-form transition densities—their discrete resampling step and finite particle budget J create distinct obstacles for parameter estimation and

for online control. Chapter 7 studies the statistical estimation problem for passive observation sequences, and Chapter 8 brings particle belief tracking into online action selection and planning.

Parameter Estimation under Partial Observability

Chapter 7, based on joint work with A. J. Abkemeier, Kunyang He, Giles Hooker, and Edward L. Ionides (Tan et al., 2024a), studies maximum likelihood and Bayesian parameter estimation in partially observed Markov processes (POMPs). Because a bootstrap particle filter produces an unbiased likelihood estimate using only a forward simulator and observation density evaluations, differentiating the filter via automatic differentiation (AD) is an attractive route to gradient-based parameter estimation. The obstacle is that discrete ancestor resampling changes discontinuously when the parameter θ is perturbed under fixed random numbers, making the raw particle-filter log-likelihood estimate a step function with zero or undefined pathwise gradient.

Existing attempts to differentiate through particle filters either stop gradients at the resampling weights (Naesseth et al., 2018; Jonschkowski et al., 2018), which discards past weight sensitivities and incurs $O(1)$ asymptotic score bias, or retain full unrescaled ancestral weight ratios (Poyiadjis et al., 2011; Ścibior and Wood, 2021), whose score variance grows linearly or quadratically with the time horizon N and typically requires evaluating latent transition densities. We introduce a smooth, plug-and-play off-parameter particle filter family parameterized by a weight discount factor $\alpha \in [0, 1]$ (Algorithm 7.1) and its single-pass differentiable counterpart at the baseline parameter (Algorithm 7.2). By coupling nearby parameter evaluations θ to a baseline filter run at ϕ with fixed resampling ancestors and discounting carried weight ratios by $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^{\alpha}$ before applying an importance-weight correction at step n, Algorithm 7.2 yields a pathwise score estimator that recovers the biased stop-gradient filter at $\alpha = 0$ and the consistent high-variance filter as $\alpha = 1$ and $J \to \infty$, while achieving an explicit bias–variance tradeoff for $\alpha \in (0, 1)$ with squared bias $O(N\alpha^{2/(1-\alpha)})$ and particle variance $O(N(1 + \log J)^2/(J(1-\alpha)^2))$ under uniform mixing assumptions.

Because stochastic gradient ascent with particle-filter gradients converges rapidly near a local mode but can stall in rough likelihood regions far from the maximum likelihood estimate, Algorithm 7.3

combines an iterated filtering warm start (Ionides et al., 2015) with subsequent discounted AD gradient optimization. We prove finite-sample and asymptotic convergence guarantees for this two-stage procedure and demonstrate substantial empirical improvements on the nonlinear Dhaka cholera model of King et al. (2008) and the 38-parameter *Daphnia* ecological panel model of Yang et al. (2025).

Control under Partial Observability

Finally, Chapter 8, based on joint work with Ziping Xu (Tan and Xu, 2026), studies online reinforcement learning and planning in partially observable Markov decision processes (POMDPs). Even when a latent dynamics model $\bar{\theta}_k$ is estimated online from interaction trajectories, computing an optimal action requires planning in belief space over future action and observation branches. Exact belief-tree search branches over all possible actions and discretized observations at every lookahead step, incurring computational complexity exponential in the planning horizon, whereas the classical QMDP approximation $Q_{\mathrm{QMDP}}(b,a) = \mathbb{E}_{s\sim b}[Q^*_{\mathrm{MDP}}(s,a)]$ averages the fully observed state–action value over current belief particles in $O(J)$ time but ignores the value of future information gathering because it assumes the latent state will become fully observed after the next transition.

Algorithm 8.1 bridges this gap by combining epoch-wise maximum likelihood model estimation, bootstrap particle filtering with J particles, and a depth-d Monte Carlo belief-tree planner (Algorithm G.1) with N rollouts per node that expands observation branches for d steps before evaluating leaf beliefs with the fully observed state–action value function $Q^{*,S}_{\bar{\theta}_k}$. Under uniform Doeblin mixing and bounded positive observation densities, we prove an end-to-end cumulative regret bound over T steps of the form

$$\frac{\mathrm{Reg}(T)}{T} \lesssim \widetilde{O}(T^{-1/4}) + C_{\mathrm{pf}} J^{-1/(2\kappa_{\mathrm{pf}})} + C_{\mathrm{mc}}(d,M) N^{-1/2} + C\gamma^{d+1}\overline{\Xi}_T, \tag{1.3}$$

which explicitly separates the statistical interaction error $\widetilde{O}(T^{-1/4})$ of online model learning from the three computational approximations: particle belief tracking error $O(J^{-1/(2\kappa_{\mathrm{pf}})})$, Monte Carlo tree rollout error $O(C_{\mathrm{mc}}(d,M)N^{-1/2})$ across M observation bins, and structural lookahead error $O(\gamma^{d+1}\overline{\Xi}_T)$ from deferring the full-observability leaf bootstrap by d steps. We validate each of

the four scaling rates separately on a mixing POMDP that satisfies the theorem's assumptions exactly, and we instantiate the depth-zero particle state-rollout planner with learned dynamics and actor–critic networks on continuous-control and mobile-health benchmarks.

Chapter 9 concludes the dissertation by synthesizing the common principles connecting statistical coverage, structural representation, and computational approximation across Chapters 2 through 8 and outlining open directions for future research.

CHAPTER 2

LEARNING FROM OFFLINE AND ONLINE DATA

[RESOLVE: CHAPTER 2 COMBINES TWO HYBRID RL PAPERS (SECTIONS 2.1-2.6: TAN AND XU 2024; SECTIONS 2.7-2.10: TAN, FAN, AND WEI 2024).]

Reinforcement learning (RL) encompasses two main paradigms distinguished by how training data are acquired: online RL and offline RL. In online RL, an agent interacts with the environment in real time and executes a sequence of adaptively chosen policies to maximize cumulative reward (Sutton and Barto, 2018; Azar et al., 2017; Jin et al., 2018; Zhang et al., 2023). In offline RL, the agent has access only to a pre-collected dataset logged by a behavior policy and seeks a near-optimal policy without further environmental interaction (Lange et al., 2012; Levine et al., 2020; Jin et al., 2021b; Li et al., 2024a). Purely online methods often require an impractically large number of interactive trajectories, whereas purely offline methods impose stringent coverage assumptions on the pre-collected dataset, typically requiring single-policy concentrability with respect to an optimal policy (Rashidinejad et al., 2021; Zhan et al., 2022; Xie et al., 2021).

Hybrid reinforcement learning combines offline datasets with online environmental interaction (Vecerik et al., 2017; Nair et al., 2020; Xie et al., 2022b; Song et al., 2023; Nakamoto et al., 2023; Wagenmaker and Pacchiano, 2023; Li et al., 2023; Ball et al., 2023; Zhou et al., 2023; Amortila et al., 2024). A central theoretical question posed by Xie et al. (2022b) is whether hybrid RL can provably improve upon both purely offline and purely online learning when the offline dataset lacks single-policy concentrability. Indeed, Xie et al. (2022b) proved that if the behavior policy already satisfies single-policy concentrability and the learner can choose the ratio of offline to online trajectories, then either purely offline learning or purely online learning is minimax optimal, leaving no statistical advantage to combining both sources. Consequently, the regime in which hybrid RL offers genuine statistical gains is precisely when the offline dataset has only partial coverage of the state-action space, so that online interaction must fill in the regions left unexplored by the behavior policy.

This chapter develops the statistical theory and algorithmic design of hybrid RL without single-policy concentrability across two settings:

1. **General function approximation via experience replay initialization (Sections 2.1–2.6):** Based on joint work with Ziping Xu (Tan and Xu, 2024), we show that warm-starting an optimistic general function approximation algorithm by appending the entire offline dataset to its experience replay buffer (Algorithm 2.1) achieves regret guarantees that adapt automatically to the best partition of the state-action space into an offline-covered subset $\mathcal{X}_{\text{off}}$ and an online-explored subset $\mathcal{X}_{\text{on}}$, without the algorithm ever knowing or estimating the partition. We further establish a general conversion recipe (Proposition 2.9) and specialize the resulting guarantees to tabular, linear, and block Markov decision processes (MDPs).

2. **Breaking sample-size barriers in linear MDPs (Sections 2.7–2.11):** While general function approximation yields broad adaptivity, its squared-Bellman-error confidence sets incur suboptimal H^5 horizon dependence when specialized to linear MDPs. Based on joint work with Wei Fan and Yuting Wei (Tan et al., 2024b), we develop two computationally efficient algorithms tailored to linear MDPs that sharpen both the dimension dependence and the horizon dependence to H^3. The first method, Algorithm 2.2, follows an online-to-offline design that uses reward-agnostic covariate collection (Wagenmaker and Jamieson, 2022) followed by variance-weighted pessimistic value iteration (Xiong et al., 2023b) to output a fixed policy with near-optimal probably approximately correct (PAC) suboptimality. The second method, Algorithm 2.3, follows an offline-to-online design that warm-starts variance-weighted optimistic value iteration (He et al., 2023) with offline parameter estimates to achieve sublinear online regret that improves upon the minimax lower bound for purely online linear MDPs whenever offline data cover a nontrivial subspace.

All proofs and supplementary experimental figures for this chapter are presented in Appendix A.

2.1. Hybrid RL with General Function Approximation

Many online RL algorithms explore by maintaining an experience replay buffer and minimizing empirical Bellman error over the collected transitions to update confidence sets or value estimates (Auer et al., 2008; Jin et al., 2021a; Xie et al., 2022a). A natural way to extend such an online algorithm to the hybrid setting is to include the offline dataset in the replay buffer from the start (Song et al., 2023; Nakamoto et al., 2023; Amortila et al., 2024). Prior analyses of this replay-initialization strategy required single-policy concentrability over the entire state-action space and discarded offline trajectories whenever $N_{\text{off}} > N_{\text{on}}$. Meanwhile, the works that avoided single-policy concentrability (Li et al., 2023; Wagenmaker and Pacchiano, 2023) relied on multi-stage reward-free Frank–Wolfe exploration subroutines rather than direct online regret minimization.

We show that simply appending the entire offline dataset to the replay buffer of an optimistic squared-Bellman-error algorithm (Jin et al., 2021a; Xie et al., 2022a) yields provable improvements over both pure online and pure offline learning even when the offline dataset has arbitrarily poor coverage. We establish this by considering arbitrary (not necessarily disjoint) partitions $\mathcal{X}_{\text{off}} \cup \mathcal{X}_{\text{on}} = \mathcal{X}$ of the state-action-horizon space, bounding the regret on $\mathcal{X}_{\text{off}}$ via a partial offline concentrability coefficient and the regret on $\mathcal{X}_{\text{on}}$ via a restricted online sequential extrapolation coefficient, and taking the infimum over all partitions $\mathcal{X}_{\text{off}}, \mathcal{X}_{\text{on}}$.

Notation. Let $\mathcal{N}_{\mathcal{F}}(\rho)$ denote the ρ-covering number of a function class $\mathcal{F}$ with respect to the supremum norm. Let N_{off} and N_{on} (with $N = N_{\text{off}} + N_{\text{on}}$) denote the number of episodes in the offline dataset and the number of online episodes, respectively. We use $T = N_{\text{on}}$ interchangeably when referring to the online horizon. For any subset $\mathcal{X}' \subset \mathcal{S} \times \mathcal{A} \times [H]$, let $\mathcal{X}'_h = \{(s,a) \in \mathcal{S} \times \mathcal{A} : (s,a,h) \in \mathcal{X}'\}$, and let $\Delta(\mathcal{Y})$ denote the set of probability distributions over $\mathcal{Y}$.

Episodic MDPs. We consider episodic Markov decision processes denoted by $\mathcal{M} = (\mathcal{S}, \mathcal{A}, H, P, R)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, H is the episode horizon, $P = \{P_h\}_{h\in[H]}$ is the collection of transition probability kernels with $P_h : \mathcal{S} \times \mathcal{A} \mapsto \Delta(\mathcal{S})$, and $R = \{R_h\}_{h\in[H]}$ is the collection of deterministic reward functions with $R_h : \mathcal{S} \times \mathcal{A} \mapsto [0,1]$. Within each episode, at step $h \in [H]$, the agent observes state $s_h \in \mathcal{S}$, selects action $a_h \in \mathcal{A}$, receives reward $r_h = R_h(s_h, a_h)$,

and transitions to $s_{h+1} \sim P_h(\cdot \mid s_h, a_h)$. A policy $\pi = \{\pi_h\}_{h\in[H]}$ maps states to distributions over actions. A value-function class $\mathcal{F} = \{\mathcal{F}_h\}_{h\in[H]}$ induces a greedy policy class $\Pi \coloneqq \{\pi^f : f \in \mathcal{F}\}$, where $\pi_h^f(s) = \arg\max_{a\in\mathcal{A}} f_h(s,a)$. Throughout Sections 2.1–2.6, we write $\mathcal{X} = \mathcal{S} \times \mathcal{A} \times [H]$.

Definition 2.1 (Occupancy Measure)**.** The occupancy measure $d^\pi = \{d_h^\pi\}_{h=1}^H$ is the collection of state-action distributions induced by executing policy π in $\mathcal{M}$, where $d_h^\pi(s,a) = \mathbb{P}(s_h = s, a_h = a \mid \pi)$. We write $\mathbb{D} = \{d^\pi : \pi \in \Pi\}$ and $\mathbb{D}_h = \{d_h^\pi : \pi \in \Pi\}$.

Hybrid RL Interaction Protocol. In the offline-to-online fine-tuning protocol, the learner is given an offline dataset $\mathcal{D}_{\text{off}} = \{\mathcal{D}_{\text{off},h}\}_{h=1}^H$ consisting of N_{off} episodes collected by a fixed behavior policy $\pi_{\text{off}} = \{\pi_{\text{off},h}\}_{h\in[H]}$ with occupancy measure $\mu = \{\mu_h\}_{h=1}^H$, and subsequently interacts with the environment for N_{on} online episodes. Denote by $s_h^{(t)}$, $a_h^{(t)}$, and $r_h^{(t)}$ the state, action, and reward at step $h \in [H]$ of online episode $t \in [N_{\text{on}}]$. For any policy π, its value function and action-value function at step h are defined as

$$V_h^\pi(s) = \mathbb{E}_\pi\left[\sum_{h'=h}^H r_{h'} \,\middle|\, s_h = s\right], \qquad Q_h^\pi(s,a) = \mathbb{E}_\pi\left[\sum_{h'=h}^H r_{h'} \,\middle|\, s_h = s, a_h = a\right], \tag{2.1}$$

with $V_{H+1}^\pi(s) \equiv 0$. We write $V^\star$ and $Q^\star$ for the optimal value and action-value functions. The cumulative regret of an online learning algorithm $\mathcal{L}$ over N_{on} episodes is

$$\mathrm{Reg}(N_{\text{on}}, \mathcal{L}) = \mathbb{E}_{\mathcal{L}}\left[\sum_{t=1}^{N_{\text{on}}} \left(V_1^\star(s_1^{(t)}) - \sum_{h=1}^H r_h^{(t)}\right)\right]. \tag{2.2}$$

Function Approximation. We approximate $Q^\star$ using a function class $\mathcal{F} = \{\mathcal{F}_h\}_{h\in[H]}$ with $\mathcal{F}_h \subseteq [0,H]^{\mathcal{S}\times\mathcal{A}}$ and $f_{H+1} \equiv 0$. For each $h \in [H]$, the Bellman optimality operator $\mathcal{T}_h$ is defined by

$$\mathcal{T}_h f_{h+1}(s,a) \coloneqq R_h(s,a) + \mathbb{E}_{s'\sim P_h(\cdot|s,a)}\left[\max_{a'\in\mathcal{A}} f_{h+1}(s',a')\right]. \tag{2.3}$$

We write $\mathcal{E}_h f = \mathcal{T}_h f_{h+1} - f_h$ for the Bellman residual of $f \in \mathcal{F}$ at step h, and for any distribution $d \in \Delta(\mathcal{S} \times \mathcal{A})$ we define the squared $L^2(d)$ Bellman error norm by

$$\|f_h - \mathcal{T}_h f_{h+1}\|_{2,d}^2 = \mathbb{E}_{(s_h,a_h)\sim d}\left[(f_h(s_h,a_h) - \mathcal{T}_h f_{h+1}(s_h,a_h))^2\right]. \tag{2.4}$$

We impose the standard Bellman completeness assumption on $\mathcal{F}$ (Liu et al., 2020; Jin et al., 2021a; Xie et al., 2022a).

Assumption 2.2 (Bellman Completeness)**.** For all $h \in [H]$ and $f_{h+1} \in \mathcal{F}_{h+1}$, we have $\mathcal{T}_h f_{h+1} \in \mathcal{F}_h$. In particular, $Q_h^\star \in \mathcal{F}_h$ for all $h \in [H]$.

2.2. Partial Offline and Online Complexity Measures

To quantify the statistical benefit of hybrid data when μ covers only part of $\mathcal{X}$, we restrict standard offline and online complexity measures to arbitrary subsets of the state-action space.

Full-Space Complexity Measures. In offline RL with general function approximation, sample complexity is governed by Bellman-error concentrability (Xie et al., 2021; Zhan et al., 2022). For a comparator policy π, the single-policy and all-policy concentrability coefficients are defined by

$$c_{\text{off}}(\mathcal{F},\pi) \coloneqq \max_{h\in[H]} \sup_{f\in\mathcal{F}} \frac{\|f_h - \mathcal{T}_h f_{h+1}\|_{2,d_h^\pi}^2}{\|f_h - \mathcal{T}_h f_{h+1}\|_{2,\mu_h}^2}, \qquad c_{\text{off}}(\mathcal{F}) \coloneqq \sup_{\pi\in\Pi} c_{\text{off}}(\mathcal{F},\pi). \tag{2.5}$$

In online RL, Xie et al. (2022a) characterize exploration complexity via the Sequential Extrapolation Coefficient (scaled here to $Q_h \in [0,H]$):

$$c_{\text{on}}(\mathcal{F},T) \coloneqq \max_{h\in[H]} \sup_{\{f^{(1)},\dots,f^{(T)}\}\subseteq\mathcal{F}} \sup_{(\pi^{(1)},\dots,\pi^{(T)})} \left\{ \sum_{t=1}^{T} \frac{\mathbb{E}_{d_h^{\pi^{(t)}}}[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}]^2}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{\pi^{(i)}}}[(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)})^2]} \right\}. \tag{2.6}$$

Partitioned Complexity Measures. Let $\mathcal{X}_{\text{off}} \cup \mathcal{X}_{\text{on}} = \mathcal{X} = \mathcal{S} \times \mathcal{A} \times [H]$ be an arbitrary (not necessarily disjoint) partition of the state-action-horizon space. We define the partial offline all-policy concentrability coefficient on $\mathcal{X}_{\text{off}}$ and the partial online sequential extrapolation coefficient

on $\mathcal{X}_{\mathrm{on}}$ by

$$c_{\mathrm{off}}(\mathcal{F}, \mathcal{X}_{\mathrm{off}}) \coloneqq \max_{h\in[H]} \sup_{\pi\in\Pi} \sup_{f\in\mathcal{F}} \frac{\|(f_h - \mathcal{T}_h f_{h+1})\mathbf{1}_{(\cdot,h)\in\mathcal{X}_{\mathrm{off}}}\|^2_{2,d_h^\pi}}{\|(f_h - \mathcal{T}_h f_{h+1})\mathbf{1}_{(\cdot,h)\in\mathcal{X}_{\mathrm{off}}}\|^2_{2,\mu_h}}, \tag{2.7}$$

$$c_{\mathrm{on}}(\mathcal{F}, \mathcal{X}_{\mathrm{on}}, T) \coloneqq \max_{h\in[H]} \sup_{\{f^{(1)},\ldots,f^{(T)}\}\subseteq\mathcal{F}} \sup_{(\pi^{(1)},\ldots,\pi^{(T)})} \left\{ \sum_{t=1}^{T} \frac{\mathbb{E}_{d_h^{\pi^{(t)}}}[(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)})\mathbf{1}_{(\cdot,h)\in\mathcal{X}_{\mathrm{on}}}]^2}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{\pi^{(i)}}}[(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)})^2\mathbf{1}_{(\cdot,h)\in\mathcal{X}_{\mathrm{on}}}]} \right\}. \tag{2.8}$$

Viewing c_{off} and c_{on} as complexity measures on the Bellman-error class $\mathcal{F}_h - \mathcal{T}_h\mathcal{F}_{h+1}$, Equations (2.7)–(2.8) restrict each Bellman-error function to vanish outside $\mathcal{X}_{\mathrm{off}}$ or $\mathcal{X}_{\mathrm{on}}$, respectively. Although $c_{\mathrm{off}}(\mathcal{F}, \mathcal{X}_{\mathrm{off}})$ takes a supremum over policies $\pi \in \Pi$ on $\mathcal{X}_{\mathrm{off}}$, the infimum over partitions $\mathcal{X}_{\mathrm{off}} \cup \mathcal{X}_{\mathrm{on}} = \mathcal{X}$ ensures that $c_{\mathrm{off}}(\mathcal{F}, \mathcal{X}_{\mathrm{off}})$ is always finite on well-chosen subsets $\mathcal{X}_{\mathrm{off}}$ (and vanishes identically when $\mathcal{X}_{\mathrm{off}} = \emptyset$), even when the full-space single-policy concentrability coefficient $c_{\mathrm{off}}(\mathcal{F}, \pi^\star)$ is infinite.

2.3. Optimistic Experience Replay with Offline Initialization

We now formalize our hybrid general function approximation method in Algorithm 2.1, which modifies the optimistic least-squares value-function elimination algorithm of Jin et al. (2021a) and Xie et al. (2022a) by including the offline transitions $\mathcal{D}_{\mathrm{off},h}$ in the empirical squared-Bellman-error loss $\mathcal{L}_h^{(t)}$ at every online episode $t \in [N_{\mathrm{on}}]$.

Theorem 2.3 shows that the regret of Algorithm 2.1 decomposes across any partition $\mathcal{X}_{\mathrm{off}}, \mathcal{X}_{\mathrm{on}}$ of $\mathcal{X}$, and therefore achieves the infimum over all partitions simultaneously.

Theorem 2.3 (Regret Bound for Algorithm 2.1)**.** *Suppose Assumption 2.2 holds. Let $\mathcal{X}_{\mathrm{off}}, \mathcal{X}_{\mathrm{on}}$ be an arbitrary partition of $\mathcal{X} = \mathcal{S} \times \mathcal{A} \times [H]$. Then with probability at least $1 - \delta$, Algorithm 2.1 satisfies*

$$\mathrm{Reg}(N_{\mathrm{on}}) = \mathcal{O}\left(\inf_{\mathcal{X}_{\mathrm{on}}, \mathcal{X}_{\mathrm{off}}} \left(\sqrt{\beta H^4 N_{\mathrm{on}} \left(\frac{N_{\mathrm{on}}}{N_{\mathrm{off}}} \right) c_{\mathrm{off}}(\mathcal{F}, \mathcal{X}_{\mathrm{off}})} + \sqrt{\beta H^4 N_{\mathrm{on}} c_{\mathrm{on}}(\mathcal{F}, \mathcal{X}_{\mathrm{on}}, N_{\mathrm{on}})} \right) \right), \tag{2.9}$$

where $\beta = c_1 \log(NH\mathcal{N}_{\mathcal{F}}(1/N)/\delta)$ for an absolute constant $c_1 > 0$ and $N = N_{\mathrm{on}} + N_{\mathrm{off}}$.

Algorithm 2.1 Optimistic Hybrid RL with General Function Approximation

Input: Offline dataset $\mathcal{D}_{\text{off}}$, sample sizes N_{on}, N_{off}, function class $\mathcal{F}$, confidence parameter $\beta > 0$.

Initialize: $\mathcal{F}^{(0)} \leftarrow \mathcal{F}$, $\mathcal{D}_h^{(0)} \leftarrow \emptyset$ for all $h \in [H]$.

for episode $t = 1, 2, \ldots, N_{\text{on}}$ **do**

Select policy $\pi^{(t)} \leftarrow \pi_{f^{(t)}}$, where $f^{(t)} \coloneqq \arg\max_{f \in \mathcal{F}^{(t-1)}} f_1(s_1^{(t)}, \pi_{f,1}(s_1^{(t)}))$.

Execute $\pi^{(t)}$ for one episode and observe trajectory $(s_1^{(t)}, a_1^{(t)}, r_1^{(t)}), \ldots, (s_H^{(t)}, a_H^{(t)}, r_H^{(t)})$.

Update online replay buffer $\mathcal{D}_h^{(t)} \leftarrow \mathcal{D}_h^{(t-1)} \cup \{(s_h^{(t)}, a_h^{(t)}, r_h^{(t)}, s_{h+1}^{(t)})\}$ for all $h \in [H]$.

Update confidence set:

$$\mathcal{F}^{(t)} \leftarrow \left\{ f \in \mathcal{F} : \mathcal{L}_h^{(t)}(f_h, f_{h+1}) - \min_{f'_h \in \mathcal{F}_h} \mathcal{L}_h^{(t)}\big(f'_h, f_{h+1}\big) \le \beta \quad \forall h \in [H] \right\},$$

where for all $f \in \mathcal{F}_h$ and $f' \in \mathcal{F}_{h+1}$,

$$\mathcal{L}_h^{(t)}\big(f, f'\big) \coloneqq \sum_{(s,a,r,s') \in \mathcal{D}_h^{(t)} \cup \mathcal{D}_{\text{off},h}} \left(f(s,a) - r - \max_{a' \in \mathcal{A}} f'\big(s', a'\big) \right)^2.$$

end for

The proof of Theorem 2.3 is given in Appendix A.1. Several features of this guarantee merit emphasis:

- **Automatic adaptation to the best partition:** Algorithm 2.1 takes no partition as input. When $N_{\text{on}} \gg N_{\text{off}}$, choosing $\mathcal{X}_{\text{on}} = \mathcal{X}$ and $\mathcal{X}_{\text{off}} = \emptyset$ yields $c_{\text{off}}(\mathcal{F}, \emptyset) = 0$, recovering the purely online regret guarantee $\tilde{\mathcal{O}}(\sqrt{\beta H^4 N_{\text{on}} c_{\text{on}}(\mathcal{F}, \mathcal{X}, N_{\text{on}})})$. Conversely, in the offline-abundant regime $N_{\text{off}} \gg N_{\text{on}}$, the first term becomes negligible on the region $\mathcal{X}_{\text{off}}$ well-covered by μ, reducing the effective online complexity from $c_{\text{on}}(\mathcal{F}, \mathcal{X}, N_{\text{on}})$ to $c_{\text{on}}(\mathcal{F}, \mathcal{X}_{\text{on}}, N_{\text{on}})$.

- **No single-policy concentrability or data discarding:** Unlike Song et al. (2023), Nakamoto et al. (2023), and Amortila et al. (2024), Theorem 2.3 does not require full-space single-policy concentrability and retains all N_{off} offline episodes even when $N_{\text{off}} > N_{\text{on}}$, allowing the offline term to vanish at rate $\sqrt{N_{\text{on}}^2 / N_{\text{off}}}$ as N_{off} grows.

2.4. Specializations to Tabular, Linear, and Block MDPs

We now instantiate Theorem 2.3 in three canonical MDP families to illustrate how state-action partitioning reduces the complexity of online exploration. Proofs for all three case studies are provided in Appendix A.2.

2.4.1. Tabular MDPs

In a tabular MDP with $|\mathcal{S}| = S < \infty$ states and $|\mathcal{A}| = A < \infty$ actions, we take the unconstrained tabular value class $\mathcal{F}_h = [0, H]^{SA}$, which has log-covering number $\log \mathcal{N}_{\mathcal{F}}(1/N) = \mathcal{O}(HSA \log N)$. For any threshold $\rho > 0$, a natural partition chooses $\mathcal{X}_{\mathrm{off}}(\rho) := \{(s, a, h) : \sup_\pi d_h^\pi(s, a)/\mu_h(s, a) \le \rho\}$ and $\mathcal{X}_{\mathrm{on}}(\rho) = \mathcal{X} \setminus \mathcal{X}_{\mathrm{off}}(\rho)$.

Proposition 2.4 (Tabular MDP Regret of Algorithm 2.1)**.** *In a tabular MDP with $\mathcal{F}_h = [0, H]^{SA}$, for any partition $\mathcal{X}_{\mathrm{off}} \cup \mathcal{X}_{\mathrm{on}} = \mathcal{X}$, we have*

$$c_{\mathrm{off}}(\mathcal{F}, \mathcal{X}_{\mathrm{off}}) \le \sup_{\pi \in \Pi} \sup_{(s,a,h) \in \mathcal{X}_{\mathrm{off}}} \frac{d_h^\pi(s, a)}{\mu_h(s, a)} = \sup_{\pi \in \Pi} \left\| \frac{d_h^\pi \mathbf{1}_{\mathcal{X}_{\mathrm{off}}}}{\mu_h} \right\|_\infty, \qquad c_{\mathrm{on}}(\mathcal{F}, \mathcal{X}_{\mathrm{on}}, N_{\mathrm{on}}) \lesssim \max_{h \in [H]} |\mathcal{X}_{\mathrm{on},h}| \log(N_{\mathrm{on}}). \tag{2.10}$$

Consequently, with probability at least $1 - \delta$, Algorithm 2.1 satisfies

$$\mathrm{Reg}(N_{\mathrm{on}}) = \tilde{\mathcal{O}}\left(\inf_{\mathcal{X}_{\mathrm{on}}, \mathcal{X}_{\mathrm{off}}} \left(\sqrt{H^5 SA N_{\mathrm{on}} \left(\frac{N_{\mathrm{on}}}{N_{\mathrm{off}}} \right) \sup_{\pi \in \Pi} \left\| \frac{d_h^\pi \mathbf{1}_{\mathcal{X}_{\mathrm{off}}}}{\mu_h} \right\|_\infty} + \sqrt{H^5 SA \max_{h \in [H]} |\mathcal{X}_{\mathrm{on},h}| N_{\mathrm{on}}} \right) \right). \tag{2.11}$$

When $N_{\mathrm{off}} \gg N_{\mathrm{on}}$, the online exploration complexity depends on $\max_h |\mathcal{X}_{\mathrm{on},h}|$, the number of state-action pairs per step left under-covered by the behavior policy μ, rather than the full cardinality SA.

2.4.2. Linear MDPs

Next, we consider linear MDPs (Yang and Wang, 2019; Jin et al., 2020), for which the linear action-value class $\mathcal{F}_h = \{\langle \phi(\cdot, \cdot), w_h \rangle : w_h \in \mathbb{R}^d, \|w_h\|_2 \le 2H\sqrt{d}\}$ satisfies Assumption 2.2.

Definition 2.5 (Linear MDP)**.** An episodic MDP is a linear MDP with known feature map $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$ if for every $h \in [H]$, there exist d unknown signed measures $\boldsymbol{\nu}_h = (\nu_h^{(1)}, \ldots, \nu_h^{(d)})$ over

$\mathcal{S}$ and an unknown vector $\boldsymbol{\theta}_h \in \mathbb{R}^d$ such that for all $(s,a) \in \mathcal{S} \times \mathcal{A}$,

$$P_h(\cdot \mid s,a) = \langle \phi(s,a), \boldsymbol{\nu}_h(\cdot) \rangle \quad \text{and} \quad R_h(s,a) = \langle \phi(s,a), \boldsymbol{\theta}_h \rangle, \tag{2.12}$$

with $\|\phi(s,a)\|_2 \le 1$ for all $(s,a) \in \mathcal{S} \times \mathcal{A}$ and $\max\{\|\boldsymbol{\nu}_h(\mathcal{S})\|_2, \|\boldsymbol{\theta}_h\|_2\} \le \sqrt{d}$ for all $h \in [H]$.

Any partition $\mathcal{X}_{\text{off}} \cup \mathcal{X}_{\text{on}} = \mathcal{X}$ induces two feature subspaces $\Phi_{\text{off}} = \text{Span}(\{\phi(s,a) : (s,a,h) \in \mathcal{X}_{\text{off}}\}) \subseteq \mathbb{R}^d$ and $\Phi_{\text{on}} = \text{Span}(\{\phi(s,a) : (s,a,h) \in \mathcal{X}_{\text{on}}\}) \subseteq \mathbb{R}^d$ of dimensions $d_{\text{off}} = \dim(\Phi_{\text{off}})$ and $d_{\text{on}} = \dim(\Phi_{\text{on}})$, respectively. Let $\mathcal{P}_{\text{off}}$ and $\mathcal{P}_{\text{on}}$ denote the orthogonal projection matrices onto Φ_{off} and Φ_{on}, and let $\lambda_k(\boldsymbol{M})$ denote the k-th largest eigenvalue of a symmetric matrix $\boldsymbol{M}$.

Proposition 2.6 (Linear MDP Regret of Algorithm 2.1)**.** *In a linear MDP (Definition 2.5), let $\phi_{\text{off}} = \mathcal{P}_{\text{off}}\phi$. For any partition $\mathcal{X}_{\text{off}} \cup \mathcal{X}_{\text{on}} = \mathcal{X}$, we have*

$$c_{\text{off}}(\mathcal{F}, \mathcal{X}_{\text{off}}) \le \max_{h \in [H]} \frac{1}{\lambda_{d_{\text{off}}}(\mathbb{E}_{\mu_h}[\phi_{\text{off}}(s,a)\phi_{\text{off}}(s,a)^\top])}, \qquad c_{\text{on}}(\mathcal{F}, \mathcal{X}_{\text{on}}, N_{\text{on}}) = \mathcal{O}(d_{\text{on}} \log(HN_{\text{on}}) \log(N_{\text{on}})). \tag{2.13}$$

Consequently, with probability at least $1-\delta$, Algorithm 2.1 satisfies

$$\text{Reg}(N_{\text{on}}) = \tilde{\mathcal{O}}\left(\inf_{\mathcal{X}_{\text{on}}, \mathcal{X}_{\text{off}}} \left(\sqrt{dH^5 N_{\text{on}} \left(\frac{N_{\text{on}}}{N_{\text{off}}}\right) \max_{h \in [H]} \frac{1}{\lambda_{d_{\text{off}}}(\mathbb{E}_{\mu_h}[\phi_{\text{off}} \phi_{\text{off}}^\top])}} + \sqrt{d_{\text{on}} dH^5 N_{\text{on}}} \right) \right). \tag{2.14}$$

Proposition 2.6 shows that on the online partition $\mathcal{X}_{\text{on}}$, the dimension dependence improves from d^2 to $d_{\text{on}}d$ because the restricted distributional Bellman–Eluder dimension on $\mathcal{X}_{\text{on}}$ scales with the subspace dimension d_{on} rather than the ambient dimension d (while the factor of d from the log-covering number of $\mathcal{F}$ remains). In Sections 2.7–2.11, we show how specialized variance-weighted linear MDP algorithms further reduce the horizon dependence from H^5 to H^3.

2.4.3. Block MDPs

A Block MDP (Dann et al., 2019; Misra et al., 2019; Du et al., 2021) has a finite unobservable latent state space $\mathcal{U}$, a finite action space $\mathcal{A}$, and a potentially infinite observable context space $\mathcal{S}$ generated via emission distributions $s_h \sim q(\cdot \mid u_h)$ with disjoint supports.

Definition 2.7 (Block Structure)**.** An episodic MDP is a Block MDP if there exists a latent decoding function $f^\star : \mathcal{S} \mapsto \mathcal{U}$ such that for every latent state $u \in \mathcal{U}$, the emission distribution $q(\cdot \mid u)$ is supported on $(f^\star)^{-1}(u)$.

Any partition $\mathcal{X}_{\text{off}} \cup \mathcal{X}_{\text{on}} = \mathcal{X}$ induces a latent partition $\bar{\mathcal{X}}_{\text{off}} = \{(f^\star(s), a, h) : (s, a, h) \in \mathcal{X}_{\text{off}}\}$ and $\bar{\mathcal{X}}_{\text{on}} = \{(f^\star(s), a, h) : (s, a, h) \in \mathcal{X}_{\text{on}}\}$, along with latent occupancy measures $\bar{\mu}_h$ and $\bar{d}_h^\pi$ on $\mathcal{U} \times \mathcal{A}$.

Proposition 2.8 (Block MDP Regret of Algorithm 2.1)**.** *In a Block MDP with a Bellman-complete value class $\mathcal{F}$, we have*

$$c_{\text{off}}(\mathcal{F}, \mathcal{X}_{\text{off}}) \le \sup_{\pi \in \Pi} \sup_{(u,a,h) \in \bar{\mathcal{X}}_{\text{off}}} \frac{\bar{d}_h^\pi(u,a)}{\bar{\mu}_h(u,a)}, \qquad c_{\text{on}}(\mathcal{F}, \mathcal{X}_{\text{on}}, N_{\text{on}}) = \mathcal{O}\left(\max_{h \in [H]} |\bar{\mathcal{X}}_{\text{on},h}| \log(N_{\text{on}}) \right). \quad (2.15)$$

Consequently, with probability at least $1 - \delta$, Algorithm 2.1 satisfies

$$\text{Reg}(N_{\text{on}}) = \tilde{\mathcal{O}}\left(\inf_{\mathcal{X}_{\text{on}}, \mathcal{X}_{\text{off}}} \left(\sqrt{\beta H^4 N_{\text{on}} \left(\frac{N_{\text{on}}}{N_{\text{off}}} \right) \sup_{\pi \in \Pi} \sup_{(u,a,h) \in \bar{\mathcal{X}}_{\text{off}}} \frac{\bar{d}_h^\pi(u,a)}{\bar{\mu}_h(u,a)}} + \sqrt{\beta H^4 N_{\text{on}} \max_{h \in [H]} |\bar{\mathcal{X}}_{\text{on},h}|} \right) \right). \quad (2.16)$$

2.5. A General Recipe for Hybridizing Online Algorithms

The state-action space partitioning technique used to analyze Algorithm 2.1 extends to any online learning algorithm whose regret decomposes into per-step error terms that are controlled in-sample over the combined replay buffer. Proposition 2.9 states this general conversion recipe, and its proof is given in Appendix A.2.

Proposition 2.9 (General Hybrid Conversion Recipe)**.** *Let $\mathcal{L}$ be an online learning algorithm initialized with an offline dataset $\mathcal{D}_{\text{off}}$ of N_{off} episodes from $\mu = \{\mu_h\}_{h=1}^H$ and run for $T = N_{\text{on}}$ online episodes, inducing occupancy measures $d_h^{(t)}$ at episode t and step h. Suppose $\mathcal{L}$ satisfies the following four conditions for a collection of random error functions $\delta_h^t : \mathcal{X} \mapsto \mathbb{R}_{\ge 0}$:*

1. ***Pointwise regret decomposition:*** $\text{Reg}_{\mathcal{L}}(T) \le \sum_{t=1}^T \sum_{h=1}^H \mathbb{E}_{(s,a) \sim d_h^{(t)}}[\delta_h^t(s,a)]$.

2. ***Combined in-sample error bound:*** *With probability at least* $1-\delta$,

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\left(N_{\mathrm{off}}\mathbb{E}_{(s,a)\sim\mu_h}[\delta_h^t(s,a)^2]+\sum_{i=1}^{t-1}\mathbb{E}_{(s,a)\sim d_h^{(i)}}[\delta_h^t(s,a)^2]\right)\le\beta(\delta,H). \tag{2.17}$$

3. ***Restricted online regret bound:*** *There exist a complexity functional* $c_{\mathrm{on}}:\mathcal{P}(\mathcal{X})\times\mathbb{N}\mapsto\mathbb{R}_{\ge0}$ *and exponents* $\xi\in(0,1)$, $\gamma\in\mathbb{Z}_{\ge0}$ *such that for every subset* $\mathcal{X}'\subset\mathcal{X}$, *with probability at least* $1-\delta$,

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{(s,a)\sim d_h^{(t)}}\left[\delta_h^t(s,a)\mathbf{1}((s,a,h)\in\mathcal{X}')\right]=\mathcal{O}\Big((c_{\mathrm{on}}(\mathcal{X}',T)H^\gamma\beta(\delta,H)T)^\xi\Big). \tag{2.18}$$

4. ***Partial offline coverage:*** *For any subset* $\mathcal{X}'\subset\mathcal{X}$, *define (with* $0/0:=0$*)*

$$c_{\mathrm{off}}(\mathcal{X}'):=\sup_{t\in[T]}\sup_{h\in[H]}\sup_{\pi\in\Pi}\frac{\mathbb{E}_{d_h^\pi}[\delta_h^t(s,a)^2\mathbf{1}((s,a,h)\in\mathcal{X}')]}{\mathbb{E}_{\mu_h}[\delta_h^t(s,a)^2\mathbf{1}((s,a,h)\in\mathcal{X}')]}. \tag{2.19}$$

Then, with probability at least $1-\delta$, *the hybrid algorithm* $\mathcal{L}$ *satisfies*

$$\mathrm{Reg}_{\mathcal{L}}(T)=\mathcal{O}\left(\inf_{\mathcal{X}_{\mathrm{on}},\mathcal{X}_{\mathrm{off}}}\left\{(c_{\mathrm{on}}(\mathcal{X}_{\mathrm{on}},T)\beta(\delta,H)H^\gamma T)^\xi+H\sqrt{\beta(\delta,H)\cdot c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}})\cdot\frac{N_{\mathrm{on}}^2}{N_{\mathrm{off}}}}\right\}\right). \tag{2.20}$$

2.6. Numerical Experiments on Partitioned Exploration

To verify empirically that warm-starting an optimistic online algorithm with an offline replay buffer directs online exploration toward the under-covered partition $\mathcal{X}_{\mathrm{on}}$, we conduct simulation studies in a tabular MDP and a linear MDP. Additional reward and visitation plots are reported in Appendix A.2.

2.6.1. Tabular MDP: Forest Management

We use the forest management environment from `pymdptoolbox` (Cordwell, 2015) with $S=4$ states, $A=2$ actions (wait or cut), and horizon $H=20$. Each year the forest grows by one age class unless

it burns down (with probability 0.1, resetting to age 0) or is harvested. We warm-start UCBVI (Azar et al., 2017) with an offline dataset of $N_{\text{off}} = 100$ trajectories collected under three behavior policies: adversarial (choosing the suboptimal action 60% of the time and a uniform action 40% of the time), uniform random, and optimal. We define $\mathcal{X}_{\text{off}}$ as the state-action pairs with empirical offline occupancy at least $1/(SA)$ and $\mathcal{X}_{\text{on}} = \mathcal{X} \setminus \mathcal{X}_{\text{off}}$. Figure 2.1 plots the partial single-policy concentrability coefficients of the online trajectories collected by hybrid UCBVI versus purely online UCBVI on $\mathcal{X}_{\text{on}}$ and $\mathcal{X}_{\text{off}}$. When the behavior policy is adversarial or uniform, the hybrid agent explores $\mathcal{X}_{\text{on}}$ substantially more rapidly than the purely online agent while spending fewer episodes re-exploring $\mathcal{X}_{\text{off}}$.

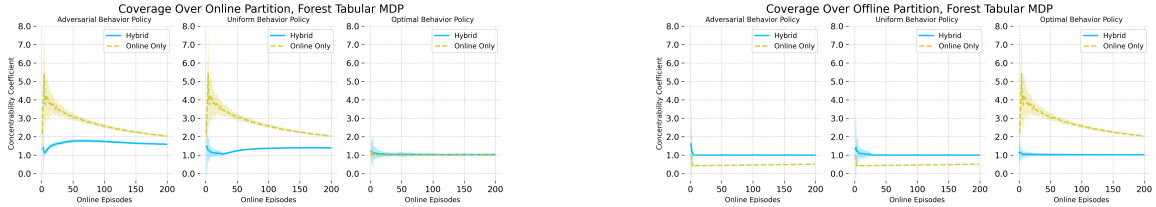


Figure 2.1: Coverage of online trajectories in the tabular forest management environment averaged over 30 trials (shaded bands indicate ±1.96 standard errors, where lower values correspond to better coverage). Left: online partition $\mathcal{X}_{\text{on}}$. Right: offline partition $\mathcal{X}_{\text{off}}$.

2.6.2. Linear MDP: Scaled-Down Tetris

We next consider a scaled-down Tetris environment with board width 6, pieces of size at most 2×2, four rotation actions $(0°, 90°, 180°, 270°)$, horizon $H = 40$, and reward equal to the negative increase in stack height above 2. Following Tan and Xu (2024), we project the 640-dimensional one-hot state-action encoding onto the 60-dimensional subspace spanned by the top 60 eigenvectors of the offline covariance matrix estimated from $N_{\text{off}} = 200$ uniform-policy trajectories, taking Φ_{off} as the top 5 eigenvectors and Φ_{on} as the remaining 55 eigenvectors. Figure 2.2 compares the full and partial all-policy concentrability coefficients of LSVI-UCB (Jin et al., 2020) with and without offline replay initialization over $N_{\text{on}} = 100$ online episodes, confirming that offline initialization accelerates exploration across both the full space and the online subspace.

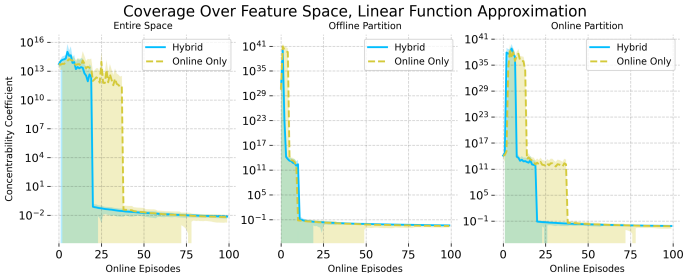


Figure 2.2: Full and partial all-policy concentrability coefficients of online samples in the linear Tetris environment over 100 online episodes (mean over 30 trials ± 1.96 standard errors, where lower is better).

2.7. Breaking Sample Size Barriers in Linear MDPs

While Algorithm 2.1 applies to any Bellman-complete function class, its specialization to d-dimensional linear MDPs in Proposition 2.6 yields a regret bound of $\tilde{\mathcal{O}}(\sqrt{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}})dH^5N_{\mathrm{on}}^2/N_{\mathrm{off}}} + \sqrt{d_{\mathrm{on}}dH^5N_{\mathrm{on}}})$. In purely online linear MDPs, the minimax-optimal regret is $\tilde{\Theta}(\sqrt{d^2H^3T})$ (Zhou et al., 2021; He et al., 2023; Hu et al., 2023; Agarwal et al., 2023), and in purely offline linear MDPs, variance-weighted pessimistic value iteration achieves the minimax-optimal instance-dependent suboptimality bound $\tilde{\mathcal{O}}(\sqrt{d}\sum_{h=1}^{H}\mathbb{E}_{\pi^\star}\|\phi(s_h,a_h)\|_{(\mathbf{\Sigma}^\star_{\mathrm{off},h})^{-1}}) \le \tilde{\mathcal{O}}(\sqrt{C^\star d^2H^4/N_{\mathrm{off}}})$ (Xiong et al., 2023b). Likewise, the reward-agnostic hybrid linear MDP algorithm of Wagenmaker and Pacchiano (2023) attains a worst-case suboptimality bound of $\tilde{\mathcal{O}}(\sqrt{d^2H^7/N})$. Thus, existing general or reward-agnostic hybrid bounds in linear MDPs lose factors of H^2 to H^4 relative to the optimal purely online and purely offline rates, as summarized in Tables 2.1 and 2.2.

Work	Function Class	Single-Policy Concentrability	Beats Pure Lower Bounds?	Guarantee
Song et al. (2023), Nakamoto et al. (2023)	General	Required	No	Regret
Amortila et al. (2024)	General	Required	No	Regret
Algorithm 2.1 (Tan and Xu, 2024)	General	Not Required	No (H^5 in linear)	Regret
Wagenmaker and Pacchiano (2023)	Linear	Not Required	No (H^7 worst-case)	PAC
Li et al. (2023)	Tabular	Not Required	Yes	PAC
Algorithms 2.2 and 2.3 (Tan et al., 2024b)	Linear	Not Required	Yes	PAC, Regret

Table 2.1: Comparison of hybrid RL approaches by function approximation class, coverage assumptions, and whether they improve upon purely offline or purely online minimax lower bounds.

In the remainder of this chapter, we study both major paradigms of hybrid RL in linear MDPs:

- **Online-to-offline PAC learning (Section 2.8):** When the goal is to output a single deter-

Setting	Upper Bound	Lower Bound
Offline Linear MDP (PAC Error)	$\sqrt{d}\sum_{h=1}^{H}\mathbb{E}_{\pi^\star}\|\phi(s_h,a_h)\|_{(\mathbf{\Sigma}^\star_{\mathrm{off},h})^{-1}} \le \sqrt{C^\star d^2H^4/N_{\mathrm{off}}}$ (Xiong et al., 2023b)	$\sqrt{C^\star d^2H^2/N_{\mathrm{off}}}$ (Xiong et al., 2023b)
Online Linear MDP (Regret)	$\sqrt{d^2H^3T}$ (He et al., 2023)	$\sqrt{d^2H^3T}$ (Zhou et al., 2021)
Hybrid Paradigm	Guarantee	
Online-to-Offline (PAC Error)	$\sqrt{d^2H^7/N}$ (Wagenmaker and Pacchiano, 2023) $\sqrt{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}})dH^3\min\{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}}),H\}/N_{\mathrm{off}}}+\sqrt{d_{\mathrm{on}}dH^3\min\{d_{\mathrm{on}},H\}/N_{\mathrm{on}}}$ (**Algorithm 2.2, Section 2.8**)	
Offline-to-Online (Regret)	$C^\star\sqrt{d^2H^6N_{\mathrm{on}}}$ (Song et al., 2023; Nakamoto et al., 2023) $\sqrt{(C^\star+c_{\mathrm{on}}(\mathcal{X}))d^3H^6N_{\mathrm{on}}}$ (Amortila et al., 2024) $\sqrt{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}})dH^5N_{\mathrm{on}}^2/N_{\mathrm{off}}}+\sqrt{d_{\mathrm{on}}dH^5N_{\mathrm{on}}}$ (**Algorithm 2.1, Section 2.1**) $\sqrt{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}})^2dH^3N_{\mathrm{on}}^2/N_{\mathrm{off}}}+\sqrt{d_{\mathrm{on}}dH^3N_{\mathrm{on}}}$ (**Algorithm 2.3, Section 2.9**)	

Table 2.2: Comparison of PAC error and cumulative regret bounds for offline, online, and hybrid RL in d-dimensional episodic linear MDPs with horizon H.

ministic policy $\widehat{\pi}$ for deployment after exploration, randomizing over an online policy history (as in regret-to-PAC conversion) is often undesirable in safety-critical applications. Moreover, reward-agnostic online exploration produces a combined dataset that can be reused across multiple downstream reward functions. Algorithm 2.2 combines reward-agnostic covariate collection (Wagenmaker and Jamieson, 2022) with variance-weighted pessimistic value iteration (Xiong et al., 2023b) to achieve an error bound of $\tilde{\mathcal{O}}(\sqrt{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}})dH^3\min\{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}}),H\}/N_{\mathrm{off}}}+\sqrt{d_{\mathrm{on}}dH^3\min\{d_{\mathrm{on}},H\}/N_{\mathrm{on}}})$.

- **Offline-to-online regret minimization (Section 2.9):** When the agent must maintain high reward throughout online interaction, Algorithm 2.3 warm-starts variance-weighted optimistic value iteration (He et al., 2023) with offline parameter and variance estimates, achieving $\mathrm{Reg}(N_{\mathrm{on}}) = \tilde{\mathcal{O}}(\inf_{\mathcal{X}_{\mathrm{off}},\mathcal{X}_{\mathrm{on}}}\{\sqrt{c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}})^2dH^3N_{\mathrm{on}}^2/N_{\mathrm{off}}}+\sqrt{d_{\mathrm{on}}dH^3N_{\mathrm{on}}}\})$ without knowing the partition.

Additional Linear MDP Notation. In a linear MDP satisfying Definition 2.5, write $\phi_{n,h} = \phi(s_h^{(n)}, a_h^{(n)})$ for the feature vector observed at episode n and step h. Let $\mathbf{\Lambda}_h = \sum_{n=1}^{N}\phi_{n,h}\phi_{n,h}^\top + \lambda\mathbf{I}$ and $\mathbf{\Lambda}_{\mathrm{off},h} = \sum_{n=1}^{N_{\mathrm{off}}}\phi_{n,h}\phi_{n,h}^\top + \lambda\mathbf{I}$ denote the unweighted ridge covariance matrices of the combined dataset and the offline dataset, respectively, and let $\mathbf{\Omega}$ denote the set of population feature covariance matrices induced by policies in Π. We consider two variance-weighted covariance matrices:

$$\mathbf{\Sigma}_h^\star = \sum_{n=1}^{N}\frac{\phi_{n,h}\phi_{n,h}^\top}{\left[\mathbb{V}_hV_{h+1}^\star\right]\left(s_h^{(n)},a_h^{(n)}\right)}+\lambda\mathbf{I}, \qquad \mathbf{\Sigma}_{t,h} = \sum_{n=1}^{N_{\mathrm{off}}+t-1}\bar{\sigma}_{n,h}^{-2}\phi_{n,h}\phi_{n,h}^\top+\lambda\mathbf{I}, \tag{2.21}$$

where $[\mathbb{V}_h V^\star_{h+1}](s,a) = \max\{1, [\mathrm{Var}_h V^\star_{h+1}](s,a)\}$ is the truncated conditional variance of the optimal value function and $\bar{\sigma}^2_{n,h}$ is the variance estimator of He et al. (2023). For any partition $\mathcal{X}_{\mathrm{on}} \cup \mathcal{X}_{\mathrm{off}} = \mathcal{X}$ with projected subspaces Φ_{off} (dimension d_{off}, projector $\mathcal{P}_{\mathrm{off}}$) and Φ_{on} (dimension d_{on}, projector $\mathcal{P}_{\mathrm{on}}$), we write the linear-MDP partial offline concentrability and partial online coverability coefficients as

$$c_{\mathrm{off}}(\mathcal{X}_{\mathrm{off}}) \coloneqq \max_{h\in[H]} \frac{1}{\lambda_{d_{\mathrm{off}}}(\mathbb{E}_{\mu_h}[(\mathcal{P}_{\mathrm{off}}\phi_h)(\mathcal{P}_{\mathrm{off}}\phi_h)^\top])}, \tag{2.22}$$

$$c_{\mathrm{on}}(\mathcal{X}_{\mathrm{on}}) \coloneqq \inf_{\pi\in\Pi} \max_{h\in[H]} \frac{1}{\lambda_{d_{\mathrm{on}}}\left(\mathbb{E}_{d^\pi_h}[(\mathcal{P}_{\mathrm{on}}\phi_h)(\mathcal{P}_{\mathrm{on}}\phi_h)^\top]\right)}. \tag{2.23}$$

2.8. Online-to-Offline PAC Learning in Linear MDPs

Given the offline dataset $\mathcal{D}_{\mathrm{off}}$ with covariance matrices $\mathbf{\Lambda}_{\mathrm{off},h}$, Algorithm 2.2 first runs the reward-agnostic covariate-collection subroutine of Wagenmaker and Jamieson (2022) (stated in full in Appendix A.3) to collect online trajectories until the maximum leverage score $\max_{\phi_h\in\Phi} \phi_h^\top(\mathbf{\Lambda}_h + \lambda\mathbf{I} + \mathbf{\Lambda}_{\mathrm{off},h})^{-1}\phi_h$ falls below a target tolerance τ. It then runs variance-weighted pessimistic value iteration (Xiong et al., 2023b) on the combined dataset $\mathcal{D}_{\mathrm{off}} \cup \mathcal{D}_{\mathrm{on}}$ to output a deterministic policy $\widehat{\pi}$.

Algorithm 2.2 Online-to-Offline Hybrid RL via Reward-Agnostic Exploration and Pessimistic Planning

1: **Input:** Offline dataset $\mathcal{D}_{\mathrm{off}}$, sample sizes N_{on}, N_{off}, feature maps ϕ_h, exploration tolerance parameter $\tau > 0$.
2: **Initialize:** $\mathcal{D}_h^{(0)} \leftarrow \emptyset$ for all $h \in [H]$, $\lambda = 1/H^2$, $\beta_2 = \tilde{O}(\sqrt{d})$.
3: **for** step $h = 1, \ldots, H$ **do**
4: Run reward-agnostic covariate collection (Wagenmaker and Jamieson, 2022) to collect online covariates $\mathbf{\Lambda}_h$ until
$$\max_{\phi_h\in\Phi} \phi_h^\top(\mathbf{\Lambda}_h + \lambda\mathbf{I} + \mathbf{\Lambda}_{\mathrm{off},h})^{-1}\phi_h \le \tau.$$
5: **end for**
6: **Output:** Policy $\widehat{\pi}$ obtained by running variance-weighted pessimistic value iteration (Xiong et al., 2023b) with parameters λ, β_2 on the combined dataset $\mathcal{D}_{\mathrm{off}} \cup \{\mathcal{D}_h^{(N_{\mathrm{on}})}\}_{h\in[H]}$.

To guarantee that the reward-agnostic exploration phase terminates, we impose a full-rank projected covariate condition analogous to that of Wagenmaker and Pacchiano (2023). This assumption is

used only for Algorithm 2.2.

Assumption 2.10 (Full-Rank Projected Covariates)**.** For any partition $\mathcal{X}_{\text{on}} \cup \mathcal{X}_{\text{off}} = \mathcal{X}$, we have $c_{\text{on}}(\mathcal{X}_{\text{on}}) < \infty$, or equivalently,

$$\sup_{\pi \in \Pi} \min_{h \in [H]} \lambda_{d_{\text{on}}} \left(\mathbb{E}_{d_h^\pi} [(\mathcal{P}_{\text{on}} \phi_h)(\mathcal{P}_{\text{on}} \phi_h)^\top] \right) = \lambda^\star_{d_{\text{on}}} > 0. \tag{2.24}$$

By applying the Kiefer–Wolfowitz theorem (Lattimore et al., 2020) on the d_{on}-dimensional projected feature space Φ_{on} and an eigenspace truncation on Φ_{off}, Lemma 2.11 bounds the partial coverability and concentrability coefficients by the subspace dimensions.

Lemma 2.11 (Bounded Partial Coverability in Linear MDPs)**.** *For any partition $\mathcal{X}_{\text{off}}, \mathcal{X}_{\text{on}}$ of $\mathcal{X}$, we have $c_{\text{on}}(\mathcal{X}_{\text{on}}) \le d_{\text{on}}$. Moreover, there always exists at least one partition such that $c_{\text{off}}(\mathcal{X}_{\text{off}}) = \mathcal{O}(d)$.*

Let $\alpha_{\text{off}} \coloneqq N_{\text{off}}/N$, $\alpha_{\text{on}} \coloneqq N_{\text{on}}/N$, and define the minimal online sample complexity required to achieve tolerance τ by

$$N^\star(\tau) \coloneqq \min \left\{ N \in \mathbb{N} : \inf_{\mathbf{\Lambda} \in \mathbf{\Omega}} \max_{\boldsymbol{\phi} \in \Phi} \boldsymbol{\phi}^\top \left(N(\mathbf{\Lambda} + \bar{\lambda}\mathbf{I}) + \mathbf{\Lambda}_{\text{off}} \right)^{-1} \boldsymbol{\phi} \le \tau \right\}. \tag{2.25}$$

Theorem 2.12 establishes both an instance-dependent suboptimality bound and a worst-case partitioned suboptimality bound for Algorithm 2.2, with the full proof given in Appendix A.4.

Theorem 2.12 (PAC Suboptimality Bound for Algorithm 2.2)**.** *Suppose Assumption 2.10 holds. For every $\delta \in (0,1)$ and any partition $\mathcal{X}_{\text{off}}, \mathcal{X}_{\text{on}}$, when $\tau \le \tilde{\mathcal{O}}(\max\{d_{\text{on}}/N_{\text{on}}, c_{\text{off}}(\mathcal{X}_{\text{off}})/N_{\text{off}}\})$ and $N \ge \max\{\alpha_{\text{on}}^4 d_{\text{on}}^{-4}, \alpha_{\text{off}}^4 c_{\text{off}}(\mathcal{X}_{\text{off}})^{-4}\} \max\{N^\star(\tau), \text{poly}(d, H, c_{\text{off}}(\mathcal{X}_{\text{off}}), \log(1/\delta))\}$, Algorithm 2.2 achieves with probability at least $1 - \delta$:*

$$V_1^\star(s) - V_1^{\widehat{\pi}}(s) \lesssim \sqrt{d} \sum_{h=1}^H \mathbb{E}_{\pi^\star} \|\phi(s_h, a_h)\|_{(\mathbf{\Sigma}^\star_{\text{off},h} + \mathbf{\Sigma}^\star_{\text{on},h})^{-1}} \le \sqrt{d} \sum_{h=1}^H \mathbb{E}_{\pi^\star} \|\phi(s_h, a_h)\|_{(\mathbf{\Sigma}^\star_{\text{off},h})^{-1}}, \tag{2.26}$$

$$V_1^\star(s) - V_1^{\widehat{\pi}}(s) \lesssim \min \left\{ \sqrt{\frac{c_{\text{off}}(\mathcal{X}_{\text{off}}) d H^4}{N_{\text{off}}}} + \sqrt{\frac{d_{\text{on}} d H^4}{N_{\text{on}}}}, \sqrt{\frac{c_{\text{off}}(\mathcal{X}_{\text{off}})^2 d H^3}{N_{\text{off}} \alpha_{\text{off}}}} + \sqrt{\frac{d_{\text{on}}^2 d H^3}{N_{\text{on}} \alpha_{\text{on}}}} \right\}. \tag{2.27}$$

When specialized to tabular MDPs with one-hot features $\phi(s,a) = \boldsymbol{e}_{s,a} \in \mathbb{R}^{SA}$, where $\pi^\star$ is deterministic, Theorem 2.12 yields an H^3 horizon dependence without the extra factor of $c_{\text{off}}(\mathcal{X}_{\text{off}})$ or d_{on} (proof in Appendix A.4).

Corollary 2.13 (Tabular Specialization of Algorithm 2.2)**.** *In a tabular MDP, for every $\delta \in (0,1)$, with probability at least $1-\delta$, Algorithm 2.2 satisfies*

$$V_1^\star(s) - V_1^{\widehat{\pi}}(s) \lesssim \sqrt{H^3 S^2 A}\left(\sqrt{\frac{c_{\text{off}}(\mathcal{X}_{\text{off}})}{N_{\text{off}}}} + \sqrt{\frac{d_{\text{on}}}{N_{\text{on}}}}\right). \tag{2.28}$$

Proof Sketch for Theorem 2.12. Equation (2.26) follows from invoking the instance-dependent suboptimality guarantee of Xiong et al. (2023b, Theorem 2) on the combined dataset $\mathcal{D}_{\text{off}} \cup \mathcal{D}_{\text{on}}$ and noting that $\boldsymbol{\Sigma}_h^\star = \boldsymbol{\Sigma}_{\text{off},h}^\star + \boldsymbol{\Sigma}_{\text{on},h}^\star \succeq \boldsymbol{\Sigma}_{\text{off},h}^\star$. To obtain Equation (2.27), we decompose the expectation under $\pi^\star$ across Φ_{on} and Φ_{off}:

$$V_1^\star(s) - V_1^{\widehat{\pi}}(s) \lesssim \sqrt{d}\sum_{h=1}^{H} \max_{\phi_h \in \Phi_{\text{on}}} \sqrt{\phi_h^\top (\boldsymbol{\Sigma}_h^\star)^{-1} \phi_h} + \sqrt{d}\sum_{h=1}^{H} \max_{\phi_h \in \Phi_{\text{off}}} \sqrt{\phi_h^\top (\boldsymbol{\Sigma}_h^\star)^{-1} \phi_h}. \tag{2.29}$$

Using $(\boldsymbol{\Sigma}_h^\star)^{-1} \preceq H^2 \boldsymbol{\Lambda}_h^{-1}$ together with Lemma 2.11 and the stopping condition $\max_{\phi_h \in \Phi} \phi_h^\top \boldsymbol{\Lambda}_h^{-1} \phi_h \lesssim \max\{c_{\text{off}}(\mathcal{X}_{\text{off}})/N_{\text{off}}, d_{\text{on}}/N_{\text{on}}\}$ immediately gives the first bound $\sqrt{c_{\text{off}}(\mathcal{X}_{\text{off}}) d H^4/N_{\text{off}}} + \sqrt{d_{\text{on}} d H^4/N_{\text{on}}}$. To sharpen the horizon dependence from H^4 to H^3, we observe that by the total variance lemma (Jin et al., 2018, Lemma C.5), the empirical average over (n,h) of $[\mathbb{V}_h V_{h+1}^\star](s_h^{(n)}, a_h^{(n)})$ is $\tilde{\mathcal{O}}(H)$ rather than H^2. In Lemma A.4, we truncate the high-variance trajectories at each step h where $[\mathbb{V}_h V_{h+1}^\star] \geq C_h \bar{\sigma}_h$, bound their cardinality via Markov's inequality, and choose C_h so that the remaining low-variance trajectories preserve at least half of the minimum eigenvalue of $\boldsymbol{\Lambda}_h$, yielding the second bound in Equation (2.27).

2.9. Offline-to-Online Regret Minimization in Linear MDPs

When the agent must minimize cumulative regret during online interaction, reward-agnostic exploration is unsuitable. Algorithm 2.3 warm-starts variance-weighted optimistic value iteration (He et al., 2023) by computing initial optimistic and pessimistic weight vectors $\widehat{\boldsymbol{w}}_{1,h}, \check{\boldsymbol{w}}_{1,h}$, action-

value functions $Q_{1,h}, \check{Q}_{1,h}$, and variance weights $\bar{\sigma}_{n,h}^2$ on the N_{off} offline episodes, and initializing the precision matrix as $\mathbf{\Sigma}_{1,h} = \sum_{n=1}^{N_{\text{off}}} \bar{\sigma}_{n,h}^{-2} \phi_{n,h} \phi_{n,h}^\top + \lambda \mathbf{I}$. The unabridged pseudocode is given in Appendix A.3.

Algorithm 2.3 Offline-to-Online Hybrid RL via Variance-Weighted Optimistic Value Iteration

Input: Offline dataset $\mathcal{D}_{\text{off}}$, sample sizes N_{on}, N_{off}, feature maps ϕ_h, regularization $\lambda > 0$, confidence radii $\beta, \bar{\beta}, \tilde{\beta}$, determinant update index $t_{\text{last}} = 0$.

Initialize: For each $h \in [H]$, estimate $\widehat{\mathbf{w}}_{1,h}, \check{\mathbf{w}}_{1,h}, Q_{1,h}, \check{Q}_{1,h}, \sigma_{n,h}, \bar{\sigma}_{n,h}$ from $\mathcal{D}_{\text{off}}$, and set

$$\mathbf{\Sigma}_{0,h} = \mathbf{\Sigma}_{1,h} = \mathbf{\Sigma}_{\text{off},h} + \lambda \mathbf{I} = \sum_{n=1}^{N_{\text{off}}} \bar{\sigma}_{n,h}^{-2} \phi_{n,h} \phi_{n,h}^\top + \lambda \mathbf{I}.$$

for episode $t = 1, \ldots, T$ **do**

  Update optimistic and pessimistic weights $\widehat{\mathbf{w}}_{t,h}, \check{\mathbf{w}}_{t,h}$ for all $h \in [H]$.

  if there exists $h' \in [H]$ such that $\det(\mathbf{\Sigma}_{t,h'}) \geq 2 \det(\mathbf{\Sigma}_{t_{\text{last}},h'})$ **then**

    Update optimistic and pessimistic action-value functions $Q_{t,h}(s,a), \check{Q}_{t,h}(s,a)$ for all $h \in [H]$ and set $t_{\text{last}} = t$.

  end if

  for step $h = 1, \ldots, H$ **do**

    Execute action $a_h^{(t)} \leftarrow \arg\max_{a \in \mathcal{A}} Q_{t,h}(s_h^{(t)}, a)$ and observe $r_h^{(t)}, s_{h+1}^{(t)}$.

    Compute variance estimate $\sigma_{t,h}$, set $\bar{\sigma}_{t,h} \leftarrow \max\{\sigma_{t,h}, \sqrt{H}, 2d^3H^2 \|\boldsymbol{\phi}(s_h^{(t)}, a_h^{(t)})\|_{\mathbf{\Sigma}_{t,h}^{-1}}^{1/2}\}$, and update $\mathbf{\Sigma}_{t+1,h} \leftarrow \mathbf{\Sigma}_{t,h} + \bar{\sigma}_{t,h}^{-2} \phi_{t,h} \phi_{t,h}^\top$.

  end for

end for

Output: Greedy policy $\widehat{\pi} = \pi^{Q_{T,h}}$ (or uniform mixture $\text{Unif}(\pi^{Q_{1,h}}, \ldots, \pi^{Q_{T,h}})$ for PAC conversion).

Theorem 2.14 establishes the cumulative regret bound of Algorithm 2.3, with the full proof given in Appendix A.4.

Theorem 2.14 (Regret Bound for Algorithm 2.3)**.** *For any $\delta \in (0,1)$, if $N_{\text{on}}, N_{\text{off}} = \tilde{\Omega}(d^{13}H^{14})$, then with probability at least $1-\delta$, Algorithm 2.3 satisfies*

$$\text{Reg}(N_{\text{on}}) \lesssim \inf_{\mathcal{X}_{\text{off}}, \mathcal{X}_{\text{on}}} \left\{ \sqrt{c_{\text{off}}(\mathcal{X}_{\text{off}})^2 dH^3 \frac{N_{\text{on}}^2}{N_{\text{off}}}} + \sqrt{d_{\text{on}} dH^3 N_{\text{on}}} \right\}. \tag{2.30}$$

Corollary 2.15 (Regret-to-PAC Conversion for Algorithm 2.3)**.** *Under the conditions of Theorem 2.14, the uniform mixture policy $\widehat{\pi} = \text{Unif}(\pi^{Q_{1,h}}, \ldots, \pi^{Q_{T,h}})$ output by Algorithm 2.3 satisfies,*

with probability at least $1-\delta$,

$$V_1^\star(s) - V_1^{\widehat{\pi}}(s) \lesssim \inf_{\mathcal{X}_{\text{off}}, \mathcal{X}_{\text{on}}} \left\{ \sqrt{\frac{c_{\text{off}}(\mathcal{X}_{\text{off}})^2 d H^3}{N_{\text{off}}}} + \sqrt{\frac{d_{\text{on}} d H^3}{N_{\text{on}}}} \right\}. \tag{2.31}$$

Comparison and Proof Sketch for Theorem 2.14. Like Algorithm 2.1, Algorithm 2.3 requires no knowledge of the partition $(\mathcal{X}_{\text{off}}, \mathcal{X}_{\text{on}})$ and automatically adapts to the infimum over all partitions. Taking $\mathcal{X}_{\text{on}} = \mathcal{X}$ and $\mathcal{X}_{\text{off}} = \emptyset$ immediately recovers the minimax-optimal purely online regret bound $\tilde{\mathcal{O}}(\sqrt{d^2 H^3 N_{\text{on}}})$ of He et al. (2023), while in the offline-abundant regime $N_{\text{off}} \gg N_{\text{on}}$ the online regret scales as $\tilde{\mathcal{O}}(\sqrt{d_{\text{on}} d H^3 N_{\text{on}}})$, breaking the $\Omega(\sqrt{d^2 H^3 N_{\text{on}}})$ purely online lower bound whenever the offline dataset covers a subspace of dimension $d_{\text{off}} = d - d_{\text{on}} \geq 1$. Compared to Proposition 2.6 for Algorithm 2.1, Theorem 2.14 sharpens the horizon dependence from H^5 to H^3.

To prove Theorem 2.14, we decompose the exploration bonuses across $\mathcal{X}_{\text{off}}$ and $\mathcal{X}_{\text{on}}$:

$$\text{Reg}(T) \lesssim \sqrt{H^3 T \log(H/\delta)} + \beta \sum_{t=1}^{T} \sum_{h=1}^{H} \|\boldsymbol{\Sigma}_{t,h}^{-1/2} \phi_h(s_h^{(t)}, a_h^{(t)}) \mathbf{1}_{\mathcal{X}_{\text{off}}}\|_2 + \beta \sum_{t=1}^{T} \sum_{h=1}^{H} \|\boldsymbol{\Sigma}_{t,h}^{-1/2} \phi_h(s_h^{(t)}, a_h^{(t)}) \mathbf{1}_{\mathcal{X}_{\text{on}}}\|_2. \tag{2.32}$$

On the offline partition $\mathcal{X}_{\text{off}}$, since $\boldsymbol{\Sigma}_{t,h} \succeq \boldsymbol{\Sigma}_{\text{off},h} + \lambda \mathbf{I}$, Lemma A.4 bounds the sum of bonuses by $\sum_{h=1}^{H} \sqrt{d(N_{\text{on}}^2/N_{\text{off}})} \sqrt{\max_{\phi_h \in \Phi_{\text{off}}} \phi_h^\top \bar{\boldsymbol{\Sigma}}_{\text{off},h}^{-1} \phi_h}$. Controlling $\bar{\boldsymbol{\Sigma}}_{\text{off},h}^{-1}$ with H^3 rather than H^4 or H^8 requires ruling out the trajectories $n \in \mathcal{I}_2 \subset [N_{\text{off}}]$ where the leverage-score floor $4d^6 H^4 \|\phi_{n,h}\|_{\boldsymbol{\Sigma}_{n,h}^{-1}}$ dominates $\bar{\sigma}_{n,h}^2$. Using the elliptical potential lemma, we show $|\mathcal{I}_2| \lesssim d^{13} H^5 \log(1 + N/d) = \tilde{o}(N_{\text{off}})$, and then apply the variance-truncation bound in Lemma A.4 together with the total variance bound $\frac{1}{N_{\text{off}}} \sum_{n,h} \sigma_{n,h}^2 = \tilde{\mathcal{O}}(H^2)$ on the remaining good trajectories $\mathcal{I}_1$ in Lemma A.4. On the online partition $\mathcal{X}_{\text{on}}$, we establish a projected subspace elliptical potential bound in Lemma A.4 using thin-SVD projection $U_{\text{on}} U_{\text{on}}^\top$ inside the matrix inverse (Lemma A.4), reducing the dimension factor in the variance-weighted bonus sum from d to d_{on} in Lemma A.4.

2.10. Numerical Experiments in Linear MDPs

We evaluate Algorithms 2.2 and 2.3 on the scaled-down linear Tetris environment described in Section 2.6 (60-dimensional SVD feature representation, $N_{\text{off}} = 200$ offline trajectories).

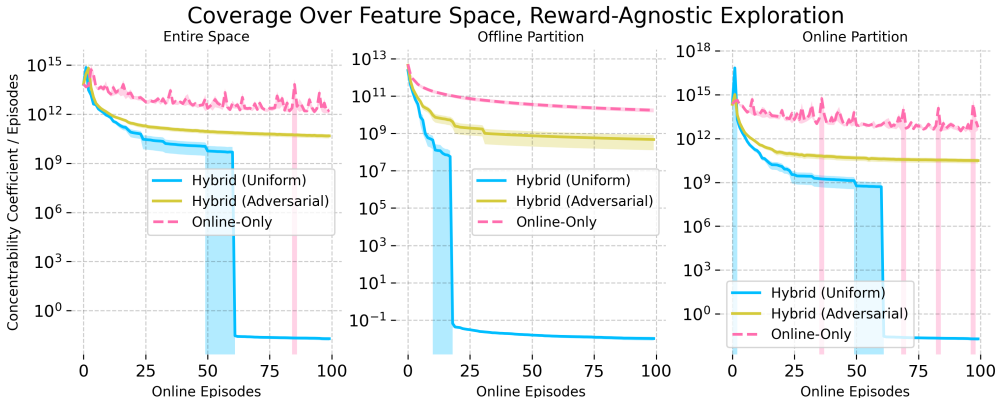


Figure 2.3: Inverse minimum eigenvalue coverage ($1/\lambda_{\min}(\mathbf{\Lambda})$, $1/\lambda_{d_{\text{off}}}(\mathbf{\Lambda}_{\text{off}})$, $1/\lambda_{d_{\text{on}}}(\mathbf{\Lambda}_{\text{on}})$) during the reward-agnostic online exploration phase (Algorithm 2.2) initialized with 200 uniform offline trajectories, 200 adversarial offline trajectories, or no offline data (mean over 30 trials ± 1.96 standard errors, where lower is better).

Figure 2.3 shows the covariate coverage achieved during the exploration phase of Algorithm 2.2 when initialized with 200 uniform-policy trajectories, 200 adversarial-policy trajectories (generated by negating the weights of a trained policy), or no offline data. Even with adversarially collected offline data, hybrid exploration achieves better conditioning across both the full space and the online subspace than purely online exploration.

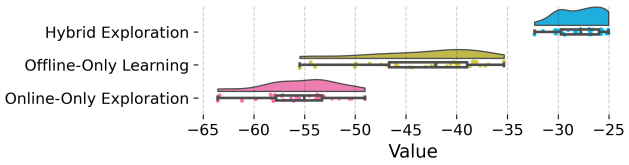


Figure 2.4: Policy value learned by pessimistic value iteration after reward-agnostic exploration (Algorithm 2.2) on the hybrid dataset (200 adversarial offline and 100 online trajectories) compared against 300 adversarial offline trajectories and 300 purely online reward-agnostic trajectories over 30 trials (higher is better).

Figure 2.4 compares the value of the policy output by Algorithm 2.2 using 200 adversarial offline trajectories and 100 online exploration trajectories against purely offline learning (300 adversarial trajectories) and purely online reward-agnostic exploration (300 online trajectories). The hybrid dataset yields substantially higher policy value than either pure baseline. Finally, Figure 2.5 compares Algorithm 2.3 against purely online LSVI-UCB++ (He et al., 2023), confirming that offline initialization yields strictly lower cumulative regret and faster reward convergence.

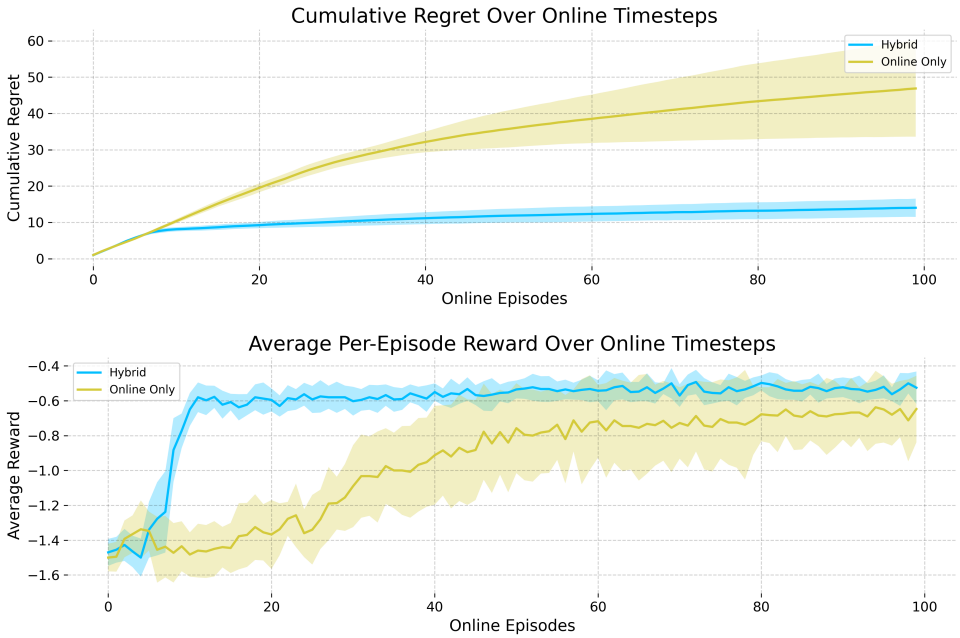


Figure 2.5: Cumulative regret (top) and per-episode reward (bottom) of offline-to-online variance-weighted optimistic value iteration (Algorithm 2.3) versus purely online LSVI-UCB++ (He et al., 2023) in the linear Tetris environment (mean ± 1 standard deviation over 10 trials).

2.11. Discussion

Together, Algorithms 2.1, 2.2, and 2.3 show that offline datasets with partial coverage can be exploited without single-policy concentrability by partitioning the state-action space into an offline-covered component $\mathcal{X}_{\text{off}}$ and an online-explored component $\mathcal{X}_{\text{on}}$. Under general function approximation, Algorithm 2.1 achieves automatic adaptivity to the optimal partition simply by retaining the offline dataset in its experience replay buffer. In linear MDPs, Algorithms 2.2 and 2.3 combine this subspace decomposition with variance-weighted regression and variance truncation to achieve H^3 horizon dependence and $d_{\text{on}}d$ online dimension dependence in both the online-to-offline PAC setting and the offline-to-online regret setting.

In the next chapter, we turn from learning a single decision process from partial-coverage offline data to learning shared low-dimensional representation structure across heterogeneous tasks or users in latent contextual bandits.

CHAPTER 3

LEARNING SHARED STRUCTURE FROM HETEROGENEOUS DATA

This chapter is based on joint work with Chinmaya Kausik and Ambuj Tewari (Kausik et al., 2024): Chinmaya Kausik, Kevin Tan, and Ambuj Tewari, "Leveraging Offline Data in Linear Latent Contextual Bandits," in Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR 235:23476–23516, 2024.

3.1. Introduction

Many sequential decision-making problems can be effectively modeled using the bandit framework. This can span domains as diverse as healthcare (Lu et al., 2021b), randomized clinical trials (Press, 2009), search and recommendation (Li et al., 2010), distributed networks (Kar et al., 2011), and portfolio design (Brochu et al., 2011). There is often a wealth of offline data in such domains, which has led to a growing interest in using offline data to accelerate online learning. However, there often also exist unobserved contexts in the population that influence the distribution of rewards, making it non-trivial to leverage offline data. In Hong et al. (2020), it is shown that this uncertainty can be modeled by a *latent bandit* (or mixture of bandits). This is a bandit where an unobserved latent state determines the reward model for the trajectory. For example, a patient's underlying genetic conditions in healthcare and a user's tastes in recommendation systems are both examples of latent states in sequential decision making. Typically, these latent states are less complex than the actual models underlying users or patients, making it valuable to reduce the online task to learning the latent state (Hong et al., 2020, 2022).

The latent bandit framework therefore has high practical value, and efficient principled algorithms are needed for using offline data to speed up online learning. Using traditional bandit algorithms in this setting does not leverage the offline data that is often available to the agent. Naturally, one also cannot treat the offline data as coming from a single bandit. For example, different user tastes or different underlying genetic conditions require modeling the offline data as coming from a latent bandit. So, we have to develop algorithms *specific* to the latent bandit setting that leverage the

offline data to improve online performance.

We note that in bandit literature, it is common to impose a structure on bandit rewards when designing algorithms, the most popular one being a linear structure (Li et al., 2010; Abbasi-Yadkori et al., 2011). In this light, we study a linear contextual bandit setting where each user has its own high-dimensional reward parameter, but reward parameters across users lie in a low-rank subspace. This is a *linear latent contextual bandit*[1], and is much more general than existing models that restrict themselves to finitely many latent states (Hong et al., 2020, 2022). We design a two-pronged algorithm to tackle this setting. First, we provide a method to approximate the low-dimensional subspace spanned by latent states from an offline dataset of unlabeled trajectories collected under some behavior policy π_b. This is non-trivial since the trajectories are unlabeled, and standard unsupervised learning methods fail. Second, we use this subspace to speed up online learning. However, since the subspace is only learnt *approximately,* we also tackle the non-trivial task of accounting for the uncertainty in the subspace. We design two methods for the latter, facing a trade-off between computational tractability and tightness of guarantees. Experiments show the efficacy of our methods. We outline our contributions below:

- **Offline method:** We present an offline moment-corrected subspace estimator (Algorithm 3.1 in Section 3.3), a novel offline method for learning low-dimensional subspaces of reward parameters with guarantees, inspired by the spectral methods in Kausik et al. (2023).

- **Tight online algorithm:** We present a subspace-constrained optimistic linear bandit algorithm (Algorithm 3.2 in Section 3.4), an online algorithm leveraging the subspace estimated offline to sharpen optimism, achieving $\tilde{O}(\min(d_A\sqrt{T}, d_K\sqrt{T}(1+\sqrt{d_AT/d_KN})))$ regret.

- **Lower Bound:** We establish a matching lower bound showing that Algorithm 3.2 is minimax optimal. To the best of our knowledge, this is the first lower bound in a hybrid (offline-online) sequential decision-making setting.

- **Tractable online algorithm:** Finally, we present a projected confidence-ball optimistic

[1]This can also be thought of as a continuous mixture of bandits.

linear bandit algorithm (Algorithm 3.3 in Section 3.6), a practical and computationally efficient online algorithm with a slightly looser regret guarantee. This also illustrates a general algorithmic idea for integrating offline subspace estimation into optimistic algorithms.

- **Experiments:** We establish the efficacy of our algorithms outlined above through a simulation study and a demonstration on a real recommendation problem with the MovieLens-1M (Harper and Konstan, 2015) dataset.

Related work. There are three main threads of related work.

- **Latent Bandits.** The line of work most relevant to us has been on latent bandits. The work of Hong et al. (2020, 2022) studies the latent bandit problem under finitely many states. However, they black-box the offline step and do not provide end-to-end guarantees, and their ideas do not extend to infinitely many states. Our work seeks to provide end-to-end guarantees for both the offline and online component under infinitely many latent states.

- **Meta learning, multi-task learning and mixture learning.** A long line of work studies learning with multiple underlying tasks or models. For example, the work of Vempala and Wang (2004); Kong et al. (2020); Anandkumar et al. (2014); Tripuraneni et al. (2022) study learning under latent variable or multi-task models in a supervised setting. The work of Kausik et al. (2023); Chen and Poor (2022) extend some of these ideas to unsupervised but purely offline learning in a time-series setting. On the other hand, work like Yang et al. (2022); Cella et al. (2022) instead focuses on the purely online setting of learning the low-rank structure while simultaneously interacting with multiple finitely many bandit instances. Finally, Zhou et al. (2024); Lu et al. (2021a) work with multiple underlying models in MDPs but in a purely offline and generative model setting respectively. Unlike these papers, we crucially combine the offline and online settings for sequential data and study the problem of using offline data to accelerate online learning in bandits.

- **Hybrid (offline-online) RL.** Work in hybrid RL studies the use of offline data to accelerate online RL, first proposed by Song et al. (2023), with extensions to linear MDPs by

Wagenmaker and Pacchiano (2023); Tan et al. (2024b). Cai et al. (2024) studies the same problem for contextual bandits. However, all work so far assumes that the offline data is generated by a single model, and does not account for latent states. Our work explores a hybrid offline-online setting while also accounting for the offline data being generated by multiple underlying models.

3.2. Linear Bandits With Latent Structure

We will first introduce latent contextual bandits, and then specialize to our linear model later in this section. A latent contextual bandit is a decision process with contexts $\mathcal{X}$, actions $\mathcal{A}$ and a random latent state θ that is sampled independently at the beginning of each trajectory. Given a sequence of contexts and actions, the rewards at all steps are independent *conditioned* on θ, and depend on the latent state θ, the context and the action. Since we often have access to an offline dataset of trajectories coming from different kinds of users or patients, it is important to account for a changing latent state θ between trajectories.

Latent bandits are a powerful and general framework for encoding uncertainty in reward models. However, this generality is both a blessing and a curse. It is hard to design concrete algorithms without further assumptions on the structure and effect of the latent state θ. We therefore focus on a linear structure here. We consider the natural generalization of the linear contextual bandit to the latent bandit setting, where we impose a linear structure on the effect of low-dimensional latent states. We further justify this by noting that in most application domains, it is reasonable to assume that a parsimonious, low-dimensional latent state affects the reward distribution. This motivates the following definition.

Definition 3.1. A *linear latent contextual bandit* is a linear bandit equipped with a feature map for context-action pairs $\phi : \mathcal{X} \times \mathcal{A} \to \mathbb{R}^{d_A}$, a latent random variable $\boldsymbol{\theta} \in \mathbb{R}^{d_K}$ with distribution $\mathcal{D}_\theta$ and a map $\mathbf{U}_\star : \mathbb{R}^{d_K} \to \mathbb{R}^{d_A}$ such that for any H and context-action sequence $((x_1, a_1), \dots (x_H, a_H))$, the rewards $(Y_1, \dots Y_H)$ are independent conditioned on $\boldsymbol{\theta}$. Moreover, $Y_h \mid \boldsymbol{\theta} \sim \phi(x_h, a_h)^\top \boldsymbol{\beta} + \epsilon$, where ϵ is subgaussian noise independent of all actions and all other observations, and $\boldsymbol{\beta} = \mathbf{U}_\star \boldsymbol{\theta}$.

Further, we note that WLOG $\mathbf{U}_\star$ is unitary. This is because for any invertible map $A : \mathbb{R}^{d_K} \to \mathbb{R}^{d_K}$, the observation distribution does not change upon replacing $\boldsymbol{\theta}$ with $A\boldsymbol{\theta}$ and $\mathbf{U}_\star$ with $\mathbf{U}_\star A^{-1}$. One can see this as a generalization of the fact that with finitely many latent states, permuting the labels and rewards together does not change observations.

Let us now assume that we have access to a dataset $\mathcal{D}_{\text{off}}$ of N *short* trajectories $\tau_n = ((x_{n,1}, a_{n,1}, r_{n,1}), ..., (x_{n,H}, a_{n,H}, r_{n,H}))$ of length H, collected by some behavior policy π_b. The trajectories are short in the sense that in most relevant domains, individual trajectories are not long enough to learn the underlying reward model. Each trajectory τ_n has a different $\boldsymbol{\beta}_n = \mathbf{U}_\star \boldsymbol{\theta}_n$. In online deployment, a single latent label $\boldsymbol{\theta}_\star$ is chosen and rewards are generated using $\boldsymbol{\beta}_\star = \mathbf{U}_\star \boldsymbol{\theta}_\star$. At each timestep t, an agent observes contexts x_t and uses both the offline data and the online data at time t to execute a policy π_t. Define the optimal action at time t by $a_t^\star := \arg\max_{a \in \mathcal{A}} \phi(x_t, a)^\top \boldsymbol{\beta}_\star$. We tackle the problem of minimizing the *frequentist* regret in linear latent contextual bandits, given by

$$\text{Reg}_T := \textstyle\sum_{t=1}^T \phi(x_t, a_t^\star)^\top \boldsymbol{\beta}_\star - \mathbb{E}_{a \sim \pi_t}[\phi(x_t, a)^\top \boldsymbol{\beta}_\star].$$

For example, in medical applications, data from short randomized controlled trials can be used to help an agent suggest treatment decisions for a new patient online. In this case, we would like the algorithm to administer the correct treatments for *each* patient. This means that the *frequentist* regret is the relevant performance metric here, and not the Bayesian regret over some prior. Additionally, any worst-case bound on the frequentist regret is a bound on the Bayesian regret for arbitrary priors.

Challenges with latent bandits. Despite the linear assumption, and the dimension reduction obtained in the common case when $d_K \ll d_A$, significant challenges remain. First, the value of the latent state $\boldsymbol{\theta}$ and the map $\mathbf{U}_\star$ are both unknown a priori, making it hard to leverage the low-dimensional structure of the problem. Second, a good choice of dimension d_K is itself unknown a priori, and must be determined from data in a principled manner. Third, even if we learn the low-dimensional structure, our learning will be *approximate*, and the online procedure must account for this uncertainty. In the following sections, we will provide a method to estimate and use latent

subspaces given offline data that allows us to overcome these challenges.

Additional Notation. We use $\mathbf{V}$ to denote regularized design matrices given by $\mu\mathbf{I} + \sum_{(x,a)} \phi(x,a)\phi(x,a)^\top$. We define $\mathbf{D}_{n,i} = I - \mu\mathbf{V}_{n,i}^{-1}$ and $\overline{\mathbf{D}}_{N,i} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{D}_{n,i}$. Denote by $\hat{\beta}_{n,1}, \hat{\beta}_{n,2}$ independent estimates of β_n from τ_n. Let $\mathbf{M}_n = \frac{1}{2}(\hat{\boldsymbol{\beta}}_{n,1}\hat{\boldsymbol{\beta}}_{n,2}^\top + \hat{\boldsymbol{\beta}}_{n,2}\hat{\boldsymbol{\beta}}_{n,1}^\top)$ and $\overline{\mathbf{M}}_N \leftarrow \frac{1}{N}\sum_{n=1}^{N} \mathbf{M}_n$.

3.3. Estimating Latent Subspaces Offline

Although we do not have access to the values of the latent states $\boldsymbol{\theta}$ or to the map $\mathbf{U}_\star$, we can still extract useful information from data. To that effect, recall that we have access to a dataset $\mathcal{D}_{\text{off}}$ of N trajectories $\tau_n = ((x_{n,h}, a_{n,h}, r_{n,h}))_{h=1}^{H}$ of length H, collected by some behavior policy π_b.

How can offline data help us in online deployment? To minimize the regret, one must learn the reward parameter $\boldsymbol{\beta}_\star$ online. However, it is much easier to search among all latent states $\boldsymbol{\theta} \in \mathbb{R}^{d_K}$ than to search among all possible reward parameters $\boldsymbol{\beta} \in \mathbb{R}^{d_A}$ since typically, $d_K \ll d_A$. So, it will help to learn some projection matrix $\hat{\mathbf{U}}^\top \approx \mathbf{U}_*^\top : \mathbb{R}^{d_A} \to \mathbb{R}^{d_K}$ offline so that for any estimate $\hat{\boldsymbol{\beta}}_t$ of $\boldsymbol{\beta}_\star$, $\hat{\mathbf{U}}^\top\hat{\boldsymbol{\beta}}_t$ is an estimate of $\hat{\boldsymbol{\theta}}_t \in \mathbb{R}^{d_K}$. This amounts to *learning a subspace* of the feature space from logged bandit data. We therefore provide a method for offline moment-corrected subspace estimation in Algorithm 3.1. Recall that since the learnt subspace is approximate, we also need to compute the uncertainty over the subspace to get a subspace confidence set that we can use online.

Algorithm 3.1 Offline Moment-Corrected Subspace Estimation

1: **Input:** Dataset $\mathcal{D}_{\text{off}}$ of collected trajectories $\tau_n = ((x_{n,1}, a_{n,1}, r_{n,1}), ..., (x_{n,H}, a_{n,H}, r_{n,H}))$ under a behavior policy π_b, dimension of latent subspace d_K.
2: **Divide** each τ_n into odd and even steps, giving trajectory halves $\tau_{n,1}$ and $\tau_{n,2}$.
3: **Estimate** reward parameters $\hat{\boldsymbol{\beta}}_{n,i} \leftarrow \mathbf{V}_{n,i}^{-1}\mathbf{b}_{n,i}$, where $\mathbf{V}_{n,i} \leftarrow \mu\mathbf{I} + \sum_{(x,a,r)\in\tau_{n,i}} \phi(x,a)\phi(x,a)^\top$ and $\mathbf{b}_{n,i} \leftarrow \sum_{(x,a,r)\in\tau_{n,i}} \phi(x,a)r$ for $i = 1, 2$.
4: **Compute** $\mathbf{M}_n \leftarrow \frac{1}{2}(\hat{\boldsymbol{\beta}}_{n,1}\hat{\boldsymbol{\beta}}_{n,2}^\top + \hat{\boldsymbol{\beta}}_{n,2}\hat{\boldsymbol{\beta}}_{n,1}^\top)$ and compute $\overline{\mathbf{M}}_N \leftarrow \frac{1}{N}\sum_{n=1}^{N} \mathbf{M}_n$.
5: **Compute** $\overline{\mathbf{D}}_{N,i} \leftarrow \frac{1}{N}\sum_{n=1}^{N}(I - \mu\mathbf{V}_{n,i}^{-1})$, $i = 1, 2$.
6: **Obtain** $\hat{\mathbf{U}}$, the top d_K eigenvectors of $\overline{\mathbf{D}}_{N,1}^{-1}\overline{\mathbf{M}}_N\overline{\mathbf{D}}_{N,2}^{-1}$. **return** Projection matrix $\hat{\mathbf{U}}\hat{\mathbf{U}}^\top$, Δ_{off} as in Theorem 3.5

On trajectory splitting and corrections. To extract the d_K-dimensional subspace, we aim to estimate $\mathbb{E}[\boldsymbol{\beta}\boldsymbol{\beta}^\top] = \mathbf{U}_\star\mathbb{E}[\boldsymbol{\theta}\boldsymbol{\theta}^\top]\mathbf{U}_\star^\top$, which has the same d_K-dimensional span as $\mathbf{U}_\star$. This cannot

be achieved by using a single estimator $\hat{\boldsymbol{\beta}}_n$ for each trajectory τ_n and averaging the outer products $\hat{\boldsymbol{\beta}}_n\hat{\boldsymbol{\beta}}_n^\top$ across all n. That is because the per-reward noise ϵ will be shared by both copies of $\hat{\boldsymbol{\beta}}_n$, and so the variance of ϵ will make $\mathbb{E}[\hat{\boldsymbol{\beta}}_n\hat{\boldsymbol{\beta}}_n^\top]$ full rank. We therefore split each trajectory τ_n to obtain two independent estimators $\hat{\boldsymbol{\beta}}_{n,1}, \hat{\boldsymbol{\beta}}_{n,2}$, compute the outer products $\hat{\boldsymbol{\beta}}_{n,1}\hat{\boldsymbol{\beta}}_{n,2}^\top$, and obtain the top d_K eigenvectors of the mean outer product across trajectories.

However, there is a further wrinkle here. We cannot simply take the top d_K eigenvectors of the mean outer product $\overline{\mathbf{M}}_N$. One can compute that $\mathbb{E}[\overline{\mathbf{M}}_N] = \mathbb{E}[\mathbf{D}_{n,1}\boldsymbol{\beta}_n\boldsymbol{\beta}_n^\top\mathbf{D}_{n,2}] = \mathbb{E}[\mathbf{D}_{n,1}\mathbf{U}_\star\boldsymbol{\theta}_n\boldsymbol{\theta}_n^\top\mathbf{U}_\star^\top\mathbf{D}_{n,2}]$. To separate $\mathbb{E}[\boldsymbol{\beta}_n\boldsymbol{\beta}_n^\top]$ from this, we need $\mathbf{D}_{n,1}, \boldsymbol{\beta}_n, \mathbf{D}_{n,2}$ to be independent. If π_b does not use $\boldsymbol{\theta}$ and contexts are generated independently of each other and of $\boldsymbol{\theta}$, then this is satisfied. Intuitively, we need the offline trajectories to be non-adaptive. In fact, we show in the lemma below that if any of these three conditions is violated, then it is in fact impossible to determine the latent subspace $\mathbf{U}^\star$ using *any* method, even with infinitely many infinitely long trajectories.

Lemma 3.2 (Contexts, $\boldsymbol{\theta}$, and π_b cannot be dependent)**.** *For each of these conditions:*

1. *Contexts in a trajectory are dependent but do not depend on $\boldsymbol{\theta}$, and π_b also does not use $\boldsymbol{\theta}$,*
2. *Contexts are generated independently using $\boldsymbol{\theta}$, while π_b does not use $\boldsymbol{\theta}$,*
3. *Contexts are generated independently without using $\boldsymbol{\theta}$, while π_b uses $\boldsymbol{\theta}$,*

there exist two different linear latent contextual bandits with orthogonal latent subspaces satisfying the condition, and a behavior policy π_b so that the offline data distributions are indistinguishable and cover all (x, a) pairs with probability at least $1/4$. Since the latent subspaces are orthogonal, an action that gives the maximum reward on one latent bandit gives reward 0 on the other.

To estimate the latent subspace, one is thus forced to make the following assumption.

Assumption 3.3 (Unconfounded Offline Actions)**.** The offline behavior policy π_b does not use $\boldsymbol{\theta}$ to choose actions, and contexts $x_{n,h}$ are stochastic and generated independently of each other and of $\boldsymbol{\theta}$.

This is satisfied when the offline data comes from randomized controlled trials or A/B testing, which are common sources of offline datasets. Even if this is not satisfied, Algorithm 3.1 can learn a good subspace whenever $\overline{\mathbf{D}}_{N,1}^{-1}\overline{\mathbf{M}}_N\overline{\mathbf{D}}_{N,2}^{-1}$ has eigenspace close to the span of $\mathbf{U}_\star$, e.g. when high-reward actions contribute heavily to $\mathbf{D}_{n,i}$. This can happen if offline trajectories were collected to maximize rewards.

Returning to our scrutiny of $\overline{\mathbf{M}}_N$, let the covariance matrix of $\boldsymbol{\theta}$ be Λ and let its mean be μ_θ. Then we have that $\mathbb{E}[\overline{\mathbf{M}}_N] = \mathbb{E}[\mathbf{D}_{n,1}\boldsymbol{\beta}_n\boldsymbol{\beta}_n^\top\mathbf{D}_{n,2}] = \mathbb{E}[\mathbf{D}_{n,1}]\mathbf{U}_\star(\Lambda + \mu_\theta\mu_\theta^\top)\mathbf{U}_\star^\top\mathbb{E}[\mathbf{D}_{n,2}]$. So, we still cannot merely consider the top d_K eigenvectors of $\overline{\mathbf{M}}_N$ without accounting for $\mathbf{D}_{n,1}$. Intuitively, $\mathbf{D}_{n,1}$ captures the distortion in reward estimation caused by regularization in ridge regression[2]. We therefore construct correction matrices $\overline{\mathbf{D}}_{N,i}$ and use them to "neutralize" the distortion from regularization. In particular, $\overline{\mathbf{D}}_{N,1}^{-1}\overline{\mathbf{M}}_N\overline{\mathbf{D}}_{N,2}^{-1}$ is an estimator for $\mathbf{U}_\star(\Lambda + \mu_\theta\mu_\theta^\top)\mathbf{U}_\star^\top$. Crucially, this allows us to aggregate information across many trajectories to overcome the challenge of learning from short trajectories. We can now take the top d_K eigenvectors of $\overline{\mathbf{D}}_{N,1}^{-1}\overline{\mathbf{M}}_N\overline{\mathbf{D}}_{N,2}^{-1}$ to estimate the subspace determined by $\mathbf{U}_\star$. To give guarantees, we must make a coverage assumption. Unlike in standard offline RL, where only coverage along actions is needed, we also need coverage along latent states.

Assumption 3.4 (Boundedness and Coverage)**.** Rewards $|r_{n,h}| \leq R$ [3] for all n, h, $\|\phi(x,a)\|_2 \leq 1$ and $\|\boldsymbol{\beta}\|_2 \leq R$. Also, $\lambda_A := \min_{i=1,2}\lambda_{\min}(\mathbb{E}[\mathbf{D}_{n,i}]) > 0$ and $\lambda_\theta := \frac{1}{R^2}\lambda_{\min}(\Lambda) > 0$.

Intuitively, λ_A measures coverage along actions, while λ_θ measures coverage along latent states $\boldsymbol{\theta}$. Both must be non-zero to expect satisfactory estimation of the subspace. We can then use confidence bounds for $\overline{\mathbf{M}}_N$ and $\overline{\mathbf{D}}_{N,i}$ to give a data-dependent confidence bound Δ_{off} for the projection matrix $\hat{\mathbf{U}}\hat{\mathbf{U}}^\top$, as in Theorem 3.5 below. In one instantiation, Propositions B and B in Appendix B derive simple data-dependent bounds for $\overline{\mathbf{M}}_N$ and $\overline{\mathbf{D}}_{N,i}$ respectively. Under this choice, we control the growth of Δ_{off} in terms of the unknown problem parameters at the end of Theorem 3.5.

[2]This is not unique to regularization. Pseudo-inverses cause an analogous problem of distortion caused by unseen actions.

[3]R-bounded rewards are automatically R-subgaussian. We can easily extend our results to more general subgaussian rewards, but stick to bounded rewards for simplicity of proofs.

Theorem 3.5 (Computing and Bounding Δ_{off})**.** *Let* $\|\overline{\boldsymbol{M}}_N - \mathbb{E}[\boldsymbol{M}_1]\|_2 \le \Delta_M$ *and* $\|\overline{\boldsymbol{D}}_{N,i} - \mathbb{E}[\boldsymbol{D}_{n,i}]\|_2 \le \Delta_D$ *for* $i = 1, 2$ *with probability* $1 - \delta/3$ *each. Then, with probability* $1 - \delta$, $\|\hat{\boldsymbol{U}}\hat{\boldsymbol{U}}^\top - \boldsymbol{U}_\star \boldsymbol{U}_\star^\top\|_2 \le \Delta_{\text{off}}$, *where for* $B_D = \max_{i=1,2} \|\overline{\boldsymbol{D}}_{N,i}^{-1}\|_2$ *and* $\hat{\lambda} := \lambda_{d_K}(\overline{\boldsymbol{M}}_N) - \lambda_{d_K+1}(\overline{\boldsymbol{M}}_N)$,

$$\Delta_{\text{off}} = \frac{2\sqrt{2d_K}}{\hat{\lambda}} \left(\frac{B_D^3(2 - B_D\Delta_D)}{(1 - B_D\Delta_D)^2}(R^2 + \Delta_M)\Delta_D + \left(\frac{B_D}{1 - B_D\Delta_D}\right)^2 \Delta_M \right).$$

Obtaining Δ_M *and* Δ_D *from Propositions B and B,* $\Delta_{\text{off}} = \widetilde{O}(\frac{1}{\lambda_\theta \lambda_A^3} N^{-1/2} \sqrt{d_K d_A \log(d_A/\delta)})$.

Estimating d_K offline. As our estimator $\overline{\mathbf{D}}_{N,1}^{-1} \overline{\mathbf{M}}_N \overline{\mathbf{D}}_{N,2}^{-1}$ is approximately rank-d_K, the number of nonzero eigenvalues of the estimator is a principled heuristic for determining d_K. See Kausik et al. (2024) for a detailed comparison showing why naive PCA and probabilistic matrix factorization (Mnih and Salakhutdinov, 2007) fail to recover the latent subspace with uncertainty guarantees.

3.4. Offline Data Sharpens Online Optimism

Here, we motivate and describe our subspace-constrained optimistic linear bandit algorithm (Algorithm 3.2), a natural algorithm that accelerates LinUCB with offline data. The core idea is **sharpening optimism** by being optimistic over the intersection of two confidence sets – one obtained using offline and online data and another purely from online data.

We geometrically motivate our update rule here, and illustrate it in Figure 3.1. After any t steps, we can construct a d_K-dimensional confidence ellipsoid for every subspace in the subspace confidence set obtained from Algorithm 3.1. The union of all these ellipsoids gives us our "offline confidence set"[4], called $\mathcal{C}_{\text{off}}^t(\boldsymbol{\beta})$. The usual d_A-dimensional ellipsoid forms our "online confidence set." We call this $\mathcal{C}_{\text{on}}^t(\boldsymbol{\beta})$. Since the true parameter lies in both confidence sets with high probability, being optimistic over their intersection allows us to sharpen or "further localize" optimism. Even though the offline confidence set uses both offline and online data, it will never shrink to a point due to the

[4]The set is not only dependent on offline data, since online data is used to construct the d_K-dimensional ellipsoids.

Algorithm 3.2 Subspace-Constrained Optimistic Linear Bandit

1: **Input:** Projection matrix $\hat{\mathbf{U}}\hat{\mathbf{U}}^\top$, confidence bound Δ_{off} from an offline uncertainty-aware method, e.g. Algorithm 3.1.
2: **Initialize** $\mathbf{V}_1 \leftarrow \mathbf{I}_{d_A}$, $\mathbf{b}_1 \leftarrow 0$, α_t
3: **for** $t = 1, \dots T$ **do**
4: **Play** action a_t and receive reward r_t according to:

$$a_t, \widetilde{\boldsymbol{\beta}}_t, \widetilde{\mathbf{U}}_t \leftarrow \arg\max_{a,\boldsymbol{\beta},\mathbf{U}} \phi(x_t, a)^\top \boldsymbol{\beta} \text{ such that}$$
$$\hat{\boldsymbol{\beta}}_{1,t} \leftarrow \mathbf{U}(\mathbf{U}^T\mathbf{V}_t\mathbf{U})^{-1}\mathbf{U}^\top\mathbf{b}_t, \|\mathbf{U}^\top(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{1,t})\|_{(\mathbf{U}^\top\mathbf{V}_t\mathbf{U})^{-1}} \le \alpha_{1,t}$$
$$\hat{\boldsymbol{\beta}}_{2,t} \leftarrow \mathbf{V}_t^{-1}\mathbf{b}_t, \|(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{2,t})\|_{\mathbf{V}_t^{-1}} \le \alpha_{2,t}$$
$$\|\boldsymbol{\beta}\|_2 \le R, \mathbf{U}^\top\mathbf{U} = \mathbf{I}_{d_K}, \mathbf{U}\mathbf{U}^\top\boldsymbol{\beta} = \boldsymbol{\beta},$$
$$\|\hat{\mathbf{U}}^\top\mathbf{U}\|_F \ge \sqrt{d_K - \Delta_{\text{off}}^2/2}$$

5: **Compute** $\mathbf{b}_{t+1} \leftarrow \mathbf{b}_t + \phi(x_t, a_t)r_t$, $\mathbf{V}_{t+1} \leftarrow \mathbf{V}_t + \phi(x_t, a_t)\phi(x_t, a_t)^\top$, α_{t+1}
6: **end for**

frozen subspace confidence set. So, we need the intersection of both sets to be sharply optimistic. This is the intuition behind Algorithm 3.2.

We formalize this intuition in Algorithm 3.2 by formulating the sharpened optimism as an optimization problem in step 4.

We provide the following guarantee for Algorithm 3.2.

Theorem 3.6 (Algorithm 3.2 Regret)**.** *Under Assumptions 3.3 and 3.4, if* $\alpha_{1,t} = R\sqrt{\mu} + CR\sqrt{d_K \log(2T/\delta)}$ *and* $\alpha_{2,t} = R\sqrt{\mu} + CR\sqrt{d_A \log(2T/\delta)}$ *for a universal constant* C*, then with probability at least* $1 - \delta$ *over offline data and online rewards, Algorithm 3.2 has regret* Reg_T *bounded by*

$$O\Big(\min\Big(Rd_A\sqrt{T}, Rd_K\sqrt{T}\Big(1 + \tfrac{1}{\lambda_\theta\lambda_A^3}\sqrt{\tfrac{Td_A}{d_K N}}\Big)\Big)\Big).$$

However, the subspace constraint $\|\hat{\mathbf{U}}^\top\mathbf{U}\|_F \ge \sqrt{d_K - \Delta_{\text{off}}^2/2}$ is *nonconvex*, making us search for $\widetilde{\mathbf{U}}_t$ over a complicated star-shaped set. So, it is unclear if Algorithm 3.2 can be made computationally

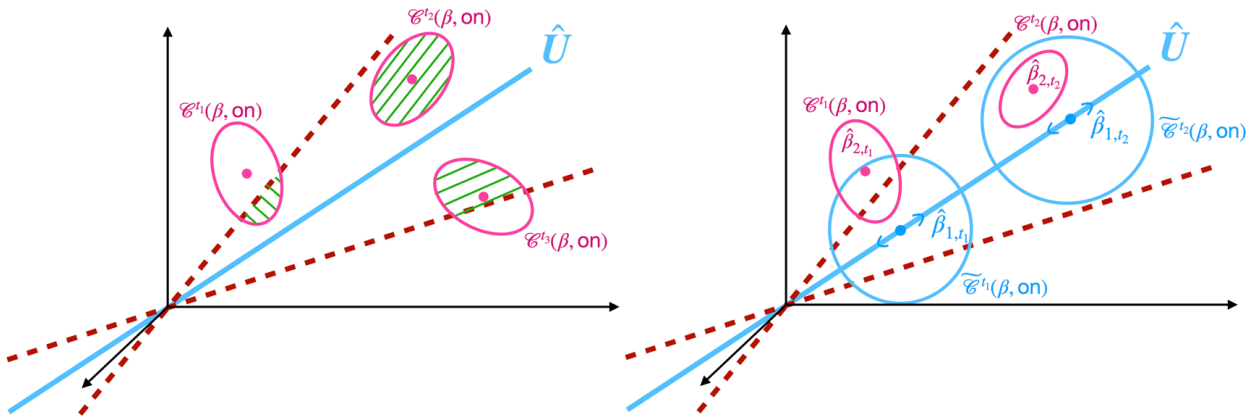


Figure 3.1: **Left:** Geometric interpretation of subspace-constrained optimism (Algorithm 3.2), showing the intersection $\mathcal{C}^t_{\text{on}}(\boldsymbol{\beta}) \cap \mathcal{C}^t_{\text{off}}(\boldsymbol{\beta})$ in green at three timepoints $t = t_1, t_2, t_3$, with dotted lines delineating the subspace confidence set. **Right:** Geometric interpretation of projected confidence-ball optimism (Algorithm 3.3). At t_1, $\mathcal{C}^{t_1}_{\text{on}}(\boldsymbol{\beta}) \not\subset \widetilde{\mathcal{C}}^{t_1}_{\text{off}}(\boldsymbol{\beta})$, so the algorithm uses the low-dimensional projection, whereas by t_2, $\mathcal{C}^{t_2}_{\text{on}}(\boldsymbol{\beta}) \subset \widetilde{\mathcal{C}}^{t_2}_{\text{off}}(\boldsymbol{\beta})$ and the algorithm switches to the full-dimensional confidence set.

efficient.

3.5. Lower Bound

We now establish that Algorithm 3.2 is in fact minimax optimal. While the full statement and proof are given in Kausik et al. (2024) (see Appendix B), we provide an informal version here.

Theorem 3.7. *There exists a family of tuples $(F, \boldsymbol{\beta})$, where F is a latent bandit with a rank d_K latent subspace and $\boldsymbol{\beta}$ is a reward parameter in its support, so that for any offline behavior policy π_b and any learner, (i) λ_θ is uniformly bounded for all F, (ii) there exists a $(F, \boldsymbol{\beta})$ such that the regret $\text{Reg}(T, \boldsymbol{\beta})$ of the learner under offline data from π_b and F and online reward parameter $\boldsymbol{\beta}$ is bounded below by*

$$\Omega\left(\min\left(d_A\sqrt{T}, d_K\sqrt{T}\left(1+\sqrt{\frac{d_A T}{d_K N}}\right)\right)\right)$$

3.6. Practical Optimism via Projected Confidence Balls

While Algorithm 3.2 is minimax-optimal, it is not computationally efficient due to the non-convex constraint discussed in Section 3.4. We address this by introducing our projected confidence-ball op-

timistic linear bandit method (Algorithm 3.3), a practical and computationally efficient relaxation of Algorithm 3.2 illustrated in Figure 3.1.

Algorithm 3.3 works in the subspace estimated by Algorithm 3.1 until the online confidence set is small enough. The algorithm maintains a low-dimensional confidence set $\widetilde{\mathcal{C}}^t_{\text{off}}(\boldsymbol{\beta})$ around the projected estimate and the standard d_A-dimensional online confidence ellipsoid $\mathcal{C}^t_{\text{on}}(\boldsymbol{\beta})$, and swaps between them to achieve acceleration. Once the cumulative error of using the low-dimensional confidence set ($\approx \Delta_{\text{off}} T$ in Algorithm 3.3) exceeds the cumulative error of using the high-dimensional confidence set ($\approx d_A\sqrt{T}$), we stop using the former. This is instantiated with LinUCB, but the same idea can be immediately applied to other algorithms like SupLinUCB or Bayesian algorithms like Thompson sampling.

Algorithm 3.3 Projected Confidence-Ball Optimistic Linear Bandit

Input: Projection matrix $\hat{\mathbf{U}}\hat{\mathbf{U}}^\top$, confidence bound Δ_{off}. Hyperparameters $\alpha_{1,t}, \alpha_{2,t}, \tau, \tau'$.
Initialize $\mathbf{V}_1 \leftarrow I$, $\mathbf{b}_1 \leftarrow 0$, $\mathbf{C}_1 \leftarrow 0$, $\kappa_1 \leftarrow 0$
for $t = 1, \dots T$ **do**
  if $\Delta_{\text{off}}\tau\sqrt{t} + \Delta_{\text{off}}\tau'\sqrt{d_K \sum_{s=1}^t \kappa_s^2/t} \le d_A$ **then**
    Compute $\hat{\boldsymbol{\beta}}_{1,t} \leftarrow \hat{\mathbf{U}}(\hat{\mathbf{U}}^\top \mathbf{V}_t \hat{\mathbf{U}})^{-1}\hat{\mathbf{U}}^\top \mathbf{b}_t$
    Play $a_t \leftarrow \arg\max_a \phi(x_t, a)^\top \hat{\mathbf{U}}\hat{\mathbf{U}}^\top \hat{\boldsymbol{\beta}}_{1,t} + \alpha_{1,t}\|\phi(x_t, a)^\top \hat{\mathbf{U}}\|_{(\hat{\mathbf{U}}^\top \mathbf{V}_t \hat{\mathbf{U}})^{-1}}$
  else
    Compute $\hat{\boldsymbol{\beta}}_{2,t} \leftarrow \mathbf{V}_t^{-1}\mathbf{b}_t$
    Play $a_t \leftarrow \arg\max_a \phi(x_t, a)^\top \hat{\boldsymbol{\beta}}_{2,t} + \alpha_{2,t}\,\|\phi(x_t, a)\|_{\mathbf{V}_t^{-1}}$
  end if
  Observe reward r_t and update $\mathbf{b}_{t+1} \leftarrow \mathbf{b}_t + \phi(x_t, a_t) r_t$, $\mathbf{V}_{t+1} \leftarrow \mathbf{V}_t + \phi(x_t, a_t)\phi(x_t, a_t)^\top$
  Update $\mathbf{C}_{t+1} \leftarrow \mathbf{C}_t + \hat{\mathbf{U}}^\top \phi(x_t, a_t)\phi(x_t, a_t)^\top$, $\kappa_{t+1} \leftarrow \|\mathbf{C}_{t+1}\|_{(\hat{\mathbf{U}}^\top \mathbf{V}_{t+1} \hat{\mathbf{U}})^{-1}}$
end for

Theorem 3.8 (Regret for Algorithm 3.3)**.** *Let* $\alpha_{1,t} = R\sqrt{\mu} + \tau' R \Delta_{\text{off}} \kappa_t + CR\sqrt{d_K \log(T/\delta)}$ *and let* $\alpha_{2,t} = R\sqrt{\mu} + CR\sqrt{d_A \log(T/\delta)}$*. Let* S *be the first timestep when Algorithm 3.3 does not play Line 6 and let* $S = T$ *if no such timestep exists. For* $\tau = \tau' = 1$ *we have that*

$$\text{Reg}_T = \widetilde{O}\left(\min\left(\text{Reg}_{on,T}, \text{Reg}_{hyb,T}\right)\right).$$

where $\text{Reg}_{on,T} = Rd_A\sqrt{T}$ *and* $\text{Reg}_{hyb,T} = Rd_K\sqrt{T}\left(1 + \frac{1}{\lambda_A^3 \lambda_\theta}\left(\sqrt{\frac{d_A T}{d_K N}} + \sqrt{\frac{d_A}{SN}\sum_{t=1}^S \kappa_t^2}\right)\right)$ *In the worst case,* $\kappa_t = O(t)$ *and so* $\frac{1}{S}\sum_{t=1}^S \kappa_t^2 = O(T^2)$*, but if all features* $\phi(x_t, a_t)$ *lie in the span of* $\hat{\boldsymbol{U}}$

for $t \le S$, *then* $\frac{1}{S}\sum_{t=1}^{S} \kappa_t^2 = O(T)$.

Because $\hat{\boldsymbol{\beta}}_{1,t}$ lies in the span of $\hat{\mathbf{U}}$ while $\boldsymbol{\beta}_\star$ might not, establishing Theorem 3.8 requires a modified self-normalized concentration bound (Lemma B in Kausik et al. (2024), which also discusses natural geometric conditions under which the favorable $\frac{1}{S}\sum_{t=1}^{S} \kappa_t^2 = O(T)$ regime holds).

3.7. Experiments

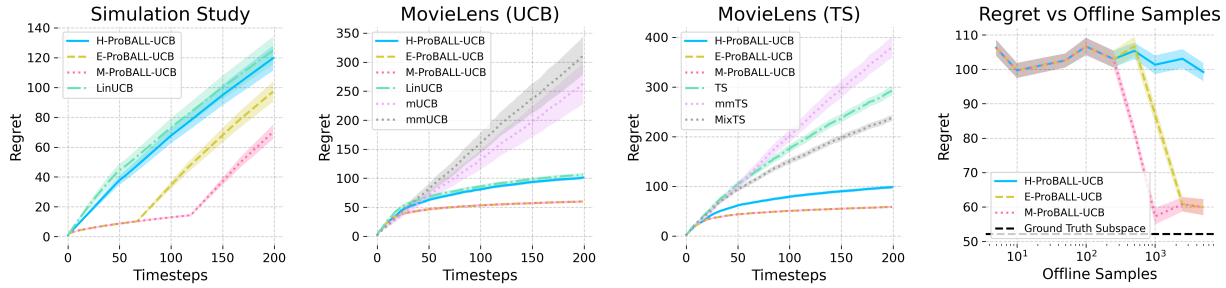


Figure 3.2: Regret and offline sample complexity in simulation and on MovieLens-1M. **First:** Synthetic linear latent bandit comparing projected confidence-ball UCB (Algorithm 3.3) against LinUCB ($\tau = 5$). **Second:** MovieLens-1M comparison of Algorithm 3.3 initialized with Algorithm 3.1 against LinUCB, mUCB, and mmUCB ($\tau = 0.1$). **Third:** MovieLens-1M comparison of projected confidence-ball Thompson sampling (Algorithm B) against TS, mmTS, and MixTS ($\tau = 0.1$). **Fourth:** Cumulative regret of Algorithm 3.3 at $T = 200$ on MovieLens-1M versus the number of offline trajectories N used by Algorithm 3.1, compared against oracle low-dimensional LinUCB. Shaded bands indicate ± 1 standard error over 30 trials with freshly drawn $\boldsymbol{\theta}$.

We now establish the practical efficacy of Algorithm 3.1 and Algorithm 3.3 for linear latent contextual bandits through a simulation study and a demonstration on the MovieLens-1M dataset. In all experiments, we obtain confidence bounds Δ_{off} using three concentration inequalities, abbreviated in the figure legends by prefixing `ProBALL` (for Algorithm 3.3): (1) Hoeffding as in Proposition B (H-ProBALL), (2) empirical Bernstein as in Proposition B (E-ProBALL), and (3) the martingale Bernstein concentration inequalities of Waudby-Smith and Ramdas (2023) (M-ProBALL). We set $\tau' = 0$ and tune τ to adjust for conservative Δ_{off}. For the MovieLens experiments, we also evaluate a Thompson sampling counterpart of Algorithm 3.3, projected confidence-ball Thompson sampling (Algorithm B, abbreviated `ProBALL-TS` in the legends). Full experimental details and ablation studies over τ, confidence bound constructions, and offline sample size N are deferred to Kausik et al. (2024) (see Appendix B).

Simulation study. We first perform a simulation study on a latent linear bandit with $d_A = 50$ and $d_K = 2$, with 5000 trajectories generated offline. The results are presented in Figure 3.2. Note that Algorithm 3.3 performs no worse than LinUCB, no matter what we choose for τ and Δ_{off}. However, we see a clear benefit from using tighter confidence bounds – as Δ_{off} gets smaller, Algorithm 3.3 chooses to utilize the projected estimate $\hat{\boldsymbol{\beta}}_{1,t}$ more often, resulting in better performance. Note that the kinks in the regret curves correspond to points where Algorithm 3.3 switches over to the higher dimensional optimism in step 7.

MovieLens dataset. In line with Hong et al. (2020), we assess the performance of our algorithms on real data using the MovieLens dataset. Like them, we filter the dataset to include only movies rated by at least 200 users and vice versa, and apply probabilistic matrix factorization (PMF) to the rating matrix to generate ground truth user preferences for online experiments. Applying PMF gives $d_K = 18$ and we choose $d_A = 200$, generating 5000 trajectories offline. For baselines, we reproduce the methods of Hong et al. (2020, 2022) and implement LinUCB with canonical hyperparameter choices. We initialize Algorithm 3.3 with a subspace estimated with an unregularized variant of Algorithm 3.1 (Algorithm B in Appendix B) that uses pseudo-inverses instead of inverses. Figure 3.2 depicts the result of this experiment. Once again, Algorithm 3.3 performs no worse than LinUCB, no matter what we choose for τ and Δ_{off}, and the benefit of using tighter confidence bounds remains. Although mUCB and mmUCB perform slightly better than Algorithm 3.3 and LinUCB at the beginning due to lower initial exploration, the model misspecification incurred by discretizing the features into d_K clusters ensures that they typically suffer linear regret in this scenario.

3.8. Discussion, Limitations and Further Work

In this chapter, we have addressed the problem of leveraging offline data to accelerate online learning in linear latent bandits. Beyond the linear setting studied in this chapter, Kausik et al. (2024) also establishes a de Finetti theorem showing that coherent, exchangeable stateless decision processes are latent bandits. Our work has a few limitations. First, while Algorithm 3.3 is practical and computationally efficient, it has a slightly weaker worst-case guarantee than Algorithm 3.2. Second,

when the data is noisy, it can be hard to tune the regularization μ. We use a pseudoinverse-based version of Algorithm 3.1 in such a case (Algorithm B in Appendix B), implemented in our code. Third, the offline uncertainty sets computed using Δ_{off} can be overly conservative, and a discount hyperparameter τ must be fine-tuned online.

Despite these limitations, our work enjoys strong theoretical guarantees and convincing empirical performance. We hope that this method opens the door for developing other efficient and scalable algorithms for sequential decision-making with continuous latent states. One can use ideas presented in this chapter to design similar algorithms for MDPs, linear MDPs, and RL or bandits with general function approximation.

CHAPTER 4

STATISTICALLY AND COMPUTATIONALLY EFFICIENT ACTOR–CRITIC METHODS

This chapter is based on joint work with Wei Fan and Yuting Wei (Tan et al., 2025) (Part I: Kevin Tan, Wei Fan, and Yuting Wei, "Actor-Critics Can Achieve Optimal Sample Efficiency," in Proceedings of the 42nd International Conference on Machine Learning (ICML), PMLR 267, 2025) and joint work with Max Qiushi Lin, Reza Asad, Haque Ishfaq, Csaba Szepesvári, and Sharan Vaswani (Lin et al., 2026) (Part II: Max Qiushi Lin, Reza Asad, Kevin Tan, Haque Ishfaq, Csaba Szepesvári, and Sharan Vaswani, "Optimistic Actor-Critic with Parametric Policies for Linear Markov Decision Processes," in Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 300, 2026).

[RESOLVE: CHAPTER 4 COMBINES TWO ACTOR-CRITIC PAPERS: PART I (TAN, FAN, AND WEI, ICML 2025) AND PART II (LIN, ASAD, TAN, ISHFAQ, SZEPESVARI, AND VASWANI, AISTATS 2026).]

Chapters 2 and 3 studied how offline datasets reduce the statistical complexity of value-based and bandit learning. In practice, however, many reinforcement learning systems maintain an explicit policy (an *actor*) updated by policy-gradient or mirror-descent steps alongside a value estimator (a *critic*). This chapter investigates both the *statistical* and *computational* efficiency of actor–critic methods in two parts:

1. **Part I (Section 4.1): Achieving optimal sample efficiency with general function approximation.** Standard online actor–critic methods update the critic at every step to track $Q^{\pi^{(t)}}$, which causes the log-covering number of the induced policy class Π_T to grow linearly with the number of episodes T unless the policy cover happens to remain small (Algorithm 4.1). We resolve this fundamental statistical obstacle in Algorithm 4.2 by designing a rare-switching optimistic critic that targets the optimal action-value function $Q^\star$ directly and

resets the actor to the uniform distribution whenever the critic switches, achieving minimax-optimal $\widetilde{O}(\sqrt{d_{\mathrm{BE}}H^3T})$ regret without policy completeness.

2. **Part II (Section 4.2): Achieving constant per-step policy evaluation complexity with parametric policies.** Even when the critic switches rarely, non-parametric mirror-descent actors sum over all historical critic estimates at deployment time, incurring $O(t)$ per-step action-selection cost at episode t. In Part II, we maintain an explicitly parameterized log-linear policy updated via weighted least-squares logit matching on a G-optimal experimental design coreset (Algorithm 4.7), paired with a computationally tractable Langevin Monte Carlo optimistic critic (Algorithm 4.8) inside an optimistic parametric actor–critic loop (Algorithm 4.6).

All proofs and supplementary experimental details for both parts of this chapter are provided in Appendix C.

4.1. Part I: Optimal Sample Efficiency via Rare-Switching Optimistic Actor–Critics

4.1.1. Introduction

Actor-critic algorithms have emerged as a foundational approach in reinforcement learning (RL), mitigating the deficiencies of both policy-based and value-based approaches (Sutton and Barto, 2018; Mnih et al., 2016; Haarnoja et al., 2018). These methods combine two components: an actor, which directly learns and improves the policy, and a critic, which evaluates the policy's quality. Given their popularity, significant recent progress has been made in understanding their theoretical underpinnings and statistical efficiency, especially in the presence of function approximation (Cai et al., 2020; Zhong and Zhang, 2023; Sherman et al., 2023; Liu et al., 2023b) – which is required in real-world applications with prohibitively large state and action spaces.

However, much existing work (Abbasi-Yadkori et al., 2019; Neu et al., 2017; Liu et al., 2023a; Bhandari and Russo, 2024; Agarwal et al., 2021; Cen et al., 2022; Gaur et al., 2024) on the convergence of actor-critic methods requires assumptions on the reachability of the state-action space or on the coverage of the sampled data. Liu et al. (2023b) remark that this implies that the state-space

is either well-explored or easy to explore. This allows the agent to bypass the need to actively explore the state-action space, making learning significantly easier.[5] Therefore, these approaches analyze actor-critic methods from an optimization perspective and do not address the problem of exploration (Efroni et al., 2020) – a salient problem that we seek to tackle, hence the need for strategic exploration.

Without reachability assumptions, policy gradient methods struggle due to a lack of strategic exploration.[6] A recent line of work utilizes optimism to address this. Efroni et al. (2020); Wu et al. (2022a) and Cai et al. (2020) achieve $\sqrt{T}$ regret within the settings of tabular and linear mixture MDPs respectively, with Wu et al. (2022a) attaining the minimax-optimal rate. Still, Zhong and Zhang (2023); Liu et al. (2023b) remark that these analyses do not generalize to more general MDPs due to the need to cover an exponentially growing policy class. Optimism is not the only way to solve this issue. For instance, Huang et al. (2024) utilize Gaussian perturbations to perform exploration in the linear-quadratic control problem. However, naive, direction-unaware perturbation-based exploration comes with a cost, achieving only $T^{3/4}$ regret in this case.

Within linear MDPs, Sherman et al. (2023) and Cassel and Rosenberg (2024) have very recently been able to obtain the optimal rate of $\sqrt{T}$ regret or $1/\epsilon^2$ sample complexity. They do so via methodological advancements (specific to linear MDPs) that let them overcome the growing policy class issue. However, the problem is unresolved with general function approximation – the best known algorithm from Liu et al. (2023b) requires at least $1/\epsilon^3$ samples, increasing to $1/\epsilon^4$ when the policy class grows exponentially.

An open problem. *No actor-critic algorithm with general function approximation is currently known to achieve* $1/\epsilon^2$ *sample complexity or* $\sqrt{T}$ *regret* in this more challenging setting where strategic exploration is necessary. Zhong and Zhang (2023) and Liu et al. (2023b) remark that a way forward to achieve the desired $1/\epsilon^2$ sample complexity remains unclear, and raise the open

[5]When coverage/reachability assumptions are made, the linear convergence of policy-based methods (Lan, 2022; Xiao, 2022; Yuan et al., 2023; Chen and Maguluri, 2022) and the gradient domination lemma (Kumar et al., 2024; Mei et al., 2022) enable the natural actor-critic algorithm to learn an ϵ-optimal policy within $1/\epsilon^2$ samples if one can access the exact NPG update, although vanilla policy gradient methods can take (super) exponential time to converge (Li et al., 2021).

[6]To illustrate, this was not solved until Agarwal et al. (2020), who required $1/\epsilon^{11}$ samples to do so in linear MDPs.

Algorithm	Sample Complexity	Regret	Remarks
Agarwal et al. (2020)	$d^3H^{15}\log\lvert\mathcal{A}\rvert/\epsilon^{11}$	None	
Zanette et al. (2021)	$H^4\log\lvert\mathcal{A}\rvert/\epsilon^2 + d^3H^{13}\log\lvert\mathcal{A}\rvert/\epsilon^3$	$\sqrt{H^4\log\lvert\mathcal{A}\rvert\log\lvert\mathcal{A}\rvert T} + \sqrt{d^3H^{13}T}$	Linear MDPs only
Zhong and Zhang (2023)	$d^3H^8\log\lvert\mathcal{A}\rvert/\epsilon^4 + d^5H^4/\epsilon^2$	$\sqrt{d^3H^8\log\lvert\mathcal{A}\rvert T} + \sqrt{d^5H^4T}$	
Sherman et al. (2023)	$d^4H^7\log\lvert\mathcal{A}\rvert/\epsilon^2$	$\sqrt{d^4H^7\log\lvert\mathcal{A}\rvert T}$	
Cassel and Rosenberg (2024)	$dH^5\log\lvert\mathcal{A}\rvert/\epsilon^2 + d^3H^4/\epsilon^2$	$\sqrt{dH^5\log\lvert\mathcal{A}\rvert T} + \sqrt{d^3H^4T}$	
Zhou et al. (2023)	$(\log\lvert\mathcal{A}\rvert + C^3_{\text{npg}} \wedge C^6_{\text{off}})\log\lvert\mathcal{F}\rvert H^{14}/\epsilon^6$	Linear	Requires offline data
Liu et al. (2023b)	$d\log\lvert\mathcal{A}\rvert\log\lvert\mathcal{F}\rvert H^6/\epsilon^3$	None	"Good" case only, $1/\epsilon^4$ generally
Algorithm 4.1 (Sec. 4.1.3)	$H^4\log\lvert\mathcal{A}\rvert/\epsilon^2 + dH^4\log\lvert\mathcal{F}\rvert/\epsilon^2$	$\sqrt{H^4\log\lvert\mathcal{A}\rvert T} + \sqrt{dH^4\log\lvert\mathcal{F}\rvert T}$	"Good" case only, vacuous generally
Algorithm 4.2 (Sec. 4.1.4)	$dH^5\log\lvert\mathcal{A}\rvert/\epsilon^2 + dH^4\log\lvert\mathcal{F}\rvert/\epsilon^2$	$\sqrt{dH^5\log\lvert\mathcal{A}\rvert T} + \sqrt{dH^4\log\lvert\mathcal{F}\rvert T}$	Holds generally

Table 4.1: Comparison of our work to existing literature. Algorithm 4.2 achieves the best known bound for actor-critic methods with general function approximation, and resolves an open problem on whether $\sqrt{T}$ regret or $1/\epsilon^2$ sample complexity is achievable in this scenario.

problem:

Can actor-critic or policy optimization algorithms achieve $1/\epsilon^2$ sample complexity or $\sqrt{T}$ regret with general function approximation and when strategic exploration is necessary?

Contributions of Part I

We resolve this open problem in the affirmative. As a warm-up, we first consider an easy case – where the complexity of the class of policies considered by the policy optimization procedure does not increase exponentially with the number of (critic) updates.[7] Then, a simple modification to the GOLF algorithm of Jin et al. (2021a) (Algorithm 4.1 in Section 4.1.3) allows one to achieve a regret of $\sqrt{H^4\log|\mathcal{A}|T} + \sqrt{dH^4\log|\mathcal{F}|\,T}$, in line with the results of Efroni et al. (2020); Cai et al. (2020) for tabular and linear mixture MDPs respectively.

However, this is not the case in many practical scenarios – for example, where one uses linear models, decision trees, neural networks, or even random forests for the critic. In this much harder setting, Algorithm 4.1 cannot achieve sublinear regret. We address this by introducing a rare-switching optimistic actor-critic with uniform policy resets (Algorithm 4.2 in Section 4.1.4), which leverages three crucial ingredients: (1) optimism, (2) off-policy learning, and (3) rare-switching critic updates that target Q^* and accompanying policy resets. Algorithm 4.2 achieves $\sqrt{dH^4\log|\mathcal{F}|T} + \sqrt{dH^5\log|\mathcal{A}|T}$ regret, requiring only a factor of $dH\log T$ more samples than Algorithm 4.1 even

[7] It was previously considered in Zhong and Zhang (2023) that the log-covering number of the policy class increases in the number of actor and critic updates. We sharpen this bound to the number of critic updates in Lemma 4.6, which may be of independent interest.

in the best case for the latter.

Extensions to hybrid RL

Zhou et al. (2023) use both offline and online data to bypass the need to perform strategic exploration in policy optimization. This corresponds to the setting of hybrid RL (Nakamoto et al., 2023; Amortila et al., 2024; Ren et al., 2024b; Wagenmaker and Pacchiano, 2023), where Song et al. (2023) show that using both offline and online data allows one to achieve $\sqrt{T}$ regret without optimism. However, the claimed $\sqrt{T}$ regret bound in Zhou et al. (2023) requires on-policy sampling of $O(T)$ samples per timestep that does not contribute to the regret, leading to a sample complexity of $1/\epsilon^6$. Their algorithm cannot achieve sublinear regret in the more common setup where each sample contributes to the regret. Furthermore, they require bounded occupancy measure ratios of the optimal policy to *any* policy.

We demonstrate that these issues can be mitigated. Specifically, we extend our optimistic algorithm to leverage both offline and online data, and show that actor-critic methods can benefit from hybrid data and achieve the provable gains in sample efficiency as observed in Li et al. (2023); Tan and Xu (2024); Tan et al. (2024b). We also provide a non-optimistic provably efficient actor-critic algorithm that only additionally requires $N_{\text{off}} \geq c^*_{\text{off}}(\mathcal{F}, \Pi) dH^4/\epsilon^2$ offline samples (with bounded single-policy concentrability) in exchange for omitting optimism. This, along with the result in Theorem 4.17, shows that hybrid RL therefore allows for sample efficiency gains with optimistic algorithms and computational efficiency gains with non-optimistic algorithms.

Notation. We write $\mathcal{N}_A(\rho)$ for the ρ-covering number over a set A, and in a small abuse of notation use $\mathcal{N}_{A,B}(\rho)$ to denote the ρ-covering number over the set $A \cup B$. We use standard big-O and little-O notation within this chapter. $O(g(n))$ denotes the set of functions $f(n)$ such that there exist positive constants C and n_0 such that $0 \leq f(n) \leq C \cdot g(n)$ for all $n \geq n_0$. When we say a function is $O(g(n))$, we say that it grows no faster than $g(n)$ asymptotically. $o(g(n))$ denotes the same, except that the inequality $0 \leq f(n) < C \cdot g(n)$ is strict, and so grows strictly slower than $g(n)$ asymptotically. $\Omega(g(n))$ involves the inequality $0 \leq C \cdot g(n) \leq f(n)$, and so denotes the set of functions $f(n)$ that grow at least as fast as $g(n)$. Finally, $\Theta(g(n))$ denotes the set of functions that

are both $O(g(n))$ and $\Omega(g(n))$.

4.1.2. Problem setting

Markov decision processes. Part I of this chapter focuses on finite horizon, episodic MDPs, represented by a tuple

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, H, \{\mathbb{P}_h\}_{h=1}^H, \{r_h\}_{h=1}^H),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, H is the horizon, $r_h : \mathcal{S} \times \mathcal{A} \to [0,1]$ is the reward function at step h and $\mathbb{P}_h : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ is the transition kernel for step h. A policy $\{\pi_h\}_{h=1}^H$ is a set of H functions, where each $\pi_h : \mathcal{S} \to \Delta(\mathcal{A})$ maps from a state on step h to a probability distribution on actions. Write Π for the class of all policies, and $\pi(s)$ as shorthand for the random variable $a \sim \pi(\cdot|s)$. Given a policy $\{\pi_h\}_{h=1}^H$ and reward function $\{r_h\}_{h=1}^H$, the state value function is defined as

$$V_h^\pi(s) = \mathbb{E}\Big[\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'})|s_h = s\Big], \tag{4.1}$$

where the expectation is taken over the randomness of $a_{h'} \sim \pi_{h'}(s_{h'})$ and $s_{h'+1} \sim \mathbb{P}_h(\cdot|s_{h'}, a_{h'})$ for any $h' \geq h$. The action value, or Q function is defined as

$$Q_h^\pi(s,a) = \mathbb{E}\Big[\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'})|s_h = s, a_h = a\Big], \tag{4.2}$$

where the expectation is taken over the similar randomness of action and state transition, with the only difference that the action randomness is only random as $h' \geq h+1$.

Without loss of generality, write s_1 for the initial state. The optimal policy is $\pi^\star = \mathrm{argmax}_{\pi\in\Pi} V_1^\pi(s_1)$. Correspondingly, we denote $V^\star = V^{\pi^\star}$ and $Q^\star = Q^{\pi^\star}$ as the optimal value and Q-functions. The Bellman operator with respect to the greedy policy and any policy π is given by

$$\mathcal{T}_h Q_{h+1}(s,a) = r_h(s,a) + \mathbb{E}_{s'\sim P_h}\Big[\max_{a'\in\mathcal{A}} Q_{h+1}(s',a')\Big], \tag{4.3a}$$

$$\text{and} \quad \mathcal{T}_h^\pi Q_{h+1}(s,a) = r_h(s,a) + \mathbb{E}_{s'\sim P_h}\Big[Q_{h+1}(s', \pi_{h+1}(s'))\Big]. \tag{4.3b}$$

The optimal Q-function $Q^\star$ is uniquely determined as the solution to the Bellman equation: $Q_h^\star(s,a) = \mathcal{T}_h Q_{h+1}^\star(s,a)$. Our goal is typically to learn an ϵ-optimal policy $\widehat{\pi}$, such that $V_1^*(s_1) - V_1^{\widehat{\pi}}(s_1) \le \epsilon$, or to obtain sublinear regret over T rounds while playing $(\pi^{(t)})_{t=1}^T$:

$$\mathrm{Reg}(T) := \sum_{t=1}^{T} \left(V_1^*(s_1) - V_1^{\pi^{(t)}}(s_1) \right) = o(T). \tag{4.4}$$

RL with function approximation. Under general function approximation, we approximate Q-functions with a function class $\mathcal{F} = \{\mathcal{F}_h\}_{h\in[H]}$, where each $f_h : \mathcal{S} \times \mathcal{A} \to [0,H]$. The Bellman error with regard to $f \in \mathcal{F}$ is $f_h - \mathcal{T}_h f_{h+1}$, and additionally with regard to $\pi \in \Pi$ as $f_h - \mathcal{T}_h^\pi f_{h+1}$. Additionally, we write $\pi_h^f(a|s) = \mathbf{1}(a' \in \arg\max_{a\in\mathcal{A}} f_h(s,a))$ for the greedy policy that plays the best action under f. We make the following routine assumptions on the richness of $\mathcal{F}$ (Jin et al., 2021a; Xie et al., 2022a; Rajaraman et al., 2020; Rashidinejad et al., 2021):

Assumption 4.1 (Realizability)**.** The function class $\mathcal{F}$ is rich enough such that for all $h \in [H]$, the function class $\mathcal{F}_h$ contains the optimal action value function $Q_h^\star$: $Q_h^* \in \mathcal{F}_h$.

Assumption 4.2 (Generalized Completeness)**.** There exists an auxiliary function class $\mathcal{G} = \mathcal{G}_1 \times ... \times \mathcal{G}_H$, where each $g_h \in \mathcal{G}_h$ satisfies $g_h : \mathcal{S} \times \mathcal{A} \to [0,H]$, that is sufficiently rich such that it contains all Bellman backups of $f \in \mathcal{F}$.

This auxiliary function class is $(\mathcal{T}^\Pi)^T\mathcal{F} = \{\mathcal{T}^{\pi^{(T)}} \cdot ... \cdot \mathcal{T}^{\pi^{(1)}} f \mid f \in \mathcal{F}, \pi^{(1)}, ..., \pi^{(T)} \in \Pi\}$ for Algorithm 4.1, and $\mathcal{T}\mathcal{F} = \{\mathcal{T}f \mid f \in \mathcal{F}\}$ for Algorithm 4.2. The former is far larger than the latter, with exceptions that we highlight in Section 4.1.3. We write $\mathcal{N}_\mathcal{F}(\rho)$ for the ρ-covering number of a function class $\mathcal{F}$.[8] To learn $f \in \mathcal{F}$ that approximates the Q-function of a policy π (we say that f targets π), it is common to minimize the temporal difference (TD) error over a dataset $\mathcal{D}$, as an

[8]The ρ-covering number of a class corresponds to the smallest cardinality of a set of points, such that every point in the class is at least ρ-close to some point in that set. See Wainwright (2019).

estimate of the Bellman error:

$$\mathcal{L}_h^{(t,\pi)}(f_h, f_{h+1}) = \sum_{(s,a,r,s')\in\mathcal{D}} (f_h(s,a) - r - f_{h+1}(s', \pi_{h+1}(s')))^2. \tag{4.5}$$

Measures of complexity. The complexity of online learning in the presence of general function approximation is governed by complexity measures such as the Bellman rank (Jiang et al., 2016), which corresponds to the intrinsic dimension in tabular, linear, and linear mixture MDPs. Another is the Bellman eluder dimension (Jin et al., 2021a), which subsumes the Bellman rank and additionally characterizes the complexity of kernel, neural, and generalized linear MDPs. We use the squared distributional version:

Definition 4.3 (Squared Distributional Bellman Eluder dimension (Jin et al., 2021a; Xiong et al., 2023a))**.** Let $\mathcal{F}$ be a function class. The distributional Bellman Eluder dimension is the largest d such that there exist measures $\{d_h^{(1)}, \ldots, d_h^{(d-1)}\}$, $d_h^{(d)}$, Bellman errors $\{\delta_h^{(1)}, \ldots, \delta_h^{(d-1)}\}$, $\delta_h^{(d)}$, and some $\epsilon' \geq \epsilon$, such that for all $t = 1, ..., d$,

$$\left|\mathbb{E}_{d_h^{(t)}}[\delta_h^{(t)}]\right| > \varepsilon^{(t)} \text{ and } \sqrt{\sum_{i=1}^{t-1} \left(\mathbb{E}_{d_h^{(i)}}[(\delta_h^{(t)})^2]\right)} \leq \varepsilon^{(t)}. \tag{4.6}$$

Jin et al. (2021a) primarily consider two types of distributions: (1) distributions $\mathcal{D}_{\mathcal{F}}$ induced by greedy policies π^f, and Dirac delta measures over state-action pairs $\mathcal{D}_{\Delta}$. They suggest an RL problem has low Bellman eluder dimension if either variant is small. Instructive examples include tabular MDPs, where this corresponds to the cardinality of the state-action space, and linear MDPs, where this is the corresponding dimension.

The sequential extrapolation coefficient (SEC) of Xie et al. (2022a) subsumes the Bellman eluder dimension:

Definition 4.4 (Sequential Extrapolation Coefficient (SEC))**.**

$$\mathsf{SEC}(\mathcal{F}, \Pi, T) \coloneqq \max_{h\in[H]} \sup_{f^{(1)},\dots,f^{(T)}\in\mathcal{F}} \sup_{\pi^{(1)},\dots,\pi^{(T)}\in\Pi} \left\{ \sum_{t=1}^{T} \frac{\mathbb{E}_{\pi^{(t)}}[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}]^2}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{\pi^{(i)}}[(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)})^2]} \right\}. \tag{4.7}$$

The SEC is always bounded by $d \log T$, but there exist MDPs that have a Bellman eluder dimension d on the order of $\sqrt{T}$, but a constant SEC (Xie et al., 2022a). We shall use these measures of complexity to characterize the regret. Algorithm 4.1 scales with the SEC, which is more general and weaker than the Bellman eluder dimension. Algorithm 4.2, as presented, has a switching cost that scales with the $\mathcal{D}_\Delta$-type Bellman eluder dimension. While this can be weakened to the more general ℓ_2 eluder condition of Xiong et al. (2023a) with nothing more than a change in notation, we present our results in the Bellman eluder dimension framework for familiarity and ease of presentation.

Policy optimization and actor-critic algorithms. Policy optimization approaches optimize directly in the space of policies, enabled by the policy gradient theorem (Sutton and Barto, 2018): $\nabla_\pi V_1^\pi(s_1) = \mathbb{E}_\pi[Q_1^\pi(s,a)\nabla_\pi \log \pi(s,a)]$. This can be done with Monte-Carlo estimates of $Q_1^\pi(s,a)$ (the REINFORCE algorithm), or a learned estimate of $Q_1^\pi(s,a)$ called a critic (actor-critic methods).

However, vanilla policy gradient methods can converge very slowly in the worst case (Li et al., 2021). It is often preferable to use other optimization algorithms, such as a second-order method in natural policy gradient (NPG) (Kakade, 2001), KL-regularized gradient ascent in trust region policy optimization (TRPO) from Schulman et al. (2015), or proximal policy optimization (PPO), which performs a similar, but easier to compute, update. These methods are closely related in the limit of small step-sizes, and are approximate versions of mirror ascent (Schulman et al., 2015; Neu et al., 2017; Rajeswaran et al., 2017).

One instance in which the NPG, TRPO, and PPO updates coincide is with softmax policies (Cai et al., 2020; Cen et al., 2022; Agarwal et al., 2021): $\pi(a|s) = \exp(g(s,a))/\sum_{a\in\mathcal{A}} \exp(g(s,a))$ for

Algorithm 4.1 Double Optimistic Actor-Critic Under Bounded Policy-Cover Growth

Input: Function class $\mathcal{F}$.
Initialize: $\mathcal{F}^{(0)} \leftarrow \mathcal{F}$, $\mathcal{D}_h^{(0)} \leftarrow \emptyset$, learning rate $\eta = \Theta(\sqrt{\log|\mathcal{A}|H^{-2}T^{-1}})$, policy $\pi^{(1)} \propto 1$, confidence width $\beta = \Theta(\log(HT\mathcal{N}_{\mathcal{F},(\mathcal{T}^{\Pi})^T\mathcal{F}}(1/T)/\delta))$.
for episode $t = 1, 2, \ldots, T$ **do**
  $f_h^{(t)}(s,a) \in \operatorname{argmax}_{f \in \mathcal{F}^{(t-1,\pi^{(t-1)})}} f_h(s,a) \; \forall s, a, h.$
  Play policy $\pi^{(t)}$ for one episode, update dataset $\mathcal{D}_h^{(t)}$.
  Compute confidence set $\mathcal{F}^{(t,\pi^{(t)})}$:

$$\mathcal{F}^{(t,\pi^{(t)})} \leftarrow \left\{ f \in \mathcal{F} : \mathcal{L}_h^{(t,\pi^{(t)})}(f_h, f_{h+1}) - \min_{f'_h \in \mathcal{F}_h} \mathcal{L}_h^{(t,\pi^{(t)})}(f'_h, f_{h+1}) \le H^2\beta \; \forall h \right\},$$
$$\mathcal{L}_h^{(t,\pi^{(t)})}(f, f') \leftarrow \sum_{(s,a,r,s') \in \mathcal{D}_h^{(t)}} \left(f(s,a) - r - f'(s', \pi_{h+1}^{(t)}(s')) \right)^2.$$

  Update $\pi_h^{(t+1)}(a|s) \propto \pi_h^{(t)}(a|s)\exp(\eta f_h^{(t)}(s,a))$.
end for

some function $g : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. In this case, the update has a closed form:

$$\pi_h^{(t+1)}(a|s) \propto \pi_h^{(t)}(a|s)\exp(\eta f_h^{(t)}(s,a)), \; f_h^{(t)} \in \mathcal{F}. \tag{4.8}$$

As mirror ascent, this update is identical to the multiplicative weights or Hedge algorithm. Like Cai et al. (2020); Zhong and Zhang (2023); Liu et al. (2023b), we exploit this equivalence to prove our regret bounds.

4.1.3. Optimistic actor-critics – The easy case

We now present an optimistic actor-critic algorithm, Algorithm 4.1, with provable guarantees under general function approximation. Algorithm 4.1 is a natural derivative of the GOLF algorithm of Jin et al. (2021a) for actor-critic approaches[9], with only two (very natural) changes. The critic targets $Q^{\pi^{(t)}}$ instead of Q^* while performing optimistic planning at every s, a pair, and we maintain a stochastic policy that is updated with Equation 4.8. Learning an optimistic critic off-policy achieves sample efficiency by reusing data while exploring efficiently.[10]

[9] Or a completely off-policy version of Liu et al. (2023b).
[10] Similarly to Cai et al. (2020) in linear mixture MDPs.

Algorithm 4.1 maintains confidence sets $\mathcal{F}^{(t,\pi^{(t)})}$ that contain all $f \in \mathcal{F}$ where the TD error $\mathcal{L}_h^{(t,\pi^{(t)})}$ with respect to the Bellman operator $\mathcal{T}_h^{\pi^{(t)}}$ is a $H^2\beta$-additive approximation of the minimizer $\min_{f'_h \in \mathcal{F}_h} \mathcal{L}_h^{(t,\pi^{(t)})}(f'_h, f_{h+1})$. Upon the start of each trajectory t, Algorithm 4.1 maximizes among all functions in the confidence set to play $f_h^{(t)}(s,a) = \sup_{f_h \in \mathcal{F}_h^{(t-1,\pi^{(t-1)})}} f_h(s,a)$ for all h, s, a. This is exactly in line with GOLF, except that the critic targets $Q^{\pi^{(t)}}$ instead of Q^*, and we perform optimistic planning with regard to every state and action like Liu et al. (2023b). We then perform a mirror ascent update $\pi_h^{(t+1)} \propto \pi_h^{(t)} \exp(\eta f_h^{(t)})$ instead of playing the greedy policy $\pi_h^{f^{(t)}}$. This algorithm satisfies the following regret/sample complexity bound:

Theorem 4.5 (Regret Bound for Algorithm 4.1)**.** *Algorithm 4.1 achieves the following regret with probability at least* $1-\delta$*:*

$$\mathrm{Reg}(T) = O\left(\sqrt{H^4 T \log|\mathcal{A}|} + \sqrt{\beta H^4 T \mathsf{SEC}(\mathcal{F}, \Pi, T)}\right),$$

where $\beta = \Theta\left(\log\left(HT\mathcal{N}_{\mathcal{F},(\mathcal{T}^\Pi)^T\mathcal{F}}(1/T)/\delta\right)\right)$*. To learn an* ϵ*-optimal policy, it therefore requires:*

$$N \geq \Omega\left(H^4 \log|\mathcal{A}|/\epsilon^2 + \beta H^4 \mathsf{SEC}(\mathcal{F}, \Pi, T)/\epsilon^2\right).$$

Proof sketch: We decompose the regret with Lemma C.1.1:

$$\begin{aligned}\mathrm{Reg}(T) = &\underbrace{\sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}(s_h,\cdot), \pi_h^*(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h)\right\rangle\right]}_{\text{Tracking error of } \pi^{(t)} \text{ w.r.t. } \pi^*\text{, bounded by mirror ascent arguments.}} \\ &\underbrace{-\sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{\pi^*}\left[\left(f_h^{(t)} - \mathcal{T}_h^{\pi^*} f_{h+1}^{(t)}\right)(s_h,a_h)\right]}_{\text{Negative Bellman error under } \pi^*\text{, bounded by optimism.}} \\ &\underbrace{+\sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)}\right)(s_h,a_h)\right]}_{\text{Bellman error under current policy occupancy, bounded by critic error.}}.\end{aligned} \tag{4.9}$$

We will then bound each term in (4.9) separately. By a mirror ascent argument in Lemma C.1.1, the first term is bounded as

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}\left(s_h,\cdot\right),\pi_h^*\left(\cdot\mid s_h\right)-\pi_h^{(t)}\left(\cdot\mid s_h\right)\right\rangle\right]\leq \eta H^3T/2+\frac{H\log|\mathcal{A}|}{\eta}, \tag{4.10}$$

Lemma C.1.1 establishes optimism: $f_h^{(t)}\geq \mathcal{T}_h^{(t-1)}f_{h+1}^{(t)}$. So we see in Lemma C.1.1 that the second term is decomposed as

$$-\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[f_h^{(t)}-\mathcal{T}_h^{\pi^*}f_{h+1}^{(t)}\right]\leq \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_{h+1}^{(t)}(s_{h+1},\cdot),\pi_{h+1}^*(\cdot|s_{h+1})-\pi_{h+1}^{(t)}(\cdot|s_{h+1})\right\rangle\right]+\eta H^3T \tag{4.11}$$

$$\leq 3\eta H^3T/2+\frac{H\log|\mathcal{A}|}{\eta}, \tag{4.12}$$

where the the second inequality holds as the first term of (4.11) is a copy of the first term in (4.9), thus can be bounded by (4.10). For the third term, which is the Bellman error under the $\pi^{(t)}$ occupancy measure, we use Lemma C.1.1 to bound it as

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)}-\mathcal{T}_h^{\pi^{(t)}}f_{h+1}^{(t)}\right)(s_h,a_h)\right]\leq\sqrt{\beta H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}+\eta H^3T, \tag{4.13}$$

where $\beta=\Theta\left(\log\left(HT\mathcal{N}_{\mathcal{F},(\mathcal{T}^{\Pi})^T\mathcal{F}}(1/T)/\delta\right)\right)$. Plugging (4.10), (4.12) and (4.13) into (4.9), we obtain that

$$\begin{aligned}\mathrm{Reg}(T)&\leq\sqrt{\beta H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}+3\eta H^3T+\frac{2H\log|\mathcal{A}|}{\eta}\\&=O\left(\sqrt{H^4T\log|\mathcal{A}|}+\sqrt{\beta H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}\right),\end{aligned} \tag{4.14}$$

where the second equality holds when we set $\eta=O(\sqrt{\log|\mathcal{A}|H^{-2}T^{-1}})$. □

A Vacuous Bound. The form of the above regret bound is appealing at first glance. However, the upper bound on the log-covering number of the policy class increases linearly with the number

of significant[11] critic updates by default, as we show below in Lemma 4.6.[12]

Lemma 4.6 (Bound on Covering Number of Policy Class, Modified Lemma B.2 from Zhong and Zhang (2023))**.** *Consider the policy class $\Pi^{(T)}$ obtained by performing T updates of the mirror ascent update as in (4.8), where the critics $f_h^{(t)} \in \mathcal{F}$ are updated at times $t_1, ..., t_K$. Then, the covering number of the policy class at time T is bounded by*

$$\mathcal{N}_{\Pi_h^{(T)}}(\rho/2H) \leq \min\left\{\prod_{k=1}^{K} \mathcal{N}_{\mathcal{F}_h^{(t_k)}}(\rho^2/16\eta k H^2 T),\ \mathcal{N}_{\mathcal{A}}(\rho/2H)\right\}. \tag{4.15}$$

The proof is deferred to Appendix C.1.1. Within Algorithm 4.1, this implies that $\log \mathcal{N}_{(\mathcal{T}^{\Pi})^T\mathcal{F}}(\rho) = O(\min\{T, H\mathcal{N}_{\mathcal{A}}(\rho)\} \log \mathcal{N}_{\mathcal{F}}(\rho))$ in general, making the bound in Theorem 4.5 vacuous.

A Good Case. However, within certain circumstances, it is possible to have $\log \mathcal{N}_{(\mathcal{T}^{\Pi})^T\mathcal{F}}(\rho) = O(\log \mathcal{N}_{\mathcal{F}}(\rho))$. This happens, for instance, when there exists a low-dimensional representation of the sum of clipped Q-functions – such as when the sum of clipped Q-functions is a scaled Q-function:

Definition 4.7 (Closure Under Truncated Sums)**.** $\mathcal{F}$ is closed under truncated sums if for any $T \in \mathbb{N}$ and $f^{(1)}, ..., f^{(T)} \in \mathcal{F}$, $T^{-1}\sum_{t=1}^{T} \min\{\max\{f^{(t)}, 0\}, H\} \in \mathcal{F}$.

As such, this is an interesting case where the log-covering number of the policy class does not blow up, with downstream implications for the regret of Algorithm 4.1:

Lemma 4.8 (Policy Class Growth)**.** *Let $\mathcal{F}$ be a function class that satisfies Definition 4.7, i.e. it is closed under truncated sums. Then,* $\log \mathcal{N}_{(\mathcal{T}^{\Pi})^T\mathcal{F}}(\rho) = O(\mathcal{N}_{\mathcal{F}}(\rho^2/16\eta T H^2))$.

Corollary 4.9 (Regret of Algorithm 4.1, The Good Case)**.** *If $\mathcal{F}$ is closed under truncated sums,*

[11] One can use an ϵ-net and collapse nearby critics when $\exp(\eta(f - f')) < \epsilon$, which can reduce the bound significantly.

[12] It was previously thought in Zhong and Zhang (2023) that the log-covering number of the policy class increases in the number of actor and critic updates. We sharpen this bound to the number of critic updates in Lemma 4.6, which may be of independent interest.

then with probability at least $1-\delta$*, Algorithm 4.1 achieves a regret of:*

$$\mathrm{Reg}(T) = O\left(\sqrt{H^4 T \log|\mathcal{A}|} + \sqrt{\beta H^4 T \mathsf{SEC}(\mathcal{F}, \Pi, T)}\right),$$

where $\beta = \Theta(\log(HT\mathcal{N}_{\mathcal{F}}(H/\log|\mathcal{A}|T^2)/\delta))$*. To learn an* ϵ*-optimal policy, it therefore requires:*

$$N \geq \Omega\left(H^4 \log|\mathcal{A}|/\epsilon^2 + H^4 \beta \mathsf{SEC}(\mathcal{F}, \Pi, T)/\epsilon^2\right).$$

We defer the proof of Lemma 4.8 to Appendix C.1.1. Algorithm 4.1 then achieves a regret that aligns with the results of Efroni et al. (2020); Cai et al. (2020) for tabular and linear mixture MDPs, respectively.[13] However, closure under truncated sums is a strong condition that is not fulfilled by many function classes, although tabular classes fulfill it. It is not fulfilled by linear models due to the clipping operator, requiring Sherman et al. (2023) and Cassel and Rosenberg (2024) to develop bespoke algorithms to get around this in the setting of linear MDPs.[14] In the case of trees, random forests, boosting, and neural networks, without further assumptions, one needs to increase the size of the function class – perhaps even on the same order as the increase in Lemma 4.6.[15]

This prompts us to explore the possibility of algorithmic modifications to Algorithm 4.1 in order to achieve the optimal regret rates in more general settings where the log-covering number of the policy class may increase linearly in the number of critic updates. We do so in the next section.

4.1.4. Optimistic actor-critics – The hard case

Given what we have seen in our analysis of Algorithm 4.1, can we simply modify Algorithm 4.1 to include rare-switching critic updates? If we can perform only $O(dH \log T)$ critic updates as in

[13] Where both the SEC and Bellman eluder dimension are SA (tabular) and d (linear).

[14] The former use reward-agnostic exploration to warm-start the critic to avoid truncation. The latter shrink features to do the same. Both employ rare-switching updates for the bonus function but not the critic, avoiding the moving target issue.

[15] For instance, with neural networks, averaging two neural networks of 12 layers results in a larger neural network with 24 layers, and averaging two random forests of 500 trees results in a random forest of 1000 trees. However, one can argue that the effective complexity does not increase too much in these cases, and as such we can expect to see sublinear regret with certain nonparametric function classes.

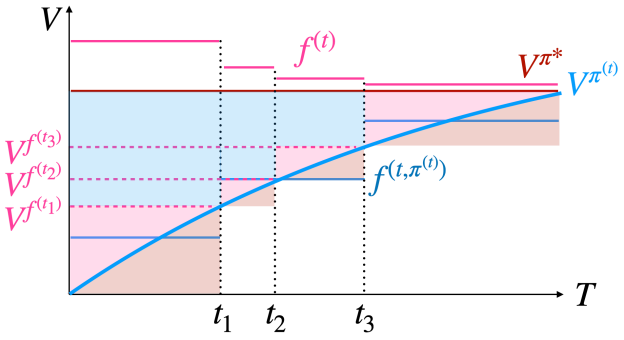


Figure 4.1: Illustration of tracking error in policy optimization, with a rare-switching critic $f^{(t)}$ that targets π^*. The {blue and pink, pink} area depicts the tracking error of $\{\pi^*$ to $\pi^{(t)}$, $\pi^{f^{(t)}}$ to $\pi^{(t)}\}$. Both incur $\sqrt{T}$ regret. In contrast, $f^{(t,\pi^{(t)})}$ is a rare-switching critic that targets $\pi^{(t)}$, and so is insufficiently optimistic as $\pi^{(t)}$ changes. The blue, pink, and rust area depicts the tracking error of π^* to $\pi^{(t_{\text{last}})}$ from insufficient optimism, which yields linear regret.

Xiong et al. (2023a), perhaps it may be possible to obtain a similar regret bound to Corollary 4.9.

Challenges and algorithm design

However, this is not the case. Intuitively, if the critic targets a rapidly changing $\pi^{(t)}$, the Bellman error with regard to the policy $\pi^{(t_{\text{last}})}$ at the last update will not be close to the Bellman error with regard to $\pi^{(t)}$. Although the former is what the critic $f^{(t_{\text{last}})} = f^{(t)}$ targets, we evaluate the latter when considering a switch at time t. Therefore, the critic updates far more often when we target $Q^{\pi^{(t)}}$ rather than Q^* – as it tries to hit a moving target.

Furthermore, this results in insufficient optimism. In Lemma C.1.1, we can only guarantee optimism with respect to the Bellman operator at the last critic update time t_{last}, $f_h^{(t)} \geq \mathcal{T}_h^{\pi^{(t_{\text{last}})}} f_{h+1}^{(t)}$. But we require $f_h^{(t)} \geq \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)}$. Attempting to work with this form of limited optimism results in the tracking error relative to π^* becoming

$$\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{\pi^*} \left[\left\langle f_h^{(t)}(s', \cdot), \pi_h^*(\cdot|s') - \pi_h^{(t_{\text{last}})}(\cdot|s') \right\rangle \right],$$

incurring linear regret. This is the shaded area in Figure 4.1.

A way forward. Having the critic target Q^* instead of $Q^{\pi^{(t)}}$ provides a solution. This ensures sufficient optimism, as we show in Lemma C.7 that $f_h^{(t)} \geq \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)}$. Furthermore, we do not need

to control $\log \mathcal{N}_{(\mathcal{T}^{\Pi})^T \mathcal{F}}(\rho)$, as $\max_a Q_{h+1}(s,a)$ is a contraction and it suffices to control $\log \mathcal{N}_{\mathcal{T}\mathcal{F}}(\rho)$, which is often approximately $\log \mathcal{N}_{\mathcal{F}}(\rho)$.

However, this introduces an additional term we need to control – the deviation of the current policy $\pi^{(t)}$ from the greedy policy $\pi^{f^{(t)}}$, depicted as the pink area in Figure 4.1. This term is difficult to control, as $\pi^{f^{(t)}}$ changes with every critic update, and the actor requires sufficient time to catch up to the critic updates. To address this issue, we introduce policy resets at every critic update, allowing us to bound this with the standard mirror descent regret bound as in Lemma C.5. The total tracking error scales with the number of critic updates – which is then resolved by performing rare-switching critic updates in line with Xiong et al. (2023a). Lastly, we increase the learning rate to reduce the regret incurred by the policy resets. This makes more aggressive policy updates to catch up with the sudden, rare, and large critic updates.

Summary. Algorithm 4.2 combines *optimism* for strategic exploration and *off-policy learning* for sample efficiency. While rare-switching critic updates are a-priori appealing, they are unstable when the critic targets $\pi^{(t)}$, due to limited optimism and the challenge of tracking a moving policy (as we describe below, and further show in Appendix C.1.3). These are resolved by *targeting* π^* and re-introducing *rare-switching critic updates* respectively. However, this introduces additional error, as the agent has to effectively unlearn the policy after each rare critic update. We therefore introduce *policy resets* to the uniform policy after each critic update – incurring minimal cost due to their rarity (as there are only $O(dH \log T)$ updates). A *more aggressive learning rate* (by a factor of $\sqrt{dH \log T}$) mitigates some of the additional regret incurred, and can be seen as making more aggressive updates to make up for lost ground from policy resets. See Appendix C.1.2 for more details.

Regret bound for rare-switching optimistic actor-critic

To control the switching cost with the framework of Xiong et al. (2023a), we concern ourselves with function classes of low D_Δ-type Bellman Eluder dimension (Jin et al., 2021a):

Assumption 4.10 (Bounded D_Δ-type Bellman Eluder Dimension)**.** Let $\mathcal{D}_\Delta := \{\mathcal{D}_{\Delta,h}\}_{h\in[H]}$, where $\mathcal{D}_{\Delta,h} = \left\{\delta_{(s,a)}(\cdot) \mid s \in \mathcal{S}, a \in \mathcal{A}\right\}$. That is, we only consider distributions that are Dirac deltas on

Algorithm 4.2 Rare-Switching Optimistic Actor-Critic with Uniform Actor Resets

Input: Function class $\mathcal{F}$.
Initialize: $\mathcal{F}^{(0)} \leftarrow \mathcal{F}$, offline dataset $\mathcal{D}_{\text{off}}$, $\mathcal{D}_h^{(0)} \leftarrow \emptyset, \forall h \in [H]$, $\eta = \Theta(\sqrt{d \log T \log |\mathcal{A}| H^{-1} T^{-1}})$, $\pi^{(1)} \propto 1$, confidence width $\beta = \Theta(\log(HT^2 \mathcal{N}_{\mathcal{F},\mathcal{TF}}(1/T)/\delta))$.
for episode $t = 1, 2, \ldots, T$ **do**
  Set $f_h^{(t)}(s,a) \in \operatorname{argmax}_{f \in \mathcal{F}^{(t_{\text{last}})}} f_h(s,a) \, \forall s, a, h$.
  Play policy $\pi^{(t)}$ for one episode, obtain trajectory, update dataset $\mathcal{D}_h^{(t)}$.
  if $\mathcal{L}_h^{(t)}(f_h^{(t)}, f_{h+1}^{(t)}) \geq \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t)}(f_h', f_{h+1}^{(t)}) + 5H^2\beta$ for some h **then**
    Compute confidence set $\mathcal{F}^{(t)}$:

$$\mathcal{F}^{(t)} \leftarrow \left\{ f \in \mathcal{F} : \mathcal{L}_h^{(t)}(f_h, f_{h+1}) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t)}(f_h', f_{h+1}) \leq H^2\beta \; \forall h \right\},$$
$$\mathcal{L}_h^{(t)}(f, f') \leftarrow \sum_{(s,a,r,s') \in \mathcal{D}_h^{(t)}} \left(f(s,a) - r - \max_{a' \in \mathcal{A}} f'(s', a') \right)^2.$$

    Reset policy $\pi^{(t)} \propto 1$.
    Set $t_{\text{last}} \leftarrow t$, $N_{\text{updates}}^{(t)} \leftarrow N_{\text{updates}}^{(t-1)} + 1$.
  else
    Set $N_{\text{updates}}^{(t)} \leftarrow N_{\text{updates}}^{(t-1)}$, $\mathcal{F}^{(t)} \leftarrow \mathcal{F}^{(t-1)}$.
  end if
  Update $\pi_h^{(t+1)}(a|s) \propto \pi_h^{(t)}(a|s) \exp(\eta f_h^{(t)}(s,a))$.
end for

a single state-action pair. We assume that $d := d_{\text{BE}}(\mathcal{F}, D_\Delta, 1/\sqrt{T}) < \infty$.

This can be weakened to the more general ℓ_2 eluder condition of Xiong et al. (2023a) with nothing more than a change in notation. However, we present our results in the Bellman eluder dimension framework for familiarity and ease of presentation. We now show the following for Algorithm 4.2:

Theorem 4.11 (Regret Bound for Algorithm 4.2)**.** *Algorithm 4.2 achieves the following regret with probability at least* $1 - \delta$*:*

$$\text{Reg}(T) = O\left(\sqrt{dH^5 T \log T \log |\mathcal{A}|} + dH^3 \log T + \sqrt{\beta H^4 T \mathsf{SEC}(\mathcal{F}, \Pi, T)} \right). \tag{4.16}$$

where $\beta = \Theta(\log\left(HT^2 \mathcal{N}_{\mathcal{F},\mathcal{TF}}(1/T)/\delta\right))$*. This implies a sample complexity (ignoring lower-order*

terms) of

$$N \geq \Omega\left(dH^5 \log T \log|\mathcal{A}|/\epsilon^2 + H^4\beta\mathsf{SEC}(\mathcal{F},\Pi,T)/\epsilon^2\right). \tag{4.17}$$

Proof sketch: By Lemma C.1, we have a slightly different regret decomposition than usual:

$$\begin{aligned}
\mathrm{Reg}(T) = & \underbrace{\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}(s_h,\cdot), \pi_h^*(\cdot\mid s_h) - \pi_h^{(t)}(\cdot\mid s_h)\right\rangle\right]}_{\text{Tracking error of } \pi^{(t)} \text{ w.r.t. } \pi^*, \text{ bounded by mirror ascent arguments.}} \\
& \underbrace{-\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left(f_h^{(t)} - \mathcal{T}_h^{\pi^*} f_{h+1}^{(t)}\right)(s_h,a_h)\right]}_{\text{Negative Bellman error under } \pi^*, \text{ bounded by optimism.}} \\
& \underbrace{+\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)(s_h,a_h)\right]}_{\text{Bellman error under current policy occupancy, bounded by critic error.}} \\
& \underbrace{+\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left\langle f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')\right\rangle\right]}_{\text{Tracking error of } \pi^{(t)} \text{ w.r.t. } \pi^{f^{(t)}}, \text{ bounded by mirror ascent and policy resets.}}.
\end{aligned} \tag{4.18}$$

We will then bound each term in (4.18) separately. We further decompose the first term as follows

$$\begin{aligned}
&\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left\langle f_h^{(t)}(s_{h+1},\cdot), \pi_{h+1}^{\star}(\cdot\mid s_{h+1}) - \pi_{h+1}^{(t)}(\cdot\mid s_h)\right\rangle\right] \\
&= \sum_{k=1}^{K}\sum_{t=t_k+1}^{t_{k+1}}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left\langle f_h^{(t)}(s_h,\cdot), \pi_h^{\star}(\cdot\mid s_h) - \pi_h^{(t)}(\cdot\mid s_h)\right\rangle\right] \\
&= \sum_{k=1}^{K}\sum_{t=t_k+1}^{t_{k+1}-1}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left\langle f_h^{(t)}(s_h,\cdot), \pi_h^{\star}(\cdot\mid s_h) - \pi_h^{(t)}(\cdot\mid s_h)\right\rangle\right] \\
&\quad + \sum_{k=1}^{K}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left\langle f_h^{(t_k)}(s_h,\cdot), \pi_h^{\star}(\cdot\mid s_h) - \pi_h^{(t_k)}(\cdot\mid s_h)\right\rangle\right].
\end{aligned} \tag{4.19}$$

For the first term of (4.19), following Lemma C.2 and summing over all $t_1, \ldots, t_K$, we have

$$\sum_{k=1}^{K} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^*} \left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^*(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h) \right\rangle \right] \leq \eta H^3 T/2 + \frac{KH \log |\mathcal{A}|}{\eta} + KH^2, \quad (4.20)$$

while for the second term, one may note that each inner product can be bounded by $2H$, therefore the summation can be bounded as

$$\sum_{k=1}^{K} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left\langle f_h^{(t_k)}(s_h, \cdot), \pi_h^\star(\cdot \mid s_h) - \pi_h^{(t_k)}(\cdot \mid s_h) \right\rangle \right] \leq 2KH^2. \quad (4.21)$$

We apply this similar strategy on the fourth term of (4.18), which leads to the following decomposition

$$\begin{aligned}
&\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left\langle f_h^{(t)}(s_{h+1}, \cdot), \pi_{h+1}^{f^{(t)}}(\cdot \mid s_{h+1}) - \pi_{h+1}^{(t)}(\cdot \mid s_h) \right\rangle \right] \\
&= \sum_{k=1}^{K} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h) \right\rangle \right] \\
&\quad + \sum_{k=1}^{K} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left\langle f_h^{(t_k)}(s_h, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t_k)}(\cdot \mid s_h) \right\rangle \right], \quad (4.22)
\end{aligned}$$

and the first term of (4.22) is bounded by Lemma C.5 with summing over all $t_1, \ldots, t_K$ as

$$\sum_{k=1}^{K} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_{h+1}^{(t)}(s', \cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \right] \leq \eta H^3 T/2 + \frac{KH \log |\mathcal{A}|}{\eta} + KH^2, \quad (4.23)$$

while the second term of (4.22) can be bounded similar to (4.21) as

$$\sum_{k=1}^{K} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left\langle f_h^{(t_k)}(s_h, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t_k)}(\cdot \mid s_h) \right\rangle \right] \leq 2KH^2. \quad (4.24)$$

It only remains to bound the second term and third term of (4.18). We note that targeting Q^* yields sufficient optimism to assert that $f_h^{(t)} \geq \mathcal{T}_h^{\pi^*} f_{h+1}^{(t)}$ in Lemma C.7, and so we argue that the

second term in (4.18) for the negative Bellman error under π^* is nonpositive in Lemma C.3,

$$-\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left(f_h^{(t)}-\mathcal{T}_h^{\pi^*}f_{h+1}^{(t)}\right)(s_h,a_h)\right]\leq 0 \tag{4.25}$$

The third term via a standard argument from Xie et al. (2022a) in Lemma C.4,

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)}-\mathcal{T}_h f_{h+1}^{(t)}\right)(s_h,a_h)\right]\leq\sqrt{\beta H^4 T\mathsf{SEC}(\mathcal{F},\Pi,T)}. \tag{4.26}$$

Plugging (4.19) – (4.26) into (4.18), we obtain that

$$\begin{aligned}\mathrm{Reg}(T)&\leq\eta H^3T+\frac{2KH\log|\mathcal{A}|}{\eta}+6KH^2+\sqrt{\beta H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}\\&\leq\eta H^3T+\frac{2dH^2\log T\log|\mathcal{A}|}{\eta}+6dH^3\log T+\sqrt{\beta H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}\\&=2\sqrt{2dH^5T\log T\log|\mathcal{A}|}+6dH^3\log T+\sqrt{\beta H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}\end{aligned} \tag{4.27}$$

where the second inequality is obtained by bounding the number of critic updates by $dH\log T$ in Lemma C.6 with the techniques of Xiong et al. (2023a), and the last equality holds when we set the learning rate $\eta=\Theta(\sqrt{d\log T\log|\mathcal{A}|H^{-1}T^{-1}})$. Here we conclude the proof. □

Quality of the regret bound. Even compared to the "good case" for Algorithm 4.1 in Corollary 4.9, Algorithm 4.2 requires only $dH\log T$ more samples to learn an ϵ-optimal policy. This is exactly the same as the switch cost, in line with what Cassel and Rosenberg (2024) observed for linear MDPs.

It is known that the Bellman eluder dimension may scale unfavorably with T in rare cases (Xie et al., 2022a) where $d=d_{\mathrm{BE}}(\mathcal{F},D_\Delta,1/\sqrt{T})=\Omega(\sqrt{T})$. Although we use the SEC of Xie et al. (2022a) to bound the online regret whenever possible, as it is never larger than $O(d\log T)$, our result still depends on d due to the switch cost. Still, function classes with low Bellman eluder dimension are ubiquitous, and so Algorithm 4.2 achieves $\sqrt{T}$ regret on a large class of problems including linear and kernel MDPs (Jin et al., 2021a). We note that it is possible to weaken the

dependence on the d_Δ-type Bellman eluder dimension to the ℓ_2 eluder condition of Xiong et al. (2023a) with nothing more than a change of notation, but we present our results in the language of the Bellman eluder dimension for ease of presentation.

Comparison with prior work. The only other method we know of that claims to have achieved $\sqrt{T}$ regret with policy optimization and general function approximation is Zhou et al. (2023). However, they allow themselves to collect $\Omega(T)$ samples *at every timestep t without incurring any additional regret*. This incurs a sample complexity of $1/\epsilon^6$, while Algorithm 4.2 enjoys a $1/\epsilon^2$ guarantee in comparison. As such, their regret bound is not sublinear in the more common setup where this sampling contributes to the regret. Liu et al. (2023b) achieve a result of $1/\epsilon^3$ by performing critic updates in batches. However, Liu et al. (2023b) do not account for the potential growth of the policy class in the $O(\sqrt{T})$ critic updates they make. Their result of $1/\epsilon^3$ sample complexity is therefore somewhat optimistic, and may very well be $1/\epsilon^4$.

Extension to other policy updates. We are able to accommodate other policy optimization updates other than the multiplicative weights update, if they satisfy the following:

Corollary 4.12. *If there exists some policy optimization oracle $\pi^{(t+1)} \leftarrow \mathrm{PO}(\pi^{(t)}, f^{(t)}, \eta)$ and some OPT^* and OPT^f such that*

$$\sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}(s,\cdot), \pi_h^*(\cdot|s) - \pi_h^{(t)}(\cdot|s)\right\rangle\right] \le \mathrm{OPT}^*, \tag{4.28a}$$

$$\sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}}\left[\left\langle f_h^{(t)}(s,\cdot), \pi_h^{f^{(t)}}(\cdot|s) - \pi_h^{(t)}(\cdot|s)\right\rangle\right] \le \mathrm{OPT}^f, \tag{4.28b}$$

then Algorithm 4.2 with this policy update obtains a regret of

$$\mathrm{Reg}(T) = O\left(\sqrt{\beta H^4 T \mathsf{SEC}(\mathcal{F}, \Pi, T)} + \mathrm{OPT}^* + \mathrm{OPT}^f\right). \tag{4.29}$$

Connections to deep actor-critic methods

Targeting Q^*. We are not the only ones who consider a critic that targets Q^* instead of $Q^{\pi^{(t)}}$. By way of illustration, Crites and Barto (1994) propose an actor-critic algorithm that mimics Q-learning

via a critic that targets Q^*. Another example is the popular DDPG algorithm (Lillicrap et al., 2015) and its successor, TD3 (Fujimoto et al., 2018), that take turns to update an approximately greedy deterministic policy that approximately maximizes an estimate of the approximately greedy deterministic policy's Q-function (and so approximately targets Q^*), while performing stochastic exploration that tracks the deterministic policy over time.

Similarity to deep deterministic policy gradients (DDPG). Several aspects of the design of Algorithm 4.2 are reminiscent of the popular DDPG algorithm of Lillicrap et al. (2015). DDPG alternates between optimizing a deterministic policy $g_h^{(t)}(s)$ and a critic $f_h^{(t)}(s,a)$:

$$g_h^{(t)}(s) \approx \arg\max_{a \in \mathcal{A}} f_h^{(t)}(s,a), \tag{4.30}$$

$$f_h^{(t)}(s,a) \approx \arg\min \mathcal{L}_h^{(t,g^{(t-k)})}(f_h, f_{h+1}^{(t-k)}), \tag{4.31}$$

while exploring according to a stochastic exploration policy that tracks the deterministic policy. In practice, the exploration policy $\pi_{\text{exp}}^{(t)}(a|s) \propto \mathbf{1}(a = g^{(t)}(s)) + \mathcal{N}_t$ is often given by the deterministic function perturbed by Gaussian noise. The critic is optimized according to slowly updating targets approximating the actor and critic from a few steps ago – in a manner reminiscent of the rare-switching critic update in Algorithm 4.2. As the deterministic policy approximates the greedy policy, and the approximately greedy policy is used when computing TD error targets, the DDPG critic approximately targets Q^*. This suggests that DDPG and its successor TD3 (Fujimoto et al., 2018) may be useful backbones for adapting algorithmic insights garnered from the design and analysis of Algorithm 4.2 for practical RL.[16]

4.1.5. Extension to hybrid RL

In this section, we demonstrate that the benefits of having access to both offline and online data are two-fold with actor-critic algorithms. With optimism, actor-critic algorithms achieve sample efficiency gains. However, if one does not wish to use an optimistic algorithm (as optimism is difficult to implement in practice with deep function approximation), having access to offline data

[16]This provides insight on why DDPG/TD3 are more sample-efficient than on-policy PPO. Algorithm 4.2 needs $1/\epsilon^2$ samples, while on-policy sampling needs at least $1/\epsilon^4$ (Liu et al., 2023b)

allows one to omit the use of optimism, achieving computational efficiency gains (e.g., Jin et al., 2021b; Li et al., 2024a; Rashidinejad et al., 2021).

Computational efficiency through optimism-free hybrid RL

The benefits of hybrid RL extend beyond sample efficiency gains. Song et al. (2023); Amortila et al. (2024); Zhou et al. (2023) show that it is possible for a hybrid RL algorithm to bypass the need for optimism altogether, as long as the offline dataset achieves sufficient coverage. In this spirit, we provide two non-optimistic hybrid actor-critic variants in Algorithms 4.3 and 4.4 that achieve $\sqrt{T}$ regret without using optimism. Algorithm 4.3 targets $\pi^{(t)}$, and follows the very natural procedure of performing a critic update via Fitted Q-Evaluation (FQE) (Munos and Szepesvári, 2008) and an actor update in every episode. It therefore requires closure of the critic function class under truncated sums as in Definition 4.7 to control the growth of the policy class as in Lemma 4.8. Algorithm 4.4, like Algorithm 4.2, circumvents this by targeting π^* and performing a rare-switching critic update. Both algorithms are fully off-policy, utilizing offline data and all collected online data without throwing any away.

We use the following form of the single-policy concentrability coefficient, tweaked from that of Zhan et al. (2022), that has some resemblance to the transfer coefficient of Song et al. (2023):

Definition 4.13 (Single-Policy Concentrability Coefficient)**.**

$$c^*_{\mathrm{off}}(\mathcal{F},\Pi) = \max_{h\in[H]} \sup_{f\in\mathcal{F}} \sup_{\pi\in\Pi} \frac{\mathbb{E}_{\pi^*}\left[f_h - \mathcal{T}^\pi_h f_{h+1}\right]^2}{\mathbb{E}_\mu[\left(f_h - \mathcal{T}^\pi_h f_{h+1}\right)^2]}, \quad c^*_{\mathrm{off}}(\mathcal{F}) = \max_{h\in[H]} \sup_{f\in\mathcal{F}} \frac{\mathbb{E}_{\pi^*}\left[f_h - \mathcal{T}_h f_{h+1}\right]^2}{\mathbb{E}_\mu[(f_h - \mathcal{T}_h f_{h+1})^2]}.$$

Notably, this is similar to the squared Bellman error variant of the single policy concentrability coefficient in the literature, with the exception that the square in the numerator is outside the expectation, and we have an additional supremum over policies that the Bellman operator is taken with respect to in the first definition.

With the definition of single-policy concentrability coefficient, we provide a guarantee for this

Algorithm 4.3 Non-Optimistic Hybrid Actor-Critic Targeting $\pi^{(t)}$

Input: Function class $\mathcal{F}$.
Initialize: $\mathcal{F}^{(0)} \leftarrow \mathcal{F}$, $\mathcal{D}_h^{(0)} \leftarrow \emptyset, \forall h \in [H]$, $\eta = \Theta(\sqrt{\log|\mathcal{A}|H^{-2}T^{-1}})$, $\pi^{(1)} \propto 1$, confidence width $\beta = \Theta(\log(HT^2\mathcal{N}_{\mathcal{F},\mathcal{TF}}(1/T)/\delta))$.
for episode $t = 1, 2, \ldots, T$ **do**
    Play policy $\pi^{(t)}$ for one episode, obtain trajectory, update dataset $\mathcal{D}_h^{(t)}$.
    Compute critic $f^{(t+1)}$ targeting $\pi^{(t)}$ via FQE:

$$f^{(t)} \leftarrow \arg\min_{f\in\mathcal{F}} \mathcal{L}_h^{(t,\pi^{(t)})}(f_h, f_{h+1}) \text{ for } h = H-1, ..., 1,$$
$$\mathcal{L}_h^{(t,\pi^{(t)})}(f, f') \leftarrow \sum_{(s,a,r,s')\in\mathcal{D}_h^{(t)}\cup\mathcal{D}_{\text{off}}} \left(f(s,a) - r - f'(s', \pi_{h+1}^{(t)}(s'))\right)^2.$$

    Update $\pi_h^{(t+1)}(a|s) \propto \pi_h^{(t)}(a|s)\exp(\eta f_h^{(t)}(s,a))$.
end for

procedure below, but defer the proof to Appendices C.1.3 and C.1.3. At a high level, the proofs proceed in a similar way to Theorems 4.5 and 4.11, with the exception that the negative Bellman error under π^* is bounded using the critic's Bellman error under the offline data.

Theorem 4.14 (Regret Guarantee of Algorithm 4.3)**.** *Let $c^*_{\text{off}}(\mathcal{F}, \Pi)$ be the single-policy concentrability coefficient defined in Definition 4.13. Algorithm 4.3 satisfies with probability at least $1-\delta$:*

$$\text{Reg}^{\pi}(T) = O\left(\sqrt{H^4T\log|\mathcal{A}|} + \sqrt{\beta^{\pi}H^4c^*_{\text{off}}(\mathcal{F},\Pi)T^2/N_{\text{off}}} + \sqrt{\beta^{\pi}H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}\right). \quad (4.32)$$

where $\beta^{\pi} = \Theta\left(\log\left(HT\mathcal{N}_{\mathcal{F},(\mathcal{T}^{\Pi})^T\mathcal{F}}(1/T)/\delta\right)\right)$.

Theorem 4.15 (Regret Guarantee of Algorithm 4.4)**.** *Let $c^*_{\text{off}}(\mathcal{F})$ be the single-policy concentrability coefficient defined in Definition 4.13. Algorithm 4.4 satisfies with probability at least $1-\delta$:*

$$\text{Reg}^{*}(T) = O\left(\sqrt{dH^5T\log T\log|\mathcal{A}|} + dH^3\log T + \sqrt{\beta^{*}H^4c^*_{\text{off}}(\mathcal{F})T^2/N_{\text{off}}} + \sqrt{\beta^{*}H^4T\mathsf{SEC}(\mathcal{F},\Pi,T)}\right), \quad (4.33)$$

where $\beta^* = \Theta(\log\left(HT^2\mathcal{N}_{\mathcal{F},\mathcal{TF}}(1/T)/\delta\right))$.

As such, in exchange for omitting optimism, we simply incur an additional $\sqrt{\beta H^4c^*_{\text{off}}(\mathcal{F})T^2/N_{\text{off}}}$

Algorithm 4.4 Non-Optimistic Hybrid Actor-Critic Targeting π^*

Input: Function class $\mathcal{F}$.

Initialize: $\mathcal{F}^{(0)} \leftarrow \mathcal{F}$, $\mathcal{D}_h^{(0)} \leftarrow \emptyset, \forall h \in [H]$, $\eta = \Theta(\sqrt{d \log T \log |\mathcal{A}| H^{-1} T^{-1}})$, $\pi^{(1)} \propto 1$, confidence width $\beta = \Theta(\log(HT^2 \mathcal{N}_{\mathcal{F},\mathcal{TF}}(1/T)/\delta))$.

for episode $t = 1, 2, \ldots, T$ **do**

  Play policy $\pi^{(t)}$ for one episode, obtain trajectory, update dataset $\mathcal{D}_h^{(t)}$.

  if $\mathcal{L}_h^{(t)}(f_h^{(t)}, f_{h+1}^{(t)}) \geq \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t)}(f_h', f_{h+1}^{(t)}) + 5H^2\beta$ for some h **then**

    Compute critic $f^{(t+1)}$ via FQE:

$$f^{(t)} \leftarrow \arg\min_{f \in \mathcal{F}} \mathcal{L}_h^{(t)}(f_h, f_{h+1}) \text{ for } h = H-1, ..., 1,$$

$$\mathcal{L}_h^{(t)}(f, f') \leftarrow \sum_{(s,a,r,s') \in \mathcal{D}_h^{(t)} \cup \mathcal{D}_{\text{off}}} \left(f(s,a) - r - \max_{a' \in \mathcal{A}} f'(s', a') \right)^2.$$

    Reset policy $\pi^{(t)} \propto 1$.

    Set $t_{\text{last}} \leftarrow t$, $N_{\text{updates}}^{(t)} \leftarrow N_{\text{updates}}^{(t-1)} + 1$.

  else

    Set $N_{\text{updates}}^{(t)} \leftarrow N_{\text{updates}}^{(t-1)}$, $f^{(t+1)} \leftarrow f^{(t)}$.

  end if

  Update $\pi_h^{(t+1)}(a|s) \propto \pi_h^{(t)}(a|s) \exp(\eta f_h^{(t)}(s,a))$.

end for

error term, which amounts to $\sqrt{T}$ regret as long as $N_{\text{off}} = \Omega(T)$.

Still, our result shows that the provable efficiency achieved without optimism by hybrid FQI-type methods (Song et al., 2023) also extends to actor-critic methods. At the same time, this also resolves the issue within Zhou et al. (2023) of needing to collect $\Theta(T)$ samples at every timestep t. We therefore achieve a true $\sqrt{T}$ regret bound and a sample complexity of $1/\epsilon^2$, in contrast to the $1/\epsilon^6$ sample complexity of Zhou et al. (2023).

Sample efficiency gains with hybrid RL

We now extend Algorithm 4.2 to leverage both offline and online data, which aligns with the framework of Tan and Xu (2024). The extension, found in Algorithm 4.5, appends N_{off} offline samples to the online data at rounds $t = 1, ..., T$ and minimizes the TD error over the combined dataset when constructing the confidence sets.

Before we proceed, it is useful to introduce the partial all-policy concentrability coefficient from

Algorithm 4.5 Rare-Switching Optimistic Actor-Critic for Hybrid RL

Input: Offline dataset $\mathcal{D}_{\text{off}}$ which can be the empty set, samples sizes T, N_{off}, function class $\mathcal{F}$ and confidence width $\beta > 0$
Initialize: $\mathcal{F}^{(0)} \leftarrow \mathcal{F}$, $\mathcal{D}_h^{(0)} \leftarrow \emptyset, \forall h \in [H]$, $\eta = \Theta(\sqrt{d \log T \log |\mathcal{A}| H^{-1} T^{-1}})$, $\pi^{(1)} \propto 1$.
for episode $t = 1, 2, \ldots, T$ **do**
  Select critic $f_h^{(t)}(s,a) := \operatorname{argmax}_{f_h \in \mathcal{F}_h^{(t_{\text{last}})}} f_h(s,a)$ for all s, a.
  Play policy $\pi^{(t)}$ for one episode and obtain trajectory $(s_1^{(t)}, a_1^{(t)}, r_1^{(t)}), \ldots, (s_H^{(t)}, a_H^{(t)}, r_H^{(t)})$.
  Update dataset $\mathcal{D}_h^{(t)} \leftarrow \mathcal{D}_h^{(t-1)} \cup \{(s_h^{(t)}, a_h^{(t)}, r_h^{(t)}, s_{h+1}^{(t)})\}, \forall h \in [H]$.
  if there exists some h such that $\mathcal{L}_h^{(t)}(f_h^{(t)}, f_{h+1}^{(t)}) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t)}(f_h', f_{h+1}^{(t)}) \geq 5H^2\beta$ **then**
    Compute confidence set $\mathcal{F}^{(t)}$:

$$\mathcal{F}^{(t)} \leftarrow \left\{ f \in \mathcal{F} : \mathcal{L}_h^{(t)}(f_h, f_{h+1}) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t)}(f_h', f_{h+1}) \leq H^2\beta \quad \forall h \in [H] \right\},$$

$$\text{where } \mathcal{L}_h^{(t)}(f, f') := \sum_{(s,a,r,s') \in \mathcal{D}_h^{(t)} \cup \mathcal{D}_{\text{off},h}} \left(f(s,a) - r - \max_{a' \in \mathcal{A}} f'(s', a') \right)^2, \forall f \in \mathcal{F}_h, f' \in \mathcal{F}_{h+1}.$$

    Reset policy $\pi^{(t)} \propto 1$.
    Set $t_{\text{last}} := t$, increment number of updates $N_{\text{updates}}^{(t)} := N_{\text{updates}}^{(t-1)} + 1$.
  else
    Set $N_{\text{updates}}^{(t)} := N_{\text{updates}}^{(t-1)}$, $\mathcal{F}^{(t)} := \mathcal{F}^{(t-1)}$.
  end if
  Select policy $\pi_h^{(t+1)} \propto \pi_h^{(t)} \exp(\eta f_h^{(t)})$.
end for

Tan and Xu (2024).

Definition 4.16 (Partial All-Policy Concentrability Coefficient)**.** For a function class $\mathcal{F}$ and a partition on the state-action space $\mathcal{X} = \mathcal{S} \times \mathcal{A} \times [H]$, where we denote the offline and online partitions by $\mathcal{X}_{\text{off}}$ and $\mathcal{X}_{\text{on}}$, respectively, the partial all-policy concentrability coefficient is given by:

$$c_{\text{off}}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{off}}}) := \max_h \sup_{f \in \mathcal{F}} \frac{\|(f_h - \mathcal{T}_h f_{h+1})\mathbf{1}_{(\cdot,h) \in \mathcal{X}_{\text{off}}}\|_{2,d_h^\pi}^2}{\|(f_h - \mathcal{T}_h f_{h+1})\mathbf{1}_{(\cdot,h) \in \mathcal{X}_{\text{off}}}\|_{2,\mu_h}^2},$$

where $\mathbf{1}_{\mathcal{X}_{\text{off}}}$ denotes the indicator variable for whether $s, a \in \mathcal{X}_{\text{on}}$.

This yields the following guarantee:

Theorem 4.17 (Hybrid RL Regret Bound for Algorithm 4.2)**.** *Let $c_{\text{off}}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{off}}})$ be the partial all-*

policy concentrability coefficient defined in Definition 4.16 and let $\mathcal{X}_{\text{off}}, \mathcal{X}_{\text{on}}$ *be an arbitrary partition over* $\mathcal{X} = \mathcal{S} \times \mathcal{A} \times [H]$*. Algorithm 4.5 satisfies with probability at least* $1 - \delta$*:*

$$\text{Reg}(T) = \mathcal{O}\left(\inf_{\mathcal{X}_{\text{on}}, \mathcal{X}_{\text{off}}} \left(\sqrt{\beta H^4 c_{\text{off}}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{off}}}) T^2 / N_{\text{off}}} + \sqrt{\beta H^4 T \mathsf{SEC}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{on}}}, \Pi, T)} + \sqrt{dH^5 T \log |\mathcal{A}|}\right)\right), \tag{4.34}$$

where $\beta = C \log\left(HN\mathcal{N}_{\mathcal{F},\mathcal{TF}}(1/N)/\delta\right)$ *for some constant* C*,* $N = N_{\text{off}} + T$*,* $\mathbf{1}_{\mathcal{X}_{\text{off}}}, \mathbf{1}_{\mathcal{X}_{\text{on}}}$ *are indicator variables for whether* $s, a \in \mathcal{X}_{\text{off}}$ *or* $\mathcal{X}_{\text{on}}$*, and* $c_{\text{off}}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{off}}})$ *is the partial all-policy concentrability coefficient (Tan and Xu, 2024).*

We defer further details to Appendix C.1.3. On the high level, the critic error is split into an offline and an online term. One uses the coverage of the offline data to bound the former and online exploration for the latter. Algorithm 4.5 is completely unaware of the partition, but the regret bound optimizes over all partitions of the state-action space. Should the offline data be plentiful and of good coverage, the regret approximately becomes $\sqrt{\beta H^4 T \mathsf{SEC}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{on}}}, \Pi, T)} + \sqrt{dH^5 T \log |\mathcal{A}|}$ for some small $\mathcal{X}_{\text{on}}$, yielding improvements over Theorem 4.11. This shows that actor-critic methods can benefit from hybrid data, achieving the provable gains in sample efficiency compared to offline-only and online-only learning observed by Li et al. (2023); Tan et al. (2024b).

This guarantee in Theorem 4.17 for Algorithm 4.5 is stronger than the result in Theorem 4.15 for Algorithm 4.4. As mentioned earlier, if the offline data is plentiful and enjoys good coverage, the regret in Theorem 4.17 primarily comes from the $\sqrt{dH^5 T \log |\mathcal{A}|}$ cost of optimizing the actor as $\mathsf{SEC}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{on}}}, \Pi, T)$ is small if $\mathcal{X}_{\text{on}}$ is small.[17] On the other hand, Algorithm 4.4 incurs a term depending on $\mathsf{SEC}(\mathcal{F}, \Pi, T)$, which can be much larger than $\mathsf{SEC}(\mathcal{F}\mathbf{1}_{\mathcal{X}_{\text{on}}}, \Pi, T)$.

4.1.6. Numerical experiments

We provide two numerical experiments to empirically verify our findings. Details on reproducing our findings are deferred to Appendix C.1.3.

[17] Which can possibly be mitigated as well by performing offline policy optimization.

Optimism in linear MDPs. The first experiment examines Algorithms 4.1 and 4.2 in a linear MDP setting, in order to validate if they indeed achieve $\sqrt{T}$ regret in practice. Accordingly, we implement optimism with LSVI-UCB bonuses (Jin et al., 2020) instead of global optimism as in GOLF.

We compare our algorithms to a rare-switching[18] version of LSVI-UCB on a linear MDP tetris task (Tan et al., 2024b; Tan and Xu, 2024). The condition required for Algorithm 4.1 to work holds here, as we do not clip the Q-function estimates. Figure 4.2 shows that Algorithm 4.1 (surprisingly) performs better than LSVI-UCB, and Figure 4.3 empirically illustrates that Algorithm 4.2 also achieves $\sqrt{T}$ regret even though it performs slightly worse than LSVI-UCB.

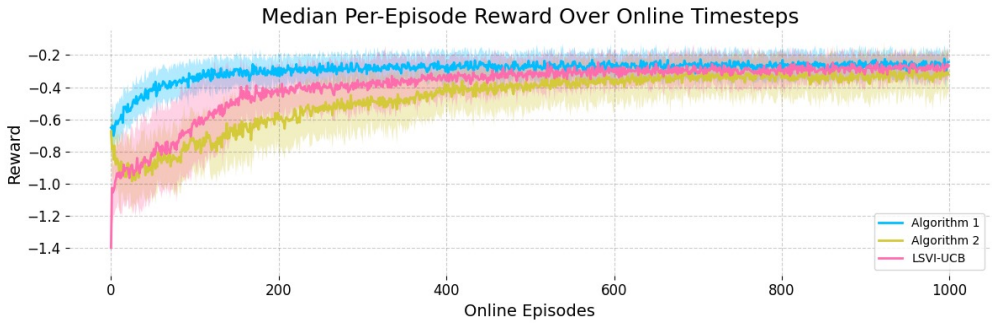


Figure 4.2: Per-episode reward of Algorithms 4.1 and 4.2 (labeled `Alg. 1` and `Alg. 2` in the legend), compared to a rare-switching version of LSVI-UCB (Jin et al., 2020) on a linear MDP Tetris task. Here, the condition required for Algorithm 4.1 holds, as we do not clip the Q-function estimates. Algorithm 4.1 outperforms LSVI-UCB, and Algorithm 4.2 catches up after some time. Results are averaged over 30 trials.

Deep hybrid RL. We compare practical deep RL variants of Algorithms 4.3 and 4.4 (denoted Algorithms 1H and 2H, matching the plot labels in Figure 4.4) against Cal-QL (Nakamoto et al., 2023). Algorithms 1H and 2H employ offline pretraining of both the actor and the critic, and the actor employs the soft actor-critic (SAC) update from Haarnoja et al. (2018). Additionally, Algorithm 2H omits the policy resets. Like Cal-QL, and unlike Algorithm 1H, Algorithm 2H takes a maximum over 10 randomly sampled actions from the policy to enhance exploration. This can be viewed as a computationally efficient version of optimism, or alternatively as approximately targeting π^*.

[18]Implemented with the doubling determinant method used in He et al. (2023). This incurs at most a $dH \log T$ switching cost.

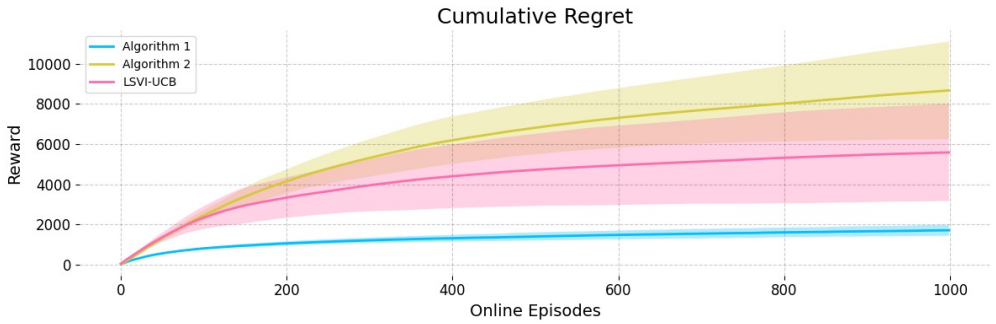


Figure 4.3: Cumulative regret of Algorithms 4.1 and 4.2 (labeled `Alg. 1` and `Alg. 2` in the legend) versus LSVI-UCB (Jin et al., 2020) on Tetris. This setting favors Algorithm 4.1, which outperforms LSVI-UCB, while Algorithm 4.2 also achieves $\sqrt{T}$ regret. Results are averaged over 30 trials.

Algorithm 2H outperforms Cal-QL, which in turn slightly outperforms Algorithm 1H. These results suggest that Algorithms 1H and 2H remain highly competitive, and may perform just as well as the state of the art in hybrid RL (Cal-QL) – even without the use of pessimism. Our results support our theoretical findings that hybrid RL allows for computationally efficient actor-critic algorithms.

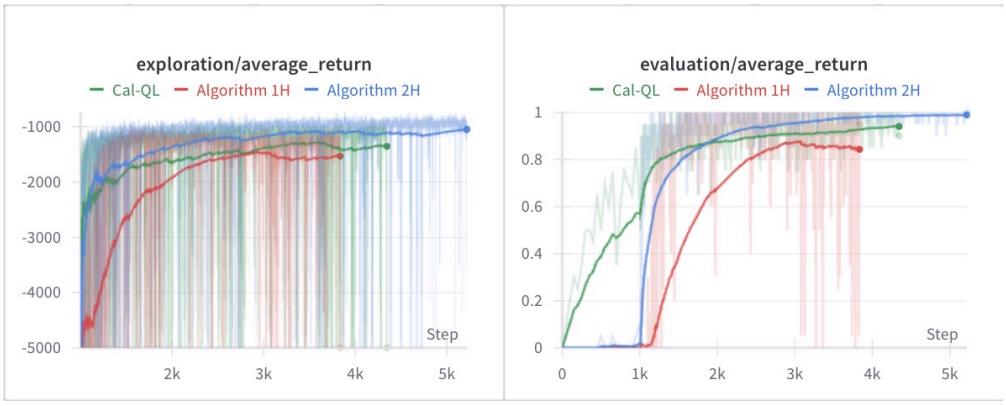


Figure 4.4: Hybrid RL experiment on the `antmaze-medium-diverse-v2` task comparing Algorithms 1H and 2H (deep RL adaptations of Algorithms 4.3 and 4.4) against Cal-QL (Nakamoto et al., 2023) with offline pretraining. Algorithm 2H uses approximate optimism via action sampling and outperforms both Cal-QL and Algorithm 1H without pessimism. Evaluation plots show offline pretraining in the first 1000 steps, with exponential smoothing.

4.1.7. Conclusions and future work

To conclude, we resolve an open problem in the online RL literature by designing an actor-critic method, Algorithm 4.2, with general function approximation that achieves $1/\epsilon^2$ (and matching $\sqrt{T}$ regret) without making any reachability or coverage assumptions. This was achieved through

several key ingredients in the algorithm design. By (1) performing optimistic exploration, (2) learning a critic off-policy without throwing away any data, (3) having the critic target Q^* instead of $Q^{\pi^{(t)}}$ to ensure sufficient optimism and enable rare-switching critic updates, and (4) performing policy resets to control the deviation of the current policy from the greedy policy, we achieve the desired result in Theorem 4.11.

We resolve another open problem in the hybrid RL literature by providing a non-optimistic provably efficient actor-critic algorithm that only additionally requires $N_{\text{off}} \geq c^*_{\text{off}}(\mathcal{F}, \Pi) dH^4/\epsilon^2$ offline samples in exchange for omitting optimism. This, along with the result in Theorem 4.17, shows that hybrid RL therefore allows for sample efficiency gains with optimistic algorithms and computational efficiency gains with non-optimistic algorithms.

While Algorithm 4.2 is generally computationally inefficient, the insights gained are promising for empirical applications. Optimism is necessary for strategic exploration, and can be implemented via linear model bonuses Agarwal et al. (2020), count-based exploration Martin et al. (2017), randomized value functions Osband et al. (2019), or random latent exploration Mahankali et al. (2024). An off-policy critic can be learned with the DDQN algorithm of van Hasselt et al. (2015), with the insight that one only needs to update when the TD error is high. Finally, an optimistic version of TD3 (Fujimoto et al., 2018) with rare-switching critic updates and policy resets when necessary is a promising approach for a practical adaptation of Algorithm 4.2 in future work. Further numerical experiments and an extensive empirical study would constitute a very welcome contribution to the literature.

One may wish to incorporate bonus function classes as in Agarwal et al. (2023); Zhao et al. (2023). With a rarely-updated bonus function class, updating the base critic every episode, one may be able to achieve sufficient optimism without targeting Q^*, but only if the sum of Q-functions is close to a scaled Q-function. Lastly, adapting the work of He et al. (2023) to see if policy optimization can be minimax-optimal in linear MDPs, like Wu et al. (2022a) do for tabular MDPs, is a welcome contribution to the literature.

4.2. Part II: Computationally Efficient Parametric Actor–Critic via Projected Natural Policy Gradient

While Part I resolved the statistical sample-efficiency barrier for actor–critic algorithms under general function approximation by combining rare-switching $Q^\star$ critics with actor resets, the resulting actor policies are represented implicitly as Boltzmann distributions over cumulative historical critic estimates. Evaluating $\pi^{(t)}(\cdot \mid s)$ when the critic does not reset (or across long inner mirror-descent loops) requires evaluating all accumulated critic functions, and computing global optimistic confidence sets in general function classes requires solving nonconvex constrained optimization problems. In Part II, based on joint work with Max Qiushi Lin, Reza Asad, Haque Ishfaq, Csaba Szepesvári, and Sharan Vaswani (Lin et al., 2026), we address both computational bottlenecks in linear MDPs by projecting natural policy gradient updates onto an explicit parametric log-linear policy class over a G-optimal design coreset (yielding $O(1)$ test-time policy evaluation cost with respect to T) and replacing explicit confidence-ellipsoid maximization with Langevin Monte Carlo critic sampling.

4.2.1. Introduction

Recent works (Cai et al., 2020; Jin et al., 2021a; Zanette et al., 2021; Zhong and Zhang, 2023; Agarwal et al., 2023; He et al., 2023; Liu et al., 2023b; Sherman et al., 2023; Cassel and Rosenberg, 2024; Tan et al., 2025) tackle strategic exploration in actor-critic methods directly. However, the algorithms analyzed are significantly different from those implemented in practice. Much of this body of work studies AC methods that use the natural policy gradient (`NPG`) update for policy optimization. However, the canonical implementation of the `NPG` update does not consider an explicit policy parameterization. Instead, the update involves constructing "implicit" policies *on the fly* using all previously stored Q-functions. This makes it computationally expensive to sample from these policies and use them to interact with the environment (see Table 4.2). In contrast, the algorithms used in practice typically employ explicitly parameterized complex models as learnable policies and optimize them with gradient descent-based methods. Therefore, we aim to address the following question:

Can we design a provably efficient actor–critic algorithm with parametric policies for linear MDPs in both on- and off-policy settings?

Algorithm	Sample Complexity (On-Policy)	Sample Complexity (Off-Policy)	Policy Param.	Clipping Q-Function	Computational Cost for Policy Inference
Liu et al. (2023b)	$\widetilde{\mathcal{O}}(\frac{1}{\epsilon^4})$	$\times$	implicit	yes	$\mathcal{O}(d_{\boldsymbol{c}}^2 H \lvert\mathcal{A}\rvert \epsilon^{-2})$
Sherman et al. (2023) Cassel and Rosenberg (2024)	$\times$	$\widetilde{\mathcal{O}}(\frac{1}{\epsilon^2})$	implicit	no	$\mathcal{O}(d_{\boldsymbol{c}}^2 H \lvert\mathcal{A}\rvert \epsilon^{-2})$
Ours	$\widetilde{\mathcal{O}}(\frac{1}{\epsilon^4})$	$\widetilde{\mathcal{O}}(\frac{1}{\epsilon^2})$	explicit	yes	$\mathcal{O}(d_{\boldsymbol{a}}^2 H \lvert\mathcal{A}\rvert)$

Table 4.2: Comparison with the state-of-the-art algorithms for episodic finite-horizon linear MDPs (with feature dimension $d_{\boldsymbol{c}}$ and horizon H). The sample complexity refers to the number of interactions required for outputting an ϵ-optimal policy, whereas the cost of policy inference refers to the per-episode cost for interacting with the environment for H steps. Our proposed algorithm matches the optimal sample efficiency in both the on- and off-policy settings. Furthermore, in contrast to existing works, our algorithm employs an explicit policy parameterization (log-linear policies with dimension $d_{\boldsymbol{a}}$), resulting in lower computational cost for policy inference.

Contributions We answer the above question affirmatively and make the following contributions.

1. General framework with an explicitly parameterized actor In Section 4.2.3, we propose a general optimistic actor-critic framework that utilizes an explicitly parameterized policy. We apply this framework in the setting of linear function approximation for both the environment (i.e., linear MDP, Jin et al., 2020) and the policy (i.e., log-linear policy class). In Section 4.2.4, we propose an actor algorithm that learns a log-linear policy by solving a specific regression problem at each iteration. This allows us to directly control the error between the explicitly parametrized policy and the implicit policy induced by `NPG`. Using this error bound in conjunction with the well-established theoretical results of `NPG` (Hazan, 2016; Szepesvári, 2022) enables us to analyze the performance of the parameterized actor. We show that the proposed algorithm benefits from a substantially improved memory complexity, while retaining similar theoretical guarantees.

2. LMC critic for practical strategic exploration In Section 4.2.5, instead of constructing `UCB` bonuses (Jin et al., 2020), which are ubiquitous within prior works (Zhong and Zhang, 2023; Liu et al., 2023b; Sherman et al., 2023; Cassel and Rosenberg, 2024), we adopt a more practical approach. We employ Langevin Monte Carlo (`LMC`) (Welling and Teh, 2011) to update the critic parameters at each episode. Unlike `UCB`-based approaches that require computing confidence sets

at every episode, LMC simply perturbs (by Gaussian noise) the gradient descent update on the critic loss. This gradient descent-based approach is both easier to implement (Ishfaq et al., 2025) and to extend to general function approximation (Ishfaq et al., 2024b). Furthermore, the LMC algorithm directly leads to an optimistic estimate of the Q-function that has similar guarantees as UCB bonuses. Nevertheless, previous works have only successfully designed provably efficient algorithms for solving multi-armed bandits (Mazumdar et al., 2020), contextual bandits (Xu et al., 2022), and linear MDPs via value-based methods (Ishfaq et al., 2024a). This work is the first to analyze an LMC-based approach in the context of policy optimization.

3. End-to-end theoretical guarantees for actor-critic In Section 4.2.6, we analyze the proposed actor-critic framework in both the on-policy and off-policy settings without making any assumptions on the mismatch ratio or data coverage. In particular, in the on-policy setting, we prove that our method requires $\widetilde{\mathcal{O}}(\epsilon^{-4})$ samples to learn an ϵ-optimal policy. This matches the result in Liu et al. (2023b) that uses an implicit NPG policy in conjunction with UCB bonuses. On the other hand, we also prove that our framework can attain a sample complexity of $\widetilde{\mathcal{O}}(\epsilon^{-2})$ in the off-policy setting. This matches the results of Sherman et al. (2023); Cassel and Rosenberg (2024), but with a far less complicated algorithm design.

We thus demonstrate that our proposed algorithm is both sample-efficient and aligned with practice.

4.2.2. Preliminaries

In this section, we introduce the finite-horizon linear MDP setting and the log-linear policy class, adopting the episodic MDP setup and value function definitions from Section 4.1.2. For any function $V : \mathcal{S} \to \mathbb{R}$, we define $[\mathbb{P}_h V_{h+1}](s,a) :- \mathbb{E}_{s' \sim \mathbb{P}_h(\cdot|s,a)} V_{h+1}(s')$.

Finite-Horizon Linear MDP We assume that both $\mathbb{P}$ and r are unknown to the agent. In order to efficiently learn these quantities, we consider the linear MDP assumption (Jin et al., 2020) where both the transition kernel and the reward function are assumed to be linear functions of given features.

Definition 4.18 (Linear MDP)**.** A finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathbb{P}, r, H)$ is a linear MDP with

a feature map $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^{d_{\boldsymbol{c}}}$ if the following holds. There exist H signed measures $\psi_h : \mathcal{S} \to \mathbb{R}^{d_{\boldsymbol{c}}}$ and $\upsilon_h : \mathbb{R}^{d_{\boldsymbol{c}}}$ such that $\mathbb{P}_h(s' \mid s, a) = \langle \phi(s,a), \psi_h(s') \rangle$ and $r_h(s,a) = \langle \phi(s,a), \upsilon_h \rangle$ for all h, s, and a. It should also satisfy the following constraints: $\|\phi(s,a)\|_2 \leq 1$ and $\|\upsilon_h\|_2 \leq \sqrt{d_{\boldsymbol{c}}}$ for all h, s, and a. Additionally, for any measurable function $V : \mathcal{S} \to [0,1]$, $\left\| \int_{s \in \mathcal{S}} V(s)\, \psi_h(s)\, \mathrm{d}s \right\|_2 \leq \sqrt{d_{\boldsymbol{c}}}$.

According to Jin et al. (2020, Proposition 2.3), for a linear MDP and any policy π, Q_h^π is a linear function of the features: for all (h, s, a), there exists a $w_h \in \mathbb{R}^{d_{\boldsymbol{c}}}$ such that $Q_h^\pi(s,a) = \langle \phi(s,a), w_h \rangle$.

Learning Objective For this linear MDP setting, we assume that only ϕ is given to the learner whereas ψ and υ are not. The agent sequentially interacts with the environment for T episodes and aims to minimize the *cumulative regret* defined as $\texttt{Reg}(T) \coloneqq \sum_{t=1}^{T} [V_1^\star(s_1) - V_1^{\pi_t}(s_1)]$, where $V_1^\star \coloneqq V_1^{\pi^\star} \coloneqq \sup_\pi V_1^\pi$ is the value function of the optimal policy $\pi^\star \coloneqq \arg\sup_\pi V_1^\pi(s_1)$. Equivalently, if $\overline{\pi}^T$ denotes the mixture policy that picks a policy among $\{\pi^1, \ldots, \pi^T\}$ uniformly randomly, we aim to learn an ϵ-optimal $\overline{\pi}^T$, i.e., its *optimality gap* ($\texttt{OG}$) is bounded such that

$$\texttt{OG}(T) \coloneqq \mathbb{E}\Big[V_1^\star(s_1) - V_1^{\overline{\pi}^T}(s_1) \Big] = \frac{\texttt{Reg}(T)}{T} \leq \widetilde{\mathcal{O}}(\epsilon)\,, \tag{4.35}$$

where the expectation is taken with respect to the randomness of the mixture policy.

Log-Linear Policy We consider a restricted policy class Π_{lin} consisting of *log-linear policies*. Log-linear policies are represented using the softmax function with linear function approximation. In particular, a log-linear policy is defined as follows: for all $h \in [H]$,

$$\pi_h(a \mid s, \theta) = \frac{\exp(z_h(s, a \mid \theta_h))}{\sum_{a' \in \mathcal{A}} \exp(z_h(s, a' \mid \theta_h))}\,, \tag{4.36}$$

where $z_h(s, a \mid \theta_h) = \langle \varphi(s,a), \theta_h \rangle$ represents the logits parameterized by θ_h, and $\varphi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^{d_{\boldsymbol{a}}}$ are policy features given to the learner. W.l.o.g, we assume that $\|\varphi(s,a)\| \leq 1$ for all s and a. For convenience, we use the shorthand $\pi(\theta) : [H] \times \mathcal{S} \to \Delta(\mathcal{A})$ to refer to the log-linear policy corresponding to the parameters θ.

Algorithm 4.6 Actor-Critic with Parametric Policies

1: **Input**: number of update steps T, data collection batch size N (only for on-policy)
2: set $\mathcal{D}^0 \leftarrow \emptyset$, $w_h^1 \leftarrow \mathbf{0}$, $\pi_h^1(\cdot \mid s) \leftarrow \text{Unif}(\mathcal{A}) \quad \forall (h, s)$
3: **for** $t = 1, \ldots, T-1$ **do**
4: $\quad \mathcal{D}^t \leftarrow \begin{cases} \text{On-Policy:} & \left\{ N \text{ fresh traj.} \overset{\text{i.i.d.}}{\sim} \pi^t \right\} \\ \text{Off-Policy:} & \mathcal{D}^{t-1} \cup \left\{ 1 \text{ traj.} \sim \pi^t \right\} \end{cases}$
5: $\quad w^{t+1} \leftarrow \texttt{CRITIC_UPDATE}(\mathcal{D}^t, \pi^t, w^t)$
6: $\quad \theta^{t+1} \leftarrow \texttt{ACTOR_UPDATE}(w^{t+1}, \theta^t)$
7: $\quad \pi^{t+1} = \pi(\theta^{t+1})$
8: **end for**
9: **Return**: mixture policy $\overline{\pi}^T$

4.2.3. A General Actor-Critic Framework with Parametric Policies

In this section, we start by introducing our general optimistic actor-critic framework as shown in Algorithm 4.6. Starting with a uniform policy π^1, at the beginning of every learning episode $t \in [T]$, the agent interacts with the environment using policy π^t (Line 4). Our framework allows for collecting data from the environment in either an on-policy or off-policy fashion. In the on-policy setting, at episode t, the agent collects N fresh trajectories $\mathcal{D}^t$ by interacting with the environment using the current policy π^t. On the other hand, in the off-policy setting, at episode t, the agent collects only 1 trajectory from the environment using π^t. However, the agent stores all the historical data collected by the previous policies, and hence, $\mathcal{D}^t$ consists of t trajectories, each collected by $\pi^1, \ldots, \pi^t$ respectively.

The *critic* uses the collected data and estimates an (optimistic) Q-function via learning the critic parameters $w^{t+1} \in [H] \times \mathbb{R}^{d_c}$ (Line 5). The *actor* then uses the estimated Q-function, and updates the parameters of $\theta^t \in [H] \times \mathbb{R}^{d_a}$ of the log-linear policy (Line 6). The updated log-linear policy is denoted by π^{t+1} (Line 7), and is used to collect data in the next episode.

Given this general framework, we will next instantiate the actor in Section 4.2.4 and the critic in Section 4.2.5.

4.2.4. Instantiating the Actor: Projected Natural Policy Gradient

In this section, we instantiate the actor using natural policy gradient (`NPG`) with parametric policies and analyze its behavior. In particular, in Section 4.2.4, we devise an algorithm that projects the standard `NPG` update onto the class of realizable policies. In Section 4.2.4, we analyze and control the errors induced by the projection step. Finally, in Section 4.2.4, we put everything together and instantiate the complete actor algorithm for the log-linear policy class.

Projected Natural Policy Gradient

At episode $t \in [T]$, given $\widehat{Q}_h^t$, the estimated Q-function, `NPG` updates the policy as: for every $(h, s) \in [H] \times \mathcal{S}$,

$$\pi_h^{t+1}(\cdot \mid s) \propto \pi_h^t(\cdot \mid s) \exp(\eta \widehat{Q}_h^t(s, \cdot)) \tag{4.37}$$

with the corresponding normalization across $\mathcal{A}$. Existing work on policy optimization in linear MDPs (Liu et al., 2023b; Sherman et al., 2023; Cassel and Rosenberg, 2024) uses `NPG` to update the actor because of its favorable theoretical properties. Importantly, *these results do not consider any explicit parameterization for the actor.* Directly implementing the update in Eq. (4.37) requires $\mathcal{O}(|\mathcal{S}||\mathcal{A}|)$ memory, and is therefore impractical with large state-action spaces.

Consequently, existing work uses the following equivalent form of the `NPG` update: for every $(h, s) \in [H] \times \mathcal{S}$,

$$\pi_h^{t+1}(\cdot \mid s) \propto \pi_h^1(\cdot \mid s) \exp\Big(\eta \textstyle\sum_{i=1}^t \widehat{Q}_h^i(s, \cdot)\Big), \tag{4.38}$$

which characterizes the policy implicitly. In particular, at episode t, for any (h, s, a), we can compute the policy *on the fly* if we have access to the sum of all the parameterized Q-functions up to episode t. However, in existing works (Liu et al., 2023b; Sherman et al., 2023; Cassel and Rosenberg, 2024), the sum of parameterized Q functions cannot be stored in a succinct manner, and they require storing *all* the parameterized Q functions. Consequently, when interacting with the environment using such an implicit policy, these works suffer from an extensive per-episode

computational cost of $\mathcal{O}(d_{\boldsymbol{c}}^2\, H\, |\mathcal{A}|\, T)$. Therefore, the resulting algorithm is far from practice that typically uses an explicit (and often sophisticated) actor parameterization.

To alleviate these issues, we aim to compute a policy that is (i) realizable by the explicit actor parameterization and (ii) provably approximates the policy induced by the `NPG` update in Eq. (4.37) (referred to as the *implicit policy*). To this end, we use a projected `NPG` update:

$$\pi_h^{t+1}(\cdot \mid s) = \mathrm{Proj}_{\Pi}\left[\frac{\pi_h^t(\cdot \mid s)\exp(\eta\, \widehat{Q}_h^t(s,\cdot))}{\sum_{a'} \pi_h^t(a' \mid s)\exp(\eta\, \widehat{Q}_h^t(s,a'))}\right], \tag{4.39}$$

where Proj is the projection operator, which will be instantiated subsequently in Section 4.2.4.

When theoretically analyzing policy optimization methods, an important intermediate result is the bound on the regret for a specific online linear optimization problem. For the standard `NPG` update in Eq. (4.37), this regret can be bounded by $\widetilde{\mathcal{O}}(\sqrt{T})$ (Hazan, 2016; Szepesvári, 2022). In the following lemma, we analyze the effect of the projection operator and bound the regret for the projected `NPG`.

Theorem 4.19. *Given a sequence of linear functions $\{\langle p^t, g^t\rangle\}_{t\in[T]}$ for a sequence of vectors $\{g^t\}_{t\in[T]}$ where for any $t \in [T]$, $p^t \in \Delta(\mathcal{A})$, $g^t \in \mathbb{R}^{|\mathcal{A}|}$, and $\|g^t\|_\infty \le H$. Consider $p^{t\in[T]}$ where p^1 is the uniform distribution, and for all $t \in [T]$,*

$$p^{t+1/2} = \arg\min_{p\in\Delta_A}\left\{\langle p, -\eta\, g^t\rangle + D_{\mathrm{KL}}(p \,\|\, p^t)\right\}, \tag{4.40}$$

$$p^{t+1} = \mathrm{Proj}_{\Pi}(p^{t+1/2}). \tag{4.41}$$

Let $\epsilon^t := D_{\mathrm{KL}}(u \,\|\, p^{t+1}) - D_{\mathrm{KL}}(u \,\|\, p^{t+1/2})$ be the projection error induced by Eq. (4.41). Then, for any comparator $u \in \Delta(\mathcal{A})$, it holds that

$$\sum_{t=1}^{T}\langle u - p^t, g^t\rangle \le \frac{\log|\mathcal{A}| + \sum_{t=1}^T \epsilon^t}{\eta} + \frac{\eta\, H^2\, T}{2}. \tag{4.42}$$

The update in Eq. (4.40) with $p^t = \pi_h^t(\cdot|s)$ is equivalent to the standard `NPG` update in Eq. (4.37) (Xiao,

2022). Using this lemma for each state s and step h, with $g^t = \widehat{Q}_h^t(s,\cdot)$ and an appropriate choice of η gives the following regret bound[19] for the projected `NPG`:

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\max_{s\in\mathcal{S}}\Big\langle \pi_h^\star(\cdot|s) - \pi_h^t(\cdot|s), \widehat{Q}_h^t(s,\cdot)\Big\rangle \le \mathcal{O}\Big(H^2\sqrt{\log|\mathcal{A}|}\,\sqrt{T} + H^2\sqrt{\bar{\epsilon}}\,T\Big), \tag{4.43}$$

where $\bar{\epsilon} :- \max_{t,s,h}\epsilon_h^t(s)$ is the largest error across all t, s, and h. For the `NPG` in Eq. (4.37) without projection, $\bar{\epsilon} = 0$ and the above result recovers the standard regret bound for `NPG`. The above lemma suggests that by choosing the projection operator carefully and controlling the projection errors, we can bound the regret.

Controlling the Projection Error for Log-Linear Policies

To bound the projection error in Theorem 4.19, one could choose that $\mathrm{Proj}_\Pi(p) = \arg\min_p D_{\mathrm{KL}}(u \,\|\, p) - D_{\mathrm{KL}}(u \,\|\, p^{t+1/2})$, and hence directly control ϵ_h^t. However, this results in a non-convex optimization problem. Consequently, we instead choose Proj to minimize the following regression loss in the logit space: $\frac{1}{2}\big\|z - (z^t + \eta\, g^t)\big\|$ where z^t is the logit corresponding to p^t such that $p^t \propto \exp(z^t)$. For the projected `NPG` with log-linear policies, we aim to minimize the sum of such regression losses (across all $(s,a) \in \mathcal{S}\times\mathcal{A}$) at episode t and step h, and obtain the loss function:

$$\ell_h^t(\theta) = \frac{1}{2}\sum_{(s,a)\in\mathcal{S}\times\mathcal{A}}\Big[\Big\langle \varphi(s,a), \theta - \widehat{\theta}_h^t\Big\rangle - \eta\,\widehat{Q}_h^t(s,a)\Big]^2, \tag{4.44}$$

where $\widehat{Q}_h^t(s,a)$ is the estimated Q-function from the critic. As a regression problem, this actor loss can be easily optimized via gradient descent-based methods.

However, note that the above actor loss requires a minimization over the entire state-action space, which may be impractical. Therefore, we propose to construct a good and preferably small subset $\mathcal{D}_{\mathrm{exp}} \subset \mathcal{S}\times\mathcal{A}$ along with a corresponding distribution $\rho_{\mathrm{exp}} \in \Delta(\mathcal{D}_{\mathrm{exp}})$ that offers good coverage of the feature space. Given $\mathcal{D}_{\mathrm{exp}}$ and ρ_{exp} (the construction of which will be detailed subsequently),

[19] This generalized regret bound holds for any other mirror descent-based policy optimization method (e.g., `SPMA`, Asad et al., 2025, in Appendix C.2), but we discuss `NPG` within the main text for the ease of exposition.

we instantiate the actor loss, $\widetilde{\ell}_h^t(\theta)$, in terms of a *logit-matching* regression:

$$\widetilde{\ell}_h^t(\theta) = \frac{1}{2} \sum_{(s,a)\in\mathcal{D}_{\text{exp}}} \rho_{\text{exp}}(s,a) \left[\left\langle \varphi(s,a), \theta - \widehat{\theta}_h^t \right\rangle - \widehat{Q}_h^t(s,a) \right]^2, \tag{4.45}$$

In order to show that optimizing the above actor loss can indeed bound the projection error, we require the following assumptions. We assume that the given policy features φ are expressive enough to control the bias when minimizing $\ell_h^t(\theta)$.

Assumption 4.20 (Bias)**.** $\min_\theta \sup_{t,h} \widetilde{\ell}_h^t(\theta) \le \epsilon_{\text{bias}}$

In practice, ϵ_{bias} can be controlled by choosing high-dimensional features (e.g., $d_{\boldsymbol{a}} \gg d_{\boldsymbol{c}}$) or a sufficiently expressive policy class (e.g., neural network).

Next, we assume the loss $\widetilde{\ell}_h^t(\theta)$ is sufficiently minimized.

Assumption 4.21 (Optimization Error)**.** Suppose θ_h^t is obtained by minimizing $\widetilde{\ell}_h^t(\theta)$ in the critic update, then

$$\sup_{t,h} \left| \widetilde{\ell}_h^t(\theta_h^t) - \min_\theta \widetilde{\ell}_h^t(\theta) \right| \le \epsilon_{\text{opt}} \,.$$

In practice, minimizing $\widetilde{\ell}_h^t(\theta)$ by K_t steps of gradient descent ensures that $\epsilon_{\text{opt}} \le \mathcal{O}(\exp(-K_t))$. Given these two mild assumptions, we can then proceed to bound the projection error. Using Theorem 4.19 for the projected `NPG` update at state s, step h, and setting $u = \pi^\star(\cdot \mid s)$, the projection error $\epsilon_h^t(s)$ can be bounded as follows.

Lemma 4.22. *Let* $\overline{\varphi}_G \coloneqq \sup_{(s,a)\in\mathcal{S}\times\mathcal{A}} \|\varphi(s,a)\|_{G^{-1}}$ *where* $G \coloneqq \sum_{(s,a)\in\mathcal{D}_{\text{exp}}} \rho_{\text{exp}}(s,a)\, \varphi(s,a)\, \varphi(s,a)^\top$. *Under Theorems 4.20 and 4.21, it holds that*

$$\left|\epsilon_h^t(s)\right| \le \overline{\epsilon} \coloneqq 2\,(\overline{\varphi}_G + 1)\,\sqrt{\epsilon_{\text{bias}}} + 2\,\sqrt{\epsilon_{\text{opt}}} \quad \forall (t,h,s)\,. \tag{4.46}$$

The above lemma is true for any choice of $\mathcal{D}_{\text{exp}}$ and ρ_{exp}, and suggests that if we can control $\overline{\varphi}_G$, the projection error can be bounded. To bound $\overline{\varphi}_G \le \mathcal{O}(d_{\boldsymbol{a}})$ with a coreset of size $|\mathcal{D}_{\text{exp}}| \le \widetilde{\mathcal{O}}(d_{\boldsymbol{a}})$, one

Algorithm 4.7 Actor: Projected NPG

Input: critic parameters w^t, policy optimization learning rate η, number of actor updates K_t, actor learning rate $\alpha_{\boldsymbol{a}}^t$, subset and distribution of the state-action space $\mathcal{D}_{\text{exp}}$ and ρ_{exp}
for $h = 1, 2, \ldots, H$ **do**
    $\widehat{Q}_h^t(\cdot, \cdot) = \min\{\langle \phi(\cdot, \cdot), w_h^t \rangle, H - h + 1\}^+$ [20]
    Define the actor loss $\widetilde{\ell}_h^t(\theta)$ using Eq. (4.45)
    for $k = 1, \ldots, K_t$ **do**
        $\theta_h^{t,k} \leftarrow \theta_h^{t,k-1} - \alpha_{\boldsymbol{a}}^t \nabla_\theta \widetilde{\ell}_h^t(\theta_h^{t,k-1})$
    end for
end for
Return: actor parameters for the policy θ^t

can construct $(\mathcal{D}_{\text{exp}}, \rho_{\text{exp}})$ offline via G-optimal experimental design using the Kiefer–Wolfowitz theorem (Kiefer and Wolfowitz, 1960) and Frank–Wolfe optimization, or online via reward-free exploration procedures such as `CoverTraj` (Wagenmaker et al., 2022b). We defer the complete coreset construction and experimental design details to Lin et al. (2026).

Putting Everything Together: Projected NPG with Log-Linear Policies

In Algorithm 4.7, we instantiate the complete actor algorithm, which uses the projected `NPG` update for log-linear policies. Unlike the standard `NPG` update, Algorithm 4.7 alleviates the necessity of storing past Q-functions, improving the memory complexity to $\mathcal{O}(d_{\boldsymbol{a}})$, while enjoying similar theoretical guarantees. Furthermore, the actor parameters are updated by using gradient descent on a properly defined surrogate loss, rendering it closer to the practical implementation of common algorithms (e.g., PPO, Schulman et al., 2017).

We remark that although we focused on the log-linear policies, our theoretical guarantees readily extend to general function approximation when Theorems 4.20 and 4.21 are satisfied, and one has access to an exploratory policy (Hao et al., 2021, Definition 1). In the next section, we instantiate the critic in Algorithm 4.6.

4.2.5. Instantiating the Critic: Langevin Monte Carlo

In this section, we use Langevin Monte Carlo (`LMC`) to instantiate the critic. We describe the resulting algorithm in Section 4.2.5, and analyze it in Section 4.2.5.

[20] $\min\{x, a\}^+ := \max\{\min\{x, a\}, 0\}$ is the clipping function that bounds the given value x to the range $[0, a]$.

The LMC approaches allow for sampling from a posterior distribution and have recently been used in sequential decision-making problems. For example, Mazumdar et al. (2020) achieves optimal instance-dependent regret bounds for multi-armed bandits using Langevin dynamics for approximate Thompson sampling. On the other hand, Xu et al. (2022) uses LMC for contextual bandits, achieving comparable theoretical results to Thompson sampling. More recently, Ishfaq et al. (2024a) leverages LMC for linear MDPs by using it to sample the Q-function from its posterior distribution, achieving the optimal $\widetilde{\mathcal{O}}(\sqrt{T})$ regret.

Nevertheless, all existing LMC-based approaches for MDPs, including those for general function approximation (Ishfaq et al., 2024b; Jorge et al., 2024) use value-based algorithms. To the best of our knowledge, such approaches have never been theoretically analyzed in the context of policy optimization. Next, we incorporate the LMC algorithm into our actor-critic framework and provide the first provable result.

LMC for Linear MDPs

At episode t, the critic uses the collected dataset $\mathcal{D}^t$ to obtain an optimistic estimate of the Q function. In order to instantiate the critic loss, we consider the dataset $\mathcal{D}^t$ as split into H disjoint subsets $\left\{\mathcal{D}_h^t\right\}_{h\in[H]}$, where $\mathcal{D}_h^t$ consists of (s_h, a_h, s_{h+1}) tuples indexed as $\left\{(s_h^i, a_h^i, s_{h+1}^i)\right\}_{i=1}^{|\mathcal{D}^t|}$ [21]. The critic loss at episode t and step h uses the estimated value function at step $h+1$, and forms the following ridge regression problem:

$$\mathcal{L}_h^t(w) = \frac{1}{2}\sum_{i=1}^{|\mathcal{D}^t|}\Big[r_h(s_h^i, a_h^i) + \widehat{V}_{h+1}^t(s_{h+1}^i) - \big\langle\phi(s_h^i, a_h^i), w\big\rangle\Big]^2 + \frac{\lambda}{2}\left\|w\right\|^2. \tag{4.47}$$

For each step h, the LMC algorithm iteratively adds Gaussian noise to the gradient descent updates on $\mathcal{L}_h^t(w)$, and aims to produce approximate samples of the critic parameters from its underlying posterior distribution (Line 6-8). In particular, for an arbitrary loss ℓ, the LMC update can be written as:

$$w^{t+1} = w^t - \alpha^t\nabla_w\ell(w^t) + \sqrt{\alpha^t/\zeta}\,\nu^t, \tag{4.48}$$

[21] $|\mathcal{D}^t|$ represents the number of trajectories in $\mathcal{D}^t$ or the number of (s_h, a_h, s_{h+1}) tuples in $\mathcal{D}_h^t$

Algorithm 4.8 Critic: LMC

1: **Input**: collected data $\mathcal{D}^t$, policy π^{t-1}, number of critic updates J_t, critic learning rate $\alpha_{\boldsymbol{c}}^{h,t}$, inverse temperature ζ, number of critic samples M
2: $\widehat{V}_{H+1}^t(\cdot) \leftarrow 0$
3: **for** $h = H, H-1, \ldots, 1$ **do**
4: Define the critic loss $\mathcal{L}_h^t(w)$ using Eq. (4.47)
5: $w_h^{t,m,0} \leftarrow w_h^{t-1,m,J_{t-1}} \quad \forall m \in [M]$
6: **for** $j = 1, \ldots, J_t$ **do**
7: $\nu_h^{t,m,j} \leftarrow \mathbf{N}(0, I) \quad \forall m \in [M]$
8: $w_h^{t,m,j} \leftarrow w_h^{t,m,j-1} - \alpha_{\boldsymbol{c}}^{h,t} \nabla_w \mathcal{L}_h^t(w_h^{t,m,j-1}) + \sqrt{\alpha_{\boldsymbol{c}}^{h,t}/\zeta}\, \nu_h^{t,m,j} \quad \forall m \in [M]$
9: **end for**
10: $\widehat{Q}_h^t(\cdot,\cdot) = \min\left\{ \max_{m \in [M]} \left\langle \phi(\cdot,\cdot), w_h^{t,m,J_t} \right\rangle, H \right\}^+$
11: $\widehat{V}_h^t(\cdot) = \mathbb{E}_{a \sim \pi^{t-1}(\cdot|s)} \widehat{Q}_h^t(\cdot, a)$
12: **end for**
13: **Return**: critic parameters for the estimated Q-function $\{w_h^{t,m,J_t}\}_{(m,h) \in [M] \times [H]}$

where α_t is the learning rate, ζ is the inverse temperature parameter, and ν_t is sampled from an isotropic Gaussian distribution. After J_t steps of the `LMC` update on the critic loss (Lines 6-8 in Algorithm 4.8), the resulting critic parameters are used to produce an optimistic sample of the Q-function (Line 9). From a theoretical perspective, we note that it is important to clip Q_h^t appropriately. In order to improve the optimism guarantees of the `LMC` algorithm, we follow the idea in Ishfaq et al. (2021), and repeat the `LMC` update M times, taking the maximum over these samples (Line 9). Iterating this procedure backwards from $h = H$ to 1, we can obtain the desired critic parameters.

Note that compared to `UCB`-based approaches, `LMC` does not require computing confidence sets at every episode. Instead, it simply perturbs gradient descent by injecting Gaussian noise, allowing for a natural extension beyond the linear function approximation setting and rendering it easier to implement in practice.

Optimism Guarantee and Error Bound

In order to theoretically analyze Algorithm 4.8, we first define the following model prediction error.

Definition 4.23. Given an estimated Q-function $\widehat{Q}^t$ and the corresponding estimated value function $\widehat{V}^t$, for all (t, h, s, a), the model prediction error is defined as: $\iota_h^t(s,a) :- r_h(s,a) + \mathbb{P}_h \widehat{V}_{h+1}^t(s,a) - \widehat{Q}_h^t(s,a)$.

The theoretical analyses in existing work (Jin et al., 2020; Zhong and Zhang, 2023; Liu et al., 2023b) that use `UCB` bonuses typically proceed by proving an upper bound of 0 on ι_h^t (*optimism*) and a lower bound of $\widetilde{\mathcal{O}}(\sqrt{T})$. The following lemma shows that `LMC` can offer similar guarantees.

Lemma 4.24. *Let $\Lambda_h^t :- \sum_{(s,a,s') \in \mathcal{D}_h^t} \phi(s,a)\,\phi(s,a)^\top + \lambda\, I$. With appropriate choices of λ, ζ, J_t, $\alpha_{\boldsymbol{c}}^t$, M and for any $\delta \in (0,1)$, Algorithm 4.6 with the* `LMC` *critic in Algorithm 4.8 ensures that in both the on-policy and off-policy settings, for all t, h, s, a and some constant $\Gamma_{\texttt{LMC}} = \widetilde{\mathcal{O}}(H\, d_{\boldsymbol{c}})$, with probability at least $1 - \delta$,*

$$-\Gamma_{\texttt{LMC}} \times \|\phi(s,a)\|_{(\Lambda_h^t)^{-1}} \leq \iota_h^t(s,a) \leq 0.$$

The exact definition of $\Gamma_{\texttt{LMC}}$ varies between the on-policy and off-policy settings, although they are both bounded by $\widetilde{O}(H\, d_{\boldsymbol{c}})$ (see Proposition C.2 for the full version of this lemma). In order to prove this result in the on-policy setting, we use the fact that all the data points in $\mathcal{D}_h^t$ are collected via independent trajectories from the same policy π^t, and are therefore independent and identically distributed. Hence, we can use the self-normalized bounds in Abbasi-Yadkori et al. (2011) to analyze the dependence in h, and prove the corresponding result. In the off-policy setting, since the data points in $\mathcal{D}_h^t$ are collected by different data-dependent policies, these samples are correlated in a complicated manner. Hence, we use the value-aware uniform concentration result from Jin et al. (2020). We remark that this result requires control over the covering number of the value function class, which is deferred to Section 4.2.6.

Therefore, we conclude that, compared to `UCB` bonuses, `LMC` offers significant practical advantages while still providing similar theoretical guarantees.

4.2.6. Sample Complexity Analysis

In this section, we analyze the sample complexity of Algorithm 4.6 with the projected `NPG` actor from Algorithm 4.7 and the `LMC` critic from Algorithm 4.8. Section 4.2.6 focuses on the on-policy setting, while Section 4.2.6 addresses the off-policy setting.

On-Policy Setting

We now present the following theorem that shows that our proposed algorithm achieves a sample complexity of $\widetilde{\mathcal{O}}(\epsilon^{-4})$ in the on-policy setting.

Theorem 4.25. *Under Theorems 4.20 and 4.21, consider Algorithm 4.6 in the on-policy setting with the projected* `NPG` *actor (Algorithm 4.7) and the* `LMC` *critic (Algorithm 4.8). Suppose $\overline{\epsilon}$ is the projection error in the actor. For an appropriate choice of the actor and critic parameters, including $N = H^4/\epsilon^2$, and any $\delta \in (0,1)$, it holds that with probability at least $1-\delta$,*

$$\texttt{OG}(T) \leq \widetilde{\mathcal{O}}\left(\frac{H^2\sqrt{d_{\mathbf{c}}^3\log|\mathcal{A}|}}{\sqrt{T}} + H^2\sqrt{\overline{\epsilon}}\right). \tag{4.49}$$

Hence, for any $\epsilon > 0$, Algorithm 4.6 with $T = d_{\mathbf{c}}^3 H^4 \log|\mathcal{A}|\,\epsilon^{-2}$ yields a $(\epsilon + H^2\sqrt{\overline{\epsilon}})$-optimal mixture policy, and therefore requires $T \times N = \widetilde{\mathcal{O}}(\epsilon^{-4})$ samples.

Proof Sketch. We decompose the difference between $V_1^{\overline{\pi}^T}(s_1)$ and $V_1^\star(s_1)$ into two terms that only depend on either the actor or the critic.

$$\mathbb{E}\Big[V_1^{\pi^\star} - V_1^{\overline{\pi}^T}(s_1)\Big] = \underbrace{\frac{1}{T}\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^\star}\Big[\Big\langle \pi_h^\star(\cdot\mid s_h) - \pi_h^t(\cdot\mid s_h), \widehat{Q}_h^t(s_h,\cdot)\Big\rangle\Big]}_{\text{policy optimization (actor) error}} \tag{4.50}$$

$$+ \underbrace{\frac{1}{T}\sum_{t=1}^{T}\sum_{h=1}^{H}\big(\mathbb{E}_{\pi^\star}[\iota_h^t(s_h,a_h)] - \mathbb{E}_{\pi^t}[\iota_h^t(s_h,a_h)]\big)}_{\text{policy evaluation (critic) error}}. \tag{4.51}$$

The policy optimization (actor) error can be bounded using Eq. (4.43), and the policy evaluation

(critic) error is bounded using Lemma C.2. In particular, in the on-policy setting, the lower-bound in Lemma C.2 can be instantiated as:

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^t}\left[-\iota_h^t(s,a)\right] \leq \mathcal{O}\left(\sqrt{d_{\boldsymbol{c}}^3\, H^4\, T\, \log^2(N/\delta)/N}\right). \tag{4.52}$$

Putting everything together with the chosen value of N leads to the stated sample complexity. □

Off-Policy Setting

Next, we show that, in the off-policy setting, Algorithm 4.6 can achieve $\widetilde{\mathcal{O}}(\epsilon^{-2})$ sample complexity.

Theorem 4.26. *Under Theorems 4.20 and 4.21, consider Algorithm 4.6 in the off-policy setting with the projected* `NPG` *actor (Algorithm 4.7) and the* `LMC` *critic (Algorithm 4.8). Suppose $\bar{\epsilon}$ is the projection error in the actor. For an appropriate choice of the actor and critic parameters and any $\delta \in (0,1)$, it holds that with probability at least $1-\delta$,*

$$\texttt{OG}(T) \leq \widetilde{\mathcal{O}}\left(\frac{H^2\sqrt{d_{\boldsymbol{c}}^3\, \max\{d_{\boldsymbol{a}}, d_{\boldsymbol{c}}\}\, \log|\mathcal{A}|}}{\sqrt{T}} + H^2\,\sqrt{\bar{\epsilon}}\right). \tag{4.53}$$

Hence, for any $\epsilon > 0$, Algorithm 4.6 with $T = d_{\boldsymbol{c}}^3\, \max\{d_{\boldsymbol{a}}, d_{\boldsymbol{c}}\}\, H^4\, \log|\mathcal{A}|\, \epsilon^{-2}$ yields a $(\epsilon + H^2\sqrt{\bar{\epsilon}})$-optimal mixture policy, and therefore requires $T \times 1 = \widetilde{\mathcal{O}}(\epsilon^{-2})$ samples.

Proof Sketch. The proof uses a similar regret decomposition to Theorem 4.25. Compared to the on-policy setting, the most significant difference is the bound on the policy evaluation (critic) errors. First, using Theorem 4.24, we have that

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^t}\left[-\iota_h^t(s,a)\right] \leq \Gamma \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^t}\|\phi(s,a)\|_{(\Lambda_h^t)^{-1}}. \tag{4.54}$$

The term, $\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^t}\|\phi(s,a)\|_{(\Lambda_h^t)^{-1}}$, can be bounded using the standard elliptical potential lemma. However, since we are in the off-policy setting, bounding $\Gamma > 0$ requires the uniform

concentration argument from Jin et al. (2020), which yields that

$$\Gamma \leq \mathcal{O}\left(H\sqrt{d_{\boldsymbol{c}}\log T + \log\left(\frac{\mathcal{N}_\Delta(\mathcal{V})}{\delta}\right)} + T\,\Delta\right). \tag{4.55}$$

This involves obtaining a bound on $\log(\mathcal{N}_\Delta(\mathcal{V}))$, the logarithm of the covering number of the value function class. The covering number is a measure of the complexity of the space of value functions. In particular, we show that for an actor with log-linear policies, we can bound the logarithm of the covering number using the following lemma.

Lemma 4.27. *Let Π_{lin} be the policy class induced by Eq. (4.36) such that $\sup_{\theta,h,s,a}\|z_h(s,a \mid \theta)\| \leq \overline{Z}$. Let $\mathcal{Q} = \left\{\min\left\{\langle\phi(\cdot,\cdot), w\rangle, H\right\}^+ \mid \|w\| \leq \overline{W}\right\}$ be the Q-function class and $\mathcal{V} = \left\{\langle Q(\cdot,\cdot), \pi(\cdot \mid \cdot,\theta)\rangle_{\mathcal{A}} \mid Q \in \mathcal{Q},\ \pi \in \Pi_{lin}\right\}$ be the corresponding value function class. Then, it holds that*

$$\log\mathcal{N}_\Delta(\mathcal{V}) \leq \mathsf{V} :- d_{\boldsymbol{c}}\,\log\left(1 + \frac{4\,\overline{W} + 4\,H\,\sqrt{2\,\overline{Z}}}{\Delta}\right) + d_{\boldsymbol{a}}\,\log\left(1 + \frac{4\,H\,\sqrt{2\,\overline{Z}}}{\Delta}\right).$$

In particular, we can show that $\overline{W} \leq \mathcal{O}(\sqrt{T})$ (Lemma C.2), and $\overline{Z} \leq \mathcal{O}(\overline{\epsilon}\,T)$ (Lemma C.2). Putting everything together and setting $\Delta = 1/T$ yields that

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^t}\left[-\iota_h^t(s,a)\right] \leq \widetilde{\mathcal{O}}\left(\sqrt{d_{\boldsymbol{c}}^3\,\max\{d_{\boldsymbol{a}}, d_{\boldsymbol{c}}\}\,H^4\,T}\right). \tag{4.56}$$

Following a proof similar to Theorem 4.25 leads to the desired sample complexity. □

Remark 4.28. In order to effectively bound the logarithm of the covering number, previous work (Sherman et al., 2023; Cassel and Rosenberg, 2024; Tan et al., 2025) has incorporated various algorithmic tweaks, including reward-free warm-ups, feature contractions, and rare-switching. On the contrary, since our algorithm projects the implicit policy onto the log-linear policy class at each iteration, the logarithm of the covering number can be bounded in a more direct manner.

4.2.7. Experiments

In this section, we evaluate our proposed algorithm in linear MDPs, consistent with our theoretical analyses.

Experiments in Linear MDPs

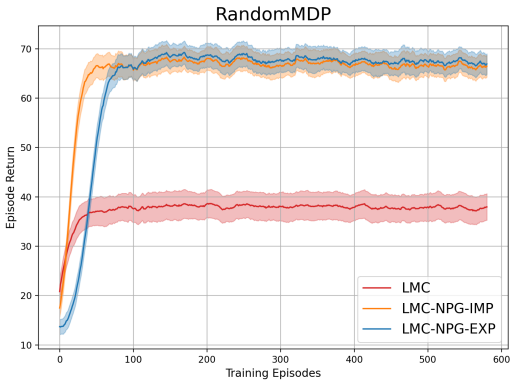


Figure 4.5: Comparison of projected NPG with LMC critic (explicit coreset) (our proposed algorithm), projected NPG with LMC critic (implicit coreset) (memory-intensive variant), and `LMC` (value-based baseline) in the Random MDP.

To demonstrate the practical value of our proposed algorithm, we evaluate it over various benchmarks. We first test our proposed algorithm in the randomly generated linear MDPs (Random MDP). We compare our proposed algorithm, projected NPG with LMC critic (explicit coreset), with the memory-intensive variant with implicit policy parameterization, projected NPG with LMC critic (implicit coreset), and the value-based baseline, `LMC` (Ishfaq et al., 2024a). As shown in Figure 4.5, our proposed algorithm can achieve comparable performance with the memory-intensive variant and better performance than `LMC`. Additional linear MDP experiments on Deep Sea, ablation studies, and deep RL evaluations on Atari benchmarks comparing against PPO and DQN are deferred to Lin et al. (2026).

4.2.8. Discussion

We proposed an optimistic actor–critic algorithm with explicitly parameterized policies and a systematic exploration mechanism. In particular, for the actor, we demonstrated that using projected `NPG` with parametric policies is not only practical, but also equipped with theoretical guarantees. For the critic, we demonstrated that `LMC` is a principled and easy-to-implement exploration scheme

for policy optimization methods. We derived theoretical guarantees in both the on-policy and off-policy settings, showcasing that the proposed actor-critic algorithm can simultaneously achieve sample efficiency and practicality.

For future work, eliminating the dependence on $\mathcal{D}_{\text{exp}}$ and ρ_{exp} will result in a more practical algorithm. We also aim to investigate the actor-critic algorithms in more realistic setups (e.g., infinite-horizon discounted MDPs) with more general function approximation schemes beyond linear models for both the environment and the policy. It would also be fruitful to further evaluate their empirical performance in more challenging large-scale deep RL applications.

CHAPTER 5

LEARNING CONTINUOUS-ACTION DIFFUSION POLICIES

This chapter is based on joint work with Zhihan Huang and Yuting Wei (Tan et al., 2026): Kevin Tan, Zhihan Huang, and Yuting Wei, "Learning Diffusion Policies with Sublinear Regret," working paper, 2026.

[RESOLVE: CHAPTER 5 IS BASED ON YOUR WORKING PAPER WITH ZHIHAN HUANG AND YUTING WEI (LEARNING DIFFUSION POLICIES WITH SUBLINEAR REGRET).]

5.1. Introduction

Originally introduced as a novel approach to policy modeling in robotics (Chi et al., 2023), diffusion policies have rapidly become a powerful and influential framework in reinforcement learning (RL) (Black et al., 2023; Ren et al., 2024a; Yang et al., 2023; Hansen-Estruch et al., 2023; Wang et al., 2023; Psenka et al., 2024; Li et al., 2024c), attracting substantial attention due to their ability to model complex, multimodal and stochastic decision rules (Li et al., 2024c; Ren et al., 2024a). This surge of interest has led to the development of a diverse set of methods for learning diffusion-based policies within both online and offline RL. In online RL, proposed approaches include policy gradient-based optimization (Ren et al., 2024a), behavior cloning (Chi et al., 2023), critic-induced greedy sampling (Yang et al., 2023; Li et al., 2024c), and score-function alignment with critic gradients (Psenka et al., 2024). Within offline RL, Wang et al. (2023) proposes a score-matching objective on the behavior policy, augmented with a Q-value maximization term incorporated directly into the training loss.

Recent work has begun to clarify the theoretical properties of diffusion policies. For instance, Psenka et al. (2024) show that the policy gradient theorem (Sutton et al., 1999) applies to diffusion model sampling, while Yang et al. (2023) establish a PAC guarantee for diffusion behavior cloning given an accurate score estimate. Kveton et al. (2024) study diffusion models as priors over unknown reward parameters in contextual bandits. More recently, Vu et al. (2026) quantify the

trade-off between diffusion-policy expressivity and statistical complexity. Their empirical objective requires evaluating candidate policies on counterfactual trajectories generated from the same sampled initial states and noise sequences, which operationally can be realized by a simulator with reusable randomness. This leaves open an online regret guarantee that accounts for exploration and adaptive sampling. We ask:

Can one obtain provable sublinear regret for diffusion policies at all, despite their iterative denoising structure, stochastic samplers, and intractable policy densities?

Specifically, we consider two representative classes of algorithms for online RL with diffusion policies. The first performs policy optimization over diffusion policies using the policy gradient theorem (Ren et al., 2024a). The key difficulty is that diffusion policies are implicit: we can sample from them, but the terminal action density is generally intractable. For vanilla policy gradient (PG), an unbiased policy-score estimate is sufficient. For natural policy gradient (NPG), the issue is more delicate, since the update is a least-squares problem in policy-score features, whose covariance is not generally recovered from the sampled denoising-path score.

The second class learns a critic and behavior-clones its greedy policy (Chi et al., 2023; Yang et al., 2023; Li et al., 2024c). This reduces policy learning to supervised action prediction (Foster et al., 2024), but introduces a different diffusion-specific issue: the diffusion sampler must match a critic-induced greedy policy, which is deterministic conditional on the state.

We show that these diffusion-specific obstacles do not preclude sublinear regret. Our analysis establishes that, under the stated regularity conditions, one does not pay an additional statistical regret-order penalty merely because the policy is implemented through diffusion sampling. Approximation and sampling errors are retained explicitly. For policy optimization, we recover usable policy-score features from the sampler. Vanilla PG uses an unbiased denoising-path score estimate for denoising diffusion probabilistic models (DDPM; Ho et al., 2020; Psenka et al., 2024) or the exact policy score for denoising diffusion implicit models (DDIM; Song et al., 2022), while our NPG analysis uses the exact DDIM score features (Lemma D.2). For NPG, we compare with a

policy of finite initial KL divergence and retain its gap to the optimal value, since a deterministic optimal policy may have infinite KL divergence relative to the initial policy. We also give a gradient-domination lemma under coverage, compatible advantage approximation, and conditioning of the positive Fisher spectrum (Agarwal et al., 2021; Mei et al., 2022). It allows redundant parameters through the Fisher pseudoinverse and retains the error in representing the advantage by policy-score features. For behavior cloning, existing diffusion model guarantees do not directly apply because the greedy target is deterministic. We handle this with a Wasserstein smoothing argument and track the initialization, sampling, and score-matching errors over K denoising steps.

The expressivity of the actor matters even before optimization: fitting a single Gaussian to two separated rewarding modes can put most of its mass between them. We also exhibit a family with exponentially many combinations of modes that a diffusion actor represents with $O(d)$ denoising units, while a flat Gaussian mixture needs exponentially many components. When the mode probabilities are unknown, the structured actor can be learned from polynomially many demonstrations (Appendix D.1). We summarize the resulting guarantees as follows.

Policy optimization. Under reset or mixing access and the stated coverage and regularity conditions, vanilla PG achieves $O(T^{3/4})$ expected regret up to approximation and comparator errors. Its mixing guarantee requires a policy-independent stationary state law, and its density-ratio factor can be exponential in the effective horizon (Agarwal et al., 2021). DDIM NPG also achieves $O(T^{3/4})$ regret with high probability, up to approximation and comparator errors, while replacing this density-ratio dependence by a transfer condition on the compatible approximation (Kakade, 2001; Song et al., 2022). For NPG, the stationary state law may depend on the policy. Hybrid NPG retains the $O(T^{3/4})$ statistical exponent when logged trajectories from a known behavior policy provide importance-weighted Monte Carlo labels with bounded score-weighted second moments. The construction preserves the offline feature distribution, so offline features can supply directions missing from online samples, without restarting at their stored states or fitting a critic.

Behavior cloning. In finite-horizon RL, critic cloning (Li et al., 2024c; Yang et al., 2023) achieves a $\widetilde{O}(\sqrt{T})$ regret bound for controlled critic complexity, up to cumulative score-matching and sampler

Method	Sampling access	Regret
Vanilla PG	Resets or fixed-state mixing	$O(T^{3/4})$ + comparison-policy and compatible-approximation errors
DDIM NPG	Resets, mixing, or hybrid data	$O(T^{3/4})$ + comparison-policy and advantage-transfer errors
Critic cloning	Online interaction, optionally with offline data	$\widetilde{O}(\sqrt{T})$ + cumulative actor error

Table 5.1: Summary of results for fixed regularity constants. PG bounds include approximation and comparator errors. The vanilla PG bound is in expectation, while NPG and critic cloning hold with high probability. The critic contribution is $O(H^2\sqrt{\beta T \mathsf{SEC}(\mathcal{F}, \Pi, T)})$, with β defined in Theorem 5.17. The table suppresses these exploration and critic-class factors.

errors, by decomposing the regret into critic-estimation and behavior-cloning errors. As in Song et al. (2023), offline data with sufficient Bellman-error coverage can remove the need for optimism. The distinction between these routes also concerns exploration. Our policy-gradient guarantees assume that the available trajectories cover the relevant state-action directions. An expressive actor alone does not ensure that it visits states needed to discover a good policy. Critic cloning instead uses optimism to choose informative actions online, or uses offline Bellman-error coverage in the hybrid case. The price is that the actor must track the changing critic well enough for its accumulated cloning error to remain small.

Finally, our experiments compare practical versions of these algorithmic families and show that several tested diffusion-policy methods exhibit sublinear empirical regret across standard Gym and Robomimic benchmarks.

5.2. Setup and Background

RL Notation. We examine two standard RL models. An infinite-horizon discounted MDP is a tuple $(\mathcal{S}, \mathcal{A}, \gamma, \mathbb{P}, r)$ with $r \in [0,1]$ and $\gamma \in (0,1)$. We take $\mathcal{A} \subseteq \mathbb{R}^{d_A}$, with $\|a\| \le R_A$ in the behavior-cloning analysis. Policy gradients use actions in $\mathbb{R}^{d_A}$ or a fixed invertible output transformation into the action domain. From a starting state distribution ρ and policy π, write

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{h=0}^\infty \gamma^h r(s_h, a_h) | s_0 = s\right], \;\; Q^\pi(s,a) = \mathbb{E}_\pi\left[\sum_{h=0}^\infty \gamma^h r(s_h, a_h) | s_0 = s, a_0 = a\right].$$

The advantage function is $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$. Write $V^\pi(\rho) = \mathbb{E}_{s\sim\rho}[V^\pi(s)]$. The optimal policy π^* maximizes V^π. We write $V^* = V^{\pi^*}$ and do the same for Q^* & A^*.

In the finite-horizon setting, an MDP is a tuple $(\mathcal{S}, \mathcal{A}, H, \{\mathbb{P}_h\}_{h=1}^H, \{r_h\}_{h=1}^H)$, with time-dependent

$$V_h^\pi(s) = \mathbb{E}\left[\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'})|s_h = s\right], \;\; Q_h^\pi(s,a) = \mathbb{E}\left[\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'})|s_h = s, a_h = a\right].$$

Q^* is the unique solution to the Bellman equation $Q_h^* = \mathcal{T}_h Q_{h+1}^*$:

$$\mathcal{T}_h Q_{h+1}(s,a) = r_h(s,a) + \mathbb{E}_{s'}[\max_{a'\in\mathcal{A}} Q_{h+1}(s',a')], \mathcal{T}_h^\pi Q_{h+1}(s,a) = r_h(s,a) + \mathbb{E}_{s'} Q_{h+1}(s', \pi_{h+1}).$$

In both settings, observed rewards lie in $[0,1]$ and have conditional mean $r(s,a)$ or $r_h(s,a)$ given the current state, action, and past. At round $t = 1, ..., T$ the agent deploys a policy $\pi^{(t)}$ parameterized by some $\theta^{(t)} \in \Theta \subseteq \mathbb{R}^p$, and incurs regret $\mathrm{Reg}(T) := \sum_{t=1}^T \left(V^*(\rho) - V^{(t)}(\rho)\right)$ where $V^{(t)} := V^{\pi^{(t)}}$.

Diffusion models. A diffusion model samples an initial noise vector $Y_K \sim N(0, I_{d_A})$, applies K denoising steps conditional on the state s, and plays the terminal action Y_0. Let $\epsilon_\theta(\cdot, k, s)$ be the trainable denoising field conditional on state s. With $\bar{\alpha}_k := \prod_{i=1}^k \alpha_i$, the DDIM-type (Song et al., 2022) and DDPM-type (Ho et al., 2020) update rules for $k = K, ..., 1$ are

$$Y_{k-1} \leftarrow \alpha_k^{-\frac{1}{2}}\left(Y_k + \eta_k^{\mathsf{ddim}} \epsilon_\theta(Y_k, k, s)\right) =: \Phi_k^\theta(Y_k). \tag{5.1}$$

$$Y_{k-1} \leftarrow \alpha_k^{-\frac{1}{2}}\left(Y_k + \eta_k^{\mathsf{ddpm}} \epsilon_\theta(Y_k, k, s) + \sigma_k^{\mathsf{ddpm}} Z_k\right) =: \Psi_k^\theta(Y_k, Z_k). \tag{5.2}$$

Here Z_k are independent standard Gaussian vectors, independent of Y_K. We write $\Phi_{k_2:k_1}(Y_{k_2}) = (\Phi_{k_1+1} \circ ... \circ \Phi_{k_2})(Y_{k_2})$ for the composition of DDIM updates from times $k_1 \le k_2$, and $\Psi^{(t)}, \Phi^{(t)}$ as shorthand for $\Psi^{\theta^{(t)}}, \Phi^{\theta^{(t)}}$ and omit the dependence on s. Here η_k^{ddim}, η_k^{ddpm} and σ_k^{ddpm} are design parameters that admit multiple alternatives. For policy-gradient sampling, we choose positive, parameter-independent DDPM transition variances at every step, including the final transition. Section 5.4 specifies the different coefficients and stopping rule used for behavior cloning. A key technical issue here is that the distribution of Y_0 is implicit: we can sample actions, but the terminal density $\pi_\theta(a \mid s)$ and its score are not generally available.

Expressivity. A policy may need to retain several good actions at a state, for example when cloning demonstrations with distinct modes. Fitting one Gaussian can place mass between these actions, where reward is low. In Appendix D.1, we give a two-mode example in which the population maximum-likelihood Gaussian loses almost all reward. Directly optimizing reward can instead choose one good mode, so the example concerns the distribution-matching objective used in cloning.

Example 5.1 (Representing and learning combinations of modes)**.** A more pronounced difference appears when each of d action coordinates has two independently chosen modes. There are 2^d joint combinations, but a denoiser can generate them coordinate by coordinate. Proposition D.1 realizes a fixed-accuracy approximation with $O(d)$ tanh units, whereas any flat Gaussian mixture attaining total variation error 0.01 needs at least $0.79 \cdot 2^d$ components. If the coordinate probabilities are unknown and lie in $[1/4, 3/4]$, Proposition D.1 learns an $O(d)$-unit actor from $O(d \log(d/\delta)/\varepsilon^2)$ independent demonstrations with total variation error at most ε, with probability at least $1-\delta$. A flat mixture still needs exponentially many components at fixed accuracy. Thus the product structure permits learning a compact actor from polynomially many demonstrations. The comparison uses the same demonstrations and distribution-matching objective for both families.

5.3. Diffusion Policies with Policy Optimization

Policy gradient methods seek to solve the optimization problem $\max_{\theta\in\Theta} V^{\pi_\theta}(\rho)$, where $\{\pi_\theta : \theta \in \Theta\}$ is a policy class parametrized by $\theta \in \Theta$, and $\rho \in \Delta(\mathcal{S})$. At round t, given some $\theta^{(t)} \in \Theta$, one rolls out a trajectory under $\pi^{(t)} = \pi_{\theta^{(t)}}$, and uses the observed state $s^{(t)}$, action $a^{(t)}$, and rewards to estimate an update direction for $\theta^{(t)}$. One candidate utilizes the policy gradient theorem (Sutton et al., 1999):

$$(1-\gamma)\nabla_\theta V^{\pi_\theta}(\rho) = \mathbb{E}_{d_\rho^{\pi_\theta}}\left[A^{\pi_\theta}(s,a)\nabla_\theta \log \pi_\theta(a|s)\right] \tag{5.3}$$

where we substitute A for Q as $\int_a \nabla_\theta \pi_\theta(a|s)\mathrm{d}a = 0$. This relies on the discounted occupancy:

$$d_\nu^\pi(ds, da) := (1-\gamma)\sum_{h\geq 0}\gamma^h \mathbb{P}_\nu^\pi(S_h \in ds, A_h \in da), \qquad S_0 \sim \nu, \quad A_h \sim \pi(\cdot \mid S_h).$$

Here ν is the initial-state law. We write $d_\nu^\pi(s,a)$ for the density of this state–action measure when it exists.

Coverage. To control the error under ρ, we have to make a coverage assumption. Options include the *reset distribution* setting (Kakade, 2001; Kakade and Langford, 2002; Bagnell et al., 2003; Agarwal et al., 2021; Zhou et al., 2023), where the agent can reset the environment to an initial state $s_0 \sim \nu$. Another is the *strong mixing* setting (Gaur et al., 2023; Wu et al., 2022b; Xu et al., 2020). Yet another is having access to an *offline dataset* $\mathcal{D}_{\text{off}}$ (Song et al., 2023; Amortila et al., 2024; Nakamoto et al., 2023).

Assumption 5.2 (Sampling access)**.** Policies are fixed within each batch. We have one of the following forms of access:

(a) independent trajectories initialized from a fixed state law ν;

(b) a continuing trajectory whose policy-induced state kernel K_θ has stationary law m_θ and satisfies $\sup_{\theta,s} \mathrm{TV}(K_\theta^h(s,\cdot), m_\theta) \le c_{\text{mix}}\kappa^h$ for $c_{\text{mix}} \ge 1$ and $0 < \kappa < 1$. Vanilla PG uses a common stationary state law $m_\theta = m$, while NPG permits dependence on θ;

(c) ordinary online interaction initialized from ρ, together with independent logged behavior trajectories whose initial state–action law is μ. The known behavior policy can be evaluated and sampled at stored states.

Otherwise, we would have to consider strategic exploration, for which incorporating policy optimization is currently a nascent area of research.[22] We will address strategic exploration in the following section on behavior cloning.

Mixing replaces resets by a burn-in period followed by complete return samples. Hybrid NPG combines online samples with logged trajectories. It does not restart the environment at recorded states. Its importance-weighted labels require absolute continuity and finite score-weighted return

[22] Most papers that consider strategic exploration in policy optimization employ the softmax mirror descent update (Zhong and Zhang, 2023; Liu et al., 2023b; Tan et al., 2025; Sherman et al., 2023; Cassel and Rosenberg, 2024). While Lin et al. (2026) considers parameterized updates, they essentially clone the softmax update using a KL projection.

moments, stated in Appendix D.3.4. Reserved independent trajectory blocks make these labels valid for adaptively chosen policies. The online part may also use the continuing sampler under the mixing condition in (b).

Write q_θ for the state–action law used to fit an update. It is $d_\nu^{\pi_\theta}$ under resets, $m_\theta \pi_\theta$ under mixing, and $(d_\rho^{\pi_\theta} + \mu)/2$ for hybrid NPG, replacing the online component by $m_\theta \pi_\theta$ for a continuing hybrid trajectory. For vanilla PG the first two laws both have the form $d_\nu^{\pi_\theta}$, with $\nu = m$ in the mixing case. For any state–action law, a subscript S denotes its state marginal. Let $\boldsymbol{\phi}^\theta(s,a) = \nabla_\theta \log \pi_\theta(a \mid s)$ and $\boldsymbol{\Lambda}^\theta = \mathbb{E}_{q_\theta}[\boldsymbol{\phi}^\theta \boldsymbol{\phi}^{\theta\top}]$ be the policy-score feature and its covariance. The next condition requires coverage only in directions these features can identify.

A continuous action density assigns zero probability to any fixed action. Consequently, if an optimal policy always chooses one action, its state–action law is not dominated by the sampled policy law, and its KL divergence from that policy is infinite. These are separate obstacles for the density-ratio argument used by vanilla PG and the KL argument used by NPG.

We therefore compare with a fixed stochastic policy, denoted π_{cmp}, and retain its value gap $V^*(\rho) - V^{\pi_{\mathrm{cmp}}}(\rho)$ in regret against the optimum. This policy need not belong to the learned class. We write $d_\rho^{\pi_{\mathrm{cmp}}}$ for its discounted occupancy and keep it fixed during learning. Example 5.11 constructs such a policy by restricting the initial actor to actions near an optimum, and shows when its value gap can be $O(T^{-1/2})$.

For now, we make an assumption that the error under ρ can be related to the error under q_θ with the following concentrability assumption:

Assumption 5.3 (Concentrability and score covariance)**.** Assume $C := \sup_\theta \|d_\rho^{\pi_{\mathrm{cmp}}}/d_\nu^{\pi_\theta}\|_\infty < \infty$ for vanilla PG, and $\lambda := \inf_\theta \lambda_{\min}^+(\boldsymbol{\Lambda}^\theta) > 0$ for both algorithms, where $\lambda_{\min}^+$ denotes the smallest positive eigenvalue and the infimum runs over the update region.

The density ratio is analogous to single-policy concentrability (Xie et al., 2022b). The score covariance is the Fisher information under on-policy action sampling. Its positive spectrum controls

identifiable score directions and allows redundant parameters, including under the hybrid mixture law. Under the reset law we also write this lower bound as $\lambda^{+}_{\min}(\mathbf{\Lambda}_\nu)$. We use $q^{(t)} = q_{\theta^{(t)}}$, and $d^{(t)} = d_\nu^{\pi^{(t)}}$ when the sampling law has that form.

Policy scores. The gradient in Equation (5.3) uses the parameter score $\nabla_\theta \log \pi_\theta(a \mid s)$. The denoising field ϵ_θ is a different object: even if it initially represents a spatial log-density gradient, a reward update need not preserve that property. Appendix D.1 gives a one-step example in which a reward-improving update breaks this property while leaving the sampling map invertible and the policy differentiable.

We therefore compute the parameter score of the policy induced by the denoising map. Write $a_K, \ldots, a_0$ for a sampled path, with only a_0 played in the environment. However, one runs into a complication when trying to apply policy gradient updates with the DDPM sampler. As Psenka et al. (2024) notes, evaluating the policy density $\pi_\theta(a_0|s)$ is intractable, involving a high-dimensional integral over the space of possible paths. Still, the policy gradient can be estimated using the transition densities along the sampled path. In Appendix B.1, Psenka et al. (2024) show the policy gradient is equal to

$$(1-\gamma)\nabla_\theta V^{\pi_\theta}(\nu) = \mathbb{E}\left[A^{\pi_\theta}(s, a_0) \sum_{k=1}^{K} \nabla_\theta \log \pi_\theta(a_{k-1} \mid a_k, s)\right], \tag{5.4}$$

where $(s, a_0) \sim d_\nu^{\pi_\theta}$ and the expectation includes its sampled denoising path. This makes the vanilla policy gradient update tractable for the DDPM sampler, though we will see later that this still remains an issue for second-order methods. Regarding the DDIM sampler, we will show in Lemma D.2 that we can indeed evaluate $\nabla_\theta \log \pi_\theta(a_0|s)$ under sufficient assumptions.

The same environment advantage weights every denoising step in Equation (5.4), because only the terminal action affects the environment reward. Additional discounting across denoising steps, as used in Ren et al. (2024a), changes this estimator. We retain the environment discount γ and analyze the undiscounted sum of denoising-path scores.

Score estimates and regularity. For the sequence $(\theta^{(t)}, s^{(t)}, a^{(t)})_{t=1}^{T}$, write $\boldsymbol{\phi}^{(t)} = \boldsymbol{\phi}^{\theta^{(t)}}(s^{(t)}, a^{(t)})$. The computable DDPM estimate is $\widehat{\boldsymbol{\phi}}^{(t)} = \sum_{k=1}^{K} \nabla_\theta \log \pi_\theta(a_{k-1}^{(t)} \mid a_k^{(t)}, s^{(t)})$, whose conditional expectation given $(s^{(t)}, a_0^{(t)})$ equals $\boldsymbol{\phi}^{(t)}$. For DDIM, the change-of-variables formula in Lemma D.1 evaluates the endpoint score exactly, and we set $\widehat{\boldsymbol{\phi}}^{(t)} = \boldsymbol{\phi}^{(t)}$.

Assumption 5.4 (Smoothness, moments, and invertibility)**.** Likelihood derivatives and expectations below may be interchanged. The following conditions hold uniformly at the iterates and along their update segments: For DDIM, $\Phi_{K:0}^{\theta}$ is a smooth diffeomorphism onto a parameter-independent open action support. Under the applicable sampling law, $\mathbb{E}_{q_\theta}\|\widehat{\boldsymbol{\phi}}\|^2 \le \beta_K^2$, including the denoising path in the DDPM expectation. For vanilla PG, $\theta \mapsto V^{\pi_\theta}(\nu)$ is L_V-smooth for fixed ν. For DDIM NPG, $\boldsymbol{\phi}^\theta$ is $L_\theta^{(K)}$-Lipschitz in θ.

The DDIM score is evaluated on its common action support. A fixed invertible output transformation may be included in the full map. For regression, write $\mathbf{b}^\theta = \mathbb{E}_{q_\theta}[\boldsymbol{\phi}^\theta A^{\pi_\theta}]$ and $\mathcal{G} = \{\boldsymbol{g} \in \mathbb{R}^p : \|\boldsymbol{g}\|_2 \le G\}$. We assume the minimum-norm population coefficient $\boldsymbol{g}^\theta = (\boldsymbol{\Lambda}^\theta)^\dagger \mathbf{b}^\theta$ lies in $\mathcal{G}$ throughout the update region. The choice $G \ge ((1-\gamma)\sqrt{\lambda})^{-1}$ suffices. Conditionally unbiased advantage labels give the same target $\mathbf{b}^\theta$.

In line with Agarwal et al. (2021), we impose the following assumption on compatible approximation. Intuitively, the policy has to be rich enough so that the score features $\boldsymbol{\phi}^\theta$ can model the advantage function. Writing $\mathcal{L}_A^d(\boldsymbol{g}, \theta) = \mathbb{E}_{s,a\sim d}[(A^{\pi_\theta}(s,a) - \langle \boldsymbol{g}, \boldsymbol{\phi}^\theta(s,a)\rangle)^2]$ for this squared residual under any state–action law d, assume:

Assumption 5.5 (Compatible Advantage Approximation)**.** For the fixed denoising depth K used by vanilla PG, assume uniformly over its update region that

$$\mathcal{L}_A^{d_\nu^{\pi_\theta}}(\boldsymbol{g}^\theta, \theta) = \mathbb{E}_{d_\nu^{\pi_\theta}}(A^{\pi_\theta} - \langle \boldsymbol{g}^\theta, \boldsymbol{\phi}^\theta\rangle)^2 \le \epsilon_{\text{comp}}.$$

The score moment β_K^2 controls the noise in the gradient estimate, while compatible approximation lets us relate a small gradient to a small value gap.

Neural denoisers and adapters. Suppose $\epsilon_\theta(\cdot, k, s)$ is smooth and L_a-Lipschitz, with $|\eta_k^{\mathsf{ddim}}|L_a < 1$ at every DDIM step. Solving for its input is then a contraction, so the step is invertible and its inverse Jacobian has norm at most $\sqrt{\alpha_k}/(1 - |\eta_k^{\mathsf{ddim}}|L_a)$. On a uniformly invertible parameter region, the width-256 tanh MLP and smooth U-Net architectures in Appendix D.1 have finite score moments when their parameters and state conditioning are bounded. The same calculation controls averaged log-likelihood curvature if the comparison policy has a finite third moment after pulling actions back through the initial actor. The weights may be independently trained, but the Fisher and advantage-approximation conditions also depend on the MDP.

For a trainable location adapter on a frozen MLP or U-Net, we can also bound the positive Fisher spectrum in terms of the backbone's Lipschitz constant and the adapter design (Appendix D.1). The bound allows redundant parameters and applies under reset or mixing sampling, where actions follow the current policy. For the hybrid sampling law, conditioning instead follows from the mixture-covariance argument in Appendix D.3.4.

5.3.1. Monte Carlo Diffusion Vanilla Policy Gradient

We are now in a position to analyze the vanilla policy gradient update on diffusion policies, where one uses Equation (5.4) for the DDPM sampler and Equation (5.3) for the DDIM sampler. The algorithm we analyze is presented below in Algorithm 5.1, for which we defer the proofs to Appendix D.2.

Algorithm and score estimate. Algorithm 5.1 is vanilla PG with the diffusion-policy score substituted for the usual log-policy score. We use L policy updates and N online return labels per update, for $T = LN$ labels in total. Algorithm D.1 returns a state, an accepted action and its denoising path, together with a conditionally unbiased advantage label. With reset access, it uses geometric acceptance to sample $(s, a) \sim d_\nu^{\pi_\theta}$. With mixing, it takes complete geometric returns along the continuing trajectory, after one burn-in per batch. Both implementations freeze the policy during a batch. The path is retained because the DDPM score estimate requires its transition likelihoods, whereas DDIM uses the terminal score.

In dealing with diffusion policies, we first have to compute and bound the score features in Lemmas

D.1 and D.1. Under resets or a stationary-start mixing batch, the estimators $\widehat{A}^{(t)}\widehat{\boldsymbol{\phi}}^{(t)}/(1-\gamma)$ and $\widehat{A}^{(t)}\boldsymbol{\phi}^{(t)}/(1-\gamma)$ are unbiased (Lemma D.3) for the policy gradient $\nabla_\theta V^{\pi_\theta}(\nu)$. For a finite burn-in, Theorem 5.7 includes the resulting coupling error. Together with the stated smoothness and second-moment conditions, this yields convergence to a stationary point: the stochastic ascent argument (Yuan et al., 2022) gives $\sum_{t=1}^{T}\mathbb{E}\|\nabla_\theta V^{(t)}(\nu)\|_2^2 = O(\sqrt{T})$.

To relate this to regret, we prove a notion of gradient domination that bounds the suboptimality gap in terms of the policy gradient, up to approximation errors. This second step is not diffusion-specific. Once the score features are controlled, gradient domination transfers convergence to stationarity into regret. The term ϵ_{comp} below is a squared advantage-regression residual. It measures how well score features fit the current advantage, whereas $V^*(\rho) - V^{\pi_{\text{cmp}}}(\rho)$ measures the loss from replacing the optimal policy by the chosen stochastic comparison policy. These errors need not be related.

Lemma 5.6 (Gradient Domination Lemma)**.** *Under Assumption 5.5, for $\rho, \nu \in \Delta(\mathcal{S})$ and θ satisfying $C_\theta := \|d_\rho^{\pi_{\text{cmp}}}/d_\nu^{\pi_\theta}\|_\infty < \infty$,*

$$V^*(\rho) - V^{\pi_\theta}(\rho) \le V^*(\rho) - V^{\pi_{\text{cmp}}}(\rho) + \frac{\sqrt{C_\theta \epsilon_{\text{comp}}}}{1-\gamma} + \sqrt{\frac{C_\theta}{\lambda^+_{\min}(\boldsymbol{\Lambda}^\theta_{d_\nu^{\pi_\theta}})}}\|\nabla_\theta V^{\pi_\theta}(\nu)\|_2.$$

Lemma 5.6 bounds the suboptimality gap of π_θ in terms of the minimum positive eigenvalue of its score feature covariance matrix and the norm of the policy gradient, up to the approximation and comparator errors. Related results have been established for tabular MDPs, softmax MDPs, linear quadratic control, regularized linear MDPs (Bhandari and Russo, 2024), and softmax neural policies (Wang et al., 2019). While used here for diffusion policies, this gradient domination lemma applies more broadly to policy classes satisfying its conditions, and so may be of independent interest. Plugging our bounds on the rate of convergence into a stationary point into Lemma 5.6 yields:

Theorem 5.7 (Vanilla Diffusion Policy Gradient Regret)**.** *Suppose Assumptions 5.2–5.5 hold under reset access or fixed-state mixing, and run Algorithm 5.1 with DDPM or DDIM, fixed denoising*

depth K, $T = LN$ return labels, and stepsize $0 < \eta \le L_V^{-1}$. Under reset access, set $\tau_{\mathrm{mix}} = 1$ and $\varepsilon_b = 0$. Under fixed-state mixing, take $\nu = m$, use b burn-in transitions per batch, and set

$$\tau_{\mathrm{mix}} = 1 + \frac{4\sqrt{c_{\mathrm{mix}}}}{1-\sqrt{\kappa}}, \qquad \varepsilon_b = c_{\mathrm{mix}}\kappa^b.$$

Then

$$\mathbb{E}[\mathrm{Reg}(T)] \le T[V^*(\rho) - V^{\pi_{\mathrm{cmp}}}(\rho)] + \frac{T\sqrt{C\epsilon_{\mathrm{comp}}} + TL\varepsilon_b}{1-\gamma} + \sqrt{\frac{C}{\lambda}}\, N\sqrt{L}\left(\frac{2[V^*(\nu) - V^{(1)}(\nu)]}{\eta} + \frac{4L_V\eta\tau_{\mathrm{mix}}(1+\gamma)\beta_K^2 L}{N(1-\gamma)^4}\right)^{1/2}.$$

For $N \le \sqrt{T}$, choosing η to balance the two terms in parentheses, subject to $\eta \le L_V^{-1}$, gives a statistical term of order $T^{3/4}$ for fixed problem constants. Positive upper bounds may be used for any constants that vanish in this choice of stepsize.

Reset access permits $N = 1$ and requires no burn-in. Mixing increases the variance by τ_{mix} and adds the coupling error $TL\varepsilon_b/(1-\gamma)$. Taking $\varepsilon_b \le T^{-2}$ and $N \asymp bT^{1/4} \le \sqrt{T}$ also keeps the regret charged to burn-in transitions of order $T^{3/4}$. Here T counts return labels. Reset sampling uses $O(T/(1-\gamma))$ environment transitions in expectation, including its occupancy search, while mixing uses $Lb + T/(1-\gamma)$. Appendices D.2 and D.2.5 give the proof, a sharper constant for the reset case, and the high-probability transition count.

If an optimal policy itself satisfies the comparison conditions, we may take $\pi_{\mathrm{cmp}} = \pi^*$ and the value-gap term vanishes. Otherwise, the displayed bound retains the cost of the stochastic approximation.

Algorithm 5.1 Vanilla Diffusion Policy Gradient

Input: reset or fixed-state mixing access, $T = LN$, batch size N, stepsize η.
Initialize: $\theta^{(1)} \in \Theta$.
for batch $i = 1, \ldots, L$ **do**
    Freeze the current policy for all N rounds of this batch, set $\mathbf{b} \leftarrow 0$, and, under mixing, run b burn-in transitions.
    for $n = 1, \ldots, N$ **do**
        Obtain $\widehat{A}, s, a, a_{1:K}$ from the reset or continuing branch of Algorithm D.1.
        Evaluate $\widehat{\boldsymbol{\phi}}$ from the DDPM path, or set $\widehat{\boldsymbol{\phi}} = \boldsymbol{\phi}$ using the exact DDIM score.
        $\mathbf{b} \leftarrow \mathbf{b} + \widehat{A}\widehat{\boldsymbol{\phi}}/(N(1-\gamma))$.
    end for
    Update $\theta \leftarrow \theta + \eta\mathbf{b}$.
end for

Algorithm 5.2 Monte Carlo Diffusion Natural Policy Gradient

Input: reset, mixing, or hybrid access, $T = LN$, stepsize η, optional T reserved offline trajectories.
Initialize: $\theta^{(1)}$ with finite comparator KL divergence.
for batch $i = 1, \ldots, L$ **do**
    Let $t_i = iN$ and write $\theta^{(t_i)}$ for the parameter frozen throughout this batch. Under mixing, first run b burn-in transitions.
    Collect N online labels with the reset or continuing branch of Algorithm D.1; hybrid initial-state interaction uses $\nu = \rho$.
    Evaluate the DDIM endpoint feature $\boldsymbol{\phi}(s,a) = \nabla_\theta \log \pi_\theta(a \mid s)$ at every sampled pair.
    if hybrid access **then**
        Reveal N unused offline trajectories. Apply the offline branch of Algorithm D.1 to each, and evaluate the same current-policy features.
        Form $\widehat{\mathcal{L}}_A$ by giving the online and offline squared losses equal total weight.
    else
        Set $\widehat{\mathcal{L}}_A(\boldsymbol{g}) = N^{-1}\sum_{n=1}^{N}(\widehat{A}_n - \langle \boldsymbol{g}, \boldsymbol{\phi}_n\rangle)^2$.
    end if
    $\widehat{\boldsymbol{g}} \leftarrow \arg\min_{\boldsymbol{g}\in\mathcal{G}} \widehat{\mathcal{L}}_A(\boldsymbol{g})$, choosing the minimum-norm minimizer.
    Update $\theta \leftarrow \theta + \eta\widehat{\boldsymbol{g}}$.
end for

However, the above result depends on $C = \sup_\theta \|d_\rho^{\pi_{\text{cmp}}}/d_\nu^{\pi_\theta}\|_\infty$, which can be exponential in the effective horizon, and depends on the state space cardinality (Agarwal et al., 2021). Natural policy gradient updates (Kakade, 2001) can yield dimension-free convergence guarantees (Agarwal et al., 2021; Cen et al., 2022), and are closely related to the local quadratic trust-region update in TRPO (Schulman et al., 2015). The NPG update possesses strong theoretical guarantees while being close to the popular PPO algorithm (Schulman et al., 2017) used in Ren et al. (2024a), and so we analyze it for diffusion policies to see if we can remove the dependence on C.[23]

Natural policy gradient. With on-policy action sampling, NPG preconditions the policy gradient by the Moore–Penrose pseudoinverse of the Fisher information. The resulting minimum-norm coefficient is $\boldsymbol{g}^{(t)} = (\boldsymbol{\Lambda}^{(t)})^\dagger \mathbf{b}^{(t)}$, with the factor $1/(1-\gamma)$ absorbed into the stepsize. Its regression form also lets us use mixing or hybrid sampling laws. Define

$$\mathcal{L}_A^{(t)}(\boldsymbol{g}, \theta^{(t)}) = \mathbb{E}_{(s,a)\sim q^{(t)}}\big[(A^{\pi^{(t)}}(s,a) - \langle \boldsymbol{g}, \boldsymbol{\phi}^{(t)}(s,a)\rangle)^2\big].$$

Then $\boldsymbol{g}^{(t)}$ is the minimum-norm minimizer over $\mathcal{G}$, and the population update is $\theta^{(t+1)} = \theta^{(t)} + \eta \boldsymbol{g}^{(t)}$. As such, performing NPG amounts to regressing the advantage function of π_θ on the policy score features. The exact NPG update is rarely available, but can be approximated with on-policy sampling (Algorithm D.1). Under hybrid sampling, the same regression uses the mixed feature covariance, so offline data can supply directions that online samples miss. We bound its error under the comparison policy to obtain a performance guarantee.

Algorithm and DDIM restriction. Algorithm 5.2 performs batched NPG. Within batch i, the policy is fixed, N samples are collected, and the update direction is obtained by regressing advantage estimates on score features. This is tractable for DDIM by Lemma D.2. For DDPM, squaring a conditionally unbiased path feature adds its conditional covariance to the regression

[23] PPO is widely used in practice, including for diffusion policies (Ren et al., 2024a). It approximates TRPO, which is strongly related to NPG. In the limit of small stepsizes and with KL-divergence regularization, TRPO and NPG coincide to first order under the corresponding local quadratic approximation (Schulman et al., 2015; Neu et al., 2017; Rajeswaran et al., 2017).

matrix. Thus unbiasedness suffices for vanilla PG but does not recover the NPG regression. We analyze the exact DDIM endpoint features.

Mixing and hybrid sampling. For mixing, a frozen policy has stationary sampling law $m_\theta \pi_\theta$. NPG uses this law as regression weights, so the proof need not differentiate a value objective through m_θ. The remaining obstacle is dependence between complete returns, handled by the batch variance bound in Appendix D.3.4.

The logged returns estimate the behavior policy's value, not the current policy's, so we correct them using importance sampling. For $Q^{\pi_\theta}(s, a)$, the recorded action a is fixed and the importance ratios begin at the next action. For $V^{\pi_\theta}(s)$, we also reweight the first action. This value estimate averages over the trajectory's first action, so it cannot also be conditioned on that action as the regression feature. Algorithm D.1 instead pairs it with an independent action drawn from the behavior policy at the stored state. Combining this branch with the Q branch gives a conditionally unbiased advantage label and preserves the feature law μ. The learner can therefore use the logged trajectories without fitting a critic or restarting at their stored states. The proof requires finite score-weighted return moments and a fresh block of logged trajectories for each update.

For the fixed stochastic comparison policy introduced above, write the initial KL cost as

$$\mathcal{K}_0 := \mathbb{E}_{s \sim d_{\rho,S}^{\pi_{\mathrm{cmp}}}} D_{\mathrm{KL}}(\pi_{\mathrm{cmp}}(\cdot \mid s) \| \pi^{(1)}(\cdot \mid s)).$$

Assumption 5.8 (NPG transfer and moments)**.** The comparator has $\mathcal{K}_0 < \infty$, and the current-policy score $\boldsymbol{\phi}^\theta$ is integrable under $d_\rho^{\pi_{\mathrm{cmp}}}$. Uniformly over the update region, the fitted population coefficient satisfies $\mathcal{L}_A^{d_\rho^{\pi_{\mathrm{cmp}}}}(\boldsymbol{g}^\theta, \theta) \le \epsilon_{\mathrm{bias}}$ and $\|\mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}} \boldsymbol{\phi}^\theta\| \le B_*$. Under reset or mixing sampling, $\mathbb{E}_{q_\theta} \|\boldsymbol{\phi}^\theta \boldsymbol{\phi}^{\theta\top} - \boldsymbol{\Lambda}^\theta\|_F^2 \le M_F$. Under hybrid sampling, the separate online and offline covariance and score-weighted label moments satisfy the conditions in Appendix D.3.4.

The transfer error measures how well the fitted advantage predicts under the comparator's occupancy, rather than only under its training law q_θ. The bound B_* on the comparison mean score

converts coefficient-estimation error into performance error. The fourth moment controls estimation of the feature covariance. Write $\boldsymbol{g}_i = \boldsymbol{g}^{\theta^{(t_i)}}$ for the batch population coefficient and $\widehat{\boldsymbol{g}}_i$ for its constrained least-squares estimate. Conditional expectation given the history before batch i is denoted by $\mathbb{E}_i$. For each sampling case, the least-squares analysis gives $\mathbb{E}_i\|\widehat{\boldsymbol{g}}_i - \boldsymbol{g}_i\| \le a_N$. Under resets it is

$$a_N = \frac{2(\sqrt{C_A}\beta_K + G\sqrt{M_F})}{\lambda\sqrt{N}} + \frac{8GM_F}{\lambda^2 N}, \qquad C_A = \frac{4(1+\gamma)}{(1-\gamma)^2}.$$

For mixing, $a_N = a_N^{\mathrm{mix}}$ is the displayed bound in Appendix D.3.4, which multiplies the sampling variances by τ_{mix} and adds $2Gc_{\mathrm{mix}}\kappa^b$. For hybrid sampling, $a_N = a_N^{\mathrm{IS}}$ is the bound in Appendix D.3.4, using the online and importance-weighted offline moments. These bounds are all of order $N^{-1/2}$ for fixed moments and mixing constants, with a sufficiently long burn-in in the mixing case.

Theorem 5.9 (DDIM NPG Regret)**.** *Under Assumptions 5.2–5.4 and 5.8, use the corresponding reset, mixing, or hybrid branch of Algorithm 5.2, with $T = LN$ and its coefficient bound a_N. For any $\eta > 0$ whose update segments satisfy the regularity assumptions, for $0 < \delta < 1$, with probability at least $1 - \delta$,*

$$\mathrm{Reg}(T) \le T[V^*(\rho) - V^{\pi_{\mathrm{cmp}}}(\rho)] + \frac{N}{1-\gamma}\left(\frac{\mathcal{K}_0}{\eta} + \frac{\eta L_\theta^{(K)} G^2 L}{2}\right) + \frac{T}{1-\gamma}(\sqrt{\epsilon_{\mathrm{bias}}} + B_* a_N) + \frac{2NB_*G}{1-\gamma}\sqrt{2L\log(1/\delta)}.$$

The expectation bound omits the last term. With $L = N = \sqrt{T}$ and $\eta = \sqrt{2\mathcal{K}_0/(LL_\theta^{(K)}G^2)}$, using positive upper bounds for zero constants, the statistical terms are $O(T^{3/4})$. The mixing case uses $c_{\mathrm{mix}}\kappa^b \le T^{-1}$ and adds Lb burn-in transitions. The hybrid case uses T independent offline trajectory records, whose expected total length is $T/(1-\gamma)$. Online return lengths are charged as in Appendix D.2.5.

The coefficient bound a_N retains the dependence on the smallest positive covariance eigenvalue λ in Assumption 5.3. Example 5.11 makes the comparison-policy term $O(\sqrt{T})$ and controls the resulting score constants for the neural denoisers verified in Appendix D.1. The advantage-transfer

error remains explicit.

This rate improves upon the $T^{5/6}$ scaling obtained by balancing the optimization and sample-estimation terms in Agarwal et al. (2021): their Corollary 33 has an $N^{-1/4}$ estimation term, whereas our covariance conditions give $N^{-1/2}$. It matches the $T^{3/4}$ regret exponent of Zhong and Zhang (2023), who analyze multi-batched optimistic PPO in episodic linear MDPs with full-information reward feedback. Unlike the result in Theorem 5.7 for vanilla diffusion policy gradients, it has no explicit density-ratio factor, but retains the comparator transfer error. Algorithm 5.2 requires only $\sqrt{T}$ NPG updates, and pays a cost for estimating the update direction. With $T = LN$, the optimization and leading estimation terms scale as $T/\sqrt{L}$ and $T/\sqrt{N}$, respectively, so $L = N = \sqrt{T}$ balances them.

Establishing this presents some additional technical challenges beyond the work we have already done for Theorem 5.7. To improve upon the $T^{5/6}$ rate within Agarwal et al. (2021), we establish a tighter bound on the update-direction error within Lemma D.5, controlling the covariance and regression coefficient together. Because action spaces with diffusion policies are continuous, and the KL divergence between a potentially deterministic optimal policy and a continuous diffusion policy can be infinite, the standard mirror descent argument does not go through directly. We therefore apply it to the finite-KL comparison policy π_{cmp}, retaining its value gap to the optimum. This gives the initial KL term in Theorem 5.9.

Remark 5.10 (Transfer error along the learning trajectory). The transfer error can also be retained batch by batch. Let $\epsilon_i \geq 0$ be measurable before batch i and satisfy $\mathcal{L}_A^{d_\rho^{\pi_{\mathrm{cmp}}}}(\boldsymbol{g}^{(t_i)}, \theta^{(t_i)}) \leq \epsilon_i$. Under the other conditions of Theorem 5.9, its high-probability bound holds with $T\sqrt{\epsilon_{\mathrm{bias}}}/(1-\gamma)$ replaced by $N\sum_{i=1}^{L}\sqrt{\epsilon_i}/(1-\gamma)$. The expectation bound uses the expectation of this sum. Taking $\epsilon_i = \epsilon_{\mathrm{bias}}$ recovers the displayed theorem. Appendix D.3.4 verifies this cumulative error for a two-layer tanh actor whose linked hidden and output weights change its distributional shape under the actual stochastic NPG update.

Example 5.11 (Comparison near optimal actions)**.** A stochastic comparison policy can approach

a deterministic optimum without incurring infinite initial KL cost. Suppose a measurable optimal action $a^*(s)$ exists, and let $B_r(s)$ be its radius-r ball. Assume that Q^* is L_Q-Lipschitz in the action on these balls and that the initial policy assigns them mass at least $c_0(r/R)^m$, for $0 < r \le R$, $0 < c_0 \le 1$, and $m \ge 0$. Restrict the initial actor to these actions:

$$\pi_{\mathrm{cmp},r}(\mathrm{d}a \mid s) = \frac{\mathbf{1}\{a \in B_r(s)\}\pi^{(1)}(\mathrm{d}a \mid s)}{\pi^{(1)}(B_r(s) \mid s)}.$$

The restriction has log density ratio $-\log \pi^{(1)}(B_r(s) \mid s)$, while its chosen actions lose at most $L_Q r$ in optimal action value. The performance-difference identity therefore gives

$$\mathcal{K}_0 \le \log(c_0^{-1}) + m\log(R/r), \qquad V^*(\rho) - V^{\pi_{\mathrm{cmp},r}}(\rho) \le \frac{L_Q r}{1-\gamma}.$$

Taking $r = R/\sqrt{T}$ makes the regret contribution of this value gap at most $L_Q R\sqrt{T}/(1-\gamma)$, with $\mathcal{K}_0 \le \log(c_0^{-1}) + (m/2)\log T$.

For the Gaussian-latent neural denoisers in Appendix D.1, conditioning on these balls increases the comparison score and averaged-curvature bounds only polylogarithmically in T. At fixed architecture, mass exponent m, and confidence level, the averaged-curvature NPG argument gives statistical terms $O(T^{3/4}(\log(2T))^{5/4})$ on a common bounded parameter region with uniform invertibility, Fisher, coefficient, and sampling-moment bounds. The calculation following Equation (D.1) establishes these rates. The advantage-transfer contribution remains $N\sum_i \sqrt{\epsilon_i}/(1-\gamma)$. Vanilla PG additionally needs control of its density ratio C, which may grow as the ball shrinks.

Pretraining can improve the mass constants c_0 and m. An effective exponent $m < d_A$ can reflect concentration near a lower-dimensional set over a range of scales. At points of finite positive density, a fixed smooth full-dimensional policy instead has ambient-dimensional scaling as $r \to 0$.

Example 5.12 (Finetuning only task-relevant coordinates)**.** Suppose the reward and controlled transitions depend on an action only through its task-relevant coordinates, and finetuning leaves the conditional law of the remaining coordinates fixed. Appendix D.3.4 shows that the score, Fisher,

compatible residual, and stochastic NPG updates then coincide with those in the reduced action space. Adding such fixed task-irrelevant coordinates therefore leaves all NPG learning constants unchanged. This supplies an intrinsic-dimensional rate when the actor and the task share this structure.

Example 5.13 (Learning with neural and multimodal actors)**.** To apply the NPG bound, the learning assumptions must hold together along the stochastic updates. Appendix D.3.4 verifies them for a two-layer tanh actor with linked hidden and output weights that change its distributional shape while learning an unknown noisy reward. Its cumulative transfer error is nonzero but controlled, yielding $T^{3/4}$ regret up to logarithmic factors. Appendix D.3.4 gives a complementary analytic two-mode actor with exact compatibility and the same regret order. This second construction verifies multimodality explicitly, whereas the first supplies a finite neural network. Both allow redundant parameters through the Fisher pseudoinverse.

5.4. Diffusion Policies with Behavior Cloning

It is difficult to achieve $\sqrt{T}$ regret for policy optimization in general (Sherman et al., 2023; Tan et al., 2025), let alone for diffusion policies. On the other hand, we analyze another approach introduced by Li et al. (2024c); Yang et al. (2023), of learning the actor via behavior cloning the critic. This lets the critic choose exploratory actions through optimism, while the diffusion model learns to execute them. We show that the critic contributes $O(H^2\sqrt{\beta T\mathsf{SEC}})$ regret and quantify the additional cost of imperfect cloning. Thus the $\widetilde{O}(\sqrt{T})$ rate is conditional on both controlled critic complexity and sufficiently accurate actor fitting. The online result obtains exploration from optimism and does not assume a covering offline distribution.

Notation and assumptions. We approximate Q-functions with a function class $\mathcal{F} = \prod_{h=1}^{H} \mathcal{F}_h$ and set $f_{H+1} = 0$. Each $f_h : \mathcal{S} \times \mathcal{A} \to [0, H]$ is also known as a critic, with quality quantified by the Bellman error $f_h - \mathcal{T}_h f_{h+1}$. Each critic induces a measurable greedy policy $\pi_h^f(s) \in \arg\max_a f_h(s, a)$, whose existence we assume. Within finite-horizon RL, we make an assumption on the richness of $\mathcal{F}$ (Jin et al., 2021a; Rashidinejad et al., 2021; Xie et al., 2022a):

Assumption 5.14 (Completeness)**.** $\mathcal{T}_h f_{h+1} \in \mathcal{F}_h$ for all $f \in \mathcal{F}, h \in [H]$. This implies $Q_h^* \in \mathcal{F}_h$.

Let Π contain every policy deployed by the algorithm, including the sampler's early stopping and projection. The sequential extrapolation coefficient (SEC) (Xie et al., 2022a), a measure of exploration complexity, captures the difficulty of learning a critic. Write $\mathsf{SEC}(\mathcal{F}, \Pi, T)$ for

$$\sup_{h, f^{(1:T)} \in \mathcal{F}, \pi^{(1:T)} \in \Pi} \sum_{t=1}^{T} \frac{\mathbb{E}_{\pi^{(t)}}[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}]^2}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{\pi^{(i)}}[(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)})^2]}.$$

Expectations under $\pi^{(t)}$ use its stage-h occupancy. A small SEC allows the critic to generalize from the states and actions already visited to those visited next.

For the actor, we need to control prediction error for the current cloning target. Write $d_{h,S}^{(t)}$ for the state marginal of the deployed policy and, conditional on s, let $a^*(s) = \pi_h^{f^{(t)}}(s)$, $v_k = 1 - \bar{\alpha}_k$, and

$$q_k(\cdot \mid s) = \mathcal{N}(\sqrt{\bar{\alpha}_k} a^*(s), v_k I), \qquad \epsilon_{\theta^*}(x, k, s) = \frac{x - \sqrt{\bar{\alpha}_k} a^*(s)}{\sqrt{v_k}}.$$

Here θ^* denotes the current target predictor, which depends on t, h and need not be realized by a single network. It predicts noise, and its spatial score is $-\epsilon_{\theta^*}/\sqrt{v_k}$.

Assumption 5.15 (Score-Matching Error)**.** For the fitted actor and critic at iteration t, there exist $\epsilon_{\text{score}}^{(t)}$ such that, for every $h \in [H]$,

$$\mathbb{E}_{s \sim d_{h,S}^{(t)}} \left[\frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{X \sim q_k(\cdot|s)} \|\epsilon_{\theta_h^{(t)}}(X, k, s) - \epsilon_{\theta^*}(X, k, s)\|_2^2 \right] \le (\epsilon_{\text{score}}^{(t)})^2.$$

These history-dependent bounds measure population error under the deployed states. A small replay-buffer training loss alone does not establish them. As in diffusion sampling analyses (Li et al., 2024b; Li and Yan, 2025), our guarantee retains this prediction error. Example 5.18 gives a training construction for a stochastic cloning target when all states are known and the pretrained sampler can be queried at each state.

We also use action regularity to translate sampling error into lost critic value.

Assumption	Purpose	Used in	Location
Assumptions 5.2, 5.3	Reset/mixing access, comparator coverage, positive Fisher spectrum	Theorems 5.7, 5.9, D.3.4	Section 5.3
Appendix D.3.4	Logged trajectories, known behavior policy, weighted-return moments, mixture positive spectrum	Theorem D.3.4	Appendix D.3.4
Assumption 5.4	Likelihood regularity, moments, result-specific smoothness	Theorems 5.7, 5.9	Section 5.3
Assumption 5.5	Compatible-approximation residual for vanilla PG	Theorem 5.7, Lemma 5.6	Section 5.3
Assumption 5.8	Comparison-policy transfer error, mean score, and covariance moments	Theorem 5.9	Section 5.3
Assumption 5.14	Bellman completeness for critic estimation	Theorem 5.17	Section 5.4
Assumption 5.15	Population prediction error under deployed states	Theorem 5.17	Section 5.4
Assumption 5.16	Action projection and selected-critic regularity	Theorem 5.17	Section 5.4

Table 5.2: Assumptions and corresponding results. The NPG comparison and moment conditions appear in Assumption 5.8, while sampler stability for critic cloning is specified in Section 5.4.

Assumption 5.16. Assume $\mathcal{A}$ is compact and convex, $\|a\|_2 \le R_A$ on $\mathcal{A}$, and each selected critic $f_h^{(t)}$ is L_f-Lipschitz in actions. We retain the dependence on L_f explicitly.

Define the actor's per-round cloning loss by

$$B_t = \max_{h\in[H]} \mathbb{E}_{s\sim d_{h,S}^{(t)}} \left[f_h^{(t)}(s, \pi_h^{f^{(t)}}(s)) - \mathbb{E}_{a\sim\pi_h^{(t)}(\cdot|s)} f_h^{(t)}(s,a) \right]. \tag{5.5}$$

Each difference is nonnegative because $\pi_h^{f^{(t)}}$ is greedy. This measures what matters for regret: the critic value lost when the deployed actor approximates the greedy action. Lemma D.11 bounds B_t using the prediction error above. For DDPM it uses exact Gaussian reverse kernels for the noised deterministic target. For DDIM it also tracks the Lipschitz constants of the learned denoising maps. In both cases we stop at Y_1 and project onto $\mathcal{A}$. Appendix D.4.3 specifies the coefficients and the resulting bounds.

Algorithm 5.3 Diffusion Policy Behavior Cloning of an Optimistic Critic

Input: $\{\pi_\theta \mid \theta \in \Theta\}$, $\mathcal{F}$, optional $\mathcal{D}^{\text{off}}$.
Initialize: $\mathcal{D}_h^{(0)} \leftarrow \emptyset$. Select $f^{(1)}$ using the appropriate critic rule below on the available data, then clone it to obtain $\pi^{(1)}$.
for episode $t = 1, 2, \ldots, T$ **do**
    Execute $\pi^{(t)}$ for one episode, collect $\tau^{(t)}$. For the cloning guarantee, stop the DDPM/DDIM chain at Y_1 and play $\text{Proj}_{\mathcal{A}}(Y_1)$.
    $\mathcal{D}_h^{(t)} \leftarrow \mathcal{D}_h^{(t-1)} \cup \{(s_h^{(t)}, a_h^{(t)}, r_h^{(t)}, s_{h+1}^{(t)})\}$ and $\widehat{\mathcal{L}}_h^{(t)}(g, u) \leftarrow \sum_{\mathcal{D}_h^{(t)} \cup \mathcal{D}_h^{\text{off}}} [g(s,a) - r - \max_{a'} u(s', a')]^2$.
    if online data only: **then**
        $\mathcal{F}^{(t)} \leftarrow \{f \in \mathcal{F} : \widehat{\mathcal{L}}_h^{(t)}(f_h, f_{h+1}) - \min_{g \in \mathcal{F}_h} \widehat{\mathcal{L}}_h^{(t)}(g, f_{h+1}) \le H^2\beta,\ \forall h\}$.
        $f^{(t+1)} \in \arg\max_{f \in \mathcal{F}^{(t)}} \mathbb{E}_{s\sim\rho} f_1(s, \pi_1^f(s))$.
    else
        Set $f_{H+1}^{(t+1)} = 0$. For $h = H, \ldots, 1$, set $f_h^{(t+1)} \in \arg\min_{g \in \mathcal{F}_h} \widehat{\mathcal{L}}_h^{(t)}(g, f_{h+1}^{(t+1)})$.
    end if
    Construct behavior cloning dataset
$\widetilde{\mathcal{D}}_h^{(t+1)} \leftarrow \{(s_h, \pi_h^{f^{(t+1)}}(s_h))\}_{s_h \in \mathcal{D}_h^{(t)} \cup \mathcal{D}_h^{\text{off}}}$.
    Match $\pi_h^{(t+1)}$ to $\pi_h^{f^{(t+1)}}$ on $\widetilde{\mathcal{D}}_h^{(t+1)}$.
end for

Algorithm design. Algorithm 5.3 alternates between critic learning and diffusion behavior cloning. With online data only, the critic is the optimistic GOLF critic of Jin et al. (2021a). Optimism encourages exploration by selecting a critic whose predicted initial value is largest among those consistent with the data. With offline and online data, sufficient Bellman-error coverage allows the confidence-set search to be replaced by backward fitted-Q iteration. As in Li et al. (2024c), critic estimation uses the original transitions, while behavior cloning uses a separate dataset whose actions are replaced by the current critic-greedy actions. This avoids the inconsistency in the DIPO algorithm of Yang et al. (2023) caused by overwriting the actions in the Bellman targets.

This decouples critic estimation and behavior cloning (Li et al., 2024c), allowing us to show the following regret bound. Here, critic learning supplies the $\sqrt{T}$ term for controlled critic complexity, while the diffusion-specific work is in controlling the cost of cloning a deterministic greedy target with a stochastic sampler.

Theorem 5.17 (Regret Bound for Algorithm 5.3)**.** *Under Assumptions 5.14, 5.15, and 5.16, the*

sampler conditions in Appendix D.4.3, and exact critic and greedy-action optimization, fix $\delta \in (0,1)$ *and use the confidence parameter* $\beta = C\log(2H(T+1)\mathcal{N}_{\mathcal{F}}(1/(T+1))^2/\delta)$*, where* C *is a sufficiently large universal constant and* $\mathcal{N}_{\mathcal{F}}$ *is a uniform-norm covering number. Given only online data, Algorithm 5.3 achieves, with probability at least* $1-\delta$*,*

$$\mathrm{Reg}(T) \le O\left(H^2\sqrt{T\mathsf{SEC}(\mathcal{F},\Pi,T)\log\frac{2H(T+1)\mathcal{N}_{\mathcal{F}}(1/(T+1))^2}{\delta}}\right) + 2H\sum_{t=1}^{T} B_t.$$

For fixed H, L_f, R_A, d_A, the schedule in Appendix D.4.3 uses $K = \Theta(\sqrt{T})$ and gives

$$2H\sum_{t=1}^{T} B_t \le \begin{cases} O(\sqrt{T} + \sqrt{\log(T+1)}\sum_t \epsilon_{\mathrm{score}}^{(t)}), & \text{DDPM,} \\ O(\sqrt{T} + \log(T+1)\sum_t \epsilon_{\mathrm{score}}^{(t)}), & \text{DDIM with nonexpansive learned maps.} \end{cases} \tag{5.6}$$

Consequently, if β and SEC grow at most polylogarithmically in T and $\sum_t \epsilon_{\mathrm{score}}^{(t)} = O(\sqrt{T})$, the total regret is $\widetilde{O}(\sqrt{T})$. For example, a $1/\sqrt{t}$ prediction-error rate for each current target under the deployed states is sufficient. Establishing this rate for general replay-based neural finetuning remains open. The DDIM conclusion additionally needs uniform nonexpansiveness, which an average Jacobian-matching error alone does not imply.

Hybrid data. Under the offline sampling and coverage conditions of Appendix D.1, backward fitted-Q iteration replaces the confidence-set search. Set $N_{\mathrm{off}} \ge 1$, $n = N_{\mathrm{off}} + T$, and use

$$\beta_{\mathrm{hyb}} = C\log\frac{2H(n+1)\mathcal{N}_{\mathcal{F}}(1/(n+1))^2}{\delta}$$

in the critic term. With probability at least $1-\delta$, the regret bound then has the additional term

$$O\left(H^2 T\sqrt{\frac{c^*_{\mathrm{off}}(\mathcal{F})}{N_{\mathrm{off}}}\log\frac{2H(n+1)\mathcal{N}_{\mathcal{F}}(1/(n+1))^2}{\delta}}\right).$$

Given access to offline data, the learner can omit the computationally inefficient search over the online confidence sets, at the cost of an additional term dependent on the offline dataset's quality.

Here, offline coverage means Bellman-error coverage rather than uniform state-action coverage: the offline distribution must place mass where Bellman errors relevant to $\pi^\star$ can be detected. This is quantified by the single-policy concentrability coefficient $c^*_{\text{off}}(\mathcal{F})$ in Definition D.1, so poor offline data degrades the bound through this coefficient rather than through raw dataset size alone. For $N_{\text{off}} = \Theta(T)$, fixed $H, c^*_{\text{off}}(\mathcal{F})$, and a polylogarithmic covering-number logarithm, this extra term is $\widetilde{O}(\sqrt{T})$.

When compared to the results in Section 5.3, we see that the critic-cloning result has a critic contribution of order $\sqrt{T}$ rather than $T^{3/4}$, at the cost of score-matching and sampler-approximation terms. When the score model is learned online from scratch, these terms can dominate early training. Policy optimization has a weaker asymptotic rate in our theory, but it does not need to repeatedly score-match a moving critic-induced greedy policy.

Proof outline. We defer the proof to Appendix D.4. The proof rests on decomposing the regret into behavior cloning error $2H \sum_t B_t$ and critic estimation error, which are bounded individually. This suggests that if the behavior cloning error is well-controlled by e.g. having access to good score estimates and controlling the sampler error, learning diffusion policies by mimicking a critic recovers essentially the same sample efficiency as the critic learning procedure. To control cloning loss, we first add a small amount of Gaussian noise to the greedy action. This incurs a Wasserstein error controlled by the critic's action Lipschitz constant. We then compare the learned sampler with this noised target using a KL bound for DDPM or a Wasserstein stability bound for DDIM. The resulting prediction, initialization, and smoothing terms give Equation 5.6.

If a pretrained policy already assigns mass to several useful actions, cloning a single greedy action discards that information. We can instead reweight the pretrained distribution by the critic and fit the resulting stochastic target.

Example 5.18 (Fitting a critic-reweighted pretrained policy)**.** For a pretrained policy π_0, take the cloning target

$$q^f_{\tau,h}(da \mid s) \propto e^{f_h(s,a)/\tau} \pi_{0,h}(da \mid s), \qquad \tau > 0.$$

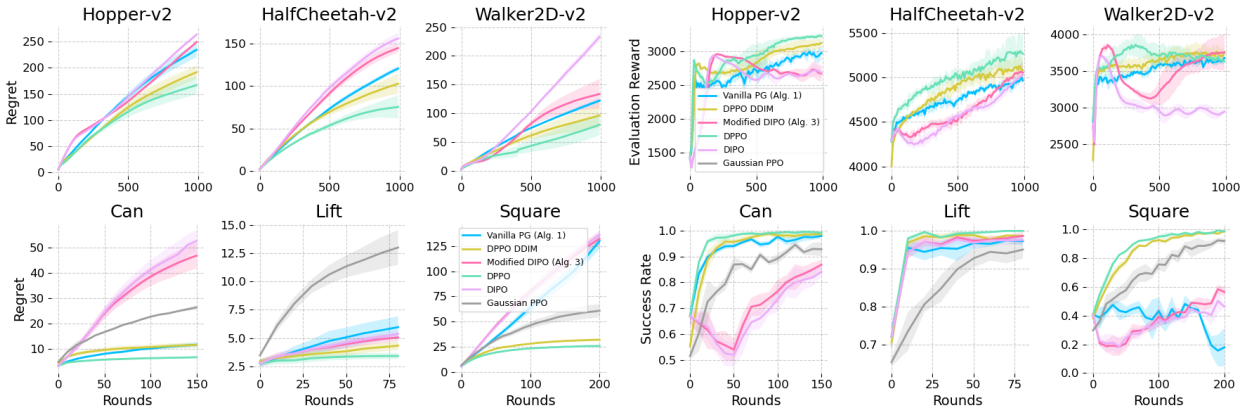


Figure 5.1: The six plots on the left column depict cumulative reference-return regret, defined in Appendix D.1, for each algorithm on each environment. The six plots on the right column depict the evaluation reward of each algorithm on each environment, where the Gym environments are evaluated on the actual reward function and the Robomimic environments are evaluated on the success rate, as in Ren et al. (2024a). 95% CI over 5 seeds.

This multiplies the relative probability of two actions by the exponential of their critic-value difference divided by τ. It therefore favors high-value actions while retaining the pretrained policy's support. If the pretrained law places at least $c_0(r/R)^m$ mass within radius r of each selected greedy action, Proposition D.5.4 bounds the per-stage target loss by $L_f r+\tau[\log(c_0^{-1})+m\log(R/r)]$. Choosing r and τ of order $T^{-1/2}$ gives $\widetilde{O}(\sqrt{T})$ cumulative target loss for fixed constants. The proposition also gives rejection sampling from this target with explicit proposal cost, which can grow exponentially in m. This changes the cloning target, while preserving the critic update and its regret analysis.

To turn this target into a learned diffusion actor, Appendix D.5.4 draws fresh synthetic actions at each state, weights their denoising losses by the critic, and searches a finite neural class. A finite softmax-network construction controls approximation error, and exhaustive optimization gives an explicit training guarantee. With per-round Wasserstein error and temperature both $t^{-1/2}$, the cumulative actor contribution is $\widetilde{O}(\sqrt{T})$ for fixed problem constants. This construction assumes all states are known and the pretrained sampler can be queried at each one. Its synthetic-sample, network-search, and denoising costs can be large, although they use no additional environment transitions. A small SEC for the deployed policies is still needed for the same total regret rate.

5.5. Numerical Experiments

To investigate how our theory holds up empirically, building on the work of Ren et al. (2024a), we provide experiments on the {Hopper, Walker2D, HalfCheetah} Gym (Brockman et al., 2016; Towers et al., 2025) and {Can, Square, Lift} Robomimic (Mandlekar et al., 2021) environments that investigate whether a representative array of diffusion policy algorithms exhibit sublinear empirical regret. Two are the baseline methods from prior work (DPPO, Ren et al., 2024a, and DIPO, Yang et al., 2023), and another is a PPO-trained Gaussian policy on Robomimic. We implement three more practical variants motivated by Algorithms 5.1–5.3. The first modifies DPPO to use the vanilla policy gradient update (with a step size an order of magnitude smaller). DPPO+DDIM uses PPO with a DDIM sampler, rather than the least-squares NPG update in Algorithm 5.2. The last modifies DIPO to use separate datasets for behavior cloning and critic estimation as described in Section 5.4. These are labeled Vanilla PG, DPPO DDIM, and Modified DIPO respectively. All experiments were run on two RTX 3090 GPUs, taking around a week. Hyperparameters are identical to Ren et al. (2024a), except for a learning rate an order of magnitude smaller for vanilla policy gradients.

The experiments are run on five seeds, $42, 43, 44, 45, 46$, in sequence from the seed used in Ren et al. (2024a). We report means and 95% confidence intervals. In line with Ren et al. (2024a), we employ a critic with generalized advantage estimation (Schulman et al., 2018) instead of using advantage estimates from Algorithm D.1. Further tables with CIs over seeds depicting the best value achieved by each algorithm, normalized regret statistics, and fits of the regret exponent T^{α} for each environment can be found in Appendix D.1.

Algorithm	Regret	Value	Sublinear	Exponent
DPPO	1.00	1.00	5/6	0.77
DPPO+DDIM	2.00	2.17	6/6	0.68
Vanilla PG (Alg. 5.1)	3.50	4.50	5/6	1.03
Modified DIPO (Alg. 5.3)	4.17	3.50	4/6	0.85
Gaussian PPO	4.33	4.33	3/3	0.51
DIPO	5.17	4.67	4/6	1.25

Table 5.3: Average rank across environments for plotted regret and best value. Number of environments rejecting $\alpha \geq 1$ after Holm correction. Largest late-half fitted exponent across environments.

Diffusion policies can exhibit sublinear empirical regret. Figure 5.1 and Table 5.3 show that most diffusion-policy methods exhibit sublinear empirical regret on most environments. The per-seed tests in Appendix D.1 reject an exponent of at least one in 15 of the 18 cells for Vanilla PG, DPPO+DDIM, and Modified DIPO. Not every method is sublinear on every environment. The modified DIPO variant improves over DIPO on most tasks by decoupling critic estimation from behavior cloning.

Policy optimization versus critic cloning. The results should not be read as contradicting Theorem 5.17. The bound contains score-matching and sampler-approximation terms. When the score model is learned online from scratch, as it is within the numerical experiments, these terms can dominate early training. Policy optimization has a weaker asymptotic rate in our theory, but it does not need to repeatedly score-match a moving critic-induced greedy policy. This may help explain why DPPO-style methods can outperform DIPO-style methods in our experiments, even though Theorem 5.17 has better dependence in T when the cloning and sampler errors are controlled. Table D.1 examines the empirical denoising losses, whose association with regret is inconclusive.

The advantage of diffusion policies. Gaussian PPO exhibits sublinear empirical regret on Robomimic, but DPPO and DPPO+DDIM attain higher reported values and lower normalized regret (Tables D.1–D.1). Their actor parameter counts are similar, with the diffusion actors using 4.27–7.10% more parameters (Table D.1). The diffusion actors also start from higher returns, so these comparisons do not isolate the effect of the policy class at matched initialization.

Practical takeaways. Our theory and experiments therefore suggest practical guidance for practitioners. We suggest that users employ policy optimization as a robust option when accurate score estimates are unavailable, and to employ behavior cloning when a good score model or a warm start allows the actor to track the changing critic, as then the improved scaling in T may become meaningful.

5.6. Conclusion, Limitations and Future Work

We provide sublinear regret guarantees for two representative diffusion-policy learning paradigms. For policy optimization, vanilla PG and DDIM NPG achieve $T^{3/4}$ regret under reset or mixing access and coverage, up to approximation and comparator errors, with NPG removing the explicit density-ratio dependence at the cost of score-feature and covariance conditions. Hybrid NPG preserves this exponent with importance-weighted Monte Carlo labels from logged behavior trajectories under finite weighted-return moments and improved mixture conditioning. For critic cloning, an optimistic critic plus diffusion behavior cloning gives a critic contribution of $O(H^2\sqrt{\beta T \mathsf{SEC}(\mathcal{F}, \Pi, T)})$ plus cumulative actor and sampler error, and offline Bellman-error coverage can replace optimism in the hybrid setting.

Extending the NPG argument to DDPM requires handling errors-in-features from the stochastic denoising path. Appendix D.1 gives a finite-sample imitation separation from flat Gaussian mixtures for structured product-mixture targets. Whether diffusion policies improve RL regret or sample efficiency over Gaussian policies and flat Gaussian mixtures under the same objective and data access remains open. More discussion can be found in Appendix D.1.

Our work leaves several compelling directions open. Matching the score estimate with the critic's action gradient, as in Psenka et al. (2024), may yield provable guarantees in settings where Q-score matching amounts to matching the softmax policy induced by the current critic. In such cases, it may be possible to integrate our analysis with that of Tan et al. (2025) to establish theoretical guarantees. An alternative option involves extending the idea of compatible function approximation (Sutton et al., 1999) using action gradients to reinterpret score-matching with respect to some policy derived from the current critic.

CHAPTER 6

STATISTICAL INFERENCE FOR BOOSTED REGRESSION AND ADDITIVE MODELS

This chapter is based on joint work with Haimo Fang and Giles Hooker (Fang et al., 2025) (Part I: Haimo Fang, Kevin Tan, and Giles Hooker, "Statistical Inference for Gradient Boosting Regression," Advances in Neural Information Processing Systems 38, 2025) and joint work with Haimo Fang, Jonathan Pipping, and Giles Hooker (Fang et al., 2026) (Part II: Haimo Fang, Kevin Tan, Jonathan Pipping, and Giles Hooker, "Statistical Inference for Explainable Boosting Machines," in Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 300, 2026).

[RESOLVE: CHAPTER 6 COMBINES THE TWO BOOSTING INFERENCE PAPERS (PART I: FANG, TAN, AND HOOKER, NEURIPS 2025; PART II: FANG, TAN, PIPPING, AND HOOKER, AISTATS 2026). PER EARLIER DISCUSSION, VERIFY WHETHER TO KEEP THIS CHAPTER AT SPOT 6 (BRIDGING ITERATIVE FUNCTIONAL APPROXIMATION AND STATISTICAL INFERENCE) OR OMIT IT IF YOU DECIDE TO RESTRICT SOLELY TO RL/BANDIT/POMDP CHAPTERS.]

Chapters 2–5 examined how function approximation, confidence sets, and sequential updates govern the statistical and computational complexity of policy learning. A closely parallel challenge arises in sequential ensemble methods for nonparametric regression and additive modeling: standard gradient boosting machines update residuals sequentially without contraction, causing the fitted trajectory to depend path-dependently on the entire sequence of trees and precluding tractable asymptotic distribution theory. While Boulevard regularization (Zhou and Hooker, 2022a) stabilizes sequential boosting via moving-average shrinkage so that predictions converge to kernel ridge regression, it compresses the recovered signal amplitude to $\frac{\lambda}{1+\lambda} f(\mathbf{x})$ and requires $O(n^3)$ matrix inversions for uncertainty quantification.

In this chapter, we study how algorithmic modifications to the sequential ensemble update alter both the limiting fixed point and the computational cost of valid statistical inference across two settings:

1. **Part I (Section 6.1): Gradient boosting regression with dropout and parallel leave-one-out updates.** We introduce two moving-average gradient boosting algorithms: Algorithm 6.1, which incorporates random tree dropout during residual construction, and Algorithm 6.2, which grows K trees in parallel per boosting round against leave-one-out residuals so that the limiting fixed point recovers $\frac{K}{K+1} f(\mathbf{x})$ (approaching full signal recovery as K increases). Coupled with Nyström and random-projection matrix sketching, these methods yield asymptotically valid confidence intervals, prediction intervals, and hypothesis tests in $O(n)$ time per query.

2. **Part II (Section 6.2): Explainable boosting machines for generalized additive models.** Extending moving-average stabilization to cyclic univariate tree updates over p features (Algorithm 6.3), we prove finite-sample contraction and componentwise asymptotic normality for each additive feature effect $f_k(x_k)$, and exploit feature quantile binning so that componentwise confidence intervals are computed in bin space with complexity independent of sample size n.

All proofs, sketching analyses, and supplementary empirical results are presented in Appendix E.

6.1. Part I: Statistical Inference for Gradient Boosting Regression

6.1.1. Introduction

Gradient boosting (Friedman, 2001), particularly through widely used implementations such as XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018), has become one of the most powerful and widely adopted methods for supervised learning, especially on tabular data. However, this flexibility and predictive accuracy come at a cost. Unlike their base learners – such as decision trees or linear models – boosting methods are typically far less interpretable, and uncertainty quantification for their predictions is far from straightforward.

While point estimates may suffice for ad hoc prediction tasks, they overlook randomness inherent in both the data and the algorithm itself. Uncertainty quantification therefore asks a central question: *if a new dataset were collected and models retrained, how different would the resulting predictions be?*

In this light, various methods have been proposed for uncertainty quantification in boosting. These include Langevin boosting (Tan et al., 2023; Ustimenko and Prokhorenkova, 2022; Malinin et al., 2021), k-nearest neighbors-based techniques (Brophy and Lowd, 2022), Gaussian graphical models (Chen and Wang, 2023), quantile regression approaches (Yin et al., 2023), and probabilistic prediction via natural gradients (Duan et al., 2020). However, most of these methods lack formal theoretical guarantees, with many relying primarily on heuristic justifications.

A principled route to uncertainty quantification is through statistical inference, e.g. via prediction intervals. However, this remains limited. In the Bayesian setting, Ustimenko et al. (2023) show randomized boosting converges to a kernel ridge regression and approximate Gaussian process posterior. Yet, they re-run the entire boosting procedure to generate even a single posterior sample. Malinin et al. (2021) suggest a virtual ensemble method, but do not provide formal guarantees.

In the frequentist setting, Zhou and Hooker (2022a) propose a regularization scheme yielding a central limit theorem for boosting via convergence to kernel ridge regression. However, they do not provide proper confidence or prediction intervals, and their method recovers at most half the true signal (rescaling is suggested, but this amplifies errors). Despite these limitations, their framework remains the only foundation for frequentist inference in boosting, which we extend here.

A broader discussion of related work connecting adaptive kernel representations of random forests (Wager et al., 2014; Athey et al., 2019; Mentch and Hooker, 2016) and gradient boosting (Zhou and Hooker, 2022a; Ustimenko et al., 2023) is provided in Fang et al. (2025). Motivated by this gap, we study frequentist inference for uncertainty quantification in boosting under supervised nonparametric regression: $y = \mathbf{f}(\mathbf{x}) + \epsilon$, with subgaussian noise σ^2. Our goal is to develop principled, statistically sound procedures for uncertainty quantification, outlined below.

Methodological contributions. In detail, we provide the following methodological contributions:

1. *Improvements via dropout:* We incorporate random dropout (Rashmi and Gilad-Bachrach, 2015) into the regularization procedure of Zhou and Hooker (2022a). This yields an improved procedure (Algorithm 6.1) that achieves provably better performance by up to a factor of 4 in the asymptotic relative efficiency (ARE), and increased signal recovery by up to a factor of 2. By tuning the dropout rate in Algorithm 6.1, we smoothly bridge the vanilla Boulevard method (Zhou and Hooker, 2022a) and a modified Random Forest (Breiman, 2001) that draws observation subsamples without replacement and considers the full covariate space at each split.

2. *Parallel boosting:* We construct a novel leave-one-out procedure for parallel boosting (Algorithm 6.2) that, when coupled with the above regularization procedure, improves in the ARE by at least a factor of 4, and enjoys increased signal recovery by at least a factor of 2.

3. *Prediction, confidence, and reproduction intervals:* We leverage the central limit theorems our procedures enjoy to construct a variety of procedures for statistical inference. First, we conduct predictive inference via constructing prediction intervals on (potentially unseen) labels $\mathbf{y}$. We then construct confidence intervals for the underlying ground truth function $\mathbf{f}$. Lastly, to quantify uncertainty within the learning algorithm, we provide reproduction intervals (Zhou and Hooker, 2022a) for another booster $\widehat{\mathbf{f}}$ trained on an independent realization of the training set.

4. *Hypothesis testing:* We construct a chi-squared hypothesis test for variable importance, extending the work of Mentch and Hooker (2016) for random forests to the boosting setting.

5. *Computational efficiency:* Unlike prior approaches to statistical inference for random forests and boosting, we utilize the Nyström approximation of Musco and Musco (2017). This allows our tests and intervals to scale linearly with the number of data points.

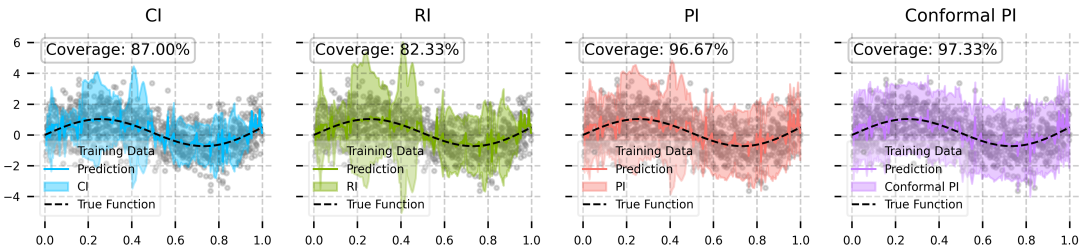


Figure 6.1: Demonstration of confidence, prediction, and reproduction intervals on $\mathbf{f}(x) = \sin 2\pi x + \frac{1}{2}x^2$. Conformal baseline. 200 trees, learning rate 0.6, depth 8, subsampling and dropout 0.6.

6.1.2. Setup

Consider boosting for nonparametric regression with squared error loss. Given random features $\mathbf{x} \in [0,1]^d$, labels $y \in \mathbb{R}$, and noise $\epsilon \sim \text{SubG}(0, \sigma^2)$, say there exists some function $\mathbf{f}$ such that $y = \mathbf{f}(\mathbf{x}) + \epsilon$. The learner is given a training set $(\mathbf{X}^{\mathcal{D}}, \mathbf{y}^{\mathcal{D}})$ of size n, and learns an estimate $\widehat{\mathbf{f}}$ of $\mathbf{f}$ by minimizing the mean squared error (MSE) $\frac{1}{n}\sum_{i=1}^{n}(\widehat{\mathbf{f}}(\mathbf{x}_i) - y_i)^2$ over the training set. It is customary to analyze how $\widehat{\mathbf{f}}$ converges to $\mathbf{f}$, or in other words, the generalization error of $\widehat{\mathbf{f}}$ on a test set $(\mathbf{X}^{\mathcal{D}_{\text{test}}}, \mathbf{y}^{\text{test}})$. Subscripts $\widehat{\mathbf{f}}_n^{(b)}$ indicate the estimated predictor trained on n datapoints for b boosting rounds, where we omit dependencies on b and n whenever possible.

Regularized stochastic gradient boosting. Zhou and Hooker (2022a) propose a regularization procedure called Boulevard that recovers a central limit theorem for boosting. They do so by showing that the resulting regularized procedure converges to a kernel ridge regression. At each iteration

$$b = 1, ..., B-1, \text{ instead of } \widehat{\mathbf{f}}^{(b+1)} \leftarrow \widehat{\mathbf{f}}^{(b)} + \lambda \mathbf{t}^{(b)}, \text{ they update } \widehat{\mathbf{f}}^{(b+1)} \leftarrow \tfrac{b-1}{b}\widehat{\mathbf{f}}^{(b)} + \tfrac{\lambda}{b}\mathbf{t}^{(b)}$$

for the current function estimate $\widehat{\mathbf{f}}^{(b)}$, built tree $\mathbf{t}^{(b)}$, and learning rate $\lambda \in [0, 1)$. As this is equivalent to updating $\widehat{\mathbf{f}}^{(b+1)} \leftarrow \frac{\lambda}{b}\sum_{i=1}^{b}\mathbf{t}^{(i)}$, the resulting ensemble is an average of trees. This observation, along with assumptions on tree adaptivity, allows them to prove that Boulevard predictions are asymptotically normal, though they do not construct confidence or prediction intervals. Additionally, this regularization comes at a cost. It turns out that $\widehat{\mathbf{f}}_n^{(b)}$ converges only to $\frac{\lambda}{1+\lambda}\mathbf{f} \leq \mathbf{f}/2$ as $n, b \to \infty$, and so *Boulevard recovers at most half the signal.* Still, this is the only method available for frequentist inference for gradient boosting, and so we seek to improve upon it in this chapter.

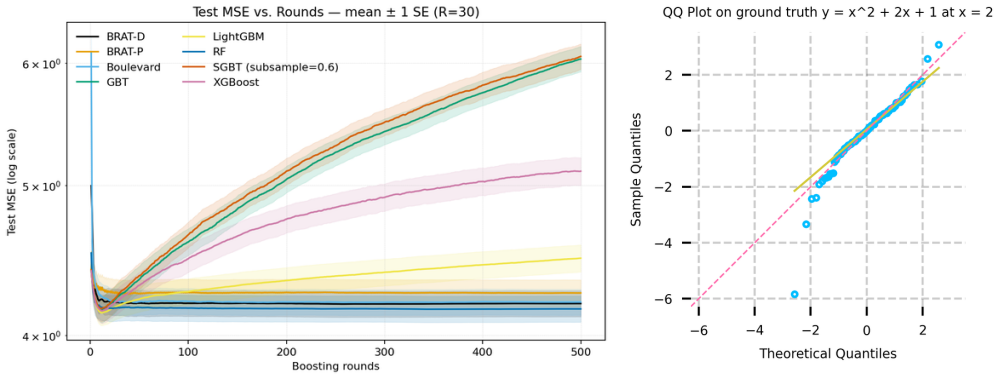


Figure 6.2: An added benefit of Boulevard regularization is that it renders early stopping unnecessary. Because the procedure converges to a fixed point, one can (in theory) boost indefinitely without encountering overfitting. It also produces asymptotically normal predictions (see the Q-Q plot).

Regression trees and tree structure. We are primarily concerned with boosting algorithms where we use regression trees as weak learners. We introduce a formulation of a regression tree below.

Definition 6.1 (Regression trees)**.** A regression tree $\mathbf{t}_n$ trained on n datapoints segments the covariate space $[0,1]^d$ into a partition of hyper-rectangles $\{A_i\}_{i=1}^m$. When $A(\mathbf{x})$ for the rectangle containing $\mathbf{x}$, and $s_{n,j}(\mathbf{x})$ is the frequency at which training point $\mathbf{x}_j$ and test point $\mathbf{x}$ share a leaf:

$$s_{n,j}(\mathbf{x}) = \frac{\mathbf{1}(\mathbf{x}_j \in A(\mathbf{x}))}{\sum_{k=1}^n \mathbf{1}(\mathbf{x}_k \in A(\mathbf{x}))}, \text{ the regression tree } \mathbf{t}_n \text{ predicts } \mathbf{t}_n(\mathbf{x}) = \sum_{j=1}^n s_{n,j}(\mathbf{x}) \cdot y_j.$$

As such, a tree can be thought of as an adaptive nearest neighbor method (Athey et al., 2019; Friedberg et al., 2021), where $s_{n,j}(\mathbf{x})$ yields the influence of datapoint $\mathbf{x}_j$ on the tree's prediction at test point $\mathbf{x}$. This motivates the definition of a structure vector and matrix from Zhou and Hooker (2022a):

Definition 6.2 (Structure vectors and matrices)**.** Let $\mathbf{t}_n$ be a tree trained on the input $(\mathbf{X}_n, \mathbf{y}_n)$. For an arbitrary point $\mathbf{x}$, let $\mathbf{s}_n(\mathbf{x}) = (s_{n,1}(\mathbf{x}), ..., s_{n,n}(\mathbf{x}))^\top$ be the structure vector of $\mathbf{x}$. As such, we say that the matrix $\mathbf{S}_n = (s_{n,j}(\mathbf{x}_i))_{i,j=1}^n$ with rows $\mathbf{s}_n(\mathbf{x}_i)^\top$ is the structure matrix of $\mathbf{t}_n$.

The structure matrix is a kernel matrix. A natural idea is to perform kernel ridge regression (KRR):

$$\widehat{\mathbf{f}}_n(\mathbf{x}) = \langle \mathbf{r}_n(\mathbf{x}), \mathbf{y}_n \rangle, \ \mathbf{k}_n(\mathbf{x}) = \mathbb{E}[\mathbf{s}_n(\mathbf{x})], \ \mathbf{K}_n = \mathbb{E}[\mathbf{S}_n], \text{ and } \mathbf{r}_n(\mathbf{x})^\top = \mathbf{k}_n(\mathbf{x})^\top (\lambda^{-1}\mathbf{I} + \mathbf{K}_n)^{-1}.$$

Zhou and Hooker (2022a) show Boulevard regularization converges to a kernel ridge regression $\widehat{\mathbf{f}}_n(\mathbf{x}) = \langle \mathbf{r}_n(\mathbf{x}), \mathbf{y}_n \rangle$ as $b \to \infty$. However, as $\lambda \in (0, 1]$, this converges to at most half the signal. We later propose two algorithms in Algorithms 6.1 and 6.2, that converge to $\langle \mathbf{r}_n^D(\mathbf{x}), \mathbf{y}_n \rangle = \mathbf{k}_n(\mathbf{x})^\top (\lambda^{-1}\mathbf{I} + q\mathbf{K}_n)^{-1}\mathbf{y}_n$ and $\langle \mathbf{r}_n^P(\mathbf{x}), \mathbf{y}_n \rangle = \mathbf{k}_n(\mathbf{x})^\top (\mathbf{I} + (K-1)\mathbf{K}_n)^{-1} K\mathbf{y}_n$ respectively, achieving increased signal recovery. As a final note, to accommodate training $\mathbf{t}_n(\mathbf{x}, \mathcal{G})$ on a subsample $\mathcal{G} \subseteq \{1, ..., n\}$, we write $s_{n,j}(\mathbf{x}; \mathcal{G}) = \mathbf{1}(\mathbf{x}_j \in A, j \in \mathcal{G}) / \sum_{k=1}^n \mathbf{1}(\mathbf{x}_k \in A, k \in \mathcal{G})$, and $\mathbf{t}_n(\mathbf{x}; \mathcal{G}) = \sum_{j=1}^n s_{n,j}(\mathbf{x}; \mathcal{G}) \cdot y_j$. We suppress the dependence on subsamples when possible.

Integrity & adaptivity. We inherit the following two assumptions from Boulevard regularization:

Assumption 6.3 (Structure-value isolation)**.** $(\mathbf{t}_n^{(b)})_{b=1}^B$ has $\mathbf{s}_n^{(b)}(\mathbf{x}) \perp\!\!\!\perp \mathbf{y}_n$ for all $b = 1, ..., B$.

Assumption 6.4 (Non-adaptivity)**.** There exists a probability measure $\mathcal{Q}_n$ on the space of tree structures and some b' so the tree structure at iteration $b = b', ..., B$ is an independent draw from $\mathcal{Q}_n$.

These are required for theoretical guarantees. In practice, the first can be implemented via a stronger form of honesty (Wager and Athey, 2017) that we call *integrity*, refitting leaf values on an independent dataset after boosting. One can also randomly sample splits, or get them from a random forest trained on an independent sample. The second only requires non-adaptivity after a finite number of iterations, which may occur naturally (Zhou and Hooker, 2022a). To enforce it, one can boost normally for a fixed number of steps and then sample tree structures from earlier iterations. Still, numerical experiments indicate that our algorithms perform well even without these adjustments.

6.1.3. Algorithms

To address the limitations of Boulevard regularization while retaining its statistical guarantees, we propose two improvements that achieve enhanced signal recovery and improved sample complexity:

1. **Algorithm 6.1:** By incorporating random dropout (Rashmi and Gilad-Bachrach, 2015) in the ensemble.

2. **Algorithm 6.2:** By growing trees in parallel within each round with a leave-one-out procedure.

The intuition is as follows. Surprisingly, increasing the dropout probability allows one to recover more of the signal. This is because each tree in the current ensemble is effectively downweighted during residual construction, and the new tree is fit on more of the signal. On the other hand, as the number of trees grown in parallel within each round grows, the resulting ensemble increasingly resembles the procedure of boosting random forests in sequence – allowing each round to recover more signal than a single tree can. We present both algorithms and explain them in detail below.

Algorithm 6.1 Moving-Average Gradient Boosting with Dropout

Input: Features $\mathbf{X}$, labels $\mathbf{y}$, dropout rate $p = 1-q$, subsample rate ξ, learning rate λ, boosting rounds B. Set $\widehat{\mathbf{f}}^{(0)}, \mathbf{t}^{(0)} \leftarrow 0$.
for $b = 1, ..., B-1$ **do**
  Subsample $\mathcal{S}_b \subseteq \{0, ..., b-1\}$ w.p. q.
  Subsample data $\mathcal{G}_b \subseteq \{1, ..., n\}$ w.p. ξ.
  $z_i \leftarrow y_i - \frac{\lambda}{b} \sum_{s \in \mathcal{S}_b} \mathbf{t}^{(s)}(\mathbf{x}_i)$.
  Construct tree $\mathbf{t}^{(b)}$ on $(\mathbf{x}_i, z_i)_{i \in \mathcal{G}_b}$.
  $\widehat{\mathbf{f}}^{(b+1)} \leftarrow \frac{b-1}{b}\widehat{\mathbf{f}}^{(b)} + \frac{\lambda}{b}\mathbf{t}^{(b)} = \frac{\lambda}{b}\sum_{i=1}^{b} \mathbf{t}^{(i)}$.
end for
Return final predictor $\frac{1+\lambda q}{\lambda}\widehat{\mathbf{f}}^{(B)}$.

Algorithm 6.2 Parallel Moving-Average Gradient Boosting with Leave-One-Out Residuals

Input: Features $\mathbf{X}$, labels $\mathbf{y}$, subsample ξ, trees/round K, rounds B. $\widehat{\mathbf{f}}^{(0)}, \mathbf{t}^{(0,k)} \leftarrow 0$.
Fit $\widehat{\mathbf{f}}^{(1)}, (\mathbf{t}^{(1,k)})_{k=1}^{K}$ with K boosting steps.
for $b = 2, ..., B-1$ **do**
  for $k = 1, ..., K$ **in parallel do**
    Subsample $\mathcal{G}_{b,k} \subseteq \{1, ..., n\}$ w.p. ξ.
    $z_{i,k} \leftarrow y_i - \frac{1}{b-1} \sum_{s=1}^{b-1} \sum_{l \neq k} \mathbf{t}^{(s,l)}(\mathbf{x}_i)$
    Construct tree $\mathbf{t}^{(b,k)}$ on $(\mathbf{x}_i, z_{i,k})_{i \in \mathcal{G}_{b,k}}$.
  end for
  Define $\widehat{\mathbf{f}}^{(b+1)} \leftarrow \frac{1}{b} \sum_{s=1}^{b} \sum_{k=1}^{K} \mathbf{t}^{(s,k)}$.
end for
Return final predictor $\widehat{\mathbf{f}}^{(B)}$.

Random dropout. Unlike vanilla boosting, Algorithm 6.1 computes residuals with both data (Friedman, 2002) and ensemble subsampling (Rashmi and Gilad-Bachrach, 2015). At each round $b = 1, ..., B-1$, each tree is subsampled with a dropout rate of $p \in [0, 1)$ (equivalently, with

probability $q = 1 - p$). We write $\mathcal{S}_b \subseteq \{0, ..., b-1\}$ for the collection of subsampled tree indices at each round b. Likewise, data indices $\mathcal{G}_b \subseteq \{1, ..., n\}$ are subsampled with probability $\xi \in (0, 1]$.

As a result, each tree $\mathbf{t}^{(b)}$ is trained on the dataset $(\mathbf{x}_i, z_i)_{i \in \mathcal{G}_b}$, where the residuals are given by $z_i = y_i - \frac{1}{b} \sum_{s \in \mathcal{S}} \mathbf{t}^{(s)}(\mathbf{x}_i)$. Within the residual computation, we intentionally divide the sum of the subsampled ensemble's predictions by b instead of $|\mathcal{S}|$. This is done so each new tree is fit on more of the signal, allowing for increased signal recovery when applying Boulevard regularization.

We later show in Section 6.1.5 that the predictions of Algorithm 6.1 converge to a kernel ridge regression that enjoys a central limit theorem. Informally, for any test point $\mathbf{x}$, we have that:

$$\widehat{\mathbf{y}}_n^{(b)} \xrightarrow[b \to \infty]{a.s.} \left(\lambda^{-1}\mathbf{I} + q\mathbb{E}\left[\mathbf{S}_n\right]\right)^{-1} \mathbb{E}\left[\mathbf{S}_n\right] \mathbf{y}_n, \quad \|\mathbf{r}_n^D(\mathbf{x})\|_2^{-1} \left(\widehat{\mathbf{f}}_n^D(\mathbf{x}) - \tfrac{\lambda}{1+\lambda q}\mathbf{f}(\mathbf{x})\right) \xrightarrow[n \to \infty]{d} \mathcal{N}(0, \sigma^2),$$

for a kernel matrix induced by the tree ensemble $\mathbf{K}_n = \mathbb{E}[\mathbf{S}_n]$, and limit $\widehat{\mathbf{f}}_n^D = \lim_{b \to \infty} \widehat{\mathbf{f}}_n^{(b)}$ where $\widehat{\mathbf{f}}_n^D(\mathbf{x}) = \langle \mathbf{r}_n^D(\mathbf{x}), \mathbf{y}_n \rangle = \mathbf{k}_n(\mathbf{x})^\top (\lambda^{-1}\mathbf{I} + q\mathbf{K}_n)^{-1}\mathbf{y}_n$. $\|\mathbf{r}_n\|_2$ is the norm of kernel weights. Algorithm 6.1 encompasses two important special cases. When $\lambda = 1$ and $p \to 1$, this yields a random forest. When $p = 0$, Algorithm 6.1 corresponds to the original Boulevard algorithm (Zhou and Hooker, 2022a). Tuning the dropout parameter p allows the user to interpolate between the two, while improving over Zhou and Hooker (2022a) by up to a factor of $\frac{1+\lambda}{1+\lambda q} \in (1, 2]$ in signal recovery.

Parallel boosting. Algorithm 6.2 integrates a novel leave-one-out procedure and Boulevard regularization into the parallel boosting framework (Long and Servedio, 2011; Karbasi and Larsen, 2024; da Cunha et al., 2024). The algorithm is warm-started by performing K vanilla boosting iterations in the first round. At each subsequent round $b = 2, ..., B-1$, we grow K trees $(\mathbf{t}^{(b,k)})_{k=1}^K$ in parallel on the K datasets $((\mathbf{x}_i, z_{i,k})_{i \in \mathcal{G}_{b,k}})_{k=1}^K$. These consist of independently subsampled data indices $\mathcal{G}_{b,k} \subseteq \{1, ..., n\}$, and residuals that are computed by leaving one "column" out (or alternatively, one tree out per round): $z_{i,k} = y_i - \frac{1}{b-1} \sum_{s=1}^{b-1} \sum_{l \neq k} \mathbf{t}^{(s,l)}(\mathbf{x}_i)$. The resulting predictor is given by the regularized boosting update: $\widehat{\mathbf{f}}^{(b+1)} \leftarrow \frac{b-1}{b}\widehat{\mathbf{f}}^{(b)} + \frac{1}{b} \sum_{k=1}^K \mathbf{t}^{(b,k)} = \frac{1}{b} \sum_{s=1}^b \sum_{k=1}^K \mathbf{t}^{(s,k)}$.

Likewise, we later show in Section 6.1.5 that the predictions of Algorithm 6.2 also converge to a

kernel ridge regression that enjoys a central limit theorem. Informally, for any test point $\mathbf{x}$:

$$\widehat{\mathbf{y}}_n^{(b)} \xrightarrow[b\to\infty]{a.s.} \left(\mathbf{I} + (K-1)\mathbb{E}\left[\mathbf{S}_n\right]\right)^{-1} K\mathbb{E}\left[\mathbf{S}_n\right]\mathbf{y}_n, \quad \|\mathbf{r}_n^P(\mathbf{x})\|_2^{-1}\left(\widehat{\mathbf{f}}_n^P(\mathbf{x}) - \mathbf{f}(\mathbf{x})\right) \xrightarrow[n\to\infty]{d} \mathcal{N}(0,\sigma^2),$$

with limit $\widehat{\mathbf{f}}_n^P = \lim_{b\to\infty}\widehat{\mathbf{f}}_n^{(b)}$ given by $\widehat{\mathbf{f}}_n^P(\mathbf{x}) = \langle \mathbf{r}_n^P(\mathbf{x}), \mathbf{y}_n\rangle = \mathbf{k}_n(\mathbf{x})^\top(\mathbf{I} + (K-1)\mathbf{K}_n)^{-1}K\mathbf{y}_n$. Algorithm 6.2 can be viewed as boosting a modified parallel backfitting algorithm (Breiman and Friedman, 1985)[24] that is warm-started with K vanilla boosting iterations.[25] When $K = 1, B \to \infty$, Algorithm 6.2 reduces to a random forest $\mathbb{E}\left[\mathbf{S}_n\right]\mathbf{y}_n$. On the other hand, Algorithm 6.2 reduces to vanilla boosting when $B = 1, K \to \infty$. Note that the CLT only holds for $1 < K < \infty$.

The attentive reader may observe that we do not divide the final predictions by K. This is because the combination of leaving one column out when fitting a tree and Boulevard averaging is sufficient to allow our procedure to stabilize.[26] Still, dividing by K yields a viable algorithm (deferred to future work) that can be thought of as boosting random forests each of K trees over B rounds, using the natural strategy of averaging all trees but the left-out column to form predictions.

6.1.4. Statistical Inference and Uncertainty Quantification for Boosting

We are now in a place to introduce our methods for conducting statistical inference for boosting, whose theoretical basis is provided below. Our methods rest on the aforementioned convergence to a kernel ridge regression and central limit theorems that we formally prove later in Theorems 6.6 and 6.11. In the meantime, we provide the following asymptotically valid procedures.

Estimation of asymptotic variance. The CLTs for Algorithms 6.1 and 6.2 imply that one can construct confidence intervals for $\mathbf{f}$ with the following normal approximations:

$$\lambda^{-1}(1+\lambda q)\widehat{\mathbf{f}}_n^D(\mathbf{x}) \overset{d}{\approx} \mathcal{N}\left(\mathbf{f}(\mathbf{x}), \lambda^{-2}(1+\lambda q)^2\|\mathbf{r}_n^D(\mathbf{x})\sigma^2\|_2^2\right), \quad \widehat{\mathbf{f}}_n^P(\mathbf{x}) \overset{d}{\approx} \mathcal{N}\left(\mathbf{f}(\mathbf{x}), \sigma^2\|\mathbf{r}_n^P(\mathbf{x})\|_2^2\right).$$

[24] This can be thought of as a *structured* variant of dropout (Zhao et al., 2022; Xin et al., 2020; Fan et al., 2020). Dropping one out of every K trees allows the regularization to scale in $1/K$.

[25] The initial K boosting iterations yield better convergence in practice, and do not hinder our theoretical arguments as the impact of the vanilla boosting iterations washes out when we take $B \to \infty$.

[26] Parallelized backfitting can be thought of as parallelized coordinate descent. Averaging over the coordinate updates is sufficient for convergence. Dividing by b, as in Boulevard averaging, is sufficient when $b \geq K$.

As such, it remains to estimate $\mathbf{r}_n^D, \mathbf{r}_n^P$, and σ^2. We first estimate $\mathbf{k}_n(\mathbf{x})$ by the sample averages

$$\widehat{\mathbf{k}}_n^D(\mathbf{x}) = \tfrac{1}{B}\sum_{b=1}^B s_{n,i}^{(b)}(\mathbf{x}, \mathcal{G}_b), \;\; \widehat{\mathbf{k}}_n^P(\mathbf{x}) = \tfrac{1}{BK}\sum_{b=1}^B \sum_{k=1}^K s_{n,i}^{(b,k)}(\mathbf{x}, \mathcal{G}_{b,k}), \tag{6.1}$$

where $s_{n,i}^{(b)}(\mathbf{x}; \mathcal{G}_b)$ is the fraction of subsampled datapoints in the same leaf $A_j^{(b)}$ as $\mathbf{x}$ over the trees. Likewise, we estimate $(\mathbf{K}_n)_{i,j}$, the fraction of trees where datapoints i and j fall in the same leaf:

$$(\widehat{\mathbf{K}}_n^D)_{i,j} = \tfrac{1}{B}\sum_{b=1}^B s_{n,j}^{(b)}(\mathbf{x}_i; \mathcal{G}_b), \;\; (\widehat{\mathbf{K}}_n^P)_{i,j} = \tfrac{1}{B}\sum_{b=1}^B s_{n,j}^{(b,k)}(\mathbf{x}_i; \mathcal{G}_{b,k}), \tag{6.2}$$

Our estimates of $\widehat{\mathbf{r}}_n$ are then given by plugging the above into the formula for $\mathbf{r}_n$:

$$\widehat{\mathbf{r}}_n^D(\mathbf{x})^\top = \widehat{\mathbf{k}}_n^D(\mathbf{x})^\top(\lambda^{-1}\mathbf{I} + q\widehat{\mathbf{K}}_n^D)^{-1}, \;\; \widehat{\mathbf{r}}_n^P(\mathbf{x})^\top = \widehat{\mathbf{k}}_n^P(\mathbf{x})^\top(\mathbf{I} + (K-1)\widehat{\mathbf{K}}_n^P)^{-1}K.$$

Lastly, we estimate σ^2 with the residuals $\widehat{\sigma}^2 = \frac{1}{n_{\text{cal}}}\sum_{i=1}^{n_{\text{cal}}}(y_i - \widehat{y}_i)^2$ on a hold-out calibration set. We describe a tweak later in Section 6.1.6 that uses the calibration set for increased robustness.

Confidence intervals for f. Putting everything together yields $1-\alpha$ confidence intervals for $\mathbf{f}$ (where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of the standard normal distribution):

$$\lambda^{-1}(1+\lambda q)\widehat{\mathbf{f}}_n^D(\mathbf{x}) \pm z_{1-\alpha/2}\lambda^{-1}(1+\lambda q)\widehat{\sigma}\|\widehat{\mathbf{r}}_n^D(\mathbf{x})\|_2, \;\; \widehat{\mathbf{f}}_n^P(\mathbf{x}) \pm z_{1-\alpha/2}\widehat{\sigma}\|\widehat{\mathbf{r}}_n^P(\mathbf{x})\|_2. \tag{6.3}$$

Prediction intervals for y. We can construct prediction intervals for $y|\mathbf{x}$ as follows:

$$\lambda^{-1}(1+\lambda q)\widehat{\mathbf{f}}_n^D(\mathbf{x}) \pm z_{1-\alpha/2}\lambda^{-1}(1+\lambda q)\widehat{\sigma}\sqrt{1+\|\widehat{\mathbf{r}}_n^D(\mathbf{x})\|_2^2}, \;\; \widehat{\mathbf{f}}_n^P(\mathbf{x}) \pm z_{1-\alpha/2}\widehat{\sigma}\sqrt{1+\|\widehat{\mathbf{r}}_n^P(\mathbf{x})\|_2^2}.$$

These prediction intervals have asymptotic pointwise coverage, conditional on the test point $\mathbf{x}$. That is, $\mathbb{P}(y \in \text{PI}(\mathbf{x})|\mathbf{x}) \to 1-\alpha$ as $n \to \infty$, which holds as a corollary of the CLTs we present in the following section. Note that this conditional coverage guarantee is a stronger guarantee than is typically possible with conformal inference (Barber et al., 2021). We compare our prediction intervals to conformal prediction intervals within our numerical experiments later in this section.

Variable importance tests. Suppose we wish to test for variable importance. Consider the possibility that there exists some subset of the feature space $[0,1]^d$ containing only $\check{d} < d$ features, and let $\boldsymbol{g} : [0,1]^{\check{d}} \to \mathbb{R}$ be the projection of $\mathbf{f}$ onto $\mathcal{L}^2([0,1]^{\check{d}})$. We test the possibility that $\mathbf{f}(\mathbf{x}) = \boldsymbol{g}(\mathbf{x})$ by proxy. Consider a split of the training dataset $(\mathbf{X}_n, \mathbf{y}_n)$ into $(\mathbf{X}_{n,1}, \mathbf{y}_{n,1}), (\mathbf{X}_{n,2}, \mathbf{y}_{n,2})$, where $\mathbf{X}_{n,1} \in \mathbb{R}^{n/2,d}, \mathbf{X}_{n,2} \in \mathbb{R}^{n/2,\check{d}}$. Let $\widehat{\mathbf{f}}_{n,1}$ be the boosting learner trained on $(\mathbf{X}_n, \mathbf{y}_n)$, and $\widehat{\mathbf{f}}_{n,2}$ be the same trained on $(\mathbf{X}_{n,2}, \mathbf{y}_{n,2})$. Given a hold-out dataset $(\mathbf{X}_m, \mathbf{y}_m)$, we test the null:

$$H_0 : \mathbf{f}(\mathbf{x}_j) = \boldsymbol{g}(\mathbf{x}_j) \text{ for all } j = 1, ..., m \text{ against } H_1 : \mathbf{f}(\mathbf{x}_j) \neq \boldsymbol{g}(\mathbf{x}_j) \text{ for some } j$$

by comparing $\widehat{\mathbf{f}}_{n,1}(\mathbf{x}_j)$ and $\widehat{\mathbf{f}}_{n,2}(\mathbf{x}_j)$. We exploit the CLTs our algorithms enjoy to do so. Write $\widehat{\mathbf{r}}_n^{D,P}(\mathbf{X}_m)^\top \in \mathbb{R}^{m \times n}$ for the matrix with rows $(\widehat{\mathbf{r}}_n^{D,P}(\mathbf{x}_l)^\top)_{l=1}^m$, where we write D, P within the superscript if the formula holds for both Algorithms 6.1 and 6.2. The difference in predictions

$$\widehat{\mathrm{d}}_m = (\widehat{\mathbf{r}}_{n,1}^{D,P}(\mathbf{X}_m) - \widehat{\mathbf{r}}_{n,2}^{D,P}(\mathbf{X}_m))^\top \mathbf{y} \xrightarrow{d} \mathcal{N}_m(\mathbf{0}, \sigma^2 \widehat{\boldsymbol{\Xi}}_n) \text{ is multivariate normal under the null,}$$

with covariance matrix $\sigma^2 \widehat{\boldsymbol{\Xi}}_n = \sigma^2 (\widehat{\mathbf{r}}_{n,1}^{D,P}(\mathbf{X}_m) - \widehat{\mathbf{r}}_{n,2}^{D,P}(\mathbf{X}_m))^\top (\widehat{\mathbf{r}}_{n,1}^{D,P}(\mathbf{X}_m) - \widehat{\mathbf{r}}_{n,2}^{D,P}(\mathbf{X}_m))$ estimated by plugging in $\widehat{\sigma}^2$. As such, we can conduct a chi-squared test with test statistic

$$\widehat{\sigma}^{-2} \widehat{\mathrm{d}}_m^\top \widehat{\boldsymbol{\Xi}}_n^{-1} \widehat{\mathrm{d}}_m \sim \chi^2_m \text{ under the null, rejecting the null if } \widehat{\sigma}^{-2} \widehat{\mathrm{d}}_m^\top \widehat{\boldsymbol{\Xi}}_n^{-1} \widehat{\mathrm{d}}_m > \chi^2_{m,1-\alpha}.$$

The test as presented runs in $O(n^3)$ time. In Appendix E.1, we present an accelerated version of the test that runs in $O(ns(r+s) + r^3)$ time, where s is the number of points subsampled for Nyström approximation and r is the number of test points subsampled.

Reproduction intervals for an independent realization of $\widehat{\mathbf{f}}$ (obtained by scaling the confidence interval half-width by $\sqrt{2}$) and Nyström matrix sketching approximations (Williams and Seeger, 2000; Musco and Musco, 2017) that reduce kernel precomputation from $O(n^3)$ to $O(ns^2)$ time and per-query inference to $O(s^2)$ time with $s = \widetilde{O}(d_{\text{eff}}^{\mu})$ (Appendix E.1) are detailed in Fang et al. (2025).

6.1.5. Theoretical Guarantees

As promised, we now formally present guarantees for convergence and asymptotic normality for Algorithms 6.1 and 6.2. Consider the (Lipschitz) nonparametric regression model introduced below:

Assumption 6.5 (Lipschitz Nonparametric Regression)**.** Let $y = f(\mathbf{x}) + \epsilon$, where $\mathbf{x}$ has density μ satisfying $0 < c_1 \leq \mu(\mathbf{x}) \leq c_2 < \infty$ on its support. Further assume f is α-Lipschitz continuous on $\text{supp}(\mu)$, and noise ϵ is sub-Gaussian with variance proxy σ^2.

Finite-sample convergence to KRR. We now show that our algorithms achieve finite-sample convergence to kernel ridge regression as the number of boosting rounds increases. At first glance, this is surprising – greedy split selection and sequential ensemble construction create a highly nonlinear and time-dependent mapping. Fortunately, the regularization procedure of Zhou and Hooker (2022a) provides a way forward. For the special case of $p = 0$ within Algorithm 6.1, they show convergence to a kernel ridge regression via the stochastic contraction framework of Almudevar (2022), utilizing Assumptions 6.3 (structure–value isolation) and 6.4 (non-adaptivity) to ensure stagewise stability. A key novelty within our analysis lies in showing that this convergence extends to Algorithms 6.1 and 6.2, even with the added complexity introduced by dropout and parallelism respectively. Despite these added sources of stochasticity and dependence, our use of the same regularization mechanism ensures that both algorithms inherit the stability required for convergence:

Theorem 6.6 (Finite Sample Convergence to KRR)**.** *For fixed* $\mathbf{X}, \mathbf{y}$*, under Assumptions 6.3, 6.4, and 6.5,*

$$\widehat{\mathbf{y}}_b \overset{a.s.}{\to} \left(\lambda^{-1}\mathbf{I} + q\mathbb{E}\left[\mathbf{S}_n\right]\right)^{-1} \mathbb{E}\left[\mathbf{S}_n\right] \mathbf{y} \text{ for Alg. 6.1, } \widehat{\mathbf{y}}_b \overset{a.s.}{\to} \left(I + (K-1)\mathbb{E}\left[\mathbf{S}_n\right]\right)^{-1} K\mathbb{E}\left[\mathbf{S}_n\right] \mathbf{y} \text{ for Alg. 6.2,}$$

as $b \to \infty$*, where* $q = 1 - p$ *is one minus the dropout probability,* K *is the number of trees grown in parallel, and* $\mathbf{S}_n$ *is the kernel matrix induced by the tree structures within the ensemble.*

The proof, deferred to Appendix E.1, departs from Zhou and Hooker (2022a) in two crucial ways. Within Algorithm 6.1, dropout injects unbounded variance into the ensemble updates. To control

this, we introduce the hard truncation function $\Gamma_M(y) = \text{sign}(y)\min\{M, |y|\}$ into each residual update, $\widehat{\mathbf{y}}_b = \frac{b-1}{b}\widehat{\mathbf{y}}_{b-1} + \frac{\lambda}{b}\mathbf{S}_b\big(\mathbf{y} - \Gamma_M(\widetilde{\mathbf{y}}_b)\big)$, and then show that the probability of the partial ensemble escaping this cap vanishes as $b \to \infty$. Within Algorithm 6.2, truncating $\widetilde{\mathbf{y}}_{b,k} = \widehat{\mathbf{y}}_{b-1,K} - \Gamma_M(\frac{1}{b-1}\sum_{g=1}^{b-1}\widehat{\mathbf{t}}_{b,k})$ alone is insufficient. We further introduce a delay mechanism, requiring that $\mathbf{t}_{b,k}$ does not rely on $\mathbf{t}_{b,k-1}$, allowing us to apply Theorem E.1 and parallelize tree training.

A central limit theorem. Having established almost-sure convergence to fixed points, we now examine the asymptotic distribution of our estimators to justify the statistical procedures introduced in Section 6.1.4. We seek a central limit theorem for the learner $\widehat{\mathbf{f}}_n^{D,P}$, trained on random $\mathbf{X}_n, \mathbf{y}_n$, demonstrating that the predictions $\widehat{\mathbf{f}}_n^{D,P}(\mathbf{x})$ are asymptotically normal with mean $\mathbf{f}(\mathbf{x})$ as $n \to \infty$. To do so, we inherit the following assumptions from Zhou and Hooker (2022a): two on the leaves to control the norm of the KRR weight vector $\mathbf{r}_n^{D,P}$, and a restriction on the tree distribution space:

Assumption 6.7 (Bounded Leaf Diameter)**.** Write $\mathsf{diam}(A) = \sup_{x,y\in A}\|x - y\|$. For any leaf A in a tree with structure $q \in Q_n$, we need $\sup_{A\in q}\mathsf{diam}(A) = O(d_n)$, where $d_n = O(n^{-1/(d+1)})$.

Assumption 6.8 (Increased Minimal Leaf Size)**.** For any $\nu > 0$, $v_n = n^{-\frac{d+1}{d+2}+\nu} < n^{-\frac{d}{d+1}} = O(d_n^d)$.

Assumption 6.9 (Restricted Tree Support)**.** The cardinality of the tree space Q_n is bounded by $O(n^{-1}\exp(0.5n^{1/(d+2)-\nu} - n^\alpha))$, for some small $\alpha > 0$.

Intuitively, Assumption 6.8 prevents leaves from becoming too small too quickly, which in turn controls the maximal coordinate of the weight vectors across random samples. Assumption 6.9, guarantees that the complexity of possible tree partitions does not explode with n. We also make assumptions on the regularity of tree splits in line with the notion of α-regularity in Athey et al. (2019):

Assumption 6.10 (Median Splitting Rules)**.** The trees in Algorithm 6.2 are split at the medians.

We defer the details to Definition E.1. Under this splitting rule, the point-to-point collision probability is well-controlled, i.e. to say none of the leaves are big enough to contain most of the points. These conditions allow for uniform control over both leaf sizes and tree complexity, yielding the

result:

Theorem 6.11 (Central Limit Theorem for Predictions)**.** *Let* $\mathbf{x} \in [0,1]^d$*,* $q \in (0,1]$*,* $K > 1$*. As* $n \to \infty$*,*

$$\left\|\mathbf{r}_n^D\right\|_2^{-1}\left(\widehat{\mathbf{f}}_n^D(\mathbf{x}) - \lambda^{-1}(1+\lambda q)\mathbf{f}(\mathbf{x})\right) \xrightarrow{d} \mathcal{N}(0,\sigma^2), \quad \left\|\mathbf{r}_n^P\right\|_2^{-1}\left(\widehat{f}_n^P(\mathbf{x}) - f(\mathbf{x})\right) \xrightarrow{d} \mathcal{N}(0,\sigma^2).$$

We prove this in two stages. In the first, with proof deferred to Appendix E.1, we show asymptotic normality of the predictions around the noiseless KRR predictions $\langle \mathbf{r}_n^{D,P}(\mathbf{x}), \mathbf{f}(\mathbf{X}_n)\rangle$, conditional on the training data $\mathbf{X}_n, \mathbf{y}_n$ and test point $\mathbf{x}$, by applying the Lindeberg-Feller CLT to $\langle \mathbf{r}_n^{D,P}(\mathbf{x}), \varepsilon\rangle$. Next, we show that $\langle \mathbf{r}_n^{D,P}(\mathbf{x}), \mathbf{f}(\mathbf{X}_n)\rangle$ converges to the underlying ground truth function at a sufficiently fast rate in Appendices E.1 and E.1. For Algorithm 6.1, although $\mathbf{r}_n^D$ now involves $[\frac{1}{\lambda}\mathbf{I} + q\mathbf{K}_n]^{-1}$, this still preserves the rate $\|\mathbf{r}_n\|_2 \asymp n^{-1/(2(d+1))}$, allowing us to show $\langle \mathbf{r}_n, f(\mathbf{X}_n)\rangle - \frac{\lambda q}{1+\lambda q}\mathbf{f}(\mathbf{x}) = o_p(\|\mathbf{r}_n\|_2)$. The Lindeberg–Feller verification then proceeds identically to the non-dropout case. However, to give a CLT for Algorithm 6.2, controlling the influence from distant training points is challenging, though we show that Assumption 6.10 is sufficient to do so.

Understanding the result. The rate of convergence of the CLT in Theorem 6.11 depends on the norm of $\mathbf{r}_n^{D,P}$. This is controlled as follows, yielding generalization bounds for Algorithms 6.1 and 6.2:

Lemma 6.12 (Rate of Convergence)**.** *Let* $\mathbf{B}_n := \{i : \|\mathbf{x} - \mathbf{x}_i\| \le d_n\}$ *be the points within distance* d_n *from test point* $\mathbf{x}$*. If* $|\mathbf{B}_n| = \Omega\left(n \cdot d_n^d\right)$*, then* $\|\mathbf{k}_n\|_2 = \Theta(n^{-\frac{1}{2}\frac{1}{d+1}})$*,* $\left\|\mathbf{r}_n^{D,P}\right\|_2 = \Theta(n^{-\frac{1}{2}\frac{1}{d+1}})$*.*

As immediate consequences of Theorem 6.11 and Lemma 6.12 (detailed in Fang et al. (2025) and Appendix E.1), both estimators achieve $O(\sigma^2 n^{-1/(d+1)})$ asymptotic risk, nonasymptotic PAC sample complexity, almost-sure $(1-\alpha)$ pointwise interval coverage as $n, b \to \infty$, and an asymptotic relative efficiency improvement over vanilla Boulevard (Zhou and Hooker, 2022a) of $(\frac{1+\lambda}{1+\lambda q})^2 \in [1,4]$ for Algorithm 6.1 and $(\frac{1+\lambda}{\lambda})^2 \in [4,\infty)$ for Algorithm 6.2.

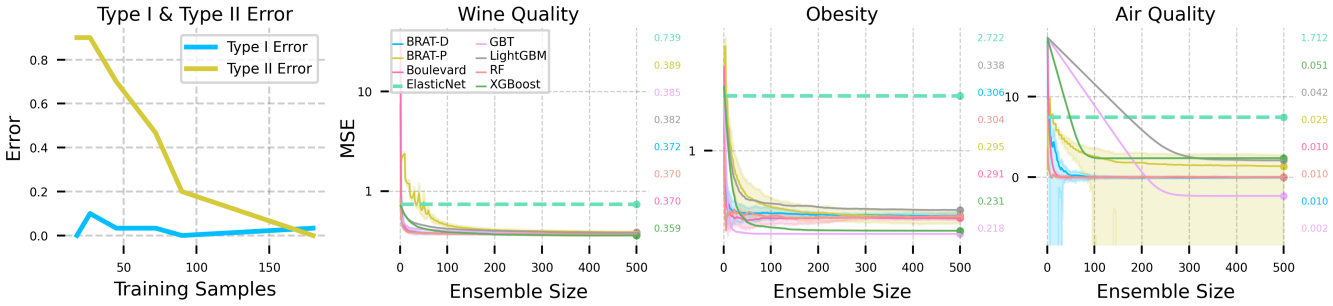


Figure 6.3: **Left:** Type I and Type II error of the hypothesis test for variable importance against training set size. Test set is of the same size. Error rates computed over 30 trials. **Right:** Comparison of MSE achieved by various machine learning algorithms on different datasets, with hyperparameters tuned for each by Optuna. Shaded area depicts two standard deviations over 5 trials.

6.1.6. Numerical Experiments

We have presented a rich theory in the previous section. Alongside convergence to kernel ridge regression and a central limit theorem for predictions, we provide asymptotic risk bounds and nonasymptotic PAC guarantees, and show asymptotic coverage for our intervals. In light of this, we present a series of numerical experiments to justify our algorithms and methods empirically.

Predictive accuracy. The first, and most natural, thing to do is to examine the performance of our algorithms against a handful of competitors in terms of test MSE on nine datasets from the UCI Machine Learning Repository in Figure 6.3. Our competitors include popular methods such as XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017), as well as random forests and vanilla boosting. We also compare to vanilla Boulevard (Zhou and Hooker, 2022a), and an elastic net regression as a baseline. All hyperparameters were tuned with Optuna (Akiba et al., 2019), reported in Appendix E.1. There is no clear winner in general, but XGBoost is a consistently strong performer, and random forests are clearly more suited to some datasets than others. Neither Algorithm 6.1 nor Algorithm 6.2 consistently outperform each other, but Algorithm 6.2 occasionally exhibits instability on some datasets. Regardless, both of our algorithms are competitive in terms of final MSE and rate of convergence, and exhibit the ability to be tuned to be closer to boosting (Wine Quality) or random forests (Air Quality), providing increased flexibility by interpolating between the two.

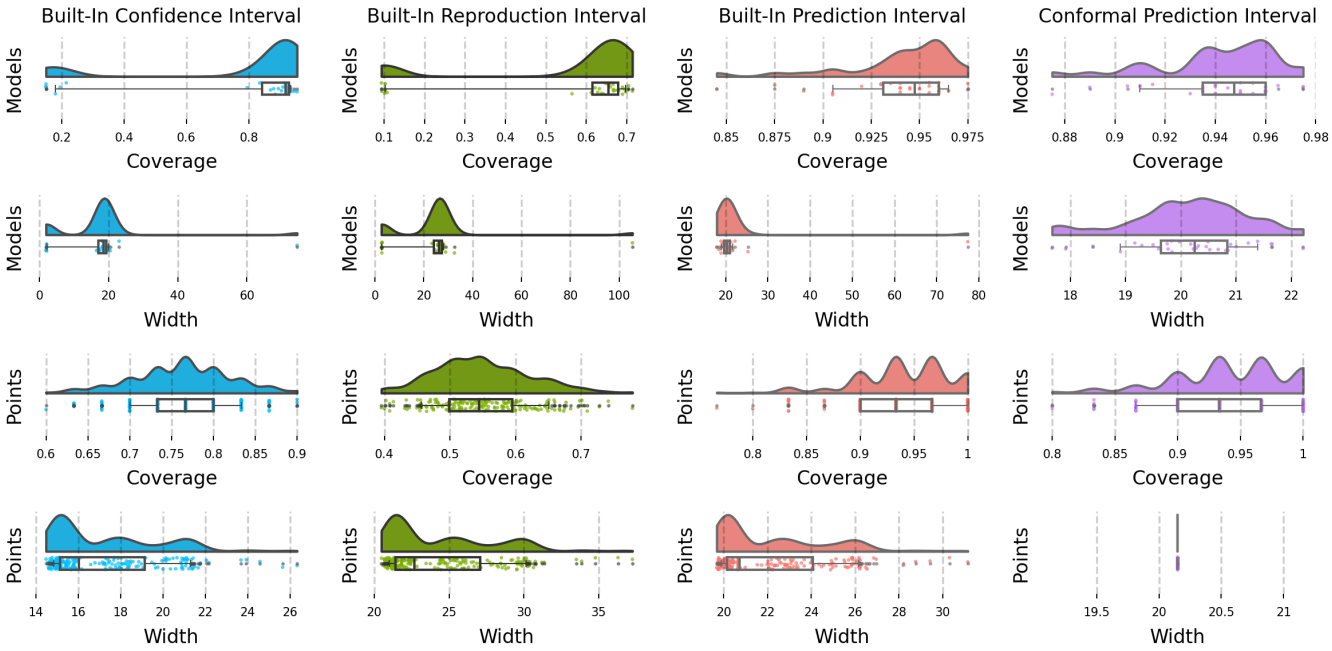


Figure 6.4: Points in the first two rows represent one of 30 models, depicting the fraction of test points that fell inside the interval generated by the model (marginal coverage) and average width for each model. Points in the last two represent one of 100 test points, depicting the fraction of models with intervals containing the test point (conditional coverage) and average width for each point. Results from Algorithm 6.1 with 200 trees, learning rate 0.6, subsampling 0.8, dropout 0.3, max depth 4.

Our results in Figure 6.3 are somewhat unfair to Algorithm 6.2, as the x-axis is in the ensemble size and not the number of boosting rounds. The number of boosting rounds that Algorithm 6.2 encounters can be as few as 31 on Abalone, in contrast to the 500 boosting rounds all other algorithms enjoy. Given the same number of boosting rounds, we would certainly expect an improvement.

Variable importance tests. We examine their performance by conducting a simulation study, testing the null $H_0 : w = 0$ on data from $\mathbf{f}(\mathbf{x}) = 4x_1 - x_2^2 + wbx_3$. We fit Algorithm 6.1 on data generated from $\mathbf{f}(\mathbf{x})$ and $\boldsymbol{g}(\mathbf{x}) = 4x_1 - x_2^2$, with 100 trees, $\lambda = 1$, subsampling rate 1, dropout rate 0.95, and a max depth of 6. The size and power of our test presented in Section 6.1.4 are depicted in the left panel within Figure 6.3. Our test performs very well, maintaining appropriate size control throughout while the Type II error decreases quickly. Empirically, increasing the dropout rate increases power.

Coverage of intervals. Other than the Nyström approximations for fast and practical interval computation in Appendix E.1, there is one more tweak that we encourage users to make in practice. We demonstrate in Section 6.1.5 that our intervals have asymptotic coverage, but this says nothing about its finite sample properties. Fortunately, Romano et al. (2020) yields a simple enhancement. Since we estimate $\widehat{\sigma}$ through computing the residuals on a hold-out calibration set, we reuse the calibration set to adaptively grow or shrink our intervals according to the empirical prediction interval coverage on the calibration set. This procedure converges quickly (amounting to a doubling trick and binary search), and empirically is very helpful in increasing robustness and finite-sample performance.

To examine the empirical performance of our intervals, we consider the Friedman function $\mathbf{f}(\mathbf{x}) = 10\sin(\pi\mathbf{x}_1\mathbf{x}_2) + 20(\mathbf{x}_3 - 0.5)^2 + 5\mathbf{x}_5 - 10$ (Friedman, 2001). This is depicted as a raincloud plot in Figure 6.4. Importantly, we consider two notions of coverage: marginal coverage, where coverage is averaged over test points, and conditional coverage, where the coverage is conditional on a test point. The first two rows depict the former, while the last two depict the latter. Our intervals perform reasonably well, attaining nominal coverage with the exception of the reproduction interval's coverage and the confidence interval's conditional coverage. In particular, the prediction interval performs very well after the adaptive coverage adjustment. Importantly, unlike the conformal benchmark that has a constant width at each point, our prediction intervals have different interval widths at each point, allowing users to identify "hard" examples. Ablation studies examining how the dropout probability p, subsample rate ξ, and tree depth affect test power and confidence interval width are reported in Fang et al. (2025) and Appendix E.1.

6.1.7. Conclusions and Further Work

We incorporate dropout and parallel boosting into the regularization procedure of Zhou and Hooker (2022a), allowing for increased signal recovery and improved asymptotic variance. With the CLTs our algorithms enjoy, we construct confidence and prediction intervals, and hypothesis tests for variable importance. Numerical experiments empirically validate our algorithms and statistical procedures.

Our theoretical guarantees depend on structure–value isolation, non-adaptivity, and tree regularity assumptions. Relaxing these – e.g., via honesty-based splitting or quantifying mild adaptivity – would broaden applicability. Extending inference beyond regression (to classification, survival, or structured outputs) also poses new challenges: non-quadratic losses, different tree behavior, and the need for new CLTs. Doing so under weaker assumptions is a welcome direction for future work.

6.2. Part II: Statistical Inference for Explainable Boosting Machines

While Part I established pointwise asymptotic normality and sketched uncertainty quantification for general multidimensional regression trees, many scientific and high-stakes decision-making domains require interpretable generalized additive decompositions $f(\mathbf{x}) = \mu + \sum_{k=1}^{p} f_k(x_k)$ together with componentwise confidence bands for each individual feature effect f_k. In Part II, based on joint work with Haimo Fang, Jonathan Pipping, and Giles Hooker (Fang et al., 2026), we analyze cyclic univariate moving-average boosting (Algorithm 6.3), establish finite-sample convergence to a block-structured backfitting fixed point and componentwise central limit theorems, and show that operating in histogram bin space reduces the per-query variance calculation from $O(n^3)$ to $O(M^3)$ where $M = \sum_{k=1}^{p} m_k \ll n$ is the total number of feature bins.

6.2.1. Introduction

Explainable Boosting Machines (EBM) (Lou et al., 2012; Nori et al., 2019) have emerged as a popular "glass-box" model that learns a Generalized Additive Model (GAM) (Hastie and Tibshirani, 1986), $\mathbf{f}(\mathbf{x}) = \sum_{k=1}^{p} \mathbf{f}^{(k)}(\mathbf{x}^{(k)})$, by fitting univariate gradient boosted regression trees to each feature. While EBMs provide a clear graphical representation for the effects of each feature, these are still point estimates, and vanilla EBMs (Lou et al., 2012) compute variance estimates via bootstrapping, which is computationally intensive and only heuristically justified. Although GAMs fitted via classical spline backfitting enjoy asymptotic confidence intervals (Wahba, 1983) and conformal prediction yields marginal prediction intervals, neither directly supplies frequentist feature-wise confidence intervals for tree-based EBM components. Extended motivation comparing black-box post-hoc explanations and glass-box additive models, together with 1D and UCI Obesity feature-effect visualizations, is provided in Fang et al. (2026).

Our contributions. Restricting the regression function to the class of GAMs yields interpretability, but the existing Boulevard theory (Zhou and Hooker, 2022a; Fang et al., 2025) does not directly apply: EBMs induce *feature-specific* structure matrices and kernels whose interaction must be controlled under centering and GAM identifiability to establish asymptotic normality for each additive component. To handle these challenges, we make the following contributions:

- **Asymptotic Theory.** We establish central limit theorems for three variations of EBMs (parallelized, backfitting-like parallelized, and sequential), showing that the Boulevard-regularized ensemble converges to a feature-wise kernel ridge regression. The resulting predictions are asymptotically normal and achieve the minimax-optimal MSE of $O(pn^{-2/3})$ for fitting Lipschitz p-dimensional GAMs (Yuan and Zhou, 2015). This avoids the curse of dimensionality, significantly improving upon the rates of Zhou and Hooker (2022a); Fang et al. (2025).

- **Frequentist Inference.** By leveraging our proposed CLT for baseline vector $\boldsymbol{\beta}$ and feature-wise predictions $\widehat{\mathbf{f}}^{(k)}$, we extend the heuristic bootstrap to a set of inference toolkits. We provide confidence intervals for each underlying function $\mathbf{f}^{(k)}$ and prediction intervals for the response. This also allows a direct test of variable importance following the methods of Mentch and Hooker (2016). We round out those propositions by numerical experiments to back up the validity of our algorithms.

- **Computational Efficiency.** The derived procedures above rely on a kernel ridge regression, with a prohibitively expensive $O(n^3)$ runtime. Fang et al. (2025) employ the Nyström approximation to produce intervals in $O(np^2)$ kernel precomputation and $O(p^2)$ query time, but the recursive implementation of Musco and Musco (2017) can be slow in practice. Given that EBMs utilize histogram trees, we instead work out the computations in histogram bin-space. This bin-based implementation runs in an additional lower-order $O(pm^2)$ time per training round, $O(pm^3)$ kernel precomputation time, and $O(pm^2)$ interval query time, where p is the number of features and m the maximum number of bins per-feature. As the number of bins is usually capped at 255 or 511, this is independent of the number of samples n!

6.2.2. Setup and Notation

Let $(\mathbf{X}_n, \mathbf{y}_n) = (\mathbf{x}_i, y_i)_{i=1}^n$ be i.i.d. samples with $\mathbf{x}_i = (\mathbf{x}_i^{(1)}, \ldots, \mathbf{x}_i^{(p)})^\top$. Let $\mathbf{J}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ denote the centering projector. We adopt the standard GAM identifiability convention that each component is centered on the training set: $\sum_{i=1}^n \mathbf{f}^{(k)}(\mathbf{x}_i^{(k)}) = 0$ for all k, so the intercept equals the sample mean. Then:

Assumption 6.13 (Additive Lipschitz Nonparametric Regression)**.** Assume that the response satisfies

$$y \;=\; \boldsymbol{\beta} \;+\; \sum_{k=1}^{p} \mathbf{f}^{(k)}(\mathbf{x}^{(k)}) \;+\; \epsilon := \mathbf{f}(\mathbf{x}) + \epsilon,$$

where $y, \boldsymbol{\beta} \in \mathbb{R}$, $\mathbf{x} = (\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(p)})^\top \in \mathbb{R}^p$, and ε are i.i.d. sub–Gaussian errors with mean zero and variance $\sigma^2 < \infty$. Assume the covariates have a bounded density on their support: $0 < c_1 \leq \mu(\mathbf{x}) \leq c_2 < \infty$. Each component function $\mathbf{f}^{(k)} : \mathbb{R} \to \mathbb{R}$ is mean–zero with respect to the marginal of $\mathbf{x}^{(k)}$, and is L-Lipschitz. The intercept $\boldsymbol{\beta}$ is defined as the unconditional mean of the response: $\boldsymbol{\beta} := \mathbb{E}_{\mathbf{x},\epsilon}[y] = \int y \, d\mu_{\mathbf{x}}(x) \, d\mu_\epsilon(\epsilon)$.

Boulevard Regularization. As introduced in Section 6.1.2, Boulevard regularization (Zhou and Hooker, 2022a) replaces the standard additive boosting update with the moving average $\widehat{\mathbf{f}}_b \leftarrow \frac{b-1}{b}\mathbf{f}_{b-1} + \frac{\lambda}{b}\mathbf{t}_b$; we extend this moving-average stabilization below to centered univariate trees across p additive components.

6.2.3. Algorithms

We adapt the regularized boosting procedure to fitting EBMs in Algorithms 6.3 and 6.4.

Parallelization and inference. Algorithm 6.3 is a minimal parallelized variation on EBMs. At each round b, the new set of p univariate trees $(\mathbf{t}_b^{(k)})_{k=1}^p$ are fitted in parallel by regressing the current residual $r_{i,k}$ on each of the p features $\mathbf{x}_i^{(k)}$. The trees are then centered and clipped, before being introduced into the ensemble as part of the average: $\mathbf{f}_b^{(k)} \leftarrow \frac{b-1}{b}\mathbf{f}_{b-1}^{(k)} + \frac{\lambda}{b}\widetilde{\mathbf{t}}_b^{(k)}$. Averaging yields a kernel ridge regression limit and a CLT, but (as in Boulevard) the fitted signal is asymptotically shrunk, requiring rescaling by $\frac{1+\lambda}{\lambda}$. Specifically, the predictions made by Algorithm 6.3 converge

Algorithm 6.3 Cyclic Univariate Moving-Average Boosting for Generalized Additive Models

Input: $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathbf{y} \in \mathbb{R}^n$, subsample rate ξ, learning rate λ, rounds B, truncation level $M > 0$.
Init: $\widehat{\boldsymbol{\beta}}_0 \leftarrow \frac{1}{n}\sum_{i=1}^n \mathbf{y}_i$, $\mathbf{f}_k^{(0)} \leftarrow 0$ for all k.
for $b = 1$ **to** B **do**
  for $k = 1$ **to** p **in parallel do**
    Sample $G_{b,k} \subset \{1, \ldots, n\}$ i.i.d. w.p. ξ.
    $r_{i,k} \leftarrow \mathbf{y}_i - (\widehat{\boldsymbol{\beta}}_{b-1} + \sum_\ell \mathbf{f}_{b-1}^{(\ell)}(\mathbf{x}_i^{(\ell)}))$.
    Fit tree $\mathbf{t}^{(b,k)}$ to $\{(\mathbf{x}_i^{(k)}, r_{i,k})\}_{i \in G_{b,k}}$.
    $\mu_{b,k} \leftarrow \frac{1}{n}\sum_{i=1}^n \mathbf{t}^{(b,k)}(\mathbf{x}_i^{(k)})$.
    $\widetilde{\mathbf{t}}_b^{(k)}(\mathbf{x}) \leftarrow t_b^{(k)}(\mathbf{x}) - \mu_{b,k}$.
    $\widetilde{\mathbf{t}}_b^{(k)}(x) \leftarrow \max\{-M, \min\{\widetilde{\mathbf{t}}_b^{(k)}(\mathbf{x}), M\}\}$.
    $\widehat{\boldsymbol{\beta}}_b \leftarrow \widehat{\boldsymbol{\beta}}_{b-1} + \mu_{b,k}$.
    $\mathbf{f}_b^{(k)} \leftarrow \frac{b-1}{b}\mathbf{f}_{b-1}^{(k)} + \frac{\lambda}{b}\widetilde{\mathbf{t}}_b^{(k)}$.
  end for
end for
Output: $\widehat{\mathbf{f}}(x) \leftarrow \widehat{\boldsymbol{\beta}}_B + \frac{1+\lambda}{\lambda}\sum_{k=1}^p \mathbf{f}_k^{(B)}(\mathbf{x}^{(k)})$.

Algorithm 6.4 Leave-One-Out Parallel Moving-Average Boosting for Generalized Additive Models

Input: $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathbf{y} \in \mathbb{R}^n$, subsample rate ξ, rounds B, truncation level $M > 0$.
Init: $\widehat{\boldsymbol{\beta}}_0 \leftarrow \frac{1}{n}\sum_{i=1}^n \mathbf{y}_i$, $\mathbf{f}_k^{(0)} \leftarrow 0$ for all k.
for $b = 1$ **to** B **do**
  for $k = 1$ **to** p **in parallel do**
    Sample $G_{b,k} \subset \{1, \ldots, n\}$ i.i.d. w.p. ξ.
    $r_{i,k} \leftarrow \mathbf{y}_i - (\widehat{\boldsymbol{\beta}}_{b-1} + \sum_{\ell \neq k} \mathbf{f}_{b-1}^{(\ell)}(\mathbf{x}_i^{(\ell)}))$.
    Fit tree $\mathbf{t}^{(b,k)}$ to $\{(\mathbf{x}_i^{(k)}, r_{i,k})\}_{i \in G_{b,k}}$.
    $\mu_{b,k} \leftarrow \frac{1}{n}\sum_{i=1}^n \mathbf{t}^{(b,k)}(\mathbf{x}_i^{(k)})$.
    $\widetilde{\mathbf{t}}_b^{(k)}(\mathbf{x}) \leftarrow t_b^{(k)}(\mathbf{x}) - \mu_{b,k}$.
    $\widetilde{\mathbf{t}}_b^{(k)}(x) \leftarrow \max\{-M, \min\{\widetilde{\mathbf{t}}_b^{(k)}(\mathbf{x}), M\}\}$.
    $\widehat{\boldsymbol{\beta}}_b \leftarrow \widehat{\boldsymbol{\beta}}_{b-1} + \mu_{b,k}$.
    $\mathbf{f}_b^{(k)} \leftarrow \frac{b-1}{b}\mathbf{f}_{b-1}^{(k)} + \frac{1}{b}\widetilde{\mathbf{t}}_b^{(k)}$.
  end for
end for
Output: $\widehat{\mathbf{f}}(x) \leftarrow \widehat{\boldsymbol{\beta}}_B + 2\sum_{k=1}^p \mathbf{f}_k^{(B)}(\mathbf{x}^{(k)})$.

almost surely as $b \to \infty$ to

$$\widehat{\mathbf{f}}_b(\mathbf{x}) \xrightarrow[b\to\infty]{a.s.} \bar{\mathbf{y}} + \sum_{k=1}^p \mathbb{E}[\mathbf{s}^{(k)}(\mathbf{x})]^\top \mathbf{J}_n[\mathbf{I} + \mathbf{J}_n\mathbf{K}]^{-1}\mathbf{y},$$

where $\mathbf{K} = \sum_{k=1}^{p} \mathbf{K}^{(k)}$. A sequential random-cyclic warmup variant that samples a single feature $k \sim \text{Unif}(\{1, \ldots, p\})$ per boosting round and converges to the same fixed point without feature-level parallelism is analyzed in Fang et al. (2026) (Appendix E.2).

Leave-one-out procedure. Because Algorithm 6.3 converges to $\frac{\lambda}{1+\lambda}\mathbf{f} \leq \mathbf{f}/2$ and requires rescaling by $\frac{1+\lambda}{\lambda}$, we provide an alternative inspired by the backfitting-like procedure of Fang et al. (2025) in Algorithm 6.4 that does not require rescaling. This hybrid backfitting-boosting procedure amounts to omitting predictions from trees trained on feature $\mathbf{x}^{(k)}$ when fitting a new tree $\mathbf{t}_b^{(k)}$ for it – instead using leave-one-feature-out residuals $\mathbf{y}_i - (\widehat{\boldsymbol{\beta}}_{b-1} + \sum_{\ell \neq k} \mathbf{f}_{b-1}^{(\ell)}(\mathbf{x}_i^{(\ell)}))$. Informally, Algorithm 6.4 converges to:

$$\bar{\mathbf{y}} + \sum_{k=1}^{p} \mathbb{E}[\mathbf{s}^{(k)}(\mathbf{x})]^\top (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^{-1} \mathbf{J}_n \Big[\mathbf{I} - (\mathbf{I} + \mathcal{K})^{-1}\mathcal{K}\Big]\mathbf{y},$$

where $\mathcal{K} := \sum_{k=1}^{p} (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^{-1} \mathbf{J}_n \mathbb{E}\mathbf{S}^{(k)}$. Detailed comparisons between Algorithms 6.3–6.4, vanilla EBMs, and multidimensional Boulevard are given in Fang et al. (2026).

6.2.4. Theory

We adopt the regression tree and structure matrix definitions from Definitions 6.1 and 6.2 in Section 6.1.2, applied to univariate partitions along each coordinate $k \in \{1, \ldots, p\}$. Following Zhou and Hooker (2022a); Fang et al. (2025), we require that the tree *structures* be separated from the leaf *values*, and stabilize after a burn-in period:

Assumption 6.14 (Integrity)**.** Assume

1. (Structure-Value Isolation) The density for $\mathbf{s}_b(\mathbf{x})$ is independent from the terminal leaf value $\mathbf{y} \in \mathbb{R}$.

2. (Non-adaptivity) For each feature k, there exists a set of probability measures $\mathcal{Q}_n^{(k)}$ for tree structures such that all tree structures $\mathbf{S}_b^{(k)}$ are independent draws from $\mathcal{Q}_n^{(k)}$ after some finite b'.

Tree Geometry. We control the geometry and size of leaves to ensure locality and stable averages, obtaining dimension-free univariate rates due to the GAM structure:

Assumption 6.15 (Bounded Leaf Diameter)**.** Write $\mathsf{diam}(A) = \sup_{\mathbf{x}_1,\mathbf{x}_2\in A} \|\mathbf{x}_1 - \mathbf{x}_2\|$. For any leaf A in a tree with structure $\Pi \in Q_n^{(k)}, \forall k = 1, \cdots, p$, we need $\sup_{A\in\Pi} \mathsf{diam}(A) = O(d_n) = O(n^{-1/3}) = o(\frac{1}{\log n})$.

Assumption 6.16 (Increased Minimal Leaf Size)**.** For any $\nu > 0$, the leaf geometric volume v_n is bounded such that $v_n = n^{-\frac{2}{3}+\nu} < n^{-\frac{1}{3}} = O(d_n)$.

Assumption 6.17 (Restricted Tree Support)**.** For a tree constructed using feature k, the cardinality of the tree space $Q_n^{(k)}$ is bounded by $O\Big(n^{-1/3}\exp(n^{2/3-\epsilon_n})\Big)$, for some small $\epsilon_n \to 0$.

Feature-specific (specialized) kernels. Unlike Zhou and Hooker (2022a) and Fang et al. (2025), which consider a single expected kernel $\mathbb{E}\mathbf{S}$, EBMs induce *feature-specific* kernels $\mathbf{K}^{(k)} := \mathbb{E}[\mathbf{S}^{(k)}]$, enabling a characterization of partial dependence and feature-wise convergence and confidence intervals. The warmup fixed-point analysis for sequential random feature updates is deferred to Fang et al. (2026).

Theory for Algorithm 6.3. To prove that Algorithm 6.3 converges with learning rate $\lambda \approx 1$ (rather than $\lambda = 1/p$), we make the following assumption:

Assumption 6.18 (Projector-like feature kernels and conditional orthogonality)**.** Let $\mathbf{K}^{(k)} = \mathbb{E}[\mathbf{S}^{(k)}]$ and $\mathbf{J}_n := \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. For each feature k, let

$$\mathcal{H}_k \;:=\; \Big\{\, \mathbf{v}\in\mathbb{R}^n :\; \exists\, g:\mathbb{R}\to\mathbb{R}, \mathbf{v}_i = g(\mathbf{x}_i^{(k)}),\; \mathbf{1}^\top\mathbf{v} = 0 \,\Big\}$$

with P_k the orthogonal projector onto $\mathcal{H}_k$. Assume:

- **Projector-like approximation.** There exists $\varepsilon_{n,k} \to 0$ with $\big\|\, \mathbf{J}_n\, \mathbf{K}^{(k)}\, \mathbf{J}_n - P_k \,\big\|_{\mathrm{op}} \;\le\; \varepsilon_{n,k}\,.$
- **Conditional orthogonality.** For $a \neq k$, the additive subspaces are orthogonal: $P_a\, P_k \;=\; \mathbf{0}\,.$

Assumption 6.18 is used only to analyze Algorithm 6.3; Algorithm 6.4 removes this requirement and needs only Assumption 6.20. Formally, the finite-sample convergence of Algorithm 6.3 is characterized in Theorem 6.19.

Theorem 6.19. *Define* $\widehat{\mathbf{y}}_b := \widehat{\boldsymbol{\beta}}_b + \sum_{a=1}^{p} \widehat{\mathbf{y}}_b^{(a)}, \mathbf{J}_n = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. *The fixed points are*

$$\widetilde{\mathbf{y}}_k^* \;=\; \mathbf{J}_n \mathbf{K}^{(k)} \left[\lambda^{-1}\mathbf{I} + \mathbf{J}_n \mathbf{K}\right]^{-1}\mathbf{y}, \qquad \mathbf{K} \;:=\; \sum_{k=1}^{p} \mathbf{K}^{(k)},$$

and $\widehat{\mathbf{y}}^* \;=\; \widehat{\boldsymbol{\beta}}^*\mathbf{1} + \sum_{k=1}^{p} \widetilde{\mathbf{y}}_k^*$, *with* $\widehat{\boldsymbol{\beta}}^* = \bar{\mathbf{y}}$. *For each* k,

$$\widehat{\mathbf{y}}_b^{(k)} \;\xrightarrow{a.s.}\; \widetilde{\mathbf{y}}^{(k)*}, \qquad \widehat{\boldsymbol{\beta}}_b \;\xrightarrow{a.s.}\; \bar{\mathbf{y}} = \tfrac{1}{n}\mathbf{1}^\top\mathbf{y},$$

and hence $\widehat{\mathbf{y}}_b \overset{a.s.}{\to} \widehat{\mathbf{y}}^*$ *almost surely.*

Theory for Algorithm 6.4. Algorithm 6.4 does not rely on Assumption 6.18, and only depends on a weaker assumption below (verified empirically in Table E.2).

Assumption 6.20 (Feature-kernel distinctness)**.** Let $\mathbf{K}^{(k)} = \mathbb{E}[\mathbf{S}^{(k)}]$. There exists $c_3 < 1$ such that, for all $a \neq k$, $\left\| \mathbb{E}[\mathbf{S}^{(a)}]\, \mathbb{E}[\mathbf{S}^{(k)}] \right\|_{\text{op}} \;\leq\; c_3, \quad (p-1)\, c_3 \;<\; 1.$

Theorem 6.21. *Denote the prediction using feature* k *at boosting round* b *as* $\widehat{\mathbf{y}}_b^{(k)} = \frac{1}{b}\sum_{s=1}^{b} t^{(s,k)}(\mathbf{x}^{(k)})$. *Define the aggregated kernel* $\mathcal{K} := \sum_{k=1}^{p}(\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^{-1}\mathbf{J}_n\mathbb{E}\mathbf{S}^{(k)}$. *Define feature-wise fixed points*

$$\widetilde{\mathbf{y}}_k^* = (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^{-1}\mathbf{J}_n\mathbb{E}\mathbf{S}^{(k)} \left[\mathbf{I} - (I + \mathcal{K})^{-1}\,\mathcal{K}\right]\mathbf{y}.$$

Then, $\widehat{\mathbf{y}}_b^{(k)} \overset{a.s.}{\to} \widetilde{\mathbf{y}}_k^*$, $\widehat{\mathbf{y}}_b \overset{a.s.}{\to} \widehat{\mathbf{y}}^*$, *and* $\widehat{\boldsymbol{\beta}}^* \overset{a.s.}{\to} \bar{\mathbf{y}} := \frac{1}{n}\mathbf{1}^\top\mathbf{y}$.

A Central Limit Theorem. Unlike Zhou and Hooker (2022a) and Fang et al. (2025), the GAM structure of our problem allows us to recover the optimal univariate minimax rate $n^{-2/3}$ (Yuan and Zhou, 2015). Below, we use the subscript $E = \{A, B\}$ to label relevant variables in Algorithm 6.3 and 6.4.

Lemma 6.22 (Rate of Convergence)**.** *Let* $\mathbf{B}_n^{(k)} := \{i : \left\|\mathbf{x}^{(k)} - \mathbf{x}_i^{(k)}\right\| \leq d_n\}$ *be the points within distance* d_n *from test point* $\mathbf{x}$ *on feature* k. *Under Assumption 6.17, if* $|\mathbf{B}_n| = \Omega(n\, d_n)$ *and* $\inf_{A \in \Pi \in Q_n^{(k)}} \sum_i \mathbf{I}(\mathbf{x}_i \in A) = \Omega(n^{\frac{2}{3}})$, *then* $\left\|\mathbf{k}^{(k)}\right\|_2, \left\|\mathbf{r}_E^{(k)}\right\|_2 = \Theta(n^{-\frac{1}{3}})$ *almost surely.*

Theorem 6.23 (Asymptotic Normality)**.** *Define* $c_A = \frac{\lambda}{1+\lambda}, c_B = 1$. *For all* $k = 1, \cdots, p$, *we have*

that:

$$\left\|\mathbf{r}_E^{(k)}(\mathbf{x})\right\|^{-1}\left(\widehat{\mathbf{f}}_E^{(k)}(\mathbf{x})-c_E\mathbf{f}^{(k)}(\mathbf{x})\right)\xrightarrow{d}\mathcal{N}(0,\sigma^2)$$

Normality of the intercept term $\hat{\boldsymbol{\beta}}$ is also established in Lemma E.2, and our CLTs yield the minimax-optimal risk bound for Lipschitz p-dimensional GAMs (Yuan and Zhou, 2015):

Corollary 6.24. $\mathbb{E}[(\frac{1}{c_E}\hat{\mathbf{f}}_E^{(k)}(\mathbf{x})-\mathbf{f}^{(k)}(\mathbf{x}))^2]\lesssim\frac{\sigma^2}{c_E^2}p\,n^{-\frac{2}{3}}$.

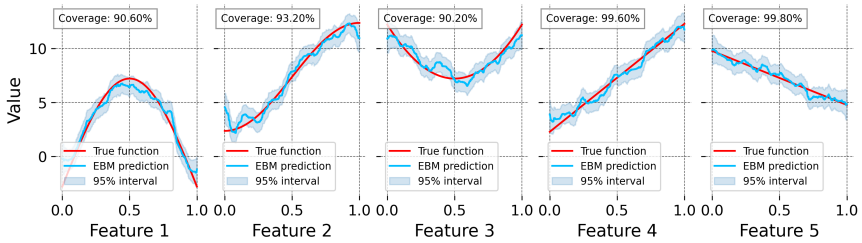


Figure 6.5: Per-feature CIs and coverages for $f(x)=-5+10\sin(\pi x^{(1)})+5\cos(\pi x^{(2)})+20(x^{(2)}-0.5)^2+10x^{(3)}-5x^{(4)}$.

6.2.5. Inference and Computation

We now construct practical intervals from Theorem 6.23 and show how to compute the required standard errors in *bin space* so that inference time is independent of n, using a consistent variance estimate $\hat{\sigma}^2$.

Confidence, Reproduction, and Prediction Intervals. We construct $(1-\alpha)$ intervals as follows:

- **Feature effect CIs:** $c_E^{-1}\widehat{\mathbf{f}}_E^{(k)}(\mathbf{x})\pm z_{1-\alpha/2}\,c_E^{-1}\hat{\sigma}\,\|\mathbf{r}_E^{(k)}(\mathbf{x})\|$ yields a $(1-\alpha)$-confidence interval for $\mathbf{f}^{(k)}(\mathbf{x}^{(k)})$ (validated in Figure 6.5).

- **Intercept CI:** By Lemma E.2, $\widehat{\boldsymbol{\beta}}\pm z_{1-\alpha/2}\,\hat{\sigma}/\sqrt{n}$.

- **Overall response CI and PI:** For $\widehat{f}_E(\mathbf{x})=\widehat{\boldsymbol{\beta}}+c_E^{-1}\sum_{k=1}^p\widehat{\mathbf{f}}_E^{(k)}(\mathbf{x})$, a $(1-\alpha)$ confidence interval is $\widehat{f}_E(\mathbf{x})\pm z_{1-\alpha/2}c_E^{-1}\hat{\sigma}\sqrt{\frac{1}{n}+\|\sum_{k=1}^p\mathbf{r}_E^{(k)}(\mathbf{x})\|_2^2}$, where $\|\sum_{k=1}^p\mathbf{r}_E^{(k)}(\mathbf{x})\|_2^2=\sum_{k=1}^p\|\mathbf{r}_E^{(k)}(\mathbf{x})\|_2^2$ under Assumption 6.18; scaling the CI width by $\sqrt{2}$ yields reproduction intervals, and replacing $\|\mathbf{r}_E^{(k)}(\mathbf{x})\|_2$ by $\sqrt{1+\|\mathbf{r}_E^{(k)}(\mathbf{x})\|_2^2}$ yields prediction intervals.

Binning. Because EBMs (Nori et al., 2019) utilize histogram trees, we compress all n samples along feature k into $m_k\ll n$ histogram bins and work entirely in *bin space* (see Appendices E.2

and E.2). Decomposing the structure matrix in bin space reduces computing $\|\mathbf{r}^{(k)}(\mathbf{x})\|_2$ to solving an $m_k \times m_k$ linear system and computing two m_k-length dot products. Denoting $m = \max_k m_k$, caching $\mathbf{M}$ takes $O(pm^3)$ time and each per-point inference query takes $O(m^2)$ time – independent of n when m is capped (e.g., at 255 or 511), outperforming the $O(nd_{\text{eff}}^2)$ Nyström runtime of Part I.

6.2.6. Numerical Experiments

Predictive Accuracy and Overfitting Resistance. Comparisons of test MSE across Optuna-tuned UCI benchmarks and overfitting trajectories on $f(x) = \sin(4\pi x)$ and the Diabetes dataset (Efron et al., 2004) demonstrating that Algorithm 6.3 and EBM resist overfitting without early stopping are reported in Fang et al. (2026).

Interval Coverage and Scalability. Figures 6.5 and 6.6 validate the empirical coverage and width of our feature-wise and overall confidence, prediction, and reproduction intervals. We further evaluate scalability in higher-dimensional correlated synthetic additive models for Algorithm 6.4 under Assumption 6.20 in Table 6.1, and on high-dimensional real datasets ($p \approx 95$–127) in Table 6.2. Bootstrap EBMs severely under-cover and are slow due to repeated refits, whereas Algorithm 6.4 attains near-nominal 95% coverage (0.943–0.956) with lower RMSE, tighter intervals, and substantially faster runtime.

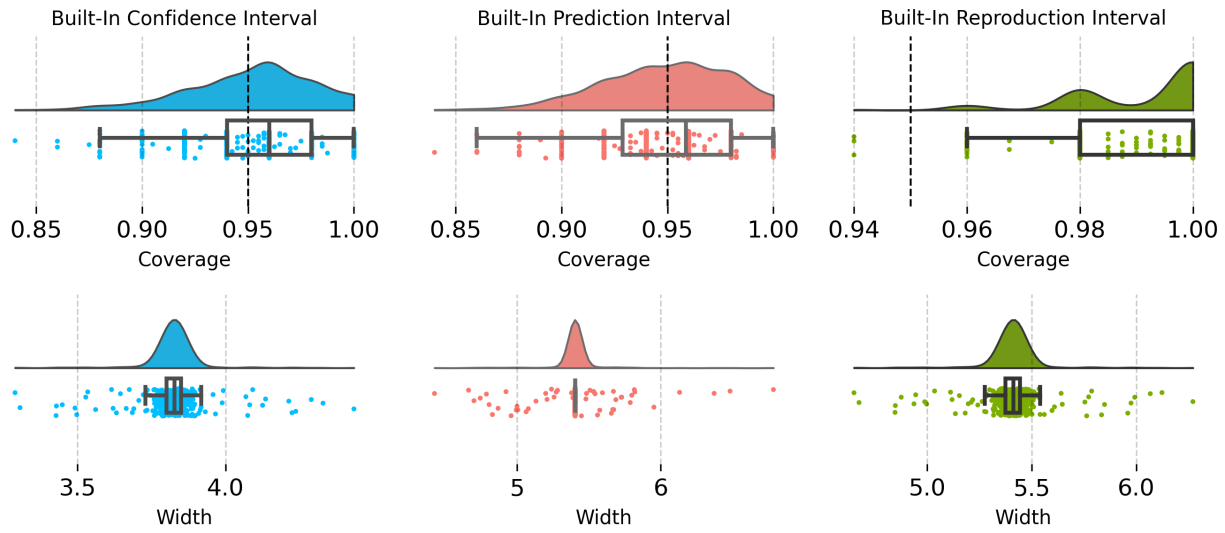


Figure 6.6: Coverage and width of our intervals over 50 trials, $n = 2000$, on the same function as Figure 6.5.

p	Method	RMSE	Coverage	Width	Time (s)
5	Bootstrap EBM	0.514	0.290	0.386	26.42
	Algorithm 6.4	0.661	0.951	2.609	3.00
	BRATD	0.611	0.956	2.899	10.41
10	Bootstrap EBM	0.537	0.373	0.545	21.24
	Algorithm 6.4	0.788	0.951	3.069	5.90
	BRATD	0.866	0.935	3.531	13.58
15	Bootstrap EBM	0.563	0.453	0.675	31.51
	Algorithm 6.4	0.900	0.945	3.520	8.77
	BRATD	1.112	0.946	4.398	10.29
20	Bootstrap EBM	0.583	0.511	0.801	43.14
	Algorithm 6.4	1.004	0.948	3.901	11.56
	BRATD	1.332	0.948	5.121	10.70

Table 6.1: Synthetic Lipschitz GAMs (bootstrap EBM vs. Algorithm 6.4 vs. BRATD). Synthetic additive model with $n = 1000$ and $p \in \{5, 10, 15, 20\}$ (all p features are structural; five base univariate functions are cycled across coordinates): $f_1(x) = 2\sin(\pi x)$, $f_2(x) = (x - 0.5)^2$, $f_3(x) = 0.5 \times \mathbf{1}\{x > 0.3\}$, $f_4(x) = \sqrt{x}$, $f_5(x) = -1.5x$. We report mean RMSE, model-level empirical coverage at nominal 95%, average interval width, and runtime (seconds) over 10 repetitions ($\alpha = 0.05$; 250 bootstrap resamples for EBM).

Dataset	n	p	RMSE (Alg. 6.4)	Cov. (Alg. 6.4)	Time (s) (Alg. 6.4)
taiwanese_bankruptcy	6819	95	0.151	0.948	18.01
myocardial_infarction	1700	111	0.309	0.943	3.53
communities_and_crime	1994	127	0.140	0.953	5.68
dota2_results	50000	116	0.142	0.956	28.78

Dataset	RMSE (EBM)	Time (s) (EBM)	RMSE (BRATD)	Time (s) (BRATD)
taiwanese_bankruptcy	0.153	197.31	0.154	24.58
myocardial_infarction	0.309	52.62	0.311	3.78
communities_and_crime	0.135	178.41	0.144	7.80
dota2_results	0.013	3736.10	0.147	82.62

Table 6.2: We report RMSE, 95% empirical coverage, and training time for Algorithm 6.4 (top), and baseline RMSE and training time for vanilla EBM and BRATD (bottom). Baseline coverage is omitted because large-scale uncertainty estimation is not computationally scalable.

6.2.7. Conclusion

Adopting Boulevard regularization within EBMs removes the need for early stopping and yields a feature-wise kernel ridge regression limit with componentwise central limit theorems, enabling asymptotically valid confidence intervals, prediction intervals, and variable-importance tests computable in histogram bin space in time independent of n. Extensions to pairwise interactions

(Lengerich et al., 2020), varying-coefficient models (Zhou and Hooker, 2022b), and non-Gaussian outcomes are promising directions for future work.

CHAPTER 7

PARAMETER ESTIMATION UNDER PARTIAL OBSERVABILITY

This chapter is based on joint work with A. J. Abkemeier, Kunyang He, Giles Hooker, and Edward L. Ionides (Tan et al., 2024a): Kevin Tan, A. J. Abkemeier, Kunyang He, Giles Hooker, and Edward L. Ionides, "Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes," working paper (revise and resubmit at Journal of the Royal Statistical Society: Series B), arXiv:2407.03085, 2024.

[RESOLVE: CHAPTER 7 MARKS THE TRANSITION INTO PARTIAL OBSERVABILITY (STATISTICAL ESTIMATION PERSPECTIVE).]

In Chapters 2–5, the learner observed the underlying state or context directly at each decision step, so that statistical uncertainty arose solely from unknown reward or transition parameters. In many scientific and engineering systems, however, the latent Markov state X_n is never observed directly and must be inferred sequentially from noisy measurements Y_n. In this chapter and Chapter 8, hidden-state filtering itself becomes part of the computational and statistical loop. Here we study parameter estimation in general nonlinear, non-Gaussian partially observed Markov processes (POMPs), where discrete resampling in bootstrap particle filters breaks differentiability of the Monte Carlo log-likelihood and causes exponential-in-horizon gradient variance when differentiating unresampled importance weights. We introduce a fixed-ancestor weight-discounted off-parameter particle filter (Algorithm 7.1), its automatic-differentiation score estimator (Algorithm 7.2), and a two-stage maximum likelihood procedure that warm-starts discounted automatic differentiation with iterated filtering (Algorithm 7.3). All proofs and supplementary experimental results are provided in Appendix F.

7.1. Introduction

Many scientific models involve highly nonlinear stochastic dynamic systems possessing significant random variation in both the process dynamics and the measurements. Commonly, the latent

system is modeled as a Markov process, giving rise to a partially observed Markov process (POMP) model, also known as a hidden Markov model or a state space model. POMP models arise in fields as diverse as optimal control (Singh et al., 2023), epidemiology (He et al., 2010; Stocks et al., 2018), ecology (Knape and de Valpine, 2012) and finance (Kim and Stoffer, 2008; Bret'o, 2014). Despite their ubiquity, estimation and inference for this broad class of models remains a challenging problem. We develop full-information plug-and-play inference for POMP models that takes advantage of automatic differentiation (AD) computation while controlling the Monte Carlo bias and variance of the resulting estimator. These desiderata have not previously been combined, and we discuss each component in turn.

Full-information methods are those which make statistically efficient use of data, typically via likelihood-based or Bayesian methods that avoid dependence on summary statistics. The particle filter, also known as sequential Monte Carlo, serves as the foundation for various full-information inference algorithms for POMP models, since it provides an unbiased estimate of the likelihood function (Del Moral, 2004), enabling Bayesian inference via particle Markov chain Monte Carlo (Andrieu et al., 2010) and likelihood-based inference via iterated filtering algorithms IF1 (Ionides et al., 2006) and IF2 (Ionides et al., 2015).

An inference algorithm for a POMP model is said to be plug-and-play if it does not need to evaluate the transition density of the latent Markov process, enabling an arbitrary model simulator to be plugged into the algorithm, a convenient property that facilitates scientific discovery (He et al., 2010). Some plug-and-play methods depend on the construction of low-dimensional summary statistics (Wood, 2010; Toni et al., 2009), sacrificing statistical efficiency for computational convenience. Here, our mission is to extend current computational capability among methods that maintain statistical efficiency and the plug-and-play property. Plug-and-play methods are sometimes called likelihood-free (Owen et al., 2015), and it may be counter-intuitive that likelihood-based inference is possible using likelihood-free methods. Basic particle filters, also known as bootstrap filters, possess the plug-and-play property. Iterated particle filtering algorithms, such as IF1 and IF2, have shown that plug-and-play likelihood maximization is practical in various scientific investigations

(King et al., 2008; Blake et al., 2014; Pons-Salort and Grassly, 2018; Subramanian et al., 2021; Fox et al., 2022; Drake et al., 2023). However, iterated filtering algorithms have slow convergence near the MLE (Doucet et al., 2013).

Advances in AD for gradient estimation using particle filters (Naesseth et al., 2018; Jonschkowski et al., 2018; Corenflos et al., 2021; Ścibior and Wood, 2021; Singh et al., 2023) have drawn attention to AD as a tool for inference in POMP models. Widespread adoption of these approaches has been limited by their bias (Naesseth et al., 2018; Jonschkowski et al., 2018), high variance (Poyiadjis et al., 2011; Ścibior and Wood, 2021), computational expense quadratic in the number of particles (Corenflos et al., 2021; Chen and Li, 2024), or by the loss of the plug-and-play property (Poyiadjis et al., 2011; Ścibior and Wood, 2021; Singh et al., 2023; Chen and Li, 2024). We present a new algorithm which (for reasons discussed later) we call a *Differentiated Measurement Off-Parameter* particle filter (Algorithm 7.2). Algorithm 7.2 computes the derivative of a smooth *Measurement Off-Parameter* variant of the particle filter (Algorithm 7.1), though Algorithm 7.2 is conveniently evaluated using AD without explicitly running the two-pass construction in Algorithm 7.1. Algorithm 7.1 has the special property that, unlike most particle filters, the gradient of the filtered log-likelihood estimate provides a consistent estimate of the gradient of the log-likelihood. Particle filters have previously been adapted to provide a smooth estimate of the likelihood function (Svensson et al., 2018; Malik and Pitt, 2011), but Algorithm 7.1 and Algorithm 7.2 differ from these by possessing the plug-and-play property.

Algorithm 7.2 trades bias against variance via a discount factor, $\alpha \in [0, 1]$. We find that Algorithm 7.2 possesses a bias-variance tradeoff that permits both low bias and low variance for suitable choices of α strictly in $(0, 1)$, both in theory (Theorem 7.5) and practice (Figure 7.1). Theorem 7.1 shows that Algorithm 7.2 coincides with the biased gradient estimator of Naesseth et al. (2018) when $\alpha = 0$. In a limit as the number of particles increases, Theorem 7.1 also shows that Algorithm 7.2 with $\alpha = 1$ matches the high-variance gradient estimate from Poyiadjis et al. (2011) and Ścibior and Wood (2021). In this case, the variance can become prohibitively high, as discovered by these researchers and others (Arya et al., 2022). However, the variance rapidly reduces as

α decreases, and so a choice of α close to one can provide a low-bias, low-variance, plug-and-play gradient estimate with computational scaling linear in the number of particles.

In practice, likelihood optimization via gradient descent with Algorithm 7.2 converges quickly within a neighborhood of the maximum but struggles to reach this neighborhood in the presence of local minima and saddle points. By contrast, iterated filtering algorithms provide a relatively fast and stable way to identify this neighborhood. We therefore build a simple but effective hybrid algorithm called *Iterated Filtering with Automatic Differentiation* (Algorithm 7.3) that warm-starts first-order or second-order gradient methods with a preliminary solution obtained from iterated filtering. We implement our algorithm using JAX (Bradbury et al., 2018), which also enables us to take advantage of graphical processing unit (GPU) hardware. Our methodology is provided in an open-source software library, Pypomp (Abkemeier et al., 2025).

Promising numerical results indicate that Algorithm 7.3 beats IF2 (and by the numerical results of Ionides et al. (2015), also IF1 and the Liu-West filter of Liu and West (2001)) on a challenging problem in epidemiology, the Dhaka cholera model of King et al. (2008). Algorithm 7.3 also outperforms marginalized panel iterated filtering on a challenging ecological panel problem, the 38-parameter *Daphnia* model of Yang et al. (2025). Empirically, the benchmark tests suggest that warm-starting discounted automatic-differentiation optimization with iterated filtering (Algorithm 7.3) is a significant methodological advance over iterated filtering (IF2) alone (Ionides et al., 2015) for plug-and-play statistical inference on POMP models. Plug-and-play methods are necessary when transition densities are not available, but can have advantages even when they are. The plug-and-play property avoids difficulties arising when the full latent transition densities at different parameter values are mutually singular (Roberts and Stramer, 2001; Karppinen et al., 2024). The plug-and-play property also facilitates the development of statistical software applicable to a broad class of POMP models.

We also demonstrate empirical advances using Algorithm 7.2 for Bayesian inference. In Section 7.6, we use the gradient estimates from Algorithm 7.2 within a No-U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014) in conjunction with an empirical Bayes-type prior estimated with IF2 to reduce

the burn-in period of particle MCMC (Andrieu et al., 2010) on the Dhaka cholera model of King et al. (2008) from the 700,000 iterations in Fasiolo et al. (2016) to just 500. To the best of our knowledge, attaining rapid mixing on this challenging problem has not previously been achieved.

7.2. Problem Setup

Consider an unobserved Markov process $\{X(t), t \geq t_0\}$, with discrete-time observations $Y_1, ..., Y_N$ realized at values $y_1^*, \ldots, y_N^*$ with times $t_1, \ldots, t_N$. The process is parameterized by an unknown parameter $\theta \in \Theta \subseteq \mathbb{R}^p$, where the state $X(t)$ takes values in the state space $\mathcal{X} \subseteq \mathbb{R}^d$, the observations Y_n take values in $\mathcal{Y}$, and we write $X_n = X(t_n)$. We suppose that the discrete-time latent process model has a density $f_{X_n|X_{n-1}}(x_n \mid x_{n-1}; \theta)$. Although we assume this density exists, plug-and-play inference algorithms do not explicitly evaluate it. Instead, they have access to a corresponding simulator, which we write as $\texttt{process}_n(x_n \mid x_{n-1}; \theta)$. We set $f_{Y_n|X_n}(y_n \mid x_n; \theta)$ to be the measurement density, which can be directly evaluated. We call $f_{X_{1:n}|Y_{1:n}}(x_{1:n}|y_{1:n}^*; \theta)$ the posterior distribution of states, and $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$ the filtering distribution at time t_n. All joint and conditional densities are defined on a probability space $(\Omega, \Sigma, \mathbb{P})$ which is also assumed to enable construction of independent replicates and all random variables defined in our algorithms. For the algorithmic interpretation of our theory, we identify the random number seed with an element of the sample space $\omega \in \Omega$. We assume that ω can be used to construct a sequence $\{\omega_k, k \in \mathbb{Z}\}$ that comprise independent uniform random variables in the mathematical formulation and a sequence of pseudo-random numbers in the equivalent computational formulation. We require that $\texttt{process}_n$ is a function of ω_n, though this dependence is not written explicitly. Mathematically and computationally, we can permit ω_n to be vector valued.

7.3. Off-Parameter Particle Filters

The *Measurement Off-Parameter with discount factor* α particle filter is defined by the pseudocode in Algorithm 7.1. Note that this algorithm has the plug-and-play property. We suppose that the simulator is a differentiable function of θ for every fixed ω. This condition is equivalent to the *reparameterization trick* (Corenflos et al., 2021): by writing the generated random numbers as a parameter-dependent transformation from a reference distribution—e.g., the quantile transform of a

uniform random variable—we can differentiate their values, and hence the simulation, with respect to parameters. The discrete resampling step of the bootstrap particle filter has been the primary point of concern for the use of AD to estimate derivatives for particle filters. The discontinuity of this resampling (as a function of θ, for fixed ω) means that the derivative of the expectation is not consistently estimated by the expectation of the derivative. Algorithm 7.1 is a mathematical construction designed to avoid this discontinuity. We begin with a filter that resamples particles with weights calculated at a baseline parameter value, ϕ, and then corrects these weights to target the filtering distribution at another parameter value, θ, yielding the name *measurement off-parameter*. The terminology is chosen by analogy with similar *off-policy* algorithms used for reinforcement learning in which a policy that was not implemented is evaluated by reweighting the rewards of the policy that was actually applied. If we fix ω and ϕ, the sample indices do not change with θ, but the re-weighting scheme does, allowing Algorithm 7.1 to be differentiated with respect to θ. When $\theta = \phi$, we have $w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta} = 1$ and Algorithm 7.1 reduces to the standard particle filter, which is the setting we use in practice, while the off-parameter construction allows us to define a derivative. One can think of Algorithm 7.1 as reconstructing local smooth approximations of the discontinuous likelihood estimates produced by a particle filter for a fixed number of particles J and fixed seed ω. These approximate the likelihood at θ in a neighborhood around ϕ, with the intent that differentiating these smooth log-likelihood reconstructions yields a score estimate.

Algorithm 7.2 computes the fixed-seed derivative of the likelihood estimate obtained by Algorithm 7.1 at $\theta = \phi$. To better understand the implementation in Algorithm 7.2, we review some background for AD. Standard AD consists of two evaluation passes: a forward pass in which the function is evaluated and a graph built of the evaluation process, followed by a backward pass where the derivative is evaluated by recursive use of the chain rule on the computation graph (Griewank and Walther, 2008). The two evaluation passes in Algorithm 7.1 at ϕ and θ correspond to the two AD evaluation passes in Algorithm 7.2. Differentiation of Algorithm 7.1 with respect to θ requires a preliminary evaluation at ϕ and subsequent differentiation with respect to θ for the second pass which we obtain through AD. In practice, however, we only work in the $\theta = \phi$ setting in which case Algorithm 7.1 corresponds to the standard particle filter and we only need to make one pass. In

order to apply AD in this setting without using two passes, we must both instantiate the weighting scheme (even though the weights are always 1) and distinguish which components of the weights correspond to which pass. This can be achieved using the *stop-gradient* capability that appears in Algorithm 7.2 where its application in the weight correction step indicates that AD should apply the chain rule to the numerator but not the denominator of the correction. Note that this construction yields weights which are always 1, but which also have non-zero derivatives.

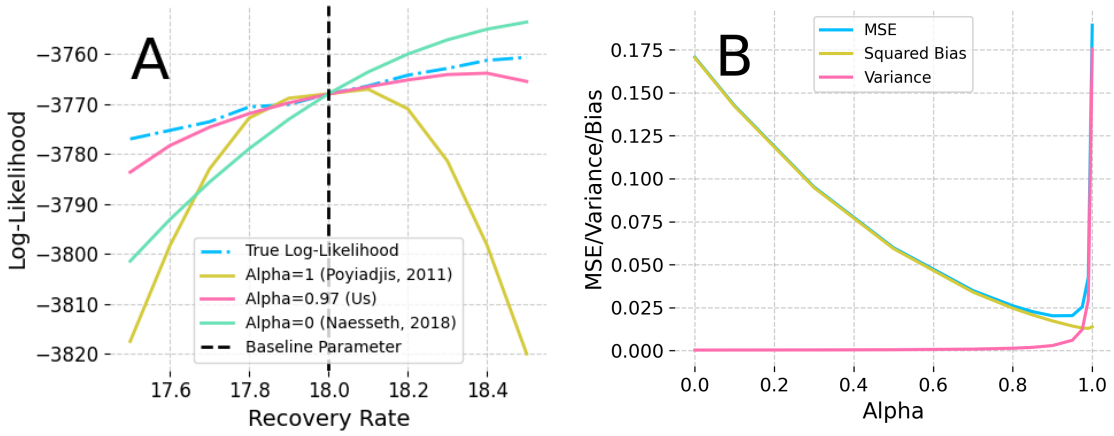


Figure 7.1: **A:** Non-differentiable likelihood estimates and smooth off-parameter approximations that hold locally around the baseline parameter ϕ. **B:** Illustration of the bias-variance tradeoff induced by the discounting hyperparameter α. We display the MSE of score estimates for the recovery rate at the MLE for the Dhaka cholera model of King et al. (2008). Our novel contribution of $\alpha \in (0, 1)$ significantly improves the quality of the likelihood approximation and the score estimates.

Algorithm 7.1 has similarities with the algorithm of Svensson et al. (2018), in which particles from a previous run at ϕ are reweighted to evaluate the likelihood at any sufficiently nearby parameter θ for 0-th order optimization. However, their method is not plug-and-play because they use transition densities to reweight particles, and the variance of their log-likelihood estimate has an exponential dependence in the horizon, N, when $\theta \neq \phi$. Our approach bypasses these issues: the first since the reparameterization trick allows us to consider particle paths dependent on θ, and the second as using AD allows us to only evaluate the likelihood and score estimates when $\theta = \phi$.

Algorithm 7.1 Fixed-Ancestor Weight-Discounted Off-Parameter Particle Filter

Input: Number of particles J, timesteps N, measurement model $f_{Y_n|X_n}(y_n^*|x_n,\theta)$, simulator $\texttt{process}_n(\cdot|x_n;\theta)$, evaluation parameter θ, baseline parameter ϕ, seed ω.
First pass: Set $\theta=\phi$ and fix ω, yielding $X_{n,j}^{P,\phi}$, $X_{n,j}^{F,\phi}$, $g_{n,j}^{\phi}$.
Second pass: Fix ω, and filter at $\theta\neq\phi$:
Initialize particles $X_{0,j}^{F,\theta}\sim f_{X_0}(\cdot\,;\theta)$, weights $w_{0,j}^{F,\theta}=1$.
For $n=1,...,N$:
Accumulate discounted weights, $w_{n,j}^{P,\theta}=\big(w_{n-1,j}^{F,\theta}\big)^{\alpha}$.
Simulate process model, $X_{n,j}^{P,\theta}\sim\texttt{process}_n\big(\cdot\,|X_{n-1,j}^{F,\theta};\theta\big)$.
Measurement density, $g_{n,j}^{\theta}=f_{Y_n|X_n}(y_n^*|X_{n,j}^{P,\theta};\theta)$.
Compute $L_n^{B,\theta,\alpha}=\sum_{j=1}^{J}g_{n,j}^{\theta}\,w_{n,j}^{P,\theta}\,/\sum_{j=1}^{J}w_{n,j}^{P,\theta}$.
Conditional likelihood under ϕ, $L_n^{\phi}=\frac{1}{J}\sum_{m=1}^{J}g_{n,m}^{\phi}$.
Select resampling indices $k_{1:J}$ with $\mathbb{P}\big(k_j=m\big)\propto g_{n,m}^{\phi}$.
Obtain resampled particles $X_{n,j}^{F,\theta}=X_{n,k_j}^{P,\theta}$.
Calculate corrected weights $w_{n,j}^{F,\theta}=w_{n,k_j}^{P,\theta}\,g_{n,k_j}^{\theta}\,/\,g_{n,k_j}^{\phi}$.
Compute $L_n^{A,\theta,\alpha}=L_n^{\phi}\cdot\sum_{j=1}^{J}w_{n,j}^{F,\theta}\,/\sum_{j=1}^{J}w_{n,j}^{P,\theta}$.
Return: likelihood estimate $\hat{\mathcal{L}}^A(\theta)=\prod_{n=1}^{N}L_n^{A,\theta,\alpha}$ or $\hat{\mathcal{L}}^B(\theta)=\prod_{n=1}^{N}L_n^{B,\theta,\alpha}$, log-likelihood $\hat{\ell}^A(\theta)=\log\big(\hat{\mathcal{L}}^A(\theta)\big)$ or $\hat{\ell}^B(\theta)=\log\big(\hat{\mathcal{L}}^B(\theta)\big)$, and filtering distributions $\{(X_{n,j}^{F,\theta},w_{n,j}^{F,\theta})\}_{n,j=1}^{N,J}$.

Algorithm 7.2 Differentiable Weight-Discounted Particle Filter

Input: Number of particles J, timesteps N, measurement model $f_{Y_n|X_n}(y_n^*|x_n,\theta)$, simulator $\texttt{process}_n(\cdot|x_n;\theta)$, parameter θ.
Forward pass: AD constructs a computation graph for the likelihood estimate, $\hat{\mathcal{L}}^A(\theta)=\prod_{n=1}^{N}L_n^{A,\theta,\alpha}$ or $\hat{\mathcal{L}}^B(\theta)=\prod_{n=1}^{N}L_n^{B,\theta,\alpha}$.
Backward pass: AD evaluates the gradient estimate, $\hat{\mathcal{D}}^A(\theta)$ or $\hat{\mathcal{D}}^B(\theta)$.
Initialize particles $X_{0,j}^{F,\theta}\sim f_{X_0}(\cdot\,;\theta)$, weights $w_{0,j}^{F,\theta}=1$.
For $n=1,...,N$:
Accumulate discounted weights, $w_{n,j}^{P,\theta}=\big(w_{n-1,j}^{F,\theta}\big)^{\alpha}$.
Simulate process model, $X_{n,j}^{P,\theta}\sim\texttt{process}_n\big(\cdot\,|X_{n-1,j}^{F,\theta};\theta\big)$.
Measurement density, $g_{n,j}^{\theta}=f_{Y_n|X_n}(y_n^*|X_{n,j}^{P,\theta};\theta)$.
Compute $L_n^{B,\theta,\alpha}=\sum_{j=1}^{J}g_{n,j}^{\theta}\,w_{n,j}^{P,\theta}\,/\sum_{j=1}^{J}w_{n,j}^{P,\theta}$.
Baseline conditional likelihood, $L_n^{\phi}=\texttt{stop_gradient}\big(\frac{1}{J}\sum_{m=1}^{J}g_{n,m}^{\theta}\big)$.
Select resampling indices $k_{1:J}$ with $\mathbb{P}\big(k_j=m\big)\propto g_{n,m}^{\theta}$.
Obtain resampled particles $X_{n,j}^{F,\theta}=X_{n,k_j}^{P,\theta}$.
Calculate corrected weights $w_{n,j}^{F,\theta}=w_{n,k_j}^{P,\theta}\,g_{n,k_j}^{\theta}\,/\,\texttt{stop_gradient}(g_{n,k_j}^{\theta})$.
Compute $L_n^{A,\theta,\alpha}=L_n^{\phi}\cdot\sum_{j=1}^{J}w_{n,j}^{F,\theta}\,/\sum_{j=1}^{J}w_{n,j}^{P,\theta}$.
Return: log-likelihood estimate $\hat{\ell}^A(\theta)=\log\big(\hat{\mathcal{L}}^A(\theta)\big)=\sum_{n=1}^{N}\log L_n^{A,\theta,\alpha}$ or $\hat{\ell}^B(\theta)=\log\big(\hat{\mathcal{L}}^B(\theta)\big)=\sum_{n=1}^{N}\log L_n^{B,\theta,\alpha}$, and gradient estimate $\hat{\mathcal{D}}^A(\theta)$ or $\hat{\mathcal{D}}^B(\theta)$.

7.3.1. Some algorithmic details

The resampling indices, which we write as $k_j \sim \text{Categorical}(g^{\phi}_{n,1}, ..., g^{\phi}_{n,J})$, are a function of ϕ for any value of θ. For the categorical distribution, we use systematic resampling (Arulampalam et al., 2002; King et al., 2016) which usually outperforms multinomial resampling. The finite-sample concentration result used for Theorem 7.2, however, is stated for multinomial resampling, and does not assert the same guarantee for the systematic-resampling implementation used in our numerical work.

In Algorithm 7.1, the conditional likelihoods are reweighted by a correction factor $w^{P,\theta}_{n,j}$ that accumulates over filtering steps to account for the resampling under ϕ. Writing $g^{\theta}_{n,j} = f_{Y_n|X_n}(y^*_n|X^{P,\theta}_{n,j};\theta)$ for the measurement density and $L^{\phi}_n = \frac{1}{J}\sum_{m=1}^{J} g^{\phi}_{n,m}$ for the conditional likelihood estimate under ϕ, we obtain two estimates of the conditional likelihood under θ,

$$L^{B,\theta,\alpha}_n = \frac{\sum_{j=1}^{J} g^{\theta}_{n,j}\, w^{P,\theta}_{n,j}}{\sum_{j=1}^{J} w^{P,\theta}_{n,j}}, \qquad L^{A,\theta,\alpha}_n = L^{\phi}_n \cdot \frac{\sum_{j=1}^{J} w^{F,\theta}_{n,j}}{\sum_{j=1}^{J} w^{P,\theta}_{n,j}}, \tag{7.1}$$

where we recursively correct the weights for each particle for the cumulative error incurred by the off-parameter resampling:

$$w^{P,\theta}_{n,j} = (w^{F,\theta}_{n-1,j})^{\alpha}, \quad w^{F,\theta}_{n,j} = w^{P,\theta}_{n,k_j}\, g^{\theta}_{n,k_j} \,/\, g^{\phi}_{n,k_j}, \quad w^{F,\theta}_{0,j} = 1. \tag{7.2}$$

The before-resampling estimate $L^{B,\theta,\alpha}_n$ is preferable in practice, as it has slightly lower variance than the after-resampling estimate $L^{A,\theta,\alpha}_n$, but the latter is useful in deriving properties of the gradient estimate of Algorithm 7.1 such as that in Theorem 7.1.

7.3.2. Discounted Weights and a Bias-Variance Tradeoff

Heuristically, α controls the exponentially decaying memory of Algorithm 7.1 and Algorithm 7.2. Importantly, α is irrelevant for Algorithm 7.1 at $\theta = \phi$ when the weights are always 1 and therefore also irrelevant for the function evaluation in the first pass of Algorithm 7.2, but crucially not in the second when the gradients are evaluated. If $\alpha = 1$, the gradient of the weights, $w^{P,\theta}_{n,j}$ and $w^{F,\theta}_{n,j}$, accumulates as n increases. This leads to numerical instability and high variance unless N is small.

For $\alpha < 1$, the weights from previous timesteps are discounted as in Equation (7.2), allowing for lower variance at the expense of bias in the gradient estimate.

When $\alpha = 1$, Algorithm 7.1 maintains complete memory of each particle's ancestral trajectory. Theorem 7.1 shows that the undiscounted score estimator (Algorithm 7.2 with $\alpha = 1$) has a similar form to the consistent but high-variance score estimate in the first algorithm of Poyiadjis et al. (2011). Unlike Poyiadjis et al. (2011), our algorithm is plug-and-play, replacing differentiation of the latent transition density in Poyiadjis et al. (2011) at $\alpha = 1$ with pathwise differentiation of the simulator.

When $\alpha = 0$, Algorithm 7.1 considers only single-step transition dynamics, and Theorem 7.1 shows that Algorithm 7.2 recovers the low-variance but asymptotically biased score estimate of Naesseth et al. (2018). Ścibior and Wood (2021) showed that this matches the gradient of a vanilla particle filter when resampling terms are dropped.

Theorem 7.1 (Endpoint Functional Forms for Algorithm 7.2)**.** *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the gradient estimate yielded by Algorithm 7.2 and use the after-resampling conditional likelihood estimate so that $\hat{\ell}(\theta) = \log \hat{\mathcal{L}}(\theta) = \log \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,*

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log f_{Y_n|X_n}\left(y_n^* | X_{n,j}^{F,\theta}; \theta\right),$$

where the derivative with respect to θ on the right hand side propagates the derivative of the particle locations, via the chain rule, for fixed ω. When $\alpha = 1$,

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* | X_{1:N,j}^{A,F,\theta}; \theta\right).$$

Here, $X_{1:N,j}^{A,F,\theta}$ is the ancestral trajectory of $X_{N,j}^{F,\theta}$ at times $t_1, \ldots, t_N$.

We defer the proof to Appendix F.1. The argument relies on a useful decomposition of the after-resampling estimate $L_n^{A,\theta,\alpha}$, repeated applications of the log-derivative identity that $\nabla_x \log(f(x)) =$

$(\nabla_x f(x))/f(x)$, and noting $\theta = \phi$ implies that $w_{n,j}^{P,\theta}$ evaluates to 1.

The result in Theorem 7.1 provides mathematical intuition for why Ścibior and Wood (2021)'s use of the stop-gradient procedure in the particle filter recovers the estimate of Poyiadjis et al. (2011). When $\theta = \phi$, the correction $g_{n,k_j}^{\theta}/g_{n,k_j}^{\phi}$ is equivalent to $g_{n,k_j}^{\theta}/\text{stop_gradient}(g_{n,k_j}^{\theta})$, and is implemented in this way in practice. As such, the use of the stop-gradient in Ścibior and Wood (2021) can be seen as the weight correction in Algorithm 7.1 for $\alpha = 1$ and $\theta = \phi$.

7.4. From Gradients to Maximum Likelihood Estimation

To design an effective procedure for likelihood maximization using Algorithm 7.2, we carry out iterated filtering then automatic differentiation (Algorithm 7.3). A basic version of this approach is presented in Algorithm 7.3. The second stage uses a symmetric positive-definite curvature matrix B_m whose eigenvalues are kept in a fixed interval $[c, C]$. A raw particle Hessian or a quasi-Newton matrix may be spectrally clipped to this interval, and the choice $B_m = I_p$ gives stochastic gradient descent (SGD). Algorithm 7.3 is convenient for theoretical analysis, but two further modifications are helpful in practice. Our best numerical results were obtained by putting the gradient estimate into the Adam optimizer (Kingma and Ba, 2015) together with learning rate decay. Numerical results are provided in Section 7.5.

Algorithm 7.3 Iterated Filtering Warm Start with Discounted Automatic-Differentiation Optimization

Input: Number of particles J, timesteps N, IF2 cooling schedule a_m, AD step size η, discount parameter α, spectral bounds $0 < c \leq C$, gradient tolerance ϵ, stopping multiplier σ, maximum AD iterations M, and initial value θ_0.
Run the initial IF2 search under cooling schedule a_m to obtain $\{\Theta_j, j = 1, ..., J\}$, and set $\theta_0 := J^{-1} \sum_{j=1}^{J} \Theta_j$.
For $m = 0, \ldots, M-1$:
  Using fresh random numbers, run Algorithm 7.2 to obtain $\hat{\ell}(\theta_m)$ and set $g_m := \nabla_\theta(-\hat{\ell}(\theta_m))$.
  If $\|g_m\|_2 \leq \sigma\epsilon$, **return** $\hat{\theta} := \theta_m$.
  Choose a symmetric raw curvature estimate $\widetilde{B}_m$ and clip its eigenvalues to $[c, C]$ to obtain B_m.
  Update $\theta_{m+1} := \theta_m - \eta B_m^{-1} g_m$.
Return $\hat{\theta} := \theta_M$.

The result below analyzes the specialization of Algorithm 7.3 that uses $\alpha = 1$, the after-resampling likelihood estimate, and multinomial resampling. These restrictions are needed for the stated finite-

sample probability bound.

The convergence of IF2 (and so the first stage of Algorithm 7.3) to a neighborhood of the MLE happens rapidly in practice. However, even when applying much larger amounts of Monte Carlo effort, IF2 fails to get much closer than 2 log-likelihood units of the maximum on this problem (Figure 7.3). For this example, IF2, without assistance from AD, struggles to reach the inferentially critical region within one log-likelihood unit of the maximum, even when additional computationally intensive techniques are used, such as repeated pruning and profiling.

For a fixed optimization budget, the AD stage has a high-probability linear convergence rate up to a neighborhood of the MLE that shrinks as the particle gradient accuracy improves, under the usual smoothness and strong convexity assumptions, as shown below.

Theorem 7.2 (Linear Convergence of Algorithm 7.3)**.** *Condition on the IF2 output θ_0 and write $f = -\ell$. Suppose f is γ-strongly convex and Γ-smooth on a convex region containing the second-stage trajectory, with unique minimizer θ^*, and that (A3) and (A5) hold uniformly there. Consider the $\alpha = 1$, after-resampling, multinomial specialization of Algorithm 7.3, using fresh particle randomness at each iteration and curvature matrices $cI_p \preceq B_m \preceq CI_p$.*

Fix $M \geq 1$, $\delta \in (0,1)$, $\epsilon > 0$, and $\beta \in (0,1)$, and choose $0 < \eta \leq c(1-\beta)/\Gamma$ and $\sigma \geq 2C/[c(1-\beta)]$. For J sufficiently large, as quantified in Appendix F.2, with conditional probability at least $1-\delta$ every nonstopping iteration up to M satisfies

$$\ell(\theta^*) - \ell(\theta_{m+1}) \leq q\{\ell(\theta^*) - \ell(\theta_m)\}, \qquad q = 1 - \frac{2\eta\beta\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \in (1/2, 1).$$

Thus, if no stop occurs, the final likelihood gap is at most $q^M\{\ell(\theta^*) - \ell(\theta_0)\}$, whereas if the algorithm stops at θ_τ, then $\|\nabla\ell(\theta_\tau)\|_2 \leq (\sigma+1)\epsilon$ and $\ell(\theta^*) - \ell(\theta_\tau) \leq (\sigma+1)^2\epsilon^2/(2\gamma)$.

The proof of Theorem 7.2 is a finite-horizon inexact-gradient argument. It retains the two-sided spectral control and the per-iteration probability accounting that are essential in the related subsampled-Newton analysis of Roosta-Khorasani and Mahoney (2016). Details are deferred to

Appendix F.2. This result shows that, over any prespecified finite optimization budget, particle gradient estimates can retain a linear rate until their Monte Carlo accuracy becomes the limiting factor. The theorem is stated for Algorithm 7.2 with $\alpha = 1$: an analogous result for $\alpha < 1$ additionally requires a high-probability bound on the discount bias relative to the exact score and is not implied merely by taking α close to one.

We therefore see that the second stage of Algorithm 7.3 converges linearly toward a Monte Carlo neighborhood of the MLE if (1) the log-likelihood surface is well-conditioned in a neighborhood of the MLE and (2) the first stage of Algorithm 7.3 successfully reaches a basin in which the theorem's assumptions hold. Within a neighborhood of the MLE that grows as more data become available (Ning et al., 2021), local quadratic approximations such as local asymptotic normality apply under general conditions (Le Cam and Yang, 2000). This suggests that the linear convergence may occur often in practice.

7.5. Application to a Cholera Transmission Model

The cholera transmission model that King et al. (2008) developed for Dhaka, Bangladesh, has been used to benchmark the performance of various POMP inference methods (Ionides et al., 2015; Fasiolo et al., 2016; Wycoff et al., 2026), and we employ it here for the same purpose. This model categorizes individuals in a population as susceptible, $S(t)$, infected, $I(t)$, and recovered, $R(t)$ and so is called an SIRS compartmental model. In this case, the compartment $R(t)$ is further subdivided into three recovered compartments $R^1(t)$, $R^2(t)$, $R^3(t)$ denoting varying degrees of cholera immunity. We write $P(t)$ for the total population, and M_n for the cholera deaths in each month. As in King et al. (2008), the transition dynamics follow a stochastic differential equation (SDE):

$$
\begin{aligned}
dS &= \big(k\epsilon R^k + \delta(P-S) - \lambda(t)S\big)\,dt + dP - (\sigma SI/P)\,dB,\\
dI &= \big(\lambda(t)S - (m+\delta+\gamma)I\big)\,dt + (\sigma SI/P)\,dB,\\
dR^1 &= \big(\gamma I - (k\epsilon+\delta)R^1\big)\,dt, \qquad \ldots\\
dR^k &= \big(k\epsilon R^{k-1} - (k\epsilon+\delta)R^k\big)\,dt,
\end{aligned}
$$

with Brownian motion $B(t)$, cholera death rate m, recovery rate γ, mean immunity duration $1/\epsilon$, standard deviation of the force of infection σ, and population death rate $\delta = 0.02$. The force of infection, λ_t, is modeled by splines $(s_j)_{j=1}^6$

$$\lambda_t = \exp\left(\beta_{\text{trend}}(t - t_0) + \sum_{j=1}^{6} \beta_j s_j(t)\right)\frac{I}{P} + \exp\left(\sum_{j=1}^{6} \omega_j s_j(t)\right),$$

where the coefficients $(\beta_j)_{j=1}^6$ model seasonality in the force of infection, β_{trend} models the trend in the force of infection, and the ω_j represent seasonality of a non-human environmental reservoir of disease. The measurement model for observed monthly cholera deaths is $Y_n \sim \mathcal{N}(M_n, \tau^2 M_n^2)$, where $M_n = m \int_{t_{n-1}}^{t_n} I(s)\, ds$ is the modeled cholera deaths in that month.

We numerically solve this SDE using Euler's method with a time step of 1/20 months, matching King et al. (2008). This example is a partially observed hypo-elliptical SDE, and methodology specific to this model class has been previously studied (Ditlevsen and Samson, 2019). Our method applies to a class of latent Markov processes larger than Gaussian SDEs, permitting the use of Lévy noise (Bhadra et al., 2011) or other discontinuities.

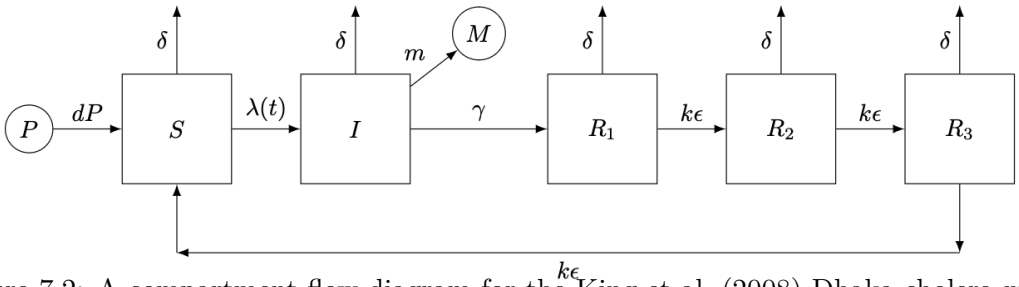


Figure 7.2: A compartment flow diagram for the King et al. (2008) Dhaka cholera model.

We tested Algorithm 7.3 with $\alpha = 0$, $\alpha = 0.97$ and $\alpha = 1$ against IF2 on a global search problem for the Dhaka cholera model using our implementation in Pypomp 1.0.5 (Abkemeier et al., 2025). For each method, we performed 100 searches, initialized with 100 starting parameter vectors drawn uniformly from the same wide bounding box used by Ionides et al. (2015). To ensure a fair comparison under comparable computational effort, we ran each method for a similar amount of time. For each method, we chose algorithmic parameters that maximized the log-likelihood well

for that method while finishing within the allotted time. We ran IF2 for 1950 iterations, while Algorithm 7.3 ($\alpha = 0.97$) is run using a short IF2 warm-start of 175 iterations followed by the Adam optimizer for 400 iterations. Algorithm 7.3 ($\alpha = 0$) and Algorithm 7.3 ($\alpha = 1$) were run for 60 IF2 iterations followed by 410 and 425 Adam iterations, respectively. The cooling rate used for IF2 was slower than that used for the warm start in Algorithm 7.3. To assess whether other methods can catch up to Algorithm 7.3 ($\alpha = 0.97$) given more time, we separately ran the IF2-only approach for 4 times as many iterations and the variants of Algorithm 7.3 with 4 times as many Adam iterations, slowing the cooling rate for each as well. A summary of the optimization settings is provided in Appendix F.7, and the complete source code to reproduce the numerical results is available at https://github.com/pypomp/dmop.

	Maximum	Re-evaluated	Median	IF2 time (s)	AD stage time (s)
Comparable computational effort					
Algorithm 7.3 ($\alpha = 0.97$)	-3743.68	-3744.26	-3744.40	92	851
Algorithm 7.3 ($\alpha = 1$)	-3744.36	-3744.40	-3746.84	35	902
Algorithm 7.3 ($\alpha = 0$)	-3748.80	-3749.02	-3751.62	36	912
IF2	-3749.00	-3748.97	-3754.24	946	-
Extended computational effort					
Algorithm 7.3 ($\alpha = 0.97$)	-3743.53	-3744.30	-3744.03	96	3557
Algorithm 7.3 ($\alpha = 1$)	-3743.96	-3744.33	-3745.39	35	3583
IF2	-3745.97	-3746.11	-3752.63	3782	-
Algorithm 7.3 ($\alpha = 0$)	-3748.32	-3748.38	-3750.37	35	3465

Table 7.1: Maximum log-likelihood found by IF2 and Algorithm 7.3 under both comparable and extended computational effort. The maximum is the largest of the 100 searches, and is biased upward by the winner's curse, while the re-evaluated column gives a fresh particle filter evaluation of that same search, which removes the selection bias at the cost of being a single noisy evaluation. Where the searches cluster tightly, as they do for Algorithm 7.3 ($\alpha = 0.97$) under extended effort, the re-evaluation regresses toward the median. Algorithm 7.3 ($\alpha = 0.97$) performs the best among all methods.

Our findings are summarized in Table 7.1 and Figures 7.3, 7.4, and 7.5. Because estimating the log-likelihoods at every iteration of IF2 with particle filtering for Figure 7.5 lengthens the computation time and thus compromises the fair comparison between methods, we obtained the times in Table 7.1 and Figure 7.5 by running the methods again without the extra calls to the particle filter. Every other value displayed for the Dhaka results comes from the runs with the extra particle filter calls.

Examining Table 7.1, Algorithm 7.3 ($\alpha = 0.97$) attained a log-likelihood of -3743.68, which surpassed IF2 even when IF2 ran for 4 times as long. Algorithm 7.3 ($\alpha = 1$) sometimes came close to Algorithm 7.3 ($\alpha = 0.97$), but its additional noise gave it worse median performance. Algorithm 7.3 ($\alpha = 1$) needed to be run for an extended time for the median log-likelihood to start catching up with Algorithm 7.3 ($\alpha = 0.97$).

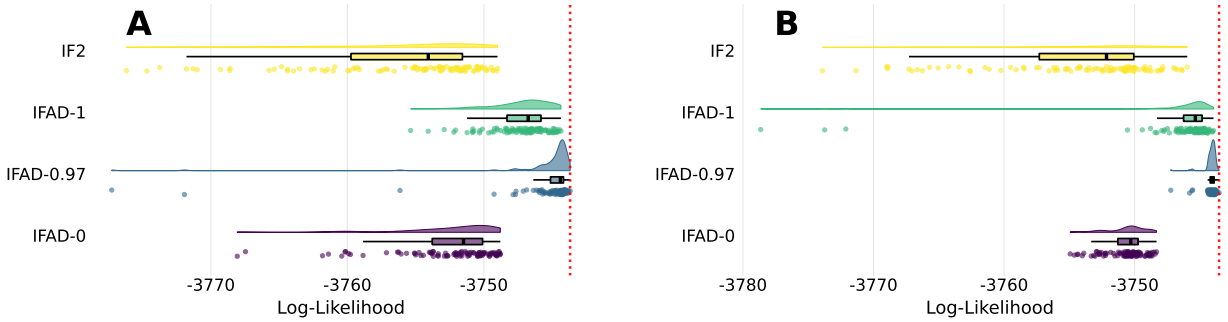


Figure 7.3: Raincloud plot of the searches. *(a)* Comparable computational effort. *(b)* Extended computational effort. Log-likelihoods lower than -3800 are omitted. Algorithm 7.3 ($\alpha = 0.97$) outperforms all other methods. The dotted red line shows the best-known maximized log-likelihood.

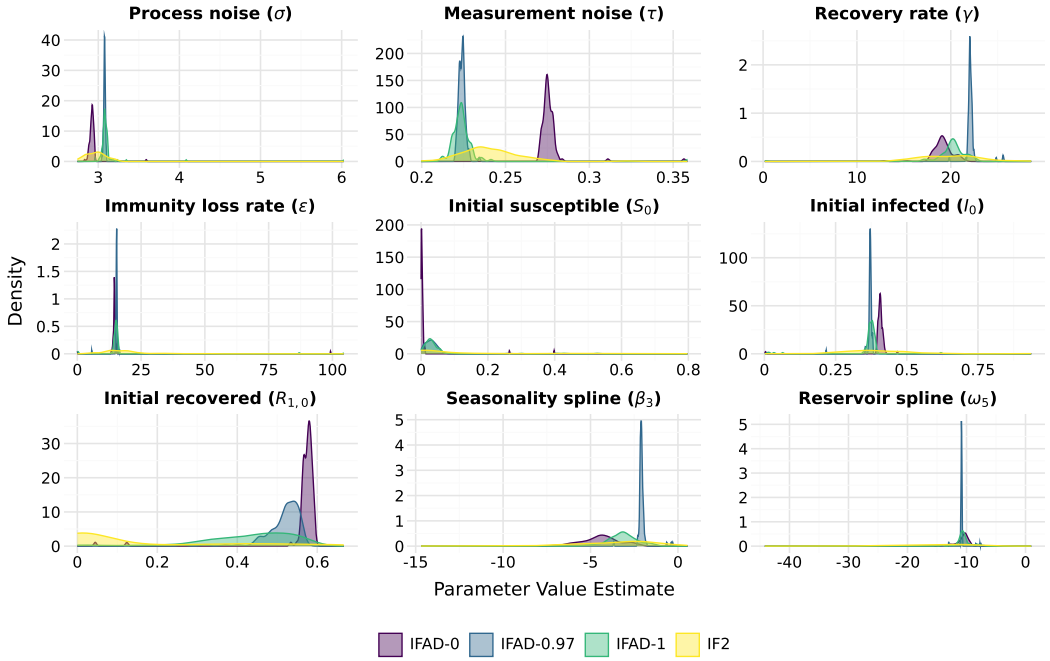


Figure 7.4: Density plots of the final parameter estimates across the 100 replicates for nine key parameters of the cholera model under extended computational effort.

Algorithm 7.3 ($\alpha = 0$) fell nearly 5 log-units short of Algorithm 7.3 ($\alpha = 0.97$) even with extended

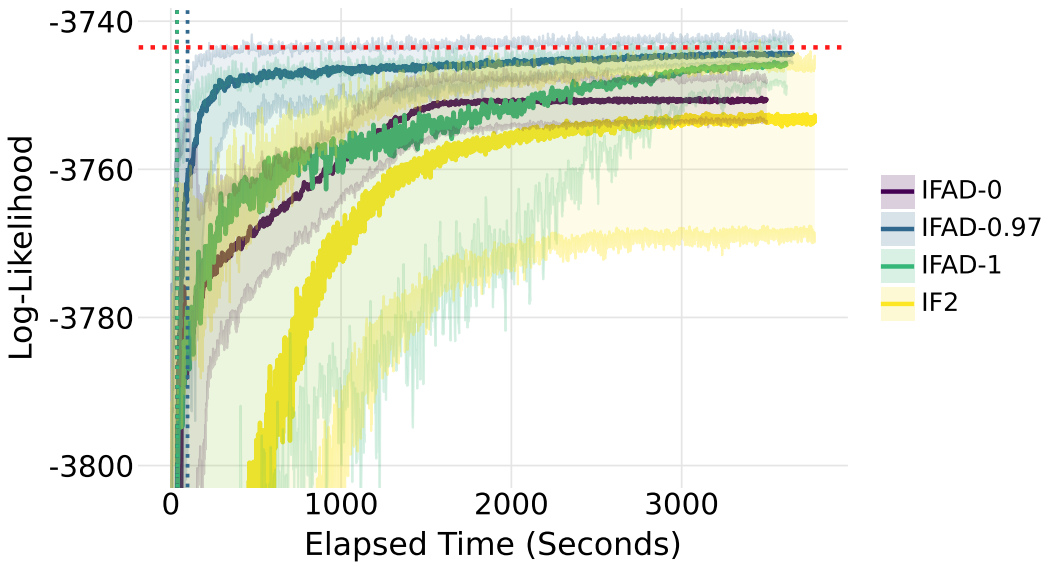


Figure 7.5: Optimization progress of Algorithm 7.3 and IF2 under extended computational effort. The shaded regions cover the area between the 100th and 10th percentiles of log-likelihood estimates at each iteration. We use a dotted red line to display the MLE. The vertical dotted lines indicate where the Algorithm 7.3 runs switch from IF2 to Adam optimization. Some log-likelihood estimates appear to be slightly above the MLE due to Monte Carlo evaluation error.

time, presumably due to its high bias. Based on Figure 7.5, Algorithm 7.3 ($\alpha = 0$) and IF2-only plateau heavily after being run for long enough and could not catch up to Algorithm 7.3 ($\alpha = 0.97$) within a reasonable timeframe.

Previously, a maximized log-likelihood of -3748.6 for this model and data was reported by King et al. (2016). This MLE was obtained with much computational effort, using many global and local IF2 searches and with the assistance of likelihood profiling, 8 years after the model was first proposed by King et al. (2008). Meanwhile, Ionides et al. (2015) only achieve a maximum log-likelihood of -3768.6, and King et al. (2008) only -3793.4. Using the improved computational efficiency of Pypomp (elaborated upon in Section 7.8), IF2 alone surpasses all of these previous results within a wall clock time of about an hour, while Algorithm 7.3 ($\alpha = 0.97$) does so in under 16 minutes and holds the record at -3743.53.

Notably, the log-likelihood and parameter estimates for Algorithm 7.3 ($\alpha = 0.97$) frequently exhibit much less Monte Carlo variance than that of the other methods, as can be seen in Figures 7.3 and 7.4.

Algorithm 7.3 in general exhibits less Monte Carlo variance than IF2, especially for the parameter estimates. Algorithm 7.3 ($\alpha = 0$) has the second-lowest Monte Carlo variance for most parameter estimates, although it has the lowest for the initial state proportions S_0, I_0 and $R_{1,0}$. That said, the choice $\alpha = 0$ is also known to have the greatest bias, and Algorithm 7.3 ($\alpha = 1$) agrees more closely with Algorithm 7.3 ($\alpha = 0.97$). Because multiple searches are employed to mitigate the effects of Monte Carlo variance, the lower Monte Carlo variance of Algorithm 7.3 ($\alpha = 0.97$) suggests that fewer searches are needed when running Algorithm 7.3 ($\alpha = 0.97$) to obtain a log-likelihood close to the maximum. Lowering the Monte Carlo variance also narrows the width of Monte-Carlo-adjusted profile confidence intervals (Ionides et al., 2017).

While IF2 effectively approaches a neighborhood of the MLE (Figure 7.5), IF2 alone ultimately fails to achieve the last few log-likelihood units, as IF2 struggles to get much closer than about 2 log-likelihood units of the MLE even when run for an extended time (Table 7.1). Conversely, we encountered many failed searches with gradient descent alone. This is a difficult, nonconvex, and noisy problem with 23 estimated parameters, and gradient descent without a warm-start gets stuck in local minima and saddle points.

Algorithm 7.3, by comparison, approaches the MLE quickly due to the IF2 warm-start (Figure 7.5) and also succeeds at refining the coarse solution found by the warm-start with gradient steps from Algorithm 7.2 to find the MLE. Algorithm 7.3 therefore successfully combines the best qualities of IF2 and Algorithm 7.2, outperforming either of them alone. Monte Carlo replication is appropriate for Algorithm 7.3, or any Monte Carlo algorithm used to solve a challenging numerical problem, but Figure 7.5 shows that Algorithm 7.3 can find higher likelihood values more quickly and more reliably than the previous state-of-the-art. Since replicated searches are inevitably needed for Monte Carlo maximization of this challenging likelihood maximization task, it is important to give extra consideration to the right tail of the distributions in Figure 7.3 when comparing the capabilities of the methods.

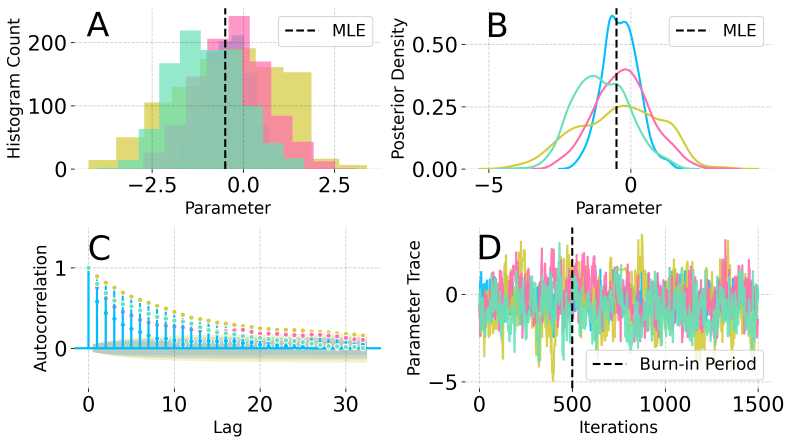


Figure 7.6: Convergence diagnostics for NUTS with 4 chains, where we plot 1000 samples from the posterior of β_{trend} within the Dhaka cholera model of King et al. (2008). **A:** Histogram of posterior samples. **B:** density estimate of the posterior. **C:** Autocorrelation plot for MCMC samples. **D:** Trace plot for MCMC samples, pre and post burn-in. NUTS mixes quickly when regularized by an empirical prior with a standard deviation of 6.3% of the MLE, and the posterior estimates from each chain share roughly the same posterior mode.

7.6. Application to Bayesian Inference

Particle MCMC (Andrieu et al., 2010) is arguably the most popular method for full-information plug-and-play Bayesian inference for POMP models. However, particle Metropolis-Hastings often mixes slowly, requiring long burn-in periods. For instance, Fasiolo et al. (2016) reported needing 700000 burn-in iterations for effective sampling in the King et al. (2008) model, even after coarsening each Euler timestep to reduce the computational burden. To alleviate this, we demonstrate a No U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014) powered by Algorithm 7.2, with a nonparametric empirical prior constructed from density estimation of the parameter swarm from the IF2 warm-start. Using NUTS with Algorithm 7.2 and this prior significantly lowers the mixing time of the Markov chain to as little as 500 iterations (Figure 7.6). NUTS has been implemented with previous differentiable particle filters (Rosato et al., 2022) but not in our plug-and-play context with control over the bias/variance tradeoff. In common with other methods using an approximate gradient within an MCMC algorithm such as NUTS, the approximation error of our algorithm propagates to affect the target distribution.

Without the IF2 prior, NUTS with the gradient estimate from Algorithm 7.2 alone tended to diverge. In contrast, particle Metropolis-Hastings, with or without said prior, fails to effectively explore the parameter space. We display these results in Appendix F.7. This speedup in the mixing time of particle MCMC by over three orders of magnitude provides hope that particle Bayesian inference might be increasingly employed for the complex scientific models commonly encountered in areas like epidemiology.

7.7. Theoretical Analysis of Algorithm 7.1 and Algorithm 7.2

Here, we will show that Algorithm 7.1 targets the filtering distribution and likelihood under θ, is consistent when $\alpha = 1$, and characterize rates for its bias and variance under different choices of α. To do so, we require the following assumptions:

(A1) **Continuity of the Likelihood.** $\ell(\theta)$ has more than two continuous derivatives in a neighborhood $\{\theta : \ell(\theta) > \lambda_1\}$ for some $\lambda_1 < \sup_\varphi \ell(\varphi)$.

(A2) **Bounded Process Model.** There exist $\underline{M}, \bar{M}$ such that $0 < \underline{M} \leq f_{X_n|X_{n-1}}(x_n|x_{n-1};\theta) \leq \bar{M} < \infty$.

(A3) **Bounded Measurement Model.** Uniformly in n, j, states, simulator draws, reference-filter resampling indices, and θ near ϕ, $0 < \underline{G} \leq f_{Y_n|X_n}(y_n^* \mid x_n;\theta) \leq \bar{G} < \infty$, $|\partial_{\theta_a} \log g_{n,j}^{A,F,\theta}| \leq G'(\theta) < \infty$, and $|\partial^2_{\theta_a\theta_b} \log g_{n,j}^{A,F,\theta}| \leq G''(\theta) < \infty$ for $1 \leq a, b \leq p$. The derivatives include the dependence of the simulated state on θ. The symbols G' and G'' denote separate derivative bounds, not derivatives of the function G in (A4).

(A4) **Bounded Gradient Estimates.** There are functions $G(\theta), H(\theta) : \Theta \to [0, \infty)$ uniformly bounded by $G^*, H^* < \infty$, such that the derivatives of Algorithm 7.1 at $\theta = \phi$ satisfy $\|\nabla_\theta \hat{\ell}_J^\alpha(\theta)\|_2 \leq G(\theta)$ and $\|\nabla^2_\theta \hat{\ell}_J^\alpha(\theta)\|_{\text{op}} \leq H(\theta)$ almost surely, uniformly in J and α. For $\alpha = 1$, these bounds hold on one probability-one event simultaneously for all J and all θ in a fixed closed ball K of positive radius centered at ϕ and contained in the interior of Θ. All estimates are coupled on one probability space, with the simulator draws and reference-filter resampling indices generated at ϕ held fixed as θ varies.

(A5) **Differentiable Density Ratios and Simulator.** For fixed ω, and any fixed collection of times, the simulated path and its log measurement densities have more than two continuous derivatives in θ.

(A6) **Measurement-sensitivity decay.** The simulator is C^1 in its incoming state. Conditioning on $X_m^\theta = x_m$, with ω fixed,

$$\left\| D_{x_m} \log f_{Y_n|X_n}\big(y_n^* \mid X_n^\theta; \theta\big) \right\|_2 \le C\kappa^{n-m}, \qquad C < \infty, \quad 0 < \kappa < 1.$$

The bound is uniform in $m \le n$, x_m, X_n^θ, ω, and θ near ϕ. The parameter derivatives of the initial state, simulator, and log measurement density are bounded, holding incoming states fixed.

(A6-J) **Alternative joint version of (A6).** The log measurement density is C^1 in (x, θ), and (A2) holds for θ near ϕ for the joint Markov process $(X_n^\theta, \partial_\theta X_n^\theta)$ relative to a common finite measure on its invariant state space.

Either (A6) or (A6-J) suffices for the particle bounds (see Appendix F.7 for the sufficient condition and example).

In particular, assumptions (A2) and (A3) imply that the POMP model satisfies a strong mixing condition, and as such are fairly strong but commonplace in the literature (Karjalainen et al., 2023; Del Moral, 2004). The fact that the likelihood estimate yielded by Algorithm 7.1 has more than two continuous derivatives in θ follows from the construction of the likelihood estimate in equations 7.1 and 7.2, as well as Assumptions (A3) and (A5).

We show that Algorithm 7.1 targets the filtering distribution under θ and is strongly consistent for the likelihood under θ when $\alpha = 1$ or $\theta = \phi$. Employing $\alpha < 1$ leads to inconsistency when θ deviates from ϕ, but this bias disappears as θ approaches ϕ.

While the result is presented here as specific to Algorithm 7.1, we prove a more general result in Appendix F.4. This is a strong law of large numbers for triangular arrays of particles resampled

arbitrarily, with resampling probabilities not proportional to the targeted distribution of interest, but reweighted to correct for the cumulative discrepancy between the resampling and the target distribution.

Theorem 7.3 (Algorithm 7.1 Targets the Filtering Distribution and Likelihood)**.** *When $\alpha = 1$ or $\theta = \phi$, Algorithm 7.1 targets the filtering distribution under θ and is strongly consistent for the likelihood under θ. That is, for $\pi_n(\theta) = f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$ and any measurable and bounded functional h and for $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$ or $\prod_{n=1}^N L_n^{B,\theta,\alpha}$, it holds that*

$$\frac{\sum_{j=1}^J h(x_{n,j}^{F,\theta})\, w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{F,\theta}} \overset{a.s.}{\to} E_{\pi_n(\theta)}\big[h(X_n)\big], \qquad \hat{\mathcal{L}}(\theta) \overset{a.s.}{\to} \mathcal{L}(\theta).$$

Proof. We provide a proof sketch here, deferring the details to Appendix F.4. When $\theta = \phi$, the ratio $g_{n,j}^\theta / g_{n,j}^\phi = 1$ regardless of α, reducing to the vanilla particle filter. When $\alpha = 1$ and $\theta \neq \phi$, suppose inductively that $\{(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta})\}_{j=1}^J$ targets $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y^*_{1:n-1};\theta)$. It can then be shown that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}_{j=1}^J$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$, that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\, g_{n,j}^\theta)\}_{j=1}^J$ targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$, and that weighting the particles by $(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}) = (X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta}\, g_{n,k_j}^\theta / g_{n,k_j}^\phi)$, resampling the k_j with probabilities proportional to $g_{n,j}^\phi$, also targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. Estimating the likelihood with the before-resampling estimator $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{B,\theta,\alpha}$, we see that the strong consistency is a direct consequence of our result that $\{\big(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\big)\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. □

We take advantage of the equivalence between Algorithm 7.2 and the derivative of Algorithm 7.1 to investigate the former by studying the latter. Although we have shown that the gradient estimate of Algorithm 7.2 at $\alpha = 1$ yields the estimate of Poyiadjis et al. (2011) and Ścibior and Wood (2021) when applied to the bootstrap filter, those authors estimate the Fisher score by $\frac{1}{J}\sum_{j=1}^J \nabla_\theta \log f_{X_{0:N},Y_{1:N}}(x_{0:n,j}^{A,F,\theta}, y^*_{1:N};\theta)$ and not $\frac{1}{J}\sum_{j=1}^J \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}(y^*_{1:N}|x_{1:n,j}^{A,F,\theta};\theta)$. It is therefore not immediately apparent that these two converge to the same thing. As such we directly show the consistency of the gradient estimate of Algorithm 7.1 at $\alpha = 1$ below.

Theorem 7.4 (Consistency of the Undiscounted Score Estimate)**.** *The gradient estimate of Algorithm 7.1 when $\alpha = 1$, $\theta = \phi$ is strongly consistent for the score: $\nabla_\theta \hat{\ell}_J^1(\theta) \overset{a.s.}{\to} \nabla_\theta \ell(\theta)$ as $J \to \infty$.*

We present a proof outline here, postponing the full argument to Appendix F.5. On the compact neighborhood K in Assumption (A4), Theorem 7.3 with $\alpha = 1$ gives pointwise almost-sure convergence of $\hat{\ell}_J^1(\theta)$ to $\ell(\theta)$, including when $\theta \neq \phi$. Intersecting these events over a countable dense subset of K, the gradient bound upgrades this to uniform convergence on K. The Hessian bound makes the gradients uniformly equicontinuous, so Arzelà–Ascoli gives uniformly convergent gradient subsequences. Passing to the limit in the fundamental theorem of calculus identifies each such limit as $\nabla_\theta \ell(\theta)$ in the interior of K. Thus the full gradient sequence converges locally uniformly there almost surely, and in particular $\nabla_\theta \hat{\ell}_J^1(\phi) \to \nabla_\theta \ell(\phi)$.

We now provide a result showing that Algorithm 7.2 has a desirable bias-variance tradeoff when $0 < \alpha < 1$. This is achieved because it combines favorable properties from the low-variance but asymptotically biased estimate of Naesseth et al. (2018) and the high-variance but consistent estimate of Poyiadjis et al. (2011). The result below bounds the particle error for $\alpha \in [0, 1]$ and the full-score MSE and population discount bias for $0 \leq \alpha < 1$.

Theorem 7.5 (MSE and Variance of Algorithm 7.2)**.** *Under (A2), (A3), (A5), either (A6) or (A6-J), fixed observations, multinomial resampling, $J \geq 2$, $\theta = \phi$, and $\alpha \in [0, 1]$, write $\nabla_\theta \ell^\alpha(\theta)$ for the deterministic population score of Algorithm 7.2 defined in Appendix F.7. Here, $\hat{\ell}^\alpha(\theta) = \sum_{n=1}^N \log L_n^{A,\theta,\alpha}$, as in Theorem 7.1. Expectations and variances are over the particle-filter randomness, and $\mathrm{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2$ for a random vector Z. Then*

$$\begin{aligned} &\max\left\{\mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2^2, \mathrm{Var}\{\nabla_\theta \hat{\ell}^\alpha(\theta)\}\right\} \\ &\lesssim pG'(\theta)^2 \left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1} \alpha^r\right)^2. \end{aligned} \tag{7.3}$$

For $0 \le \alpha < 1$, the full-score MSE satisfies

$$
\begin{aligned}
&\mathbb{E}\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2^2 \\
&\lesssim pG'(\theta)^2 \Bigg[\left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2 \\
&\qquad + \frac{(1-\sqrt{2(1-\varrho)})^2(1-\alpha)^2}{(1+\sqrt{2(1-\varrho)})^2} \left\{ N - 1 + \sum_{q=1}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho - 1}{3 - 2\varrho} \right)^q \right\}^2 \Bigg] \qquad (7.4) \\
&\lesssim pG'(\theta)^2 \Bigg[\frac{N(1+\log J)^2}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} \\
&\qquad + N^2 \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)} \right\}^2 \Bigg].
\end{aligned}
$$

Here $\varrho = 1 - \underline{M}/(2\bar{M}) \in [\frac{1}{2}, 1)$. The sum over q is empty for $N \le 2$, and the population terms vanish at $\varrho = \frac{1}{2}$. The constants suppressed by $\lesssim$ depend on the model and stability bounds, independently of N, J, α. The finite geometric sum is $(1-\alpha^N)/(1-\alpha)$ for $\alpha < 1$ and is N at $\alpha = 1$. The population part of the fixed-observation bias is bounded by

$$
\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2 \le 2(N-1)\sqrt{p}\, G'(\theta) \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}.
$$

For each fixed $\alpha < 1$ and $J \ge N$, the leading particle term is $NpG'(\theta)^2(1+\log J)^2/J$, multiplied by a constant depending on α.

We use the notation of Appendix F.7, where the full proof is given. Here, h_s is the time-s measurement gradient along an ancestral path, $h_{s,i}$ is its ith component, $m(\xi_t)$ is the empirical measure of the filtered simulator histories at time t, and ρ_t is the corresponding posterior law. The scores $\nabla_\theta \hat{s}_k^1(\theta)$ and $\nabla_\theta s_k^1(\theta)$ are the particle and population scores truncated at lag k, respectively. Assumptions (A3) and (A5) justify differentiation under finite-horizon expectations. Posterior backward-kernel contraction then bounds the population lag coefficients. For the particle term, the lag-truncated score $\nabla_\theta \hat{s}_k^1(\theta)$ telescopes by creation time. Under (A6), expand the time-s gradient by the chain rule and enlarge $G'(\theta)$ to bound the parameter derivatives in (A6) coordinatewise. Each age-r contribution is bounded by $CG'(\theta)\kappa^r$ and depends on a window starting at $\max(0, s-r-1)$.

At time $s+q$, this window contains at most $q+r+2$ times. Apply Theorems 3.3 and 3.5 of Chan et al. (2018) to these finite prediction genealogies. The Bayes-ratio expansion and conditional multinomial resampling give the filtered-average bounds. Lemma 2.3 of Karjalainen et al. (2023) controls the prediction cloud at the start of each window. Sum over ages using

$$\sum_{r\geq 0}\kappa^r\sqrt{q+r+2}\leq C_\kappa\sqrt{q+1},\qquad \sum_{r\geq 0}\kappa^r(q+r+2)\leq C_\kappa(q+1).$$

Under (A6-J), h_s is a bounded function of the current joint state. Either condition gives

$$\mathbb{E}|m(\xi_{s+q})(h_{s,i})-\rho_{s+q}(h_{s,i})|^2\leq\frac{C(q+1)G'(\theta)^2}{J},\qquad |\mathbb{E}m(\xi_{s+q})(h_{s,i})-\rho_{s+q}(h_{s,i})|\leq\frac{C(q+1)G'(\theta)}{J}.$$

Minkowski's inequality bounds the terminal second moment and bias by $C(k+1)^3G'(\theta)^2/J$ and $C(k+1)^2G'(\theta)/J$, respectively. Use Cauchy–Schwarz for overlapping windows and Davydov's inequality for separated windows, with gap $t-s-k-r-2$ under (A6) or $t-s-k-1$ under (A6-J). Summing covariances and then ages gives

$$\mathrm{Var}\{\nabla_\theta\hat{s}_k^1(\theta)\}\lesssim\frac{NpG'(\theta)^2(k+1)^2(1+\log J)^2}{J},\qquad \|\mathbb{E}\nabla_\theta\hat{s}_k^1(\theta)-\nabla_\theta s_k^1(\theta)\|_2\lesssim\frac{N\sqrt{p}\,G'(\theta)(k+1)}{J}.$$

Finally, set

$$\beta_k=(1-\alpha)\alpha^k\quad(0\leq k<N-1),\qquad \beta_{N-1}=\alpha^{N-1}.$$

Then

$$\nabla_\theta\hat{\ell}^\alpha(\theta)=\sum_{k=0}^{N-1}\beta_k\nabla_\theta\hat{s}_k^1(\theta)$$

and $\sum_k\beta_k(k+1)=\sum_{r=0}^{N-1}\alpha^r$ give the particle bounds by Minkowski's inequality.

Theorem 7.5 provides theoretical understanding of the empirically favorable bias-variance tradeoff enjoyed by Algorithm 7.2 in Figure 7.1. The MSE bound in Equation 7.4 separates particle approximation error from the squared population discount bias. The particle upper bound grows with the finite geometric sum, while the bound on the population bias tends to zero as α tends to 1.

The particle upper bound is

$$CpG'(\theta)^2 \left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2 .$$

Suppressing the common factor $pG'(\theta)^2$, every fixed $\alpha < 1$, including Algorithm 7.2 ($\alpha = 0$), has variance of order at most $N(1+\log J)^2/J$ and, for $J \geq N$, the same particle-MSE order. The constant increases with the discount memory. At $\alpha = 1$, the terminal second-moment calculation gives the possibly loose upper bound $\min\{N^2, N^3/J\}$, and the population discount bias is zero.

Theorem 1 of Poyiadjis et al. (2011) gives a quadratic lower bound for the central-limit asymptotic variance of their unnormalized additive path estimator. A separate generic norm-one path-space RMSE bound of order $N/\sqrt{J}$ becomes $N^2/\sqrt{J}$, and hence N^4/J in MSE, after rescaling by the $O(N)$ oscillation of the full unnormalized score. This norm-rescaling upper bound is separate from the Poyiadjis lower bound. The latter does not automatically transfer to Algorithm 7.2 at $\alpha = 1$, whose reparameterization contributions can differ from the likelihood-ratio contributions particle by particle. The assumptions give no MSE lower bound for Algorithm 7.2 at $\alpha = 0$. Appendix F.7 controls each bounded one-time contribution before summing lags, so its bounds require no further N^2 rescaling.

7.8. Computational Efficiency

Algorithm 7.2 and Algorithm 7.3 are fast algorithms, both in theory and practice: the cheap gradient principle of Kakade and Lee (2018) suggests that a gradient estimate from Algorithm 7.2 should take no more than 6 times that of the runtime of the particle filter. We found the empirical value of this factor to be 3.75, using the implementation of our methods in the Pypomp library (Abkemeier et al., 2025). Algorithm 7.2 and Algorithm 7.3 therefore share the same $O(NJ)$ time complexity as the particle filter, unlike the $O(NJ^2)$ complexity of Corenflos et al. (2021) and the $O(NJ^2)$ particle smoother of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Implementation of the particle filter, IF2, and Algorithm 7.2 in JAX (Bradbury et al., 2018) takes advantage of just-in-time compilation and GPU acceleration. While CPU-based implementations

of the particle filtering methods can effectively leverage multiple cores by parallelizing over independent replications of a given method, Pypomp can effectively leverage tens of thousands of GPU cores by vectorizing over both particles and independent method replications. To examine the performance improvement on the Dhaka model, we ran 1 instance of IF2 with 60 iterations and 5000 particles using 1 CPU core and compared this with running 100 replications on a GPU. The results in Table 7.2 show that, despite running 100 times as many replications, the GPU finishes 26.0 times faster, giving it a throughput 2597.2 times greater than the single CPU core.

Metric	`pomp` (King et al., 2016)	Pypomp (Abkemeier et al., 2025)
Language	R (C simulator)	Python (JAX)
Hardware	3.0 GHz Intel Xeon Gold 6154 (1 core)	NVIDIA RTX6000 Pro Blackwell
IF2 Replications	1	100
Latency (s)	935	36

Table 7.2: Comparison of execution latency for running IF2 (60 iterations, 5000 particles, 1 versus 100 replications) between `pomp` (CPU-only, R and compiled C) and Pypomp (GPU-accelerated, compiled Python).

7.9. Discussion

Practical likelihood-based data analysis involves many likelihood optimizations (King et al., 2008; Blake et al., 2014; Pons-Salort and Grassly, 2018; Subramanian et al., 2021; Fox et al., 2022; Drake et al., 2023). This occurs with profile likelihood confidence intervals and model selection. A careful scientist should consider many model variations, to see whether the conclusions of the study are sensitive to alternative model specifications. Coding model variations is relatively simple when using plug-and-play inference methodology. However, assessing the scientific potential of these variations requires likelihood optimization. Monte Carlo adjusted profile methods can help to accommodate maximization and evaluation error (Ionides et al., 2017; Ning et al., 2021), but these methods are most effective when the maximized log-likelihood can be obtained within a few units. Improving the process of model fitting has practical consequences for the ability of scientists to evaluate hypotheses. We have demonstrated, for the first time, a plug-and-play maximum likelihood approach which provides a high-quality log-likelihood maximization in the complex benchmark model of King et al. (2008). The *Daphnia* application in Appendix F.7 demonstrates that these gains also extend to

complex panel data. We anticipate that this will promote scientific advances in all domains where complex POMP models arise.

If the simulator is discontinuous in θ, Algorithm 7.2 no longer applies. We are working on a variation of Algorithm 7.2 applicable to this case. Specifically, differentiable transition density ratios can be used instead of a differentiable simulator. The off-parameter treatment of the measurement model is extended to an off-parameter simulation for the dynamic process. In that case, one requires access to the transition densities, or at least their ratios. This algorithm may be better suited to large discrete state spaces than either Algorithm 7.2 or differentiation of the Baum-Welch algorithm.

Discounting the weights by some $\alpha \in [0, 1]$ is not the only way to interpolate between the estimators of Naesseth et al. (2018) and Poyiadjis et al. (2011). As the proof of Theorem 7.5 implies, we can also truncate the weights at a fixed lag, corresponding to the fixed-lag $(1, k)$ variant of Algorithm 7.1 analyzed in Appendix F.7. The analysis is similar, with comparable rates but with a slightly less convenient implementation.

CHAPTER 8

CONTROL UNDER PARTIAL OBSERVABILITY

This chapter is based on joint work with Ziping Xu (Tan and Xu, 2026): Kevin Tan and Ziping Xu, "From POMDP Theory to Deep RL with Particle Filters," in Advances in Neural Information Processing Systems 39 (NeurIPS), 2026.

[RESOLVE: CHAPTER 8 COMPLETES THE PARTIAL-OBSERVABILITY PROGRESSION WITH CONTROL AND PLANNING (TAN AND XU, NEURIPS 2026).]

8.1. Introduction

Partially observable Markov decision processes (POMDPs) provide a standard framework for sequential decision making under latent state uncertainty (Kaelbling et al., 1998). They are a natural model for problems in robotics, healthcare, recommendation, and other settings (Hauskrecht and Fraser, 2000; Pineau et al., 2003) where the learner must act using only indirect and noisy observations of the underlying system. In principle, optimal control in a POMDP can be reduced to planning in the belief-state MDP, where the belief is the posterior distribution over latent states given the observation history (Cassandra et al., 1994; Kaelbling et al., 1998). This makes model-based RL especially natural in POMDPs: to act under a belief, the learner must already learn a model rich enough to perform filtering, and this same model can be used for planning.

However, the belief state reduction does not remove the main computational and learning difficulties. An online model-based decision maker must simultaneously (i) learn the unknown model, (ii) maintain a posterior over latent states online, and (iii) plan over this continuous belief space. These three tasks—*learning, inference, and planning*—interact in a way that makes exact online control in POMDPs intractable in both computation and sample complexity in general (Papadimitriou and Tsitsiklis, 1987) without special structure, motivating a large literature on *approximate* methods for each component individually.

Existing methods typically make one component of the above POMDP pipeline tractable while leav-

ing the others either computationally expensive or theoretically uncontrolled. For instance, on the planning side, Monte Carlo tree search has led to scalable online POMDP solvers such as Partially Observable Monte Carlo Planning (POMCP), sparse tree search, and progressive widening methods (Silver and Veness, 2010; Lim et al., 2021; Sunberg and Kochenderfer, 2018). However, these are primarily planning algorithms under a known generative model, and do not natively provide end-to-end learning guarantees for unknown latent dynamics. A more aggressive approximation is QMDP, which replaces the belief-MDP continuation value by the fully observed MDP value averaged under the current belief (Littman et al., 1995). QMDP is computationally efficient but can introduce substantial bias. Existing theory gives limited guidance on when this bias is small or how it can be mitigated. Regarding inference, particle filters, a.k.a. sequential Monte Carlo (SMC), provide a flexible approximation to Bayesian filtering in nonlinear non-Gaussian systems (Crisan and Doucet, 2002; Whiteley, 2013), but SMC theory for POMDPs (Lim et al., 2021, 2023) has not been fully integrated with approximate online planning and model learning. On learning, recent theoretical work has made significant progress on sample-efficient model estimation for POMDPs. Liu et al. (2022, 2023b) develop Optimistic MLE as a generic model-based algorithm with polynomial sample complexity for a broad class of POMDPs. These results establish statistical feasibility under exact or abstract planning oracles. They do not specify how to perform approximate filtering or planning with the learned model in a computationally practical way.

A separate practical line of work uses recurrent policies or differentiable planning modules. Recurrent model-free deep RL methods, such as GRU or LSTM-based PPO and Q-learning variants, can perform well empirically under partial observability, but their hidden states are difficult to interpret as beliefs and are hard to analyze theoretically. QMDP-net takes a more structured approach by embedding approximate Bayesian filtering and QMDP planning into a differentiable architecture (Karkus et al., 2017). This is closer in spirit to our goal: combine model learning, belief state inference, and planning in a way that resembles modern deep learning practice, while maintaining provable guarantees. However, the approximation error of learned variational or differentiable filtering is difficult to control. Our view is that using SMC as the filtering primitive yields a more principled bridge: it is compatible with learned latent dynamics and has well-developed stability

and approximation theory (Crisan and Doucet, 2002; Whiteley, 2013).

In summary, existing work advances individual components—planning, inference, or learning—but no current method provides a unified pipeline that combines all three with both computational practicality and end-to-end theoretical guarantees. This leaves open the central question

Can we design a practical model-based deep RL pipeline for POMDPs that still admits end-to-end theoretical guarantees?

8.1.1. Our Contributions

This chapter develops a practical and analyzable route from POMDP theory to model-based deep RL. We answer the central question by developing a model-based learning-inference-planning pipeline for POMDPs (Algorithm 8.1) that is compatible with deep RL practice. Our contributions are:

1. **A practical inference-learning-planning pipeline.** We propose an online algorithm that learns a latent dynamics model, tracks the current belief with particles, and plans under this belief. Each is modular and can be instantiated with standard deep RL primitives.

2. **A controlled alternative to full belief-tree planning.** Instead of treating the particle belief as the planning state, we plan particle-wise and average action-values of the fully observable MDP. This gives a practical middle ground between expensive belief-tree search and cheap but potentially biased QMDP. Our analysis shows that the bias decays geometrically after d steps of belief-MDP lookahead. For cheaper state-level planners, we show that QMDP is exact for the belief-linear component of the value and fails only through an uncertainty-sensitive residual, clarifying when the approximation is small.

3. **An end-to-end regret guarantee.** We prove a regret bound that decomposes performance loss into model estimation error, particle filtering error, Monte Carlo rollout error, and QMDP structural bias. The bound makes explicit how sample size, particle budget, rollout budget, and planning depth jointly control performance.

4. **A bridge to deep RL practice.** Compared with theoretical POMDP learning algorithms that rely on exact POMDP solvers or posterior sampling over parameters, our approach uses MPC-style online planning, approximate filtering, and critic bootstrapping. These are standard primitives in model-based deep RL, while remaining explicit enough to support theoretical analysis. We instantiate our algorithm in an MPC-based deep RL implementation and evaluate it on a variety of POMDP environments, demonstrating its practicality.

Overall, this chapter argues that particle filtering can serve as a practical bridge between POMDP theory and deep RL. It gives a belief representation that is explicit enough to analyze, flexible enough for nonlinear latent dynamics, and compatible with MPC-style planning and learned critics. This yields a route to online POMDP learning that is closer to modern model-based RL than exact belief-space planning, while providing a transparent theoretical account that lets users trace each source of approximation error and make informed design choices.

The rest of this chapter is organized as follows. Section 8.2 introduces the online learning problem and the regret criterion in the belief-state MDP. Section 8.3 presents Algorithm 8.1, describing the finite-depth belief-tree planning subroutine (Algorithm G.1) and the full online pipeline. Section 8.4 develops the error analysis: we decompose the regret into four traceable terms, analyze the structural QMDP bias and its variance interpretation, bound each error source, and combine them into the end-to-end regret guarantee. We conclude with case studies on LQG, contextual MDPs, and smooth-value POMDPs that illustrate when the QMDP approximation is benign.

8.2. Problem Formulation

We study online reinforcement learning in a partially observable environment with unknown dynamics. Since the latent state is hidden, the natural state variable for control is the belief, the posterior distribution of the latent state given the history. We begin by introducing the POMDP model and the associated belief-state dynamics. Figure 8.1 shows a particle approximation of such a belief for a partially observed double integrator (Section 8.5).

Parameterized POMDP model. For each parameter $\theta \in \Theta$, let $\mathcal{M}_\theta = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P_\theta, Z_\theta, r, \gamma, b_0)$ be an infinite-horizon discounted POMDP, where $\mathcal{S}$ is the latent state space, $\mathcal{A}$ is the action space,

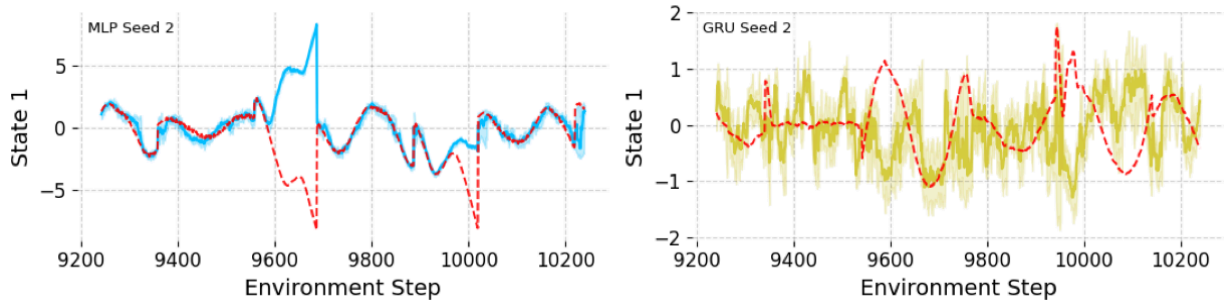


Figure 8.1: Particle beliefs for the unobserved velocity of a double integrator LQR over the last 1000 steps. Mean, 10th and 90th quantiles visualized. Beliefs track the state when the system is stable.

and $\mathcal{O}$ is the observation space. The controlled state transition kernel is $P_\theta(\cdot \mid s, a) \in \Delta(\mathcal{S})$, the observation kernel is $Z_\theta(\cdot \mid s') \in \Delta(\mathcal{O})$, the reward function is $r : \mathcal{S} \times \mathcal{A} \to [0, R_{\max}]$, the discount factor is $\gamma \in (0, 1)$, and $b_0 \in \Delta(\mathcal{S})$ is the initial belief. The true environment is specified by an unknown true parameter $\theta^\star$, and we write $\mathcal{M}^\star := \mathcal{M}_{\theta^\star}$. Here, for a measurable space $\mathcal{X}$, $\Delta(\mathcal{X})$ denotes the set of probability measures on $\mathcal{X}$, and beliefs $b \in \Delta(\mathcal{S})$ are written as measures $b(\mathrm{d}s)$ throughout.

Generative process and belief states. At each time $t = 0, 1, 2, \ldots$, the latent state, observation, and reward evolve according to

$$S_{t+1} \sim P_{\theta^\star}(\cdot \mid S_t, A_t), \qquad O_{t+1} \sim Z_{\theta^\star}(\cdot \mid S_{t+1}), \qquad R_t = r(S_t, A_t). \tag{8.1}$$

The learner does not observe S_t. Instead, decisions are based on the entire history up to time t denoted by $H_t := (O_{1:t}, A_{0:t-1}, R_{0:t-1})$. Here O_t is available before choosing A_t. A possibly randomized policy is a sequence $\pi = \{\pi_t\}_{t \geq 0}$ such that $\pi_t(\cdot \mid H_t) \in \Delta(\mathcal{A})$.

Belief states and Bayes filtering. Fix a candidate parameter θ. The corresponding belief state at time t is the posterior distribution $b_t^\theta(\cdot) := \mathbb{P}_\theta(S_t \in \cdot \mid H_t)$. When $\theta = \theta^\star$, we write $b_t := b_t^{\theta^\star}$ for the true belief. Given a belief $b \in \Delta(\mathcal{S})$ and action $a \in \mathcal{A}$, define the one-step predictive state distribution and the corresponding predictive observation law respectively by

$$\tilde{b}_\theta(\cdot \mid b, a) := \int_{\mathcal{S}} P_\theta(\cdot \mid s, a)\, b(\mathrm{d}s), \qquad p_\theta(\cdot \mid b, a) := \int_{\mathcal{S}} Z_\theta(\cdot \mid s')\, \tilde{b}_\theta(\mathrm{d}s' \mid b, a). \tag{8.2}$$

After observing $o \in \mathcal{O}$, the posterior is updated through the Bayes filter (which we later approximate)

$$\tau_\theta(b, a, o)(\mathrm{d}s) := \frac{Z_\theta(o \mid s)\, \tilde{b}_\theta(\mathrm{d}s \mid b, a)}{\int_{\mathcal{S}} Z_\theta(o \mid s')\, \tilde{b}_\theta(\mathrm{d}s' \mid b, a)}. \tag{8.3}$$

Belief-MDP value functions. The reward associated with belief b and action a is $r(b, a) := \mathbb{E}_{s\sim b}[r(s, a)]$. We assume throughout that $\mathcal{A}$ is finite and that $|r(s, a)| \le R_{\max}$ for all $(s, a) \in \mathcal{S} \times \mathcal{A}$.

For each θ, the POMDP induces a fully observed MDP on $\Delta(\mathcal{S})$. For any bounded measurable function $V : \Delta(\mathcal{S}) \to \mathbb{R}$, define the belief-MDP Bellman operator

$$(\mathcal{T}_\theta V)(b) := \max_{a\in\mathcal{A}} \left\{ r(b, a) + \gamma \int_{\mathcal{O}} V\big(\tau_\theta(b, a, o)\big)\, p_\theta(o \mid b, a)\, \mathrm{d}o \right\}. \tag{8.4}$$

Let $V_\theta^{\star,B}$ denote the unique fixed point of $\mathcal{T}_\theta$. The optimal belief-state action-value function is

$$Q_\theta^{\star,B}(b, a) := r(b, a) + \gamma \int_{\mathcal{O}} V_\theta^{\star,B}\big(\tau_\theta(b, a, o)\big)\, p_\theta(o \mid b, a)\, \mathrm{d}o, \text{ and } V_\theta^{\star,B}(b) = \max_{a\in\mathcal{A}} Q_\theta^{\star,B}(b, a). \tag{8.5}$$

More generally, for a policy π acting on histories, let $V_\theta^{\pi,B}(b)$ denote the corresponding discounted value in the belief-MDP when the initial belief is b. Furthermore, we let $Q_\theta^{\star,S}$ and $V_\theta^{\star,S}$ denote the value function of the fully observable MDP.

8.3. Online Particle Filtering and Belief-Tree Planning

We present Algorithm 8.1, a model-based online learning algorithm that integrates model learning, belief inference, and planning into a modular pipeline. We first divide time into doubling epochs $E_k = [2^{k-1}, 2^k)$, $k = 1, \dots, \lceil \log_2 T \rceil$, using the standard doubling trick (Besson and Kaufmann, 2018). At the start of each epoch, the algorithm (i) updates the MLE estimator $\bar{\theta}_k$ using all past data and freezes it for the epoch, (ii) maintains a particle approximation b_t^J of the current belief via sequential predict-reweight-resample, and (iii) plans under $\bar{\theta}_k$ and b_t^J using finite-depth Monte-Carlo belief-tree search with QMDP leaf evaluation. The doubling-epoch structure prevents the model from updating too frequently. We discuss each component respectively.

Learning oracle. Let $\mathcal{L}_t(\theta)$ denote the log-likelihood of the observed history H_t under model θ. At the start of each epoch E_k, the oracle ModelUpdate returns a β-approximate maximizer using all data up to T_k:

$$\bar{\theta}_k \in \big\{\theta \in \Theta : \mathcal{L}_{T_k}(\theta) \geq \max_{\theta' \in \Theta} \mathcal{L}_{T_k}(\theta') - \beta\big\}, \tag{8.6}$$

for a tolerance $\beta \geq 0$. The model $\bar{\theta}_k$ is frozen throughout epoch E_k. When $\beta = 0$ this is exact MLE, but we accommodate a β-suboptimal solver. In practice, recent work has produced a wide variety of methods for estimating the MLE in partially observed Markov process (POMP) models via gradient ascent, applicable here (Naesseth et al., 2018; Tan et al., 2024a; Chen et al., 2025; Ionides et al., 2015; Corenflos et al., 2021).

Filtering update. After playing a_t and observing o_{t+1}, particles are propagated and reweighted under the current epoch's model $\bar{\theta}_{k(t)}$:

$$\hat{s}_j^{(t+1)} \sim P_{\bar{\theta}_{k(t)}}(\cdot \mid s_j^{(t)}, a_t), \qquad w_j^{(t+1)} \propto Z_{\bar{\theta}_{k(t)}}(o_{t+1} \mid \hat{s}_j^{(t+1)}), \tag{8.7}$$

followed by multinomial resampling $s_j^{(t+1)} \sim \text{Categorical}(w_1^{(t+1)}, \ldots, w_J^{(t+1)})$ for $j = 1, \ldots, J$. This is the bootstrap sampling-importance-resampling (SIR) filter, and any SMC scheme satisfying the L^p error bound of Appendix G.3.3 can be substituted.

Approximate planning. For a given current belief, Algorithm G.1 runs a Monte Carlo tree search on the belief tree for the first d steps and applies the QMDP approximation at the leaf nodes. This gives a practical middle ground between expensive belief-tree search and the cheap but potentially biased pure QMDP approximation.

For a belief $b \in \Delta(\mathcal{S})$, define the QMDP leaf value and the depth-d belief tree value as

$$\bar{V}_\theta^{\star,S}(b) := \max_{a \in \mathcal{A}} \int_{\mathcal{S}} Q_\theta^{\star,S}(s,a)\, b(\mathrm{d}s), \quad \text{and} \quad \bar{V}_{\theta,d}^{\star,S} := \mathcal{T}_\theta^d\, \bar{V}_\theta^{\star,S}, \qquad d \geq 0, \tag{8.8}$$

where $Q_\theta^{\star,S}$ is the optimal action-value function of the fully observable MDP induced by θ, and $\mathcal{T}_\theta$ is the Bellman operator of the belief-MDP from (8.4). The corresponding action-value function is

Algorithm 8.1 Online Particle Filtering with Finite-Depth Monte Carlo Belief-Tree Planning. The belief-tree planner (Algorithm G.1) takes the particle belief b_t^J and returns an estimate $\widehat{Q}_t(b_t^J, a)$ of $Q_{\bar{\theta}_k,d}(b_t^J, a)$ for every root action $a \in \mathcal{A}$.

Require: particles J; depth d; rollouts N; horizon T; tolerance β; model class Θ; exploration parameter $\eta > 0$.

Sample $s_1^{(1)}, \dots, s_J^{(1)} \sim b_0$.
Set epochs $E_k = [2^{k-1}, 2^k)$ for $k = 1, \dots, K := \lceil \log_2 T \rceil$, with $\varepsilon_k := \eta/\sqrt{|E_k|}$.
for epoch $k = 1, 2, \dots, K$ **do**
    $\bar{\theta}_k \leftarrow \textsc{ModelUpdate}(H_{T_k})$ via (8.6); compute $\widehat{Q}^{\mathrm{MDP}}_{\bar{\theta}_k} \approx Q^{\star,S}_{\bar{\theta}_k}$.
    for $t \in E_k$ **do**
        $\{\widehat{Q}_t(b_t^J, a)\}_{a\in\mathcal{A}} \leftarrow \mathrm{MCTQ}\big(\bar{\theta}_k,\, b_t^J,\, d,\, N,\, \widehat{Q}^{\mathrm{MDP}}_{\bar{\theta}_k}\big)$ ▷ Algorithm G.1
        Play $a_t = \arg\max_a \widehat{Q}_t(b_t^J, a)$ w.p. $1 - \varepsilon_k$; else $a_t \sim \mathrm{Unif}(\mathcal{A})$.
        Observe (o_{t+1}, r_t).
        Propagate, reweight, and resample particles via (8.7) under $\bar{\theta}_k$.
    end for
end for

defined recursively as

$$Q_{\theta,d}(b,a) := r(b,a) + \gamma \int_{\mathcal{O}} \bar{V}^{\star,S}_{\theta,d-1}\big(\tau_\theta(b,a,o)\big)\, p_\theta(o \mid b, a)\, \mathrm{d}o, \qquad d \geq 1. \tag{8.9}$$

When $d = 0$, $\bar{V}^{\star,S}_{\theta,0}(b) = \max_a \int Q^{\star,S}_\theta(s,a)\, b(\mathrm{d}s)$ and Algorithm G.1 reduces exactly to QMDP. As $d \to \infty$, $\bar{V}^{\star,S}_{\theta,d} \to V^{\star,B}_\theta$ (see Proposition G.1 in Appendix G.3.1 for the proof).

Particle belief and MCTS approximation. In practice, the current belief is represented by a particle approximation $b_t^J = \frac{1}{J}\sum_{j=1}^J \delta_{s_t^{(j)}}$. Starting from b_t^J, Algorithm G.1 builds a search tree of depth d with N Monte Carlo simulations per root action, inspired by the POMCP framework (Silver and Veness, 2010). Each simulation samples a particle from b_t^J, descends to depth d using uniform random action selection at internal nodes, expands nodes as it passes through them, and records its state in the particle set $B(v)$ of every node v it visits. Each depth-d leaf is evaluated by the QMDP value of its pooled particles, $\max_{a'} |B(v)|^{-1} \sum_{s\in B(v)} Q^{\star,S}_\theta(s,a')$, and a bottom-up dynamic program then computes $\widehat{Q}_t(b_t^J, a)$ for each root action. Uniform selection is a proof device rather than a performance claim: the analysis only needs the internal rule to read no statistic of the current tree and to give every action probability at least $\pi_{\min} > 0$, so a frozen learned proposal also qualifies, whereas UCB1 does not (Appendix G.1).

Observation discretization. With continuous $\mathcal{O}$, every simulated observation is distinct, so tree nodes below the root are never revisited and a depth-d tree is not well defined. Algorithm G.1 therefore indexes children by a binned observation $\phi(o)$, where $\phi : \mathcal{O} \to \bar{\mathcal{O}}$ has $M := |\bar{\mathcal{O}}|$ bins of diameter $\mathrm{diam}(\phi)$. For finite $\mathcal{O}$ we take $\phi = \mathrm{id}$ and $M = |\mathcal{O}|$, and for $d = 0$ no binning is needed. The resulting bias and its trade-off against N are made explicit in Theorem 8.2.

8.4. Error Analysis and Regret Guarantee

We measure performance by the expected cumulative regret defined as:

$$\mathrm{Reg}_T := \mathbb{E}\left[\sum_{t=1}^{T} \mathrm{regret}_t\right] := \mathbb{E}\left[\sum_{t=1}^{T} \big(V_{\theta^\star}^{\star,B}(b_t) - V_{\theta^\star}^{\pi_t,B}(b_t)\big)\right], \tag{8.10}$$

where π_t is the policy deployed from round t, and the expectation is over the trajectory and algorithmic randomness. A key contribution is an end-to-end regret bound for Algorithm 8.1, enabled by the following novel regret decomposition. Let $k(t)$ denote the epoch containing round t, and $\bar{\theta}_{k(t)}$ be the frozen model during that epoch:

$$\mathrm{regret}_t = \underbrace{V_{\theta^\star}^{\star,B}(b_t) - V_{\bar{\theta}_{k(t)}}^{\star,B}(b_t)}_{\text{(A) optimal-value model error}} + \underbrace{V_{\bar{\theta}_{k(t)}}^{\star,B}(b_t) - V_{\bar{\theta}_{k(t)}}^{\pi_t,B}(b_t)}_{\text{(B) planning + state-id error}} + \underbrace{V_{\bar{\theta}_{k(t)}}^{\pi_t,B}(b_t) - V_{\theta^\star}^{\pi_t,B}(b_t)}_{\text{(C) fixed-policy model error}}, \tag{8.11}$$

Terms (A) and (C) capture model estimation error: the gap between $\bar{\theta}_{k(t)}$ and $\theta^\star$. Term (B) captures planning approximation under the learned model, which is further decomposed into three sources:

$$\text{(B)} \;\le\; \underbrace{\big|V_{\bar{\theta}_{k(t)}}^{\star,B}(b_t) - V_{\bar{\theta}_{k(t)}}^{\star,B}(b_t^J)\big|}_{\text{(B1) belief approx.}} + \underbrace{V_{\bar{\theta}_{k(t)}}^{\star,B}(b_t^J) - V_{\bar{\theta}_{k(t)}}^{\pi_t,B}(b_t^J)}_{\text{(B2) planning suboptimality}} + \underbrace{\big|V_{\bar{\theta}_{k(t)}}^{\pi_t,B}(b_t^J) - V_{\bar{\theta}_{k(t)}}^{\pi_t,B}(b_t)\big|}_{\text{(B1}') \text{ belief approx.}}. \tag{8.12}$$

Terms (B1) and (B1′) measure the cost of evaluating value functions at the particle belief b_t^J instead of the true posterior b_t, each bounded by $O(1/J^{1/(2\kappa_{\mathrm{pf}})})$ via particle-filter concentration. The planning gap (B2) further splits into Monte Carlo error from finite MCTS rollouts ($O(1/\sqrt{N})$) and structural error from QMDP bootstrapping at depth d.

Three objects. The analysis separates the belief-MDP optimum $V_\theta^{\star,B}$, the deterministic *planning*

target $\bar{V}^{\star,S}_{\theta,d} = \mathcal{T}^d_\theta \bar{V}^{\star,S}_\theta$ of (8.8), and the randomized estimate $\widehat{Q}_t(b^J_t, a)$ that Algorithm G.1 computes from pooled particle sets. The target differs from the optimum by the structural error, which decays geometrically in d and involves no particles, whereas the estimate differs from the target by the Monte Carlo error, which grows with d, so for a fixed N the bound is U-shaped in the depth. Each component enters only through an interface (a β-approximate MLE oracle, an SMC filter with an L^p bound, a planner with a Monte Carlo error bound), so the decomposition survives substituting, e.g., POMCP-style planners or variational filters.

8.4.1. Assumptions and main result

Modern SMC analyses (Del Moral, 2004; Chopin and Papaspiliopoulos, 2020; Karjalainen et al., 2023) impose rather unavoidable but stringent conditions on the transition kernel and observation density. We inherit these in the form of the following Doeblin and bounded observation density assumption (Assumption 8.1) on the latent dynamics. This choice aligns well with our aim to provide an end-to-end regret bound that decomposes error across the learning–inference–planning pipeline, interpolating between theory and practice, rather than to develop optimal exploration methods for POMDP model estimation, as Assumption 8.1 ensures that simple exploration strategies like ε-greedy have sufficient coverage. At the same time, the Doeblin condition resembles commonplace strong mixing assumptions within the RL literature (Gaur et al., 2024; Wu et al., 2022b; Xu et al., 2020; Kausik et al., 2023) and the α-revealing condition commonplace in the POMDP literature plays a similar role to the bounded observation density condition (Liu et al., 2022, 2023b).

Assumption 8.1 (Uniform mixing and bounded weights)**.**

1. *Uniform Doeblin condition.* There exist $\varepsilon_{\mathrm{mix}} > 0$ and a probability measure $\nu \in \Delta(\mathcal{S})$ such that $P_\theta(\cdot \mid s, a) \geq \varepsilon_{\mathrm{mix}}\, \nu(\cdot)$ for all $(\theta, s, a) \in \Theta \times \mathcal{S} \times \mathcal{A}$.

2. *Bounded observation density.* There exist $0 < g_{\min} \leq g_{\max} < \infty$ such that $g_{\min} \leq Z_\theta(o \mid s) \leq g_{\max}$ for all $(\theta, s, o) \in \Theta \times \mathcal{S} \times \mathcal{O}$.

Let $\mathcal{N}_\Theta(\varepsilon)$ denote the ε-covering number of Θ, i.e. the minimum number of ℓ_1-balls of radius ε (in the metric defined as trajectory level likelihood) needed to cover Θ. Define the effective model

complexity $\beta_T := \beta + \log\big[\mathcal{N}_\Theta(1/T)\,T/\delta\big]$, where β is the MLE tolerance from (8.6) and $\delta \in (0,1)$ is a confidence parameter.

Theorem 8.2 (End-to-end regret of Algorithm 8.1)**.** *Suppose Assumption 8.1 holds. Algorithm 8.1, run with exploration parameter* $\eta \asymp T^{1/4}(\beta_T/\log T)^{1/4}$ *and observation binning* ϕ *with* M *bins of diameter* $\operatorname{diam}(\phi) = \omega_Z^{-1}\big((1-\gamma)g_{\min}/(2\gamma g_{\max} V_{\max}\sqrt{N})\big)$ *($\phi = \mathrm{id}$ and $M = |\mathcal{O}|$ when $\mathcal{O}$ is finite), and, if $d \geq 2$, N above the threshold of Remark G.14, and* $\kappa_{\mathrm{pf}} := \max\big(1, \log\big(2g_{\max}^2/g_{\min}^2\big)/\log(1/\gamma)\big)$ *satisfies*

$$\mathrm{Reg}_T \;\leq\; \underbrace{\widetilde{O}\!\left(\frac{C_{\mathrm{est}}\,R_{\max}}{(1-\gamma)^{3/2}}\,T^{3/4}\right)}_{\textit{model + exploration + filter}} + \underbrace{\frac{C_{\mathrm{pf}}\,R_{\max}}{(1-\gamma)^2}\cdot\frac{T}{J^{1/(2\kappa_{\mathrm{pf}})}}}_{\textit{particle approx.}} + \underbrace{\frac{C_{\mathrm{mc}}\,R_{\max}}{(1-\gamma)^2}\cdot\frac{T}{\sqrt{N}}}_{\textit{Monte Carlo}} + \underbrace{\frac{2\gamma^{d+1}}{(1-\gamma)^2}\sum_{t=1}^{T}\Xi_{\bar\theta_{k(t)}}}_{\textit{QMDP structural bias}},$$

Here $\widetilde{O}$ hides factors polylogarithmic in T, ω_Z is the modulus of continuity of the observation likelihood ratio (Appendix G.3.2), $M = |\bar{\mathcal{O}}|$ is the number of observation bins, and $\Xi_{\bar\theta_{k(t)}}$ is the QMDP structural bias of (8.13). The coefficients are

$$C_{\mathrm{est}} = \mathrm{Poly}(|\mathcal{A}|, g_{\max}/g_{\min}, \epsilon_{\mathrm{mix}}^{-1}, \log(1/\gamma), \log\mathcal{N}_\Theta(1/T), \beta), \quad C_{\mathrm{mc}} = O(d\gamma^d|\mathcal{A}|^{(d+3)/2}M^{d/2}),$$
$$C_{\mathrm{pf}} = \widetilde{O}\big(V_{\max}^2\, g_{\max}/(\varepsilon_{\mathrm{mix}}^3\, g_{\min})\cdot(\log J)^{3/2}\sqrt{\log(\mathcal{N}_\Theta(1/T)\,|\mathcal{A}|)}\big) \quad (\text{See (G.43)}).$$

Reading the bound. Theorem 8.2 bounds cumulative regret with four major terms. The first term captures model estimation error, exploration cost, and filter mismatch between the true posterior $b_t^{\theta^\star}$ and the particle belief $b_t^{\bar\theta_{1:K}}$. The $\widetilde{O}(T^{3/4})$ rate arises from the coverage transfer needed to bound the optimal-value model error (term (A) in (8.11)) via ε-greedy exploration. Terms 2, 3, and 4 capture the total approximate planning errors that may scale linearly in T. They can be diminished by increasing the number of particles J, number of Monte Carlo rollouts N, and the planning depth d. The particle error decays as $J^{-1/(2\kappa_{\mathrm{pf}})}$ at cost $O(J)$, the Monte Carlo error as $N^{-1/2}$ at cost $O(N)$, and the structural error as γ^d at a tree size of up to $(|\mathcal{A}|M)^d$ nodes. The bound therefore exposes a **sample-vs.-computational efficiency trade-off** through the

marginal return of each budget, but a concrete allocation requires knowledge of κ_{pf} and Ξ_θ and is not an optimality statement. Section 8.5.1 empirically validates each of these four scaling laws on a Doeblin-mixing POMDP satisfying Assumption 8.1 and demonstrates how the decomposition guides the choice of (J, N, d) under a target per-term error tolerance. With continuous observations, the binning keeps the discretization bias at the Monte Carlo order $O(V_{\max}/\sqrt{N})$, and the price is paid in the constant: M grows as $N^{d_o/2}$ for $\mathcal{O} \subseteq \mathbb{R}^{d_o}$, so C_{mc} grows exponentially in $d \cdot d_o$, whereas balancing bin width against N instead gives the rate $\widetilde{O}(N^{-1/(d\,d_o+2)})$ (Remark G.15). This cost is absent only at $d = 0$, the regime used in our deep RL experiments.

Technical challenges. Bounding each source in (8.12) under online learning raises three difficulties. First, controlling (B1) requires working with a test-function class defined directly over $\mathcal{S}$, rather than only the parametric family $\{Q_\theta^{\star,B}\}_\theta$. This is because the belief-space value functions may lack the regularity needed for uniform particle-filter error control. In fact, the planning-induced class over multiple steps is effectively uncountable over the possible model parameters, so we control it through truncation. The required truncation level depends on the desired particle approximation accuracy: as the particle error target decreases, the tail must be made smaller, which enlarges the truncated function class. This creates a tradeoff between the complexity of the truncated class and the particle ensemble size, leading to a slower $(1/J^{1/2\kappa_{\mathrm{pf}}})$, essentially nonparametric, rate that emphasizes the technical challenge of the problem.

Second, the filtering analysis must account for the fact that the data are generated online by a changing policy. Classical particle-filter stability assumes a fixed generative model, whereas here the filter receives actions from a non-stationary, improving policy. We extend time-uniform L^p bounds (Del Moral, 2004) to this setting by exploiting the Doeblin condition uniformly over actions. Third, since Doeblin coverage applies only to states and not actions, we further ensure action coverage via ε-greedy exploration. This enables us to transfer the MLE concentration result across policies, controlling both terms (A) and (C), and to control the Monte Carlo error in (B2).

Remark 8.3 (Where the hardness of POMDPs goes). POMDP sample complexity may in the worst case scale exponentially in the effective horizon $1/(1-\gamma)$. Our bound is polynomial in $1/(1-\gamma)$

for two reasons. First, under the Doeblin condition the Bayes filter forgets its initialization at a geometric rate $\rho = 1 - \varepsilon_{\text{mix}} g_{\min}/g_{\max} < 1$ uniformly over actions, so errors add up over time rather than compound. Second, bounded observation densities play the role of the α-revealing condition (Liu et al., 2022): they bound importance weights, so data transfer across policies without an $(|\mathcal{A}|/\varepsilon)^H$ blow-up, and particle weights stay non-degenerate. This does not circumvent the lower bound: $\varepsilon_{\text{mix}}^{-1}$ and $g_{\max}/g_{\min}$ can be exponential in the effective horizon for hard instances, and C_{est} and C_{pf} depend polynomially on them. Appendix G.4 provides controlled stress tests illustrating the mechanisms by which violations of $g_{\min} > 0$ and large multi-step information-gathering gaps $\Xi(\theta_\star)$ affect filtering and shallow planning.

Remark 8.4 (Bypassing exploration with offline coverage). The $\widetilde{O}(T^{3/4})$ rate is driven by the coverage transfer for term (A), which requires ε-greedy exploration to ensure model accuracy at the optimal policy. If instead the agent has access to N_{off} offline trajectories from a behavior policy π_b with single-policy concentrability $c^\star_{\text{off}}$ (the worst-case ratio of trajectory ℓ_1 distances between $\pi^\star$ and π_b), then greedy exploitation ($\varepsilon_t = 0$) achieves regret $\widetilde{O}(\sqrt{T}/(1-\gamma)) + O(c^\star_{\text{off}} T/(\sqrt{N_{\text{off}}}(1-\gamma)))$ plus computational terms. When $N_{\text{off}} \geq (c^\star_{\text{off}})^2 T$, the total regret is $\widetilde{O}(\sqrt{T})$ without any additional online exploration. The formal statement and proof are in Appendix G.4.

8.4.2. Understanding the structural bias

It remains to discuss the QMDP approximation error, which is governed by the *Jensen gap*

$$\Xi_\theta := \sup_{b \in \Delta(\mathcal{S})} \left(\int_{\mathcal{S}} V_\theta^{\star,S}(s)\, b(\mathrm{d}s) - \max_{a \in \mathcal{A}} \int_{\mathcal{S}} Q_\theta^{\star,S}(s,a)\, b(\mathrm{d}s) \right), \tag{8.13}$$

which measures the value lost by choosing a single action for all states (QMDP planning) rather than the state-dependent optimal action. We show (Proposition G.1 in Appendix G.3.1) that the depth-d belief-tree structural error of Algorithm G.1 satisfies $\|\bar{V}_{\theta,d}^{\star,S} - V_\theta^{\star,B}\|_\infty \leq \frac{\gamma^{d+1}}{1-\gamma}\Xi_\theta$, decaying geometrically with depth d.

The structural quantity Ξ_θ admits a natural interpretation in terms of posterior *variance* of value-relevant features. Let $g : \mathcal{S} \to \mathbb{R}^k$ be a measurable embedding, and write $m_g(b) := \mathbb{E}_{S\sim b}[g(S)]$ and

$\Sigma_g(b) := \mathrm{Cov}_{S\sim b}(g(S))$.

Assumption 8.5 (Value-relevant embedding)**.** There exist twice continuously differentiable maps $F_\theta^a : \mathbb{R}^k \to \mathbb{R}$ for each $a \in \mathcal{A}$ such that $Q_\theta^{\star,S}(s,a) = F_\theta^a(g(s))$ for all $(s,a) \in \mathcal{S} \times \mathcal{A}$, with curvature bounds, where $F_\theta(x) := \max_a F_\theta^a(x)$ and $V_\theta^{\star,S}(s) = F_\theta(g(s))$, of

$$\sup_{a\in\mathcal{A}} \sup_{x\in\mathrm{conv}(g(\mathcal{S}))} \left\|\nabla^2 F_\theta^a(x)\right\|_{\mathrm{op}} \le L_\theta \qquad \text{and} \qquad \sup_{x\in\mathrm{conv}(g(\mathcal{S}))} \left\|\nabla^2 F_\theta(x)\right\|_{\mathrm{op}} \le L_\theta,$$

where $\|\cdot\|_{\mathrm{op}}$ is the spectral norm.

In practice, the condition holds when either (i) a single action is optimal throughout $\mathrm{conv}(g(\mathcal{S}))$, or (ii) F_θ is a single smooth function of s (as in linear-quadratic-Gaussian (LQG) systems, where it is a quadratic form and both conditions hold with the same L_θ).

Proposition 8.6 (Curvature bound on Ξ_θ)**.** *Under Assumption 8.5, for every belief $b \in \Delta(\mathcal{S})$,*

$$\Xi_\theta(b) := \widetilde{V}_\theta^\star(b) - \bar{V}_\theta^{\star,S}(b) \;\le\; L_\theta \,\mathrm{tr}\big(\Sigma_g(b)\big), \tag{8.14}$$

where $\widetilde{V}_\theta^\star(b) := \int_{\mathcal{S}} V_\theta^{\star,S}(s)\, b(\mathrm{d}s)$ is the linear QMDP surrogate. In particular, $\Xi_\theta = \sup_b \Xi_\theta(b)$.

The proof is given in Appendix G.4. That QMDP is accurate once the belief concentrates is expected; Proposition 8.6 is sharper in two ways. First, only the posterior covariance of the value-relevant features g matters, so uncertainty in directions to which the value is insensitive costs nothing and concentration of the full belief is not necessary. Second, the curvature L_θ converts this covariance into a value gap, which then enters Theorem 8.2 as $\gamma^{d+1}\Xi_\theta/(1-\gamma)$ and is traded off against planning depth. In discounted LQG, for instance, the Kalman covariance converges to a strictly positive steady state and never vanishes, yet the QMDP policy is belief-optimal, and $\Xi_\theta = 0$ when the control does not enter the dynamics (Appendix G.4).

Appendix G.4 instantiates the structural analysis on linear-quadratic Gaussian systems, where Ξ_θ has a closed form in terms of the Riccati solution and the steady-state Kalman covariance,

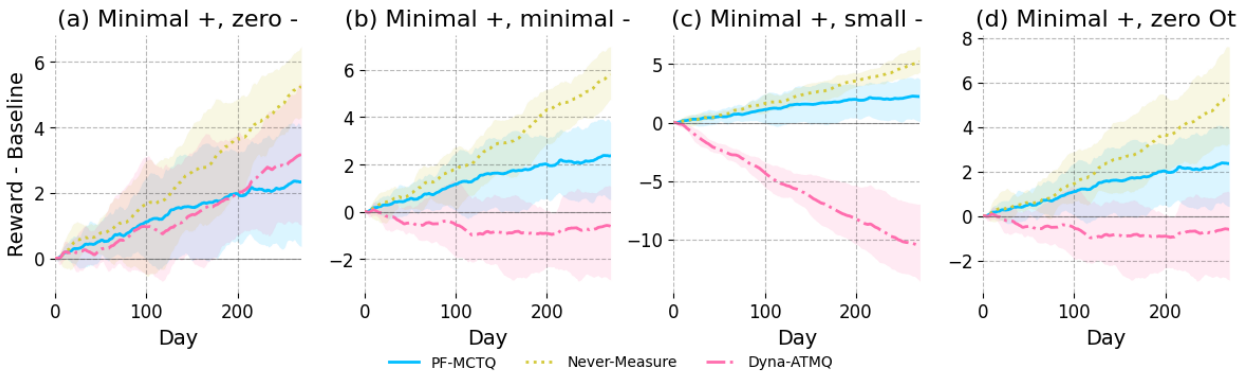


Figure 8.2: Cumulative reward above the baseline rollout across four HeartSteps scenarios, where the positive intervention effect and the delayed negative measurement effect or proxy O_t informativeness are varied. Mean and 2.5% and 97.5% quantiles displayed on five seeds.

and on contextual MDPs with identifiable latent context, where $\Xi_{\theta,t} \leq 2V_{\max}\rho_t$ for a context-misidentification probability ρ_t, which turns "eventually identified" into the rate condition $\sum_t \rho_t = o(\sqrt{T})$.

8.5. Experiments

Our experiments test whether particle filtering and model-based planning can be used as a practical layer on top of modern deep reinforcement learning under partial observability. Here, our goal is not to obtain state-of-the-art asymptotic performance. Instead, we aim to evaluate whether explicit particle beliefs can improve low-sample decision making with modest additional machinery. Table 8.2 states which experiments fall under the theory: the deep RL experiments demonstrate that the pipeline is practical and are not evidence for the constants of Theorem 8.2.

8.5.1. Empirical Validation of the Regret Decomposition

To verify the individual scaling rates predicted by Theorem 8.2 in an environment where all theoretical assumptions hold with known constants, we evaluate Algorithm 8.1 on a Doeblin-mixing modification of the Tiger POMDP (listening transition $0.95\mathbf{I} + 0.05\mathbf{U}$, uniform reset upon opening, rewards scaled to $[0, 1]$, $\varepsilon_{\text{mix}} = 0.05$, $0.15 \leq Z \leq 0.85$, and discount factor $\gamma = 0.80$). Figure 8.3 and Table 8.1 report the empirical decay of each isolated error term in the regret decomposition as we sweep the particle count $J \in \{8, \ldots, 512\}$, the rollout count $N \in \{4, \ldots, 512\}$, the belief-tree depth $d \in \{0, \ldots, 12\}$, and the online trajectory count T. Each empirical slope closely tracks its theoret-

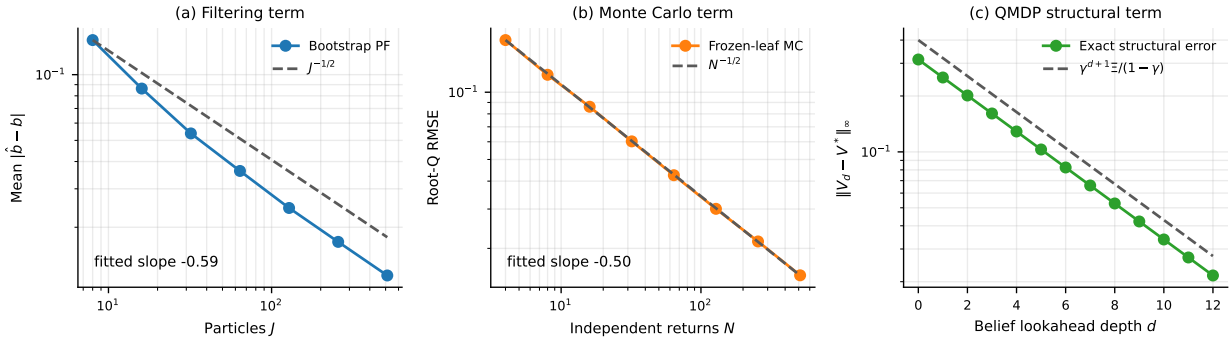


Figure 8.3: Empirical audit of the isolated filtering (J), Monte Carlo rollout (N), and structural planning-depth (d) error terms of Theorem 8.2 on the Doeblin-mixing Tiger POMDP.

Table 8.1: Empirical versus theoretical scaling rates for the four terms of Theorem 8.2 on the Doeblin-mixing Tiger POMDP.

Decomposition Term	Empirical Scaling	Theoretical Rate
Particle filtering ($J = 8 \to 512$)	$J^{-0.59}$	$J^{-1/2}$
Monte Carlo rollouts ($N = 4 \to 512$)	$N^{-0.50}$	$N^{-1/2}$
Structural depth bias ($d = 0 \to 12$)	0.80^d per level	$\gamma^d = 0.80^d$
Online MLE estimation ($\sum_{t=1}^{T} \varepsilon_{\text{opt},t}$)	$T^{0.57}$	$\widetilde{O}(\sqrt{T})$

ical counterpart. Moreover, evaluating the isolated single-term errors yields a concrete empirical budgeting rule: the smallest tested triple (J, N, d) keeping the filtering, Monte Carlo rollout, and structural depth terms each below 0.10, 0.05, and 0.025 is $(16, 16, 6)$, $(64, 64, 9)$, and $(128, 256, 12)$, respectively. Complete experimental details are provided in Appendix G.4.

We instantiate Algorithm 8.1 using the `pypomp` (Abkemeier et al., 2026) package for particle-filtering primitives with weight-discounted particle filter (Algorithm 7.2 in Chapter 7; Tan et al., 2024a) or VSMC (Naesseth et al., 2018) gradient estimators. This allows us to employ highly optimized JAX implementations of the particle filter on GPU. We learn a neural dynamics model online, maintain a particle approximation to the latent state posterior, and use an actor and a critic trained with recurrent SAC-style updates (Ni et al., 2021) in the same vein as Han et al. (2026). In the experiments we use belief-tree depth $d = 0$: each particle is rolled out at the state level under a learned actor and bootstrapped with a learned critic, rather than planning over a tree of particle beliefs. With continuous actions and observations a depth-d tree requires binning at a cost of up

Table 8.2: Which experiments satisfy the assumptions of Theorem 8.2. ✓: holds; ×: fails; ∼: approximately.

Environment	Doeblin	$g_{\min} > 0$	Finite $\mathcal{A}$	Bounded r	Role
Mixing Tiger	✓	✓	✓	✓	theory-valid audit
HeartSteps	∼	∼	✓	✓	approximately valid
Double-integrator LQR	✓	×	×	×	filtering diagnostic
MuJoCo	×	×	×	✓	heuristic demonstration

to $(|\mathcal{A}|M)^d$ nodes, so this is the main practical gap from the theory. Rollout planning corresponds to the QMDP end of the depth axis, whereas deeper belief planning reduces the structural bias. Appendix G.4 evaluates the analyzed belief-tree planner at depths $d > 0$ on finite active-sensing POMDPs against POMCP and Sparse-PFT. The practical deep RL instantiation is presented in Algorithm G.2. All deep RL experiments were run on two RTX 3090 GPUs.

[RESOLVE: CAMERA-READY ITEM E3/E4: CONFIRM THAT THE PRACTICAL DEEP RL INSTANTIATION AND THE HYPERPARAMETER TABLES DISTINGUISH STATE-LEVEL ROLLOUT HORIZON L FROM BELIEF-TREE DEPTH d=0, AND VERIFY PLANNER COMPARISON CODE AGAINST REVISED DP-BACKUP PSEUDOCODE.]

Double-integrator LQR. We first study a partially observed double-integrator LQR. The state contains position and velocity, but the observation reveals only position. The latent velocity is therefore not identifiable from the current observation alone. Different velocities induce the same observation and can only be distinguished using history. This environment isolates the filtering problem. The posterior diagnostics in Figure 8.1 show that particle beliefs can track the latent state without access to memory, while failures occur when the system is unstable. The GRU backbone improves this behavior relative to a memoryless MLP, suggesting that particle filtering and recurrent state representations are complementary. More details and figures can be found in Appendix G.2.

HeartSteps. We next evaluate the HeartSteps benchmark of Gao et al. (2025) as an actively observable MDP with a latent reward state, where the learner can choose to measure the state

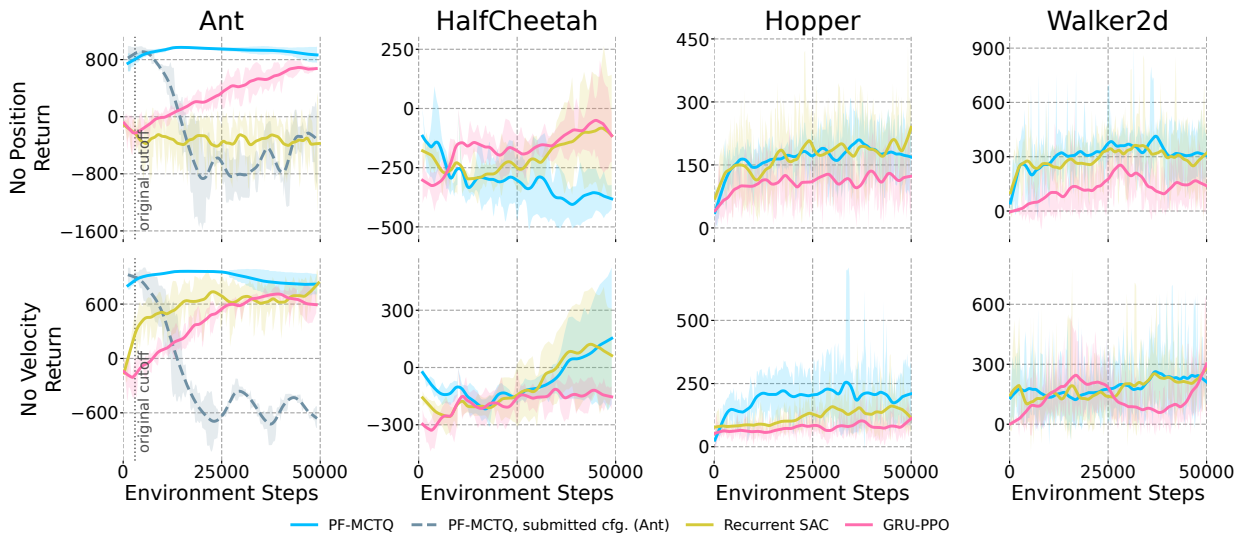


Figure 8.4: Partial-observation MuJoCo learning curves in the low-sample regime across Ant, HalfCheetah, Hopper, and Walker2d with either position or velocity hidden. Non-Ant panels and baselines show smoothed per-episode returns across 5 seeds (mean $\pm$ 1.96 SD/$\sqrt{n}$). The Ant panels plot the 10-episode rolling training return over the full 49.2k-step horizon for the stabilized update-to-data configuration (solid, 3 seeds) alongside the initial high-UTD configuration (dashed, 2 complete seeds, which peaks near 3k steps indicated by the vertical dotted line before collapsing).

at a cost. Algorithm 8.1 is run in the never-measure regime. It never directly measures the latent reward and must infer it from passive observations. The strongest benchmark is RLSVI-SMC/never-measure, which is specifically tailored to the linear-Gaussian HeartSteps structure and uses hand-specified linear features. Consequently, RLSVI-SMC performs best in most scenarios. Algorithm 8.1 typically improves over the baseline rollout but trails RLSVI-SMC because it must learn a deep model, actor, and critic from data. Dyna-ATMQ is not directly comparable as a no-measure method. It actively measures the latent reward roughly half the time, which gives it an informational advantage, but this advantage can be offset when measurement has delayed negative effects on future engagement. More details and figures can be found in Appendix G.2.

MuJoCo partial-observation control. Finally, we evaluate Ant, HalfCheetah, Hopper, and Walker2d with either position or velocity components removed from the observation. We compare Algorithm 8.1 to Recurrent SAC and GRU-PPO baselines from the recurrent model-free POMDP benchmarking literature (Ni et al., 2021). These are strong comparators: they are designed to

Table 8.3: Measured wall-clock runtime on a single RTX 3090 GPU (MuJoCo Ant, position hidden, matched end-to-end collection and gradient updates).

Method	Warm-Start Init (s)	Train Time (s / 1k steps)	Throughput (steps / s)
Algorithm G.2 (stabilized config)	132.1	58.7	17.0
Algorithm G.2 (initial config)	139.6	50.2	19.9
Recurrent SAC	12.1	52.0	19.2
GRU-PPO	2.6	4.4	228.5

solve POMDPs by learning a recurrent history representation directly from reward. Figure 8.4 shows that Algorithm 8.1 is competitive in the ultra-low-sample regime. On Ant, the initial high-update-frequency configuration reaches a high 10-episode rolling return near 3k steps (908 with position hidden and 897 with velocity hidden) but subsequently destabilizes and collapses under continued training (-448 and -648 at 49.2k steps). Regularizing the update schedule (reducing the policy update-to-data ratio to 1/10 with online model updates and weight decay, detailed in Appendix G.4) preserves rapid early learning (825 and 882 at 3k steps, rising to 960 and 962 at 20k steps) while maintaining stable returns through the end of the 49.2k-step horizon (868 and 816 rolling return at 49.2k steps, with final 10-episode evaluation returns of 862 and 767). **[RESOLVE: CAMERA-READY ITEM E4: VERIFY THE EXACT CONFIG DIFFERENCE BETWEEN THE INITIAL AND STABILIZED ANT CONFIGS (UTD 1/10 PLUS ONLINE MODEL UPDATES AND WEIGHT DECAY) IN THE CONTROL REPO LOGS.]** On Hopper and Walker2d, Algorithm 8.1 is broadly competitive with Recurrent SAC and usually stronger than GRU-PPO early in training. HalfCheetah is more mixed: Algorithm 8.1 performs well when velocity is hidden but struggles when position is hidden, suggesting that model bias and rollout error can dominate when the omitted coordinates are crucial for short-horizon prediction.

Table 8.3 reports the measured wall-clock training cost on a single RTX 3090 GPU for the position-hidden Ant benchmark. The stabilized instantiation of Algorithm G.2 incurs approximately 13% higher per-step training time than Recurrent SAC (58.7 s versus 52.0 s per 1000 environment steps) in addition to a 132.1 s warm-start initialization, whereas on-policy GRU-PPO runs substantially faster per step (4.4 s per 1000 steps) at the cost of much lower sample efficiency.

Overall, these results show that particle filtering and model-based planning can be implemented on top of existing deep RL algorithms and can yield sample-efficiency gains in partially observed control. The results should not be read as a claim of asymptotic dominance over recurrent model-free RL. Rather, they show that the theoretical learning–filtering–planning pipeline can be made operational in standard deep RL benchmarks.

8.6. Discussion

We presented an online model-based POMDP pipeline that combines particle filtering, latent-dynamics learning, and approximate planning with an end-to-end regret bound. We decompose regret into model estimation error, particle approximation error, Monte Carlo planning error, and QMDP structural bias, clarifying how statistical and computational budgets affect performance. The resulting framework provides a practical bridge between POMDP theory and modern model-based deep RL. Experiments with a simplified rollout-based implementation show that this theoretically motivated pipeline can be made operational and competitive in low-sample partially observed control.

Limitations. Our analysis relies on the Doeblin mixing condition and bounded observation densities, which ensure passive state coverage and obviate the need for strategic exploration. These assumptions fail in many POMDPs. In item search with exact observations the observation density is not bounded below, and a particle set of size J misses the true location among K with probability $(1 - 1/K)^J$, while in Light–Dark-type problems, where information must be actively gathered, Ξ_θ is large and shallow planning never gathers it (see the controlled stress tests in Appendix G.4). Second, the bounded reward assumption excludes settings such as LQG. Third, the deep RL experiments run at depth $d = 0$ with a learned critic and actor, so they sit at the QMDP end of the analyzed algorithm, and our guarantees cover non-adaptive (e.g., uniform) rather than UCT-style action selection inside the tree. In Appendix G.4, we confirm empirically on multi-step active-sensing POMDPs that UCB1 action selection within the belief tree outperforms uniform exploration at a matched simulation budget, demonstrating that uniform exploration is purely a theoretical proof device. Natural extensions are strategic exploration to relax the assumptions on

the dynamics, unbounded rewards, and guarantees for adaptive tree search.

CHAPTER 9

CONCLUSION

[RESOLVE: REVIEW SUMMARY OF CONTRIBUTIONS AND FUTURE DIRECTIONS.]

9.1. Synthesis of Contributions

Across the seven research chapters of this dissertation, we have examined how the representation and use of information govern both the statistical sample complexity of learning from sequential data and the computational complexity of turning those data into inferences and decisions. Rather than treating statistical efficiency and computational tractability as separate problems, each chapter traced how structural choices—how historical data are combined with new interaction, how policies and ensembles are parameterized, and how latent states and future information are approximated—create explicit tradeoffs between sample size, approximation bias, and computation.

9.1.1. Using Existing Data to Reduce Exploration Complexity

Chapters 2 and 3 studied two complementary mechanisms by which offline data reduce the statistical burden of online exploration:

- **Direct state–action coverage in a shared environment (Chapter 2):** When historical transitions come from the same MDP, their utility is determined by which state–action pairs or feature directions the behavior policy covered. Under general function approximation, incorporating offline transitions into an optimistic Bellman confidence set (Algorithm 2.1) allows the learner to rely on partial offline coverage where it exists and spend online exploration on the remaining Bellman eluder dimension without knowing the coverage boundary in advance. In linear MDPs, targeting poorly covered feature directions during reward-agnostic online exploration before pessimistic offline policy fitting (Algorithm 2.2) augments the offline information matrix from $\Sigma^*_{\mathrm{off},h}$ to $\Sigma^*_{\mathrm{off},h} + \Sigma^*_{\mathrm{on},h}$, while optimistic offline-to-online value iteration (Algorithm 2.3) reduces the online exploration dimension from d to the unspanned subspace dimension d_{on} in the large-sample regime.

- **Shared low-dimensional representation across heterogeneous tasks (Chapter 3):** When historical logs come from many distinct users or tasks with short trajectories, the offline dataset does not directly cover the reward parameter of a new user, but instead reveals the r-dimensional subspace in which task parameters reside. By correcting the empirical cross-moment matrix for adaptive action selection and observation noise using paired observations per user (Algorithm 3.1), we recover the latent subspace at rate $\widetilde{O}(\sqrt{d/N_{\text{off}}})$ from constant-length trajectories and reduce online contextual bandit regret for a new user from $\widetilde{O}(d\sqrt{T})$ to $\widetilde{O}(r\sqrt{T} + T\sqrt{d/N_{\text{off}}})$ (Algorithms 3.2 and 3.3).

9.1.2. Representing Policies and Sequential Ensembles Tractably

Chapters 4, 5, and 6 examined how the functional representation of a sequentially updated policy or regression ensemble affects both statistical concentration and computational cost:

- **Controlling implicit and explicit actor complexity (Chapter 4):** In actor–critic reinforcement learning, unrolling natural policy updates accumulates all past critics into the policy representation, causing the policy class covering number to grow with the number of critic updates and destabilizing policy-evaluation Bellman targets. Targeting the optimal value function Q^* with a rare-switching optimistic critic and resetting the actor to a uniform distribution at each switch (Algorithm 4.2) decouples the Bellman target operator from actor drift and achieves optimal $\widetilde{O}(\sqrt{T})$ regret and $\widetilde{O}(\epsilon^{-2})$ sample complexity under general function approximation. Projecting each actor update onto a fixed-dimensional log-linear policy class via weighted least-squares logit regression on a G-optimal design coreset, combined with Langevin Monte Carlo critic exploration (Algorithms 4.6–4.8), makes per-episode policy evaluation constant-time while preserving polynomial sample efficiency in linear MDPs.

- **Optimizing continuous multimodal policies via diffusion flows (Chapter 5):** When actions are continuous and multimodal, diffusion policies represent complex action distributions through K-step denoising paths whose marginal action density is intractable. We showed how vanilla diffusion policy gradients (Algorithm 5.1), deterministic-flow DDIM natural policy gradients under reset, mixing, or hybrid logged-trajectory access (Algorithm 5.2), and

denoising score-matching behavior cloning of an optimistic or hybrid critic (Algorithm 5.3) each trade off sampling access assumptions against comparison-policy, advantage-transfer, score-matching, and sampler errors.

- **Asymptotic normality in sequentially boosted ensembles (Chapter 6):** In supervised gradient boosting and explainable boosting machines, sequential residual fitting couples every tree to all previous trees. Combining moving-average shrinkage with random tree dropout (Algorithm 6.1), parallel structured dropout (Algorithm 6.2), or cyclic univariate feature updates (Algorithm 6.3) yields contraction mappings whose fixed points suffer less shrinkage bias toward zero while admitting explicit central limit theorems for confidence intervals, prediction intervals, and hypothesis tests.

9.1.3. Inference and Control under Partial Observability

Chapters 7 and 8 relaxed the assumption that the learner observes the underlying state directly, making sequential Monte Carlo filtering part of the estimation and control loop:

- **Plug-and-play differentiable particle filtering for parameter estimation (Chapter 7):** To resolve the discontinuity of resampled particle filters without sacrificing the plug-and-play property or suffering linear-in-N particle weight-variance growth, our fixed-ancestor weight-discounted particle filter (Algorithm 7.1) and its single-pass differentiable counterpart (Algorithm 7.2) introduce a discount factor $\alpha \in [0, 1]$ on carried log-weight derivatives, achieving $O(N\alpha^{2/(1-\alpha)})$ squared score bias and $O(N(1 + \log J)^2/(J(1 - \alpha)^2))$ particle variance under uniform mixing. Pairing iterated filtering warm starts with discounted automatic-differentiation optimization (Algorithm 7.3) combines global mode discovery with rapid local gradient refinement on high-dimensional epidemiological and ecological dynamical systems.

- **Decomposing estimation, filtering, rollout, and lookahead errors in POMDP control (Chapter 8):** For online decision making in POMDPs, Algorithm 8.1 integrates online trajectory maximum likelihood estimation, J-particle bootstrap filtering, and depth-d Monte Carlo belief-tree planning (Algorithm G.1) with N rollouts and a state-value QMDP leaf

bootstrap. The resulting regret bound isolates the statistical model-learning rate $\widetilde{O}(T^{-1/4})$ from the particle filtering error $O(J^{-1/(2\kappa_{\mathrm{pf}})})$, rollout estimation error $O(C_{\mathrm{mc}}(d, M)N^{-1/2})$, and structural lookahead error $O(\gamma^{d+1}\overline{\Xi}_T)$, clarifying when deeper belief-tree lookahead is needed to capture the value of information-gathering actions and when a depth-zero particle state-rollout approximation suffices.

9.2. Future Directions

The results in this dissertation point toward several concrete open problems at the interface of statistical learning theory, stochastic filtering, and scalable sequential decision making:

1. **Closing sample-size threshold and horizon gaps in hybrid and actor–critic RL:** In Chapter 2, the offline-to-online regret guarantee for Algorithm 2.3 in linear MDPs requires a large-sample burn-in threshold of order $\widetilde{\Omega}(d^{13}H^{14})$ to align the empirical variance-weighted covariances with their population counterparts. Similarly, in Chapter 4, while Algorithm 4.2 achieves the optimal ϵ^{-2} accuracy exponent under general function approximation, its horizon dependence remains $\widetilde{O}(dH^5 \log|\mathcal{A}|\epsilon^{-2} + dH^4 \log|\mathcal{F}|\epsilon^{-2})$. Developing variance-aware actor–critic and hybrid updates that achieve minimax horizon dependence without large burn-in thresholds remains an important theoretical question.

2. **Adaptive tree structure and growing-depth inference in boosted ensembles:** The central limit theorems in Chapter 6 rely on finite-partition regularity conditions to establish contraction of the expected kernel operator as the number of boosting rounds grows. Extending valid frequentist confidence intervals to adaptive splitting rules with growing tree depth and honesty across ancestral sample reuse is a natural next step for nonparametric ensemble inference.

3. **Joint gradient-based estimation and control in high-dimensional partially observed systems:** Chapter 7 developed low-MSE pathwise score estimators for passive partially observed Markov processes using discounted automatic differentiation, while Chapter 8 analyzed online POMDP control using trajectory maximum likelihood estimation and boot-

strap particle belief planning. Combining discounted differentiable particle filtering directly with online policy optimization and adaptive belief-tree allocation—particularly in high-dimensional spatiotemporal systems where localized or block particle resampling is needed to prevent weight collapse—promises to further unify statistical inference and sequential control under partial observability.

APPENDIX A

PROOFS AND SUPPLEMENTARY DETAILS FOR CHAPTER 2

This appendix contains the complete proofs, unabridged algorithmic descriptions, technical lemmas, and supplementary experimental figures for Chapter 2, organized into two parts corresponding to Sections 2.1–2.6 (Tan and Xu, 2024) and Sections 2.7–2.11 (Tan et al., 2024b).

A.1. Proof of Theorem 2.3

Let $\mathcal{X}_{\text{on}}, \mathcal{X}_{\text{off}}$ be an arbitrary (not necessarily disjoint) partition of $\mathcal{X} = \mathcal{S} \times \mathcal{A} \times [H]$. We bound the regret for this fixed partition, which immediately yields the infimum over all partitions in Theorem 2.3.

Following Xie et al. (2022a), we write $\delta_h^{(t)} \coloneqq f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}$ for the Bellman residual at online episode t and step h, and define the cumulative in-sample occupancy measures (without and with the offline dataset) by

$$\bar{d}_h^{(t)} \coloneqq \sum_{i=1}^{t-1} d_h^{(i)}, \qquad \tilde{d}_h^{(t)} \coloneqq N_{\text{off}} \mu_h + \sum_{i=1}^{t-1} d_h^{(i)}, \tag{A.1}$$

where $d_h^{(t)} = d_h^{\pi^{(t)}}$ is the occupancy measure induced by the greedy policy $\pi^{(t)} = \pi_{f^{(t)}}$. We further write

$$\mathbb{E}_{\bar{d}_h^{(t)}}[g] = \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}}[g], \qquad \mathbb{E}_{\tilde{d}_h^{(t)}}[g] = \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}}[g] + N_{\text{off}} \mathbb{E}_{\mu_h}[g]. \tag{A.2}$$

We first state the high-probability confidence-set and in-sample squared Bellman error guarantee, which extends Lemma 15 of Xie et al. (2022a) and Lemma 44 of Jin et al. (2021a) to include the N_{off} offline episodes.

Lemma A.1 (High-Probability Confidence Set and In-Sample Error)**.** *With probability at least* $1 - \delta$*, for all* $t \in [N_{\text{on}}]$ *and all* $h \in [H]$*,*

$$\textit{(i)}\ Q_h^\star \in \mathcal{F}_h^{(t)}, \qquad \textit{(ii)}\ \mathbb{E}_{\bar{d}_h^{(t)}}\left[(\delta_h^{(t)})^2\right] \le \mathcal{O}(H^2 \beta), \tag{A.3}$$

where $\beta = c_1 \left(\log\left[NH\mathcal{N}_{\mathcal{F}}(\rho)/\delta\right] + N\rho\right)$ *for an absolute constant* $c_1 > 0$ *and* $N = N_{\text{off}} + N_{\text{on}}$.

Proof. Lemma 44 of Jin et al. (2021a) establishes that for any sequence of N episodes, with probability at least $1-\delta$, (i) the optimal value function $Q^\star$ belongs to the confidence set at every episode $\tau \in [N]$, and (ii) any function $\bar{f}^{(\tau)}$ in the confidence set at episode τ has cumulative expected squared Bellman error $\sum_{i=1}^{\tau-1} \mathbb{E}_{d_h^{(i)}}[(\bar{f}_h^{(\tau)} - \mathcal{T}_h \bar{f}_{h+1}^{(\tau)})^2] \le \mathcal{O}(H^2\beta)$ over the distributions generating the first $\tau - 1$ episodes. Ordering the N_{off} offline episodes arbitrarily as episodes $\tau = 1, \ldots, N_{\text{off}}$ (each drawn from μ) and indexing the N_{on} online episodes as $\tau = N_{\text{off}} + 1, \ldots, N_{\text{off}} + N_{\text{on}}$ with $\bar{f}^{(N_{\text{off}}+t)} = f^{(t)} \in \mathcal{F}^{(t-1)}$, the conclusion holds simultaneously for all $\tau \in [N]$, and in particular for all online episodes $t \in [N_{\text{on}}]$. □

Proof of Theorem 2.3. Conditioned on the good event of Lemma A.1 (so that $Q^\star \in \mathcal{F}^{(t-1)}$ and optimism $f_1^{(t)}(s_1^{(t)}, \pi_1^{(t)}(s_1^{(t)})) \ge V_1^\star(s_1^{(t)})$ holds for every $t \in [T]$), the standard performance-difference regret decomposition (Xie et al., 2022a, Lemma 3) yields

$$\mathrm{Reg}(T) \le \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{(s,a)\sim d_h^{(t)}} \left[\delta_h^{(t)}(s,a)\right]. \tag{A.4}$$

Using the partition $\mathcal{X}_{\text{on}} \cup \mathcal{X}_{\text{off}} = \mathcal{X}$ and writing $\delta_{h,\text{on}}^{(t)}(s,a) \coloneqq \delta_h^{(t)}(s,a)\mathbf{1}_{(s,a,h)\in\mathcal{X}_{\text{on}}}$ and $\delta_{h,\text{off}}^{(t)}(s,a) \coloneqq \delta_h^{(t)}(s,a)\mathbf{1}_{(s,a,h)\in\mathcal{X}_{\text{off}}}$, Equation (A.4) decomposes as

$$\mathrm{Reg}(T) \le \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{(s,a)\sim d_h^{(t)}} \left[\delta_{h,\text{on}}^{(t)}(s,a)\right] + \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{(s,a)\sim d_h^{(t)}} \left[\delta_{h,\text{off}}^{(t)}(s,a)\right]. \tag{A.5}$$

For the online partition term, applying the Cauchy–Schwarz inequality and Lemma A.1 gives

$$\begin{aligned}
\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\delta_{h,\text{on}}^{(t)}\right] &= \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\delta_{h,\text{on}}^{(t)} \left(\frac{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}}[(\delta_{h,\text{on}}^{(t)})^2]}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}}[(\delta_{h,\text{on}}^{(t)})^2]}\right)^{1/2}\right] \\
&\le \sqrt{\sum_{t=1}^{T} \sum_{h=1}^{H} \frac{\mathbb{E}_{d_h^{(t)}}[\delta_{h,\text{on}}^{(t)}(s,a)]^2}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}}[(\delta_{h,\text{on}}^{(t)})^2]}} \sqrt{\sum_{t=1}^{T} \sum_{h=1}^{H} \left(H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}}[(\delta_{h,\text{on}}^{(t)})^2]\right)}
\end{aligned}$$

$$\leq \sqrt{H \cdot c_{\mathrm{on}}(\mathcal{F}, \mathcal{X}_{\mathrm{on}}, T)} \cdot \sqrt{HT \cdot \mathcal{O}(H^2\beta)}$$
$$\lesssim H^2 \sqrt{\beta \cdot c_{\mathrm{on}}(\mathcal{F}, \mathcal{X}_{\mathrm{on}}, T) \cdot T}. \tag{A.6}$$

For the offline partition term, we similarly apply the Cauchy–Schwarz inequality with respect to the combined in-sample measure $\tilde{d}_h^{(t)}$:

$$\sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}}\left[\delta_{h,\mathrm{off}}^{(t)}\right] = \sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{(s,a)\sim d_h^{(t)}}\left[\delta_{h,\mathrm{off}}^{(t)}(s,a)\left(\frac{N_{\mathrm{off}}\mathbb{E}_{\mu_h}[(\delta_{h,\mathrm{off}}^{(t)})^2] + \sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_{h,\mathrm{off}}^{(t)})^2]}{N_{\mathrm{off}}\mathbb{E}_{\mu_h}[(\delta_{h,\mathrm{off}}^{(t)})^2] + \sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_{h,\mathrm{off}}^{(t)})^2]}\right)^{1/2}\right]$$
$$\leq \sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H} \frac{\mathbb{E}_{(s,a)\sim d_h^{(t)}}\left[\left(\delta_{h,\mathrm{off}}^{(t)}(s,a)\right)^2\right]}{N_{\mathrm{off}}\mathbb{E}_{\mu_h}[(\delta_{h,\mathrm{off}}^{(t)})^2] + \sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_{h,\mathrm{off}}^{(t)})^2]}} \cdot \sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H}\left(N_{\mathrm{off}}\mathbb{E}_{\mu_h}[(\delta_{h,\mathrm{off}}^{(t)})^2] + \sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_{h,\mathrm{off}}^{(t)})^2]\right)}. \tag{A.7}$$

By definition of $c_{\mathrm{off}}(\mathcal{F}, \mathcal{X}_{\mathrm{off}})$, the first factor is bounded by

$$\sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H} \frac{\mathbb{E}_{d_h^{(t)}}[(\delta_{h,\mathrm{off}}^{(t)})^2]}{N_{\mathrm{off}}\mathbb{E}_{\mu_h}[(\delta_{h,\mathrm{off}}^{(t)})^2] + \sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_{h,\mathrm{off}}^{(t)})^2]}} \leq \sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H} \frac{\mathbb{E}_{d_h^{(t)}}[(\delta_{h,\mathrm{off}}^{(t)})^2]}{N_{\mathrm{off}}\mathbb{E}_{\mu_h}[(\delta_{h,\mathrm{off}}^{(t)})^2]}}$$
$$\leq \sqrt{\frac{HT}{N_{\mathrm{off}}} c_{\mathrm{off}}(\mathcal{F}, \mathcal{X}_{\mathrm{off}})}. \tag{A.8}$$

By Lemma A.1, the second factor is bounded by

$$\sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H} \mathbb{E}_{\tilde{d}_h^{(t)}}[(\delta_h^{(t)})^2]} \lesssim H\sqrt{HT\beta}. \tag{A.9}$$

Combining the online and offline bounds and setting $\rho = 1/N$ yields Theorem 2.3. □

A.2. Additional Proofs and Figures for Sections 2.2–2.6

The restricted sequential extrapolation coefficient bounds, proofs for the tabular, linear, and block MDP case studies (Propositions 2.4–2.8) and the general hybrid recipe (Proposition 2.9), and

supplementary experimental figures for Section 2.6 appear in Appendices B–F of Tan and Xu (2024) (arXiv:2403.09701).

A.3. Unabridged Descriptions of Hybrid Linear MDP Algorithms

In Algorithms A.1 and A.2, we present the unabridged descriptions of Algorithm 2.2 (online-to-offline hybrid PAC learning) and Algorithm 2.3 (offline-to-online hybrid regret minimization) in linear MDPs.

A.4. Proofs and Technical Lemmas for Sections 2.8–2.9

Building on the proof sketches in Sections 2.8 and 2.9, the complete proofs of Lemma 2.11, Theorem 2.12, Corollary 2.13, and Theorem 2.14, together with the supporting projected elliptical potential and variance-truncation lemmas, appear in Appendices C–F of Tan et al. (2024b) (arXiv:2408.04526).

Algorithm A.1 Unabridged Online-to-Offline Hybrid RL (Algorithm 2.2)

Input: Offline dataset $\mathcal{D}_{\text{off}}$, sample sizes N_{on}, N_{off}, feature maps ϕ_h, exploration tolerance $\tau > 0$.

Initialize: $\mathcal{D}_h^{(0)} \leftarrow \emptyset$ for all $h \in [H]$, $\lambda = 1/H^2$, $\beta_2 = \tilde{O}(\sqrt{d})$. Define smooth potential objectives

$$f_i(\mathbf{\Lambda}) = \eta_i^{-1} \log\left(\sum_{\phi\in\Phi} \exp\Big(\eta_i \|\boldsymbol{\phi}\|^2_{\mathbf{A}_i(\mathbf{\Lambda})^{-1}}\Big)\right), \qquad \mathbf{A}_i(\mathbf{\Lambda}) = \mathbf{\Lambda} + (T_i K_i)^{-1}(\mathbf{\Lambda}_{0,i} + \mathbf{\Lambda}_{\text{off}})$$

with $\mathbf{\Lambda}_{0,i} \succeq \mathbf{\Lambda}_0$ and $\eta_i = 2^{2i/5}$.

Exploration Phase: For each $h \in [H]$, collect covariates $\mathbf{\Lambda}_h$ until $\max_{\phi_h\in\Phi} \phi_h^\top(\mathbf{\Lambda}_h + \lambda\mathbf{I} + \mathbf{\Lambda}_{\text{off},h})^{-1}\phi_h \le \tau$:

for $i = 1, 2, 3, \ldots$ **do**

  Set $T_i \leftarrow 2^i$, $K_i \leftarrow 2^i$.

  Execute an arbitrary policy for K_i episodes to collect covariates $\mathbf{\Gamma}_0$ and data $\mathfrak{D}_0$, and set $\mathbf{\Lambda}_1 \leftarrow \mathbf{\Gamma}_0/K_i$.

  for $t = 1, \ldots, T_i$ **do**

    if $\sum_{j=1}^{i} T_j K_j \ge N_{\text{on}}$ **then break**

    end if

    Run a regret-minimizing subroutine (Wagenmaker et al., 2022a) on the synthetic exploration reward $g_h^{(t)}(s,a) \propto \operatorname{tr}(-\nabla_{\mathbf{\Lambda}} f_i(\mathbf{\Lambda})|_{\mathbf{\Lambda}=\mathbf{\Lambda}_t}\phi(s,a)\phi(s,a)^\top)$ for K_i episodes to collect covariates $\mathbf{\Gamma}_t$ and data $\mathfrak{D}_t$.

    Frank–Wolfe update: $\mathbf{\Lambda}_{t+1} \leftarrow (1 - \frac{1}{t+1})\mathbf{\Lambda}_t + \frac{1}{t+1}\mathbf{\Gamma}_t/K_i$.

  end for

  Set $\widehat{\mathbf{\Lambda}}_{i,h} \leftarrow \mathbf{\Lambda}_{T_i+1}$, $\mathfrak{D}_i \leftarrow \bigcup_{t=0}^{T_i} \mathfrak{D}_t$, $\mathbf{\Lambda}_h = \widehat{\mathbf{\Lambda}}_{i,h}$, $\mathcal{D}_{\text{on}} = \mathfrak{D}_i$.

  if $f_i(\widehat{\mathbf{\Lambda}}_{i,h}) \le K_i T_i \tau$ **then break**

  end if

end for

Planning Phase: Split $\mathcal{D}_{\text{off}} \cup \{\mathcal{D}_h^{(N_{\text{on}})}\}_{h\in[H]}$ into $\mathcal{D}$ and $\mathcal{D}'$. On $\mathcal{D}'$, compute reference variance estimators (Xiong et al., 2023b):

$$\widetilde{\beta}_{h,2} = \arg\min_{\beta\in\mathbb{R}^d} \sum_{\tau\in\mathcal{D}'} \Big[\langle\phi(s_h^\tau, a_h^\tau), \beta\rangle - \big(\widehat{V}'_{h+1}(s_{h+1}^\tau)\big)^2\Big]^2 + \lambda\|\beta\|_2^2,$$

$$\widetilde{\beta}_{h,1} = \arg\min_{\beta\in\mathbb{R}^d} \sum_{\tau\in\mathcal{D}'} \Big[\langle\phi(s_h^\tau, a_h^\tau), \beta\rangle - \widehat{V}'_{h+1}(s_{h+1}^\tau)\Big]^2 + \lambda\|\beta\|_2^2,$$

$$\widehat{\sigma}_h^2(s,a) \coloneqq \max\left\{1, \big[\phi(s,a)^\top\widetilde{\beta}_{h,2}\big]_{[0,H^2]} - \big[\phi(s,a)^\top\widetilde{\beta}_{h,1}\big]^2_{[0,H]} - \tilde{O}\left(\frac{dH^3}{\sqrt{N\kappa}}\right)\right\}.$$

for $h = H, H-1, \ldots, 1$ **do**

  $\mathbf{\Sigma}_h \leftarrow \sum_{\tau\in\mathcal{D}} \phi(s_h^\tau, a_h^\tau)\phi(s_h^\tau, a_h^\tau)^\top/\widehat{\sigma}_h^2(s_h^\tau, a_h^\tau) + \lambda\mathbf{I}_d$.

  $\widehat{w}_h \leftarrow \mathbf{\Sigma}_h^{-1}\left(\sum_{\tau\in\mathcal{D}} \phi(s_h^\tau, a_h^\tau)\frac{r_h^\tau + \widehat{V}_{h+1}(s_{h+1}^\tau)}{\widehat{\sigma}_h^2(s_h^\tau, a_h^\tau)}\right)$.

  $\Gamma_h(\cdot,\cdot) \leftarrow \beta_2\|\phi(\cdot,\cdot)\|_{\mathbf{\Sigma}_h^{-1}}$ and $\widehat{Q}_h(\cdot,\cdot) \leftarrow \big\{\phi(\cdot,\cdot)^\top\widehat{w}_h - \Gamma_h(\cdot,\cdot)\big\}_{[0,H-h+1]}$.

  $\widehat{\pi}_h(\cdot \mid \cdot) \leftarrow \arg\max_{\pi_h}\langle\widehat{Q}_h(\cdot,\cdot), \pi_h(\cdot \mid \cdot)\rangle_{\mathcal{A}}$ and $\widehat{V}_h(\cdot) \leftarrow \langle\widehat{Q}_h(\cdot,\cdot), \widehat{\pi}_h(\cdot \mid \cdot)\rangle_{\mathcal{A}}$.

end for

Output: $\widehat{\pi} = \{\widehat{\pi}_h\}_{h=1}^H$.

Algorithm A.2 Unabridged Offline-to-Online Hybrid RL (Algorithm 2.3)

Input: Offline dataset $\mathcal{D}_{\text{off}}$, sample sizes N_{on}, N_{off}, feature maps ϕ_h, regularization $\lambda > 0$, confidence radii $\beta, \bar{\beta}, \tilde{\beta}$, $t_{\text{last}} = 0$.

Initialize: For $h \in [H]$, compute $\widehat{\mathbf{w}}_{1,h}, \check{\mathbf{w}}_{1,h}, Q_{1,h}, \check{Q}_{1,h}, \sigma_{n,h}, \bar{\sigma}_{n,h}$ from $\mathcal{D}_{\text{off}}$ using the recursion below, and set $\mathbf{\Sigma}_{0,h} = \mathbf{\Sigma}_{1,h} = \mathbf{\Sigma}_{\text{off},h} + \lambda \mathbf{I} = \sum_{n=1}^{N_{\text{off}}} \bar{\sigma}_{n,h}^{-2} \phi_{n,h} \phi_{n,h}^\top + \lambda \mathbf{I}$.

for episode $t = 1, \ldots, T$ **do**

  Observe initial state $s_1^{(t)}$.

  for step $h = H, H-1, \ldots, 1$ **do**

    $\widehat{\mathbf{w}}_{t,h} = \mathbf{\Sigma}_{t,h}^{-1} \sum_{i=1}^{N_{\text{off}}+t-1} \bar{\sigma}_{i,h}^{-2} \boldsymbol{\phi}(s_h^{(i)}, a_h^{(i)}) V_{t,h+1}(s_{h+1}^{(i)})$.

    $\check{\mathbf{w}}_{t,h} = \mathbf{\Sigma}_{t,h}^{-1} \sum_{i=1}^{N_{\text{off}}+t-1} \bar{\sigma}_{i,h}^{-2} \boldsymbol{\phi}(s_h^{(i)}, a_h^{(i)}) \check{V}_{t,h+1}(s_{h+1}^{(i)})$.

    if there exists $h' \in [H]$ such that $\det(\mathbf{\Sigma}_{t,h'}) \geq 2 \det(\mathbf{\Sigma}_{t_{\text{last}},h'})$ **then**

      $Q_{t,h}(s,a) = \min\Big\{ r_h(s,a) + \widehat{\mathbf{w}}_{t,h}^\top \boldsymbol{\phi}(s,a) + \beta \sqrt{\boldsymbol{\phi}(s,a)^\top \mathbf{\Sigma}_{t,h}^{-1} \boldsymbol{\phi}(s,a)}, Q_{t-1,h}(s,a), H \Big\}$.

      $\check{Q}_{t,h}(s,a) = \max\Big\{ r_h(s,a) + \check{\mathbf{w}}_{t,h}^\top \boldsymbol{\phi}(s,a) - \bar{\beta} \sqrt{\boldsymbol{\phi}(s,a)^\top \mathbf{\Sigma}_{t,h}^{-1} \boldsymbol{\phi}(s,a)}, \check{Q}_{t-1,h}(s,a), 0 \Big\}$.

      Set $t_{\text{last}} = t$.

    else

      $Q_{t,h}(s,a) = Q_{t-1,h}(s,a)$, $\check{Q}_{t,h}(s,a) = \check{Q}_{t-1,h}(s,a)$.

    end if

    $V_{t,h}(s) = \max_{a \in \mathcal{A}} Q_{t,h}(s,a)$, $\check{V}_{t,h}(s) = \max_{a \in \mathcal{A}} \check{Q}_{t,h}(s,a)$.

  end for

  for step $h = 1, \ldots, H$ **do**

    Play action $a_h^{(t)} \leftarrow \arg\max_{a \in \mathcal{A}} Q_{t,h}(s_h^{(t)}, a)$ and observe $r_h^{(t)}, s_{h+1}^{(t)}$.

    Compute $\sigma_{t,h} = \sqrt{\big[\overline{\mathbb{V}}_{t,h} V_{t,h+1}\big]\Big(s_h^{(t)}, a_h^{(t)}\Big) + E_{t,h} + D_{t,h} + H}$, where

$$E_{t,h} = \min\left\{ \widetilde{\beta} \left\| \mathbf{\Sigma}_{t,h}^{-1/2} \boldsymbol{\phi}(s_h^{(t)}, a_h^{(t)}) \right\|_2, H^2 \right\} + \min\left\{ 2H\bar{\beta} \left\| \mathbf{\Sigma}_{t,h}^{-1/2} \boldsymbol{\phi}(s_h^{(t)}, a_h^{(t)}) \right\|_2, H^2 \right\},$$

$$D_{t,h} = \min\left\{ 4d^3H^2 \left(\widehat{\mathbf{w}}_{t,h}^\top \boldsymbol{\phi}(s_h^{(t)}, a_h^{(t)}) - \check{\mathbf{w}}_{t,h}^\top \boldsymbol{\phi}(s_h^{(t)}, a_h^{(t)}) + 2\bar{\beta} \sqrt{\boldsymbol{\phi}(s_h^{(t)}, a_h^{(t)})^\top \mathbf{\Sigma}_{t,h}^{-1} \boldsymbol{\phi}(s_h^{(t)}, a_h^{(t)})} \right), d^3H^3 \right\}.$$

    Set $\bar{\sigma}_{t,h} \leftarrow \max\left\{ \sigma_{t,h}, \sqrt{H}, 2d^3H^2 \left\| \boldsymbol{\phi}\Big(s_h^{(t)}, a_h^{(t)}\Big) \right\|_{\mathbf{\Sigma}_{t,h}^{-1}}^{1/2} \right\}$.

    Update $\mathbf{\Sigma}_{t+1,h} = \mathbf{\Sigma}_{t,h} + \bar{\sigma}_{t,h}^{-2} \boldsymbol{\phi}\Big(s_h^{(t)}, a_h^{(t)}\Big) \boldsymbol{\phi}\Big(s_h^{(t)}, a_h^{(t)}\Big)^\top$.

  end for

end for

Output: Greedy policy $\widehat{\pi} = \pi^{Q_{T,h}}$ or uniform mixture $\text{Unif}(\pi^{Q_{1,h}}, \ldots, \pi^{Q_{T,h}})$.

APPENDIX B

PROOFS AND SUPPLEMENTARY DETAILS FOR CHAPTER 3

Full proofs of Theorems 3.5, 3.6, 3.7, and 3.8, the de Finetti theorem for stateless and transition-agnostic contextual decision processes, pseudoinverse and Thompson sampling extensions, and additional simulation and MovieLens-1M ablation studies are provided in the supplementary material of Kausik et al. (2024) (`arXiv:2405.17324`).

APPENDIX C

PROOFS AND SUPPLEMENTARY DETAILS FOR CHAPTER 4

This appendix presents the flagship regret proof for Theorem 4.11 from Part I of Chapter 4, while deferring auxiliary proofs, hybrid RL extensions, and Part II supplementary material to Tan et al. (2025) and Lin et al. (2026).

C.1. Proofs and Extensions for Part I

C.1.1. Proofs for Theorem 4.5

The proof of Theorem 4.5 for Algorithm 4.1 decomposes regret into mirror descent actor tracking error, negative Bellman error under $\pi^\star$ controlled via $Q^{\pi^{(t)}}$-critic optimism, and on-policy Bellman error bounded via the Sequential Extrapolation Coefficient (SEC). Complete proofs of Theorem 4.5, Lemma 4.6, and Lemma 4.8 are deferred to Tan et al. (2025).

C.1.2. Proofs for Theorem 4.11

Recall the motivation for this solution:

1. **Optimism** allows one to perform strategic exploration, addressing the issue of exploration vs. exploitation, and allowing us to avoid making reachability or coverage assumptions.

2. **Off-policy learning** avoids throwing away any samples, ensuring that no samples are wasted.

3. **Do rare-switching critic updates work?** A-priori, introducing rare-switching critic updates as in Xiong et al. (2023a) offers an appealing solution to the covering number issue. However, the Bellman operator with respect to $\pi^{(t)}$ has a very limited form of optimism. Furthermore, it is difficult to control the number of rare-switching updates in the context of general function approximation, where we make an update when the Bellman error with respect to the current policy is large, as the current policy keeps changing and so we track a moving target.

4. **Letting the critic target Q^* and not Q^π** ensures sufficient optimism, as the Bellman

operator is now the same at every iteration. Further, as the critic targets Q^*, we do not need to control $\log \mathcal{N}_{(\mathcal{T}^\Pi)^T \mathcal{F}}(\rho)$, as the Bellman operator for the greedy policy is a contraction.

5. **Rare-switching critic updates now work.** However, this introduces an additional term, where we need to bound the deviation of the current policy from its target, the greedy policy with regard to the current critic. Re-introducing rare-switching critic updates resolves this, as now we allow the actor sufficient time to catch up to the critic updates. Controlling the number of critic updates is not an issue when the critic targets Q^*, as the Bellman operator is now the same at every iteration. This is reminiscent of the delayed target Q-function trick common in deep RL (Lillicrap et al., 2015; Fujimoto et al., 2018).

6. **Policy resets.** However, going through the mirror descent proof to control the additional error results in an additional term bounded by $\sum_{k=1}^{N_{\text{updates}}} \log(1/\pi^{(t_k)})$. This term can be controlled by resetting the policy to the uniform policy upon every critic update. As critic updates are rare, on the order of $dH \log T$, the additional error incurred is very small. This trick was adopted independently by Cassel and Rosenberg (2024) for the same reason.

7. **Increased learning rate.** We increase the learning rate accordingly by a factor of $\sqrt{dH \log T}$, exactly the square root of the number of critic updates/policy resets. This is done to mitigate the increase in regret incurred by the policy resets by a factor of $\sqrt{dH \log T}$. This can be seen as making more aggressive updates to make up for the lost ground due to policy resets when the critic makes a rare but large update.

Given the above, we are now in a position to continue our analysis below.

Regret decomposition

We employ the following regret decomposition below. This is a slightly different regret decomposition than that of Cai et al. (2020) and Zhong and Zhang (2023), as our critic targets Q^*.

Lemma C.1 (Regret Decomposition For Q^*-Targeting Actor-Critics)**.**

$$\begin{aligned}\mathrm{Reg}(T) =& \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}\left(s_h,\cdot\right),\pi_h^*\left(\cdot\mid s_h\right)-\pi_h^{(t)}\left(\cdot\mid s_h\right)\right\rangle\right]-\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left(f_h^{(t)}-\mathcal{T}_h^{\pi^*}f_{h+1}^{(t)}\right)\left(s_h,a_h\right)\right]\\ &+\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)}-\mathcal{T}_h f_{h+1}^{(t)}\right)\left(s_h,a_h\right)\right]+\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left\langle f_{h+1}^{(t)}(s_{h+1},\cdot),\pi_{h+1}^{f^{(t)}}(\cdot|s_{h+1})-\pi_{h+1}^{(t)}(\cdot|s_{h+1})\right\rangle\right]\end{aligned}$$

Proof. By adding and subtracting $f_1^{(t)}(s_1^{(t)},\pi_1^{(t)}(s_1^{(t)}))$, we obtain

$$\begin{aligned}\mathrm{Reg}(T) &= \sum_{t=1}^{T}\left(V_1^*(s_1^{(t)})-V_1^{\pi^{(t)}}(s_1^{(t)})\right)\\ &= \sum_{t=1}^{T}\left(V_1^*(s_1^{(t)})-f_1^{(t)}(s_1^{(t)},\pi_1^{(t)}(s_1^{(t)}))\right)+\sum_{t=1}^{T}\left(f_1^{(t)}(s_1^{(t)},\pi_1^{(t)}(s_1^{(t)}))-V_1^{\pi^{(t)}}(s_1^{(t)})\right).\end{aligned}\tag{C.1}$$

To further decompose the two terms in (C.1), we apply the value difference lemma/generalized policy difference lemma in Lemma C.1.3 (Cai et al., 2020; Efroni et al., 2020) with $f^{(t)}$ as the Q-function, $\pi'=\pi^*$, and $\pi=\pi^{(t)}$ to find that the first term can written as

$$\begin{aligned}&\sum_{t=1}^{T}\left(V_1^*(s_1^{(t)})-f_1^{(t)}(s_1^{(t)},\pi_1^{(t)}(s_1^{(t)}))\right)\\ &=\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}\left(s_h,\cdot\right),\pi_h^*\left(\cdot\mid s_h\right)-\pi_h^{(t)}\left(\cdot\mid s_h\right)\right\rangle\right]-\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left(f_h^{(t)}-\mathcal{T}_h^{\pi^*}f_{h+1}^{(t)}\right)\left(s_h,a_h\right)\right].\end{aligned}\tag{C.2}$$

Another application of Lemma C.1.3 with $\pi=\pi'=\pi^{(t)}$ and with $f^{(t)}$ as the Q-function yields

$$\begin{aligned}&\sum_{t=1}^{T}\left(f_1^{(t)}(s_1^{(t)},\pi_1^{(t)}(s_1^{(t)}))-V_1^{\pi^{(t)}}(s_1^{(t)})\right)\\ &=\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left\langle f_h^{(t)}\left(s_h,\cdot\right),\pi_h^{(t)}\left(\cdot\mid s_h\right)-\pi_h^{(t)}\left(\cdot\mid s_h\right)\right\rangle\right]+\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)}-\mathcal{T}_h^{\pi^{(t)}}f_{h+1}^{(t)}\right)\left(s_h,a_h\right)\right]\end{aligned}$$

$$= \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left(f_h^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right) (s_h, a_h) \right]. \tag{C.3}$$

We now need to relate the Bellman operator with respect to the current policy $\mathcal{T}_h^{\pi^{(t)}}$ to the Bellman operator $\mathcal{T}_h$. Observe that

$$\begin{aligned}
&\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left(f_h^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right) (s_h, a_h) \right] \\
&= \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right) (s_h, a_h) \right] + \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{\pi^{(t)}} \left[\left(\mathcal{T}_h f_{h+1}^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right) (s_h, a_h) \right]. \quad \text{(C.4)}
\end{aligned}$$

We further decompose the second term in the following way

$$\begin{aligned}
&\mathbb{E}_{\pi^{(t)}} \left[\left(\mathcal{T}_h f_{h+1}^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right) (s_h, a_h) \right] \\
&= \mathbb{E}_{\pi^{(t)}} \left[r_h(s_h, a_h) + f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{f^{(t)}}(s_{h+1}) \right) - r_h(s_h, a_h) - f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{(t)}(s_{h+1}) \right) \right] \\
&= \mathbb{E}_{\pi^{(t)}} \left[f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{f^{(t)}}(s_{h+1}) \right) - f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{(t)}(s_{h+1}) \right) \right] \\
&= \mathbb{E}_{\pi^{(t)}} \left[\left\langle f_{h+1}^{(t)}(s_{h+1}, \cdot), \pi_{h+1}^{f^{(t)}}(\cdot | s_{h+1}) - \pi_{h+1}^{(t)}(\cdot | s_{h+1}) \right\rangle \right]. \quad \text{(C.5)}
\end{aligned}$$

Replacing the original terms in (C.1) by (C.2), (C.4) and (C.5) concludes the proof. □

Each term within the regret decomposition is dealt with differently. We bound the first term via the standard mirror descent analysis, the second term with optimism, the third term with the GOLF regret decomposition and the SEC of Xie et al. (2022a), and the fourth term via a modified mirror descent analysis.

The attentive reader will note that the Bellman error is defined as $f - \mathcal{T} f$ in our setup and that of Liu et al. (2023b), and $\mathcal{T} f - f$ in that of Zhong and Zhang (2023).

Bounding the tracking error

Lemma C.2 (Mirror Descent Tracking Error for Algorithm 4.2)**.** *Let t_k and t_{k+1} be switch times within Algorithm 4.2, where we use the convention that $\pi^{(t_k)} \propto 1$ is post-policy reset and $\pi^{(t_{k+1})} \not\propto 1$*

is pre-policy reset. The tracking error with respect to the optimal policy is then bounded by:

$$\sum_{t=t_k+1}^{t_{k+1}-1}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}\left(s_h,\cdot\right),\pi_h^*\left(\cdot\mid s_h\right)-\pi_h^{(t)}\left(\cdot\mid s_h\right)\right\rangle\right]\leq \eta H^3(t_{k+1}-t_k)/2+\frac{H\log|\mathcal{A}|}{\eta}+H^2.$$

Proof. Note that for any t such that $t_k+1\leq t\leq t_{k+1}-1$, as we do not reset the policy during these timesteps as the critic does not change, we have

$$\pi_{h+1}^{(t+1)}(\cdot|s')=\frac{\pi_{h+1}^{(t)}(\cdot|s')\exp(\eta f_{h+1}^{(t)}(s',\cdot))}{\sum_{a\in\mathcal{A}}\pi_{h+1}^{(t)}(a|s')\exp(\eta f_{h+1}^{(t)}(s',a))}=Z^{-1}\pi_{h+1}^{(t)}(\cdot|s')\exp(\eta f_{h+1}^{(t)}(s',\cdot)). \tag{C.6}$$

Rearranging this yields

$$\eta f_{h+1}^{(t)}(s',\cdot)=\log Z+\log\pi_{h+1}^{(t+1)}(\cdot|s')-\log\pi_{h+1}^{(t)}(\cdot|s'),$$

where $\log Z$ is

$$\log Z=\log\left(\sum_{a\in\mathcal{A}}\pi_{h+1}^{(t)}(a|s')\exp(\eta f_{h+1}^{(t)}(s',a))\right).$$

We can now bound, noting that $\sum_{a\in\mathcal{A}}\left(\pi_{h+1}^*(\cdot|s')-\pi_{h+1}^{(t+1)}(\cdot|s')\right)=0$, that

$$\begin{aligned}
&\left\langle \eta f_{h+1}^{(t)}(s',\cdot),\pi_{h+1}^*(\cdot|s')-\pi_{h+1}^{(t+1)}(\cdot|s')\right\rangle\\
&=\left\langle \log Z+\log\pi_{h+1}^{(t+1)}(\cdot|s')-\log\pi_{h+1}^{(t)}(\cdot|s'),\pi_{h+1}^*(\cdot|s')-\pi_{h+1}^{(t+1)}(\cdot|s')\right\rangle\\
&=\left\langle \log\pi_{h+1}^{(t+1)}(\cdot|s')-\log\pi_{h+1}^{(t)}(\cdot|s'),\pi_{h+1}^*(\cdot|s')-\pi_{h+1}^{(t+1)}(\cdot|s')\right\rangle\\
&=\mathrm{KL}\left(\pi_{h+1}^*(\cdot|s')\mid\mid\pi_{h+1}^{(t)}(\cdot|s')\right)-\mathrm{KL}\left(\pi_{h+1}^*(\cdot|s')\mid\mid\pi_{h+1}^{(t+1)}(\cdot|s')\right)-\mathrm{KL}\left(\pi_{h+1}^{(t+1)}(\cdot|s')\mid\mid\pi_{h+1}^{(t)}(\cdot|s')\right),
\end{aligned} \tag{C.7}$$

where the last line follows from Lemma C.1.3 with $\pi_1=\pi_{h+1}^*(\cdot|s')$, $\pi_2=\pi_{h+1}^{(t)}(\cdot|s')$ and $\pi=$

$\pi_{h+1}^{(t+1)}(\cdot|s')$. So it must hold that

$$\begin{aligned}
&\left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{*}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \\
&= \left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{*}(\cdot|s') - \pi_{h+1}^{(t+1)}(\cdot|s') \right\rangle - \left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{(t)}(\cdot|s') - \pi_{h+1}^{(t+1)}(\cdot|s') \right\rangle \\
&\leq \mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{(t+1)}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) \\
&\qquad + \eta H ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1. \qquad \text{(C.8)}
\end{aligned}$$

Summing up t and h, we can then derive that

$$\begin{aligned}
&\sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^*}\left[\left\langle f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{*}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle\right] \\
&= \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^*}\left[\left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{*}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle\right] \\
&\leq \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^*}\Big[\mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right) \\
&\qquad\qquad -\mathrm{KL}\left(\pi_{h+1}^{(t+1)}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) + \eta H ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1\Big]. \qquad \text{(C.9)}
\end{aligned}$$

Here, we apply Pinsker's inequality on the last line of (C.9), it follows that

$$\begin{aligned}
&\sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^*}\left[\left\langle f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{*}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle\right] \\
&\leq \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^*}\Big[\mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right) \\
&\qquad\qquad -||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1^2/2 + \eta H ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1\Big] \\
&\leq \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{\pi^*}\Big[\mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{*}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right) + \eta^2 H^2/2\Big], \\
&\qquad\qquad \text{(C.10)}
\end{aligned}$$

where we use the fact that $\max_{x\in\mathbb{R}}\left\{-x^2/2 + \eta H x\right\} = \eta^2 H^2/2$ in the last line. Continuing to

simplify (C.10) yields

$$\begin{aligned}
&\sum_{t=t_k+1}^{t_{k+1}-1}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_{h+1}^{(t)}(s',\cdot),\pi_{h+1}^*(\cdot|s')-\pi_{h+1}^{(t)}(\cdot|s')\right\rangle\right]\\
&\leq\sum_{t=t_k+1}^{t_{k+1}-1}\sum_{h=1}^{H}\left(\eta H^2/2+\mathbb{E}_{\pi^*}\left[\frac{\mathrm{KL}\left(\pi_{h+1}^*(\cdot|s')\;||\;\pi_{h+1}^{(t)}(\cdot|s')\right)-\mathrm{KL}\left(\pi_{h+1}^*(\cdot|s')\;||\;\pi_{h+1}^{(t+1)}(\cdot|s')\right)}{\eta}\right]\right)\\
&\leq\eta H^3(t_{k+1}-t_k)/2+\sum_{h=1}^{H}\frac{\mathrm{KL}\left(\pi_{h+1}^*(\cdot|s')\;||\;\pi_{h+1}^{(t_k+1)}(\cdot|s')\right)-\mathrm{KL}\left(\pi_{h+1}^*(\cdot|s')\;||\;\pi_{h+1}^{(t_{k+1})}(\cdot|s')\right)}{\eta}\\
&\leq\eta H^3(t_{k+1}-t_k)/2+\frac{H\log|\mathcal{A}|}{\eta}+H^2,
\end{aligned}\tag{C.11}$$

where the last inequality follows from the fact that the KL-divergence is non-negative as well as the policy reset of setting $\pi_h^{(t_k)}$ back to the uniform policy after the critic update. Note that we use the convention that $\pi_h^{(t_k)}$ is after the reset, and $\pi_h^{(t_{k+1})}$ is before the reset. This means that

$$\frac{1}{|\mathcal{A}|\exp(\eta H)}\leq\frac{\exp(0)}{\sum_{a\in\mathcal{A}}\exp(\eta H)}\leq\pi_{h+1}^{(t_k+1)}(\cdot|s')=\frac{\exp(\eta f_{h+1}^{(t)}(s',\cdot))}{\sum_{a\in\mathcal{A}}\exp(\eta f_{h+1}^{(t)}(s',a))}\leq\frac{\exp(\eta H)}{\sum_{a\in\mathcal{A}}\exp(0)}=\frac{\exp(\eta H)}{|\mathcal{A}|},$$

and so we can conclude that

$$\mathrm{KL}\left(\pi_{h+1}^*(\cdot|s')\;||\;\pi_{h+1}^{(t_k+1)}(\cdot|s')\right)\leq\log\left(\frac{1}{\pi_{h+1}^{(t_k+1)}(\cdot|s')}\right)\leq\log|\mathcal{A}|+\eta H.\tag{C.12}$$

□

Asserting optimism

Lemma C.3 (Negative Bellman Error For Algorithm 4.2)**.** *Within Algorithm 4.2, we have that*

$$-\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[f_h^{(t)}-\mathcal{T}_h^{\pi^*}f_{h+1}^{(t)}\right]\leq 0.$$

Proof. Applying Lemma C.7, we note that $f_h^{(t)}\geq\mathcal{T}_h^{\pi^*}f_{h+1}^{(t)}$. The result then follows. □

Bounding the Bellman error under the learned policies

We now turn our attention to the Bellman error with respect to $\mathcal{T}_h$ under the current policy's occupancy measure.

Lemma C.4 (Sum of Bellman Errors Under Algorithm 4.2)**.** *Within Algorithm 4.2, the sum of Bellman errors with respect to $\mathcal{T}$ under the occupancy measure of $\pi^{(t)}$ can be bounded by:*

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)}-\mathcal{T}_h f_{h+1}^{(t)}\right)(s_h,a_h)\right]\leq\sqrt{\beta H^4 T\mathsf{SEC}(\mathcal{F},\Pi,T)},$$

where $\beta=\Theta\left(\log\left(HT^2\mathcal{N}_{\mathcal{F},\mathcal{T}\mathcal{F}}(1/T)/\delta\right)\right)$.

Proof. We can now perform the same Cauchy-Schwarz and change of measure argument as in Xie et al. (2022a) to find that

$$\begin{aligned}\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)}-\mathcal{T}_h f_{h+1}^{(t)}\right)(s_h,a_h)\right]&=\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}[\delta_h^{(t)}]\\&=\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}[\delta_h^{(t)}]\left(\frac{H^2\vee\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_h^{(t)})^2]}{H^2\vee\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_h^{(t)})^2]}\right)^{1/2}\\&\leq\sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H}\frac{\mathbb{E}_{d_h^{(i)}}[\delta_h^{(t)}]^2}{H^2\vee\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_h^{(t)})^2]}}\sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H}H^2\vee\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}[(\delta_h^{(t)})^2]}.\end{aligned}\tag{C.13}$$

Within the last inequality, the first term can be bounded by H times the SEC of Xie et al. (2022a), by Definition 4.4. The second term is bounded by Lemma C.7. Therefore, (C.13) can be bounded as

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^{(t)}}\left[\left(f_h^{(t)}-\mathcal{T}_h f_{h+1}^{(t)}\right)(s_h,a_h)\right]\leq\sqrt{H\mathsf{SEC}(\mathcal{F},\Pi,T)}\sqrt{\beta H^3 T}\leq\sqrt{\beta H^4 T\mathsf{SEC}(\mathcal{F},\Pi,T)}.\tag{C.14}$$

□

Lemma C.5 (Greedy Policy Tracking Error For Algorithm 4.2)**.** *Let t_k and t_{k+1} be switch times within Algorithm 4.2, where we use the convention that $\pi^{(t_k)} \propto 1$ is post-policy reset and $\pi^{(t_{k+1})} \not\propto 1$ is pre-policy reset. The tracking error with respect to the greedy policy corresponding to the current critic is then bounded by:*

$$\sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{a_h^{(t)}} \left[\left\langle f_{h+1}^{(t)}(s', \cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \right] \leq \eta H^3 (t_{k+1} - t_k)/2 + \frac{H \log |\mathcal{A}|}{\eta} + H^2.$$

Proof. Again note that for any t such that $t_k + 1 \leq t \leq t_{k+1}$, we do not reset the policy during these timesteps as the critic does not change. We therefore have

$$\pi_{h+1}^{(t+1)}(\cdot|s') = \frac{\pi_{h+1}^{(t)}(\cdot|s') \exp(\eta f_{h+1}^{(t)}(s', \cdot))}{\sum_{a \in \mathcal{A}} \pi_{h+1}^{(t)}(a|s') \exp(\eta f_{h+1}^{(t)}(s', a))} = Z_t^{-1} \pi_{h+1}^{(t)}(\cdot|s') \exp(\eta f_{h+1}^{(t)}(s', \cdot)), \tag{C.15}$$

and rearranging this yields

$$\eta f_{h+1}^{(t)}(s', \cdot) = \log Z_t + \log \pi_{h+1}^{(t+1)}(\cdot|s') - \log \pi_{h+1}^{(t)}(\cdot|s'),$$

where $\log Z_t$ is

$$\log Z_t = \log \left(\sum_{a \in \mathcal{A}} \pi_{h+1}^{(t)}(a|s') \exp(\eta f_{h+1}^{(t)}(s', a)) \right) = \log \pi_{h+1}^{(t)}(\cdot|s') - \log \pi_{h+1}^{(t+1)}(\cdot|s') + \eta f_{h+1}^{(t)}(s', \cdot).$$

Noting that $\sum_{a \in \mathcal{A}} \left(\pi_{h+1}(\cdot|s') - \pi'_{h+1}(\cdot|s') \right) = 0$ for any two policies $\pi, \pi' \in \Pi$, we can now bound

$$\begin{aligned}
&\left\langle \eta f_{h+1}^{(t)}(s', \cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t+1)}(\cdot|s') \right\rangle \\
&= \left\langle \log Z + \log \pi_{h+1}^{(t+1)}(\cdot|s') - \log \pi_{h+1}^{(t)}(\cdot|s'), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t+1)}(\cdot|s') \right\rangle \\
&= \left\langle \log \pi_{h+1}^{(t+1)}(\cdot|s') - \log \pi_{h+1}^{(t)}(\cdot|s'), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t+1)}(\cdot|s') \right\rangle \\
&= \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s') \right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s') \right) - \mathrm{KL}\left(\pi_{h+1}^{(t+1)}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s') \right),
\end{aligned} \tag{C.16}$$

where the last line follows from Lemma C.1.3. So it satisfies that

$$\begin{aligned}
&\left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \\
&= \left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t+1)}(\cdot|s') \right\rangle - \left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{(t)}(\cdot|s') - \pi_{h+1}^{(t+1)}(\cdot|s') \right\rangle \\
&\leq \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{(t+1)}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) \\
&\quad + \eta H ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1. \qquad \text{(C.17)}
\end{aligned}$$

Sum up with t and h, we can then derive

$$\begin{aligned}
&\sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \right] \\
&= \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle \eta f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \right] \\
&\leq \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \Big[\mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right) \\
&\qquad\qquad - \mathrm{KL}\left(\pi_{h+1}^{(t+1)}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) + \eta H ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1 \Big]. \qquad \text{(C.18)}
\end{aligned}$$

When applying Pinsker's inequality in the last line of (C.18), we can show that

$$\begin{aligned}
&\sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_{h+1}^{(t)}(s',\cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \right] \\
&\leq \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \Big[\mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right) \\
&\qquad\qquad - ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1^2/2 + \eta H ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1 \Big]. \qquad \text{(C.19)}
\end{aligned}$$

Here, we use the fact that $\max_{x\in\mathbb{R}} \left\{-x^2/2 + \eta H x\right\} = \eta^2 H^2/2$, and obtain that

$$-||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1^2/2 + \eta H ||\pi_{h+1}^{(t+1)}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s')||_1 \leq \eta^2 H^2/2.$$

We now can continue through the following. Note that $f^{(t_k+1)} = ...f^{(t)} = ... = f^{(t_{k+1})}$ due to rare-switching. With a direct calculation, we obtain that

$$\begin{aligned}
&\sum_{t=t_k+1}^{t_{k+1}-1} \mathbb{E}_{d_h^{(t)}} \left[\frac{\mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right)}{\eta} \right] \\
&= \sum_{t=t_k+1}^{t_{k+1}-1} \mathbb{E}_{d_h^{(t)}} \left[\frac{\mathrm{KL}\left(\pi_{h+1}^{f^{(t_k+1)}}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t_k+1)}}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right)}{\eta} \right] \\
&= \mathbb{E}_{d_h^{(t)}} \left[\frac{\mathrm{KL}\left(\pi_{h+1}^{f^{(t_k+1)}}(\cdot|s') \,||\, \pi_{h+1}^{(t_k+1)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t_k+1)}}(\cdot|s') \,||\, \pi_{h+1}^{(t_{k+1})}(\cdot|s')\right)}{\eta} \right]. \qquad \text{(C.20)}
\end{aligned}$$

Furthermore, there exists some $a^*(s')$ for each s' such that

$$a^*(s) = \arg\max_{a' \in \mathcal{A}} f_{h+1}^{(t)}(s', a') \text{ for all } t_k + 1 \le t \le t_{k+1},$$

hence that $\pi_h^f(a'|s) = \mathbf{1}\left(a' = \arg\max_{a \in \mathcal{A}} f_h(s,a)\right)$. This yields

$$\begin{aligned}
&\mathbb{E}_{d_h^{(t)}} \left[\frac{\mathrm{KL}\left(\pi_{h+1}^{f^{(t_k+1)}}(\cdot|s') \,||\, \pi_{h+1}^{(t_k+1)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t_k+1)}}(\cdot|s') \,||\, \pi_{h+1}^{(t_{k+1})}(\cdot|s')\right)}{\eta} \right] \\
&= \frac{1}{\eta} \mathbb{E}_{d_h^{(t)}} \left[1 \cdot \log\left(\frac{1}{\pi_{h+1}^{(t_k+1)}(a^*(s')|s')} \right) - 1 \cdot \log\left(\frac{1}{\pi_{h+1}^{(t_{k+1})}(a^*(s')|s')} \right) \right] \\
&= \frac{1}{\eta} \mathbb{E}_{d_h^{(t)}} \left[\log\left(\frac{\pi_{h+1}^{(t_{k+1})}(a^*(s')|s')}{\pi_{h+1}^{(t_k+1)}(a^*(s')|s')} \right) \right]. \qquad \text{(C.21)}
\end{aligned}$$

For this term, we note that $\pi_h^{(t_k)}$ is back to the uniform policy after the critic update. Note that we use the convention that $\pi_h^{(t_k)}$ is before the reset, and $\pi_h^{(t_k+1)}$ is after the reset. This means that

$$\frac{1}{|\mathcal{A}| \exp(\eta H)} \le \frac{\exp(0)}{\sum_{a \in \mathcal{A}} \exp(\eta H)} \le \pi_{h+1}^{(t_k+1)}(\cdot|s') = \frac{\exp(\eta f_{h+1}^{(t)}(s', \cdot))}{\sum_{a \in \mathcal{A}} \exp(\eta f_{h+1}^{(t)}(s', a))} \le \frac{\exp(\eta H)}{\sum_{a \in \mathcal{A}} \exp(0)} = \frac{\exp(\eta H)}{|\mathcal{A}|},$$

and so we can conclude that

$$\sum_{t=t_k+1}^{t_{k+1}-1} \mathbb{E}_{d_h^{(t)}} \left[\frac{\mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t)}(\cdot|s')\right) - \mathrm{KL}\left(\pi_{h+1}^{f^{(t)}}(\cdot|s') \,||\, \pi_{h+1}^{(t+1)}(\cdot|s')\right)}{\eta} \right]$$
$$= \frac{1}{\eta} \log \left(\frac{\pi_{h+1}^{(t_{k+1})}(a^*(s')|s')}{\pi_{h+1}^{(t_k+1)}(a^*(s')|s')} \right) \leq \frac{1}{\eta} \log \left(\frac{1}{\pi_{h+1}^{(t_k+1)}(a^*(s')|s')} \right) \leq \frac{1}{\eta} \left(\log |\mathcal{A}| + \eta H \right). \tag{C.22}$$

Therefore, we obtain that

$$\sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_{h+1}^{(t)}(s', \cdot), \pi_{h+1}^{f^{(t)}}(\cdot|s') - \pi_{h+1}^{(t)}(\cdot|s') \right\rangle \right] \leq \frac{1}{\eta} \sum_{t=t_k+1}^{t_{k+1}-1} \sum_{h=1}^{H} (\eta^2 H^2/2) + \frac{1}{\eta} \sum_{h=1}^{H} \left(\log |\mathcal{A}| + \eta H \right)$$
$$\leq \eta H^3 (t_{k+1} - t_k)/2 + \frac{H \log |\mathcal{A}|}{\eta} + H^2. \tag{C.23}$$

□

Auxiliary lemmas

Bound on rare-switching update frequency We bound the rare-switching update frequency via an argument similar to that of Xiong et al. (2023a).

Lemma C.6 (Switching Costs)**.** *Consider a procedure where the critic is updated only when there exists some h such that*

$$\mathcal{L}_h^{(t)}(f_h^{(t)}, f_{h+1}^{(t)}) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t)}(f_h', f_{h+1}^{(t)}) \geq 5H^2\beta.$$

This performs no more than $N_{updates,h}(T) \leq O(d \log(T))$ Q-function updates for each $h \in [H]$, and no more than $N_{updates}(T) \leq O(dH \log(T))$ Q-function updates in total.

Proof. To show this result, we control the number of switches induced by the Q-function class targeting π^* by upper and lower bounding the cumulative squared Bellman error under the observed states and actions. Fix some $h \in [H]$ for now. For simplicity, write $K_h = N_{\text{updates},h}(T)$ for the total number of updates, and $t_{1,h}, ..., t_{K_h,h}$ the update times for $f_h^{(t)}$, with $t_{0,h} = 0$. By definition,

at every $t_{k,h}$,

$$\mathcal{L}_h^{(t_{k,h})}\left(f_h^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t_{k,h})}\left(f_h', f_{h+1}^{(t_{k,h})}\right) \geq 5H^2\beta. \tag{C.24}$$

An application of Lemma C.1.3 yields

$$0 \leq \mathcal{L}_h^{\left(t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{\left(t_{k,h}\right)}\left(f_h', f_{h+1}^{(t_{k,h})}\right) \leq H^2\beta,$$

$$\min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{\left(t_{k,h}\right)}\left(f_h', f_{h+1}^{(t_{k,h})}\right) \geq \mathcal{L}_h^{\left(t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) - H^2\beta,$$

$$\mathcal{L}_h^{(t_{k,h})}\left(f_h^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) - \mathcal{L}_h^{\left(t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) \geq 4H^2\beta.$$

From the above, one can obtain

$$\begin{aligned}
&\mathcal{L}_h^{(t_{k-1,h}+1:t_{k,h})}\left(f_h^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) - \mathcal{L}_h^{\left(t_{k-1,h}+1:t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) \\
&= \mathcal{L}_h^{(t_{k-1,h}+1:t_{k,h})}\left(f_h^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right) - \mathcal{L}_h^{\left(t_{k-1,h}+1:t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right) \\
&= \mathcal{L}_h^{(t_{k,h})}\left(f_h^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right) - \mathcal{L}_h^{\left(t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right) \\
&\quad - \left(\mathcal{L}_h^{(t_{k-1,h})}\left(f_h^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right) + \mathcal{L}_h^{\left(t_{k-1,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right)\right) \\
&= \mathcal{L}_h^{(t_{k,h})}\left(f_h^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) - \mathcal{L}_h^{\left(t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k,h})}, f_{h+1}^{(t_{k,h})}\right) \\
&\quad - \left(\mathcal{L}_h^{(t_{k-1,h})}\left(f_h^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right) + \mathcal{L}_h^{\left(t_{k-1,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t_{k-1,h}+1)}, f_{h+1}^{(t_{k-1,h}+1)}\right)\right) \\
&\geq 4H^2\beta - H^2\beta = 3H^2\beta.
\end{aligned} \tag{C.25}$$

Therefore, for any t such that $t_{k-1,h} < t \leq t_{k,h}$, this argument and noting that $f_h^{(t_{k-1,h}+1)} = ... = f_h^{(t)} = ... = f_h^{(t_{k,h})}$ yields

$$\mathcal{L}_h^{(t_{k-1,h}+1:t_{k,h})}\left(f_h^{(t)}, f_{h+1}^{(t)}\right) - \mathcal{L}_h^{\left(t_{k-1,h}+1:t_{k,h}\right)}\left(\mathcal{T}_h f_{h+1}^{(t)}, f_{h+1}^{(t)}\right) \geq 3H^2\beta. \tag{C.26}$$

An application of Lemma C.1.3 while noting that $f_h^{(t_{k-1,h}+1)} = ... = f_h^{(t)} = ... = f_h^{(t_{k,h})}$ leads to

$$\sum_{i=t_{k-1,h}+1}^{t_{k,h}} \left(f_h^{(i)} - \mathcal{T}_h f_{h+1}^{(i)}\right)^2 (s_h^{(i)}, a_h^{(i)}) = \sum_{i=t_{k-1,h}+1}^{t_{k,h}} \left(f_h^{(t_{k,h})} - \mathcal{T}_h f_{h+1}^{(t_{k,h})}\right)^2 (s_h^{(i)}, a_h^{(i)}) \geq 2H^2\beta. \quad \text{(C.27)}$$

Now summing over all $t_{1,h}, ..., t_{K,h}$ yields

$$\sum_{t=1}^{T} \left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2 (s_h^{(t)}, a_h^{(t)}) = \sum_{k=1}^{K_h} \sum_{i=t_{k-1,h}+1}^{t_{k,h}} \left(f_h^{(i)} - \mathcal{T}_h f_{h+1}^{(i)}\right)^2 (s_h^{(i)}, a_h^{(i)}) \geq 2(K_h - 1)H^2\beta. \quad \text{(C.28)}$$

By Lemma C.7, we have that

$$\sum_{i=1}^{t-1} \left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2 (s_h^{(i)}, a_h^{(i)}) \leq O(H^2\beta). \quad \text{(C.29)}$$

Invoking the squared distributional Bellman eluder dimension definition, as in Xiong et al. (2023a), yields

$$\sum_{t=1}^{T} \left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2 (s_h^{(t)}, a_h^{(t)}) \leq O(dH^2\beta \log T). \quad \text{(C.30)}$$

So we have established that

$$2(K_h - 1)H^2\beta \leq \sum_{t=1}^{T} \left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2 (s_h^{(t)}, a_h^{(t)}) \leq O(dH^2\beta \log T). \quad \text{(C.31)}$$

The number of updates for each h must therefore be bounded as

$$K_h \leq d \log T, \text{ yielding } N_{\text{switch}}(T) \leq dH \log T.$$

□

Showing optimism for critics targeting Q^* The following lemma applies to Algorithms 4.2 and 4.4. As Algorithm 4.2 is optimistic, both properties apply to it, while only the second applies

to Algorithm 4.4.

Lemma C.7 (Optimism and in-sample error control for critics targeting Q^*)**.** *With probability at least $1-\delta$, for all $t \in [T]$, we have that for all $h = 1, \ldots, H$, an optimistic critic targeting Q^* in the same way as defined in Algorithm 4.2 achieves*

$$\textit{(i)}\ \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \in \mathcal{F}_h^{(t)},\ \textit{and}\ f_h^{(t)} \geq \mathcal{T}_h^{\pi} f_{h+1}^{(t)}\ \textit{for all}\ \pi,$$

Similarly, a critic targeting Q^* as in Algorithms 4.2 and 4.4 achieves

$$\textit{(ii)}\ \sum_{i=1}^{t-1} \mathbb{E}_{d^{\pi^{(i)}}} \left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right)^2 \right] \leq O\left(H^2 \beta\right),$$

by choosing $\beta = c_1 \left(\log \left[HT \mathcal{N}_{\mathcal{F}, \mathcal{TF}}(\rho)/\delta \right] \right)$ for some constant c_1.

Proof. By Lemma C.1.3 applied to the greedy policy $\pi^{f^{(t)}}$, for any $h \in [H]$ and $t \in [T]$, we have with probability $1-\delta$ that

$$0 \leq \mathcal{L}_h^{(t)}(\mathcal{T}_h f_{h+1}^{(t)}, f_{h+1}^{(t)}) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t)}(f_h', f_{h+1}^{(t)}) \leq H^2 \beta. \tag{C.32}$$

We construct the confidence sets only at timesteps t_k where switches occur:

$$\mathcal{F}_h^{(t_{\text{last}})} := \left\{ f \in \mathcal{F} : \mathcal{L}_h^{(t_{\text{last}})}(f_h, f_{h+1}) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t_{\text{last}})}(f_h', f_{h+1}) \leq H^2 \beta \ \ \forall h \in [H] \right\}.$$

We can now show the first result. As we defined $f_h^{(t)}(s,a) := \operatorname{argmax}_{f_h \in \mathcal{F}_h^{(t_{\text{last}})}} f_h(s,a)$, for all π:

$$\mathcal{T}_h^{\pi} f_{h+1}^{(t)}(s,a) = r_h(s,a) + \mathbb{E}_{s'} \left[f_{h+1}^{(t)}(s', \pi_{h+1}(s')) \right] \leq r_h(s,a) + \mathbb{E}_{s'} \left[\max_{a' \in \mathcal{A}} f_{h+1}^{(t)}(s', a') \right] = \mathcal{T}_h f_{h+1}^{(t)}(s,a). \tag{C.33}$$

We further note that $\mathcal{T}_h f_{h+1}^{(t)}(s,a) \leq f_h^{(t)}(s,a)$, as $\mathcal{T}_h f_{h+1}^{(t)} \in \mathcal{F}_h^{(t_{\text{last}})}$ and by the definition of $f_h^{(t)}(s,a)$.

The second result can now be shown using by Lemma C.1.3 using a similar argument to the proof of Theorem 1 in Xiong et al. (2023a), which in turn takes inspiration from the proofs of Lemmas 39 and 40 in Jin et al. (2021a). We elaborate accordingly.

Consider two cases, one where we perform an update at episode $t-1$ and one where we do not. If we perform an update at episode $t-1$, then by the choice of $f^{(t)}$ to be near-optimal (in fact, with Algorithm 4.4, it is optimal and this is zero) it must hold that

$$\mathcal{L}_h^{(t-1)}\left(f_h^{(t)}, f_{h+1}^{(t)}\right) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t-1)}\left(f_h', f_{h+1}^{(t)}\right) \leq 5H^2\beta,$$

and by Lemma C.1.3 it must also hold that

$$0 \leq \mathcal{L}_h^{(t-1)}\left(\mathcal{T}_h f_h^{(t)}, f_{h+1}^{(t)}\right) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t-1)}\left(f_h', f_{h+1}^{(t)}\right) \leq H^2\beta.$$

One can then see that

$$\mathcal{L}_h^{(t-1)}\left(f_h^{(t)}, f_{h+1}^{(t)}\right) - \mathcal{L}_h^{(t-1)}\left(\mathcal{T}_h f_h^{(t)}, f_{h+1}^{(t)}\right) \leq 6H^2\beta. \tag{C.34}$$

The same holds for the other case where we do not perform an update at episode $t-1$. Observe that because we did not perform an update,

$$\mathcal{L}_h^{(t-1)}\left(f_h^{(t-1)}, f_{h+1}^{(t-1)}\right) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t-1)}\left(f_h', f_{h+1}^{(t-1)}\right) \leq 5H^2\beta.$$

From Lemma C.1.3, it also holds that

$$\mathcal{L}_h^{(t-1)}\left(\mathcal{T}_h f_h^{(t-1)}, f_{h+1}^{(t-1)}\right) - \min_{f_h' \in \mathcal{F}_h} \mathcal{L}_h^{(t-1)}\left(f_h', f_{h+1}^{(t-1)}\right) \leq H^2\beta.$$

Putting the above two statements together and using that $f^{(t)} = f^{(t-1)}$ again yields

$$\mathcal{L}_h^{(t-1)}\left(f_h^{(t)}, f_{h+1}^{(t)}\right) - \mathcal{L}_h^{(t-1)}\left(\mathcal{T}_h f_h^{(t)}, f_{h+1}^{(t)}\right) = \mathcal{L}_h^{(t-1)}\left(f_h^{(t-1)}, f_{h+1}^{(t-1)}\right) - \mathcal{L}_h^{(t-1)}\left(\mathcal{T}_h f_h^{(t-1)}, f_{h+1}^{(t-1)}\right) \leq 6H^2\beta. \tag{C.35}$$

An application of Lemma C.1.3 to both cases then yields in either case that

$$\sum_{i=1}^{t-1}\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2 (s_h^{(i)}, a_h^{(i)}) \leq 7H^2\beta, \tag{C.36}$$

and also that

$$\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}\left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2\right] \leq 7H^2\beta. \tag{C.37}$$

□

C.1.3. Deferred Proofs, Auxiliary Lemmas, and Experimental Details for Part I

The proofs for the hybrid RL extensions in Section 4.1.5 (Theorem 4.14, Theorem 4.15, and Theorem 4.17), the Freedman concentration inequalities for the empirical squared Bellman loss, the analysis of rare-switching critics targeting $Q^{\pi^{(t)}}$, standard covering number and performance difference lemmas, and supplementary experimental details for Figure 4.3 and Figure 4.4 are deferred to Tan et al. (2025).

C.2. Proofs and Supplementary Details for Part II

Complete proofs and supplementary details for Part II of Chapter 4—including the projected online mirror descent and SPMA actor regret bounds (Theorems 4.19 and 4.22), the Kiefer–Wolfowitz G-optimal experimental design and reward-free coreset constructions, the Langevin Monte Carlo critic optimism and prediction error bounds (Theorem 4.24), the on-policy and off-policy sample complexity proofs (Theorems 4.25 to 4.27), and additional experimental details and ablations on Random MDP, Deep Sea, and Atari—are deferred to Lin et al. (2026).

APPENDIX D

PROOFS AND SUPPLEMENTARY DETAILS FOR CHAPTER 5

This appendix provides the complete numerical experiment tables, coverage examples, rollout sampler specification, hybrid RL extension, score regularity lemmas, and regret proofs for Algorithm 5.1, Algorithm 5.2, and Algorithm 5.3 in Chapter 5.

D.1. Additional Remarks, Examples, and Regularity Verifications

Extended numerical experiment tables, expressivity and coverage examples, the diffusion policy rollout sampler specification, the hybrid fitted Q-iteration regret statement, policy score moment bounds, and concrete regularity verifications for standard MLP and U-Net denoisers appear in Appendices A–E of Tan et al. (2026).

D.2. Analysis of Vanilla Diffusion Policy Optimization

For the proofs, abbreviate the comparison-policy value gap by $\epsilon_* := V^*(\rho) - V^{\pi_{\mathrm{cmp}}}(\rho)$. This is distinct from the squared advantage residual ϵ_{comp}. We also write $\Delta := V^*(\nu) - V^{(1)}(\nu)$, $C_A := 4(1+\gamma)/(1-\gamma)^2$, and $M_g^2 := C_A \beta_K^2/(1-\gamma)^2$ for the quantities substituted explicitly in Theorem 5.7.

D.2.1. Proof of Lemma D.1

Lemma D.1 (Features for DDPM)**.** *Consider the diffusion model features* $\widehat{\boldsymbol{\phi}}^\theta := \sum_{k=1}^K \nabla_\theta \log \pi_\theta(a_{k-1}|a_k^{(t)}, s)$, *and the ground-truth gradient of the log-policy* $\boldsymbol{\phi}^\theta = \nabla_\theta \log \pi_\theta(a|s)$. *For positive transition variances,*

$$\widehat{\boldsymbol{\phi}}^\theta(s, a_{0:K}) = \sum_{k=1}^K \frac{\eta_k^{\mathsf{ddpm}}}{\sigma_k^{\mathsf{ddpm}}} \left[\nabla_\theta \epsilon_\theta\left(a_k, k, s\right)\right]^\top Z_k.$$

Proof. Define $\mu_\theta(a, k, s) = \frac{1}{\sqrt{\alpha_k}}\big(a + \eta_k^{\mathsf{ddpm}} \epsilon_\theta(a, k, s)\big)$.

$$\nabla_\theta \log \pi_{\theta^{(t)}}(a_{k-1}|a_k, s) = \frac{\alpha_k}{(\sigma_k^{\mathrm{ddpm}})^2} \left[\nabla_\theta \mu_\theta(a_k, k, s)\right]^\top \left(a_{k-1} - \mu_\theta(a_k, k, s)\right)$$

$$= \alpha_k \cdot \left[\frac{\nabla_\theta \mu_\theta(a_k, k, s)}{\sigma_k^{\text{ddpm}}}\right]^\top \cdot \left[\frac{a_{k-1} - \mu_\theta(a_k, k, s)}{\sigma_k^{\text{ddpm}}}\right]$$
$$= \frac{\eta_k^{\mathsf{ddpm}}}{\sigma_k^{\mathsf{ddpm}}} [\nabla_\theta \epsilon_\theta(a_k, k, s)]^\top Z_k,$$

where Z_k's are i.i.d. standard Gaussian noise used in DDPM update rule (5.2). The likelihood calculation requires $\sigma_k^{\mathsf{ddpm}} > 0$ at every step, including the final step. □

Lemma D.2 (Exact Endpoint Features for DDIM)**.** *Under Assumption 5.4, the sampled latent a_K is uniquely determined by (s, a_0). Thus the DDIM feature equals the endpoint score $\widehat{\boldsymbol{\phi}} = \boldsymbol{\phi} = \nabla_\theta \log \pi_\theta(a_0 \mid s)$ and is given by Lemma D.1, including the spatial derivative of $\log|\det \mathbf{J}_\theta|$.*

Proof. By Assumption 5.4, the map $\Phi_{K:0}^{(t)}$ is invertible. We therefore have that for any action $a^{(t)} = a_{\cdot,0}^{(t)} \in \mathcal{A}$, there exists a $a_{\cdot,K}^{(t)} \in \mathbb{R}^d$ that is a realization of $\mathcal{N}(0, \mathbf{I}_d)$ such that

$$a_{\cdot,0}^{(t)} = \Phi_{K:0}^{(t)}(a_{\cdot,K}^{(t)}), \qquad a_{\cdot,K}^{(t)} = \left(\Phi_{K:0}^{(t)}\right)^{-1}(a_{\cdot,0}^{(t)}).$$

The Jacobian with regard to this map is given by the derivative with regard to the action, not the parameter:

$$\mathbf{J}_{\theta^{(t)}}(a_{\cdot,K}^{(t)}) = \nabla_z \Phi_{K:0}^{(t)}(z)\big|_{z=a_{\cdot,K}^{(t)}}, \text{ which, by Assumption 5.4, is invertible.}$$

By the change of variables formula, we have that

$$\pi_{\theta^{(t)}}(a_{\cdot,0}^{(t)}) = \frac{1}{(2\pi)^{d/2}} \exp\left\{-\frac{||a_{\cdot,K}^{(t)}||_2^2}{2}\right\} \left|\det\left(\mathbf{J}_{\theta^{(t)}}(a_{\cdot,K}^{(t)})\right)\right|^{-1},$$

$$\log \pi_{\theta^{(t)}}(a_{\cdot,0}^{(t)}) = -\frac{d\log(2\pi)}{2} - \frac{||a_{\cdot,K}^{(t)}||_2^2}{2} - \log\left|\det\left(\mathbf{J}_{\theta^{(t)}}(a_{\cdot,K}^{(t)})\right)\right|,$$

Differentiation here holds $a_{\cdot,0}^{(t)}$ fixed. Write $z = a_{\cdot,K}^{(t)}$ and $\ell_\theta(z) = \log|\det \mathbf{J}_\theta(z)|$. Then

$$\partial_{\theta_i} \log \pi_\theta(a_0 \mid s) = -z^\top \partial_{\theta_i} z - \operatorname{tr}(\mathbf{J}_\theta^{-1} \partial_{\theta_i} \mathbf{J}_\theta) - (\nabla_z \ell_\theta)^\top \partial_{\theta_i} z.$$

Regarding $\nabla_\theta a_{\cdot,K}^{(t)} = \nabla_\theta \left(\Phi_{K:0}^{(t)}\right)^{-1} (a_{\cdot,0}^{(t)})$, we can write

$$\varphi(\theta, z) = \Phi_{K:0}^{\theta}(z) - a_{\cdot,0}^{(t)}, \text{ where } a_{\cdot,0}^{(t)} \text{ is fixed, and } \varphi(\theta^{(t)}, a_{\cdot,K}^{(t)}) = \Phi_{K:0}^{(t)}(a_{\cdot,K}^{(t)}) - a_{\cdot,0}^{(t)} = 0,$$

as a function $\varphi : \Theta \times \mathbb{R}^d \to \mathcal{A}$ of the parameter and noise. By the implicit function theorem, there exists an open set $U \subseteq \Theta$ containing $\theta^{(t)}$ and some function $\psi : U \to \mathbb{R}^d$ such that $\psi(\theta^{(t)}) = a_{\cdot,K}^{(t)}$, and $\varphi(\theta, \psi(\theta)) = 0$ for all $\theta \in U$. Moreover, by the implicit function theorem as well, we can write

$$\nabla_\theta \psi(\theta) = -\left(\nabla_a \varphi(\theta, \psi(\theta))\right)^{-1} \nabla_\theta \varphi(\theta, \psi(\theta)).$$

This implies that

$$\begin{aligned}
\nabla_\theta a_{\cdot,K}^{(t)} &= \nabla_\theta \psi(\theta^{(t)}) \\
&= -\left(\nabla_z \varphi(\theta^{(t)}, a_{\cdot,K}^{(t)})\right)^{-1} \nabla_\theta \varphi(\theta^{(t)}, a_{\cdot,K}^{(t)}) \\
&= -\left(\nabla_z \Phi_{K:0}^{(t)}(a_{\cdot,K}^{(t)})\right)^{-1} \nabla_\theta \Phi_{K:0}^{(t)}(a_{\cdot,K}^{(t)}) \\
&= -\left(\mathbf{J}_{\theta^{(t)}}(a_{\cdot,K}^{(t)})\right)^{-1} \nabla_\theta \Phi_{K:0}^{(t)}(a_{\cdot,K}^{(t)}).
\end{aligned}$$

Substituting $\partial_{\theta_i} z = -w_i(z)$ gives

$$\boldsymbol{\phi}_i^\theta = z^\top w_i(z) - \operatorname{tr}(\mathbf{J}_\theta^{-1} \partial_{\theta_i} \mathbf{J}_\theta) + (\nabla_z \log|\det \mathbf{J}_\theta|)^\top w_i(z).$$

Given (s, a_0), the inverse map supplies a unique z. We can therefore evaluate the endpoint score from the sampled DDIM path without averaging over additional latent paths. □

D.2.2. Proof of Unbiased Gradient Estimates

Lemma D.3 (Unbiased Policy Gradient Estimator)**.** *Let $\widehat{A}^{\pi_\theta}, s, a_0, ..., a_K$ be a sample from Algorithm D.1 run from a distribution ν at parameter θ. Then, when $\widehat{\boldsymbol{\phi}}^\theta = \sum_{k=1}^K \nabla_\theta \log \pi_\theta(a_{k-1}|a_k, s)$, we have that $\widehat{A}^{\pi_\theta}\widehat{\boldsymbol{\phi}}^\theta/(1-\gamma)$ is an unbiased estimator for the policy gradient starting from ν:*

$$\nabla_\theta V^{\pi_\theta}(\nu) = \frac{1}{1-\gamma}\mathbb{E}_{s,a_0,...,a_K\sim d_\nu^{\pi_\theta}}\left[\widehat{A}^{\pi_\theta}(s,a_0)\cdot\widehat{\boldsymbol{\phi}}^\theta(s,a_{0:K})\right].$$

Also, when $\boldsymbol{\phi}^\theta = \nabla_\theta \log \pi_\theta(a_0|s)$, $\widehat{A}^{\pi_\theta}\boldsymbol{\phi}^\theta/(1-\gamma)$ is also an unbiased estimator for the policy gradient starting from ν:

$$\nabla_\theta V^{\pi_\theta}(\nu) = \frac{1}{1-\gamma}\mathbb{E}_{s,a_0,...,a_K\sim d_\nu^{\pi_\theta}}\left[\widehat{A}^{\pi_\theta}(s,a_0)\cdot\boldsymbol{\phi}^\theta(s,a_{0:K})\right].$$

Furthermore, conditional on the accepted state and denoising path, $\mathbb{E}[\widehat{A} \mid s, a_{0:K}] = A^{\pi_\theta}(s,a_0)$ and $\mathbb{E}[\widehat{A}^2 \mid s, a_{0:K}] \le C_A = 4(1+\gamma)/(1-\gamma)^2$. Consequently either normalized gradient estimate has second moment at most $M_g^2 = C_A\beta_K^2/(1-\gamma)^2$.

Proof. The claim that $\widehat{A}^{\pi_\theta}$ is unbiased and $s, a_0 \sim d_\nu^{\pi_\theta}$ follows from Agarwal et al. (2021) in their construction of their Algorithm 3 (our Algorithm D.1 when adapted to the special case of diffusion policies). As such, in an abuse of notation, we can now write $d_\nu^{\pi_\theta}$ for a measure over the state-action space and denoising actions yielded by Algorithm D.1 here.

Unbiasedness of the estimator then follows from plugging this into Equation 5.4:

$$\begin{aligned}
&\mathbb{E}_{s,a_0,...,a_K\sim d_\nu^{\pi_\theta}}\left[\widehat{A}^{\pi_\theta}(s,a_0)\widehat{\boldsymbol{\phi}}^\theta(s,a_{0:K})\right] && \text{(D.1)}\\
&= \mathbb{E}_{s,a_0,...,a_K\sim d_\nu^{\pi_\theta}}\left[\mathbb{E}[\widehat{A}^{\pi_\theta}(s,a_0)\cdot\widehat{\boldsymbol{\phi}}^\theta(s,a_{0:K}) \mid s, a_{0:K}]\right] && \text{(D.2)}\\
&= \mathbb{E}_{s,a_0,...,a_K\sim d_\nu^{\pi_\theta}}\left[\mathbb{E}[\widehat{A}^{\pi_\theta}(s,a_0) \mid s, a_0]\cdot\widehat{\boldsymbol{\phi}}^\theta(s,a_{0:K})\right] && \text{(D.3)}\\
&= \mathbb{E}_{s,a_0,...,a_K\sim d_\nu^{\pi_\theta}}\left[A^{\pi_\theta}(s,a_0)\cdot\widehat{\boldsymbol{\phi}}^\theta(s,a_{0:K})\right] && \text{(D.4)}\\
&= (1-\gamma)\nabla_\theta V^{\pi_\theta}(\nu). && \text{(D.5)}
\end{aligned}$$

The first equality follows from the tower property, the second from the observation that the fact that $\widehat{\boldsymbol{\phi}}^\theta(s, a_{0:K})$ is a deterministic measurable function of $s, a_0, ..., a_K$ allows us to pull it out of the conditional expectation, the third from the unbiasedness of the estimate $\mathbb{E}[\widehat{A}^{\pi_\theta}(s,a)|s,a] = A^{\pi_\theta}(s,a)$, and the fourth from Equation 5.4.

The case for $\boldsymbol{\phi}^\theta = \nabla_\theta \log \pi_\theta(a_0|s)$ follows identically, but by plugging the resulting expression into Equation 5.3 instead of Equation 5.4:

$$\mathbb{E}_{s,a_0\sim d_\nu^{\pi_\theta}}\left[\widehat{A}^{\pi_\theta}(s,a_0)\boldsymbol{\phi}^\theta(s,a_0)\right] \tag{D.6}$$

$$= \mathbb{E}_{s,a_0\sim d_\nu^{\pi_\theta}}\left[\mathbb{E}[\widehat{A}^{\pi_\theta}(s,a_0)\cdot\boldsymbol{\phi}^\theta(s,a_0) \mid s,a_0]\right] \tag{D.7}$$

$$= \mathbb{E}_{s,a_0\sim d_\nu^{\pi_\theta}}\left[\mathbb{E}[\widehat{A}^{\pi_\theta}(s,a_0) \mid s,a_0]\cdot\boldsymbol{\phi}^\theta(s,a_0)\right] \tag{D.8}$$

$$= \mathbb{E}_{s,a_0\sim d_\nu^{\pi_\theta}}\left[A^{\pi_\theta}(s,a_0)\cdot\boldsymbol{\phi}^\theta(s,a_0)\right] \tag{D.9}$$

$$= (1-\gamma)\nabla_\theta V^{\pi_\theta}(\nu). \tag{D.10}$$

The return rollout uses fresh randomness conditional on the accepted state and path. Its horizon H includes its first reward and obeys $\mathbb{P}(H = h) = (1-\gamma)\gamma^{h-1}$, so $\mathbb{E}H^2 = (1+\gamma)/(1-\gamma)^2$. Since $0 \le r \le 1$, the coin estimator satisfies $|\widehat{A}| \le 2H$ in either branch, proving its conditional second-moment bound. Multiplying by the score moment proves the gradient bound. One sampler execution uses an expected $(1+\gamma)/(1-\gamma)$ environment transitions, counting occupancy search and the return rollout. □

D.2.3. Gradient Domination, Proof of Lemma 5.6

Now we provide a counterpart of the gradient domination lemma proved for tabular and softmax policies from Agarwal et al. (2020). We establish a variant for a smooth policy class with controlled compatible advantage error, finite comparator coverage, and a positive lower bound on the nonzero Fisher spectrum. While we utilize this lemma in Lemma 5.6 for diffusion policy optimization, it does not rely on diffusion-specific properties, and as such is potentially of independent interest.

Proof. By the performance difference lemma in Lemma D.16, as well as Cauchy-Schwarz and a

change of measure, we see that

$$V^{\pi_{\mathrm{cmp}}}(\rho) - V^{\pi_\theta}(\rho) = \frac{1}{1-\gamma}\mathbb{E}_{s,a\sim d_\rho^{\pi_{\mathrm{cmp}}}}\left[A^{\pi_\theta}(s,a)\right] \tag{D.11}$$

$$\leq \frac{1}{1-\gamma}\sqrt{\left\|\frac{d_\rho^{\pi_{\mathrm{cmp}}}}{d_\nu^{\pi_\theta}}\right\|_\infty \mathbb{E}_{s,a\sim d_\nu^{\pi_\theta}}\left[A^{\pi_\theta}(s,a)^2\right]}. \tag{D.12}$$

By the policy gradient theorem,

$$\begin{aligned}\nabla_\theta V^\theta(\nu) &= \frac{1}{1-\gamma}\,\mathbb{E}_{(s,a)\sim d_\nu^{\pi_\theta}}\left[A^{\pi_\theta}(s,a)\,\nabla_\theta \log \pi_\theta(a \mid s)\right] \\ &= \frac{1}{1-\gamma}\,\mathbb{E}_{(s,a)\sim d_\nu^{\pi_\theta}}\left[A^{\pi_\theta}(s,a)\,\boldsymbol{\phi}^\theta(s,a)\right].\end{aligned}$$

We therefore require an upper bound on $\mathbb{E}_{s,a\sim d_\nu^{\pi_\theta}}\left[A^{\pi_\theta}(s,a)^2\right]$ in terms of the norm of

$$\mathbb{E}_{s,a\sim d_\nu^{\pi_\theta}}\left[A^{\pi_\theta}(s,a)\boldsymbol{\phi}^\theta(s,a)\right].$$

Write $\boldsymbol{\Lambda} = \mathbb{E}_{d_\nu^{\pi_\theta}}[\boldsymbol{\phi}\boldsymbol{\phi}^\top]$ and $\mathbf{b} = \mathbb{E}_{d_\nu^{\pi_\theta}}[A\boldsymbol{\phi}]$. If $v \in \ker \boldsymbol{\Lambda}$, then $\mathbb{E}(v^\top\boldsymbol{\phi})^2 = 0$, hence $v^\top\mathbf{b} = 0$. Thus $\mathbf{b} \in \mathrm{range}(\boldsymbol{\Lambda})$ and $\boldsymbol{g} = \boldsymbol{\Lambda}^\dagger\mathbf{b}$ solves the normal equations. Recall that the NPG algorithm updates θ in this direction. The lemma holds regardless of the update rule: we are seeking to bound the suboptimality gap by the policy-gradient norm. The residual $e = A - \boldsymbol{\phi}^\top\boldsymbol{g}$ is orthogonal in $L^2(d_\nu^{\pi_\theta})$ to every score coordinate. Expanding the square therefore gives

$$\mathbb{E}A^2 = \mathbf{b}^\top\boldsymbol{\Lambda}^\dagger\mathbf{b} + \mathbb{E}e^2 \leq \frac{\|\mathbf{b}\|^2}{\lambda_{\min}^+(\boldsymbol{\Lambda})} + \mathbb{E}e^2.$$

This also shows $\|\boldsymbol{g}\|^2 \leq \mathbb{E}A^2/\lambda_{\min}^+(\boldsymbol{\Lambda})$, which gives the stated sufficient choice of G since $|A| \leq (1-\gamma)^{-1}$. Under Assumption 5.5, $\mathbb{E}e^2 \leq \epsilon_{\mathrm{comp}}$. Substituting $\mathbf{b} = (1-\gamma)\nabla_\theta V^{\pi_\theta}(\nu)$ into the performance-difference bound above yields

$$V^*(\rho) - V^{\pi_\theta}(\rho) \leq \epsilon_* + \frac{\sqrt{C_\theta\epsilon_{\mathrm{comp}}}}{1-\gamma} + \sqrt{\frac{C_\theta}{\lambda_{\min}^+(\boldsymbol{\Lambda})}}\|\nabla_\theta V^{\pi_\theta}(\nu)\|.$$

Under exact compatibility the residual term vanishes, recovering gradient domination with the

positive Fisher spectrum. □

D.2.4. Proof of Theorem 5.7

We first prove the common bound for reset batches. Appendix D.2.5 replaces independence by the successive-return variance bound and adds the burn-in coupling error.

Proof. Write $J(\theta) = V^{\pi_\theta}(\nu)$ and let $\widehat{g}_{i,n} = \widehat{A}_{i,n}\widehat{\boldsymbol{\phi}}_{i,n}/(1-\gamma)$ be the normalized gradient output of sample n in batch i, using the DDPM path score or the DDIM endpoint score as appropriate. With reset access, the N outputs are independent conditional on the history before the batch. For their mean $\bar{g}_i = N^{-1}\sum_{n=1}^{N}\widehat{g}_{i,n}$, Lemma D.3 gives

$$\mathbb{E}_i\bar{g}_i = \nabla J(\theta_i), \qquad \mathbb{E}_i\|\bar{g}_i - \nabla J(\theta_i)\|^2 \le M_g^2/N.$$

Smoothness along the update segment therefore implies, for $\eta \le L_V^{-1}$,

$$\begin{aligned}\mathbb{E}_i[J(\theta_{i+1}) - J(\theta_i)] &\ge \eta\left(1 - \frac{L_V\eta}{2}\right)\|\nabla J(\theta_i)\|^2 - \frac{L_V\eta^2 M_g^2}{2N} \\ &\ge \frac{\eta}{2}\|\nabla J(\theta_i)\|^2 - \frac{L_V\eta^2 M_g^2}{2N}.\end{aligned}$$

Summing over the L updates yields

$$\sum_{i=1}^{L}\mathbb{E}\|\nabla J(\theta_i)\|^2 \le \frac{2\Delta}{\eta} + \frac{L_V\eta M_g^2 L}{N}.$$

Each policy is used for N labels. Applying Lemma 5.6 and Cauchy–Schwarz gives

$$\mathbb{E}\mathrm{Reg}(T) \le T\epsilon_* + \frac{T\sqrt{C\epsilon_{\mathrm{comp}}}}{1-\gamma} + \sqrt{\frac{C}{\lambda}}\,N\sqrt{L}\left(\frac{2\Delta}{\eta} + \frac{L_V\eta M_g^2 L}{N}\right)^{1/2},$$

which is the theorem with $\tau_{\mathrm{mix}} = 1$ and $\varepsilon_b = 0$.

For either access model, the stepsize that balances the two terms is

$$\eta = \min\left\{L_V^{-1}, \sqrt{\frac{2\Delta N}{L_V \tau_{\text{mix}} M_g^2 L}}\right\}.$$

For this choice, the statistical term is at most

$$\sqrt{\frac{C}{\lambda}}\left[2^{3/4}(\Delta L_V \tau_{\text{mix}} M_g^2)^{1/4} T^{3/4} + 2\sqrt{\Delta L_V N T}\right].$$

Indeed, when the cap is inactive the first term is the exact value of the bound. When the cap is active, $\tau_{\text{mix}} M_g^2 L/N < 2\Delta L_V$, so the second term bounds it. Thus $N \le \sqrt{T}$ gives the stated rate. If a tuning constant vanishes, choose a positive upper bound and use it throughout these estimates.

When $N = 1$, a direct second-moment calculation gives a sharper constant and does not require the stepsize cap. Take $\eta = \sqrt{2\Delta/(L_V M_g^2 T)}$, again using positive upper bounds for zero constants. Let $\widehat{g}_t = \widehat{A}^{(t)}\widehat{\boldsymbol{\phi}}^{(t)}/(1-\gamma)$, with the score chosen as in Algorithm 5.1. Conditional on the history, Lemma D.3 gives $\mathbb{E}_t \widehat{g}_t = \nabla V^{(t)}(\nu)$ and $\mathbb{E}_t \|\widehat{g}_t\|^2 \le M_g^2$. By L_V-smoothness along the update segment,

$$\mathbb{E}_t V^{(t+1)}(\nu) \ge V^{(t)}(\nu) + \eta \|\nabla V^{(t)}(\nu)\|^2 - \frac{L_V \eta^2 M_g^2}{2}.$$

Summing and using $V^{(T+1)}(\nu) \le V^*(\nu)$ yields

$$\sum_{t=1}^{T} \mathbb{E}\|\nabla V^{(t)}(\nu)\|^2 \le \frac{\Delta}{\eta} + \frac{L_V \eta M_g^2 T}{2} = \sqrt{2\Delta L_V M_g^2 T}. \tag{D.13}$$

By the gradient domination lemma in Lemma 5.6, we see that

$$\mathbb{E}\text{Reg}(T) - T\epsilon_* - \frac{T\sqrt{C\epsilon_{\text{comp}}}}{1-\gamma} \le \sqrt{C/\lambda^+_{\min}(\boldsymbol{\Lambda}_\nu)} \sum_{t=1}^{T} \mathbb{E}\left\|\nabla_\theta V^{(t)}(\nu)\right\|_2.$$

By Cauchy-Schwarz and Equation D.13,

$$\sum_{t=1}^{T} \mathbb{E}\left\|\nabla_\theta V^{(t)}(\nu)\right\|_2 \le \sqrt{T \sum_{t=1}^{T} \mathbb{E}\left\|\nabla_\theta V^{(t)}(\nu)\right\|_2^2}$$

$$\leq T^{3/4}(2\Delta L_V M_g^2)^{1/4}.$$

Putting it all together,

$$\mathbb{E}\mathrm{Reg}(T) \leq T[V^*(\rho) - V^{\pi_{\mathrm{cmp}}}(\rho)] + \frac{T\sqrt{C\epsilon_{\mathrm{comp}}}}{1-\gamma} + \sqrt{\frac{C}{\lambda}}\left[\frac{8[V^*(\nu) - V^{(1)}(\nu)]L_V(1+\gamma)\beta_K^2}{(1-\gamma)^4}\right]^{1/4} T^{3/4}.$$

This gives the sharper reset bound for one label per update. □

D.2.5. Policy gradients with mixing access

State mixing and complete return samples. Write K_θ for the state transition kernel under π_θ. Assume its stationary state law m_θ satisfies

$$\sup_{\theta,s} \mathrm{TV}(K_\theta^h(s,\cdot), m_\theta) \leq c_{\mathrm{mix}}\kappa^h, \qquad c_{\mathrm{mix}} \geq 1, \quad 0 < \kappa < 1.$$

For vanilla PG, $m_\theta = m$ is fixed, so $d_m^{\pi_\theta} = m\pi_\theta$ and the objective is $J(\theta) = V^{\pi_\theta}(m)$. The fixed state law avoids the additional derivative of a policy-dependent stationary distribution. For NPG the state law can depend on the frozen policy. The joint regression law is $q_\theta = m_\theta\pi_\theta$. Unlike vanilla PG, the NPG proof uses a performance-difference argument and the stationary law only as the compatible-function-approximation distribution, so it does not differentiate through m_θ. A fixed stationary joint law would force policies to agree on its state support, so the fixed object in the vanilla assumption is the state law.

In the mixing case, we do not burn in before every call to Algorithm D.1. Instead, we use an epochwise implementation. At epoch i, freeze the policy at $\theta^{(t_i)}$, run b burn-in transitions under $\pi_{\theta^{(t_i)}}$, collect N complete return samples, and update. At each sample's starting state draw an action and its denoising path and execute one geometric-return branch of Algorithm D.1, omitting its reset and occupancy search. In the value branch the accepted action is replaced before any action is executed. Only one trajectory is run and no reset to a visited state is needed. Start the next sample at the terminal state of that trajectory. Every return uses $H \geq 1$ transitions with $\mathbb{P}(H = h) = (1-\gamma)\gamma^{h-1}$ independently of its trajectory.

Lemma D.4 (Second moments of successive return samples)**.** *Fix a policy and start its first sample at stationarity. Let Y_n be a Hilbert-space-valued output of its nth complete sample, including the accepted action, denoising path and return. If $\sigma_Y^2 = \mathbb{E}\|Y_n - \mathbb{E}Y_n\|^2 < \infty$, then*

$$\mathbb{E}\left\| N^{-1}\sum_{n=1}^{N} Y_n - \mathbb{E}Y_1 \right\|^2 \le \tau_{\mathrm{mix}}\sigma_Y^2/N, \qquad \tau_{\mathrm{mix}} = 1 + 4\sqrt{c_{\mathrm{mix}}}/(1-\sqrt{\kappa}).$$

Proof. The states X_n at sample boundaries have kernel

$$R_\theta = \sum_{h\ge 1}(1-\gamma)\gamma^{h-1}K_\theta^h.$$

Both return branches execute the current policy after averaging over their action choice. Hence this identity also holds when the value branch replaces the accepted action. The kernel has stationary law m_θ. Its jth power is a mixture of K_θ^h with $h \ge j$, giving $\sup_x \mathrm{TV}(R_\theta^j(x,\cdot), m_\theta) \le c_{\mathrm{mix}}\kappa^j$. For centered Hilbert-valued $f \in L^2(m_\theta)$, weighted Cauchy–Schwarz and the total variation bound give

$$\|R_\theta^j f(x)\|^2 \le 2c_{\mathrm{mix}}\kappa^j \int \|f(y)\|^2 |R_\theta^j(x,dy) - m_\theta(dy)|.$$

Integrating in $m_\theta(dx)$, using $|R_\theta^j - m_\theta| \le R_\theta^j + m_\theta$ and stationarity, gives

$$\|R_\theta^j f\|_{L^2(m_\theta)}^2 \le 4c_{\mathrm{mix}}\kappa^j \|f\|_{L^2(m_\theta)}^2.$$

For $j = 0$ the same loose bound holds because $c_{\mathrm{mix}} \ge 1$. Set $g(x) = \mathbb{E}[Y_n - \mathbb{E}Y_n \mid X_n = x]$. Then g is centered and $\|g\|_{L^2(m_\theta)}^2 \le \sigma_Y^2$. Conditional on the complete nth sample, future samples depend on it only through X_{n+1}. Therefore for $h \ge 1$,

$$|\mathbb{E}\langle Y_n - \mathbb{E}Y_n, Y_{n+h} - \mathbb{E}Y_{n+h}\rangle| = \left|\mathbb{E}\langle Y_n - \mathbb{E}Y_n, R_\theta^{h-1} g(X_{n+1})\rangle\right| \le 2\sqrt{c_{\mathrm{mix}}}\kappa^{(h-1)/2}\sigma_Y^2.$$

Expanding the squared norm of the average and summing this geometric bound proves the result. Thus the bound includes dependence of complete returns. □

Vanilla PG. For a stationary-start epoch use the normalized gradient output $Y = \widehat{A}\widehat{\boldsymbol{\phi}}/(1-\gamma)$, with the DDPM path or DDIM endpoint score. Its conditional mean is $\nabla J(\theta_i)$ and second moment at most M_g^2. The batch mean therefore has conditional variance at most $\tau_{\mathrm{mix}} M_g^2/N$. Smoothness and $\eta \le L_V^{-1}$ imply

$$\mathbb{E}_i[J(\theta_{i+1}) - J(\theta_i)] \ge \frac{\eta}{2}\|\nabla J(\theta_i)\|^2 - \frac{L_V \eta^2 \tau_{\mathrm{mix}} M_g^2}{2N}.$$

Thus the original telescoping step gives

$$\sum_{i=1}^{L} \mathbb{E}\|\nabla J(\theta_i)\|^2 \le 2\Delta/\eta + L_V \eta \tau_{\mathrm{mix}} M_g^2 L/N.$$

Let $T = LN$, $\mathfrak{e} = \epsilon_* + \sqrt{C\epsilon_{\mathrm{comp}}}/(1-\gamma)$ and $\mathfrak{c} = \sqrt{C/\lambda}$. The original gradient-domination and Cauchy–Schwarz steps give

$$\mathbb{E}\,\mathrm{Reg}(T) \le T\mathfrak{e} + \mathfrak{c}N\sqrt{L(2\Delta/\eta + L_V \eta \tau_{\mathrm{mix}} M_g^2 L/N)} + \frac{TL\varepsilon_b}{1-\gamma}.$$

The last term couples the full adaptive run to stationary-start epochs. The conditional initial-state coupling fails in at most $L\varepsilon_b$ probability, and regret is at most $T/(1-\gamma)$. Choose

$$\eta = \min\{L_V^{-1}, \sqrt{2\Delta N/(L_V \tau_{\mathrm{mix}} M_g^2 L)}\}.$$

For $N \le \sqrt{T}$, the nonapproximation term is $O(T^{3/4})$. For example $\varepsilon_b \le T^{-2}$ and $N \asymp bT^{1/4} \le \sqrt{T}$ preserve the original epochwise rate while making the additive burn-in regret $Lb/(1-\gamma) = O(T^{3/4})$. Positive upper bounds may replace zero smoothness or gap constants.

Environment interactions. There are Lb burn-in transitions and T complete geometric returns, for expected transition count $Lb + T/(1-\gamma)$. With probability $1-\delta$ every return uses at most

$$h = \lceil \log(T/\delta)/\log(1/\gamma) \rceil$$

transitions, so the total is at most $Lb + Th$. Charging each transition its policy's value gap gives regret at most $h\,\mathrm{Reg}(T) + Lb/(1-\gamma)$ on this event. The stated T therefore counts labels, and the

interaction accounting also includes return lengths. This is a discounted-value pseudo-regret, not a claim about undiscounted reward regret of the continuing chain.

D.3. Analysis of DDIM NPG

D.3.1. NPG Update Concentration

Lemma D.5 (NPG Update Direction Bound for Oracle Features)**.** *Condition on the history $\mathcal{F}_i$ before batch i. Under the reset, positive-spectrum, and moment conditions of Theorem 5.9, let $\widehat{\boldsymbol{g}}^{(t_i)}$ be the minimum-norm minimizer of $\widehat{\mathcal{L}}_A^{(t_i)}$ over $\mathcal{G}$ and $\boldsymbol{g}^{(t_i)} = (\boldsymbol{\Lambda}^{(t_i)})^\dagger \mathbf{b}^{(t_i)}$. Then*

$$\mathbb{E}[\|\widehat{\boldsymbol{g}}^{(t_i)} - \boldsymbol{g}^{(t_i)}\| \mid \mathcal{F}_i] \le a_N.$$

Proof. Let $\mathcal{S}_\theta = \mathrm{range}(\boldsymbol{\Lambda})$ for the frozen policy. If P projects onto $\mathcal{S}_\theta^\perp$, then $\mathbb{E}_i\|P\boldsymbol{\phi}\|^2 = \mathrm{tr}(P\boldsymbol{\Lambda}P) = 0$, so every sampled feature lies in $\mathcal{S}_\theta$ almost surely. Projecting a candidate coefficient onto the span of the sampled features preserves its empirical predictions and cannot increase its norm. The minimum-norm empirical minimizer therefore lies in that span, and hence in $\mathcal{S}_\theta$, as does $\boldsymbol{g} = \boldsymbol{\Lambda}^\dagger \mathbf{b}$.

Define $\widehat{\boldsymbol{\Lambda}} = N^{-1}\sum_n \boldsymbol{\phi}_n\boldsymbol{\phi}_n^\top$ and $\widehat{\mathbf{b}} = N^{-1}\sum_n \widehat{A}_n\boldsymbol{\phi}_n$, suppressing the batch index. Due to the multi-batched updating rule, we do not update the policy within each batch, and as such the feature vectors are i.i.d. in the reset case conditional on $\mathcal{F}_i$, allowing us to continue our analysis. We address the mixing case in Appendix D.3.4. The sample averages have means $\boldsymbol{\Lambda}$ and $\mathbf{b}$. The moment assumptions and the sampler give

$$\mathbb{E}_i\|\widehat{\boldsymbol{\Lambda}} - \boldsymbol{\Lambda}\|_F^2 \le M_F/N, \qquad \mathbb{E}_i\|\widehat{\mathbf{b}} - \mathbf{b}\|^2 \le C_A\beta_K^2/N.$$

Let $\mathcal{E} = \{\|\widehat{\boldsymbol{\Lambda}} - \boldsymbol{\Lambda}\|_{\mathrm{op}} \le \lambda/2\}$. Markov's inequality yields $\mathbb{P}_i(\mathcal{E}^c) \le 4M_F/(\lambda^2 N)$. On $\mathcal{E}$, the sample covariance restricted to $\mathcal{S}_\theta$ is bounded below by $\lambda/2$. Since $\boldsymbol{g} = \boldsymbol{\Lambda}^\dagger\mathbf{b}$ lies in $\mathcal{G}$, constrained optimality gives

$$\langle \widehat{\boldsymbol{\Lambda}}\widehat{\boldsymbol{g}} - \widehat{\mathbf{b}}, \boldsymbol{g} - \widehat{\boldsymbol{g}}\rangle \ge 0.$$

Writing $h = \widehat{\boldsymbol{g}} - \boldsymbol{g} \in \mathcal{S}_\theta$ and using $\boldsymbol{\Lambda}\boldsymbol{g} = \mathbf{b}$,

$$\frac{\lambda}{2}\|h\|^2 \le h^\top \widehat{\boldsymbol{\Lambda}} h \le \|h\|\big(\|\widehat{\mathbf{b}} - \mathbf{b}\| + G\|\widehat{\boldsymbol{\Lambda}} - \boldsymbol{\Lambda}\|_{\mathrm{op}}\big).$$

Thus on $\mathcal{E}$, $\|h\|$ is at most $2/\lambda$ times the expression in parentheses. On its complement, both coefficients remain in the radius G ball, so $\|h\| \le 2G$. Taking conditional expectations and applying Cauchy–Schwarz proves

$$\mathbb{E}_i\|h\| \le \frac{2(\sqrt{C_A}\beta_K + G\sqrt{M_F})}{\lambda\sqrt{N}} + \frac{8GM_F}{\lambda^2 N} = a_N.$$

The positive spectrum controls the coefficient error on $\mathcal{E}$, while the regression constraint controls the remaining batches. This is why the argument remains valid even when an empirical covariance is singular. □

Lemma D.6 (NPG Prediction Error for Oracle Features)**.** *Let $\mathcal{F}_i$ be the pre-batch history and*

$$\mathrm{err}^{(t_i)} = \mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}[A^{(t_i)} - \langle \widehat{\boldsymbol{g}}^{(t_i)}, \boldsymbol{\phi}^{(t_i)} \rangle].$$

Then $\mathbb{E}[\mathrm{err}^{(t_i)} \mid \mathcal{F}_i] \le \sqrt{\epsilon_{\mathrm{bias}}} + B_* a_N$. Furthermore, with probability at least $1-\delta$,

$$\sum_{i=1}^{L} \mathrm{err}^{(t_i)} \le L(\sqrt{\epsilon_{\mathrm{bias}}} + B_* a_N) + 2B_* G\sqrt{2L\log(1/\delta)}.$$

Proof. Consider

$$\begin{aligned}
\mathrm{err}^{(t)} &= \mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}\left[A^{(t)}(s,a) - \left\langle \widehat{\boldsymbol{g}}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle\right] && \text{(D.14)}\\
&= \mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}\left[A^{(t)}(s,a) - \left\langle \boldsymbol{g}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle\right] + \mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}\left[\left\langle \boldsymbol{g}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle - \left\langle \widehat{\boldsymbol{g}}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle\right] && \text{(D.15)}\\
&= \mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}\left[A^{(t)}(s,a) - \left\langle \boldsymbol{g}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle\right] + \mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}\left[\left\langle \boldsymbol{g}^{(t)} - \widehat{\boldsymbol{g}}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle\right]. && \text{(D.16)}
\end{aligned}$$

The first term, corresponding to the asymptotic bias of the least-squares solution, can be bounded by an application of Cauchy-Schwarz and the comparator transfer condition of Theorem 5.9 as in Agarwal et al. (2021):

$$\mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}\left[A^{(t)}(s,a) - \left\langle \boldsymbol{g}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle\right] \le \sqrt{\mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}}\left[\left(A^{(t)}(s,a) - \left\langle \boldsymbol{g}^{(t)}, \boldsymbol{\phi}^{(t)}(s,a)\right\rangle\right)^2\right]} \tag{D.17}$$

$$= \sqrt{\mathcal{L}_A^{d_\rho^{\pi_{\mathrm{cmp}}}}(\boldsymbol{g}^{(t)}, \theta^{(t)})} \le \sqrt{\epsilon_{\mathrm{bias}}}. \tag{D.18}$$

Let $\mu_i = \mathbb{E}_{d_\rho^{\pi_{\mathrm{cmp}}}} \boldsymbol{\phi}^{(t_i)}$, which is $\mathcal{F}_i$-measurable. The second term is $\mu_i^\top(\boldsymbol{g}^{(t_i)} - \widehat{\boldsymbol{g}}^{(t_i)})$, so its conditional mean is at most $B_* a_N$ by Lemma D.5. For the probability bound, the centered prediction error is exactly

$$\mathrm{err}^{(t_i)} - \mathbb{E}_i \mathrm{err}^{(t_i)} = -\mu_i^\top(\widehat{\boldsymbol{g}}^{(t_i)} - \mathbb{E}_i \widehat{\boldsymbol{g}}^{(t_i)}).$$

Its absolute value is at most $2B_*G$ since the fitted direction is constrained to $\mathcal{G}$. These centered errors form a martingale difference sequence. Azuma's inequality proves the asserted sum bound. For the batch-dependent transfer condition following Theorem 5.9, the same Cauchy–Schwarz step bounds the conditional bias in batch i by $\sqrt{\epsilon_i}$. The centered martingale is unchanged, so the proof gives $\sum_i \sqrt{\epsilon_i}$ in place of $L\sqrt{\epsilon_{\mathrm{bias}}}$. Substituting this bound in Lemma D.7 proves the cumulative-transfer variant without changing the algorithm. □

D.3.2. NPG Analysis

Lemma D.7 (Modified NPG Regret Lemma (Agarwal et al., 2021, Lemma 6.2))**.** *Let π_{cmp} and $\mathcal{K}_0$ be the comparison policy and initial KL cost in Theorem 5.9, and write $\epsilon_* := V^*(\rho) - V^{\pi_{\mathrm{cmp}}}(\rho)$. For L batches of N samples and bounded updates $\|\widehat{\boldsymbol{g}}^{(t_i)}\| \le G$, the score-smoothness and finite-KL conditions imply*

$$\sum_{t=1}^{T}(V^*(\rho) - V^{(t)}(\rho)) \le T\epsilon_* + \frac{N}{1-\gamma}\left(\frac{\mathcal{K}_0}{\eta} + \frac{\eta L_\theta^{(K)} G^2 L}{2} + \sum_{i=1}^{L}\mathrm{err}^{(t_i)}\right).$$

Here $\mathrm{err}^{(t_i)}$ is defined using $d_\rho^{\pi_{\mathrm{cmp}}}$ as in Lemma D.6.

Proof. The proof is similar to that of Lemma 6.2 within Agarwal et al. (2021). However, to accommodate the diffusion setting, there is some additional bookkeeping needed to account for estimation error and batched updates. We therefore provide a proof here.

By Assumption 5.4, we know that for all $s, a \in \mathcal{S} \times \mathcal{A}$,

$$\left\|\nabla_\theta \log \pi_{\theta^{(t_i+1)}}(a|s) - \nabla_\theta \log \pi_{\theta^{(t_i)}}(a|s)\right\|_2 \leq L_\theta^{(K)} \left\|\theta^{(t_i+1)} - \theta^{(t_i)}\right\|_2. \tag{D.19}$$

By Taylor's theorem, we then have that

$$\left|\log \pi_{\theta^{(t_i+1)}}(a|s) - \log \pi_{\theta^{(t_i)}}(a|s) - \left(\nabla_\theta \log \pi_{\theta^{(t_i)}}(a|s)\right)^\top \left(\theta^{(t_i+1)} - \theta^{(t_i)}\right)\right| \tag{D.20}$$

$$\leq \frac{L_\theta^{(K)}}{2} \left\|\theta^{(t_i+1)} - \theta^{(t_i)}\right\|_2^2. \tag{D.21}$$

This in turn implies

$$\log\left(\frac{\pi_{\theta^{(t_i+1)}}(a|s)}{\pi_{\theta^{(t_i)}}(a|s)}\right) \geq \left(\nabla_\theta \log \pi_{\theta^{(t_i)}}\right)^\top \left(\theta^{(t_i+1)} - \theta^{(t_i)}\right) - \frac{L_\theta^{(K)}}{2} \left\|\theta^{(t_i+1)} - \theta^{(t_i)}\right\|_2^2 \tag{D.22}$$

$$= \eta \left\langle \boldsymbol{\phi}^{(t_i)}(a|s), \widehat{\boldsymbol{g}}^{(t_i)} \right\rangle - \frac{\eta^2 L_\theta^{(K)}}{2} \left\|\widehat{\boldsymbol{g}}^{(t_i)}\right\|_2^2. \tag{D.23}$$

Integrate Equation D.23 under the fixed comparator occupancy $d_\rho^{\pi_{\text{cmp}}}$. Define

$$\mathcal{K}_{i-1} = \mathbb{E}_{s \sim d_{\rho,S}^{\pi_{\text{cmp}}}} D_{\text{KL}}(\pi_{\text{cmp}}(\cdot \mid s) \| \pi^{(t_i)}(\cdot \mid s)).$$

Here $\mathcal{K}_{i-1}$ is the potential before batch i, and $\mathcal{K}_L$ uses the policy after the last update. The Taylor bound above makes each log-policy ratio integrable under the comparator, since its score features are integrable and the update direction is bounded. Starting from $\mathcal{K}_0 < \infty$, the log-ratio identity therefore implies that all potentials $\mathcal{K}_0, \ldots, \mathcal{K}_L$ are finite.

By the performance difference lemma from Kakade and Langford (2002) in Lemma D.16 and

Equation D.23, we find that

$$\mathbb{E}_{d_\rho^{\pi_{\text{cmp}}}}\left[\log\left(\frac{\pi_{\theta^{(t_i+1)}}(a|s)}{\pi_{\theta^{(t_i)}}(a|s)}\right)\right] \tag{D.24}$$

$$\geq \eta \cdot \mathbb{E}_{d_\rho^{\pi_{\text{cmp}}}}\left[\left\langle \boldsymbol{\phi}^{(t_i)}(s,a), \widehat{\boldsymbol{g}}^{(t_i)}\right\rangle\right] - \frac{\eta^2 L_\theta^{(K)}}{2}\left\|\widehat{\boldsymbol{g}}^{(t_i)}\right\|_2^2 \tag{D.25}$$

$$= \eta \cdot \mathbb{E}_{d_\rho^{\pi_{\text{cmp}}}}\left[A^{\pi_{\theta^{(t_i)}}}(s,a)\right] - \frac{\eta^2 L_\theta^{(K)}}{2}\left\|\widehat{\boldsymbol{g}}^{(t_i)}\right\|_2^2 + \eta \cdot \mathbb{E}_{d_\rho^{\pi_{\text{cmp}}}}\left[\left\langle \boldsymbol{\phi}^{(t_i)}(s,a), \widehat{\boldsymbol{g}}^{(t_i)}\right\rangle - A^{(t_i)}(s,a)\right] \tag{D.26}$$

$$= \eta(1-\gamma)\left(V^{\pi_{\text{cmp}}}(\rho) - V^{(t_i)}(\rho)\right) - \frac{\eta^2 L_\theta^{(K)}}{2}\left\|\widehat{\boldsymbol{g}}^{(t_i)}\right\|_2^2 - \eta \cdot \text{err}^{(t_i)}. \tag{D.27}$$

The expected log-policy ratio is $\mathcal{K}_{i-1} - \mathcal{K}_i$. It is now time to account for the batched update. Summing over all $t = 1, ..., T$, noting that $\pi^{(t_{i-1}+1)} = ... = \pi^{(t_i)}$, we see that

$$\sum_{t=1}^{T}\left(V^{\pi_{\text{cmp}}}(\rho) - V^{(t)}(\rho)\right) \tag{D.28}$$

$$= N\sum_{i=1}^{L}\left(V^{\pi_{\text{cmp}}}(\rho) - V^{(t_i)}(\rho)\right) \tag{D.29}$$

$$\leq N\sum_{i=1}^{L}\frac{1}{1-\gamma}\left(\frac{\mathcal{K}_{i-1} - \mathcal{K}_i}{\eta} + \frac{\eta L_\theta^{(K)} G^2}{2} + \text{err}^{(t_i)}\right). \tag{D.30}$$

However, the sum of KL-divergences below telescopes:

$$\sum_{i=1}^{L}(\mathcal{K}_{i-1} - \mathcal{K}_i) \tag{D.31}$$

$$= \mathcal{K}_0 - \mathcal{K}_L \tag{D.32}$$

$$\leq \mathcal{K}_0. \tag{D.33}$$

This then yields:

$$\sum_{t=1}^{T}\left(V^{\pi_{\text{cmp}}}(\rho) - V^{(t)}(\rho)\right) \leq \frac{N\mathcal{K}_0}{\eta(1-\gamma)} + \frac{\eta L_\theta^{(K)} G^2 NL}{2(1-\gamma)} + \frac{N}{1-\gamma}\sum_{i=1}^{L}\text{err}^{(t_i)}. \tag{D.34}$$

Adding $T\epsilon_*$ gives the result. □

To obtain the pretrained comparator described after Theorem 5.9, write $E_r(s) = \{a : \|a - a^*(s)\| \le r\}$ and $Z_r(s) = \pi^{(1)}(E_r(s) \mid s)$. We condition the initial policy on this set:

$$\pi_{\mathrm{cmp},r}(da \mid s) = \frac{\mathbf{1}_{E_r(s)}(a)\pi^{(1)}(da \mid s)}{Z_r(s)},$$
$$D_{\mathrm{KL}}(\pi_{\mathrm{cmp},r}(\cdot \mid s)\|\pi^{(1)}(\cdot \mid s)) = -\log Z_r(s) \le \log(c_0^{-1}) + m\log(R/r).$$

Averaging under $d_{\rho,S}^{\pi_{\mathrm{cmp},r}}$ bounds $\mathcal{K}_0$. The performance-difference identity and local Lipschitz condition give

$$V^*(\rho) - V^{\pi_{\mathrm{cmp},r}}(\rho) = \frac{1}{1-\gamma}\mathbb{E}_{s,a\sim d_\rho^{\pi_{\mathrm{cmp},r}}}[Q^*(s,a^*(s)) - Q^*(s,a)] \le \frac{L_Q r}{1-\gamma}.$$

For $r = R/\sqrt{T}$, set $\overline{\mathcal{K}}_0(T) = 1 + \log(c_0^{-1}) + (m/2)\log T$. With $L = N = \sqrt{T}$ and $\eta = \sqrt{2\overline{\mathcal{K}}_0(T)/(LL_\theta^{(K)}G^2)}$, the comparator and optimization terms in Theorem 5.9 are at most

$$\frac{L_Q R\sqrt{T} + G\sqrt{2L_\theta^{(K)}\overline{\mathcal{K}}_0(T)}\,T^{3/4}}{1-\gamma}.$$

The remaining estimation, transfer, and high-probability terms are unchanged, with their constants evaluated for this comparator and the resulting updates. The comparator is only used in the analysis and need not be represented by the actor or known to the learner.

D.3.3. Proof of Theorem 5.9

The reset proof below uses Lemma D.5. For mixing and hybrid access, the coefficient bounds in Appendices D.3.4 and D.3.4 replace a_N under the corresponding regression law. The prediction-error and KL-telescope steps are otherwise the same.

Proof. Substitute Lemma D.6 into Lemma D.7. This gives

$$\mathrm{Reg}(T) \le T\epsilon_* + \frac{N}{1-\gamma}\left(\frac{\mathcal{K}_0}{\eta} + \frac{\eta L_\theta^{(K)}G^2L}{2}\right) + \frac{T}{1-\gamma}(\sqrt{\epsilon_{\mathrm{bias}}} + B_* a_N) + \frac{2NB_*G}{1-\gamma}\sqrt{2L\log(1/\delta)}. \tag{D.35}$$

Taking expectations in the two lemmas instead removes the last term. For $\mathcal{K}_0 > 0$, the stated stepsize balances $\mathcal{K}_0/\eta$ with $\eta L_\theta^{(K)} G^2 L/2$. With $L = N = \sqrt{T}$, this term and the $N^{-1/2}$ coefficient-estimation term scale as $T^{3/4}$, while the N^{-1} term is $O(\sqrt{T})$. The two approximation terms account for the comparison policy's value gap and the transferred advantage error. □

D.3.4. Extensions and Concrete Neural Actor Examples

The extensions of DDIM natural policy gradient to frozen nuisance coordinates, ergodic state mixing, and importance-weighted offline trajectories, together with the complete proofs for the stochastic multimodal actor (Proposition D.3.4) and the two-layer neural actor (Proposition D.3.4), appear in Appendix G of Tan et al. (2026).

D.3.5. Miscellaneous Lemmas

Lemma D.8 (Diffusion Policy Score)**.** *For a DDPM path with a parameter-independent Gaussian initial latent and positive transition densities,*

$$\nabla_\theta \log \pi_\theta(a_0 \mid s) = \mathbb{E}\left[\sum_{k=1}^{K} \nabla_\theta \log \pi_\theta(a_{k-1} \mid a_k, s) \,\middle|\, s, a_0\right].$$

The expectation is under the posterior law of the denoising path conditional on the endpoint, not its unconditional forward sampling law.

Proof. Let $p_\theta(a_{0:K} \mid s) = \varphi(a_K) \prod_{k=1}^{K} \pi_\theta(a_{k-1} \mid a_k, s)$, where φ is the standard Gaussian density. Recall that we can write $\pi_\theta(a_0 \mid s)$ as the equivalent integral over all possible paths, integrating p_θ against Lebesgue measure over $a_{1:K}$. Under the differentiation and integrability conditions of Assumption 5.4, using the log-derivative rule, we obtain that

$$\begin{aligned}\nabla_\theta \log \pi_\theta(a_0 \mid s) &= \frac{1}{\pi_\theta(a_0 \mid s)} \int p_\theta(a_{0:K} \mid s) \nabla_\theta \log p_\theta(a_{0:K} \mid s) \, \mathrm{d}a_{1:K} \\ &= \int p_\theta(a_{1:K} \mid s, a_0) \sum_{k=1}^{K} \nabla_\theta \log \pi_\theta(a_{k-1} \mid a_k, s) \, \mathrm{d}a_{1:K}.\end{aligned}$$

This is the stated conditional expectation. The initial Gaussian contributes zero to the parameter

derivative. □

D.4. Analysis of Optimistic Critic Behavior Cloning (Algorithm 5.3)

In this section, we analyze Algorithm 5.3 by decomposing the regret of the algorithm into two parts: diffusion policy behavior cloning error and critic estimation error. Then, diffusion policy behavior cloning error and critic estimation error are analyzed separately and combined into the final regret bound. Initial-state expressions below are averaged over $s_1^{(t)} \sim \rho$, so that the regret agrees with the initial-distribution definition. The critic $f^{(t)}$ uses data from episodes $i < t$. Its confidence set is therefore $\mathcal{F}^{(t-1)}$, with $\mathcal{F}^{(0)} = \mathcal{F}$.

D.4.1. Regret Decomposition

Lemma D.9 (Regret Decomposition for Algorithm 5.3)**.** *The following regret decomposition holds for Algorithm 5.3:*

$$\begin{aligned} \mathit{Reg}(T) &= \sum_{t=1}^{T} \left(V_1^*(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)}) \right) \\ &\le \underbrace{2 \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h) \right\rangle \right]}_{\textit{Diffusion policy behavior cloning error.}} + \underbrace{\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right]}_{\textit{Critic estimation error.}}. \end{aligned}$$

Proof. We first add and subtract $f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}))$, obtaining

$$\begin{aligned} \mathrm{Reg}(T) &= \sum_{t=1}^{T} \left(V_1^*(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)}) \right) \\ &= \sum_{t=1}^{T} \left(V_1^*(s_1^{(t)}) - f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)})) \right) + \sum_{t=1}^{T} \left(f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)}) \right). \end{aligned}$$

The first term is non-positive by optimistic selection and $Q^* \in \mathcal{F}^{(t-1)}$ from Lemma D.12. It remains to address the second term. Applying the generalized policy difference lemma from Efroni et al. (2020); Cai et al. (2020) (presented in Lemma D.15 for completeness) with $\pi = \pi^{f^{(t)}}, \pi' = \pi^{(t)}$, and

with $f^{(t)}$ as the Q-function, followed by $\mathcal{T}_h^{\pi^{f^{(t)}}} f_{h+1}^{(t)} \geq \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)}$, yields the inequality:

$$\begin{aligned}
&\sum_{t=1}^{T} \left(f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)}) \right) \\
&\leq \underbrace{\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h) \right\rangle \right]}_{\text{Diffusion policy behavior cloning error.}} + \underbrace{\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[f_h^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right]}_{\text{Critic estimation error.}}.
\end{aligned}$$

The first term corresponds to the behavior cloning error, while the second corresponds to critic estimation error. However, we are not done yet. The second term, at least in this specific form, is difficult to deal with in general without making a very strong value completeness-like assumption on the function class (Liu et al., 2023b). We therefore further decompose the term:

$$\begin{aligned}
&\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[f_h^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right] \\
&= \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right] + \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\mathcal{T}_h f_{h+1}^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right] \\
&= \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right] + \sum_{t=1}^{T} \sum_{h=2}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h) \right\rangle \right],
\end{aligned}$$

where expectations involving s_{h+1} include the transition from (s_h, a_h). The last line follows from the observation that

$$\begin{aligned}
&\mathbb{E}_{\pi^{(t)}} \left[\left(\mathcal{T}_h f_{h+1}^{(t)} - \mathcal{T}_h^{\pi^{(t)}} f_{h+1}^{(t)} \right)(s_h, a_h) \right] \\
&= \mathbb{E}_{d_h^{(t)}} \left[r_h(s_h, a_h) + f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{f^{(t)}}(s_{h+1}) \right) - r_h(s_h, a_h) - f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{(t)}(s_{h+1}) \right) \right] \\
&= \mathbb{E}_{d_h^{(t)}} \left[f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{f^{(t)}}(s_{h+1}) \right) - f_{h+1}^{(t)} \left(s_{h+1}, \pi_{h+1}^{(t)}(s_{h+1}) \right) \right] \\
&= \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_{h+1}^{(t)}(s_{h+1}, \cdot), \pi_{h+1}^{f^{(t)}}(\cdot | s_{h+1}) - \pi_{h+1}^{(t)}(\cdot | s_{h+1}) \right\rangle \right].
\end{aligned}$$

Each greedy-action gap is nonnegative. Thus the sum over $h = 2, \ldots, H$ is at most the sum over

$h = 1, \ldots, H$, and combining the bounds gives:

$$\sum_{t=1}^{T} \left(f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)}) \right)$$
$$\leq \underbrace{2 \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h) \right\rangle \right]}_{\text{Diffusion policy behavior cloning error.}} + \underbrace{\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right]}_{\text{Critic estimation error.}}.$$

□

D.4.2. Critic Estimation Error

Lemma D.10 (Critic Estimation Error)**.** *With probability at least* $1 - \delta$,

$$\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right)(s_h, a_h) \right] \leq O(\sqrt{\beta H^4 T \cdot \mathsf{SEC}(\mathcal{F}, \Pi, T)}).$$

Proof. Write $\delta_h^{(t)} = f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}$. The argument follows the proof of Theorem 12 in Xie et al. (2022a):

$$\sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} \left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)} \right)(s_h, a_h) \right]$$
$$= \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} [\delta_h^{(t)}]$$
$$= \sum_{t=1}^{T} \sum_{h=1}^{H} \mathbb{E}_{d_h^{(t)}} [\delta_h^{(t)}] \left(\frac{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}} [(\delta_h^{(t)})^2]}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}} [(\delta_h^{(t)})^2]} \right)^{1/2}$$
$$\leq \sqrt{\sum_{t=1}^{T} \sum_{h=1}^{H} \frac{\mathbb{E}_{d_h^{(t)}} [\delta_h^{(t)}]^2}{H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}} [(\delta_h^{(t)})^2]}} \sqrt{\sum_{t=1}^{T} \sum_{h=1}^{H} \left(H^2 \vee \sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}} [(\delta_h^{(t)})^2] \right)}$$
$$\overset{(a)}{\lesssim} \sqrt{H \cdot \mathsf{SEC}(\mathcal{F}, \Pi, T)} \sqrt{\beta H^3 T}$$
$$\lesssim \sqrt{\beta H^4 \mathsf{SEC}(\mathcal{F}, \Pi, T) \cdot T}.$$

In (a), the first term is bounded by the definition of SEC and the second by Lemma D.12. □

D.4.3. Diffusion Policy Behavior Cloning Error

The analysis of diffusion policy behavior cloning error mainly relies on the following lemma on sampling error of diffusion models. The greedy policy is deterministic, whereas the deployed sampler retains noise. We compare them through the projected noised target $\widetilde{\pi}_h^{f^{(t)}}$, which appears only in the analysis. Denote $W_1(\cdot,\cdot)$ as the Wasserstein-1 distance between two distributions, i.e.

$$W_1(P,Q) = \inf_{\Gamma\in\mathcal{C}(P,Q)} \int \|x-y\|_2 \, d\Gamma(x,y),$$

where $\mathcal{C}(P,Q)$ is the set of couplings of P and Q. We first specify the sampler coefficients and constants used in the comparison.

For this cloning result, take $K \geq 2$ and $0 < \alpha_k < 1$. In Equation 5.2 use

$$\eta_k^{\mathsf{ddpm}} = -\frac{1-\alpha_k}{\sqrt{v_k}}, \qquad (\sigma_k^{\mathsf{ddpm}})^2 = \frac{\alpha_k(1-\alpha_k)v_{k-1}}{v_k}, \quad k = K,\dots,2.$$

These coefficients give the exact Gaussian reverse kernels of the conditional target. Stop at Y_1 and project to $\mathcal{A}$. Define the smoothing, initialization, and schedule terms

$$b_K = (1-\sqrt{\alpha_1})R_A + \sqrt{d_A(1-\alpha_1)},$$
$$I_K = \tfrac{1}{2}[\bar{\alpha}_K R_A^2 + d_A(-\bar{\alpha}_K - \log(1-\bar{\alpha}_K))], \qquad \kappa_K = K \max_{2\leq k\leq K} \frac{(\eta_k^{\mathsf{ddpm}})^2}{2(\sigma_k^{\mathsf{ddpm}})^2}.$$

For DDIM use $\eta_k^{\mathsf{ddim}} = \sqrt{\alpha_k v_{k-1}} - \sqrt{v_k}$ and the same stopping/projection rule. Assume its learned maps are ℓ_k-Lipschitz, uniformly in states and iterates. Put

$$J_K = \sqrt{\bar{\alpha}_K}R_A + (1-\sqrt{v_K})\sqrt{d_A}, \quad P_K = \prod_{k=2}^{K} \ell_k, \quad C_K = \left[K \sum_{k=2}^{K} \frac{(\eta_k^{\mathsf{ddim}})^2}{\alpha_k} \left(\prod_{j=2}^{k-1} \ell_j\right)^2\right]^{1/2}.$$

Empty products equal one. These DDIM factors measure uniform stability of the learned maps. The $\widetilde{O}(\sqrt{T})$ specialization in Section 5.4 assumes nonexpansiveness, $\ell_k \leq 1$. The Wasserstein

argument below uses this stability and prediction error instead of a separate average Jacobian-matching error. An average Jacobian-error bound alone does not control P_K or C_K. Lemma D.11 bounds the per-stage actor error by

$$B_t^{\mathsf{ddpm}} = L_f b_K + H\sqrt{[I_K + \kappa_K(\epsilon_{\text{score}}^{(t)})^2]/2}, \qquad B_t^{\mathsf{ddim}} = L_f[b_K + P_K J_K + C_K \epsilon_{\text{score}}^{(t)}].$$

The two bounds explain how the sampler affects cloning accuracy. For DDPM, the matched Gaussian reverse kernels let us accumulate prediction error through the KL chain rule. For DDIM, we follow how each prediction error propagates through the remaining denoising maps, which introduces the factors P_K and C_K. In both cases, the choice of K and its noise schedule controls how closely the initial Gaussian matches the noised target and how closely the final output approximates the greedy action.

Lemma D.11 (Diffusion Sampler Error)**.** *Let $\widetilde{\pi}_h^{f^{(t)}}(\cdot \mid s)$ be the projection to $\mathcal{A}$ of $q_1(\cdot \mid s)$, with the sampler choices and constants above. Then*

$$W_1(\pi_h^{f^{(t)}}(\cdot \mid s), \widetilde{\pi}_h^{f^{(t)}}(\cdot \mid s)) \le b_K.$$

For DDPM, the state-averaged sampling error satisfies

$$\mathbb{E}_{d_{h,S}^{(t)}} d_{\text{TV}}(\widetilde{\pi}_h^{f^{(t)}}(\cdot \mid s), \pi_h^{(t)}(\cdot \mid s)) \le \sqrt{[I_K + \kappa_K(\epsilon_{score}^{(t)})^2]/2}.$$

For DDIM, under the stated learned-map Lipschitz bounds,

$$\mathbb{E}_{d_{h,S}^{(t)}} W_1(\widetilde{\pi}_h^{f^{(t)}}(\cdot \mid s), \pi_h^{(t)}(\cdot \mid s)) \le P_K J_K + C_K \epsilon_{score}^{(t)}.$$

Consequently, in either case,

$$\mathbb{E}_{d_h^{(t)}} \langle f_h^{(t)}(s, \cdot), \pi_h^{f^{(t)}}(\cdot \mid s) - \pi_h^{(t)}(\cdot \mid s) \rangle \le B_t^{\mathsf{sampler}}.$$

Since this holds at every stage, $B_t \le B_t^{\mathsf{sampler}}$ for the cloning loss in Equation 5.5.

The proof of Lemma D.11 is provided below. To translate the sampling bounds into behavior cloning error, we add and subtract the intermediate measure:

$$\begin{aligned}\langle f, \pi^f - \pi\rangle &= \langle f, \pi^f - \widetilde{\pi}^f\rangle + \langle f, \widetilde{\pi}^f - \pi\rangle \\ &\le L_f W_1(\pi^f, \widetilde{\pi}^f) + H d_{\mathrm{TV}}(\widetilde{\pi}^f, \pi).\end{aligned}$$

The last inequality uses Lipschitzness and $0 \le f \le H$, with $d_{\mathrm{TV}}(P, Q) = \sup_B |P(B) - Q(B)|$. For DDIM we instead bound the second term by $L_f W_1(\widetilde{\pi}^f, \pi)$. Averaging over the deployed states and applying the lemma gives

$$2\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}\langle f_h^{(t)}, \pi_h^{f^{(t)}} - \pi_h^{(t)}\rangle \le 2H\sum_{t=1}^{T} B_t \le 2H\sum_{t=1}^{T} B_t^{\mathsf{sampler}}.$$

Combining this bound with Lemmas D.9 and D.10 proves Theorem 5.17. The smoothing term quantifies the value lost by replacing the deterministic greedy action with its slightly noised version, with sensitivity controlled by L_f.

Proof of Lemma D.11. Fix t, h, s and suppress these indices. Couple the deterministic greedy action $a^* = \pi_h^{f^{(t)}}(s)$ with $X_1 = \sqrt{\alpha_1}a^* + \sqrt{1-\alpha_1}Z$. Projection fixes a^* and is nonexpansive, so

$$\begin{aligned}W_1(\pi^f, \widetilde{\pi}^f) &\le \mathbb{E}\|a^* - \mathrm{Proj}_{\mathcal{A}}(X_1)\|_2 \\ &\le (1-\sqrt{\alpha_1})R_A + \sqrt{1-\alpha_1}\mathbb{E}\|Z\|_2 \le b_K. \end{aligned} \tag{D.36}$$

Thus the smoothing error decreases as the final noise variance $1 - \alpha_1$ decreases.

DDPM score and initialization errors. The conditional forward chain $X_0 = a^*$, $X_k = \sqrt{\alpha_k}X_{k-1} + \sqrt{1-\alpha_k}Z_k$ is Gaussian. Its marginal is q_k, and Gaussian conditioning gives for $k \ge 2$

$$X_{k-1} \mid X_k = x \sim \mathcal{N}\left(\frac{x + \eta_k^{\mathsf{ddpm}}\epsilon_{\theta^*}(x, k, s)}{\sqrt{\alpha_k}}, \frac{(\sigma_k^{\mathsf{ddpm}})^2}{\alpha_k}I\right).$$

Thus the coefficients specified above give the exact reverse kernels for this deterministic conditional target. Let Q_s denote the reversed forward-path law on $X_{K:1}$ and $P_{\theta,s}$ the learned chain on $Y_{K:1}$ initialized from $N(0, I)$, with the same positive covariances. The KL chain rule gives

$$D_{\mathrm{KL}}(Q_s \| P_{\theta,s}) = D_{\mathrm{KL}}(q_K(\cdot \mid s) \| N(0, I)) + \sum_{k=2}^{K} \frac{(\eta_k^{\mathsf{ddpm}})^2}{2(\sigma_k^{\mathsf{ddpm}})^2} \mathbb{E}_{q_k(\cdot|s)} \|\epsilon_{\theta_h^{(t)}} - \epsilon_{\theta^*}\|_2^2. \tag{D.37}$$

The factors $1/\alpha_k$ in the mean difference and covariance cancel. The initial KL is exactly

$$\tfrac{1}{2}[\bar{\alpha}_K \|a^*\|_2^2 + d_A(-\bar{\alpha}_K - \log(1 - \bar{\alpha}_K))] \le I_K.$$

Data processing through the endpoint and projection, followed by Pinsker's inequality, bounds the conditional TV distance by $\sqrt{D_{\mathrm{KL}}(Q_s \| P_{\theta,s})/2}$. Average over $s \sim d_{h,S}^{(t)}$ and use Jensen's inequality and Assumption 5.15. The weighted error is at most $\kappa_K (\epsilon_{\mathrm{score}}^{(t)})^2$, giving the DDPM claim. Averaging over the deployed states matches the measure appearing in the regret decomposition. Since the conditional target is Gaussian, the chosen reverse kernels are exact, and their error separates into initialization and prediction terms. Stopping at time one keeps the reverse covariances positive, with the remaining distance to the greedy action accounted for by the smoothing bound.

For the DDPM rate in Equation 5.6, take $q = T + 1$, $K = \lceil \sqrt{q} \rceil + 1$, and $r_k = v_k/(1 - v_k) = q^{-1+2(k-1)/(K-1)}$. Direct substitution gives

$$\frac{(\eta_k^{\mathsf{ddpm}})^2}{2(\sigma_k^{\mathsf{ddpm}})^2} = \frac{1}{2}\left(\frac{r_k}{r_{k-1}} - 1\right), \qquad \kappa_K = \frac{K}{2}\left(q^{2/(K-1)} - 1\right) \le 8 \log q.$$

Here $2 \log q/(K-1) \le 4/e$ and $e^x - 1 \le e^{4/e} x$ on this interval. Also $v_1 = \bar{\alpha}_K = 1/(q+1)$, so

$$b_K \le \frac{R_A}{q+1} + \sqrt{\frac{d_A}{q+1}}, \qquad I_K \le \frac{R_A^2}{2(q+1)} + \frac{3 d_A}{8(q+1)^2}.$$

Substituting these bounds into $2H \sum_t B_t^{\mathsf{ddpm}}$ gives the stated DDPM rate.

DDIM sampling error. Write $m_k = \sqrt{\bar{\alpha}_k} a^*$ and define the exact Gaussian transport

$$T_k(x) = m_{k-1} + \sqrt{v_{k-1}/v_k}(x - m_k).$$

It maps q_k to q_{k-1} and equals the DDIM update with the stated coefficient and exact predictor. Couple $X_K = m_K + \sqrt{v_K} Z$ with $Y_K = Z$, so that $\mathbb{E}\|X_K - Y_K\|_2 \le J_K$. Evolve $X_{k-1} = T_k(X_k)$ and $Y_{k-1} = \Phi_k^\theta(Y_k)$. The learned-map Lipschitz bound gives

$$\mathbb{E}\|Y_{k-1} - X_{k-1}\|_2 \le \ell_k \mathbb{E}\|Y_k - X_k\|_2 + \frac{|\eta_k^{\mathsf{ddim}}|}{\sqrt{\alpha_k}} \mathbb{E}_{q_k} \|\epsilon_{\theta_h^{(t)}} - \epsilon_{\theta^*}\|_2.$$

Iterate from K to 2 and project both endpoints. Averaging states and applying Cauchy–Schwarz to the weighted sum over k yields $P_K J_K + C_K \epsilon_{\text{score}}^{(t)}$. Finally use the smoothing bound and the actor-error inequalities above. For this deterministic conditional target, the transport comparison controls cloning error directly through prediction accuracy and the stability of the learned maps, without requiring inverse maps or a separate Jacobian-matching error.

The DDIM bound also gives an explicit square-root sampler contribution under a sufficient stability condition. Suppose that every learned denoising map is nonexpansive, so that $\ell_k \le 1$, and use the preceding geometric noise-odds schedule. Put $q = T + 1$, $n = K - 1 = \lceil \sqrt{q} \rceil$, and $u = q^{1/n}$. Since $v_k/(1 - v_k) = u^2 v_{k-1}/(1 - v_{k-1})$, we have

$$\frac{|\eta_k^{\mathsf{ddim}}|}{\sqrt{\alpha_k}} = \sqrt{v_k/\alpha_k} - \sqrt{v_{k-1}} = (u - 1)\sqrt{v_{k-1}} \le u - 1.$$

Consequently,

$$P_K \le 1, \qquad C_K \le \sqrt{K(K-1)}(u-1) \le \sqrt{3/2}\, e^{2/e} \log q.$$

The last inequality uses $K/(K-1) \le 3/2$ for $q \ge 2$, $\log q/n \le 2/e$, and $e^x - 1 \le e^{2/e} x$ on this interval. Moreover,

$$J_K \le \frac{R_A}{\sqrt{q+1}} + \frac{\sqrt{d_A}}{q+1}, \qquad b_K \le \frac{R_A}{q+1} + \sqrt{\frac{d_A}{q+1}}.$$

Substitution yields

$$2H \sum_{t=1}^{T} B_t^{\mathsf{ddim}} = O\left(\sqrt{T} + \log(T+1) \sum_t \epsilon_{\text{score}}^{(t)} \right)$$

for fixed H, L_f, R_A, d_A. Nonexpansiveness is a genuine additional condition on the learned maps. The exact conditional Gaussian maps satisfy it because $\mathrm{Lip}(T_k) = \sqrt{v_{k-1}/v_k} < 1$. A sufficient perturbation condition is

$$\mathrm{Lip}_a(\epsilon_{\theta_h^{(t)}} - \epsilon_{\theta^*}) \le \frac{\sqrt{\alpha_k}}{|\eta_k^{\mathsf{ddim}}|} \left(1 - \sqrt{v_{k-1}/v_k}\right), \qquad k \ge 2.$$

This follows by writing the learned map as the exact Gaussian map plus $\eta_k^{\mathsf{ddim}}(\epsilon_{\theta_h^{(t)}} - \epsilon_{\theta^*})/\sqrt{\alpha_k}$ and adding their Lipschitz bounds. An average Jacobian error alone does not imply this uniform condition. □

D.4.4. Optimism and In-Sample Error Control

Lemma D.12 (Optimism and In-sample Error Control, Analogue of Lemmas 39 and 40 of Jin et al. (2021a))**.** *With probability at least* $1-\delta$*, simultaneously for all* $t \in [T]$ *and* $h \in [H]$*,*

(i) $Q^* \in \mathcal{F}^{(t-1)}$*,*

(ii) $\sum_{i=1}^{t-1} \mathbb{E}_{d_h^{(i)}}[(f_h - \mathcal{T}_h f_{h+1})^2] \le O(\beta H^2)$ *for all* $f \in \mathcal{F}^{(t-1)}$*.*

Proof. The proof follows analogously to Lemmas 39 and 40 of Jin et al. (2021a). The states and actions are drawn from the deployed diffusion policies, so the concentration statement must be uniform over the critic and its target. For $g \in \mathcal{F}_h$, $u \in \mathcal{F}_{h+1}$, put $m = \mathcal{T}_h u$, $Y_i = r_h^{(i)} + \max_a u(s_{h+1}^{(i)}, a)$, and

$$X_i(g,u) = [g(s_h^{(i)}, a_h^{(i)}) - Y_i]^2 - [m(s_h^{(i)}, a_h^{(i)}) - Y_i]^2.$$

Given the history before episode i, its mean is $\mu_i = \mathbb{E}_{d_h^{(i)}}(g-m)^2$, $|X_i| \le 4H^2$, and $\mathbb{E}[X_i^2 \mid \mathcal{H}_{i-1}] \le 16H^2\mu_i$. Here $g, m \in [0, H]$ by completeness and $Y_i \in [0, H+1]$. For $D_i = X_i - \mu_i$ and $\lambda = (32H^2)^{-1}$,

$|\lambda D_i| \le 1/4$ and

$$\mathbb{E}[e^{\pm\lambda D_i} \mid \mathcal{H}_{i-1}] \le \exp(16\lambda^2 H^2 \mu_i).$$

Multiplying these bounds and applying Markov's inequality, then a union bound over times, stages, and a uniform-norm net of pairs (g, u), gives

$$\left|\sum_{i<t} X_i(g,u) - \sum_{i<t}\mu_i\right| \le \tfrac{1}{2}\sum_{i<t}\mu_i + C_0 H^2 \ell, \quad \ell = \log\left(\frac{2H(T+1)\mathcal{N}_{\mathcal{F}}(1/(T+1))^2}{\delta}\right).$$

The bound is simultaneous for all pairs: extending from the net changes the accumulated losses and squared residuals by $O(TH/(T+1))$, absorbed into the constant. Choose $\beta \ge 2C_0\ell$. For target $u = Q^*_{h+1}$, completeness gives $m = Q^*_h$, and the lower bound places Q^* within the empirical confidence threshold. For a selected critic, completeness permits comparison against $m = \mathcal{T}_h f^{(t)}_{h+1}$ in the empirical minimum, and the same lower bound gives the claimed in-sample error. Optimistic maximization over $\mathcal{F}^{(t-1)}$ completes the argument. □

D.5. Analysis of Critic Behavior Cloning with Hybrid Data (Algorithm 5.3)

The analysis framework we use is similar to the online-data analysis, where we decompose the regret into behavior cloning error and critic estimation error. The behavior cloning error analysis is the same as before, so we omit it here and focus on critic estimation error.

D.5.1. Regret Decomposition

All initial-state expressions are averaged over ρ, as in the online analysis.

Lemma D.13 (Hybrid Regret Decomposition for Algorithm 5.3)**.** *The following regret decomposition holds for Algorithm 5.3 with hybrid data:*

$$\begin{aligned} \mathit{Reg}(T) &= \sum_{t=1}^{T}\left(V_1^*(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)})\right) \\ &\le \underbrace{2\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}\left[\left\langle f_h^{(t)}(s_h,\cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h)\right\rangle\right]}_{\textit{Diffusion policy behavior cloning error.}} \end{aligned}$$

$$+\underbrace{\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\mathcal{T}_h f_{h+1}^{(t)} - f_h^{(t)}\right] + \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}\left[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right]}_{\textit{Critic estimation error.}}.$$

Proof. As in the proof of Lemma D.9, we first add and subtract $f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}))$, obtaining

$$\mathrm{Reg}(T) = \sum_{t=1}^{T}\left(V_1^*(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)})\right) \tag{D.38}$$

$$= \sum_{t=1}^{T}\left(V_1^*(s_1^{(t)}) - f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)})\right) + \sum_{t=1}^{T}\left(f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)})\right). \tag{D.39}$$

Unlike the proof of Lemma D.9, the first term in Equation D.39 as-is is not necessarily non-positive as we do not have optimism. Still, we can address this by applying the generalized policy difference lemma from Efroni et al. (2020); Cai et al. (2020) (presented in Lemma D.15 for completeness) with $\pi = \pi^{f^{(t)}}, \pi' = \pi^*$, and with $f^{(t)}$ as the Q-function, yielding the decomposition:

$$\sum_{t=1}^{T}\left(V_1^*(s_1^{(t)}) - f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)})\right) \tag{D.40}$$

$$= \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^*(\cdot \mid s_h) - \pi_h^{f^{(t)}}(\cdot \mid s_h)\right\rangle\right] + \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\mathcal{T}_h f_{h+1}^{(t)} - f_h^{(t)}\right]. \tag{D.41}$$

Here $\mathcal{T}_h^{\pi^{f^{(t)}}} f_{h+1}^{(t)} = \mathcal{T}_h f_{h+1}^{(t)}$ because $\pi^{f^{(t)}}$ is greedy. However, by definition,

$$f_h^{(t)}(s, \pi_h^*(s)) \le \max_a f_h^{(t)}(s, a) = f_h^{(t)}(s, \pi_h^{f^{(t)}}(s)),$$

so we have that the first term in Equation D.41 is non-positive:

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left\langle f_h^{(t)}(s_h, \cdot), \pi_h^*(\cdot \mid s_h) - \pi_h^{f^{(t)}}(\cdot \mid s_h)\right\rangle\right] \le 0. \tag{D.42}$$

We therefore have that

$$\sum_{t=1}^{T}\left(V_1^*(s_1^{(t)}) - f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}))\right) \leq \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\mathcal{T}_h f_{h+1}^{(t)} - f_h^{(t)}\right] \tag{D.43}$$

It remains to address the original second term in Equation D.39. Repeating the same arguments in the proof of Lemma D.9 yields:

$$\sum_{t=1}^{T}\left(f_1^{(t)}(s_1^{(t)}, \pi_1^{f^{(t)}}(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)})\right) \tag{D.44}$$

$$\leq \underbrace{2\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}\left[\left\langle f_h^{(t)}(s_h,\cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h)\right\rangle\right]}_{\text{Diffusion policy behavior cloning error.}} + \underbrace{\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}\left[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right]}_{\text{Critic estimation error.}}. \tag{D.45}$$

Putting Equations D.43 and D.45 together yields the final regret decomposition:

$$\text{Reg}(T) = \sum_{t=1}^{T}\left(V_1^*(s_1^{(t)}) - V_1^{\pi^{(t)}}(s_1^{(t)})\right) \tag{D.46}$$

$$\leq \underbrace{2\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}\left[\left\langle f_h^{(t)}(s_h,\cdot), \pi_h^{f^{(t)}}(\cdot \mid s_h) - \pi_h^{(t)}(\cdot \mid s_h)\right\rangle\right]}_{\text{Diffusion policy behavior cloning error.}} \tag{D.47}$$

$$+ \underbrace{\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\mathcal{T}_h f_{h+1}^{(t)} - f_h^{(t)}\right] + \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{d_h^{(t)}}\left[f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right]}_{\text{Critic estimation error.}}. \tag{D.48}$$

□

D.5.2. Controlling Error Under π^*

We control the diffusion policy behavior cloning error (Lemma D.11) and the critic estimation error under $d_h^{(t)}$ (Lemma D.10) in the exact same way as we do for Algorithm 5.3. The only difference is that we use Lemma D.14 within the proof of Lemma D.10 instead of Lemma D.12, but otherwise the arguments are verbatim the same. We therefore omit the proofs for Lemmas D.11 and Lemma D.10 in the case of Algorithm 5.3 with hybrid data to avoid redundancy.

It remains to control the error under π^*. To do so, we use the offline data. The argument is verbatim the same as the proof of Theorem 3 in Tan et al. (2025).

Concretely, using the squared signed Bellman-error expectation in Definition D.1,

$$
\begin{aligned}
&\sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)(s_h,a_h)\right]\\
&\le \sum_{t=1}^{T}\sum_{h=1}^{H}\mathbb{E}_{\pi^*}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)\right]\left(\frac{N_{\mathrm{off}}\mathbb{E}_{\mu_h}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]+\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]}{N_{\mathrm{off}}\mathbb{E}_{\mu_h}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]+\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]}\right)^{1/2}\\
&\le \sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H}\frac{\mathbb{E}_{\pi^*}\left[\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right]^2}{N_{\mathrm{off}}\mathbb{E}_{\mu_h}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]+\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]}}\\
&\quad\cdot\sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H}\left(N_{\mathrm{off}}\mathbb{E}_{\mu_h}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]+\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]\right)}\\
&\le \sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H}\frac{\mathbb{E}_{\pi^*}\left[\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right]^2}{N_{\mathrm{off}}\mathbb{E}_{\mu_h}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]}}\\
&\quad\cdot\sqrt{\sum_{t=1}^{T}\sum_{h=1}^{H}\left(N_{\mathrm{off}}\mathbb{E}_{\mu_h}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]+\sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]\right)}\\
&\lesssim \sqrt{c^*_{\mathrm{off}}(\mathcal{F})HT/N_{\mathrm{off}}}\sqrt{\beta H^3 T}\\
&\lesssim \sqrt{H^4\beta c^*_{\mathrm{off}}(\mathcal{F})T^2/N_{\mathrm{off}}}.
\end{aligned}
$$

We use the single-policy concentrability coefficient to bound the squared residual ratio

$$
\frac{\mathbb{E}_{\pi^*}\left[\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right]^2}{\mathbb{E}_{\mu_h}\left[\left(\mathcal{T}_h f_{h+1}^{(t)}-f_h^{(t)}\right)^2\right]}\le\max_{h\in[H]}\sup_{f\in\mathcal{F}}\frac{\mathbb{E}_{\pi^*}\left[f_h-\mathcal{T}_h f_{h+1}\right]^2}{\mathbb{E}_{\mu_h}[(f_h-\mathcal{T}_h f_{h+1})^2]}=c^*_{\mathrm{off}}(\mathcal{F}),
$$

The sign change disappears after squaring. Terms with zero denominator are interpreted using the convention in Definition D.1.

D.5.3. In-Sample Error Control

Lemma D.14 (In-Sample Error Control for Algorithm 5.3 with Hybrid Data)**.** *With probability at least $1-\delta$, simultaneously for all $t \in [T]$ and $h \in [H]$, Algorithm 5.3 with hybrid data enjoys the guarantee that*

$$N_{\text{off}}\mathbb{E}_{\mu_h}\left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2\right] + \sum_{i=1}^{t-1}\mathbb{E}_{d_h^{(i)}}\left[\left(f_h^{(t)} - \mathcal{T}_h f_{h+1}^{(t)}\right)^2\right] \le O(H^2\beta),$$

for the β and offline sampling model in the hybrid regret theorem.

Proof. The proof is analogous to that of Lemma 39 in Jin et al. (2021a). We append the independent offline episodes to the start of the sequence of online episodes, and let $\mathcal{H}_{i-1}$ be the history before episode i in this combined sequence. For a fixed stage h and pair $g \in \mathcal{F}_h$, $u \in \mathcal{F}_{h+1}$, let $m = \mathcal{T}_h u$ and $Y_i = r_h^{(i)} + \max_a u(s_{h+1}^{(i)}, a)$. As in Lemma D.12, define

$$X_i(g,u) = [g(s_h^{(i)}, a_h^{(i)}) - Y_i]^2 - [m(s_h^{(i)}, a_h^{(i)}) - Y_i]^2. \tag{D.49}$$

Now, this has conditional expectation

$$\mathbb{E}[X_i(g,u) \mid \mathcal{H}_{i-1}] = \mathbb{E}[(g-m)^2(s_h^{(i)}, a_h^{(i)}) \mid \mathcal{H}_{i-1}],$$

and variance bounded by

$$\begin{aligned}\operatorname{Var}(X_i(g,u) \mid \mathcal{H}_{i-1}) &\le \mathbb{E}[X_i(g,u)^2 \mid \mathcal{H}_{i-1}] \\ &\le 16H^2\mathbb{E}[X_i(g,u) \mid \mathcal{H}_{i-1}].\end{aligned}$$

For a fixed pair, offline conditional expectations are under μ_h, while the j-th online episode has conditional state–action law $d_h^{(j)}$. Applying the concentration argument in Lemma D.12 and a

union bound over times, stages, and a net of pairs with radius $(N_{\text{off}} + T + 1)^{-1}$ gives

$$\left|\sum X_i(g,u) - B(g,u)\right| \le \tfrac{1}{2}B(g,u) + C_0 H^2 \ell,$$

where $B(g,u) = N_{\text{off}}\mathbb{E}_{\mu_h}(g - \mathcal{T}_h u)^2 + \sum_{i<t}\mathbb{E}_{d_h^{(i)}}(g - \mathcal{T}_h u)^2$ and ℓ is the logarithmic factor in the hybrid theorem. Here the sum of X_i includes all offline episodes and the first $t-1$ online episodes. The uniform concentration event is simultaneous in both g and u, so it applies to targets fitted backwards on the same data.

This is where the proof differs. As we choose the empirical risk minimizer, and completeness places $\mathcal{T}_h f_{h+1}^{(t)}$ in $\mathcal{F}_h$,

$$\sum X_i(f_h^{(t)}, f_{h+1}^{(t)}) = \widehat{\mathcal{L}}_h^{(t-1)}(f_h^{(t)}, f_{h+1}^{(t)}) - \widehat{\mathcal{L}}_h^{(t-1)}(\mathcal{T}_h f_{h+1}^{(t)}, f_{h+1}^{(t)}) \le 0.$$

The lower concentration bound therefore gives $B(f_h^{(t)}, f_{h+1}^{(t)}) \le 2C_0 H^2 \ell \le O(H^2\beta)$. Reusing the offline data over iterations is permitted because this event is uniform over all pairs. □

D.5.4. Critic-Tilted Pretrained Policy and Denoising Regression

The analysis of cloning a critic-tilted pretrained Gibbs target and learning the resulting denoising predictor over a finite neural approximation family appears in Appendix I of Tan et al. (2026).

D.6. Miscellaneous Lemmas

Lemma D.15 (Value Difference/Generalized Policy Difference Lemma, (Cai et al., 2020; Efroni et al., 2020))**.** *Let π, π' be two policies and $f \in \mathcal{F}$ with $f_{H+1} = 0$. We interpret $f_h(s, \pi_h(s))$ as $\mathbb{E}_{a\sim\pi_h(\cdot|s)} f_h(s,a)$. Then*

$$\begin{aligned} &f_1(s_1, \pi_1(s_1)) - V_1^{\pi'}(s_1) \\ &\quad = \sum_{h=1}^{H} \mathbb{E}_{\pi'}\left[\left\langle f_h(s_h, \cdot), \pi_h(\cdot \mid s_h) - \pi_h'(\cdot \mid s_h)\right\rangle\right] + \sum_{h=1}^{H} \mathbb{E}_{\pi'}\left[f_h(s_h, a_h) - \mathcal{T}_h^{\pi} f_{h+1}(s_h, a_h)\right]. \end{aligned}$$

Lemma D.16 (Performance Difference Lemma (Kakade and Langford, 2002))**.** *For all policies* π, π'*, and states* $s_0 \in \mathcal{S}$*,*

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{d^{\pi}_{s_0}}\left[A^{\pi'}(s,a)\right].$$

APPENDIX E

PROOFS AND SUPPLEMENTARY DETAILS FOR CHAPTER 6

E.1. Supplementary Details for Part I

Full technical proofs and supplementary empirical results for Part I (Section 6.1)—including Nyström and random-projection matrix sketching derivations and complexity bounds, finite-sample stochastic contraction proofs for Theorem 6.6, conditional and random-design central limit theorem proofs for Theorem 6.11, risk and coverage rate proofs, Optuna-tuned hyperparameter tables across UCI benchmarks, and additional raincloud and 1D coverage visualizations—are provided in the appendices of Fang et al. (2025) (`arXiv:2509.23127`).

E.2. Supplementary Details for Part II

Full technical proofs and supplementary empirical results for Part II (Section 6.2)—including notation summary tables, the bin-level structure matrix decomposition and Woodbury compression algorithm, discretization error bounds under histogram binning, fixed-point and stochastic contraction proofs for Theorems 6.19 and 6.21, empirical verification of feature-kernel distinctness (Assumption 6.20), conditional and unconditional central limit theorem proofs for Theorem 6.23 and the intercept estimator, and full per-feature effect plots on the UCI Obesity dataset—are provided in the appendices of Fang et al. (2026) (`arXiv:2601.18857`).

APPENDIX F

PROOFS AND SUPPLEMENTARY DETAILS FOR CHAPTER 7

This appendix contains the complete proofs of Theorems 7.1–7.5, the Feynman–Kac triangular-array strong law of large numbers and central limit theorem, the sharper bias–variance expansion, supplementary Bayesian posterior figures, algorithmic hyperparameter tables, and the *Daphnia* population dynamics case study for Chapter 7.

F.1. Proof of Theorem 7.1 (Algorithm 7.2 Functional Forms)

Theorem 7.1 follows immediately as a consequence of the following results, Lemmas F.1 and F.2. Here, an A superscript is used to denote ancestral trajectories, so that $x_{1:n,j}^{A,P,\theta}$ contains the history of the prediction particle $x_{n,j}^{P,\theta}$ at observation times $1{:}n$. Similarly, $g_{i,j}^{A,P,\theta}$ is the ancestral weight of $x_{n,j}^{P,\theta}$ at observation time i.

Lemma F.1. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the Algorithm 7.2 gradient estimate (or, equivalently, the gradient of Algorithm 7.1 when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^{N} L_n^{A,\theta,\alpha}$. When $\alpha = 1$,*

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:N,j}^{A,F,\theta}; \theta\big), \tag{F.1}$$

with the same undiscounted path-space score target as the bootstrap filter estimators of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider Algorithm 7.1 with $\alpha = 1$. Hold ϕ, the simulator draws, and the reference-filter resampling indices fixed when differentiating, then evaluate at $\theta = \phi$. Write $k_{n,j}$ for k_j at time n. Algorithm 7.1 gives

$$w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta}, \qquad w_{n,j}^{F,\theta} = w_{n,k_{n,j}}^{P,\theta} \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = w_{n-1,k_{n,j}}^{F,\theta} \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}}. \tag{F.2}$$

The after-resampling likelihood estimate therefore telescopes:

$$\begin{aligned}\hat{\mathcal{L}}^1(\theta) &= \prod_{n=1}^{N} L_n^{A,\theta,1} = \prod_{n=1}^{N} \left(L_n^{\phi} \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}} \right) \\ &= \left(\prod_{n=1}^{N} L_n^{\phi} \right) \prod_{n=1}^{N} \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n-1,j}^{F,\theta}} = \left(\frac{1}{J} \sum_{j=1}^{J} w_{N,j}^{F,\theta} \right) \prod_{n=1}^{N} L_n^{\phi}. \end{aligned} \tag{F.3}$$

Here each intermediate weight sum cancels, and the initial sum is J because $w_{0,j}^{F,\theta} = 1$. The log-derivative identity gives

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\nabla_\theta \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{J^{-1} \sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta} \prod_{n=1}^{N} L_n^{\phi}}{J^{-1} \sum_{j=1}^{J} w_{N,j}^{F,\theta} \prod_{n=1}^{N} L_n^{\phi}} = \frac{\sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta}}{\sum_{j=1}^{J} w_{N,j}^{F,\theta}}. \tag{F.4}$$

We next expand the filtered weight along its ancestry. Using the existing A convention, let $g_{i,j}^{A,F,\theta}$ denote the measurement density at time i along the filtered trajectory $x_{1:n,j}^{A,F,\theta}$; the endpoint n is implicit. At time n, this density is $g_{n,k_{n,j}}^{\theta}$. At earlier times, the trajectory agrees with that of the selected parent $x_{n-1,k_{n,j}}^{F,\theta}$. Starting from $w_{0,j}^{F,\theta} = 1$ and repeatedly substituting the filtered-weight recursion therefore gives

$$w_{n,j}^{F,\theta} = w_{n-1,k_{n,j}}^{F,\theta} \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = \left(\prod_{i=1}^{n-1} \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}} \right) \frac{g_{n,j}^{A,F,\theta}}{g_{n,j}^{A,F,\phi}} = \prod_{i=1}^{n} \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}. \tag{F.5}$$

The product through $n-1$ is empty when $n = 1$; for $n > 1$, it is the product already accumulated by the selected parent. The reference factors and ancestor indices are fixed under differentiation, so

$$\begin{aligned}\frac{\nabla_\theta w_{n,j}^{F,\theta}}{w_{n,j}^{F,\theta}} &= \nabla_\theta \log w_{n,j}^{F,\theta} = \nabla_\theta \log \left(\prod_{i=1}^{n} \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}} \right) \\ &= \nabla_\theta \sum_{i=1}^{n} \left(\log g_{i,j}^{A,F,\theta} - \log g_{i,j}^{A,F,\phi} \right) = \sum_{i=1}^{n} \nabla_\theta \log g_{i,j}^{A,F,\theta}. \end{aligned} \tag{F.6}$$

For the terminal filtered trajectory at time N, conditional independence of the observations gives

$$\begin{aligned}\sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,F,\theta} &= \nabla_\theta \log \left(\prod_{n=1}^{N} g_{n,j}^{A,F,\theta} \right) \\ &= \nabla_\theta \log \left(\prod_{n=1}^{N} f_{Y_n|X_n}\left(y_n^* \mid x_{n,j}^{A,F,\theta}; \theta\right) \right) \\ &= \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta}; \theta\right). \end{aligned} \tag{F.7}$$

The derivative includes the dependence of the simulated particle locations on θ through the chain rule. Multiplying the log-weight derivative by $w_{N,j}^{F,\theta}$ yields

$$\nabla_\theta w_{N,j}^{F,\theta} = w_{N,j}^{F,\theta} \sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,F,\theta} = w_{N,j}^{F,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta}; \theta\right). \tag{F.8}$$

Finally, at $\theta = \phi$, every ratio in Equation (F.5) equals one, so $w_{N,j}^{F,\theta} = 1$ and $\sum_j w_{N,j}^{F,\theta} = J$. Substituting into Equation (F.4) after differentiating gives

$$\begin{aligned}\nabla_\theta \hat{\ell}^1(\theta)\Big|_{\theta=\phi} &= \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta}\Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta}; \theta\right)\Big|_{\theta=\phi}, \end{aligned} \tag{F.9}$$

as claimed. This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle. □

For $\theta \neq \phi$, the same calculation instead gives the normalized weighted path score

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\displaystyle\sum_{j=1}^{J} w_{N,j}^{F,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta}; \theta\right)}{\displaystyle\sum_{j=1}^{J} w_{N,j}^{F,\theta}}. \tag{F.10}$$

Growth of the unnormalized weight products alone does not establish a variance rate for this normalized expression. When $\theta = \phi$, a generic normalized path-functional inequality gives the coarse upper bound $O(N^4/J)$ after rescaling to the full unnormalized score. Section F.7 gives an exact constant-weight calculation with order N^2/J at $\alpha = 1$ and explains why this example does not by itself provide a general finite-J upper bound.

Lemma F.2. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the Algorithm 7.2 gradient estimate (or, equivalently, the gradient of Algorithm 7.1 when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,*

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_\theta \log \left(f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)\right), \tag{F.11}$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{g_{n,k_{n,j}}^\theta}{g_{n,k_{n,j}}^\phi} \tag{F.12}$$

as shorthand for the selected measurement density ratios, with the reference parameter and resampling indices held fixed. Observe that, when $\alpha = 0$, the likelihood estimate becomes

$$\begin{aligned}
\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} &= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right) && \text{(F.13)} \\
&= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) && \text{(F.14)} \\
&= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}\left(y_n^* | x_{n,k_{n,j}}^{P,\theta}; \theta\right)}{f_{Y_n|X_n}\left(y_n^* | x_{n,k_{n,j}}^{P,\phi}; \phi\right)} \right). && \text{(F.15)}
\end{aligned}$$

We lose the nice telescoping property observed in the Algorithm 7.1 ($\alpha = 1$) case, but this expression

still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^{N} \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right) \tag{F.16}$$

$$= \sum_{n=1}^{N} \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right)} \tag{F.17}$$

$$= \sum_{n=1}^{N} \frac{\sum_{j=1}^{J} \nabla_\theta s_{n,j}}{\sum_{j=1}^{J} s_{n,j}} \tag{F.18}$$

$$= \sum_{n=1}^{N} \frac{1}{J} \sum_{j=1}^{J} \frac{\nabla_\theta f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} \tag{F.19}$$

$$= \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log \left(f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right) \right), \tag{F.20}$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. □

F.2. Proof of Theorem 7.2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter $\theta = \phi$. Let Z_n contain the complete ancestral path through time n, the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^\theta(Z_n) = f_{Y_n|X_n}(y_n^* \mid X_n^\theta; \theta), \qquad W_\theta(Z_N) = \prod_{n=1}^{N} G_n^\theta(Z_n), \tag{F.21}$$

$$\psi_\theta(Z_N) = \nabla_\theta \log W_\theta(Z_N), \qquad \Xi_\theta(Z_N) = \nabla_\theta^2 \log W_\theta(Z_N). \tag{F.22}$$

Differentiate through X_n^θ in $G_n^\theta(Z_n)$, holding the simulator draws and resampling indices fixed.

Write $\rho_N^J = J^{-1} \sum_{j=1}^{J} \delta_{Z_{N,j}}$ for the empirical measure of the after-resampling genealogical particles,

and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_\theta(Z_N)F(Z_N)\}}{\mathbb{E}\{W_\theta(Z_N)\}}. \tag{F.23}$$

Assumptions ((A3)) and ((A5)) give $|\partial_{\theta_i} W_\theta| \leq N\bar{G}^N G'(\theta)$, so dominated convergence and the log-derivative identity yield

$$\nabla_\theta \ell(\theta) = \rho_N(\psi_\theta), \qquad \nabla_\theta \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_\theta). \tag{F.24}$$

The second equality is Lemma F.1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let Z_0 be the augmented prediction path at observation 1; for $q = 1, \ldots, N-1$, let Z_q be the prediction path at observation $q+1$; and let Z_N be the terminal path after the final resampling. For $0 \leq q \leq N$, write $\rho_q^J = J^{-1}\sum_{j=1}^J \delta_{\mathsf{Z}_{q,j}}$, where the $\mathsf{Z}_{0,j}$ are independent draws from the unweighted prediction law of Z_0 and $\mathsf{Z}_{N,j} = Z_{N,j}$. Write $V_q^\theta(\mathsf{Z}_q) = G_{q+1}^\theta(\mathsf{Z}_q)$ for $q = 0, \ldots, N-1$. Let the locally indexed augmented kernel $\mathcal{M}_{q+1}^\theta$ append the next independent simulator innovation and its gradients for $0 \leq q \leq N-2$, and put $\mathcal{M}_N^\theta = \mathrm{Id}$ for the final selection. The one-step population map is

$$\Phi_{q+1}^\theta(\mu)(dz') = \frac{\int \mu(dz) V_q^\theta(z) \mathcal{M}_{q+1}^\theta(z, dz')}{\mu(V_q^\theta)}, \qquad 0 \leq q < N. \tag{F.25}$$

Conditionally on the current cloud, the J children are independent with common law $\Phi_{q+1}^\theta(\rho_q^J)$. The final observation is handled by the $q = N-1$ transition, namely multinomial selection with potential $V_{N-1}^\theta = G_N^\theta$ followed by the identity mutation. Thus ρ_N^J is exactly the terminal empirical measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For $0 \le q \le N$, let

$$Q^{\theta}_{q,N}F(z_q) = \mathbb{E}\left[F(\mathsf{Z}_N)\prod_{r=q}^{N-1} V^{\theta}_r(\mathsf{Z}_r)\,\middle|\,\mathsf{Z}_q = z_q\right], \tag{F.26}$$

$$P^{\theta}_{q,N}F = \frac{Q^{\theta}_{q,N}F}{Q^{\theta}_{q,N}1}, \qquad g_{q,N} = \sup_{z,z'}\frac{Q^{\theta}_{q,N}1(z)}{Q^{\theta}_{q,N}1(z')}, \qquad b_{q,N} = \sup_{\mathrm{osc}(F)\le 1}\mathrm{osc}(P^{\theta}_{q,N}F), \tag{F.27}$$

where $Q^{\theta}_{N,N}F = F$ and $\mathrm{osc}(F) = \sup_{z,z'}|F(z) - F(z')|$. Thus $0 \le b_{q,N} \le 1$. With

$$R_G = \frac{\bar{G}}{\underline{G}}, \qquad S_N = \sum_{q=0}^{N} g^3_{q,N} b_{q,N}, \qquad K_N = 32 S_N, \qquad K^{\mathrm{pf}}_N = 32\sum_{r=0}^{N} R^{3r}_G, \tag{F.28}$$

the potential bounds give, term by term,

$$\underline{G}^{N-q} \le Q^{\theta}_{q,N}1(z_q) \le \bar{G}^{N-q}, \qquad g_{q,N} \le R^{N-q}_G, \qquad K_N \le K^{\mathrm{pf}}_N. \tag{F.29}$$

In particular, the constant used in the theorem is the closed-form quantity

$$K^{\mathrm{pf}}_N = \begin{cases} 32(N+1), & R_G = 1, \\ 32\dfrac{R^{3(N+1)}_G - 1}{R^3_G - 1}, & R_G > 1. \end{cases} \tag{F.30}$$

Here is the constant calculation behind the tail bound used below. Let r^{DM}_N and β^{DM}_N denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r^{\mathrm{DM}}_N \le 4S_N, \qquad (\beta^{\mathrm{DM}}_N)^2 \le 4\sum_{q=0}^{N} g^2_{q,N} b^2_{q,N} \le 4S_N, \tag{F.31}$$

where the last inequality uses $g_{q,N} \ge 1$ and $0 \le b_{q,N} \le 1$. If $S_N = 0$, the particle error is identically zero. Hence suppose $S_N > 0$. For $u \in (0, 1/2]$, set $x = Ju^2/(32S_N)$. If $x \le \log 6$, then $6e^{-x} \ge 1$ and the desired probability bound is trivial. If $x > \log 6$, use $h^{-1}(x) \le 2x + 2\sqrt{x}$ for

$h(t) = \{t - \log(1+t)\}/2$ and $1 + 2x + 2\sqrt{x} \le 8x$ to obtain

$$\frac{r_N^{\mathrm{DM}}}{J}\{1 + h^{-1}(x)\} \le \frac{u^2}{8x}\{1 + 2x + 2\sqrt{x}\} \le u^2 \le \frac{u}{2},$$
$$\beta_N^{\mathrm{DM}}\sqrt{\frac{2x}{J}} \le 2\sqrt{S_N}\sqrt{\frac{2x}{J}} = \frac{u}{2}.$$

Theorem 1.2 applied to both F and $-F$, after centering and scaling F to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr\left(|(\rho_N^J - \rho_N)(F)| > \mathrm{osc}(F)u\right) \le 6\exp\left(-\frac{Ju^2}{K_N}\right) \le 6\exp\left(-\frac{Ju^2}{K_N^{\mathrm{pf}}}\right). \tag{F.32}$$

for every F with $\mathrm{osc}(F) > 0$ and every $u \in (0, 1/2]$. If F is constant, $(\rho_N^J - \rho_N)(F) = 0$ identically. Equations F.28– F.32 replace all unnamed model-dependent concentration constants.

Lemma F.3 (Explicit concentration of the Algorithm 7.2 ($\alpha = 1$) gradient)**.** *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1 \le i \le p} \mathrm{osc}(\psi_{\theta,i}), \qquad u_g(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6p}{\delta}}. \tag{F.33}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6p/\delta)$, then, with probability at least $1-\delta$,

$$\|\nabla_\theta \hat{\ell}_J^{A,1}(\theta) - \nabla_\theta \ell(\theta)\|_2 \le A_N(\theta)\sqrt{p}\, u_g(J,\delta). \tag{F.34}$$

Moreover, Assumption ((A3)) gives $A_N(\theta) \le 2NG'(\theta)$. Consequently, the explicit condition

$$J \ge K_N^{\mathrm{pf}}\log\frac{6p}{\delta}\max\left\{4, \frac{4pN^2G'(\theta)^2}{\epsilon^2}\right\} \tag{F.35}$$

implies that the left side of (F.34) *is at most ϵ with probability at least $1-\delta$.*

Proof. The lower bound on J makes $u_g(J,\delta) \le 1/2$. Apply (F.32) to each normalized coordinate $\psi_{\theta,i}/A_N(\theta)$; a constant coordinate has zero error and needs no normalization. A union bound over

the p coordinates gives

$$\max_{1\le i\le p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \le A_N(\theta) u_g(J,\delta)$$

with probability at least $1-\delta$. Now use $\|v\|_2 \le \sqrt{p}\|v\|_\infty$ and (F.24). Finally, $|\psi_{\theta,i}| \le NG'(\theta)$, so $A_N(\theta) \le 2NG'(\theta)$; substituting this inequality and solving $2NG'(\theta)\sqrt{p}\, u_g(J,\delta) \le \epsilon$ gives (F.35). □

The next result treats the actual particle Hessian. It retains the outer-product and empirical-mean terms created by differentiating the logarithm of the terminal empirical average.

Lemma F.4 (Explicit concentration of the Algorithm 7.2 ($\alpha = 1$) particle Hessian)**.** *Suppose in addition that Assumption ((A3)) gives the entrywise bound $|\Xi_{\theta,ab}(Z_N)| \le NG''(\theta)$ for $1 \le a, b \le p$. Define*

$$d_p = \frac{p(p+3)}{2}, \qquad H_N(\theta) = 2p\{NG''(\theta) + 3N^2 G'(\theta)^2\}, \qquad u_H(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J} \log \frac{6d_p}{\delta}}. \tag{F.36}$$

If $J \ge 4K_N^{\mathrm{pf}} \log(6d_p/\delta)$, then, with probability at least $1-\delta$,

$$\|\nabla_\theta^2 \hat{\ell}_J^{A,1}(\theta) - \nabla_\theta^2 \ell(\theta)\|_{\mathrm{op}} \le H_N(\theta) u_H(J,\delta). \tag{F.37}$$

Fix $0 < \kappa < \gamma$. If

$$\gamma I_p \preceq -\nabla_\theta^2 \ell(\theta) \preceq \Gamma I_p \quad \textit{and} \quad J \ge K_N^{\mathrm{pf}} \log \frac{6d_p}{\delta} \max\left\{4, \frac{H_N(\theta)^2}{\kappa^2}\right\}, \tag{F.38}$$

then, with probability at least $1-\delta$,

$$(\gamma - \kappa) I_p \preceq -\nabla_\theta^2 \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma + \kappa) I_p. \tag{F.39}$$

In particular, the raw particle curvature is then positive definite and invertible.

Proof. For fixed reference-filter random numbers and resampling indices, the telescoping identity

in the proof of Lemma F.1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_\phi + \log\left\{\frac{1}{J}\sum_{j=1}^{J}\exp(a_j(\theta))\right\}, \qquad a_j(\theta) = \log w_{N,j}^{F,\theta}, \qquad a_j(\phi) = 0,$$

where C_ϕ is constant in θ. Twice differentiating this log-sum-exp at $\theta = \phi$ gives the exact finite-particle identity

$$\nabla_\theta^2 \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_\theta + \psi_\theta\psi_\theta^T) - \rho_N^J(\psi_\theta)\rho_N^J(\psi_\theta)^T. \tag{F.40}$$

Assumption ((A3)) also gives the entrywise dominating bound

$$|\partial^2_{\theta_a\theta_b} W_\theta| = W_\theta|\Xi_{\theta,ab} + \psi_{\theta,a}\psi_{\theta,b}| \le \bar{G}^N\{NG''(\theta) + N^2G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating $\log \mathbb{E}W_\theta$ gives the population identity

$$\nabla_\theta^2 \ell = \rho_N(\Xi_\theta + \psi_\theta\psi_\theta^T) - \rho_N(\psi_\theta)\rho_N(\psi_\theta)^T. \tag{F.41}$$

Let $A_1 = 2NG'(\theta)$ and $A_2 = 2\{NG''(\theta) + N^2G'(\theta)^2\}$. The p scalar functions ψ_a have oscillation at most A_1, and the symmetric entries of $\Xi + \psi\psi^T$ have oscillation at most A_2. Apply (F.32) jointly to these functions, using $d_p = p + p(p+1)/2$ scalar events. On the resulting event,

$$\begin{aligned}\|(\rho_N^J - \rho_N)(\psi)\|_2 &\le A_1\sqrt{p}\, u_H(J,\delta),\\ \|(\rho_N^J - \rho_N)(\Xi + \psi\psi^T)\|_{\mathrm{op}} &\le pA_2 u_H(J,\delta).\end{aligned}$$

Both $\|\rho_N^J(\psi)\|_2$ and $\|\rho_N(\psi)\|_2$ are at most $\sqrt{p}\, NG'(\theta)$. Therefore

$$\begin{aligned}&\|\rho_N^J(\psi)\rho_N^J(\psi)^T - \rho_N(\psi)\rho_N(\psi)^T\|_{\mathrm{op}}\\ &\qquad \le 2\sqrt{p}\, NG'(\theta)\, \|(\rho_N^J - \rho_N)(\psi)\|_2 \le 4pN^2G'(\theta)^2 u_H(J,\delta).\end{aligned}$$

Combining the last three displays with (F.40) and (F.41) proves (F.37). Condition (F.38) makes

its right side at most κ; Weyl's eigenvalue inequality then gives (F.39). $\square$

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If $\widetilde{B} = Q\,\mathrm{diag}(\lambda_1, \dots, \lambda_p)Q^T$ is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q\,\mathrm{diag}([\lambda_1]_c^C, \dots, [\lambda_p]_c^C)Q^T, \qquad [x]_c^C = \min\{C, \max\{c, x\}\}. \tag{F.42}$$

For every $v \in \mathbb{R}^p$, direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \le v^T B v \le C\|v\|_2^2, \qquad C^{-1}\|v\|_2^2 \le v^T B^{-1} v \le c^{-1}\|v\|_2^2. \tag{F.43}$$

Thus $cI_p \preceq B \preceq CI_p$ and $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$. Lemma F.4 shows that the unclipped particle Hessian has these properties with $c = \gamma - \kappa$ and $C = \Gamma + \kappa$ under (F.38); (F.42) guarantees the chosen c, C bounds for every realization.

Proof of Theorem 7.2. Let $\mathcal{D}$ be the convex region in Theorem 7.2, and set

$$G'_{\mathcal{D}} = \sup_{\theta \in \mathcal{D}} G'(\theta), \qquad J_0 = \left\lceil K_N^{\mathrm{pf}} \log\!\left(\frac{6pM}{\delta}\right) \max\left\{4, \frac{4pN^2(G'_{\mathcal{D}})^2}{\epsilon^2}\right\}\right\rceil. \tag{F.44}$$

Thus it is enough to take $J \ge J_0$. With $g_m = -\nabla_\theta \hat{\ell}_J^{A,1}(\theta_m)$, define

$$\tau = \min\left\{m \in \{0, \dots, M-1\} : \|g_m\|_2 \le \sigma\epsilon\right\}, \qquad \min \varnothing = M, \tag{F.45}$$

and

$$\mathcal{E}_M = \bigcap_{m=0}^{\min\{\tau, M-1\}} \left\{\|g_m - \nabla f(\theta_m)\|_2 \le \epsilon\right\}. \tag{F.46}$$

If $\mathcal{F}_m$ is the history before the mth particle-filter call, then θ_m and $\{m \le \min(\tau, M-1)\}$ are

$\mathcal{F}_m$-measurable. Lemma F.3, with failure probability δ/M, gives

$$\begin{aligned}\Pr(\mathcal{E}_M^c \mid \theta_0) &\le \sum_{m=0}^{M-1} \mathbb{E}\left[\mathbf{1}_{\{m\le \min(\tau, M-1)\}} \Pr\left(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m\right) \middle| \theta_0\right] \\ &\le \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta.\end{aligned} \tag{F.47}$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on $\mathcal{E}_M$ and write $p_m = -B_m^{-1} g_m$. If $\|g_m\|_2 \le \sigma\epsilon$, then

$$\|\nabla f(\theta_m)\|_2 \le \|g_m\|_2 + \epsilon \le (\sigma+1)\epsilon. \tag{F.48}$$

Strong convexity implies $f(\theta) - f(\theta^*) \le \|\nabla f(\theta)\|_2^2/(2\gamma)$; therefore

$$f(\theta_m) - f(\theta^*) \le \frac{(\sigma+1)^2\epsilon^2}{2\gamma}. \tag{F.49}$$

Suppose instead that $\|g_m\|_2 > \sigma\epsilon$. Since $g_m = -B_m p_m$ and $cI_p \preceq B_m \preceq CI_p$,

$$c\|p_m\|_2 \le \|g_m\|_2 \le C\|p_m\|_2, \qquad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \tag{F.50}$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned}f(\theta_m + \eta p_m) - f(\theta_m) &\le \eta \nabla f(\theta_m)^T p_m + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2 \\ &\le -\eta p_m^T B_m p_m + \eta\epsilon\|p_m\|_2 + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2.\end{aligned} \tag{F.51}$$

The choices of σ and η imply

$$\frac{C}{\sigma}\|p_m\|_2^2 \le \frac{c(1-\beta)}{2}\|p_m\|_2^2, \qquad \frac{\Gamma\eta}{2}\|p_m\|_2^2 \le \frac{c(1-\beta)}{2}\|p_m\|_2^2.$$

Using $p_m^T B_m p_m \geq c\|p_m\|_2^2$ in (F.51) now gives

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta\beta p_m^T B_m p_m. \tag{F.52}$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon < \frac{\sigma+1}{\sigma}\|g_m\|_2. \tag{F.53}$$

Consequently, (F.43) and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\geq \frac{1}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\geq \frac{2\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \tag{F.54}$$

Combining (F.52) and (F.54) proves

$$f(\theta_{m+1}) - f(\theta^*) \leq q\{f(\theta_m) - f(\theta^*)\}, \qquad q = 1 - \frac{2\eta\beta\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2. \tag{F.55}$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1-\beta)\frac{c\gamma}{C\Gamma} \leq \frac{1}{2}, \qquad \text{so} \qquad \frac{1}{2} < q < 1.$$

Iteration gives the q^m conclusion in the theorem, while (F.49) gives its stopping conclusion after substituting $f = -\ell$. □

The result is deliberately restricted to Algorithm 7.2 ($\alpha = 1$). For $\alpha < 1$, the population limit is a discounted gradient. A theorem for that case must add a finite-sample bound for the discount bias relative to the exact score; choosing α close to one does not by itself provide such a bound.

F.3. Feynman–Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 7.3, 7.4, and 7.5. Let $(\eta_n)_{n=1}^N$ and $(\pi_n)_{n=1}^N$ be the prediction and filtering probability measures on the current state space $\mathcal{X}$. The posterior measure ρ_n is on simulator histories through time n, including the base variables, as defined in Equation (F.7). Its state-path marginal is $f_{X_{0:n}|Y_{1:n}}$. These are the measures we seek to approximate with the particle filter. For any measurable bounded functional h, we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \to [0, 1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x, and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x. If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A.

Markov kernels and measures: For any measure η, any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h \, d\eta = \int h(x)\eta(dx), \tag{F.56}$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \tag{F.57}$$

$$(Mh)(x) = \int M(x, dy)h(y). \tag{F.58}$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1 M_2)(x, A) = \int M_1(x, dy) M_2(y, A). \tag{F.59}$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{\|h\|_\infty \leq 1/2} |\mu(\phi) - \nu(\phi)| = \sup_{\mathrm{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \tag{F.60}$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\mathrm{TV}}(M) = \sup_{x,y \in \mathcal{X}} \big\| M(x, \cdot) - M(y, \cdot) \big\|_{\mathrm{TV}} = \sup_{\mu,\nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}. \tag{F.61}$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \to [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}\big(y_n^* | x_{n,j}^P; \theta\big) = G_n(x_{n,j}^P)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^* | \, \cdot \,)$ is the conditional density of the observed measurement at time n, evaluated at a state in $\mathcal{X}$. At the filtered particle, $G_n(x_{n,j}^F) = g_{n,k_{n,j}}$.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0, a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation of Chapter 7, this corresponds to the starting distribution f_{X_0}, the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \tag{F.62}$$

- The map from the prediction distribution at timestep n to timestep $n+1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n) M_{n+1}. \tag{F.63}$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ ... \circ \Phi_{k+1}. \tag{F.64}$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^P}, \tag{F.65}$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^{J} g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^{J} g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^F}. \tag{F.66}$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogeneous Markov process evolving on $\mathcal{X}^J$. For $\theta = \phi$ and multinomial resampling, the corresponding Markov transition kernel is

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \left(\Phi_n\left(\eta_{n-1}^J\right)\right)^{\otimes J} = \left(\Phi_n\left(\frac{1}{J}\sum_{j=1}^{J} \delta_{x_{n-1,j}^P}\right)\right)^{\otimes J}, \tag{F.67}$$

and the composition of Markov kernels on particles from timestep n to timestep $n+k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \tag{F.68}$$

The kernel acts on the full prediction cloud. Conditional on the preceding prediction cloud, its law is

$$\text{Law}\left(x^P_{n,1:J} \mid x^P_{n-1,1:J}\right) = \mathbf{M}_n(x^P_{n-1,1:J}, \cdot). \tag{F.69}$$

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of Algorithm 7.1. That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}\left(\mathbf{M}_{n,n+k}\right) \leq (1-\epsilon)^{\lfloor k/(O(\log J))\rfloor}, \tag{F.70}$$

for some ϵ dependent on $\bar{G}, \underline{\text{G}}, \bar{M}, \underline{\text{M}}$ in Assumptions ((A3)) and ((A2)). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of Algorithm 7.1 itself.

F.4. Proof of Theorem 7.3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 7.3 by deriving a more general result, from which it will be clear that Algorithm 7.1 targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition F.5 (Off-Parameter Resampled Particle Filters)**.** An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model $\left(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N\right)$, where

we inductively define given some $x^F_{n-1,j}$, $w^F_{n-1,j}$ comprising the filtering measure approximation at time $n-1$:

1. Conditional on the filtered particles and weights at time $n-1$, draw independently $x^P_{n,j} \sim M_n(x^F_{n-1,j}, \cdot)$ using fresh simulator draws. Retain the corresponding parent weight: $w^P_{n,j} = w^F_{n-1,j}$.

2. The prediction measure at timestep n is given by

$$\eta^J_n = \frac{\sum_{j=1}^J w^P_{n,j}\, \delta_{x^P_{n,j}}}{\sum_{j=1}^J w^P_{n,j}}. \tag{F.71}$$

3. The particles are resampled at every timestep n, yielding indices k_j according to some arbitrary probabilities $(p_{n,j})_{j=1}^J$.

4. The filtering measure at timestep n is given by

$$\pi^J_n = \frac{\sum_{j=1}^J \delta_{x^P_{n,j}}\, w^P_{n,j}\, g_{n,j}}{\sum_{j=1}^J w^P_{n,j}\, g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x^P_{n,k_j}}\, w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}}{\sum_{j=1}^J w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}} = \frac{\sum_{j=1}^J w^F_{n,j}\, \delta_{x^F_{n,j}}}{\sum_{j=1}^J w^F_{n,j}}. \tag{F.72}$$

5. The prediction weights at the next timestep are given by $w^P_{n+1,j} = w^F_{n,j} = w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}$.

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition F.6 (Targeting)**.** A pair of random vectors (X, W) drawn from some measure g **targets** another measure π if, for any measurable and bounded functional h,

$$E_g\big[h(X) \cdot W\big] = E_\pi\big[h(X)\big]. \tag{F.73}$$

A particle approximation is a triangular array of pairs of random vectors $(X^J_j, W^J_j), j = 1, 2, \ldots, J$.

The particle approximation **targets** π if, for any measurable and bounded functional h,

$$\frac{\sum_{j=1}^{J} h(X_j^J)\, W_j^J}{\sum_{j=1}^{J} W_j^J} \xrightarrow{a.s.} E_\pi\big[h(X)\big] \tag{F.74}$$

as $J \to \infty$.

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma F.7 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma F.7 (Change of Weight Measure)**.** *Suppose that* $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Now, let* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *be a multinomial sample with indices* k_j *drawn from* $\{(\tilde{X}_j^J, U_j^J)\}$ *where* $(\tilde{X}_j^J, U_j^J)$ *is represented, on average, proportional to* $\pi_j^J J$ *times, with* $\pi_j^J > 0$*. Write*

$$(Y_j^J, V_j^J) = \big(\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J\big). \tag{F.75}$$

If the importance sampling weights U_j/π_j *are bounded uniformly in* j, J *after normalizing* $\sum_m U_m^J = 1$*, then* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *targets* f_X*.*

Proof. Normalize $\sum_j U_j^J = 1$, which leaves both weighted empirical measures unchanged. Note that as the Y_j^J are a subsample from X_j^J, h can be a function of Y as well as it is one for X. We

then expand

$$\frac{\sum_j h(Y_j^J)\, V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \tag{F.76}$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[h(X)\big]. \tag{F.77}$$

We want to show

$$\frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} 0. \tag{F.78}$$

For this, it is sufficient to show that

$$\frac{1}{J}\sum_j h(\tilde{X}_{k_j}^J)\frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J)\frac{U_j^J}{\pi_j^J}\pi_j^J \overset{a.s.}{\to} 0 \tag{F.79}$$

since an application of this result with $h(x) = 1$ provides almost sure convergence to 1 of J^{-1} times the denominator in (F.78). Write $g_j^J = h(\tilde{X}_j^J)\frac{U_j^J}{\pi_j^J}$. We therefore need to show that

$$\frac{1}{J}\sum_j Z_j^J := \frac{1}{J}\sum_j \left(g_{k_j}^J - \sum_{m=1}^J g_m^J \pi_m^J \right) \overset{a.s.}{\to} 0. \tag{F.80}$$

Because the functional h and importance sampling weights $U_{k_j}^J/\pi_{k_j}^J$ are bounded, we have that $\sup_{J,j} \mathbb{E}\big[(Z_j^J)^4\big] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z_j^J are conditionally independent, identically distributed, and centered given the $\tilde{X}_j^J$, U_j^J, and π_j^J,

$$\begin{aligned}
\mathbb{E}\left[\left(\sum_j Z_j^J \right)^4 \middle| (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J \right] &= J\,\mathbb{E}\left[(Z_1^J)^4 | (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J \right] \\
&\quad + 3J(J-1)\left(\mathbb{E}\big[(Z_1^J)^2 \big| (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J \big] \right)^2 \qquad \text{(F.81)} \\
&\leq CJ^2, \qquad \text{(F.82)}
\end{aligned}$$

for some $C > 0$ independent of J. Taking expectations on both sides yields

$$\mathbb{E}\left[\left(\sum_j Z_j^J\right)^4\right] \leq CJ^2 \tag{F.83}$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P}\left(\left|\frac{1}{J}\sum_{j=1}^{J} Z_j^J\right| > \epsilon\right) \leq \frac{1}{\epsilon^4 J^4}\mathbb{E}\left[\left(\sum_{j=1}^{J} Z_j^J\right)^4\right] \leq \frac{C}{\epsilon^4 J^2}, \tag{F.84}$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\frac{1}{J}\sum_j h(\tilde{X}_{k_j}^J)\,\frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J)\,\frac{U_j^J}{\pi_j^J}\,\pi_j^J \overset{a.s.}{\to} 0. \tag{F.85}$$

Similarly, we also have that

$$\sum_j \frac{U_j^J}{\pi_j^J}\pi_j^J - \frac{1}{J}\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J} \overset{a.s.}{\to} 0, \tag{F.86}$$

and the result follows by taking the ratio, since $\sum_j U_j^J = 1$.

□

Remark: Note that Lemma F.7 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma F.8 (Particle Marginals)**.** *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ targets f_X. Also suppose that $\tilde{Z}_j^J \sim f_{Z|X}(\cdot|\tilde{X}_j^J)$ where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z. Then, if the U_j^J are bounded, $\{(\tilde{Z}_j^J, U_j^J)\}$ targets f_Z.*

Proof. We want to show that, for any measurable bounded h,

$$\frac{\sum_j h(\tilde{Z}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}\big[\mathbb{E}_{f_{Z|X}}[h(Z)|X]\big]. \tag{F.87}$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}[g(X)]. \tag{F.88}$$

Let $\bar{U}_j^J = \frac{J U_j^J}{\sum_j U_j^J}$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} = \frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J}{J^{-1}\sum_j \bar{U}_j^J}. \tag{F.89}$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J, \tag{F.90}$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\begin{aligned}
\lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J\right] &= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\Big[h(\tilde{Z}_j^J)\, \bar{U}_j^J \Big| \tilde{X}_j^J, \bar{U}_j^J\Big]\right] &\text{(F.91)}\\
&= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\Big[h(Z)\Big| X = \tilde{X}_j^J\Big]\, \bar{U}_j^J\right]. &\text{(F.92)}
\end{aligned}$$

Applying (F.88) with $g(x) = \mathbb{E}\big[h(Z)|X = x\big]$, the average on the right hand side converges almost surely to $\mathbb{E}\big\{\mathbb{E}[h(Z)|X]\big\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)]. \tag{F.93}$$

□

Lemma F.9 (Particle Posteriors)**.** *Suppose that* $\{(X_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Also suppose that* $(X_j'^J, U_j'^J) = \big(X_j^J, U_j^J\, f_{Z|X}(z^*|X_j^J)\big)$*. Then, if* $U_j^J\, f_{Z|X}(z^*|X_j^J)$ *and* $U_j^J\, f_{Z|X}(z^*|X_j^J)/f_Z(z^*)$ *are bounded,* $\{(X_j'^J, U_j'^J)\}$ *targets* $f_{X|Z}(\cdot|z^*)$*.*

Proof. Again, we want to show that

$$\frac{\sum_j h(X_j^J) \cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \xrightarrow{a.s.} \mathbb{E}_{f_{X|Z}}\big[h(X)\big|z^*\big]. \tag{F.94}$$

We already have that for any measurable bounded g,

$$\frac{\sum_j g(X_j^J)\, U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[g(X)\big]. \tag{F.95}$$

Consider the following:

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \times \left(\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J}\right)^{-1}. \tag{F.96}$$

We will apply Equation (F.95) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j u_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big], \tag{F.97}$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[f_{Z|X}(z^*|X)\big] = f_Z(z^*). \tag{F.98}$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big]}{f_Z(z^*)} \;=\; \mathbb{E}_{f_X}\left[\frac{h(X)\, f_{Z|X}(z^*|X)}{f_Z(z^*)}\right] \tag{F.99}$$

$$= \mathbb{E}_{f_X}\left[h(X)\frac{f_{X|Z}(X|z^*)}{f_X(X)}\right] = \mathbb{E}_{f_{X|Z}}\left[h(X)|z^*\right], \tag{F.100}$$

to give the desired result.

□

Theorem F.10 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution)**.** *The off-parameter resampled particle filter as outlined in Definition F.5 targets the filtering distribution.*

Proof. Recursively applying Lemmas F.7, F.8, and F.9, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\left\{\left(X^F_{n-1,j}, w^F_{n-1,j}\right)\right\}$ targets π_{n-1}. Then, Lemma F.8 tells us that $\left\{\left(X^P_{n,j}, w^P_{n,j}\right)\right\}$ targets η_n. Lemma F.9 tells us that $\left\{\left(X^P_{n,j}, w^P_{n,j} g^\theta_{n,j}\right)\right\}$ therefore targets π_n. Lemma F.7 guarantees that the resampling rule, given by

$$\left(X^F_{n,j}, w^F_{n,j}\right) = \left(X^P_{n,k_j}, w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}\right), \tag{F.101}$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n. □

Proposition F.11 (Algorithm 7.1 ($\alpha = 1$) Targets the Filtering Distribution)**.** *When $\alpha = 1$ or $\phi = \theta$, Algorithm 7.1 targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α, the ratio $\frac{g^\theta_{n,j}}{g^\phi_{n,j}} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of Theorem F.10. Recursively applying Lemmas F.7, F.8, and F.9, we obtain that the Algorithm 7.1 ($\alpha = 1$) filter targets the posterior. Specifically, suppose inductively that $\left\{\left(X^{F,\theta}_{n-1,j}, w^{F,\theta}_{n-1,j}\right)\right\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y^*_{1:n-1};\theta)$. Then, Lemma F.8 tells us that $\left\{\left(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j}\right)\right\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. Lemma F.9 tells us that $\left\{\left(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j}\, g^\theta_{n,j}\right)\right\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. Lemma F.7 guar-

antees that the resampling rule, given by

$$\left(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}\right) = \left(X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta}\, g_{n,k_j}^{\theta} / g_{n,k_j}^{\phi}\right), \tag{F.102}$$

with resampling weights proportional to $g_{n,j}^{\phi}$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$. □

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma F.12 (Likelihood Proper Weighting)**.** *$f_{Y_n|Y_{1:n-1}}(y_n^*|y_{1:n-1}^*;\theta)$ is strongly consistently estimated by either the before-resampling estimate,*

$$L_n^{B,\theta} = \frac{\sum_{j=1}^{J} g_{n,j}^{\theta}\, w_{n,j}^{P,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}}, \tag{F.103}$$

or by the after-resampling estimate,

$$L_n^{A,\theta} = L_n^{\phi} \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}}. \tag{F.104}$$

where L_n^{ϕ} is as defined in the various algorithms.

Here, (F.103) is a direct consequence of our earlier result that $\{\left(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\right)\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*;\theta)$. To see (F.104), define the corrected weights before resampling and the reference resampling probabilities by

$$w_{n,j}^{FC,\theta} = w_{n,j}^{P,\theta} \frac{g_{n,j}^{\theta}}{g_{n,j}^{\phi}}, \qquad p_{n,j} = \frac{g_{n,j}^{\phi}}{J L_n^{\phi}}, \qquad \sum_{j=1}^{J} p_{n,j} = 1. \tag{F.105}$$

The numerator of (F.103) can then be written as

$$\sum_{j=1}^{J} w_{n,j}^{P,\theta} g_{n,j}^{\theta} = J L_n^{\phi} \sum_{j=1}^{J} p_{n,j} w_{n,j}^{FC,\theta}. \tag{F.106}$$

Resampling selects $w_{n,j}^{F,\theta} = w_{n,k_j}^{FC,\theta}$. With the pre-resampling particles, weights, and reference densities fixed, the expectation over multinomial resampling satisfies

$$\mathbb{E}_{\text{res}}\left[L_n^\phi \sum_{j=1}^{J} w_{n,j}^{F,\theta}\right] = JL_n^\phi \sum_{m=1}^{J} p_{n,m} w_{n,m}^{FC,\theta}$$
$$= \sum_{m=1}^{J} w_{n,m}^{P,\theta} g_{n,m}^{\theta}. \tag{F.107}$$

The same conditional triangular-array argument as in Lemma F.7 gives consistency of this resampling approximation under its boundedness conditions. Dividing by the unchanged prediction-weight sum gives (F.104). Using Lemma F.12, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^{N} \left(L_n^\phi \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}} \right). \tag{F.108}$$

Since $w_{n,j}^{F,\theta} = w_{n+1,j}^{P,\theta}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^{J} w_{0,j}^{F,\theta} = J$ on the denominator and $\sum_{j=1}^{J} w_{N,j}^{F,\theta}$ on the numerator. This derives the Algorithm 7.1 ($\alpha = 1$) likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 7.1.

F.5. Proof of Theorem 7.4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now show that, under sufficient local regularity conditions, one can interchange differentiation and the particle limit. We retain the Arzelà–Ascoli argument, working on a compact neighborhood of the evaluation parameter. Consistency is used for the estimates themselves: differentiating a weighted particle estimate also differentiates its particles, weights, and normalizer, so boundedness alone does not make its derivative an empirical average to which Theorem F.10 applies.

Theorem F.13 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals)**.** *Let K be a closed ball of positive radius centered at ϕ and contained in the*

interior of $\Theta \subset \mathbb{R}^p$. Let $F_J(\theta)$ be scalar particle estimates, such as $\eta_n^J(h_\theta)$, defined on a common probability space. On one probability-one event, suppose every F_J has two continuous derivatives on a neighborhood of K, with

$$\sup_{J\geq 1}\sup_{\theta\in K}\|\nabla_\theta F_J(\theta)\|_2 \leq G^*, \qquad \sup_{J\geq 1}\sup_{\theta\in K}\|\nabla_\theta^2 F_J(\theta)\|_{\text{op}} \leq H^*$$

for finite constants G^*, H^*. Suppose $F_J(\theta) \to h_\theta^*$ almost surely for each fixed $\theta \in K$, where h^* is deterministic. Then h^* is continuously differentiable in the interior of K, and almost surely

$$\nabla_\theta F_J(\theta) \longrightarrow \nabla_\theta h_\theta^*$$

uniformly on every compact subset of that interior, in particular at $\theta = \phi$.

Proof. For each fixed pair $x, y \in K$, pointwise almost-sure convergence and the gradient bound give $|h_y^* - h_x^*| \leq G^*\|y - x\|_2$. Thus the deterministic function h^* is Lipschitz, and in particular continuous, on K.

Choose a countable dense subset D of K. Intersect the probability-one event of the derivative bounds with the events of convergence at each $\theta \in D$, and fix ω in this intersection. Suppress ω below. By the gradient bound, the functions F_J are uniformly Lipschitz on the convex set K. Convergence on D, compactness of K, and continuity of h^* therefore imply $F_J \to h^*$ uniformly on K: for any $\epsilon > 0$, choose a sufficiently fine finite net in D, use convergence at its finitely many points, and use the common Lipschitz bound and uniform continuity of h^* between those points.

The gradients are uniformly bounded and, by the Hessian bound, uniformly equicontinuous. Explicitly, for $\theta, \theta' \in K$,

$$\|\nabla_\theta F_J(\theta) - \nabla_\theta F_J(\theta')\|_2 \leq H^*\|\theta - \theta'\|_2. \tag{F.109}$$

Thus, by Arzelà–Ascoli, every subsequence of gradients has a further subsequence converging uniformly on K to a continuous function g. We claim that every such limit is $\nabla_\theta h^*$ in the interior of

K. For $x, y \in K$, the fundamental theorem of calculus gives

$$F_J(y) - F_J(x) = \int_0^1 \nabla_\theta F_J\big(x + t(y-x)\big)^\mathsf{T}(y-x)\,dt. \tag{F.110}$$

Passing to the limit along the convergent subsequence yields

$$h_y^* - h_x^* = \int_0^1 g\big(x + t(y-x)\big)^\mathsf{T}(y-x)\,dt. \tag{F.111}$$

Taking $y = x + te_i$ for each coordinate direction and letting $t \to 0$ shows that $\partial_{\theta_i} h_x^* = g_i(x)$ for every interior x. Since g is continuous, h^* is continuously differentiable there. Every subsequential gradient limit is therefore the same on the interior, and hence on K by continuity. Relative compactness and uniqueness of this limit imply convergence of the full gradient sequence. In particular, almost surely, for interior θ,

$$\lim_{J\to\infty} \nabla_\theta F_J(\theta) = \nabla_\theta \lim_{J\to\infty} F_J(\theta) = \nabla_\theta h_\theta^*, \tag{F.112}$$

with convergence uniform on every compact subset of the interior, as required. □

For $F_J(\theta) = \eta_n^J(h_\theta)$, the conclusion is convergence to $\nabla_\theta \eta_n(h_\theta)$. No identity with $\eta_n(\nabla_\theta h_\theta)$ is asserted, since the target distribution can itself depend on θ.

Theorem 7.4 follows by taking $F_J(\theta) = \hat{\ell}_J^1(\theta)$ and $h_\theta^* = \ell(\theta)$. Keep the reference parameter ϕ fixed throughout K. For $\alpha = 1$, the off-parameter consistency result and Lemma F.12 give $\hat{\mathcal{L}}_J^1(\theta) \to \mathcal{L}(\theta) > 0$ almost surely for each fixed $\theta \in K$, not only at $\theta = \phi$. The continuous mapping theorem gives the corresponding convergence of the log likelihoods. The neighborhood form of Assumption ((A4)) supplies their uniform gradient and Hessian bounds, and Assumption ((A5)) supplies their pathwise smoothness with the simulator draws and reference-filter resampling indices held fixed. Applying the theorem to these log likelihoods establishes score consistency at ϕ directly.

F.6. Proof of Theorem 7.5 (Bias–Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of Algorithm 7.2. The earlier truncation argument is retained below for comparison. The Algorithm 7.2 recursion discounts a time-s derivative contribution by its actual age. We prove the Algorithm 7.2 ($\alpha = 0$) result below and retain the former direct extension to $\alpha > 0$ for comparison. Section F.7 obtains the sharper result by first applying the finite-lag variance calculation and then recovering geometric discounting through an exact mixture identity.

First, we relate cloud contraction to the strong-mixing coefficient used in the covariance bounds below.

Lemma F.14. *Let $\mathcal{F}_m = \sigma(x^P_{0,1:J}, \dots, x^P_{m,1:J})$ and $\mathcal{G}_n = \sigma(x^P_{n,1:J}, \dots, x^P_{N,1:J})$ be the past and future sigma-fields of the Markov particle cloud. For $1 \le \ell \le N$, define*

$$a(\ell) = \sup_{0 \le m \le N-\ell} \sup_{A \in \mathcal{F}_m,\, B \in \mathcal{G}_{m+\ell}} \big| \Pr(A \cap B) - \Pr(A) \Pr(B) \big|. \tag{F.113}$$

Then

$$a(\ell) \le \sup_{0 \le m \le N-\ell} \beta_{\mathrm{TV}}(\boldsymbol{M}_{m,m+\ell}). \tag{F.114}$$

The same statement applies to the joint state–derivative cloud under (A6-J), using its transition kernel.

Proof. Fix $A \in \mathcal{F}_m$ and $B \in \mathcal{G}_n$, where $n = m + \ell$. By the Markov property, with $h(x) = \Pr(B \mid x^P_{n,1:J} = x) \in [0, 1]$,

$$\Pr(B \mid \mathcal{F}_m) = (\mathbf{M}_{m,n} h)(x^P_{m,1:J}). \tag{F.115}$$

The expectation of this conditional probability is $\Pr(B)$, so

$$\begin{aligned}\big|\Pr(A\cap B)-\Pr(A)\Pr(B)\big| &= \big|\mathbb{E}\left[\mathbf{1}_A\left\{(\mathbf{M}_{m,n}h)(x^P_{m,1:J})-\Pr(B)\right\}\right]\big| \\ &\le \operatorname{osc}(\mathbf{M}_{m,n}h)\le \beta_{\mathrm{TV}}(\mathbf{M}_{m,n})\operatorname{osc}(h) \\ &\le \beta_{\mathrm{TV}}(\mathbf{M}_{m,n}). \end{aligned} \tag{F.116}$$

Taking the suprema proves the claim. The coefficient is defined through sigma-fields, so the argument does not require joint Lebesgue densities. □

The same argument bounds dependence between past and future windows of simulation and resampling updates. A past window ending after resampling at time b is absorbed into the next prediction cloud at $b+1$; condition there and at the prediction cloud starting the future window, and use the intervening cloud kernel. Fresh update variables are conditionally independent of earlier updates given the incoming cloud. Under (A6), apply this to the physical cloud and the derivative-age windows in Equation (F.7); under (A6-J), use the joint cloud. The number of intervening transitions determines the lag in $a(\ell)$.

As a warm-up, we consider a variance bound for Algorithm 7.2 ($\alpha = 0$). Note that we made Assumptions ((A2)) and ((A3)) to leverage the results from Karjalainen et al. (2023) on the forgetting of the particle filter. The $O\{N(1+\log J)^2/J\}$ variance bound for Algorithm 7.2 ($\alpha = 0$) additionally uses either (A6) or (A6-J), stated in Section F.7. Under (A6-J) the derivative contribution is a bounded current function of the joint state–gradient pair; under (A6) we first expand its simulator history. The former truncation extension to $\alpha > 0$ is retained for comparison. We start by decomposing the Algorithm 7.1 estimator as follows, for the case $\theta = \phi$,

$$\hat{\mathcal{L}}(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N}\left(L_n^{\phi}\cdot\frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}}\right) = \prod_{n=1}^{N}\left(L_n^{\phi}\cdot\frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J}(w_{n-1,j}^{F,\theta})^{\alpha}}\right). \tag{F.117}$$

Now, to illustrate the proof strategy for the general case in an easier context, we first analyze the special case when $\alpha = 0$. We now prove the variance bound for this case, presented below. To our

knowledge, this result for the variance of the gradient estimate of Naesseth et al. (2018) is novel.

Theorem F.15 (Variance of Algorithm 7.1 ($\alpha = 0$) Gradient Estimate)**.** *Suppose the observations are fixed, $J \geq 2$, $\theta = \phi$, Assumptions ((A2)), ((A3)), and ((A5)) hold, and multinomial resampling is used. Suppose also that either (A6) or (A6-J) holds. Then the variance of Algorithm 7.2 ($\alpha = 0$), i.e. the algorithm of Naesseth et al. (2018), is*

$$\mathrm{Var}\big(\nabla_\theta \hat{\ell}^0(\theta)\big) \lesssim \frac{Np\, G'(\theta)^2 (1 + \log J)^2}{J}. \tag{F.118}$$

Proof. First assume (A6-J). In this calculation, a current particle means the pair $(X_n, \partial_\theta X_n)$, and the particle-cloud transition is the one induced by its joint kernel. When $\alpha = 0$, we have

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,0} = \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta,0}}{\sum_{j=1}^J w_{n,j}^{P,\theta,0}} \right) \tag{F.119}$$

$$= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) = \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{g_{n,k_{n,j}}^\theta}{g_{n,k_{n,j}}^\phi} \right). \tag{F.120}$$

Similarly to the proof of Lemma F.2, we define

$$s_{n,j} = \frac{g_{n,k_{n,j}}^\theta}{g_{n,k_{n,j}}^\phi}, \tag{F.121}$$

and from the exact same arguments conclude that its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \tag{F.122}$$

$$= \sum_{n=1}^N \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \tag{F.123}$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} \tag{F.124}$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_\theta f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{F,\theta}; \theta\big)}{f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{F,\phi}; \phi\big)} \tag{F.125}$$

$$= \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J}\nabla_\theta \log\Big(f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\theta};\theta\big)\Big), \tag{F.126}$$

where we use the derivative of the logarithm, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$, and use the derivative of the logarithm where $\theta = \phi$ again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all N and J:

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J}\nabla_\theta \log\Big(f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\theta};\theta\big)\Big). \tag{F.127}$$

Therefore, for a given θ_i in the parameter vector θ,

$$\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0(\theta)\right) = \frac{1}{J^2}\mathrm{Var}\left(\sum_{n=1}^{N}\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\theta};\theta\big)\Big)\right) \tag{F.128}$$

$$\begin{aligned}
&= \frac{1}{J^2}\sum_{n=1}^{N}\mathrm{Var}\left(\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta};\theta)\Big)\right)\\
&\quad + 2\sum_{m<n}\mathrm{Cov}\left(\frac{1}{J}\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_m|X_m}(y_m^*|x_{m,j}^{F,\theta};\theta)\Big), \frac{1}{J}\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta};\theta)\Big)\right)\\
&= \sum_{n=1}^{N}\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right) + 2\sum_{m<n}\mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right).
\end{aligned} \tag{F.129}$$

Here (A6-J) and ((A3)) ensure strong mixing of the joint model. We know from Theorem 3.1 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k-step Markov operator from timestep n and $\beta_{\mathrm{TV}}(M) = \sup_{x,y\in E}\|M(x,\cdot) - M(y,\cdot)\|_{\mathrm{TV}} = \sup_{\mu,\nu\in\mathcal{P},\mu\neq\nu}\frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu-\nu\|_{\mathrm{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\mathrm{TV}}(\mathbf{M}_{n,n+k}) \leq (1-\epsilon)^{\lfloor k/(c\log(J))\rfloor}, \tag{F.130}$$

i.e. the mixing time of this joint particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}_D, \underline{\mathrm{M}}_D, \bar{G}, \underline{\mathrm{G}}$. The physical particle cloud satisfies a bound of the same form under (A2), with its own constants; that version is used under (A6) below.

Under (A6-J), apply Lemma 2.3 of Karjalainen et al. (2023) to the joint prediction cloud. Conditional resampling and the Bayes-ratio calculation give the centered L^4 bound below. Theorem 3.1 and Lemma F.14 bound the strong-mixing coefficient $a(k)$ of the score sequence. The shift by one accounts for the endpoint resampling. At $\alpha = 0$ these results give

$$\left\| \frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) - \mathbb{E}\left[\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) \right] \right\|_{L^4} \leq \frac{CG'(\theta)}{\sqrt{J}},$$
$$a(k) \leq (1-\epsilon)^{\lfloor (k-1)/[c\{1+\log J\}] \rfloor}. \tag{F.131}$$

Davydov's inequality must be applied to the centered variables. Since covariance is unchanged by centering, Equation (F.131) gives

$$\mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) \right) \leq (1-\epsilon)^{\frac{1}{2}\lfloor \frac{|n-m|-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J}. \tag{F.132}$$

The geometric sum is bounded by $C(1+\log J)$. Putting it all together, we see that

$$\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}^0(\theta) \right) = \sum_{n=1}^{N} \mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m<n} \mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) \right) \tag{F.133}$$
$$\leq \frac{NG'(\theta)^2}{J} + 2 \sum_{n=1}^{N-1} (N-n)(1-\epsilon)^{\frac{1}{2}\lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J} \tag{F.134}$$
$$\leq \frac{G'(\theta)^2}{J} \left(N + 2 \sum_{n=1}^{N-1} (N-n)(1-\epsilon)^{\frac{1}{2}\lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \right) \tag{F.135}$$
$$\lesssim \frac{G'(\theta)^2 N(1+\log J)}{J}. \tag{F.136}$$

It then follows that as $\theta \in \Theta \subseteq \mathbb{R}^p$,

$$\mathrm{Var}\big(\nabla_\theta \hat{\ell}^0(\theta)\big) \lesssim \frac{Np\, G'(\theta)^2 (1+\log J)}{J}. \tag{F.137}$$

Under (A6), apply the age decomposition and covariance calculation in Steps 1a–1d of Lemma F.7

with $k = 0$. In particular, Equation (F.7) gives

$$\mathrm{Var}\{\nabla_\theta \hat{\ell}^0(\theta)\} = \mathrm{Var}\{\nabla_\theta \hat{s}_0^1(\theta)\} \lesssim NpG'(\theta)^2(1+\log J)^2/J.$$

That argument uses the physical-cloud mixing bound above and bounded finite-history inputs, not mixing of the full gradient cloud under (A6). □

The factor of N comes from using the unnormalized gradient. Dividing by $\sqrt{N}$ gives a variance bound independent of the horizon. Dividing by N gives a centered L^2 error bound of order $(1+\log J)/\sqrt{NJ}$.

The preceding calculation gives the order $NpG'(\theta)^2(1+\log J)^2/J$ for the Algorithm 7.2 ($\alpha = 0$) baseline. It is not asserted to be a lower bound for every model or every value of α.

Theorem F.16 (MSE and variance bounds for Algorithm 7.2)**.** *Suppose the observations are fixed, $J \geq 2$, $\alpha \in [0,1]$, $\theta = \phi$, Assumptions ((A2)), ((A3)), and ((A5)) hold, and multinomial resampling is used. Suppose also that either (A6) or (A6-J), stated in Section F.7, holds. Then*

$$\max\left\{\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2, \mathrm{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\}\right\} \lesssim pG'(\theta)^2\left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2. \tag{F.138}$$

For $0 \leq \alpha < 1$,

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\
&\quad\lesssim pG'(\theta)^2\Bigg[\left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2 \\
&\qquad + \frac{(1-\sqrt{2(1-\varrho)})^2(1-\alpha)^2}{(1+\sqrt{2(1-\varrho)})^2}\left\{N-1+\sum_{q=1}^{N-2}(N-1-q)\left(\alpha\frac{2\varrho-1}{3-2\varrho}\right)^q\right\}^2\Bigg]
\end{aligned}$$

$$\lesssim pG'(\theta)^2\Bigg[\frac{N(1+\log J)^2}{J(1-\alpha)^2}+\frac{N^2}{J^2(1-\alpha)^2} \tag{F.139}$$
$$+N^2\left\{\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}\right\}^2\Bigg].$$

For the same range of α, the population component of the fixed-observation bias satisfies

$$\|\nabla_\theta \ell^\alpha(\theta)-\nabla_\theta \ell(\theta)\|_2 \le 2(N-1)\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}. \tag{F.140}$$

Here $\varrho = 1-\underline{M}/(2\bar{M}) \in [\frac{1}{2},1)$ is the same constant as in Equation (F.7). The sum beginning at $q=1$ is empty for $N\le 2$; at $\varrho=\frac{1}{2}$ the displayed population terms are zero. The constants suppressed by $\lesssim$ depend only on the fixed model bounds and not on N, J, or α. The constants may additionally depend on (A6) or (A6-J). The endpoints satisfy Equations (F.7) and (F.7); in particular,

$$\mathbb{E}\|\nabla_\theta \hat{\ell}^1(\theta)-\nabla_\theta \ell(\theta)\|_2^2 \lesssim pG'(\theta)^2\min\{N^2, N^3/J\}.$$

The population inequality itself needs neither additional condition. Section F.7 gives the complete proof. The particle argument first bounds each finite lag truncation and then uses the exact geometric-mixture identity; it does not replace Algorithm 7.2 by a hard-cutoff estimator.

Proof. Lemma F.7 proves Equation (F.138). Theorem F.7 proves Equation (F.139), and Lemma F.7 proves Equation (F.140). □

F.7. Sharper Bias–Variance Expansion and Additional Empirical Details

The sharper bias–variance expansion proving Lemma F.7, Lemma F.7, and Theorem F.7 via lag-wise first-derivative forgetting and full-score summation-by-parts identities, together with the supplementary Bayesian inference figures, algorithmic hyperparameters, and the *Daphnia* population model application, appear in Appendices G–J of Tan et al. (2024a) (`arXiv:2407.03085`).

APPENDIX G

PROOFS AND SUPPLEMENTARY DETAILS FOR CHAPTER 8

This appendix contains the complete belief-tree planner pseudocode (Algorithms G.1 and G.2), the practical state-rollout deep RL instantiation (Algorithm G.2), the proofs of the four-way regret decomposition and special-case QMDP structural bias bounds, environment specifications and posterior diagnostics for Double-Integrator LQR and HeartSteps, hyperparameter tables, and the camera-ready validation and ablation studies for Chapter 8.

G.1. Finite-Depth Belief-Tree Planning Subroutine (Algorithm G.1)

We describe the full finite-depth Monte Carlo belief-tree planning subroutine with QMDP leaves (Algorithm G.1) referenced in Algorithm 8.1. Algorithm G.1 estimates the depth-d belief-tree action-value $Q_{\theta,d}(b^J, a)$ defined in (8.9) by Monte Carlo tree search on the particle belief. It is inspired by the POMCP framework (Silver and Veness, 2010), with three departures: internal actions are drawn uniformly at random rather than by UCB1, every simulation descends to the full depth d, and values are computed by a bottom-up dynamic programming (DP) backup after all simulations complete.

Overview. Given the current particle belief $b^J = \frac{1}{J}\sum_{j=1}^{J} \delta_{s_j}$, model θ, search depth d, simulation budget N per root action, and a precomputed fully observable Q-function $Q_\theta^{\star,S}$ (e.g., from value iteration), Algorithm G.1 estimates $Q_{\theta,d}(b^J, a)$ for each root action $a \in \mathcal{A}$ separately (round-robin), so that every root action receives exactly N simulations. For each root action a, each simulation samples a particle s_j from b^J, takes the fixed root action a to obtain a child state $s' \sim P_\theta(\cdot \mid s_j, a)$ and observation $o \sim Z_\theta(\cdot \mid s')$, and then descends the search tree T_a from the child node (a, o) down to depth d, selecting actions uniformly at random at each internal node. Nodes are created the first time a simulation passes through them, and every node on the path records the simulated state. After all N simulations complete, a bottom-up DP backup computes value estimates: at each depth-d leaf, the QMDP value of its accumulated particles; at each internal node, $\hat{V}(v) = \max_{a'} \hat{Q}(v, a')$, where $\hat{Q}(v, a')$ is the empirical mean of $r + \gamma \hat{V}(\text{child})$ over the simulations that selected a' at v.

The root estimate $\widehat{Q}(b^J, a)$ is the same empirical mean at the root.

Like POMCP (Silver and Veness, 2010), Algorithm G.1 indexes its search trees by action-observation histories $h = (a_1, o_1, \ldots, a_k, o_k)$ rather than by explicit beliefs, and represents the belief at each node implicitly by the collection of states simulated through that node. The round-robin over root actions is needed because the regret analysis (Lemma G.6) requires uniformly good estimates $\widehat{Q}(b^J, a)$ for *all* $a \in \mathcal{A}$. The analysis (Proposition G.8) uses two properties of the internal action-selection rule π, i.e., the rule that picks actions at nodes below the root within one planning call:

(S1) *Non-adaptivity.* π may depend on the node's history, on θ, on the root particle set b^J, and on fresh randomness, but not on any statistic accumulated by the current tree.

(S2) *Full support.* $\pi(a' \mid \cdot) \geq \pi_{\min} > 0$ for every $a' \in \mathcal{A}$.

Uniform selection satisfies both with $\pi_{\min} = 1/|\mathcal{A}|$. So does a learned proposal that is frozen during the planning call and mixed with the uniform distribution. UCB1 violates (S1), because its choice at a node reads the visit counts and value estimates accumulated by the current tree. Uniform selection is therefore a proof device, not a performance claim. Together with the absence of an early return (every simulation descends to depth d), (S1) makes the path of each simulation a function of its own random seed alone (Lemma G.9). The simulation seeds are i.i.d. given b^J, but the simulation returns are not, because the leaf evaluator pools particles across simulations and the DP backup couples them. The analysis therefore treats $\widehat{Q}(b^J, a)$ as a bounded-difference function of the i.i.d. seeds rather than as an average of independent returns.

Tree structure and belief particles. Each search tree T_a (one per root action a) has the particle belief b^J at its root (depth 0), the root action fixed to a, internal nodes at depths $1, \ldots, d-1$, and leaves at depth d. Each node v is identified with the action-observation history that leads to it from the root, so a given history always refers to the same node. A child of node v under action a' and observation o is denoted $\textsc{Child}(v, a', o)$; it is created on demand when first visited.

Each node v maintains a *particle set* $B(v)$: the collection of all states that have been simulated through v across all simulations. The empirical distribution $\hat{b}_v := \frac{1}{|B(v)|}\sum_{s\in B(v)}\delta_s$ is the particle approximation of the belief at v, and $N(v) := |B(v)|$ is the number of simulations that have visited v. In addition, each internal node stores:

- $N(v, a')$: the number of simulations that selected action a' at v;
- $\bar{r}(v, a')$: the running mean of the immediate rewards $r(s, a')$ over those simulations.

These statistics are accumulated during the simulation phase and used in the post-simulation DP backup.

QMDP leaf evaluation. At depth-d leaf nodes (remaining depth $\ell = 0$), the QMDP value of the accumulated belief is

$$\bar{V}_\theta^{\star,S}(B(v)) := \max_{a'\in\mathcal{A}} \frac{1}{|B(v)|}\sum_{s\in B(v)} Q_\theta^{\star,S}(s, a'). \tag{G.1}$$

This takes the max *outside* the average over states, choosing a single action for all particles—exactly the QMDP approximation $\bar{V}_\theta^{\star,S}$ from (8.8). This is distinct from the per-state evaluation $\frac{1}{|B(v)|}\sum_s \max_{a'} Q_\theta^{\star,S}(s, a')$, which would compute the linear surrogate $\widetilde{V}_\theta^\star$ instead. Using (G.1) ensures the algorithm matches the theoretical target $\bar{V}_{\theta,d}^{\star,S} = \mathcal{T}_\theta^d \bar{V}_\theta^{\star,S}$ analyzed in Section 8.4.

Simulation procedure. For each root action a, each of the N simulations proceeds as follows. A particle s_j is sampled uniformly from b^J, and the root transition is simulated: $s' \sim P_\theta(\cdot \mid s_j, a)$, $o \sim Z_\theta(\cdot \mid s')$, with immediate reward $r(s_j, a)$. Starting from the child node $\textsc{Child}(\text{root}, a, o)$ with remaining depth $\ell = d-1$, the $\textsc{Simulate}$ subroutine (Algorithm G.2) descends the tree recursively:

1. *Record.* Add the current state s to $B(v)$.
2. *QMDP leaf ($\ell = 0$).* Stop. The leaf is evaluated by (G.1) on its final particle set once all simulations are complete.
3. *Expansion (new internal node).* If v is not yet expanded, initialize $N(v, a') \leftarrow 0$ and $\bar{r}(v, a') \leftarrow$

0 for all $a' \in \mathcal{A}$, and continue. There is no early return: a simulation that creates a node keeps descending.

4. *Uniform selection.* Select
$$a' \sim \mathrm{Unif}(\mathcal{A}). \tag{G.2}$$

5. *Simulate and recurse.* Draw $s'' \sim P_\theta(\cdot \mid s, a')$ and $o' \sim Z_\theta(\cdot \mid s'')$, update $N(v, a')$ and $\bar{r}(v, a')$ with the reward $r(s, a')$, and recurse into $\textsc{Child}(v, a', o')$ with remaining depth $\ell - 1$.

Every simulation therefore visits exactly one node at each depth $1, \ldots, d$. After all N simulations for root action a, the DP backup (Phase 2 of Algorithm G.1) computes, from the leaves upward,

$$\begin{aligned} \hat{V}(v) &= \bar{V}_\theta^{\star,S}(B(v)) && \text{(leaf } v\text{)}, \\ \hat{Q}(v, a') &= \bar{r}(v, a') + \gamma \sum_{w \in \mathrm{Ch}(v,a')} \frac{N(w)}{N(v, a')} \hat{V}(w) && \text{(internal } v,\ N(v, a') > 0\text{)}, \\ \hat{V}(v) &= \max_{a' :\, N(v,a') > 0} \hat{Q}(v, a') && \text{(internal } v\text{)}, \end{aligned} \tag{G.3}$$

where $\mathrm{Ch}(v, a')$ is the set of children of v under action a'. Equivalently, $\hat{Q}(v, a')$ is the average of $r(s, a') + \gamma \hat{V}(\text{child})$ over the simulations that selected a' at v. The root estimate is

$$\widehat{Q}(b^J, a) = \frac{1}{N} \sum_{n=1}^{N} \big[r(s_{j_n}, a) + \gamma\, \hat{V}(v_{n,1}) \big],$$

where s_{j_n} is the particle sampled by simulation n and $v_{n,1}$ is the depth-1 node it enters. When $d = 0$ no tree is built and $\widehat{Q}(b^J, a) = \frac{1}{J} \sum_{j=1}^{J} Q_\theta^{\star,S}(s_j, a)$.

The full procedure is given in Algorithms G.1 and G.2.

Convergence. As $N \to \infty$, two things happen. First, by Lemma G.9 the events "simulation n reaches v and selects a'" are i.i.d. across n with positive probability for every reachable node v and every a', so every such node and action is visited infinitely often. Second, the states entering a node v are i.i.d. draws from the (binned) conditional belief b_v at v, so $\hat{b}_v \to b_v$ by the law of large

Algorithm G.1 Finite-Depth Monte Carlo Belief-Tree Planning with QMDP Leaf Bootstrapping

Require: model θ; particle belief $b^J = \frac{1}{J}\sum_{j=1}^{J} \delta_{s_j}$; depth d; simulations per action N; MDP Q-function $Q_\theta^{\star,S}$

if $d = 0$ **then** ▷ No tree: exact QMDP
  return $\left\{\frac{1}{J}\sum_{j=1}^{J} Q_\theta^{\star,S}(s_j, a)\right\}_{a\in\mathcal{A}}$
end if
for each action $a \in \mathcal{A}$ **do** ▷ Round-robin: N simulations per root action
  Initialize empty search tree T_a
  for $n = 1, \ldots, N$ **do** ▷ Phase 1: tree construction
    Sample $j_n \sim \mathrm{Unif}([J])$
    Sample $s' \sim P_\theta(\cdot \mid s_{j_n}, a)$, $o \sim Z_\theta(\cdot \mid s')$
    $v_{n,1} \leftarrow \textsc{Child}(\mathrm{root}(T_a), a, o)$ ▷ Child of root for (a, o)
    $\textsc{Simulate}(s',\, d-1,\, \theta,\, v_{n,1})$ ▷ Alg. G.2
  end for
  ▷ Phase 2: DP backup (G.3)
  $\hat{V}(v) \leftarrow \max_{a'\in\mathcal{A}} \frac{1}{|B(v)|}\sum_{s\in B(v)} Q_\theta^{\star,S}(s, a')$ at each leaf v of T_a ▷ Pooled QMDP leaf
  for each internal node v of T_a, from depth $d-1$ up to depth 1 **do**
    $\hat{Q}(v, a') \leftarrow \bar{r}(v, a') + \gamma \sum_{w\in \mathrm{Ch}(v,a')} \frac{N(w)}{N(v,a')} \hat{V}(w)$ for each a' with $N(v, a') > 0$
    $\hat{V}(v) \leftarrow \max_{a':\, N(v,a')>0} \hat{Q}(v, a')$
  end for
  $\widehat{Q}(b^J, a) \leftarrow \frac{1}{N}\sum_{n=1}^{N} \left[r(s_{j_n}, a) + \gamma\, \hat{V}(v_{n,1})\right]$
end for
return $\{\widehat{Q}(b^J, a)\}_{a\in\mathcal{A}}$

numbers. At each depth-d leaf, $\bar{V}_\theta^{\star,S}(B(v)) \to \bar{V}_\theta^{\star,S}(b_v)$, and the DP backup then gives, level by level, $\hat{Q}(v, a') \to Q_{\theta,\ell}^{\phi}(b_v, a')$ and $\hat{V}(v) \to \max_{a'} Q_{\theta,\ell}^{\phi}(b_v, a') = \bar{V}_{\theta,\ell}^{\star,S}(b_v)$, where ℓ is the remaining depth at v. Since the root action is fixed and states are sampled from b^J, the backed-up estimate $\widehat{Q}(b^J, a) \to Q_{\theta,d}^{\phi}(b^J, a)$ for each $a \in \mathcal{A}$. Proposition G.8 quantifies the rate.

When $d = 0$, no tree search is performed and the planner returns $\max_a \frac{1}{J}\sum_j Q_\theta^{\star,S}(s_j, a)$—exactly the QMDP value $\bar{V}_\theta^{\star,S}(b^J)$. When $d \to \infty$, the tree-search value converges to $Q_\theta^{\star,B}(b^J, a)$.

Computational cost. For each root action, each of the N simulations traverses a path of exactly d nodes in the tree, requiring $O(d)$ generative model calls and $O(1)$ work for uniform action sampling at each internal node. The post-simulation DP backup adds $O(|\mathcal{A}|)$ work per node but is performed once, not per simulation. The QMDP leaf evaluation (G.1) can be maintained incrementally by storing running sums $\sum_{s\in B(v)} Q_\theta^{\star,S}(s, a')$ for each action a', reducing the per-simulation leaf cost to $O(|\mathcal{A}|)$. The total cost per planning step is $O(|\mathcal{A}|^2 N d)$ model evaluations across all root actions.

Algorithm G.2 SIMULATE: Recursive uniform-exploration descent to full depth

Require: state s; remaining depth ℓ; model θ; tree node v

$B(v) \leftarrow B(v) \cup \{s\}$ ▷ Record state; $N(v) = |B(v)|$

if $\ell = 0$ **then** ▷ Leaf: evaluated in Phase 2 of Alg. G.1

  return

end if

if v is not yet expanded **then** ▷ Expansion: initialize and continue (no early return)

  Expand v: $N(v, a') \leftarrow 0$, $\bar{r}(v, a') \leftarrow 0$ for all $a' \in \mathcal{A}$

end if

Sample $a' \sim \mathrm{Unif}(\mathcal{A})$ ▷ Uniform, reads no tree statistic

Sample $s' \sim P_\theta(\cdot \mid s, a')$, $o \sim Z_\theta(\cdot \mid s')$

$N(v, a') \leftarrow N(v, a') + 1$; $\bar{r}(v, a') \leftarrow \bar{r}(v, a') + \frac{r(s,a') - \bar{r}(v,a')}{N(v,a')}$ ▷ Incremental mean

SIMULATE(s', $\ell - 1$, θ, CHILD(v, a', o))

Relation to the error analysis. The Monte Carlo error $\epsilon_{\mathrm{mc}}(t, d) = \sup_a |\widehat{Q}_t(b_t^J, a) - Q_{\theta,d}(b_t^J, a)|$ arises from the finite simulation budget N per root action. The simulation seeds are i.i.d. given b_t^J, and under (S1) with no early return each simulation's path is a function of its own seed (Lemma G.9). The estimate $\widehat{Q}_t(b_t^J, a)$ is then a bounded-difference function of the seeds, so the Efron–Stein and McDiarmid inequalities control its fluctuation at rate $O(\Lambda_d/\sqrt{N})$, while the DP-backup bias vanishes at rate $O(\Phi_d/\sqrt{N})$ (Proposition G.8). The simulation returns themselves are not independent, and the argument does not require them to be. The proof is written for uniform selection; for a rule satisfying (S1)–(S2), it goes through with $|\mathcal{A}|$ replaced by $1/\pi_{\min}$ wherever it bounds the fraction of a node's visits that select a given action.

G.2. Practical Instantiation and Benchmark Specifications

Extended related work on belief-space planning and particle filtering, the practical deep reinforcement learning instantiation (Algorithm G.2), and full environment specifications, posterior diagnostics, and hyperparameter tables for the Double-Integrator LQR, HeartSteps, and MuJoCo benchmarks appear in Appendices A and C of Tan and Xu (2026).

G.3. Detailed Error Analysis and Proofs

Throughout, $|r(s, a)| \le R_{\max}$ and $V_{\max} := R_{\max}/(1 - \gamma)$. We define the per-round error quantities:

$$\epsilon_{\mathrm{mc}}(t, d) := \sup_{a \in \mathcal{A}} \left| \widehat{Q}_t(b_t^J, a) - Q_{\theta,d}(b_t^J, a) \right|, \qquad \text{(Monte Carlo error)}$$

$$\epsilon_{\text{struct}}(t,d) := \sup_{a\in\mathcal{A}} \big| Q_\theta^{\star,B}(b_t^J, a) - Q_{\theta,d}(b_t^J, a) \big|. \qquad \text{(structural error)}$$

G.3.1. Structural Error of Leaf Value Bootstrapping

Define the *linear QMDP surrogate*

$$\widetilde{V}_\theta^\star(b) := \int_{\mathcal{S}} V_\theta^{\star,S}(s)\, b(\mathrm{d}s). \tag{G.4}$$

By Jensen's inequality, $\widetilde{V}_\theta^\star(b) \geq \bar{V}_\theta^{\star,S}(b)$ and the gap $\Xi_\theta := \sup_b[\widetilde{V}_\theta^\star(b) - \bar{V}_\theta^{\star,S}(b)] \geq 0$.

Proposition G.1 (Structural Error of the QMDP Leaf Value Approximation)**.** *For every $\theta \in \Theta$ and $d \geq 0$,*

$$\|\bar{V}_{\theta,d}^{\star,S} - V_\theta^{\star,B}\|_\infty \;\leq\; \frac{\gamma^{d+1}}{1-\gamma}\,\Xi_\theta. \tag{G.5}$$

In particular, $\epsilon_{\text{struct}}(u,d) \leq \frac{\gamma^{d+1}}{1-\gamma}\Xi_{\bar{\theta}_u}$.

Proof of Proposition G.1. Write

$$\begin{aligned} Q_\theta^{\star,B}(b,a) &= r(b,a) + \gamma \int_{\mathcal{O}} V_\theta^{\star,B}(\tau_\theta(b,a,o))\, p_\theta(o \mid b,a)\, \mathrm{d}o, \\ Q_{\theta,0}(b,a) &= r(b,a) + \gamma \int_{\mathcal{S}} V_\theta^{\star,S}(s')\, \tilde{b}_\theta(\mathrm{d}s' \mid b,a) = r(b,a) + \gamma\, \widetilde{V}_\theta^\star(\tilde{b}_\theta(\cdot \mid b,a)). \end{aligned}$$

By Bayes consistency and linearity of $\widetilde{V}_\theta^\star$:

$$\widetilde{V}_\theta^\star(\tilde{b}_\theta(\cdot \mid b,a)) = \int_{\mathcal{O}} \widetilde{V}_\theta^\star(\tau_\theta(b,a,o))\, p_\theta(o \mid b,a)\, \mathrm{d}o.$$

Subtracting and using $\widetilde{V}_\theta^\star = \bar{V}_\theta^{\star,S} + (\widetilde{V}_\theta^\star - \bar{V}_\theta^{\star,S})$ with $0 \leq \widetilde{V}_\theta^\star - \bar{V}_\theta^{\star,S} \leq \Xi_\theta$:

$$\|Q_\theta^{\star,B} - Q_{\theta,0}\|_\infty \;\leq\; \gamma \|V_\theta^{\star,B} - \bar{V}_\theta^{\star,S}\|_\infty + \gamma\, \Xi_\theta.$$

Since $\|V_\theta^{\star,B} - \bar{V}_\theta^{\star,S}\|_\infty = \sup_b |\max_a Q_\theta^{\star,B}(b,a) - \max_a Q_{\theta,0}(b,a)| \leq \|Q_\theta^{\star,B} - Q_{\theta,0}\|_\infty$ (using $|\max f -$

$\max g| \le \max|f-g|$), rearranging gives $\|V_\theta^{\star,B} - \bar{V}_\theta^{\star,S}\|_\infty \le \frac{\gamma}{1-\gamma}\Xi_\theta$ (the $d=0$ case). For general $d \ge 0$: $\bar{V}_{\theta,d}^{\star,S} = \mathcal{T}_\theta^d \bar{V}_\theta^{\star,S}$ and $V_\theta^{\star,B} = \mathcal{T}_\theta V_\theta^{\star,B}$ (fixed point), so

$$\|\bar{V}_{\theta,d}^{\star,S} - V_\theta^{\star,B}\|_\infty = \|\mathcal{T}_\theta^d \bar{V}_\theta^{\star,S} - \mathcal{T}_\theta^d V_\theta^{\star,B}\|_\infty \le \gamma^d \|\bar{V}_\theta^{\star,S} - V_\theta^{\star,B}\|_\infty \le \frac{\gamma^{d+1}}{1-\gamma}\Xi_\theta. \qquad \square$$

G.3.2. Planning and State-Identification Error

Definition G.2 (Q-belief distance)**.** For $b, b' \in \Delta(\mathcal{S})$, define $d_Q(b,b') := \sup_{\theta\in\Theta,\, a\in\mathcal{A}} |Q_\theta^{\star,B}(b,a) - Q_\theta^{\star,B}(b',a)|$.

Lemma G.3 ($Q^{\star,B}$-difference via test functions)**.** *Fix $\theta \in \Theta$, $a \in \mathcal{A}$, and truncation depth $H \ge 1$. Recall the Bayesian update $\tau(b,a,o) \in \Delta(\mathcal{S})$, the observation marginal $p(o \mid b,a) = \int Z_\theta(o \mid s') P_\theta(s' \mid s,a)\, b(ds)\, ds'$, and define*

$$\psi_{o,a}(s) := \int Z_\theta(o \mid s') P_\theta(s' \mid s,a)\, ds', \qquad \tilde{f}_{o,a}(s) := \int f(s') Z_\theta(o \mid s') P_\theta(s' \mid s,a)\, ds', \tag{G.6}$$

so that $p(o \mid b,a) = b(\psi_{o,a})$ and $\tau(b,a,o)(f) = b(\tilde{f}_{o,a})/b(\psi_{o,a})$.

Under Assumption 8.1, for all $b, b' \in \Delta(\mathcal{S})$:

$$|Q_\theta^{\star,B}(b,a) - Q_\theta^{\star,B}(b',a)| \;\le\; \sum_{\ell=0}^{H-1} \gamma^\ell\, C_{\mathrm{BU}}^\ell \sup_{f\in\mathcal{F}_\ell(\theta,a)} |b(f) - b'(f)| \;+\; 2\gamma^H V_{\max}, \tag{G.7}$$

where $C_{\mathrm{BU}} := 2g_{\max}/g_{\min}^2$ and $\mathcal{F}_\ell(\theta,a)$ is a class of functions $\mathcal{S} \to [-g_{\max}^\ell V_{\max},\, g_{\max}^\ell V_{\max}]$ with $\mathcal{N}_{[]}(\mathcal{F}_\ell, \epsilon) \le |\mathcal{A}|^\ell \mathcal{N}_\Theta(\epsilon/(\ell V_{\max}))$, defined recursively:

- $\mathcal{F}_0(\theta,a) = \{r(\cdot,a') : a' \in \mathcal{A}\} \cup \{Q_\theta^{\star,S}(\cdot,a') : a' \in \mathcal{A}\}$,
- $\mathcal{F}_{\ell+1}(\theta,a) = \big\{\tilde{f}_{o,a''} :\, f \in \mathcal{F}_\ell(\theta,a''),\, a'' \in \mathcal{A}\big\} \cup \big\{\psi_{o,a''} :\, a'' \in \mathcal{A}\big\}$.

Proof. We prove by induction on h: for any $\Phi(b) = \mathbb{E}_\theta^\pi[\sum_{u=0}^{h-1} \gamma^u r_u \mid b_0 = b]$ with $h \le H$, $|\Phi(b) - \Phi(b')|$ admits the decomposition (G.7). Taking $\Phi = Q_\theta^{\star,B}(\cdot,a)$ with $h = H$ and bounding the tail by $2\gamma^H V_{\max}$ yields the result.

Base case ($h = 1$). $|\Phi(b) - \Phi(b')| = |\sup_a[b(r_a) - b'(r_a)]| \le \sup_{f\in\mathcal{F}_0} |b(f) - b'(f)|$.

Inductive step ($h \to h+1$). For a fixed action a, write

$$\Phi(b) = b(r_a) + \gamma \int_{\mathcal{O}} \Phi_{h-1}\big(\tau(b,a,o)\big)\, p(o \mid b,a)\, do,$$

where Φ_{h-1} is the $(h-1)$-step continuation. The continuation difference decomposes as:

$$\begin{aligned}
&\left| \int \Phi_{h-1}(\tau(b,a,o))\, p(o|b,a)\, do - \int \Phi_{h-1}(\tau(b',a,o))\, p(o|b',a)\, do \right| \\
&\le \underbrace{\|\Phi_{h-1}\|_\infty \int |(b-b')(\psi_{o,a})|\, do}_{\text{(I) obs. distribution shift}} + \underbrace{\int |\Phi_{h-1}(\tau(b,a,o)) - \Phi_{h-1}(\tau(b',a,o))|\, p(o|b',a)\, do}_{\text{(II) belief propagation}},
\end{aligned} \tag{G.8}$$

using $p(o \mid b,a) - p(o \mid b',a) = (b-b')(\psi_{o,a})$.

Term (I) contributes $\psi_{o,a} \in \mathcal{F}_{\ell+1}$.

Term (II) requires bounding $|\tau(b,a,o)(f) - \tau(b',a,o)(f)|$ for $\|f\|_\infty \le 1$. Since $\tau(b,a,o)(f) = b(\tilde{f}_{o,a})/b(\psi_{o,a})$, the quotient rule gives

$$\frac{b(\tilde{f}_{o,a})}{b(\psi_{o,a})} - \frac{b'(\tilde{f}_{o,a})}{b'(\psi_{o,a})} \;=\; \frac{(b-b')(\tilde{f}_{o,a})}{b(\psi_{o,a})} - \frac{b'(\tilde{f}_{o,a})}{b(\psi_{o,a})\, b'(\psi_{o,a})}\, (b-b')(\psi_{o,a}). \tag{G.9}$$

Since $b(\psi_{o,a}), b'(\psi_{o,a}) \ge g_{\min}$ and $|b'(\tilde{f}_{o,a})| \le g_{\max}$:

$$|\tau(b,a,o)(f) - \tau(b',a,o)(f)| \;\le\; \frac{|(b-b')(\tilde{f}_{o,a})|}{g_{\min}} + \frac{g_{\max}}{g_{\min}^2}\, |(b-b')(\psi_{o,a})|. \tag{G.10}$$

Both $\tilde{f}_{o,a}$ and $\psi_{o,a}$ are in $\mathcal{F}_{\ell+1}$, with per-update amplification

$$C_{\mathrm{BU}} = \frac{1}{g_{\min}} + \frac{g_{\max}}{g_{\min}^2} \;\le\; \frac{2\, g_{\max}}{g_{\min}^2},$$

By the inductive hypothesis, $|\Phi_{h-1}(\tau(b,a,o)) - \Phi_{h-1}(\tau(b',a,o))|$ decomposes into evaluations of $\tau(b,a,o)$ vs. $\tau(b',a,o)$ on $\mathcal{F}_\ell$. Substituting (G.10) reduces each to evaluations of b vs. b' on $\mathcal{F}_{\ell+1}$, with one factor of C_{BU} per level. Taking $\sup_a$ over the action set and iterating across H levels

yields (G.7). □

Lemma G.4 ($Q^{\star,B}$ is Lipschitz in TV)**.** *For any $\theta \in \Theta$, $a \in \mathcal{A}$, and $b, b' \in \Delta(\mathcal{S})$,*

$$|Q_\theta^{\star,B}(b,a) - Q_\theta^{\star,B}(b',a)| \;\le\; 2V_{\max}\,\|b - b'\|_{\mathrm{TV}}.$$

Proof. $|r(b,a) - r(b',a)| \le 2R_{\max}\|b - b'\|_{\mathrm{TV}}$. The trajectory law under b and b' differs in TV by at most $\|b - b'\|_{\mathrm{TV}}$ (data processing), so the $V_{\max}$-bounded continuation differs by at most $2V_{\max}\|b - b'\|_{\mathrm{TV}}$. Using $R_{\max} + \gamma V_{\max} = V_{\max}$ gives the result. □

Lemma G.5 (Performance decomposition)**.** *Fix $\theta \in \Theta$. Let π_t denote the stationary policy induced by the planner with frozen parameters (model $\bar{\theta}_{k(t)}$, belief b_t^J) at time t, and let $\delta_u := V_\theta^{\star,B}(b_u) - Q_\theta^{\star,B}(b_u, A_u)$ where $A_u \sim \pi_t(\cdot \mid H_u)$. Then*

$$V_\theta^{\star,B}(b_t) - V_\theta^{\pi_t,B}(b_t) \;\le\; \mathbb{E}\left[\sum_{u=t}^{\infty} \gamma^{u-t}\delta_u \;\middle|\; H_t\right].$$

Proof. Unroll the Bellman recursion $V_\theta^{\star,B}(b_u) - V_\theta^{\pi_t,B}(b_u) = \delta_u + \gamma\,\mathbb{E}[V_\theta^{\star,B}(b_{u+1}) - V_\theta^{\pi_t,B}(b_{u+1}) \mid H_u]$ from $u = t$ to $u = t + n$, using $|V_\theta^{\star,B} - V_\theta^{\pi_t,B}| \le 2V_{\max}$ as the tail bound, and send $n \to \infty$. □

Lemma G.6 (Per-step bound on δ_u)**.** *For every u and any $d \ge 0$,*

$$\delta_u \;\le\; 2\,d_Q(b_u, b_u^J) + 2\,\epsilon_{\mathrm{mc}}(u,d) + 2\,\epsilon_{\mathrm{struct}}(u,d).$$

Proof. Let $a_u^\star \in \arg\max_a Q_\theta^{\star,B}(b_u, a)$ and $\hat{a}_u \in \arg\max_a \widehat{Q}_u(b_u^J, a)$. Then

$$\begin{aligned}
\delta_u &= Q_\theta^{\star,B}(b_u, a_u^\star) - Q_\theta^{\star,B}(b_u, \hat{a}_u) \\
&\le \underbrace{|Q_\theta^{\star,B}(b_u, a_u^\star) - Q_\theta^{\star,B}(b_u^J, a_u^\star)| + |Q_\theta^{\star,B}(b_u^J, \hat{a}_u) - Q_\theta^{\star,B}(b_u, \hat{a}_u)|}_{\le 2\,d_Q(b_u, b_u^J) \text{ by Definition G.2}} \\
&\quad + Q_\theta^{\star,B}(b_u^J, a_u^\star) - Q_\theta^{\star,B}(b_u^J, \hat{a}_u).
\end{aligned}$$

For the last line, add and subtract $Q_{\theta,d}$ and $\widehat{Q}_u$, using optimality of $\hat{a}_u$ for $\widehat{Q}_u$:

$$Q_{\theta,d}(b_u^J, a_u^\star) \le \widehat{Q}_u(b_u^J, a_u^\star) + \epsilon_{\text{mc}} \le \widehat{Q}_u(b_u^J, \hat{a}_u) + \epsilon_{\text{mc}} \le Q_{\theta,d}(b_u^J, \hat{a}_u) + 2\epsilon_{\text{mc}},$$

so $Q_\theta^{\star,B}(b_u^J, a_u^\star) - Q_\theta^{\star,B}(b_u^J, \hat{a}_u) \le 2\epsilon_{\text{struct}} + 2\epsilon_{\text{mc}}$. □

Theorem G.7 (Round-t planning + state-identification bound)**.** *For each $t \ge 1$ and any depth d,*

$$V_{\bar{\theta}_{k(t)}}^{\star,B}(b_t) - V_{\bar{\theta}_{k(t)}}^{\pi_t,B}(b_t) \;\le\; \mathbb{E}\left[\sum_{u=t}^{\infty} \gamma^{u-t}\big(2\, d_Q(b_u, b_u^J) + 2\,\epsilon_{\text{mc}}(u,d) + 2\,\epsilon_{\text{struct}}(u,d)\big) \;\middle|\; H_t\right].$$

Proof. Substitute the per-step bound of Lemma G.6 into Lemma G.5. □

Observation discretization. For continuous $\mathcal{O}$, the tree never revisits observation-children (each sampled o is unique), so every node below the root holds a single particle. At $d = 1$ each leaf value (G.1) then reduces to $\max_{a'} Q_\theta^{\star,S}(s', a')$ at a single state, which targets the linear surrogate rather than $Q_{\theta,1}$, and for $d \ge 2$ the tree cannot build depth beyond 1. We discretize: let $\phi : \mathcal{O} \to \bar{\mathcal{O}}$ with $|\bar{\mathcal{O}}| = M$ and bin diameter $\operatorname{diam}(\phi) := \max_{\bar{o}} \sup_{o,o' \in \phi^{-1}(\bar{o})} \|o - o'\|$. Algorithms G.1 and G.2 are modified so that $\textsc{Child}(v, a', \phi(o'))$ replaces $\textsc{Child}(v, a', o')$. The bin-averaged posterior is $\tau_\theta^\phi(b, a, \bar{o}) := \int_{\phi^{-1}(\bar{o})} \tau_\theta(b, a, o)\, p_\theta(o \mid b, a)\, \mathrm{d}o \big/ \int_{\phi^{-1}(\bar{o})} p_\theta(o \mid b, a)\, \mathrm{d}o$, and we write $Q_{\theta,d}^\phi$ for the target (8.9) with $\tau_\theta(b, a, o)$ replaced by $\tau_\theta^\phi(b, a, \phi(o))$ at every level. When $|\mathcal{O}| < \infty$, take $\phi = \mathrm{id}$, $M = |\mathcal{O}|$, $\operatorname{diam}(\phi) = 0$. When $d = 0$, no tree is built and no discretization is needed.

Seeds and the estimator. Fix a round t and a root action a, and write $\theta := \bar{\theta}_{k(t)}$ and $b^J := b_t^J$. All probabilities and expectations below are conditional on b_t^J, and we take $d \ge 1$. Simulation $n \in [N]$ is driven by a *seed*

$$\xi_n := \big(j_n,\, u_{n,0},\, (a'_{n,\ell}, u_{n,\ell})_{\ell=1}^{d-1}\big), \tag{G.11}$$

where $j_n \sim \mathrm{Unif}([J])$ is the sampled particle index, $a'_{n,\ell} \sim \mathrm{Unif}(\mathcal{A})$ is the action drawn at depth ℓ, and $u_{n,\ell}$ is the noise that generates the next state and observation at depth ℓ from P_θ and Z_θ. The seeds $\xi_1, \ldots, \xi_N$ are i.i.d. Running Algorithm G.1 on them produces the tree T_a, and we write the

root estimate as a deterministic function of all seeds,

$$\widehat{Q}(b^J, a) = f_a(\xi_1, \dots, \xi_N) := \frac{1}{N} \sum_{n=1}^{N} \big[r(s_{j_n}, a) + \gamma \, \hat{V}(v_{n,1}) \big], \tag{G.12}$$

where $v_{n,\ell}$ is the depth-ℓ node on the path of simulation n and $\hat{V}$ is the DP backup (G.3). The function f_a is not an average of independent terms, since $\hat{V}(v_{n,1})$ depends on every simulation that passes through $v_{n,1}$. For $S \subseteq [N]$, let $f_a^{(-S)}$ be the same expression computed from the seeds $(\xi_m)_{m \notin S}$ alone, with the sum running over $n \notin S$ and the normalizer $1/N$ retained.

Balancedness. For $\rho \geq 1$, let $\mathcal{E}_\rho$ be the event that every internal node v of T_a satisfies

$$N(v) \leq \rho \, |\mathcal{A}| \, N(v, a') \qquad \text{for all } a' \in \mathcal{A}, \tag{G.13}$$

so that every action receives at least a $1/(\rho|\mathcal{A}|)$ fraction of the visits of every internal node. Define the depth factor

$$\Lambda_d := \sum_{\ell=0}^{d-1} (\rho\gamma|\mathcal{A}|)^\ell, \tag{G.14}$$

so that $\Lambda_0 = 0$ and $\Lambda_1 = 1$. Let $p_\rho := \max_{a \in \mathcal{A}} \Pr(\mathcal{E}_\rho^c \mid b_t^J)$, where the event refers to the tree T_a, and let

$$q_{\min} := \inf_{s \in \mathcal{S},\, a'' \in \mathcal{A},\, \bar{o} \in \bar{\mathcal{O}}} \int_{\mathcal{S}} Z_\theta\big(\phi^{-1}(\bar{o}) \mid s'\big) \, P_\theta(\mathrm{d}s' \mid s, a'')$$

be the smallest one-step bin probability. Under the bounded observation density of Assumption 8.1, $q_{\min} \geq g_{\min} \min_{\bar{o}} \lambda(\phi^{-1}(\bar{o}))$, where λ is the reference measure of Z_θ; for finite $\mathcal{O}$ with the counting measure, $q_{\min} \geq g_{\min}$.

Proposition G.8 (Monte Carlo error)**.** *Run Algorithm G.1 with uniform internal selection* (G.2)*, N simulations per root action, observation discretization ϕ ($|\bar{\mathcal{O}}| = M$), and returns in $[-V_{\max}, V_{\max}]$, and let $\rho \geq 1$. Let*

$$\mathrm{bias}(N, d) := \sup_a \big| \mathbb{E}[\widehat{Q}(b^J, a) \mid b_t^J] - Q_{\bar{\theta}_{k(t)}, d}(b^J, a) \big| \tag{G.15}$$

and $B_d := 2V_{\max}\big(1 + 2(1+\gamma)\Lambda_d\big)$. For every $\delta \in (0,1)$, with probability at least $1 - \delta - |\mathcal{A}| p_\rho$

conditionally on b_t^J,

$$\epsilon_{\mathrm{mc}}(t,d) \;\le\; \underbrace{2(1+\gamma)V_{\max}\Lambda_d\sqrt{\frac{2\log(2|\mathcal{A}|/\delta)}{N}}}_{\textit{fluctuation}} + \mathrm{bias}(N,d) \;+\; B_d\, p_\rho, \tag{G.16}$$

and, in second moment,

$$\mathbb{E}\big[\epsilon_{\mathrm{mc}}(t,d)^2 \,\big|\, b_t^J\big] \;\le\; \frac{8(1+\gamma)^2\,|\mathcal{A}|\,V_{\max}^2\,\Lambda_d^2}{N} \;+\; 2\,|\mathcal{A}|\,\mathrm{bias}(N,d)^2 \;+\; 3\,|\mathcal{A}|\,B_d^2\,p_\rho. \tag{G.17}$$

The bias satisfies

$$\mathrm{bias}(N,d) \;\le\; \underbrace{\frac{c_0\,\Phi_d\,V_{\max}\sqrt{|\mathcal{A}|}}{\sqrt{N}}}_{\textit{DP backup estimation error}} + \underbrace{\frac{2\gamma(1-\gamma^d)}{1-\gamma}\cdot\frac{g_{\max}}{g_{\min}}\,V_{\max}\,\omega_Z(\mathrm{diam}(\phi))}_{\textit{discretization error}}, \tag{G.18}$$

where $c_0 > 0$ *is a constant independent of* N, M *and* δ,

$$\Phi_d := \sum_{\ell=1}^{d}(\gamma\sqrt{|\mathcal{A}|M})^\ell, \tag{G.19}$$

and $\omega_Z(\delta) := \sup_{\theta,s}\sup_{\|o-o'\|\le\delta}|Z_\theta(o\mid s)/Z_\theta(o'\mid s)-1|$. *If* $d \le 1$, *then* $p_\rho = 0$. *If* $d \ge 2$, $q_{\min} > 0$ *and* $|\mathcal{A}|M \ge 2$, *then* $p_2 \le 2|\mathcal{A}|(|\mathcal{A}|M)^d\exp\big(-N(q_{\min}/|\mathcal{A}|)^{d-1}/32\big)$. *When* $d = 0$, $\widehat{Q}(b^J,a) = \frac{1}{J}\sum_j Q_\theta^{\star,S}(s_j,a)$ *is exact and every term vanishes (*$\Lambda_0 = \Phi_0 = 0$*). When* $\phi = \mathrm{id}$ *(*$M = |\mathcal{O}|$, $\mathrm{diam}(\phi) = 0$*), the discretization error vanishes.*

No stationarity assumption on the rewards at internal nodes is required, and the simulation returns need not be independent. The proof uses only that the seeds are i.i.d., that the path of each simulation depends on its own seed alone, and that the estimator has bounded differences on $\mathcal{E}_\rho$. It rests on the following four results; throughout, we fix a root action a and $d \ge 1$.

Lemma G.9 (Path decoupling)**.** *There is a deterministic map* ν *such that, whatever the other seeds are, the path of simulation* m *(its states, actions, binned observations, and nodes* $v_{m,1},\dots,v_{m,d}$*) equals* $\nu(\xi_m)$.

Proof. We induct on depth. Simulation m starts from s_{j_m} with the root action a, and $u_{m,0}$ generates its next state and observation, so $v_{m,1} = (a, \phi(o))$ is a function of ξ_m. Suppose the state $s_{m,\ell}$ and the node $v_{m,\ell}$ at depth $\ell < d$ are functions of ξ_m. Whether or not an earlier simulation expanded $v_{m,\ell}$, Algorithm G.2 only initializes statistics and continues, and it selects the action $a'_{m,\ell}$, a component of ξ_m that reads no tree statistic. The next state and observation are generated from $(s_{m,\ell}, a'_{m,\ell}, u_{m,\ell})$, and $v_{m,\ell+1}$ is the history of $v_{m,\ell}$ extended by $a'_{m,\ell}$ and the binned observation. Hence $s_{m,\ell+1}$ and $v_{m,\ell+1}$ are functions of ξ_m, and the simulation stops at depth d regardless of the other seeds. □

Both ingredients of the lemma are needed. Under UCB1, the action at $v_{m,\ell}$ depends on the statistics left by simulations $1, \dots, m-1$. Under a POMCP-style early return ("if v is not yet expanded, expand it and return"), simulation m stops at the first node that no earlier simulation created, so its depth depends on $\xi_1, \dots, \xi_{m-1}$.

Corollary G.10 (Localized influence)**.** *Let $S \subseteq [N]$ and let P_S be the set of nodes on the paths $\nu(\xi_m)$, $m \in S$. Every node $v \notin P_S$ has the same particle set $B(v)$, counts $N(v)$ and $N(v, a')$, mean rewards $\bar{r}(v, a')$, and DP value $\hat{V}(v)$ in the runs on $(\xi_m)_{m \in [N]}$ and on $(\xi_m)_{m \notin S}$. In particular, replacing one seed ξ_n by ξ'_n changes these quantities only at nodes on the two root-to-leaf paths $\nu(\xi_n)$ and $\nu(\xi'_n)$.*

Proof. By Lemma G.9, in either run the simulations through v are those m with $v \in \nu(\xi_m)$ among the seeds present. For $v \notin P_S$ no $m \in S$ is among them, so $B(v)$, $N(v)$, $N(v, a')$ and $\bar{r}(v, a')$ coincide. A descendant of v lies on a path in P_S only if v does, so the whole subtree of v avoids P_S, and backward induction on depth with (G.3) gives the same $\hat{V}(v)$. For the last claim, compare both runs with the run on $(\xi_m)_{m \neq n}$. □

Lemma G.11 (Bounded difference)**.** *On $\mathcal{E}_\rho$, for every $S \subseteq [N]$,*

$$\left| f_a(\xi_1, \dots, \xi_N) - f_a^{(-S)} \right| \leq |S| \cdot \frac{2(1+\gamma) V_{\max} \Lambda_d}{N}. \tag{G.20}$$

Consequently, if two seed vectors ξ, ξ' differ in k coordinates and both of their runs lie in $\mathcal{E}_\rho$, then

$|f_a(\xi) - f_a(\xi')| \le k\, c$, *where*

$$c := \frac{4(1+\gamma)V_{\max}\Lambda_d}{N}. \tag{G.21}$$

Proof. Write $K(v) := N(v)$ in the full run, $k(v)$ for the number of simulations $m \in S$ whose path passes through v, and $\Delta(v) := |\hat{V}(v) - \hat{V}^{(-S)}(v)|$, where $\hat{V}^{(-S)}$ is the DP value in the reduced run. If $k(v) = K(v)$, node v is absent from the reduced run, and we set $\hat{V}^{(-S)}(v) := 0$. By Corollary G.10, $\Delta(v) = 0$ unless $k(v) \ge 1$. Define the weighted influence $I(v) := (K(v) - k(v))\,\Delta(v)$ and the sequence

$$y_d := 1, \qquad y_\ell := \rho|\mathcal{A}|\,(1 + \gamma y_{\ell+1}) \quad (1 \le \ell \le d-1).$$

We show by backward induction on depth that every node v at depth $\ell \in [1, d]$ satisfies

$$I(v) \le 2V_{\max}\, y_\ell\, k(v). \tag{G.22}$$

Leaves ($\ell = d$). If $k(v) = K(v)$, then $I(v) = 0$. Otherwise, for each a', removing $k(v)$ of the $K(v)$ states in $B(v)$ moves the average of $Q_\theta^{\star,S}(\cdot, a')$ by

$$\frac{k(v)}{K(v)}\,\big|\text{mean over removed states} - \text{mean over remaining states}\big| \le \frac{2V_{\max}\,k(v)}{K(v)}.$$

The maximum over a' in (G.1) is 1-Lipschitz in the sup norm, so $\Delta(v) \le 2V_{\max}k(v)/K(v)$ and $I(v) \le 2V_{\max}k(v)$.

Internal nodes ($1 \le \ell \le d-1$). Fix a', and write $K' := N(v, a')$ and k' for the number of simulations in S that select a' at v. If $k' = 0$, then $\hat{Q}(v, a')$ is unchanged, since no child of v under a' carries a simulation of S. If $1 \le k' < K'$, let Σ and Σ' be the sums of $r(s, a') + \gamma\hat{V}(\text{child})$ over the simulations that select a' at v in the full and in the reduced run. Each removed simulation contributes a term of absolute value at most $V_{\max}$, because $|r| \le (1-\gamma)V_{\max}$ and $|\hat{V}| \le V_{\max}$. Each of the $K(w) - k(w)$

remaining simulations entering a child w changes by $\gamma\Delta(w)$. Hence

$$|\Sigma - \Sigma'| \le k' V_{\max} + \gamma \sum_{w \in \mathrm{Ch}(v,a')} I(w).$$

Since $\hat{Q}(v,a') = \Sigma/K'$, the reduced-run value is $\Sigma'/(K'-k')$, and $|\Sigma'| \le (K'-k')V_{\max}$,

$$\Big|\frac{\Sigma}{K'} - \frac{\Sigma'}{K'-k'}\Big| \le \frac{|\Sigma - \Sigma'|}{K'} + \frac{k'\,|\Sigma'|}{K'(K'-k')} \le \frac{1}{K'}\Big(2k' V_{\max} + \gamma \sum_{w \in \mathrm{Ch}(v,a')} I(w)\Big).$$

If $k' = K'$, action a' is unvisited in the reduced run and drops out of the maximum in (G.3). A maximum of values in $[-V_{\max}, V_{\max}]$ changes by at most the largest change of its retained arguments, or by at most $2V_{\max} = 2k'V_{\max}/K'$ if its maximizer drops out. Writing $k'_{a'}$ for the k' of action a', in every case

$$\Delta(v) \le \max_{a' \in \mathcal{A}} \frac{1}{N(v,a')}\Big(2k'_{a'} V_{\max} + \gamma \sum_{w \in \mathrm{Ch}(v,a')} I(w)\Big).$$

Multiplying by $K(v)$, using (G.13), and bounding the maximum by the sum,

$$I(v) \le K(v)\,\Delta(v) \le \rho|\mathcal{A}| \sum_{a' \in \mathcal{A}} \Big(2k'_{a'} V_{\max} + \gamma \sum_{w \in \mathrm{Ch}(v,a')} I(w)\Big).$$

By the induction hypothesis (G.22) at depth $\ell+1$ and $\sum_{a'} k'_{a'} = \sum_{a'} \sum_{w \in \mathrm{Ch}(v,a')} k(w) = k(v)$,

$$I(v) \le \rho|\mathcal{A}|\,\big(2V_{\max} + 2\gamma V_{\max}\, y_{\ell+1}\big)\, k(v) = 2V_{\max}\, y_\ell\, k(v).$$

Root. Each simulation enters exactly one depth-1 node, and the normalizer $1/N$ is the same in both runs, so

$$\big|f_a - f_a^{(-S)}\big| \le \frac{1}{N} \sum_{m \in S} \big|r(s_{j_m}, a) + \gamma \hat{V}(v_{m,1})\big| + \frac{\gamma}{N} \sum_{w:\,\text{depth } 1} I(w) \le \frac{|S|}{N}\, V_{\max}\,(1 + 2\gamma y_1).$$

Unrolling the recursion,

$$y_1 = \rho|\mathcal{A}| \sum_{i=0}^{d-2} (\rho\gamma|\mathcal{A}|)^i + (\rho\gamma|\mathcal{A}|)^{d-1}.$$

By (G.14) and $(\rho\gamma|\mathcal{A}|)^{d-1} \le \Lambda_d$,

$$1 + 2\gamma y_1 = 2\Lambda_d - 1 + 2\gamma(\rho\gamma|\mathcal{A}|)^{d-1} \le 2(1+\gamma)\Lambda_d.$$

This proves (G.20). For the second claim, let S be the set of coordinates where ξ and ξ' differ. Both runs have the same reduced run on $(\xi_m)_{m\notin S}$, so (G.20) applied to each and the triangle inequality give $|f_a(\xi) - f_a(\xi')| \le 2k \cdot 2(1+\gamma)V_{\max}\Lambda_d/N = kc$. □

Lemma G.12 (Fluctuation up to a bad event)**.** *Let $p := \Pr(\mathcal{E}_\rho^c \mid b_t^J)$ for the tree T_a, and let $B_d := 2V_{\max} + cN$ with c from* (G.21)*. For every $\delta \in (0,1)$, with probability at least $1 - \delta - p$,*

$$\big|f_a - \mathbb{E}f_a\big| \le 2(1+\gamma)V_{\max}\Lambda_d\sqrt{\frac{2\log(2/\delta)}{N}} + B_d\, p,$$

and for every $q \in [-V_{\max}, V_{\max}]$,

$$\mathbb{E}\big[(f_a - q)^2\big] \le \frac{8(1+\gamma)^2 V_{\max}^2\Lambda_d^2}{N} + 2\,(\mathbb{E}f_a - q)^2 + 3B_d^2\, p.$$

Proof. Let G be the set of seed vectors whose run lies in $\mathcal{E}_\rho$. If $\Pr(G) = 0$, both claims are trivial, so assume otherwise. Let H denote the Hamming distance and define the extension

$$g(\xi) := \inf_{\eta\in G}\big\{f_a(\eta) + c\,H(\xi,\eta)\big\}.$$

First, $g = f_a$ on G: for $\xi \in G$, Lemma G.11 gives $f_a(\eta) + cH(\xi,\eta) \ge f_a(\xi)$ for every $\eta \in G$, with equality at $\eta = \xi$. Second, g changes by at most c when one coordinate of ξ changes, because each $H(\cdot,\eta)$ does. Third, $-V_{\max} \le g \le V_{\max} + cN$, so $|f_a - g| \le B_d$ everywhere and

$$|\mathbb{E}f_a - \mathbb{E}g| \le \mathbb{E}\big[|f_a - g|\,\mathbf{1}_{G^c}\big] \le B_d\, p.$$

The seeds are independent, so McDiarmid's bounded-differences inequality applies to g:

$$\Pr\big(|g - \mathbb{E}g| \geq s\big) \leq 2\exp\Big(-\frac{2s^2}{Nc^2}\Big).$$

Setting the right-hand side to δ gives $s = c\sqrt{N\log(2/\delta)/2} = 2(1+\gamma)V_{\max}\Lambda_d\sqrt{2\log(2/\delta)/N}$. On $G \cap \{|g - \mathbb{E}g| < s\}$, an event of probability at least $1 - \delta - p$,

$$|f_a - \mathbb{E}f_a| \leq |g - \mathbb{E}g| + |\mathbb{E}g - \mathbb{E}f_a| < s + B_d\, p.$$

For the second moment, the Efron–Stein inequality gives $\mathrm{Var}(g) \leq Nc^2/2 = 8(1+\gamma)^2 V_{\max}^2\Lambda_d^2/N$. Since $f_a = g$ on G and $|f_a - q| \leq 2V_{\max}$,

$$\mathbb{E}\big[(f_a - q)^2\big] \leq \mathbb{E}\big[(g - q)^2\big] + 4V_{\max}^2\, p = \mathrm{Var}(g) + (\mathbb{E}g - q)^2 + 4V_{\max}^2\, p.$$

Finally, $(\mathbb{E}g - q)^2 \leq 2(\mathbb{E}f_a - q)^2 + 2B_d^2p^2$, and $2B_d^2p^2 + 4V_{\max}^2 p \leq 3B_d^2 p$. The infimum defining g is universally measurable, which suffices for both inequalities. □

Lemma G.13 (Balancedness under uniform selection)**.** *If $d \leq 1$, then $\mathcal{E}_\rho$ holds surely. If $d \geq 2$, $q_{\min} > 0$ and $|\mathcal{A}|M \geq 2$, then*

$$\Pr(\mathcal{E}_2^c \mid b_t^J) \leq 2|\mathcal{A}|\,(|\mathcal{A}|M)^d \exp\Big(-\frac{N}{32}\Big(\frac{q_{\min}}{|\mathcal{A}|}\Big)^{d-1}\Big).$$

Proof. For $d \leq 1$ the tree has no internal node. Let $d \geq 2$ and fix a potential internal node v of T_a at depth $\ell \in [1, d-1]$, i.e., a history with ℓ binned observations and $\ell - 1$ internal actions. By Lemma G.9, the indicators that simulation m reaches v, and that it reaches v and selects a' there, are functions of ξ_m. They are therefore i.i.d. Bernoulli across m, with parameters p_v and $p_v/|\mathcal{A}|$. Each binned observation along the history has conditional probability at least $q_{\min}$ and each internal action has probability $1/|\mathcal{A}|$, so

$$\frac{p_v}{|\mathcal{A}|} \geq \Big(\frac{q_{\min}}{|\mathcal{A}|}\Big)^{\ell} \geq \pi := \Big(\frac{q_{\min}}{|\mathcal{A}|}\Big)^{d-1}.$$

The multiplicative Chernoff bounds give

$$\Pr\big(N(v) \ge \tfrac{3}{2} N p_v\big) \le e^{-N p_v/12} \le e^{-N\pi/32},$$
$$\Pr\big(N(v,a') \le \tfrac{3}{4} N p_v/|\mathcal{A}|\big) \le e^{-N p_v/(32|\mathcal{A}|)} \le e^{-N\pi/32}.$$

Outside these events, $N(v) < \frac{3}{2} N p_v = 2|\mathcal{A}| \cdot \frac{3}{4} N p_v/|\mathcal{A}| < 2|\mathcal{A}|\, N(v,a')$, which is (G.13) with $\rho = 2$. A union bound over the $|\mathcal{A}|$ actions and over the at most $\sum_{\ell=1}^{d-1} M^\ell |\mathcal{A}|^{\ell-1} \le (|\mathcal{A}|M)^d$ potential internal nodes gives the claim, since $1 + |\mathcal{A}| \le 2|\mathcal{A}|$. □

Proof of Proposition G.8. When $d = 0$ the planner is exact, so let $d \ge 1$ and write $q_a := Q_{\theta,d}(b^J, a) \in [-V_{\max}, V_{\max}]$.

Step 1: Fluctuation. Apply Lemma G.12 to each root action with $\delta/|\mathcal{A}|$ in place of δ, and take a union bound over $a \in \mathcal{A}$. With probability at least $1 - \delta - |\mathcal{A}| p_\rho$, every a satisfies

$$\begin{aligned} |\widehat{Q}(b^J, a) - q_a| &\le |\widehat{Q}(b^J, a) - \mathbb{E}\widehat{Q}(b^J, a)| + |\mathbb{E}\widehat{Q}(b^J, a) - q_a| \\ &\le 2(1+\gamma) V_{\max} \Lambda_d \sqrt{\frac{2\log(2|\mathcal{A}|/\delta)}{N}} + B_d\, p_\rho + \mathrm{bias}(N, d), \end{aligned}$$

and taking the supremum over a gives (G.16). For the second moment, Lemma G.12 with $q = q_a$ gives

$$\mathbb{E}\big[(\widehat{Q}(b^J, a) - q_a)^2\big] \le \frac{8(1+\gamma)^2 V_{\max}^2 \Lambda_d^2}{N} + 2\,\mathrm{bias}(N, d)^2 + 3 B_d^2\, p_\rho.$$

Summing over $a \in \mathcal{A}$ through $\epsilon_{\mathrm{mc}}(t, d)^2 \le \sum_a (\widehat{Q}(b^J, a) - q_a)^2$ gives (G.17). The bounds on p_ρ are Lemma G.13.

Step 2: DP backup estimation error. The simulations follow the uniform policy at internal nodes, and the DP backup $\hat{V}(v) = \max_{a'} \hat{Q}(v, a')$ recovers the optimal internal actions. We bound the bias $\mathrm{bias}_{\mathrm{DP}}(N, d) := \sup_a |\mathbb{E}\widehat{Q}(b^J, a) - Q^{\phi}_{\theta,d}(b^J, a)|$ of the resulting estimate by backward induction on depth, at the level of orders of magnitude, without any independence assumption on the returns. Fix a node v at depth $\ell \in [1, d]$ with $n_v := N(v)$ visits, and condition on the set of simulations that

reach v and on their states there. By Lemma G.9, the continuation of each such simulation below v is driven by the components of its own seed at depths $\geq \ell$, which are independent across simulations and independent of the conditioning. Conditionally, the states in $B(v)$ are therefore independent draws from the binned belief b_v at v, and each $\hat{Q}(v, a')$ is formed from $n_{v,a'} := N(v, a')$ independent continuations, with $n_{v,a'} \geq n_v/(\rho|\mathcal{A}|)$ on $\mathcal{E}_\rho$. Concentration of these conditional averages gives

$$\left|\hat{Q}(v, a') - Q^{\phi}_{\theta,d-\ell}(b_v, a')\right| \;\leq\; O\!\left(\frac{V_{\max}}{\sqrt{n_{v,a'}}}\right) \;\leq\; O\!\left(\frac{V_{\max}\sqrt{|\mathcal{A}|}}{\sqrt{n_v}}\right).$$

At a leaf ($\ell = d$) the same bound holds for the pooled averages $\frac{1}{|B(v)|}\sum_{s\in B(v)} Q^{\star,S}_\theta(s, a')$ around $\int Q^{\star,S}_\theta(s, a')\, b_v(\mathrm{d}s)$. This level must be included, because the maximum over a' in (G.1) is biased upward at every finite $|B(v)|$. Under the branching structure, a depth-ℓ node receives $n_v = \Omega(N/(|\mathcal{A}|M)^\ell)$ visits in expectation when the bins have comparable probabilities. Taking the maximum over estimated Q-values, the backup error at depth ℓ contributes at most $O(V_{\max}\sqrt{|\mathcal{A}|^{\ell+1}M^\ell/N})$, discounted by γ^ℓ. The complement of $\mathcal{E}_\rho$ changes the conditional mean by at most $2V_{\max}p_\rho$, which is absorbed into the constant once $p_\rho \leq N^{-1/2}$ (Remark G.14). Summing over depths, leaves included,

$$\mathrm{bias}_{\mathrm{DP}}(N, d) \;\leq\; \sum_{\ell=1}^{d} \gamma^\ell \cdot \frac{c_1 V_{\max}\sqrt{|\mathcal{A}|}}{\sqrt{N/(|\mathcal{A}|M)^\ell}} \;=\; \frac{c_0\,\Phi_d\, V_{\max}\sqrt{|\mathcal{A}|}}{\sqrt{N}}. \tag{G.23}$$

Step 3: Discretization error. For o, o' in the same bin, the Bayes update sensitivity (Douc et al., 2014, Section 4.3) gives

$$\|\tau_\theta(b, a, o) - \tau_\theta(b, a, o')\|_{\mathrm{TV}} \;\leq\; \frac{g_{\max}}{g_{\min}}\,\omega_Z(\mathrm{diam}(\phi)).$$

Since τ^ϕ_θ is a convex combination of $\{\tau_\theta(b, a, o')\}_{o'\in\phi^{-1}(\bar{o})}$, the same bound holds for $\|\tau^\phi_\theta(b, a, \bar{o}) - \tau_\theta(b, a, o)\|_{\mathrm{TV}}$. By Lipschitz continuity of $\bar{V}^{\star,S}_{\theta,d-1}$ in TV (constant $2V_{\max}$, cf. Lemma G.4), each of

the d tree levels contributes $\frac{2g_{\max}}{g_{\min}} V_{\max}\, \omega_Z(\mathrm{diam}(\phi))$ discounted by γ:

$$\sup_{b,a} \left| Q^{\phi}_{\theta,d}(b,a) - Q_{\theta,d}(b,a) \right| \;\le\; \frac{2\gamma(1-\gamma^d)}{1-\gamma} \cdot \frac{g_{\max}}{g_{\min}}\, V_{\max}\, \omega_Z(\mathrm{diam}(\phi)). \tag{G.24}$$

Combining Steps 2 and 3 gives $\mathrm{bias}(N,d) \le \mathrm{bias}_{\mathrm{DP}}(N,d) + \mathrm{bias}_{\mathrm{disc}}(d)$, yielding (G.18). □

Remark G.14 (Constants, a verification-free variant, and the choice of depth). *(a) Order of the constants.* Compared with an analysis that treats the simulation returns as i.i.d., the only changes are the depth factor $(1+\gamma)\Lambda_d$ in the fluctuation term and the exponentially small remainder in p_ρ. Since $\Lambda_d \le d\max(1, \rho\gamma|\mathcal{A}|)^{d-1}$ and $\Phi_d \ge (\gamma\sqrt{|\mathcal{A}|M})^d$, the fluctuation term of (G.17) is at most of the same exponential-in-d order as the DP-backup term whenever $M = \Omega(\rho^2|\mathcal{A}|)$. Taking square roots in (G.17), with $\mathrm{diam}(\phi)$ chosen as in Remark G.15, then gives $\mathbb{E}[\epsilon_{\mathrm{mc}}(t,d) \mid b_t^J] \le C_{\mathrm{mc}} V_{\max}/\sqrt{N}$ with C_{mc} of the order stated in Theorem 8.2, whose order is therefore unchanged. For coarse binning, $M \ll |\mathcal{A}|$, the fluctuation dominates and $C_{\mathrm{mc}} \propto \sqrt{|\mathcal{A}|}\,\Lambda_d$. By Lemma G.13, the remainder $3|\mathcal{A}|B_d^2 p_2$ is at most $V_{\max}^2/N$ as soon as $N \ge 32(|\mathcal{A}|/q_{\min})^{d-1}\log\big(24N|\mathcal{A}|^2(|\mathcal{A}|M)^d(1+4\Lambda_d)^2\big)$, and it vanishes for $d \le 1$. At $d=0$, $\Lambda_0 = 0$ and the fluctuation vanishes identically. For an internal rule satisfying (S1)–(S2) in place of uniform selection, the same proof applies with $|\mathcal{A}|$ replaced by $1/\pi_{\min}$ in (G.13), (G.14) and Lemma G.13.

(b) A verification-free variant. Sample splitting restores exactly conditionally i.i.d. returns. Use $N/2$ seeds to build the tree, populate the particle sets, and run the DP backup, then freeze the leaf values and the greedy internal actions $\arg\max_{a'} \hat{Q}(v,a')$. Value each root action with $N/2$ fresh seeds that follow the frozen greedy actions and stop at the frozen leaf values; a simulation that leaves the frozen tree stops and is evaluated by $\max_{a'} Q_\theta^{\star,S}(s,a')$ at its current state s. Conditionally on the first half, the second-half returns are i.i.d. and bounded by $V_{\max}$, so Hoeffding's inequality applies to them verbatim at the cost of a factor 2 in N, and the bias analysis of Step 2 applies to the first half. Evaluating each leaf at its single incoming state instead of (G.1) removes the pooling but not the coupling through the maximum in the DP backup, so it restores independent returns only at $d=1$, where it also changes the leaf target to the linear surrogate $\widetilde{V}_\theta^\star$.

(c) Choice of depth. Propositions G.1 and G.8 concern different objects. The former bounds the planning target, $\|\bar{V}^{\star,S}_{\theta,d} - V^{\star,B}_{\theta}\|_\infty \le \gamma^{d+1}\Xi_\theta/(1-\gamma)$, which involves no particles and decays geometrically in d. The latter bounds the estimator $\widehat{Q}$ around $Q_{\theta,d}$, and its fluctuation $2(1+\gamma)V_{\max}\Lambda_d\sqrt{2\log(2|\mathcal{A}|/\delta)/N}$ and DP-backup bias $c_0\Phi_d V_{\max}\sqrt{|\mathcal{A}|/N}$ grow geometrically in d. Their sum is U-shaped in d: depth buys structural bias at the price of Monte Carlo error, so each budget N has a finite best depth.

Remark G.15 (Optimal discretization resolution). The discretization error in (G.18) is a fixed bias $O(\omega_Z(\mathrm{diam}(\phi)))$, while the DP backup estimation error vanishes as $N \to \infty$ at rate $\Phi_d/\sqrt{N}$. To make the discretization bias $O(1/\sqrt{N})$, set $\mathrm{diam}(\phi) = \omega_Z^{-1}\big((1-\gamma)g_{\min}/(2\gamma g_{\max}V_{\max}\sqrt{N})\big)$. When $Z_\theta(o \mid s)$ is L_Z-Lipschitz in o uniformly (so $\omega_Z(\delta) \le L_Z\delta/g_{\min}$) and $\mathcal{O} \subseteq \mathbb{R}^{d_o}$ with a uniform grid:

$$M = \left\lceil \left(\frac{2\gamma\, L_Z\, g_{\max}\, V_{\max}\, \mathrm{diam}(\mathcal{O})\sqrt{N}}{(1-\gamma)\, g_{\min}^2} \right)^{d_o} \right\rceil . \tag{G.25}$$

With this choice, M grows as $N^{d_o/2}$, and the simulation budget threshold for the DP backup error to be $O(1/\sqrt{N})$ is $N \ge c_0^2|\mathcal{A}|\Phi_d^2$, which is exponential in $d \cdot d_o$ via the dependence of Φ_d on M.

Balancing the two bias terms instead gives an explicit rate. Take $\mathcal{O} \subseteq [0, D]^{d_o}$ with a cubic grid of mesh h, so that $M = (D/h)^{d_o}$ and $\mathrm{diam}(\phi) = \sqrt{d_o}\, h$, and assume $\omega_Z(\delta) \le L_Z\delta/g_{\min}$ as above. For $\gamma\sqrt{|\mathcal{A}|M} \ge 2$ we have $\Phi_d \le 2(\gamma\sqrt{|\mathcal{A}|M})^d$, so the DP-backup and discretization terms of (G.18) are of order $h^{-d\,d_o/2}N^{-1/2}$ and h, respectively, with constants depending on d, γ, $|\mathcal{A}|$, D, L_Z, $g_{\max}/g_{\min}$ and $V_{\max}$. Balancing them gives

$$h^\star \asymp N^{-1/(d\,d_o+2)}, \qquad \epsilon_{\mathrm{mc}}(t,d) = \widetilde{O}\big(N^{-1/(d\,d_o+2)}\big), \tag{G.26}$$

since the fluctuation term of (G.16) does not depend on h and is of lower order. The $N^{-1/2}$ rate is recovered exactly when $d\,d_o = 0$, i.e., at $d = 0$. At $d = 1$, (G.26) is the familiar rate $N^{-1/(d_o+2)}$ of a histogram estimate, here of the leaf value as a function of the observation. The balancedness condition of Lemma G.13 holds at this mesh for large N: with the Lebesgue reference measure $q_{\min} \ge g_{\min}h^{d_o}$, so $N(q_{\min}/|\mathcal{A}|)^{d-1} = \Omega(N^{(d_o+2)/(d\,d_o+2)}) \to \infty$. The computational cost is $N|\mathcal{A}|$

simulations over at most $(|\mathcal{A}|M)^d$ nodes.

When $d = 0$ or $|\mathcal{O}| < \infty$, this issue does not arise: set $\phi = \mathrm{id}$ and $M = |\mathcal{O}|$.

G.3.3. Particle Filter Stability

We work under Assumption 8.1 throughout. Write $b_t := b_t^{\theta^\star}$ for the true posterior and $b_t^J := b_t^{J,(\bar{\theta}_{k(s)})_{s\le t}}$ for the particle belief filtered under the epoch-frozen model sequence. Since the model sequence is determined by the algorithm, we abbreviate the latter as b_t^J.

Proposition G.16 (Particle filter stability in d_Q)**.** *Under Assumption 8.1, there exist constants $C_{\mathrm{pf}}, C_{\mathrm{filt}} < \infty$ and a forgetting rate $\rho := 1 - \varepsilon_{\mathrm{mix}}\,(g_{\min}/g_{\max}) \in (0,1)$ such that for all $t \ge 1$,*

$$\mathbb{E}\big[d_Q(b_t, b_t^J)\big] \;\le\; \underbrace{C_{\mathrm{pf}} \cdot J^{-\frac{1}{2\kappa_{\mathrm{pf}}}}}_{\textit{particle approx.}} \;+\; \underbrace{2V_{\max}\, C_{\mathrm{filt}} \sum_{s=1}^{t} \rho^{t-s}\, d_{\mathrm{model}}^{a_s}(\bar{\theta}_s, \theta^\star)}_{\textit{filter mismatch}}, \tag{G.27}$$

where $d_{\mathrm{model}}^{a}(\theta,\theta') := \sup_{s'} \|P_\theta(\cdot \mid s', a) - P_{\theta'}(\cdot \mid s', a)\|_{\mathrm{TV}} + \sup_{s',o} |Z_\theta(o \mid s')/Z_{\theta'}(o \mid s') - 1|$, $C_{\mathrm{filt}} := g_{\max}/g_{\min}$, and κ_{pf} is the particle filter exponent:

$$\kappa_{\mathrm{pf}} := \max\!\left(1,\; \frac{\log(2g_{\max}^2/g_{\min}^2)}{\log(1/\gamma)}\right). \tag{G.28}$$

The truncation depth H_{pf} in Lemma G.3 is chosen to balance the sum and tail:

$$H_{\mathrm{pf}} := \left\lceil \frac{\log(4V_{\max}\sqrt{J})}{\kappa_{\mathrm{pf}} \log(1/\gamma)} \right\rceil = \left\lceil \frac{\log(4V_{\max}\sqrt{J})}{\log \max(1/\gamma,\; 2g_{\max}^2/g_{\min}^2)} \right\rceil, \tag{G.29}$$

and C_{pf} depends polynomially on $V_{\max}$, $g_{\max}/g_{\min}$, $1/\varepsilon_{\mathrm{mix}}$, $\log \mathcal{N}_\Theta(1/T)$, and H_{pf}. When $\kappa_{\mathrm{pf}} = 1$ (i.e., $2\gamma g_{\max}^2/g_{\min}^2 \le 1$), the rate recovers $J^{-1/2}$. This H_{pf} is distinct from the trajectory truncation horizon H in Section G.3.4.

Proof of Proposition G.16. By the triangle inequality and Lemma G.4:

$$d_Q(b_t^{\theta^\star},\, b_t^{J,\bar{\theta}_{1:t}}) \;\le\; \underbrace{d_Q(b_t^{\theta^\star},\, b_t^{\bar{\theta}_{1:t}})}_{\text{(I) filter mismatch}} + \underbrace{d_Q(b_t^{\bar{\theta}_{1:t}},\, b_t^{J,\bar{\theta}_{1:t}})}_{\text{(II) particle approx.}}. \tag{G.30}$$

Term (I) involves two exact (continuous) posteriors, so the TV bound of Lemma G.4 gives $d_Q \leq 2V_{\max}\|b_t^{\theta^\star} - b_t^{\bar{\theta}_{1:t}}\|_{\mathrm{TV}}$, which is bounded by the filter mismatch analysis below.

Term (II): test-function bound via Feynman-Kac. Conditional on $(a_{1:T}, o_{1:T})$, the particle filter is a Feynman-Kac system with kernels $M_s(\cdot \mid s') := P_{\bar{\theta}_s}(\cdot \mid s', a_s)$ and potentials $G_s(s') := Z_{\bar{\theta}_s}(o_s \mid s')$. Under Assumption 8.1, the kernels satisfy $M_s(s' \mid x) \geq \varepsilon_{\mathrm{mix}}\,\nu(s')$ and the potentials satisfy $g_{\min} \leq G_s(x) \leq g_{\max}$ uniformly. We invoke the following time-uniform particle-filter bound.

Lemma G.17 ((Karjalainen et al., 2023, Lemma 2.3); see also (Del Moral, 2004, Theorem 7.4.4))**.** *Let $(\eta_n)_{n\geq 0}$ and $(\eta_n^N)_{n\geq 0}$ be the exact and N-particle Feynman-Kac measures on a measurable space E. Suppose the Markov kernels $(M_\ell)_{\ell\geq 1}$ admit densities w.r.t. a common measure λ satisfying $\underline{M} \leq M_\ell(x,y) \leq \overline{M}$ for all ℓ, x, y, and the potentials satisfy $\underline{G} \leq G_\ell(x) \leq \overline{G}$ for all ℓ, x. Then for all $n \geq 0$, $N \geq 2$, $p \geq 1$, and φ with $\mathrm{osc}(\varphi) \leq 1$:*

$$\left(\mathbb{E}\left[|\eta_n^N(\varphi) - \eta_n(\varphi)|^p\right]\right)^{1/p} \;\leq\; \frac{2\,d(p)^{1/p}}{\sqrt{N}}\,\frac{(\overline{M}/\underline{M})^3\,\overline{G}}{\underline{G}}, \tag{G.31}$$

where $d(p)^{1/p} = O(\sqrt{p})$. The bound is uniform over n.

Our Assumption 8.1 implies the conditions of Lemma G.17 with $\underline{M} = \varepsilon_{\mathrm{mix}}$, $\overline{M} = 1$, $\underline{G} = g_{\min}$, $\overline{G} = g_{\max}$.

Proposition G.18 (Particle filter L^p bound under Assumption 8.1)**.** *Under Assumption 8.1, for every measurable $f : \mathcal{S} \to \mathbb{R}$ with $\|f\|_\infty \leq M$, all $p \geq 1$, and all $t \geq 0$:*

$$\left(\mathbb{E}\left[|\eta_t^J(f) - \eta_t(f)|^p\right]\right)^{1/p} \;\leq\; \frac{4\,d(p)^{1/p}\,M\,g_{\max}}{\varepsilon_{\mathrm{mix}}^3\,g_{\min}\,\sqrt{J}}, \tag{G.32}$$

where $d(p)^{1/p} = O(\sqrt{p})$, so $\eta_t^J(f) - \eta_t(f)$ is sub-Gaussian with parameter $\sigma^2 = O(M^2 g_{\max}^2/(\varepsilon_{\mathrm{mix}}^6 g_{\min}^2 J))$. The bound is uniform over t.

Proof. By Lemma G.17 with $\underline{M} = \varepsilon_{\mathrm{mix}}$, $\overline{M} = 1$, $\underline{G} = g_{\min}$, $\overline{G} = g_{\max}$, for any φ with $\mathrm{osc}(\varphi) \le 1$:

$$\big(\mathbb{E}\big[|\eta_t^J(\varphi) - \eta_t(\varphi)|^p\big]\big)^{1/p} \;\le\; \frac{2\,d(p)^{1/p}\,g_{\max}}{\varepsilon_{\mathrm{mix}}^3\,g_{\min}\,\sqrt{J}},$$

uniformly over t. For f with $\|f\|_\infty \le M$, applying this to $\varphi := f/(2M)$ (which has $\mathrm{osc}(\varphi) \le 1$) and rescaling by $2M$ yields (G.32). Sub-Gaussianity follows from the standard moment-to-tail conversion. □

Lemma G.19 (Unconditional bound under adaptive actions)**.** *Let $(a_s, o_s)_{s\ge1}$ be an adapted action-observation process where $a_s = a_s(b_s^J)$ may depend on the particle system. Under Assumption 8.1, for every measurable $f : \mathcal{S} \to \mathbb{R}$ with $\|f\|_\infty \le M$, all $p \ge 1$, and all $t \ge 0$:*

$$\big(\mathbb{E}\big[|b_t^{J,\bar\theta_{1:t}}(f) - b_t^{\bar\theta_{1:t}}(f)|^p\big]\big)^{1/p} \;\le\; \frac{4\,d(p)^{1/p}\,M\,g_{\max}}{\varepsilon_{\mathrm{mix}}^3\,g_{\min}\,\sqrt{J}}. \tag{G.33}$$

Proof. Let $\mathcal{H}_T := (a_{1:T}, o_{1:T}, \bar\theta_{1:T})$. Conditional on $\mathcal{H}_T$, the Feynman-Kac kernels and potentials are deterministic, so Proposition G.18 gives

$$\mathbb{E}\big[|b_t^J(f) - b_t(f)|^p \;\big|\; \mathcal{H}_T\big] \;\le\; \left(\frac{4\,d(p)^{1/p}\,M\,g_{\max}}{\varepsilon_{\mathrm{mix}}^3\,g_{\min}\,\sqrt{J}}\right)^p =: C_p^p.$$

Since C_p is independent of $\mathcal{H}_T$, the tower property yields $\mathbb{E}[|b_t^J(f) - b_t(f)|^p] \le C_p^p$. □

By Lemma G.3 with truncation depth H_{pf}:

$$d_Q(b_t^{\bar\theta_{1:t}}, b_t^{J,\bar\theta_{1:t}}) \;\le\; \sup_{\theta,a} \sum_{\ell=0}^{H_{\mathrm{pf}}-1} \gamma^\ell C_{\mathrm{BU}}^\ell \sup_{f\in\mathcal{F}_\ell(\theta,a)} |b_t(f) - b_t^J(f)| + 2\gamma^{H_{\mathrm{pf}}} V_{\max}. \tag{G.34}$$

We optimize H_{pf} to balance the sum and tail below.

Pointwise bound. For $f \in \mathcal{F}_\ell(\theta, a)$ with $\|f\|_\infty \le g_{\max}^\ell V_{\max}$, Lemma G.19 gives

$$\big(\mathbb{E}\big[|b_t^J(f) - b_t(f)|^p\big]\big)^{1/p} \;\le\; \frac{4\,d(p)^{1/p}\,g_{\max}^{\ell+1}\,V_{\max}}{\varepsilon_{\mathrm{mix}}^3\,g_{\min}\,\sqrt{J}} =: \sigma_\ell\,d(p)^{1/p}. \tag{G.35}$$

Since $d(p)^{1/p} \le C_0\sqrt{p}$ (Corollary G.18), $b_t^J(f) - b_t(f)$ is sub-Gaussian with parameter σ_ℓ^2.

Covering-number argument. For N sub-Gaussian random variables with common parameter σ^2:

$$\mathbb{E}\big[\max_{i\le N}|X_i|\big] \;\le\; \sigma\sqrt{2\log(2N)}. \tag{G.36}$$

At depth ℓ, $\bigcup_{\theta,a}\mathcal{F}_\ell(\theta,a)$ is covered by $N_\ell := \mathcal{N}_\Theta(1/T)\cdot|\mathcal{A}|^\ell$ functions (after discretizing Θ at resolution $1/T$, approximation error $O(1/T)$). Applying (G.36) and summing over depths with weights $\gamma^\ell C_{\mathrm{BU}}^\ell$:

$$\mathbb{E}\big[d_Q(b_t^{\bar\theta_{1:t}}, b_t^{J,\bar\theta_{1:t}})\big] \le \sum_{\ell=0}^{H_{\mathrm{pf}}-1}\gamma^\ell C_{\mathrm{BU}}^\ell\,\sigma_\ell\sqrt{2\log(2N_\ell)} + 2\gamma^{H_{\mathrm{pf}}}V_{\max} + O(1/T) \tag{G.37}$$

Define $\alpha := 2\gamma g_{\max}^2/g_{\min}^2$, $r := \alpha/\gamma = 2g_{\max}^2/g_{\min}^2$, and $c_1 := 4g_{\max}V_{\max}/(\varepsilon_{\mathrm{mix}}^3 g_{\min})$, so that $\sigma_\ell = c_1\,g_{\max}^\ell/\sqrt{J}$ and the ℓ-th term equals

$$\frac{c_1}{\sqrt{J}}\,\alpha^\ell\,\sqrt{2\log\big(2\mathcal{N}_\Theta(1/T)\,|\mathcal{A}|^\ell\big)}.$$

Since $\sqrt{\log(ab)} \le \sqrt{\log a} + \sqrt{\log b}$, the sum in (G.37) is bounded by

$$\mathbb{E}\big[d_Q^{(\mathrm{II})}\big] \;\le\; \frac{c_1\sqrt{2}}{\sqrt{J}}\left(\sqrt{\log(2\mathcal{N}_\Theta(1/T))}\sum_{\ell=0}^{H_{\mathrm{pf}}-1}\alpha^\ell + \sqrt{\log|\mathcal{A}|}\sum_{\ell=0}^{H_{\mathrm{pf}}-1}\alpha^\ell\sqrt{\ell}\right) + 2\gamma^{H_{\mathrm{pf}}}V_{\max}. \tag{G.38}$$

We bound the two sums in two cases.

Case 1: $\alpha \ge 1$. Choose

$$H_{\mathrm{pf}} = \left\lceil\frac{\log(4V_{\max}\sqrt{J})}{\log r}\right\rceil = \left\lceil\frac{\log(4V_{\max}\sqrt{J})}{\log(2g_{\max}^2/g_{\min}^2)}\right\rceil.$$

Using $\sum_{\ell=0}^{H-1}\alpha^\ell \le H\alpha^{H-1}$ and $\sum_{\ell=0}^{H-1}\alpha^\ell\sqrt{\ell} \le H^{3/2}\alpha^{H-1}$ (since $\alpha \ge 1$):

$$\text{sum} \;\le\; \frac{c_1\sqrt{2}\,H_{\mathrm{pf}}^{3/2}\,\alpha^{H_{\mathrm{pf}}-1}}{\sqrt{J}}\cdot\sqrt{\log\big(2\mathcal{N}_\Theta(1/T)\,|\mathcal{A}|^{H_{\mathrm{pf}}}\big)}. \tag{G.39}$$

Writing $\alpha^{H_{\mathrm{pf}}-1} = \alpha^{H_{\mathrm{pf}}}/\alpha = \gamma^{H_{\mathrm{pf}}} r^{H_{\mathrm{pf}}}/\alpha$ and substituting $r^{H_{\mathrm{pf}}} = 4V_{\max}\sqrt{J}$:

$$\text{sum} \;\le\; \frac{4\sqrt{2}\, c_1\, V_{\max}\, H_{\mathrm{pf}}^{3/2}}{\alpha} \cdot \sqrt{\log\bigl(2\mathcal{N}_\Theta(1/T)\,|\mathcal{A}|^{H_{\mathrm{pf}}}\bigr)} \cdot \gamma^{H_{\mathrm{pf}}}. \tag{G.40}$$

For the tail:

$$\gamma^{H_{\mathrm{pf}}} = \gamma^{\log(4V_{\max}\sqrt{J})/\log r} = (4V_{\max}\sqrt{J})^{-\log(1/\gamma)/\log r} = (4V_{\max})^{-1/\kappa_{\mathrm{pf}}} \cdot J^{-\frac{1}{2\kappa_{\mathrm{pf}}}}, \tag{G.41}$$

where $\kappa_{\mathrm{pf}} = \log r/\log(1/\gamma) > 1$. Substituting $c_1 = 4g_{\max}V_{\max}/(\varepsilon_{\mathrm{mix}}^3 g_{\min})$, $H_{\mathrm{pf}} = \lceil \log(4V_{\max}\sqrt{J})/\log r \rceil$, and using $\log(2\mathcal{N}_\Theta(1/T)\,|\mathcal{A}|^{H_{\mathrm{pf}}}) \le \log(2\mathcal{N}_\Theta(1/T)) + H_{\mathrm{pf}} \log|\mathcal{A}|$, the total (sum + tail) is

$$\begin{aligned}\mathbb{E}\bigl[d_Q^{(\mathrm{II})}\bigr] \;\le\; \Biggl(&\frac{16\sqrt{2}\, g_{\max}\, V_{\max}^2}{\alpha\, \varepsilon_{\mathrm{mix}}^3\, g_{\min}} \cdot \left(\frac{\log(4V_{\max}\sqrt{J})}{\log r}\right)^{3/2} \sqrt{\log\bigl(2\mathcal{N}_\Theta(1/T)\bigr) + \frac{\log(4V_{\max}\sqrt{J})}{\log r}\log|\mathcal{A}|} \\ &+ 2V_{\max}\Biggr) \cdot (4V_{\max})^{-1/\kappa_{\mathrm{pf}}} \cdot J^{-\frac{1}{2\kappa_{\mathrm{pf}}}}.\end{aligned} \tag{G.42}$$

Case 2: $\alpha < 1$. Choose

$$H_{\mathrm{pf}} = \left\lceil \frac{\log(4V_{\max}\sqrt{J})}{\log(1/\gamma)} \right\rceil.$$

The geometric series converges: $\sum_{\ell\ge 0}\alpha^\ell = 1/(1-\alpha)$ and $\sum_{\ell\ge 0}\alpha^\ell\sqrt{\ell} \le c_2/(1-\alpha)^{3/2}$ for a universal $c_2 > 0$. Hence

$$\text{sum} \;\le\; \frac{c_1\, c_2\, \sqrt{2}}{(1-\alpha)^{3/2}\,\sqrt{J}} \cdot \sqrt{\log\bigl(2\mathcal{N}_\Theta(1/T)\bigr) + \log|\mathcal{A}|} \;=\; O\Bigl(J^{-1/2}\Bigr).$$

For the tail: $\gamma^{H_{\mathrm{pf}}} = (4V_{\max}\sqrt{J})^{-1} = O(J^{-1/2})$. Here $\kappa_{\mathrm{pf}} = 1$, and both terms are $O(J^{-1/2})$.

Unified bound. The choice (G.29), $H_{\mathrm{pf}} = \lceil \log(4V_{\max}\sqrt{J})/\log\max(1/\gamma, r) \rceil$, covers both cases and yields

$$\mathbb{E}\bigl[d_Q^{(\mathrm{II})}\bigr] \;\le\; C_{\mathrm{pf}} \cdot J^{-\frac{1}{2\kappa_{\mathrm{pf}}}}, \qquad \kappa_{\mathrm{pf}} = \max\left(1, \frac{\log r}{\log(1/\gamma)}\right),$$

where

$$C_{\mathrm{pf}} := \frac{c_3\, V_{\max}\, g_{\max}\, H_{\mathrm{pf}}^{3/2}}{\varepsilon_{\mathrm{mix}}^3\, g_{\min}} \cdot \sqrt{\log\bigl(2\mathcal{N}_\Theta(1/T)\,|\mathcal{A}|^{H_{\mathrm{pf}}}\bigr)}, \tag{G.43}$$

with c_3 depending on α and $g_{\max}/g_{\min}$. In particular, the $|\mathcal{A}|$ dependence enters through $\sqrt{H_{\mathrm{pf}} \log |\mathcal{A}|}$.

Term (I): filter mismatch. By Lemma G.4,

$$d_Q(b_t^{\theta^\star}, b_t^{\bar\theta_{1:t}}) \;\le\; 2V_{\max} \|b_t^{\theta^\star} - b_t^{\bar\theta_{1:t}}\|_{\mathrm{TV}}.$$

Define $\delta_s := \sup_{s'} \|P_{\bar\theta_s}(\cdot \mid s', a_s) - P_{\theta^\star}(\cdot \mid s', a_s)\|_{\mathrm{TV}} + \sup_{s'} |Z_{\bar\theta_s}(o_s \mid s')/Z_{\theta^\star}(o_s \mid s') - 1|$ and the predict-update kernel

$$K_\theta^{a,o}(s, ds') := \frac{Z_\theta(o \mid s') P_\theta(s' \mid s, a)}{\int Z_\theta(o \mid s'') P_\theta(s'' \mid s, a)\, ds''}\, ds'.$$

Under Assumption 8.1, bounding the numerator below and denominator above:

$$K_\theta^{a,o}(s, ds') \;\ge\; \frac{g_{\min}\varepsilon_{\mathrm{mix}}}{g_{\max}}\, \nu(s')\, ds' \qquad \forall\, s \in \mathcal{S}, \tag{G.44}$$

giving Dobrushin coefficient $\rho := 1 - \varepsilon_{\mathrm{mix}} g_{\min}/g_{\max}$. Standard Dobrushin contraction (Douc et al., 2014, Theorem 4.3.1) under different models $\theta^\star, \bar\theta_s$ yields

$$\|b_s^{\theta^\star} - b_s^{\bar\theta_{1:s}}\|_{\mathrm{TV}} \;\le\; \rho\, \|b_{s-1}^{\theta^\star} - b_{s-1}^{\bar\theta_{1:s-1}}\|_{\mathrm{TV}} + C_{\mathrm{filt}}\, \delta_s, \qquad C_{\mathrm{filt}} := g_{\max}/g_{\min}. \tag{G.45}$$

Unrolling and using $\delta_s \le d_{\mathrm{model}}^{a_s}(\bar\theta_s, \theta^\star)$:

$$\|b_t^{\theta^\star} - b_t^{\bar\theta_{1:t}}\|_{\mathrm{TV}} \;\le\; C_{\mathrm{filt}} \sum_{s=1}^{t} \rho^{t-s}\, d_{\mathrm{model}}^{a_s}(\bar\theta_s, \theta^\star).$$

Combining. Substituting into (G.30):

$$\mathbb{E}\, d_Q(b_t, b_t^J) \;\le\; \frac{C'_{\mathrm{pf}}}{\sqrt{J}} + 2V_{\max}\, C_{\mathrm{filt}} \sum_{s=1}^{t} \rho^{t-s}\, d_{\mathrm{model}}^{a_s}(\bar\theta_s, \theta^\star).$$

When summed over t, exchanging summation order gives $\sum_{t=1}^{T}\sum_{s=1}^{t}\rho^{t-s}\,d_{\text{model}}^{a_s} \le \frac{1}{1-\rho}\sum_{s=1}^{T} d_{\text{model}}^{a_s}$.

□

G.3.4. Model Estimation Error (Terms A + C) and Filter Mismatch

Truncation. Define $V_\theta^{\pi,B,H}(b) := \mathbb{E}_\theta^\pi[\sum_{u=0}^{H-1}\gamma^u r_u \mid b_0 = b]$. Since $|V_\theta^{\pi,B} - V_\theta^{\pi,B,H}| \le \gamma^H V_{\max}$:

$$(\mathrm{A})_t + (\mathrm{C})_t \;\le\; |V_{\theta^\star}^{\star,B,H}(b_t) - V_{\bar\theta_{k(t)}}^{\star,B,H}(b_t)| + |V_{\bar\theta_{k(t)}}^{\pi_t,B,H}(b_t) - V_{\theta^\star}^{\pi_t,B,H}(b_t)| + \frac{4\gamma^H R_{\max}}{1-\gamma}. \tag{G.46}$$

Choose $H = \lceil \log(2R_{\max}T/(1-\gamma))/\log(1/\gamma)\rceil$ so the tail contributes $O(1)$.

Confidence sets. Let $\Pr_{\theta,H}^\pi$ denote the length-H trajectory distribution under (θ,π). Define

$$\mathcal{C}_k := \big\{\theta\in\Theta : \mathcal{L}_{T_k}(\theta) \ge \mathcal{L}_{T_k}(\bar\theta_k) - \beta\big\}, \tag{G.47}$$

where $\mathcal{L}_{T_k}$ is the log-likelihood over epochs $1,\dots,k-1$. By construction $\bar\theta_k \in \mathcal{C}_k$, and $\theta^\star \in \mathcal{C}_k$ for all k w.h.p. when $\beta \ge c_0 \log[\mathcal{N}_\Theta(1/T)T/\delta_{\text{opt}}]$ (Liu et al., 2023b, Prop. B.1).

Proposition G.20 (Likelihood-to-ℓ_1 concentration)**.** *There exists $c>0$ such that w.p. $\ge 1-\delta$, simultaneously for all $n\in[T]$ and all $\theta\in\Theta$ with $\mathcal{L}_n(\theta) \ge \mathcal{L}_n(\theta^\star) - \beta$:*

$$\sum_{i=1}^{n} \big\|\Pr_{\theta,H}^{\pi^i} - \Pr_{\theta^\star,H}^{\pi^i}\big\|_1^2 \;\le\; c\,\beta_T, \qquad \beta_T := \beta + \log[\mathcal{N}_\Theta(1/T)\,T/\delta]. \tag{G.48}$$

Proof. The episodic version is given by Liu et al. (2023b, Prop. B.2). We transfer to the online setting via a blocking argument: partition the trajectory into $\lfloor T/(2H)\rfloor$ blocks of length $2H$, each split into a burn-in half and a data half. Choose H so that

$$\rho^H = (1-\varepsilon_{\text{mix}} g_{\min}/g_{\max})^H \;\le\; 1/T^2, \qquad \text{i.e.,}\quad H \;\ge\; \frac{2\log T}{\log(1/\rho)}.$$

Our choice $H = O(\log T/\log(1/\gamma))$ satisfies this when $\varepsilon_{\text{mix}} g_{\min}/g_{\max} = \Omega(1-\gamma)$; otherwise, increase H by a constant factor. With $\rho^H \le 1/T^2$, the data halves are approximately independent (joint TV perturbation $O(1/T)$). Applying the episodic bound to the data halves gives (G.48) up to a

constant factor. □

Corollary G.21 (Epoch-level MLE concentration)**.** *Under the same conditions as Proposition G.20, w.p. $\geq 1-\delta$, simultaneously for all epochs $k \in [K]$:*

$$\sum_{i=1}^{T_{k+1}-1} \left\| \Pr^{\pi^i}_{\bar{\theta}_k,H} - \Pr^{\pi^i}_{\theta^\star,H} \right\|_1^2 \;\leq\; c\,\beta_T, \tag{G.49}$$

where $c > 0$ is a universal constant (at most twice the constant in Proposition G.20).

Proof. Fix $k \in [K]$ and write $\| \Pr^{\pi}_{\theta} \|_1 := \| \Pr^{\pi}_{\theta,H} - \Pr^{\pi}_{\theta^\star,H} \|_1$.

Historical data ($i \leq T_k$). By MLE optimality, $\bar{\theta}_k$ lies in $\mathcal{C}_k$, so Proposition G.20 at $n = T_k$ gives

$$\sum_{i=1}^{T_k} \left\| \Pr^{\pi^i}_{\bar{\theta}_k} \right\|_1^2 \;\leq\; c\,\beta_T. \tag{G.50}$$

Within-epoch data ($i \in E_k$). Since $\bar{\theta}_k$ is $\mathcal{F}_{T_k}$-measurable, we use a self-normalized argument for the fixed hypothesis $\bar{\theta}_k$. Define

$$L_n^{(k)} := \sum_{i=T_k+1}^{T_k+n} \big[\ell_i(\theta^\star) - \ell_i(\bar{\theta}_k)\big], \qquad M_n^{(k)} := \sum_{i=T_k+1}^{T_k+n} \Big[\mathrm{KL}(\Pr^{\pi^i}_{\theta^\star} \,\|\, \Pr^{\pi^i}_{\bar{\theta}_k}) - \big(\ell_i(\theta^\star) - \ell_i(\bar{\theta}_k)\big)\Big],$$

where $\ell_i(\theta) := \log p_\theta(o_i \mid s_i, a_i)$. By Pinsker's inequality and blocking:

$$\frac{1}{2} \sum_{i \in E_k} \left\| \Pr^{\pi^i}_{\bar{\theta}_k} \right\|_1^2 \;\leq\; L^{(k)}_{|E_k|} + M^{(k)}_{|E_k|}. \tag{G.51}$$

$M_n^{(k)}$ is a supermartingale (each increment has non-positive conditional expectation). The maximal inequality and a covering-number union bound give

$$\Pr\left(\sup_{n \leq |E_k|} M_n^{(k)} \geq c_1 \log\big[\mathcal{N}_\Theta(1/T)/\delta\big] \right) \leq \delta/K.$$

For $L^{(k)}_{|E_k|}$: each $|\ell_i(\theta^\star) - \ell_i(\bar\theta_k)| \le \log(g_{\max}/g_{\min})$ and $\mathbb{E}[\ell_i(\theta^\star) - \ell_i(\bar\theta_k) \mid \mathcal{F}_{i-1}] \ge 0$. By Azuma-Hoeffding:

$$\Pr\Big(L^{(k)}_{|E_k|} \ge c_2\sqrt{|E_k|\log(1/\delta)}\log(g_{\max}/g_{\min})\Big) \le \delta/K.$$

Since $\sqrt{T}\log(g_{\max}/g_{\min})$ is absorbed into β_T:

$$\sum_{i\in E_k} \big\|\Pr^{\pi^i}_{\bar\theta_k}\big\|_1^2 \;\le\; c\,\beta_T. \tag{G.52}$$

Combining. Summing (G.50) and (G.52), redefining $c \leftarrow 2c$, and taking a union bound over $k \in [K]$ gives (G.49). □

Transferring model error from historical to arbitrary policies requires action-space coverage via ε-greedy exploration.

Lemma G.22 (Trajectory coverage under Doeblin mixing and ε-greedy)**.** *Under Assumption 8.1, for any two models $\theta, \theta' \in \Theta$, any policy π, and any policy π_ε satisfying $\pi_\varepsilon(a \mid h) \ge \varepsilon/|\mathcal{A}|$ for all a, h (with $\varepsilon \in (0,1]$),*

$$\big\|\Pr^{\pi}_{\theta,H} - \Pr^{\pi}_{\theta',H}\big\|_1 \;\le\; C_{\mathrm{cov}}\,\big\|\Pr^{\pi_\varepsilon}_{\theta,H} - \Pr^{\pi_\varepsilon}_{\theta',H}\big\|_1, \qquad C_{\mathrm{cov}} := \frac{H\,|\mathcal{A}|}{\varepsilon}\cdot\frac{g_{\max}^2}{\varepsilon_{\mathrm{mix}}\,g_{\min}^2}. \tag{G.53}$$

Proof. Step 1: Telescoping. Let $\Pr^{(k)}$ be the hybrid using θ for steps $1,\dots,k$ and θ' for steps $k{+}1,\dots,H$, under policy π. Let $\Delta_k(s,a) := \|P_\theta(\cdot \mid s,a) - P_{\theta'}(\cdot \mid s,a)\|_{\mathrm{TV}}$ and $\Delta^Z(s) := \sup_o |Z_\theta(o \mid s)/Z_{\theta'}(o \mid s) - 1|$. Then

$$\|\Pr^{\pi}_{\theta,H} - \Pr^{\pi}_{\theta',H}\|_1 \;\le\; \sum_{k=0}^{H-1} \|\Pr^{(k+1)} - \Pr^{(k)}\|_1, \tag{G.54}$$

$$\|\Pr^{(k+1)} - \Pr^{(k)}\|_1 \;\le\; \mathbb{E}_{\mu_k^\pi}\big[\sup_a \Delta_k(S_k,a) + \Delta^Z(S_k)\big], \tag{G.55}$$

where μ_k^π is the state marginal at step k under the hybrid.

Step 2: State-marginal ratio. By the Doeblin decomposition $\tilde\mu_k^\pi \ge \varepsilon_{\mathrm{mix}}\nu$ and $\tilde\mu_k^{\pi'} \ge \varepsilon_{\mathrm{mix}}\nu$, for any

$h : \mathcal{S} \to [0,1]$:

$$\mathbb{E}_{\tilde{\mu}_k^{\pi}}[h] \;\le\; \varepsilon_{\mathrm{mix}}\mathbb{E}_{\nu}[h] + (1-\varepsilon_{\mathrm{mix}}) \;\le\; \frac{1}{\varepsilon_{\mathrm{mix}}}\mathbb{E}_{\tilde{\mu}_k^{\pi'}}[h].$$

Including the Bayes update normalization ($g_{\max}/g_{\min}$):

$$\mathbb{E}_{\mu_k^{\pi}}[h] \;\le\; \frac{g_{\max}}{g_{\min}\varepsilon_{\mathrm{mix}}}\,\mathbb{E}_{\mu_k^{\pi'}}[h].$$

Step 3: Action coverage. Since $\pi_\varepsilon(a \mid h) \ge \varepsilon/|\mathcal{A}|$:

$$\sup_a \Delta_k(s,a) \;\le\; \frac{|\mathcal{A}|}{\varepsilon}\,\mathbb{E}_{A\sim\pi_\varepsilon}[\Delta_k(s,A)].$$

Step 4: Assembly. Combining Steps 1–3 term-by-term:

$$\|\Pr^{(k+1)} - \Pr^{(k)}\|_1 \;\le\; \frac{|\mathcal{A}|}{\varepsilon}\cdot\frac{g_{\max}^2}{\varepsilon_{\mathrm{mix}}\, g_{\min}^2}\cdot\|\Pr_{\pi_\varepsilon}^{(k+1)} - \Pr_{\pi_\varepsilon}^{(k)}\|_1.$$

Summing over $k = 0,\dots,H-1$ yields (G.53). □

Lemma G.23 (Simulation lemma)**.** *For any $\theta,\theta' \in \Theta$, π, and b:*

$$\left|V_\theta^{\pi,B,H}(b) - V_{\theta'}^{\pi,B,H}(b)\right| \;\le\; \frac{R_{\max}}{1-\gamma}\left\|\Pr_{\theta,H}^{\pi} - \Pr_{\theta',H}^{\pi}\right\|_1.$$

Proof. Telescope the H-truncated value difference into one-step terms, each bounded by $R_{\max}$. □

Proposition G.24 (Control of (A) + (C) + filter mismatch)**.** *Under Assumption 8.1, consider the doubling-epoch variant of Algorithm 8.1 in which time is partitioned into epochs $E_k = [T_k, T_{k+1})$ with $T_k = 2^{k-1}$, $k = 1,\dots,K := \lceil \log_2 T \rceil$. Within each epoch E_k, the model $\bar{\theta}_k$ is frozen (computed from data in epochs $1,\dots,k-1$) and the agent uses ε_k-greedy exploration with a constant rate ε_k.*

Let $K_{\mathrm{cov}} := H|\mathcal{A}|\, g_{\max}^2/(\varepsilon_{\mathrm{mix}}\, g_{\min}^2)$. *Then with probability* $\geq 1-\delta_{\mathrm{opt}}$,

$$\sum_{t=1}^{T} (C)_t \;\leq\; \frac{R_{\max}}{1-\gamma}\sqrt{c\,\beta_T}\sum_{k=1}^{K}\sqrt{|E_k|} + \frac{2\gamma^H R_{\max} T}{1-\gamma}, \tag{G.56}$$

$$\sum_{t=1}^{T} (A)_t \;\leq\; \frac{R_{\max}\, K_{\mathrm{cov}}}{1-\gamma}\sqrt{c\,\beta_T}\sum_{k=2}^{K}\frac{|E_k|}{\sqrt{\sum_{j=1}^{k-1}\varepsilon_j^2|E_j|}} + \frac{2\gamma^H R_{\max} T}{1-\gamma}, \tag{G.57}$$

$$\sum_{t=1}^{T} d_{\mathrm{model}}^{a_t}(\bar\theta_{k(t)}, \theta^\star) \;\leq\; C_{\mathrm{mix}}\sqrt{c\,\beta_T}\sum_{k=1}^{K}\sqrt{|E_k|}, \tag{G.58}$$

where $d_{\mathrm{model}}^{a}(\theta,\theta') := \sup_{s\in\mathcal{S}} \|P_\theta(\cdot \mid s,a) - P_{\theta'}(\cdot \mid s,a)\|_{\mathrm{TV}} + \sup_{s\in\mathcal{S},o\in\mathcal{O}}\big|Z_\theta(o \mid s)/Z_{\theta'}(o \mid s) - 1\big|$ *is the per-action model discrepancy, and* C_{mix} *depends on* $\varepsilon_{\mathrm{mix}}, g_{\min}, g_{\max}, H$.

Proof. By Corollary G.21, equation (G.49) holds for all $k \leq K$. For brevity, write $\|\Pr_\theta^\pi\|_1 := \|\Pr_{\theta,H}^\pi - \Pr_{\theta^\star,H}^\pi\|_1$ throughout this proof.

Term (C). By Lemma G.23,

$$(\mathrm{C})_t \;\leq\; \frac{R_{\max}}{1-\gamma}\big\|\Pr_{\bar\theta_{k(t)}}^{\pi^t}\big\|_1 + \frac{2\gamma^H R_{\max}}{1-\gamma}.$$

Since $\bar\theta_k$ is fixed within E_k, Cauchy-Schwarz and (G.49) give

$$\sum_{t\in E_k}\big\|\Pr_{\bar\theta_k}^{\pi^t}\big\|_1 \;\leq\; \sqrt{|E_k|\sum_{t\in E_k}\big\|\Pr_{\bar\theta_k}^{\pi^t}\big\|_1^2} \;\leq\; \sqrt{|E_k|\cdot c\,\beta_T}.$$

Summing across epochs: $\sum_{k=1}^{K}\sqrt{|E_k|\cdot c\beta_T} = \sqrt{c\beta_T}\sum_k 2^{(k-1)/2} = O(\sqrt{T\beta_T})$.

Term (A). By Lemmas G.23 and G.22:

$$(\mathrm{A})_t \;\leq\; \frac{R_{\max}}{1-\gamma}\sup_\pi\big\|\Pr_{\bar\theta_{k(t)}}^{\pi}\big\|_1 + \frac{2\gamma^H R_{\max}}{1-\gamma}, \qquad \big\|\Pr_{\bar\theta_k}^{\pi}\big\|_1 \;\leq\; \frac{K_{\mathrm{cov}}}{\varepsilon_j}\big\|\Pr_{\bar\theta_k}^{\pi^j}\big\|_1 \quad \forall\, j<k.$$

Squaring, multiplying by ε_j^2, and summing over prior epochs using (G.49):

$$\big\|\Pr_{\bar\theta_k}^{\pi}\big\|_1^2 \sum_{j<k} \varepsilon_j^2 |E_j| \;\le\; K_{\mathrm{cov}}^2 \sum_{j<k} \sum_{t\in E_j} \big\|\Pr_{\bar\theta_k}^{\pi^t}\big\|_1^2 \;\le\; K_{\mathrm{cov}}^2\, c\,\beta_T \quad\Longrightarrow\quad \sup_{\pi} \big\|\Pr_{\bar\theta_k}^{\pi}\big\|_1 \;\le\; \frac{K_{\mathrm{cov}}\sqrt{c\beta_T}}{\sqrt{\sum_{j<k}\varepsilon_j^2|E_j|}}.$$

Multiplying by $|E_k|$ and summing gives (G.57).

Filter mismatch. By the telescoping in Lemma G.22 (Steps 1–2 only, without the $|\mathcal{A}|/\varepsilon$ action-coverage factor since we evaluate at the executed action a_t):

$$d_{\mathrm{model}}^{a_t}(\bar\theta_{k(t)}, \theta^\star) \;\le\; C_{\mathrm{mix}} \big\|\Pr_{\bar\theta_{k(t)}}^{\pi^t}\big\|_1, \qquad C_{\mathrm{mix}} := \frac{H\, g_{\max}^2}{\varepsilon_{\mathrm{mix}}\, g_{\min}^2}.$$

Cauchy-Schwarz and (G.49) give $\sum_{t\in E_k} \|\Pr_{\bar\theta_k}^{\pi^t}\|_1 \;\le\; \sqrt{|E_k|\cdot c\,\beta_T}$. Summing across epochs yields (G.58). □

G.3.5. Proof of the Main Regret Bound (Theorem 8.2)

Proof of Theorem 8.2. Use doubling epochs $E_k = [T_k, T_{k+1})$, $T_k = 2^{k-1}$, $K = \lceil \log_2 T \rceil$, with ε_k-greedy exploration within each epoch. Decompose:

$$\mathrm{Reg}_T \;\le\; \underbrace{\frac{2R_{\max}}{1-\gamma}\sum_k |E_k|\varepsilon_k}_{\text{exploration}} + \mathbb{E}\sum_{t=1}^{T}\big[(\mathrm{A})_t + (\mathrm{B})_t + (\mathrm{C})_t\big].$$

Term (C). By (G.56) (choosing $\delta_{\mathrm{opt}} = 1/T$):

$$\mathbb{E}\sum_t (\mathrm{C})_t \;\le\; \frac{R_{\max}\sqrt{c\beta_T}}{1-\gamma}\sum_{k=1}^{K}\sqrt{|E_k|} + O(1) \;=\; \widetilde{O}\left(\frac{\sqrt{T}}{1-\gamma}\right).$$

Term (A). By (G.57) with $\varepsilon_k = \eta/\sqrt{|E_k|}$ (so $\varepsilon_k^2|E_k| = \eta^2$):

$$\mathbb{E}\sum_t (\mathrm{A})_t \;\le\; \frac{R_{\max} K_{\mathrm{cov}}\sqrt{c\beta_T}}{(1-\gamma)\eta}\sum_{k=2}^{K}\frac{|E_k|}{\sqrt{k-1}} + O(1).$$

Since $\sum_{k=2}^{K} 2^{k-1}/\sqrt{k-1} = \Theta(T/\sqrt{\log T})$, this scales as $K_{\mathrm{cov}}\sqrt{\beta_T} \cdot T/(\eta\sqrt{\log T}(1-\gamma))$.

Term (B). By Theorem G.7, Propositions G.1, G.8, G.16:

$$\mathbb{E}\sum_t (\mathrm{B})_t \;\le\; \frac{2C_{\mathrm{pf}}\,T}{(1-\gamma)} \cdot J^{-\frac{1}{2\kappa_{\mathrm{pf}}}} + \frac{C_{\mathrm{mc}}V_{\max}T}{(1-\gamma)\sqrt{N}} + \frac{2\gamma^{d+1}}{(1-\gamma)^2}\sum_t \Xi_{\bar{\theta}_{k(t)}} + O(\sqrt{T\beta_T}).$$

The $O(\sqrt{T\beta_T})$ absorbs the filter mismatch via (G.58).

Optimization. Exploration cost $= O(\eta\sqrt{T}R_{\max}/(1-\gamma))$. Balancing with term (A) gives $\eta^\star \sim K_{\mathrm{cov}}^{1/2}\,T^{1/4}(\beta_T/\log T)^{1/4}$ and total $\widetilde{O}(K_{\mathrm{cov}}^{1/2}T^{3/4}/(1-\gamma)^{3/2})$. Term (C) achieves $\widetilde{O}(\sqrt{T})$; the $T^{3/4}$ rate is driven by the coverage transfer for term (A). □

G.3.6. Optimal Exploration Schedule

Set $\varepsilon_k = \eta/\sqrt{|E_k|}$ so that $\varepsilon_k^2|E_k| = \eta^2$ and $\sum_{j<k}\varepsilon_j^2|E_j| = (k-1)\eta^2$. The exploration-dependent terms scale as:

$$\begin{aligned}
\text{Exploration cost:}\quad & \textstyle\sum_k |E_k|\varepsilon_k = \eta\sum_k\sqrt{|E_k|} = O(\eta\sqrt{T}),\\
\text{Term (A):}\quad & O\big(K_{\mathrm{cov}}\sqrt{\beta_T}\cdot T/(\eta\sqrt{\log T}(1-\gamma))\big),\\
\text{Term (C):}\quad & O\big(\sqrt{T\beta_T}/(1-\gamma)\big),\quad \text{independent of } \eta.
\end{aligned}$$

Balancing (A) and exploration:

$$\eta^\star \sim \left(\frac{K_{\mathrm{cov}}\sqrt{T\beta_T}}{\sqrt{\log T}}\right)^{1/2}, \qquad (\mathrm{A}) + \text{exploration} = \widetilde{O}\left(\frac{K_{\mathrm{cov}}^{1/2}T^{3/4}}{(1-\gamma)}\right).$$

The total regret is:

$$\mathrm{Reg}_T = \widetilde{O}\left(\frac{K_{\mathrm{cov}}^{1/2}\,R_{\max}\,T^{3/4}}{(1-\gamma)^{3/2}}\right) + \frac{C_{\mathrm{pf}}\,R_{\max}\,T}{(1-\gamma)^2}\cdot J^{-\frac{1}{2\kappa_{\mathrm{pf}}}} + \frac{C_{\mathrm{mc}}\,R_{\max}\,T}{(1-\gamma)^2\sqrt{N}} + \frac{2\gamma^{d+1}}{(1-\gamma)^2}\sum_t \Xi_{\bar{\theta}_{k(t)}}. \quad \text{(G.59)}$$

The $T^{3/4}$ rate is driven by the $1/\varepsilon$ coverage transfer for term (A). Using optimistic model selection ($\tilde{\theta}_t := \arg\max_{\theta\in\mathcal{C}_t} V_\theta^{\star,B}$) eliminates term (A) and recovers $\widetilde{O}(\sqrt{T})$ from term (C) alone, at the cost

of solving an optimistic planning problem (Liu et al., 2023b).

A constant ε gives the same $\widetilde{O}(T^{3/4})$ rate after optimization and may be preferable when T is unknown.

G.4. Offline Coverage, Structural Bias Bounds, Case Studies, and Ablations

The greedy-exploitation regret analysis under offline single-policy concentrability (Proposition G.4), the posterior-variance curvature bound on the QMDP leaf value bias (Proposition G.4), the closed-form LQG and contextual MDP case studies (Corollary G.4), and the empirical regret-decomposition audits, stress tests, and depth-d planner ablations appear in Appendices D.7–D.9 and E of Tan and Xu (2026).

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